

Algorithmic Non-commutative Class Field Theory

**Hecke operators, sandpile groups and operator algebras on trees and buildings,
with applications**

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Sources, code, data and figures:
<https://github.com/Ruqing1963/anccft>

Preface

This book collects, in a single logical order, the twenty papers of the series *Algorithmic non-commutative class field theory* (September 2026) and the eleven applied notes that grew out of them. The papers were written in the order in which the questions arose; the chapters are arranged in the order in which the mathematics is best learned. Each chapter is a paper, lightly edited: the “task specification” framing of the originals has been left in place where it records a correction, because those corrections are part of the record, and every chapter keeps its own verification battery, its own “Solved / open / impossible” remark and its own provenance paragraph. Nothing has been re-proved and nothing has been quietly upgraded.

Three conventions govern the whole book.

Every number is machine-verified. Each chapter ends with a table of keyed checks, each labelled *exact* (integer or symbolic arithmetic), *sampled* (a statistical test with its p -value) or *numeric* (floating point to a stated tolerance). The scripts that produce the tables, the data they produce, the figures, the L^AT_EX sources of every chapter and of this book, and the archived outputs are in the public repository

<https://github.com/Ruqing1963/anccft>

(`code/`, `data/`, `figures/`, `papers/`, `applications/`, `monograph/`); a frozen copy of each release is archived on Zenodo under the DOI given in the repository’s README. The chapters cite one another as chapters, never as “preprints”: nothing in this corpus has been published elsewhere, and the bibliography lists only the external literature.

Proof boundaries are stated. Every result is one of: proved here; quoted from a named source; verified computationally on named instances; or conjectural. The status paragraphs at the head of each chapter say which. Where an earlier chapter’s claim was later found to be wrong or imprecise, the correction is recorded in the later chapter and flagged in the closing chapter of this book.

The negative results are results. A large part of the programme consists of showing that the obvious next step does not exist: no non-negative matrix realizes the Ihara companion; no commutative Iwasawa algebra governs the congruence tower; no separable C^* -algebra has the pro-module as its K_0 ; no cyclic cocycle computes the L -invariant; no pairwise invariant of a chord diagram computes the arrangement group of a multi-copy repeat. These are stated as theorems with witnesses, and the closing chapter lists what remains genuinely open.

The concordance that follows the table of contents maps paper numbers to chapters. Inside a chapter, “Theorem 3.7” means Theorem $n.3.7$ of that chapter n ; a reference “[Ch. 1, Thm. 3.7]” from another chapter points to the same statement.

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Concordance

The chapters are the papers of the series, re-ordered by subject. A reference of the form “[Ch. n , Thm. $a.b$]” points to Theorem $n.a.b$ of this book; the papers’ own section-wise numbering is preserved inside each chapter. References to the applied notes (Phys 1, Bio 1–5, ...) are by the labels below.

paper	folder	chapter	title
I	Paper 1	2	the Langlands–thermodynamic bridge over Bruhat–Tits trees
II	Paper 2	3	the trace obstruction and the K-homological bulk
VI	Paper 6	4	the shadow algebra — a functorial Kirchberg realization of the critical group
III	Paper 3	5	the Iwahori–Hecke tower and the type III dissolution of the trace obstruction
IV	Paper 4	6	the limit module of abelian towers and the characteristic identity
IX	Paper 9	7	exceptional zeros, the graph $\mathcal{L}$ -invariant, and the operator-algebraic Mazur–Tate–Teitelbaum formula
XIV	Paper 14	8	the graph $\mathcal{L}$ -invariant as spectral action, Bloch effective mass and Albanese length, and what the cyclic Chern character sees
V	Paper 5	9	the degeneracy structure of the Iwahori tower and the Euler correction
VII	Paper 7	10	the snake factorization of the transfer and the evaluated characteristic form
XIII	Paper 13	11	universal localization of the degeneracy algebra, the K_1 -characteristic class, and the Eisenstein boundary
VIII	Paper 8	12	the tower-level Hecke–Kirchberg limit and the non-commutative Iwasawa K -module
XVI	Paper 16	13	the out-covering correspondence of the head map, its K_0 -index η_* , and the KK -category of the Iwahori tower
XVII	Paper 17	14	the tail norm at composite q , and the splitting of the limit K -module
XIX	Paper 19	15	the odd cyclic class that does not exist, and what the exceptional zero sees instead

paper	folder	chapter	title
X	Paper 10	16	$\tilde{A}_2$ complexes, higher simplicial Laplacians, and the fate of the trace obstruction in rank two
XI	Paper 11	17	the PGL_3 positive Hecke-dynamical system: shift equivalence over $\mathbb{Z}$ and the geodesic edge flow
XII	Paper 12	18	$\tilde{A}_2$ towers — no abelian sector, the covering norm theorem, and the two digraph towers
XV	Paper 15	19	the parahoric factorization of the geodesic edge flow, the unitarity of the remainder, and the chamber companion
XVIII	Paper 18	20	the unitary remainder in rank d
XX	Paper 20	21	the invariant subspace in rank d , and one flag count that produces it
Phys1	Phys 1	22	Effective mass from spanning trees, and the absence of Bloch phases on rank-two hyperbolic lattices
Holo1	Holo 1	23	What the boundary algebra of p -adic AdS/CFT does not see
Net1	Net 1	24	The sign of λ : an algebraic certificate for how a graph tower is generated
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Bio1	Bio 1	26	Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs
Bio2	Bio 2	27	The sandpile group as a coordinate system on genome reconstructions
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Bio6	Bio 6	31	Double strands without a new lemma: the reverse-complement double cover of a de Bruijn graph
Bio7	Bio 7	32	The reverse-complement quotient is a signed graph: what Zaslavsky's theory gives, and what it cannot

Chapter 1

Introduction: one group, two channels, and what towers do to it

1.1 The object

Everything in this book starts from a finite $(q + 1)$ -regular graph Y — in the arithmetic cases a quotient $\Gamma \backslash X$ of the Bruhat–Tits tree X of PGL_2 over a non-archimedean local field with residue field of order q — and from three integer matrices attached to it: the adjacency (Hecke) operator T_p , the Laplacian $\Delta_p = (q + 1)I - T_p$, and the non-backtracking (Hashimoto) matrix A^{nb} on directed edges. The finite abelian group

$$\mathrm{Jac}(Y) = \mathrm{Tor} \, \mathrm{coker} \, \Delta_p,$$

the *sandpile* or *critical* group, has order $\kappa(Y)$, the number of spanning trees, and its structure — not just its order — is read off from the Smith normal form of Δ_p . On an arithmetic quotient, $\kappa(Y) = \frac{1}{|V|} \prod_f (p + 1 - a_p(f))$ is a product of inverse local L -values over the eigenforms f of level Y : Frobenius traces precipitate as torsion. This single identity (Chapter 2, Theorem 1.3.5) is the seed of the programme; the word “algorithmic” in the title refers to the fact that the group, and everything built on it, is computed exactly, by integer elimination, on every instance we discuss.

1.2 The two channels

There are two natural non-commutative algebras over Y . The *boundary channel* is the crossed product $C(\partial X) \rtimes \Gamma$, equivalently the Cuntz–Krieger algebra of A^{nb} ; it carries a Satake-twisted KMS dynamics whose partition function is exactly the local L -factor. The *Laplacian channel* is the $\mathbb{Z}[u^{\pm 1}]$ -module $\mathrm{coker}(1 - uT_p + qu^2)$ whose Fitting ideal is the Ihara zeta function and whose specialization at $u = 1$ has torsion $\mathrm{Jac}(Y)$.

The first principle of the book is that *these two channels do not see the same thing*. The boundary algebra is arithmetically blind: its K -theory is $\mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$ with r the rank of Γ , whatever the Hecke data (Chapter 2, Theorem 1.3.7); and its KMS *states* see only the Perron branch of the spectrum, so tempered Satake parameters survive only as functionals (Theorem 1.2.5, the weight-gap theorem 1.2.11). The Laplacian channel sees the arithmetic but is a module, not an algebra. The obvious repair — realize the Ihara companion matrix $C = \begin{pmatrix} T_p & -qI \\ I & 0 \end{pmatrix}$ by a non-negative matrix, hence by a Cuntz–Krieger algebra — is impossible: $\mathrm{Tr} \, C^2 = -n(q - 1) < 0$, and shift equivalence preserves the nonzero spectrum, so no dilation of any size works (Chapter 3).

What *can* be done is to realize the group $\mathbb{Z} \oplus \text{Jac}(Y)$ as K_0 of a Kirchberg algebra by a $2n \times 2n$ non-negative “shadow” matrix (Chapter 4); this shares only Bowen–Franks data, not spectrum, with the pencil, which is why it evades the obstruction. In the projective limit the obstruction dissolves into a type $\text{III}_{1/q}$ factor with no trace, but so does K_0 (Chapter 5).

1.3 Towers

The second principle is that new content appears only along towers. For a finite graph the Hecke–Dirac Fredholm module is linear algebra in $M_n(\mathbb{Z})$; along a tower $Y_0 \leftarrow Y_1 \leftarrow \cdots$ the groups $\text{Jac}(Y_k)$ grow, and the growth law is where the arithmetic lives.

Abelian towers (Part II, Chapters 6, 7, 8). For a $\mathbb{Z}/\ell^k$ voltage tower the inverse limit of the Laplacian cokernels is $\Lambda^n/D(1+T)\Lambda^n$, $\Lambda = \mathbb{Z}_\ell[[T]]$, with exact control and characteristic ideal generated by $\Theta(T) = \det D(1+T)$; the Iwasawa invariants are read off the Newton polygon. Θ has an exceptional zero of order exactly two at the trivial character, and its second derivative is $-\kappa(Y_0)\|h_\alpha\|^2$, where h_α is the harmonic representative of the voltage class: a graph Mazur–Tate–Teitelbaum formula whose L -invariant $-\|h_\alpha\|^2$ is a rational number. This L -invariant is, at once, the effective mass of the bottom Bloch band of the $\mathbb{Z}$ -cover, the diffusion constant of the accumulated voltage under simple random walk, and the Albanese length of the class; it is *not* a cyclic-cohomology index, and Chapter 8 proves that no such index exists on the commutative triple.

The Iwahori tower (Chapters 9, 10, 11). The tower of iterated line digraphs (non-backtracking paths of length k) has no deck group. Its sandpile orders obey $\text{ord}_p |\text{Jac}(Y_k)| = \mu p^k - k + \nu_0$ with $\lambda = -1$, which no $\mathbb{Z}_p[[T]]$ -module can produce; the $-k$ is an accumulated Euler characteristic of a two-term degeneracy complex. The tower is governed instead by two degeneracy maps τ (tail) and η (head) whose incidence calculus $\tau_*\eta^* = A_k$, $\eta^*\tau_* = A_{k+1}$, $\tau_*\tau^* = qI$ replaces the group ring. The upper Hecke operator factors through the tail norm and therefore kills its kernel, which makes the “arc determinant” exactly $q^{n_k(q-1)}$ and turns the growth law into a snake-lemma identity. A universal (Cohn) localization of the degeneracy algebra carries a K_1 -class whose level evaluations are $\pm n_k \kappa(Y_k)$ and whose boundary is a non-split extension of $\text{Jac}(Y_k)$ by $\mathbb{Z}/n_k$; the analytic side of a “main conjecture” for this class is the central open problem of the book.

Operator algebras of the tower (Part III). The tail map is an in-covering of shadow graphs and induces unital embeddings $\mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$ of the level Kirchberg algebras, whose inductive limit is a Kirchberg algebra with $K_1 \cong \mathbb{Z}$ and $\text{Tor } K_0$ the Pontryagin dual of the sandpile pro-module. For $q = p$ prime the kernel of the tail norm on $\text{Jac}(Y_{k+1})$ is exactly the p -torsion, so the p -part of the pro-module is torsion-free of infinite rank — a third certificate that no commutative Iwasawa mechanism is at work. The head map is an out-covering and induces no homomorphism, but a proper C^* -correspondence whose K_0 -index is η_* ; the two Kasparov round trips are the Hecke operators.

1.4 Rank two

Part IV replaces the tree by $\tilde{A}_2$ buildings. The cubic Hecke pencil $P(u) = I - uA_1 + qu^2A_2 - q^3u^3$ carries the arithmetic, but the $\mathbb{Z}/3$ type grading kills the trace obstruction of rank one, and the realization problem becomes an instance of the (open) weak Generalized Spectral Conjecture over $\mathbb{Z}$. Thick quotients have $b_1 = 0$, so there are no abelian towers and no rank-two Iwasawa theory: the theory is non-commutative from the start. The geodesic edge flow L_E — the rank-two

Hashimoto matrix — carries the whole pencil through an explicit integer intertwiner, and its complement is $\sqrt{q}$ times an orthogonal matrix, so the “parahoric” zeros of the edge zeta lie on the critical circle for *every* complex, Ramanujan or not; the Ramanujan property lives entirely in the vertex pencil (Chapter 19).

1.5 Applications

Part V applies the sandpile/tower machinery outside number theory: the effective-mass reading of the L -invariant becomes a statement about tree lattices in condensed matter; the boundary-blindness theorem becomes a statement about what the boundary algebra of p -adic AdS/CFT cannot see; the sign of the Iwasawa λ certifies whether a graph tower is generated by a normal subgroup; the trace obstruction gives power-sum lower bounds on hidden eigenvalues in positive realizations; and, most substantially, the critical group of a de Bruijn graph is shown to measure genome assembly ambiguity beyond branch statistics, to split exactly along repeats, and — for two-copy repeats — to be the critical group of a one-vertex map, with a real phage genome as the running example.

1.6 How to read the book

Chapter 2 is required for everything. Part II needs Chapter 3 §5 (the towers) and Chapter 5 §2 (the growth law). Part III needs Chapters 9–10 and 4. Part IV needs Chapter 3 for the obstruction and Chapter 4 for the shadow construction, and is otherwise self-contained. The applied chapters need only the definitions of Chapter 2, §3 and, for the biology, nothing beyond the sandpile group of a digraph. The closing chapter collects every “solved / open / impossible” verdict in one place, with the corrections that later chapters made to earlier ones.

Part I

The bridge over the tree: two channels and one obstruction

Chapter 2

Algorithmic non-commutative class field theory, I: the Langlands–thermodynamic bridge over Bruhat–Tits trees

Summary. We open the non-abelian, algorithmic half of the programme: the localized Bost–Connes formalism is transported from the one-dimensional valuation space to the Bruhat–Tits tree of GL_2 over a non-archimedean local field, and the arithmetic input is upgraded from Frobenius classes in an abelian Galois group to Satake parameters of unramified automorphic representations.

Section 2.1 constructs the state space Ω (the space of lattice chains, equivalently the tree boundary $\partial X \cong \mathbb{P}^1(F)$), the truncated Hecke groupoids $\mathcal{G}^{(N)}$ and their limit $\mathcal{G} = \bigcup_N \mathcal{G}^{(N)}$, and the tail equivalence relation $\mathcal{R}$ with its Hecke-degree cocycle; the groupoid is étale, amenable and essentially principal, and its algebra is a Cuntz algebra locally and a Cuntz–Krieger algebra $\mathcal{O}_{A_Y^{\mathrm{nb}}} \simeq C(\partial X) \rtimes \Gamma$ globally.

Section 2.2 inserts the Satake parameters into the Radon–Nikodym cocycle. The partition function of the twisted system is *exactly* the local L -factor, $Z_p(\beta) = L_p(\beta, \pi_p)$, an identity of the Hecke recursion, and the abscissa of convergence $\beta_c(\pi_p) = \log_q \max_i |\alpha_{p,i}|$ is the largest pole of L_p . Two structural facts then follow, and they cut in opposite directions. Positivity (Perron–Frobenius) shows that KMS *states* see only the Perron branch of the unramified spectrum, so cuspidal Satake data survive only as KMS *functionals*; and the Kakutani dichotomy for the associated conformal measures has obstruction series $\Xi_\beta(\pi, \pi'; S) = \sum_{p \in S} p^{-\beta} \|\cdot\|^2$ governed by Rankin–Selberg, giving a measure-theoretic form of strong multiplicity one. Combining them yields the *weight gap theorem*: equilibrium requires $\beta > \beta_c(S) + w/2$ while the obstruction is confined to $\beta \leq \beta_c(S)$, where w is the motivic weight; the two ranges meet if and only if $w = 0$. The abelian theory is the case $w = 0$; Paper XXI’s universal gap of width 1 is the Eisenstein case $w = 2$; tempered weight two gives $w = 1$ and a residual gap of exactly $\frac{1}{2}$, the Ramanujan exponent.

Section 2.3 is the algorithmic bridge, and it is where the higher-weight arithmetic survives. Pimsner’s exact sequence for the Hecke correspondence, the non-commutative replacement of the equivariant Pimsner–Voiculescu sequence, identifies K_* with the kernel and cokernel of $1 - uT_p + qu^2$ over $\mathbb{Z}[u^{\pm 1}]$ — the *Ihara module*, whose Fitting ideal is the Ihara zeta function by Bass’s identity. Specializing $u = 1$ gives the Hecke–Laplacian $\Delta_p = (q + 1)I - T_p$ and the structure theorem

$$K_0 \cong \mathbb{Z} \oplus \mathrm{Jac}(Y), \quad K_1 \cong \mathbb{Z}, \quad |\mathrm{Jac}(Y)| = \frac{1}{|V|} \prod_f (p + 1 - a_p(f)) = \frac{1}{|V|} \prod_f \#E_f(\mathbb{F}_p),$$

the torsion being computed exactly, as a group and not merely as an order, by the Smith normal form of Δ_p : Frobenius traces precipitate as K -theoretic torsion. By contrast the boundary crossed product is arithmetically blind, $K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$ with r the rank of Γ , a theorem proved here by collapsing the reversal relation $[f] + [\bar{f}] = c_{o(f)}$ to a single class of order $|E_Y| - |V_Y|$. Eleven independent machine computations confirm both channels. Along Benjamini–Schramm convergent families the torsion density converges exactly to Lyons’ tree entropy, $\frac{1}{n} \log |\text{Tors } K_0| \rightarrow q \log q - \frac{q-1}{2} \log(q^2 - 1)$, and the Ramanujan bound $2 \log(1 + \sqrt{q})$, while unconditional, is strict and exceeds it by $2q^{-1/2} + O(q^{-1})$ as $q \rightarrow \infty$ — a Taylor expansion, not a fit — because it replaces a Kesten–McKay integral by its spectral edge. The Laplacian channel is realised as a $\mathbb{Z}[u^{\pm 1}]$ -module and as the index of a Hecke–Dirac operator, not as a C^* -algebra: the class $T_p - qI$ is a legitimate KK -class but not a correspondence, and closing that gap is the paper’s main open problem.

Introduction, and the status of each statement

The thirty-nine papers of *Localized Bost–Connes systems* settle the GL_1 problem: the KMS_β simplex of $\mathcal{A}_{K,S}$ is $\text{Prob}(G_S/\Xi_\beta^\perp)$, the obstruction group Ξ_β is cut out by a Dirichlet series in the Frobenius classes, the dichotomy underneath is Kakutani’s, and the integral invariants are Smith normal forms of $\text{Np} \cdot 1 - \Lambda^r \Pi$. Every ingredient of that theory has a GL_2 counterpart, and the counterparts are not formal analogies: valuations become lattice chains, the profinite completion becomes the tree boundary, Frobenius classes become Satake parameters, Dirichlet series become automorphic L -functions, and the Smith normal form of a norm-minus-multiplication matrix becomes the Smith normal form of a Hecke–Laplacian. This paper writes the dictionary and proves the entries that can be proved.

It is written against a negative result of the earlier series, and we state at once how it stands with respect to it. Paper XXI of that series proves that the untwisted GL_2 system has a *universal gap of width one*: the automorphic obstruction can be non-trivial only for $\beta \leq \beta_c(S)$, Gibbs states exist only for $\beta > 1 + \beta_c(S)$, and no temperature does both. Nothing below contradicts that. What changes here is *where* the automorphic parameter is inserted — into the dynamics, not into the state — and the effect of the change is quantitative and exact: the gap is $w/2$, where w is the motivic weight of the representation (Theorem 2.2.11). Paper XXI is the case $w = 2$; the abelian theory of the monograph is the case $w = 0$, where the gap closes and the classification succeeds; tempered weight two is $w = 1$ and the residual gap is the Ramanujan exponent $\frac{1}{2}$. The thermodynamic channel therefore classifies exactly the weight-zero, that is Artin, part of the Langlands correspondence. That is a sharp statement about what thermodynamics can and cannot do, and it is the reason Section 2.3 exists: K -theory carries no positivity constraint and no temperature, and it does see the higher-weight data.

Status. We label every numbered statement in one of three ways. *Theorem/Proposition*: proved here or an immediate assembly of quoted results, with the assembly given. *Verified*: an integral statement established by exact machine computation on a stated finite list of cases, with a proof sketch but no proof; these are marked [V] and the verification table is Table 2.1; in the present version no numbered statement carries this label, Theorem 2.3.7 having been promoted to a proof, and Table 2.1 now serves as independent confirmation rather than as evidence. *Conjecture*: neither. No statement below is asserted at a higher level than it is entitled to. In particular Theorem 2.2.4 is a theorem, its extension to ramified and to GL_n data is Conjecture 2.2.12, and the identification of the full extremal simplex with the polar divisor of an L -function — the form

in which such a correspondence is usually stated informally — is *false* as stated, for the reason isolated in Theorem 2.2.5.

2.1 The geometric state space over Bruhat–Tits trees

2.1.1 The local field and its tree

Throughout, F is a non-archimedean local field with ring of integers $\mathcal{O}$, maximal ideal $\mathfrak{m} = (\varpi)$, uniformiser ϖ , normalised valuation $v: F^\times \rightarrow \mathbb{Z}$, and residue field $k_F = \mathcal{O}/\mathfrak{m}$ of cardinality q . In the arithmetic applications $F = \mathbb{Q}_p$ and $q = p$. Write $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(\mathcal{O})$, $Z \cong F^\times$ for the centre.

Definition 2.1.1 (Lattices and the tree). An $\mathcal{O}$ -lattice in $V = F^2$ is a free $\mathcal{O}$ -submodule $L \subset V$ of rank 2. Two lattices are *homothetic*, $L \sim L'$, if $L' = cL$ for some $c \in F^\times$. The Bruhat–Tits tree $X = X_F$ of $\mathrm{PGL}_2(F)$ has

$$X^{(0)} = \{\text{lattices}\} / \sim, \quad [L] \text{ adjacent to } [L'] \iff \exists L \in [L], L' \in [L'] \text{ with } \varpi L \subsetneq L' \subsetneq L.$$

X is a tree, it is $(q+1)$ -regular, G acts on it by $g \cdot [L] = [gL]$ with ZK the stabiliser of the standard vertex $x_0 = [\mathcal{O}^2]$, and the vertex set is $X^{(0)} \cong G/ZK$.

Definition 2.1.2 (Boundary). A *ray* from x_0 is a sequence $\xi = (x_0, x_1, x_2, \dots)$ of vertices with $x_i \sim x_{i+1}$ and $x_{i+1} \neq x_{i-1}$ (*non-backtracking*). The *boundary* ∂X is the set of rays from x_0 , topologised by the cylinder sets $C(x_0, \dots, x_n)$. Then ∂X is a compact totally disconnected space without isolated points — a Cantor set — and the map sending a ray to the corresponding line yields a G -equivariant homeomorphism

$$\partial X \xrightarrow{\cong} \mathbb{P}^1(F) = \mathbb{P}^1(\mathcal{O}) / \sim.$$

The cylinder $C(x_0, \dots, x_n)$ has $q+1$ children at level 1 and exactly q children at every later level.

The two branching numbers $q+1$ and q are the whole geometric content of the section, and they are the reason the boundary and not the lattice space is the correct state space; see Remark 2.1.4.

Definition 2.1.3 (Chain space). Let

$$\tilde{\Omega} := \left\{ (L_n)_{n \geq 0} : L_0 = \mathcal{O}^2, L_{n+1} \subsetneq L_n, [L_n : L_{n+1}] = q \ \forall n \right\},$$

the space of *descending lattice chains of step one*, with the inverse-limit topology. Since the sublattices of index q in a lattice L are in bijection with $\mathbb{P}^1(k_F)$, which has $q+1$ elements, $\tilde{\Omega}$ is homeomorphic to the boundary of the rooted $(q+1)$ -ary tree, hence a Cantor set, and $[L_0], [L_1], [L_2], \dots$ is a walk in X issuing from x_0 . The subspace

$$\Omega := \{(L_n) \in \tilde{\Omega} : [L_{n+1}] \neq [L_{n-1}] \ \forall n \geq 1\}$$

of *geodesic* chains is closed and $\Omega \cong \partial X \cong \mathbb{P}^1(F)$ by Definition 2.1.2. We call $\tilde{\Omega}$ the *Toeplitz* (or full) state space and Ω the *boundary* state space.

Remark 2.1.4. $\tilde{\Omega}$ is the GL_2 analogue of the space Y_S of the abelian theory: its level- n set is the set of sublattices of index q^n in $\mathcal{O}^2$, of cardinality $\sigma_1(q^n) = 1 + q + \dots + q^n$, and the associated Dirichlet series is the Solomon zeta $\zeta_{M_2(\mathcal{O})}(\beta) = (1 - q^{-\beta})^{-1}(1 - q^{1-\beta})^{-1}$ of Paper XXI. The passage $\tilde{\Omega} \rightsquigarrow \Omega$ removes exactly the excess $d_p = q+1 > q$ that Paper XXI, Theorem 1.2 identifies as the source of its off-by-one: on Ω the branching is q per step and matches the norm.

2.1.2 The Hecke monoid and the truncated Hecke groupoids

Let $\mathcal{H}(G, K) = C_c(K \backslash G / K)$ be the spherical Hecke algebra with convolution, and

$$T := [K \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix} K], \quad R^{\pm 1} := [K \begin{pmatrix} \varpi^{\pm 1} & 0 \\ 0 & \varpi^{\pm 1} \end{pmatrix} K], \quad T_n := [K \begin{pmatrix} \varpi^n & 0 \\ 0 & 1 \end{pmatrix} K] \quad (n \geq 0),$$

so that $\mathcal{H}(G, K) = \mathbb{C}[T, R^{\pm 1}]$ and the Satake isomorphism is $\text{Sat}: \mathcal{H}(G, K) \xrightarrow{\cong} \mathbb{C}[\alpha_1^{\pm 1}, \alpha_2^{\pm 1}]^{S_2}$ with $\text{Sat}(T) = q^{1/2}(\alpha_1 + \alpha_2)$ and $\text{Sat}(R) = \alpha_1 \alpha_2$. Acting on functions on $X^{(0)}$, T is the adjacency (nearest-neighbour) operator and R acts trivially.

Lemma 2.1.5 (Hecke recursion). *On X one has $T_1 = T$, $TT_1 = T_2 + (q+1)T_0$ and, for $n \geq 2$,*

$$T T_n = T_{n+1} + q T_{n-1}, \quad \text{equivalently} \quad \sum_{n \geq 0} T_n u^n = (1 - Tu + qu^2)^{-1}$$

as an identity of formal power series with coefficients in $\mathcal{H}(G, K)$ normalised by $\text{Sat}(R) = q$. Consequently, for a character λ of $\mathcal{H}(G, K)$ with Satake parameters (α_1, α_2) , $\alpha_1 \alpha_2 = q$,

$$\sum_{n \geq 0} \lambda(T_n) u^n = \frac{1}{(1 - \alpha_1 u)(1 - \alpha_2 u)}. \quad (2.1.1)$$

The pencil $1 - Tu + qu^2$ of Lemma 2.1.5 is the single object that reappears in every section below: in Section 2.2 as the inverse of a partition function, in Section 2.3 as the presentation matrix of a K -group.

Definition 2.1.6 (Hecke monoid and degree). Let $S_{\varpi} := (M_2(\mathcal{O}) \cap G) / \mathcal{O}^{\times}$, a monoid acting on lattices by $L \mapsto gL$, and let $\deg: S_{\varpi} \rightarrow \mathbb{Z}_{\geq 0}$, $\deg(g) = v(\det g)$. The K -double cosets of S_{ϖ} are indexed by elementary divisors, and $\{\deg = n\}$ is the union of the double cosets of $T_a R^b$ with $a + 2b = n$.

Definition 2.1.7 (Truncated Hecke groupoids). Let $\Gamma \leq \text{PGL}_2(F)$ be a discrete, torsion-free, cocompact subgroup; such Γ exist and are free of finite rank, and $Y := \Gamma \backslash X$ is a finite $(q+1)$ -regular graph. Write E_Y for its set of oriented edges, $\bar{e}$ for the reversal of e , and

$$A_{ef}^{\text{nb}} = \begin{cases} 1, & t(e) = o(f) \text{ and } f \neq \bar{e}, \\ 0, & \text{otherwise,} \end{cases} \quad e, f \in E_Y,$$

for the *non-backtracking* (edge) matrix. Let $\Sigma_Y := \{\omega \in E_Y^{\mathbb{N}} : A_{\omega_i \omega_{i+1}}^{\text{nb}} = 1\}$ with shift σ . For $N \in \mathbb{N}$ set

$$\mathcal{G}^{(N)} := \{(\omega, i - j, \omega') \in \Sigma_Y \times \mathbb{Z} \times \Sigma_Y : 0 \leq i, j \leq N, \sigma^i \omega = \sigma^j \omega'\}, \quad (2.1.2)$$

and $\mathcal{G}_{\Gamma} := \bigcup_{N \geq 0} \mathcal{G}^{(N)}$, the *Hecke groupoid*, with the composition $(\omega, k, \omega')(\omega', k', \omega'') = (\omega, k + k', \omega'')$, the topology generated by the sets $Z(U, i, j, V) = \{(\omega, i - j, \omega') : \omega \in U, \omega' \in V, \sigma^i \omega = \sigma^j \omega'\}$ for U, V cylinders, and the *Hecke degree cocycle*

$$c: \mathcal{G}_{\Gamma} \longrightarrow \mathbb{Z}, \quad c(\omega, k, \omega') = k.$$

The same formula with Σ_Y replaced by $\tilde{\Omega}$ and σ by the one-sided shift on the full $(q+1)$ -ary tree defines the *Toeplitz Hecke groupoid* $\tilde{\mathcal{G}}$.

The truncation index N is the number of Hecke moves allowed; $\mathcal{G}^{(N)}$ is an open subgroupoid, $\mathcal{G}^{(N)} \subset \mathcal{G}^{(N+1)}$, and $C^*(\mathcal{G}_\Gamma) = \varinjlim_N C^*(\mathcal{G}^{(N)})$. This inductive limit is what carries the thermodynamic limit of Section 2.2: a measure class on Ω is a compatible family of quasi-invariant classes for the finite-level relations $\mathcal{R}^{(N)}$ below.

Definition 2.1.8 (Tail equivalence). The *tail (Hecke) equivalence relation* is the image of $\mathcal{G}_\Gamma$ under (r, s) ,

$$\mathcal{R} := \{(\omega, \omega') \in \Omega \times \Omega : \exists i, j \geq 0, \sigma^i \omega = \sigma^j \omega'\},$$

with the finite-level subrelations $\mathcal{R}^{(N)}$ obtained by restricting $i, j \leq N$, and the *degree-zero* subrelation $\mathcal{R}_0 := \{(\omega, \omega') : \exists i, \sigma^i \omega = \sigma^i \omega'\}$. A probability measure μ on Ω is β -*quasi-invariant* for $\mathcal{R}$ if the measure class is preserved and the Radon–Nikodym cocycle of $\mathcal{G}_\Gamma$ is $e^{-\beta c}$; $\mathcal{R}_0$ is a hyperfinite (AF) subrelation, and $\mathcal{R}/\mathcal{R}_0 \cong \mathbb{Z}$ through c .

Proposition 2.1.9. $\mathcal{G}_\Gamma$ is a second countable, locally compact, Hausdorff, étale groupoid with unit space Ω ; it is amenable, minimal and essentially principal, and

- (i) $\mathcal{G}_\Gamma$ is canonically isomorphic to the transformation groupoid $\partial X \rtimes \Gamma$ under $\Omega \cong \partial X$, the Hecke degree c corresponding to the Busemann (horocycle) cocycle of Γ acting on ∂X ;
- (ii) $C^*(\mathcal{G}_\Gamma) = C_r^*(\mathcal{G}_\Gamma)$ is the Cuntz–Krieger algebra $\mathcal{O}_{A_Y^{\text{pb}}}$, simple, purely infinite, nuclear, and Morita equivalent to $C(\partial X) \rtimes \Gamma$;
- (iii) in the rooted local model, $C^*(\tilde{\mathcal{G}}) \cong \mathcal{O}_{q+1}$ and its boundary (geodesic) reduction is stably isomorphic to $\mathcal{O}_q$, so that

$$K_0(C^*(\tilde{\mathcal{G}})|_\Omega) \cong \mathbb{Z}/(q-1), \quad K_1(C^*(\tilde{\mathcal{G}})|_\Omega) = 0.$$

Remark 2.1.10. Part (iii) is the exact GL_2 analogue of the unit-class torsion $\mathbb{Z}/(\text{Np}^h - 1)$ of Paper XXXVIII: locally the boundary quotient contributes $\mathbb{Z}/(q-1)$ and nothing else. All arithmetic beyond the residue cardinality is therefore invisible at a single place, and must be sought either in the global assembly (Section 2.2) or in the quotient graph Y (Section 2.3). This is the structural reason the GL_2 theory cannot be purely local, and it has no counterpart in the GL_1 theory, where the local boundary already carries the class group.

2.2 The thermodynamic–automorphic correspondence

2.2.1 Satake weights in the Radon–Nikodym cocycle

Fix an unramified irreducible admissible representation π_p of G with Satake parameters $(\alpha_{p,1}, \alpha_{p,2})$ normalised arithmetically, $\alpha_{p,1}\alpha_{p,2} = q^w$ with w the motivic weight ($w = 0$ for Artin parameters, $w = k - 1$ for weight- k forms; $w = 1$ in the classical weight-two case), and local L -factor

$$L_p(s, \pi_p) = \prod_{i=1,2} (1 - \alpha_{p,i} q^{-s})^{-1}.$$

Definition 2.2.1 (Satake potential and twisted dynamics). Let $h_\pi: X^{(0)} \rightarrow \mathbb{C}^\times$ be a Hecke eigenfunction, $Th_\pi = \lambda_\pi h_\pi$ with $\lambda_\pi = \alpha_{p,1} + \alpha_{p,2}$, normalised by $h_\pi(x_0) = 1$. Define the *Satake potential* on oriented edges and the associated *Ruelle–Hecke transfer operator* on $C(\Omega)$ by

$$\varphi_\pi(e) := \log \frac{h_\pi(t(e))}{h_\pi(o(e))}, \quad (\mathcal{L}_{\beta, \pi} f)(\omega) := \sum_{\sigma(\eta) = \omega} q^{-\beta c(\eta)} e^{\varphi_\pi(\eta_1)} f(\eta).$$

The *twisted dynamics* $\sigma^{(\pi)}$ on $C^*(\mathcal{G}_\Gamma)$ is the one-parameter group determined on $C_c(\mathcal{G}_\Gamma)$ by

$$\sigma_t^{(\pi)}(f)(\gamma) = e^{it(\beta\text{-independent})c_\pi(\gamma)} f(\gamma), \quad c_\pi(\gamma) := c(\gamma) \log q - \varphi_\pi(\gamma),$$

so that the Radon–Nikodym cocycle of a KMS_β state is

$$\frac{d\mu \circ \gamma}{d\mu}(\omega) = e^{-\beta c_\pi(\gamma)} = q^{-\beta c(\gamma)} \left(\frac{h_\pi(t\gamma)}{h_\pi(o\gamma)} \right)^\beta. \quad (2.2.1)$$

Equation (2.2.1) is the precise sense in which the Satake parameter enters the Radon–Nikodym derivative: it is the Doob h -transform of the geometric cocycle by the Hecke eigenfunction. The untwisted system of Paper XXI is $h_\pi \equiv 1$.

2.2.2 The partition function is the local L -factor

Theorem 2.2.2 (Partition function). *Let $Z_p(\beta; \pi) := \sum_{n \geq 0} \lambda_\pi(T_n) q^{-n\beta}$ be the partition function of the truncated tower $(\mathcal{G}^{(N)})_N$ with the dynamics of Definition 2.2.1. Then the series converges absolutely if and only if $\beta > \beta_c(\pi_p)$, where*

$$\beta_c(\pi_p) := \log_q \max_i |\alpha_{p,i}|,$$

and in that range

$$Z_p(\beta; \pi) = \prod_{i=1,2} (1 - \alpha_{p,i} q^{-\beta})^{-1} = L_p(\beta, \pi_p).$$

Moreover $\beta_c(\pi_p)$ is exactly the largest real part of a pole of $s \mapsto L_p(s, \pi_p)$, so the critical inverse temperature of the twisted system is the abscissa of convergence of the local L -factor. In particular $\beta_c = w/2$ when π_p is tempered (Ramanujan), and $\beta_c = w$ when π_p is the Eisenstein point $(\alpha_1, \alpha_2) = (1, q^w)$.

Proof. Substitute $u = q^{-\beta}$ in (2.1.1) of Lemma 2.1.5; the radius of convergence of the right-hand side is $\min_i |\alpha_{p,i}|^{-1}$. The poles of $L_p(s, \pi_p)$ are at $q^{-s} = \alpha_{p,i}^{-1}$, i.e. at $\Re s = \log_q |\alpha_{p,i}|$. $\square$

Remark 2.2.3. For the trivial (Eisenstein) parameter $w = 2$, $(\alpha_1, \alpha_2) = (1, q)$, one recovers $Z_p(\beta) = (1 - q^{-\beta})^{-1}(1 - q^{1-\beta})^{-1}$, the Solomon zeta of $M_2(\mathcal{O})$ and the local factor of $\zeta(\beta)\zeta(\beta-1)$: Theorem 2.2.2 contains the partition function of the Connes–Marcolli GL_2 -system, and of Paper XXI, as its $h_\pi \equiv 1$ specialisation.

2.2.3 Equilibrium states, and what they can see

Theorem 2.2.4 (KMS classification, local). *Let $Y = \Gamma \backslash X$ be connected and non-bipartite, and let $A_{\beta,\pi}^{\text{nb}}$ be the non-backtracking matrix weighted by $e^{-\beta c_\pi}$ as in (2.2.1). Then:*

- (i) $\text{KMS}_\beta(C^*(\mathcal{G}_\Gamma), \sigma^{(\pi)}) \neq \emptyset$ if and only if the spectral radius satisfies $\rho(A_{\beta,\pi}^{\text{nb}}) = 1$, i.e. if and only if the Bowen–Ruelle pressure equation $P(-\beta c_\pi) = 0$ holds;
- (ii) in that case the KMS_β simplex is a single point, the state associated with the Perron eigenvector, and the corresponding measure $\mu_{\beta,\pi}$ is the unique $e^{-\beta c_\pi}$ -conformal measure for $\mathcal{R}$;

- (iii) the associated von Neumann factor is of type $\text{III}_{q^{-\beta}}$ if $c_\pi(\mathcal{G}_\Gamma) \subseteq \mathbb{Z} \log q$, and of type III_1 if the closed group generated by $e^{-\beta c_\pi(\mathcal{G}_\Gamma)}$ is $\mathbb{R}_{>0}$;
- (iv) for the untwisted potential the equation in (i) reads $q^{1-\beta} = 1$, so the transition is at $\beta_c = 1$ on the boundary state space Ω , against $\beta_c = 2$ on the Toeplitz state space $\tilde{\Omega}$.

Part (iv) is the promised removal of Paper XXI’s off-by-one: it is bought by the passage from $\tilde{\Omega}$ to Ω , i.e. by the boundary quotient, and by nothing else. The next theorem says what the price is.

Theorem 2.2.5 (Perron obstruction: states cannot see cuspidal parameters). *Let $\sigma^{(\pi)}$ be as in Definition 2.2.1. A KMS_β state exists only if the eigenfunction h_π can be chosen strictly positive on $X^{(0)}$. On a $(q+1)$ -regular tree a positive T -eigenfunction with eigenvalue λ exists if and only if $|\lambda| \geq 2\sqrt{q}$. Consequently:*

- (i) if π_p is tempered ($|\lambda_\pi| < 2\sqrt{q}$, i.e. $|\alpha_{p,i}| = q^{w/2}$), the twisted system has no KMS_β state at any β ; the Satake data survive only as the KMS_β functional $\phi_{\beta,\pi}$ obtained by analytic continuation in λ , whose partition function is $L_p(\beta, \pi_p)$ by Theorem 2.2.2;
- (ii) the extremal KMS_β states of the untwisted system are parameterised by the Perron branch alone — the Eisenstein/complementary-series locus $\lambda \geq 2\sqrt{q}$ of the unramified spectrum — and their partition functions are the corresponding L -factors $L_p(\beta, 1 \boxplus |\cdot|^w)$;
- (iii) therefore the assertion “the extremal KMS_β simplex is parameterised by the poles of the automorphic L -function” is true on the Eisenstein locus and false on the cuspidal locus; the correct general statement replaces states by functionals, and the simplex by the polar divisor of a meromorphic family.

Remark 2.2.6 (Thermodynamic ghost states). We propose the name *thermodynamic ghost states* for the non-positive continuations $\phi_{\beta,\pi}$ of Theorem 2.2.5(i): they are $\sigma^{(\pi)}$ -KMS functionals with the correct modular property and the correct partition function $L_p(\beta, \pi_p)$, but they are not states, having no positive extension to the algebra and hence no GNS representation and no thermal equilibrium interpretation. The terminology is borrowed, with the usual caution, from the negative-norm states of gauge quantisation: as there, the ghosts are indispensable to the bookkeeping and absent from the physical spectrum. The physical content of Theorem 2.2.5 is then this. The metric geometry of the tree fixes a threshold, $2\sqrt{q}$, which is exactly the tempered bound; a Satake parameter satisfying the Ramanujan estimate lies *below* it, and a positive Hecke eigenfunction — the Gibbs weight of the would-be equilibrium — then simply does not exist on X . Cuspidality and thermal equilibrium are mutually exclusive on the Bruhat–Tits tree. The arithmetic of the cuspidal spectrum is therefore not suppressed but displaced: it survives as analytic data off the state space, and, as Section 2.3 shows, as integral data in K -theory, where no positivity is required.

Remark 2.2.7. Theorem 2.2.5 is the operator-algebraic shadow of a familiar fact: positivity of a state is a Perron–Frobenius condition, and Perron–Frobenius selects the trivial representation. It is the same phenomenon that makes the constant function the only positive ℓ^2 -eigenfunction on a finite graph, and the same one that in Paper XXVII forces a KMS state to be tracial on the coefficient algebra and thereby erases inner twists. Any programme that hopes to read cuspidal data off an equilibrium *state* must first defeat this, and we do not know how to.

2.2.4 The Kakutani obstruction and Rankin–Selberg

We now globalise. Let S be a set of rational primes, $\beta_c(S)$ the convergence exponent of $\sum_{p \in S} p^{-\beta}$, and for unramified $\pi = \otimes_p \pi_p$ let

$$\Omega_S := \prod_{p \in S} \Omega_p, \quad \mu_\beta^\pi := \bigotimes_{p \in S} \mu_{\beta, \pi_p},$$

the product of the local conformal measures of Theorem 2.2.4(ii), taken on the locus where they exist, that is on the Eisenstein and complementary-series range $|\lambda_p| \geq 2\sqrt{q}$, or of their h -transform representatives there.

Theorem 2.2.8 (Kakutani dichotomy for Satake data). *For any two such π, π' the measures μ_β^π and $\mu_\beta^{\pi'}$ are either equivalent or mutually singular, and they are equivalent if and only if*

$$\Xi_\beta(\pi, \pi'; S) := \sum_{p \in S} p^{-\beta} |\widehat{\lambda}_p(\pi) - \widehat{\lambda}_p(\pi')|^2 < \infty, \quad \widehat{\lambda}_p := \frac{\lambda_p}{2\sqrt{p}} = \cos \theta_p, \quad (2.2.2)$$

the equivalence of the Hellinger criterion with (2.2.2) holding with absolute implied constants depending only on the temperedness bound. The series (2.2.2) is the exact GL_2 analogue of the obstruction series $\sum_{\mathfrak{p}} \mathrm{Np}^{-\beta} |1 - \chi(\mathrm{Frob}_{\mathfrak{p}})|^2$ of the abelian theory, with the Frobenius character replaced by the normalised Satake trace.

Corollary 2.2.9 (Measure-theoretic strong multiplicity one). *Let π, π' be cuspidal automorphic representations of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$, unramified outside a finite set, and S the set of all but finitely many primes, so $\beta_c(S) = 1$. Expanding (2.2.2) into Rankin–Selberg logarithms,*

$$\Xi_\beta(\pi, \pi'; S) = \log L_S(\beta, \pi \times \widetilde{\pi}) + \log L_S(\beta, \pi' \times \widetilde{\pi}') - 2\Re \log L_S(\beta, \pi \times \widetilde{\pi}') + O(1),$$

so that, by the pole of $L(s, \pi \times \widetilde{\pi})$ at $s = 1$ and the regularity and non-vanishing there of $L(s, \pi \times \widetilde{\pi}')$ for $\pi \not\cong \pi'$ (Jacquet–Shalika), one has $\Xi_1(\pi, \pi'; S) = \infty$ and hence

$$\pi \not\cong \pi' \implies \mu_1^\pi \perp \mu_1^{\pi'}.$$

Distinct automorphic representations occupy mutually singular thermodynamic phases.

Remark 2.2.10. The Rankin–Selberg L -function appears here legitimately, and its appearance is not in conflict with Paper XXI, which shows that the *one-representation* obstruction $\Xi_\beta(\rho, S)$ is linear in $\mathrm{Tr} \rho(\mathrm{Frob}_{\mathfrak{p}})$ and therefore involves only the standard L -function, the Sym^2 -term being spurious. The quantity (2.2.2) is a Hellinger distance *between two* representations; it is genuinely quadratic, and the second moment of Satake traces is by definition Rankin–Selberg.

2.2.5 The weight gap

Theorem 2.2.11 (Weight gap). *Let π have motivic weight w at almost every place, i.e. $|\alpha_{p,i}| = p^{w/2}$ for tempered π_p . Then, for every set S of primes:*

$$\text{Gibbs states exist} \iff \beta > \beta_c(S) + \frac{w}{2}, \quad \Xi_\beta(\pi, \pi'; S) = \infty \text{ possible} \iff \beta \leq \beta_c(S).$$

The two ranges are disjoint with a gap of width exactly $w/2$, and they meet if and only if $w = 0$. Consequently:

- (i) $w = 0$ (Artin parameters, weight-one forms, the abelian case): the gap closes, and one recovers the classification of the localized Bost–Connes theory, namely $\mathrm{KMS}_\beta \cong \mathrm{Prob}(G_S/\Xi_\beta^\perp)$;
- (ii) $w = 1$ (tempered weight two): the residual gap is $\frac{1}{2}$, the Ramanujan exponent;
- (iii) $w = 2$ (the Eisenstein/untwisted GL_2 system): the gap is 1, which is Paper XXI’s universal off-by-one.

Proof sketch. By Theorem 2.2.2, the equilibrium range is the domain of absolute convergence of the Euler product of the local factors $L_p(\beta, \pi_p)$, $p \in S$, whose abscissa is $\beta_c(S) + \beta_c(\pi_p)$; in the tempered case this equals $\beta_c(S) + w/2$. For the obstruction range, note that $|\hat{\lambda}_p| \leq 1$, so the series (2.2.2) is bounded above by $4 \sum_{p \in S} p^{-\beta}$ and is therefore finite for every $\beta > \beta_c(S)$. Disjointness and the width follow $\square$

Theorem 2.2.11 is the main negative–positive result of the section. It says that the thermodynamic channel is exactly a *weight-zero* instrument: it classifies the Artin part of the Langlands correspondence and is blind, by an amount equal to half the motivic weight, to everything above it. This is a sharpening, not a refutation, of Paper XXI: the twist moves the equilibrium threshold from $\beta_c(S) + 1$ down to $\beta_c(S) + w/2$, and for weight-one forms it closes the gap altogether.

Conjecture 2.2.12 (GL_n and ramified data). *Let F be local, $G_n = \mathrm{GL}_n(F)$, X_n the Bruhat–Tits building of $\mathrm{PGL}_n(F)$, and π_p unramified with Satake parameters $(\alpha_{p,1}, \dots, \alpha_{p,n})$. Then the partition function of the Hecke groupoid of the chamber boundary of X_n , twisted as in Definition 2.2.1, equals $L_p(\beta, \pi_p) = \prod_i (1 - \alpha_{p,i} q^{-\beta})^{-1}$, the critical inverse temperature is $\max_i \log_q |\alpha_{p,i}|$, the weight gap theorem holds verbatim with w the motivic weight, and for ramified π_p the same statements hold with Ω replaced by the boundary of the building with level structure and L_p by the ramified local factor. For $n = 2$ and unramified data this is Theorems 2.2.2–2.2.11.*

2.3 Algorithmic K -theory via Hecke operators

Throughout this section Γ is torsion-free and cocompact, $Y = \Gamma \backslash X$ is a finite connected $(q+1)$ -regular graph with $n = |V_Y|$ vertices and $m = |E_Y|$ unoriented edges, so that

$$r := \mathrm{rank} \Gamma = m - n + 1, \quad m = \frac{(q+1)n}{2}, \quad r - 1 = m - n = \frac{(q-1)n}{2}.$$

$T_p \in M_n(\mathbb{Z})$ denotes the Hecke (adjacency) operator of Y and $\Delta_p := (q+1)I_n - T_p$ its *Hecke–Laplacian*. By Ihara’s theorem and the Eichler–Shimura and Jacquet–Langlands correspondences, when Γ is the unit group of an Eichler order the spectrum of T_p is

$$\mathrm{Spec} T_p = \{q+1\} \cup \{a_p(f) : f \text{ a weight-two newform of the relevant level}\},$$

so the entries of Δ_p are Frobenius traces in disguise. This section turns that disguise into an algorithm.

2.3.1 The Hecke correspondence and the Pimsner sequence

Definition 2.3.1 (Hecke correspondence). Let $A := C(V_Y) \cong \mathbb{C}^n$ and let $\mathcal{E}_p$ be the A – A correspondence with underlying space $C(E_Y)$, right action $(\xi \cdot a)(e) = \xi(e)a(t(e))$, left action

$(a \cdot \xi)(e) = a(o(e))\xi(e)$, and inner product $\langle \xi, \eta \rangle(v) = \sum_{t(e)=v} \overline{\xi(e)}\eta(e)$. Then $\mathcal{E}_p$ is finitely generated projective, full, with injective left action, and its class in $KK^0(A, A) = M_n(\mathbb{Z})$ is

$$[\mathcal{E}_p] = T_p.$$

Let $\tilde{A} := C_0(V_Y \times \mathbb{Z})$ with the degree shift S , and let $\tilde{\mathcal{E}}_p$ be the corresponding correspondence over $\tilde{A}$ recording the valuation of the determinant, so that $[\tilde{\mathcal{E}}_p] = u T_p$ on $K_0(\tilde{A}) = \mathbb{Z}[u^{\pm 1}]^n$, $u := [S]$.

Theorem 2.3.2 (Hecke–Pimsner sequence; equivariant PV). *For any A – A correspondence $\mathcal{E}$ as above, the Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{E}}$ sits in the six-term exact sequence*

$$\begin{array}{ccccc} K_0(A) & \xrightarrow{1-[\mathcal{E}]} & K_0(A) & \longrightarrow & K_0(\mathcal{O}_{\mathcal{E}}) \\ \uparrow & & & & \downarrow \\ K_1(\mathcal{O}_{\mathcal{E}}) & \longleftarrow & K_1(A) & \xleftarrow{1-[\mathcal{E}]} & K_1(A) \end{array}$$

which for $A = \mathbb{C}^n$ collapses to the short exact sequence

$$0 \longrightarrow K_1(\mathcal{O}_{\mathcal{E}}) \longrightarrow \mathbb{Z}^n \xrightarrow{(1-[\mathcal{E}])^t} \mathbb{Z}^n \longrightarrow K_0(\mathcal{O}_{\mathcal{E}}) \longrightarrow 0. \quad (2.3.1)$$

When $\mathcal{E}$ is the correspondence of a single automorphism, (2.3.1) is the Pimsner–Voiculescu sequence; when A carries an action of a finite group C and $\mathcal{E}$ is C -equivariant, the same sequence holds with $\mathbb{Z}^n$ replaced by the $R(C)$ -module $K_0^C(A)$ and $1 - [\mathcal{E}]$ by its $R(C)$ -linear extension.

Definition 2.3.3 (Ihara module). The Ihara module of (Γ, p) is the $\mathbb{Z}[u^{\pm 1}]$ -module

$$M_{\Gamma}(p) := \mathbb{Z}[u^{\pm 1}]^n / (1 - u T_p + q u^2) \mathbb{Z}[u^{\pm 1}]^n,$$

the presentation matrix being the Hecke pencil of Lemma 2.1.5.

Proposition 2.3.4 (Bass’s identity; the two channels). *With A_Y^{nb} as in Definition 2.1.7,*

$$\det(I_{2m} - u A_Y^{\text{nb}}) = (1 - u^2)^{m-n} \det((1 + q u^2)I_n - u T_p), \quad (2.3.2)$$

so the 0-th Fitting ideal of $M_{\Gamma}(p)$ is generated by the reciprocal $Z_Y(u)^{-1}(1 - u^2)^{-(m-n)}$ of the Ihara zeta function of Y . Identity (2.3.2) has been verified symbolically for all graphs of Table 2.1. Specialising $u = 1$:

$$(1 - u T_p + q u^2)|_{u=1} = (q + 1)I_n - T_p = \Delta_p.$$

Proposition 2.3.4 isolates the two channels available for computation. The *boundary channel* uses A_Y^{nb} and computes the K -theory of $C(\partial X) \rtimes \Gamma$ directly. The *Laplacian channel* uses Δ_p , that is the $u = 1$ specialisation of the Ihara module. They are related by (2.3.2), and — this is the point of the section — they carry completely different arithmetic.

2.3.2 The structure theorem

Theorem 2.3.5 (Hecke–Laplacian structure theorem). *Let $\Delta_p = (q + 1)I_n - T_p$ and let $\text{SNF}(\Delta_p) = \text{diag}(d_1 \mid d_2 \mid \cdots \mid d_{n-1}, 0)$ be its Smith normal form over $\mathbb{Z}$. Then:*

- (i) $\ker \Delta_p = \mathbb{Z}$ and $\text{coker } \Delta_p \cong \mathbb{Z} \oplus \bigoplus_{i=1}^{n-1} \mathbb{Z}/d_i \mathbb{Z} = \mathbb{Z} \oplus \text{Jac}(Y)$, where $\text{Jac}(Y)$ is the Jacobian (critical, sandpile) group of Y ;

(ii) $\text{coker } \Delta_p$ and $\ker \Delta_p$ are the $u = 1$ specialisation

$$M_\Gamma(p) \otimes_{\mathbb{Z}[u^{\pm 1}]} \mathbb{Z}_{u=1} \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad \text{Tor}_1^{\mathbb{Z}[u^{\pm 1}]}(M_\Gamma(p), \mathbb{Z}_{u=1}) \cong \mathbb{Z},$$

of the Ihara module of Definition 2.3.3, and they are the integral index groups of the Hecke–Dirac operator of Y : choosing an orientation E_Y^+ and writing $\partial: \mathbb{Z}^{E_Y^+} \rightarrow \mathbb{Z}^{V_Y}$, $\partial f = t(f) - o(f)$, for the simplicial boundary map, the self-adjoint operator

$$D := \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \quad \text{on} \quad \ell^2(V_Y) \oplus \ell^2(E_Y^+), \quad D^2 = \Delta_p \oplus \partial^t \partial,$$

has $\ker D = \mathbb{Z} \oplus H_1(Y; \mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Z}^r$, and the integral cokernel of the even part $D^2|_{\ell^2(V_Y)} = \Delta_p$ is $\mathbb{Z} \oplus \text{Jac}(Y)$. No existence claim for a C^* -algebra is made here; see Remark 2.3.6;

(iii) the order of the torsion is a product of Frobenius data: by Kirchhoff’s matrix-tree theorem and Ihara’s theorem,

$$|\text{Tors } K_0(\mathcal{O})| = |\text{Jac}(Y)| = \kappa(Y) = \frac{1}{n} \prod_{\lambda \in \text{Spec } T_p \setminus \{q+1\}} (q+1-\lambda) = \frac{1}{n} \prod_f (p+1-a_p(f)), \quad (2.3.3)$$

f running over the weight-two newforms occurring in Y ; and if E_f is the elliptic curve attached to f by Eichler–Shimura, then $p+1-a_p(f) = \#E_f(\mathbb{F}_p)$, so

$$|\text{Tors } K_0(\mathcal{O})| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p).$$

The Smith normal form computes not merely the order but the group: the elementary divisors $d_1 \mid \cdots \mid d_{n-1}$ are the invariant factors of $\text{Jac}(Y)$.

Proof sketch. (i) is the standard theory of the critical group: Δ_p is the Laplacian of the $(q+1)$ -regular graph Y , symmetric with kernel the constants, and its cokernel splits as $\mathbb{Z} \oplus K(Y)$ with $K(Y)$ the critical group. (ii) is Theorem 2.3.2 applied with index map Δ_p . In (iii), Kirchhoff gives $\kappa(Y) = \frac{1}{n} \prod_{i \geq 2} \mu_i$ for the non-zero Laplacian eigenvalues $\mu_i = (q+1) - \lambda_i$, $|K(Y)| = \kappa(Y)$, and Ihara–Eichler–Jacquet–Langlands identifies $\text{Spec } T_p \setminus \{q+1\}$ with the $a_p(f)$. The last identity is the definition of a_p . $\square$

Remark 2.3.6 (Bulk and boundary: what is, and is not, an algebra). Since $A = \mathbb{C}^n$ we have $KK^0(A, A) \cong M_n(\mathbb{Z})$, so

$$x_p := [\mathcal{E}_p] - q \cdot 1_A = T_p - qI_n \in KK^0(A, A)$$

is a bona fide Kasparov class, and $1 - x_p = \Delta_p$. What fails is not the class but its *realisation*: a class is the class of a C^* -correspondence only if its matrix has non-negative entries, and $T_p - qI_n$ has $-q$ on the diagonal. Hence x_p is a formal difference of correspondences, no Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{E}}$ has index map Δ_p , and we assert none. The asymmetry between the two channels is therefore not only arithmetic but ontological:

	realisation	invariant
boundary channel	a C^* -algebra, $\mathcal{O}_{A_Y^{\text{nb}}} \simeq C(\partial X) \rtimes \Gamma$	$\mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$, no arithmetic
bulk (Laplacian) channel	a $\mathbb{Z}[u^{\pm 1}]$ -module and an index	$\mathbb{Z} \oplus \text{Jac}(Y)$, all the arithmetic

Everything asserted in Theorem 2.3.5 is unconditional at the level of the Ihara module, of the index of D , and of the integers; what is missing is an honest C^* -algebra whose K_0 is $\mathbb{Z} \oplus \text{Jac}(Y)$. Producing one — equivalently, realising the virtual correspondence x_p by a genuine, possibly non-unital or $\mathbb{Z}/2$ -graded, construction — is the main open problem of this paper.

There is a purely algebraic first step, and it isolates exactly where the difficulty lives. The Ihara pencil $1 - uT_p + qu^2$ of Definition 2.3.3 is a quadratic matrix polynomial, and it linearises in the standard way over $A^{\oplus 2}$, $A = \mathbb{C}^n$: setting

$$C := \begin{pmatrix} T_p & -qI_n \\ I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{Z}), \quad \det(I_{2n} - uC) = \det(I_n - uT_p + qu^2I_n),$$

a companion-matrix identity verified directly by block Gaussian elimination on $I - uC$. Thus C is a single, honest $\mathbb{Z}$ -linear class in $KK^0(A^{\oplus 2}, A^{\oplus 2})$ representing the whole Ihara pencil at once, with $[\mathcal{E}_p] = T_p$ visible in its upper left block; the obstruction of the previous paragraph has been moved, not removed, since C still has the non-positive block $-qI_n$ in its upper right corner and so is still not the matrix of a C^* -correspondence over $A^{\oplus 2}$.

The problem left for the sequel is accordingly sharp: *dilate* C , in the sense of dilation theory, to a genuinely non-negative matrix over a larger vertex algebra $\tilde{A} \supset A^{\oplus 2}$ — for instance by the Drinen–Tomforde desingularisation of a directed graph carrying C as a signed incidence matrix, with the negative entries reinterpreted as sinks absorbing mass at rate q per vertex — in such a way that the resulting Cuntz–Pimsner (or graph) algebra $\mathcal{O}_{\tilde{C}}$ receives $A^{\oplus 2}$ as a full corner and has $K_0(\mathcal{O}_{\tilde{C}}) \cong \mathbb{Z} \oplus \text{Jac}(Y)$ by Morita reduction back to Δ_p . We do not carry this out here; it is the intended content of a sequel to this paper, and the companion linearisation above is offered as its starting point.

Theorem 2.3.7 (Boundary channel is arithmetically blind). *Let Y be any finite connected $(q+1)$ -regular graph and Γ the corresponding torsion-free cocompact lattice, $r = \text{rank } \Gamma = m - n + 1 \geq 2$. Then*

$$K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)\mathbb{Z}, \quad K_1(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r, \quad r = \text{rank } \Gamma = \frac{(q-1)n}{2} + 1.$$

The torsion of the boundary quotient therefore depends only on the Euler characteristic $r-1 = |E_Y| - |V_Y|$ of Y , hence only on the isomorphism class of Γ as an abstract free group: no Hecke eigenvalue is visible in the boundary channel.

Proof. Write $A^{\text{nb}} = T^t S - J$, where $S, T \in M_{n \times 2m}(\mathbb{Z})$ are the incidence matrices $S_{v,e} = [o(e) = v]$, $T_{v,e} = [t(e) = v]$ and J is the edge-reversal involution, and put $M := I_{2m} - A^{\text{nb}} = I + J - T^t S$. Since $\text{SNF}(M) = \text{SNF}(M^t)$ it suffices to compute $\text{coker } M = \mathbb{Z}^{E_Y} / M\mathbb{Z}^{E_Y}$, and by Theorem 2.3.2 this is K_0 , with $K_1 = \ker M^t$.

Reading off the columns of M , the presentation of $\text{coker } M$ has generators $[e]$, $e \in E_Y$, and one relation for each $f \in E_Y$,

$$[f] + [\bar{f}] = c_{o(f)}, \quad \text{where } c_v := \sum_{t(e)=v} [e]. \quad (2.3.4)$$

Applying (2.3.4) to f and to $\bar{f}$ gives $c_{o(f)} = c_{t(f)}$ for every edge, so $c_v =: c$ is independent of v because Y is connected. Fix an orientation E_Y^+ , use $[\bar{f}] = c - [f]$ to eliminate the reversed generators, and let $d^-(v) := \#\{f \in E_Y^+ : o(f) = v\}$. Expanding c_v in the surviving generators turns the n remaining relations into

$$R_v : \quad (\partial^t\text{-column at } v) : \quad \sum_{f \in E_Y^+, t(f)=v} [f] - \sum_{f \in E_Y^+, o(f)=v} [f] = (1 - d^-(v)) c.$$

Hence $\text{coker } M \cong \text{coker } B$ for the $(m+1) \times n$ integer matrix

$$B = \begin{pmatrix} \partial^t \\ -\lambda^t \end{pmatrix}, \quad \lambda_v := 1 - d^-(v),$$

on generators $E_Y^+ \cup \{c\}$. Projecting away the last coordinate gives the exact sequence

$$0 \longrightarrow \langle c \rangle \longrightarrow \text{coker } B \longrightarrow \text{coker}(\partial^t) = H^1(Y; \mathbb{Z}) \cong \mathbb{Z}^r \longrightarrow 0,$$

H^1 of a graph being free of rank r . Finally $tc = 0$ in $\text{coker } B$ if and only if $(0, \dots, 0, t) \in \text{im } B$, i.e. if and only if there is $x \in \mathbb{Z}^{V_Y}$ with $\partial^t x = 0$ and $-\lambda \cdot x = t$; the first condition forces $x = a\mathbf{1}$, and then

$$-\lambda \cdot x = -a \sum_v (1 - d^-(v)) = -a(n - m) = a(r - 1),$$

so c has order exactly $r - 1$. The quotient $\mathbb{Z}^r$ being free, the sequence splits and $\text{coker } M \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$; $\ker M^t$ is free of rank $2m - \text{rank } M = r$. $\square$

Remark 2.3.8 (Where the spectrum goes). The proof shows exactly how the Hecke data disappear, and it is worth recording that the mechanism is *not* the one suggested by Bass's identity (2.3.2): at $u = 1$ both factors of (2.3.2) vanish, so (2.3.2) degenerates to $0 = 0$ and carries no integral information there. What kills the spectrum is the local relation (2.3.4): it forces the n a priori distinct classes c_v to collapse to a single class c , after which the only surviving numerical invariant is $\sum_v (1 - d^-(v)) = n - m$. The orientation-dependent quantities $d^-(v)$ do not survive individually — only their sum does — and the sum is the Euler characteristic. The adjacency matrix T_p never enters the computation at all.

Remark 2.3.9 (The dichotomy). Theorems 2.3.5 and 2.3.7 together are the main structural finding of this paper. The boundary quotient — the natural non-commutative object, the simple purely infinite Cuntz–Krieger algebra of the tail relation — has K -theory of purely topological content. The arithmetic sits one specialisation away, in the Hecke pencil $1 - uT_p + qu^2$ at $u = 1$; equivalently, the arithmetic is the *first-order* datum of the Ihara zeta function at its trivial zero. This is the exact non-commutative analogue of the classical statement that $\zeta_Y(u)$ has a trivial factor $(1 - u^2)^{m-n}$ and that the interesting arithmetic is in the reduced factor.

Corollary 2.3.10 (Frobenius traces as torsion). *Let Y carry a single non-trivial Hecke eigenvalue a_p with multiplicity $n - 1$. Then $\text{Jac}(Y) \cong (\mathbb{Z}/(p + 1 - a_p))^{n-2}$ whenever the Laplacian is of the form $cI - J$, and in general $|\text{Jac}(Y)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ exactly. In particular the Frobenius trace a_p is recovered from K_0 by*

$$a_p = p + 1 - (n \cdot |\text{Tors } K_0|)^{1/(n-1)}$$

in the eigenvalue-homogeneous case, and in general from the elementary divisors by the inverse of the Smith algorithm.

Corollary 2.3.11 (Ramanujan bound; strict, and never attained). *If Y is Ramanujan, i.e. $|\lambda| \leq 2\sqrt{q}$ for every $\lambda \in \text{Spec } T_p \setminus \{\pm(q+1)\}$, then*

$$|\text{Tors } K_0| \leq \frac{1}{n} (q + 1 + 2\sqrt{q})^{n-1} = \frac{1}{n} (1 + \sqrt{q})^{2(n-1)},$$

so that, with $B(q) := 2\log(1 + \sqrt{q})$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |\text{Tors } K_0| \leq B(q).$$

The bound is unconditional on the Ramanujan locus, but it is never an equality, and the deficit is exponential in n : see Theorem 2.3.12 and Remark 2.3.14.

Theorem 2.3.12 (Benjamini–Schramm asymptotic torsion). *Let $(Y_j)_{j \geq 1}$ be finite connected $(q+1)$ -regular graphs, $n_j = |V_{Y_j}| \rightarrow \infty$, which Benjamini–Schramm converge to the $(q+1)$ -regular tree T_{q+1} — for instance any family of girth tending to infinity, such as the Lubotzky–Phillips–Sarnak graphs $X^{p,\ell}$ at fixed $p = q$. Let $\mathcal{O}_j$ be the associated Hecke–Laplacian invariant of Theorem 2.3.5. Then the torsion density converges, and its limit is Lyons’ tree entropy of T_{q+1} :*

$$\lim_{j \rightarrow \infty} \frac{1}{n_j} \log |\text{Tors } K_0(\mathcal{O}_j)| = h(q) := q \log q - \frac{q-1}{2} \log(q^2 - 1) = \log \frac{q^q}{(q^2 - 1)^{(q-1)/2}}, \quad (2.3.5)$$

equivalently $h(q) = \int_{-2\sqrt{q}}^{2\sqrt{q}} \log(q + 1 - x) d\mu_{\text{KM}}(x)$ against the Kesten–McKay measure

$$d\mu_{\text{KM}}(x) = \frac{(q+1)\sqrt{4q-x^2}}{2\pi((q+1)^2-x^2)} dx,$$

the spectral measure of the simple random walk on T_{q+1} [2, 3]. Moreover $h(q) < B(q)$ for every q , with

$$\frac{B(q)}{h(q)} = 2.1061, 1.3889, 1.2340, 1.0957, 1.0089 \quad \text{for } q = 2, 5, 9, 29, 999.$$

Corollary 2.3.13 (The deficit, to two orders). *As $q \rightarrow \infty$,*

$$B(q) - h(q) = 2q^{-1/2} - \frac{3}{2}q^{-1} + O(q^{-3/2}), \quad (2.3.6)$$

an asymptotic equality, not a numerical fit. Consequently, on graphs with n vertices, the Ramanujan bound overstates $|\text{Tors } K_0|$ by the factor

$$\exp((B(q) - h(q))n) = \exp(2nq^{-1/2} + O(nq^{-1}));$$

at $q = 2$, $n = 2000$ this factor is e^{1852} .

Proof. Write $h(q) = \log q - \frac{q-1}{2} \log(1 - q^{-2})$ and expand $-\log(1 - q^{-2}) = q^{-2} + \frac{1}{2}q^{-4} + O(q^{-6})$, so

$$h(q) = \log q + \frac{q-1}{2}(q^{-2} + O(q^{-4})) = \log q + \frac{1}{2}q^{-1} - \frac{1}{2}q^{-2} + O(q^{-3}).$$

Write $B(q) = \log q + 2\log(1 + q^{-1/2})$ and expand $\log(1 + q^{-1/2}) = q^{-1/2} - \frac{1}{2}q^{-1} + \frac{1}{3}q^{-3/2} + O(q^{-2})$, so

$$B(q) = \log q + 2q^{-1/2} - q^{-1} + \frac{2}{3}q^{-3/2} + O(q^{-2}).$$

Subtracting, the $\log q$ terms cancel and $B(q) - h(q) = 2q^{-1/2} - q^{-1} - \frac{1}{2}q^{-1} + O(q^{-3/2}) = 2q^{-1/2} - \frac{3}{2}q^{-1} + O(q^{-3/2})$, which is (2.3.6). (A direct computer-algebra expansion confirms the next two orders as well, and that the q^{-2} coefficient vanishes identically — the $-\frac{1}{2}q^{-2}$ term present in $B(q)$ and in $h(q)$ separately cancels exactly on subtraction: $B(q) - h(q) = 2q^{-1/2} - \frac{3}{2}q^{-1} + \frac{2}{3}q^{-3/2} + \frac{2}{5}q^{-5/2} + O(q^{-3})$.) The numerical values of Remark 2.3.14 and of the ratios above were computed independently of this expansion and agree with it: at $q = 10^4$, (2.3.6) predicts $0.02 - 0.00015 = 0.01985$ against the quadrature value 0.019852. $\square$

Proof. By Theorem 2.3.5(iii), $|\text{Tors } K_0(\mathcal{O}_j)| = \kappa(Y_j)$, so (2.3.5) is Lyons' theorem on the asymptotics of the number of spanning trees along Benjamini–Schramm convergent sequences of bounded degree [1], applied to the limit T_{q+1} , together with Kirchhoff's formula in the form $\frac{1}{n} \log \kappa = \frac{1}{n} [\sum_{i \geq 2} \log(q+1 - \lambda_i) - \log n]$ and the convergence of the empirical spectral measures of T_p to μ_{KM} . The closed form and the integral representation of $h(q)$ agree to six decimals for $q+1 \in \{3, 4, 6, 8, 14, 30\}$ by numerical quadrature. Strictness of $h < B$ is the strict concavity of $\log$ together with the fact that μ_{KM} is not a point mass at the spectral edge. $\square$

Remark 2.3.14 (Why the Ramanujan bound is the wrong instrument). Corollary 2.3.11 replaces the whole spectrum by its extreme point $2\sqrt{q}$. But μ_{KM} vanishes like a square root at $\pm 2\sqrt{q}$ and puts almost no mass there, so the bound is a supremum where the truth is an integral; this, and not any slack in the choice of family, is the source of the exponential deficit of Theorem 2.3.12.

The deeper point is that the Ramanujan property is neither necessary nor sufficient for the limit (2.3.5), and that the invariant which does control the torsion density is Benjamini–Schramm convergence. Both halves are visible in exact computation. Below, λ_2 is the largest non-trivial eigenvalue in absolute value, KS the Kolmogorov–Smirnov distance from the empirical spectral distribution of T_p to μ_{KM} , and the last two columns compare the measured torsion density with $h(q)$ and with the Ramanujan bound $B(q)$.

Y	n	q	λ_2	Ram. ?	KS	ratio to $h(q)$	ratio to $B(q)$
random 3-regular	200	2	2.8315	no	0.0142	0.9774	0.4641
random 3-regular	1000	2	2.8261	yes	0.0032	0.9937	0.4719
random 3-regular	2000	2	2.8239	yes	0.0021	0.9964	0.4731
$\text{PG}(2, 5)$	62	5	2.2361	yes	0.2661	0.9669	0.6962
LPS $X^{5,13}$	2184	5	4.2497	yes	0.0375	0.9982	0.7187
LPS $X^{5,17}$	4896	5	4.3089	yes	0.0307	0.9991	0.7194

Two rows falsify the spectral-gap reading. The random 3-regular graph on 200 vertices has $\lambda_2 = 2.8315 > 2\sqrt{2} = 2.8284$ and is therefore *not* Ramanujan, yet its torsion density is already within 2.3% of $h(2)$. The incidence graph of $\text{PG}(2, 5)$ is Ramanujan with room to spare, $\lambda_2 = \sqrt{5} = 2.2361$ against the permitted $2\sqrt{5} = 4.4721$, yet it has girth 6, its spectral distribution is far from Kesten–McKay (KS = 0.27), and its torsion density falls 3.3% *below* $h(5)$. The KS column and the h -ratio column move together throughout; the λ_2 column does not track either.

One further datum belongs here, because it bears on Remark 2.2.6 and on the statistics of arithmetic spectra generally. The graph $X^{5,13}$ has 2184 vertices but only 55 distinct eigenvalues, of maximal multiplicity 96, whereas a random 3-regular graph on 2000 vertices has 2000 distinct eigenvalues. The degeneracy is forced by the $\text{PGL}_2(\mathbb{F}_{13})$ representation theory through which the Hecke operator factors, and it is why the empirical spectral distribution of an LPS graph is a coarse staircase whose KS distance decays more slowly than that of a random graph of half the size. It is also the mechanism behind the Poisson — not Gaussian-unitary — level statistics observed in arithmetic quantum chaos: large Hecke symmetry produces large degeneracy, and degeneracy is the opposite of level repulsion.

2.3.3 The algorithm, and its verification

Algorithm 2.3.15 (Hecke $\rightarrow$ torsion). **Input:** a prime p (so $q = p$), a torsion-free cocompact $\Gamma \leq \text{PGL}_2(\mathbb{Q}_p)$ presented by its quotient graph $Y = \Gamma \backslash X$, i.e. by the integer matrix $T_p \in M_n(\mathbb{Z})$.

Y	q	n	$\text{Jac}(Y) = \text{Tors coker } \Delta_p$	$\kappa(Y)$	boundary Tors K_0	$r - 1$
K_4	2	4	$(\mathbb{Z}/4)^2$	16	$\mathbb{Z}/2$	2
K_5	3	5	$(\mathbb{Z}/5)^3$	125	$\mathbb{Z}/5$	5
K_6	4	6	$(\mathbb{Z}/6)^4$	1296	$\mathbb{Z}/9$	9
$K_{3,3}$	2	6	$\mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$	81	$\mathbb{Z}/3$	3
Q_3 (cube)	2	8	$\mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$	384	$\mathbb{Z}/4$	4
prism $C_4 \times K_2$	2	8	$\mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$	384	$\mathbb{Z}/4$	4
Wagner V_8	2	8	$\mathbb{Z}/7 \oplus \mathbb{Z}/56$	392	$\mathbb{Z}/4$	4
prism $C_5 \times K_2$	2	10	$\mathbb{Z}/19 \oplus \mathbb{Z}/95$	1805	$\mathbb{Z}/5$	5
Möbius V_{10}	2	10	$\mathbb{Z}/11 \oplus \mathbb{Z}/165$	1815	$\mathbb{Z}/5$	5
Petersen	2	10	$\mathbb{Z}/2 \oplus (\mathbb{Z}/10)^3$	2000	$\mathbb{Z}/5$	5
Heawood	2	14	$(\mathbb{Z}/7)^4 \oplus \mathbb{Z}/21$	50421	$\mathbb{Z}/7$	7

Table 2.1: Exact Smith-normal-form computations. Columns 4–5: the Laplacian channel, Theorem 2.3.5; the order always equals $\frac{1}{n} \prod_{\lambda \neq q+1} (q+1-\lambda)$. Columns 6–7: the boundary channel, Theorem 2.3.7; the torsion is always cyclic of order $r-1 = (q-1)n/2$, independently of the spectrum, confirming the proof given there. Bass’s identity (2.3.2) was verified symbolically in the first five rows.

- (1) Form $\Delta_p := (q+1)I_n - T_p \in M_n(\mathbb{Z})$.
- (2) Compute the Smith normal form over $\mathbb{Z}$, $U\Delta_p W = \text{diag}(d_1, \dots, d_{n-1}, 0)$ with $d_1 \mid \dots \mid d_{n-1}$ (cost $O(n^3)$ ring operations with the usual bit-length control).
- (3) Output $K_0 = \mathbb{Z} \oplus \bigoplus_i \mathbb{Z}/d_i$, $K_1 = \mathbb{Z}$, and $|\text{Tors } K_0| = \prod_i d_i = \kappa(Y)$.
- (4) *Certificate.* Independently compute $\frac{1}{n} \prod_{\lambda \neq q+1} (q+1-\lambda)$ from the characteristic polynomial of T_p , and $\frac{1}{n} \prod_f (p+1-a_p(f))$ from the Hecke eigenvalues; the three numbers must agree. Optionally verify Bass’s identity (2.3.2) symbolically as a check on the graph data.

Output: the torsion subgroup of K_0 as an explicit finite abelian group, together with its factorisation into Frobenius traces.

Example 2.3.16 (Level 37, $p = 2$: two elliptic curves, one torsion group). The space $S_2(\Gamma_0(37))$ is two-dimensional, spanned by the newforms of the elliptic curves $37a : y^2 + y = x^3 - x$ (rank one) and $37b : y^2 + y = x^3 + x^2 - 23x - 50$ (rank zero), with $a_2(37a) = -2$ and $a_2(37b) = 0$. Correspondingly

$$\#E_{37a}(\mathbb{F}_2) = 2 + 1 - (-2) = 5, \quad \#E_{37b}(\mathbb{F}_2) = 2 + 1 - 0 = 3,$$

and the Hecke–Laplacian of the associated 3-regular Hecke system has non-zero eigenvalues 5 and 3, so (2.3.3) gives $|\text{Tors } K_0| = \frac{1}{3} \cdot 5 \cdot 3 = 5$. The Frobenius traces of two distinct elliptic curves precipitate into a single cyclic group of order five.

Example 2.3.17 (Level 11, $p = 3$). For the definite quaternion algebra of discriminant 11 the Brandt matrix at $p = 3$ is $B(3) = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$, with spectrum $\{4, -1\}$: the Eisenstein eigenvalue $q+1 = 4$ and $a_3(11a) = -1$, matching $\eta(z)^2 \eta(11z)^2 = q - 2q^2 - q^3 + 2q^4 + \dots$. Then $q+1-a_3 = 5 = \#E_{11a}(\mathbb{F}_3) = |E_{11a}(\mathbb{Q})_{\text{tors}}|$.

Remark 2.3.18 (Weighted quotients). Examples 2.3.16 and 2.3.17 are stated at the level of the eigenvalue identity $q + 1 - a_p = \#E(\mathbb{F}_p)$, which is unconditional. The passage to the critical group in Theorem 2.3.5(iii) requires Γ torsion-free, so that Y is a genuine $(q + 1)$ -regular graph; Brandt and supersingular isogeny graphs are quotients by groups with torsion, i.e. graphs of groups with vertex weights $w_i = |\text{Aut}|/2$, and for them Theorem 2.3.5 holds in its weighted form, with Δ_p the weighted Laplacian $w_i B_{ij} = w_j B_{ji}$ and Kirchhoff's theorem replaced by its weighted version. Congruence subgroups of sufficiently high level are torsion-free and remove the issue at the cost of enlarging n .

The structurally correct home for the weighted case is orbifold, i.e. groupoid, K -theory. A graph of groups (Y, Γ_v, Γ_e) in the sense of Bass–Serre is the same datum as an étale groupoid $\mathcal{G}_Y$ with unit space V_Y and isotropy Γ_v ; the object to compute is $K_*(C^*(\mathcal{G}_Y))$, and the Bass–Serre Mayer–Vietoris sequence

$$\cdots \rightarrow \bigoplus_{e \in E_Y} K_*(C^*(\Gamma_e)) \xrightarrow{\partial} \bigoplus_{v \in V_Y} K_*(C^*(\Gamma_v)) \rightarrow K_*(C^*(\mathcal{G}_Y)) \rightarrow \cdots$$

replaces $\mathbb{Z}^{E_Y} \rightarrow \mathbb{Z}^{V_Y}$ by a sequence of representation rings $R(\Gamma_v) = K_0(C^*(\Gamma_v))$, of ranks $|\text{Irr } \Gamma_v|$. The vertex weights are then not an ad hoc normalisation but the ranks of these isotropy contributions: the composite $\partial \partial^t$ is the weighted Laplacian $w_i B_{ij} = w_j B_{ji}$ with $w_i = |\text{Aut}|/2$, the Kirchhoff count acquires the factor $\prod_v w_v^{-1}$ of the Eichler mass formula, and the apparently fractional indices — the value $\frac{1}{2}(4 - a_3)$ of Example 2.3.17 is not an integer — become integral once read in the equivariant group of Proposition 2.3.19 rather than in $\mathbb{Z}$. The GL_2 analogue of Paper XXXVIII's entanglement law belongs here, and we do not develop it in this paper.

2.3.4 The equivariant refinement

Proposition 2.3.19 (Equivariant Hecke K -theory). *Let C be a finite group acting on Y by graph automorphisms commuting with T_p — in the arithmetic case, $C = \text{Gal}(H/K)$ acting through the class group, as in Paper XXXVIII. Then $K_0^C(A) = R(C)^{n_C}$ for the orbit decomposition, the index map of Theorem 2.3.2 is the $R(C)$ -linear extension of Δ_p , and*

$$K_0^C(\mathcal{O}) \cong \text{coker}_{R(C)}(\Delta_p \otimes_{\mathbb{Z}} R(C)), \quad K_1^C(\mathcal{O}) \cong \ker_{R(C)}(\Delta_p \otimes_{\mathbb{Z}} R(C)).$$

The entanglement law of Paper XXXVIII transfers verbatim: free $\mathbb{Z}[C]$ -strata of the vertex module contribute once by Shapiro's lemma, while strata with trivial or sign action are doubled or sign-twisted by $R(C)$; the orders of the invariant factors are unchanged, only their representation labels.

2.3.5 What the two channels say about Langlands

We summarise the bridge as a table of correspondences, each entry of which is a theorem or a definition above, not an analogy.

Automorphic side	Operator-algebraic side	Reference
unramified π_p , Satake $(\alpha_{p,1}, \alpha_{p,2})$	Radon–Nikodym cocycle $e^{-\beta c_\pi}$	Def. 2.2.1
local L -factor $L_p(\beta, \pi_p)$	partition function $Z_p(\beta; \pi)$	Thm 2.2.2
largest pole of L_p	critical inverse temperature $\beta_c(\pi_p)$	Thm 2.2.2
temperedness (Ramanujan) at p	$\beta_c(\pi_p) = w/2$	Thm 2.2.2
Eisenstein locus	the only locus carrying KMS <i>states</i>	Thm 2.2.5
Rankin–Selberg $L(s, \pi \times \tilde{\pi}')$	Kakutani/Hellinger obstruction Ξ_β	Thm 2.2.8
strong multiplicity one	mutual singularity of phases	Cor. 2.2.9
motivic weight w	width of the equilibrium/obstruction gap	Thm 2.2.11
Hecke recursion $TT_n = T_{n+1} + qT_{n-1}$	Ihara module $\mathbb{Z}[u^\pm]^n / (1 - uT_p + qu^2)$	Def. 2.3.3
Ihara zeta $Z_Y(u)$	Fitting ideal of the Ihara module	Prop. 2.3.4
Frobenius trace $a_p(f)$, $\#E_f(\mathbb{F}_p)$	invariant factors of $\text{SNF}(\Delta_p)$	Thm 2.3.5
Euler characteristic of Y (no arithmetic)	torsion of the boundary quotient	Thm 2.3.7

The reading of the table is the thesis of the paper. Along the top half — the thermodynamic rows — the correspondence is exact but weight-zero: it reproduces class field theory and Artin data, and Theorem 2.2.11 measures precisely how far it falls short above weight zero. Along the bottom half — the K -theoretic rows — the correspondence is insensitive to temperature and to positivity, and it does carry weight-two data, but only after the specialisation $u = 1$ that separates the Laplacian channel from the boundary channel. An algorithmic non-commutative class field theory for GL_2 is therefore not one bridge but two, and Bass’s identity (2.3.2) is the only thing that connects them.

Acknowledgement of provenance and caveats

This is an unrefereed preprint and no priority check against the literature has been carried out. The following are used as known and are not proved here: the Satake isomorphism and the Hecke recursion; Ihara’s theorem and Bass’s determinant identity; Kirchhoff’s matrix-tree theorem and the theory of the critical group; the Eichler–Shimura and Jacquet–Langlands correspondences; Kakutani’s dichotomy for product measures; Jacquet–Shalika non-vanishing for Rankin–Selberg L -functions; Renault’s theory of groupoid C^* -algebras and the correspondence between KMS states and conformal measures; Cuntz–Krieger and Cuntz–Pimsner K -theory and Pimsner’s exact sequence. The computations of Table 2.1, and the spectral and torsion-density computations of Remark 2.3.14, are exact or numerically controlled and were carried out independently of the statements they confirm. Theorem 2.3.12 rests on Lyons’ spanning-tree asymptotics [1], which we quote and do not prove, and Remark 2.3.14 on the Kesten–McKay spectral distribution [2, 3]. The one substantive gap left open is the one isolated in Remark 2.3.6: the Laplacian channel is a module and an index, not yet a C^* -algebra.

Chapter 3

Algorithmic non-commutative class field theory, II: the trace obstruction and the K-homological bulk

Summary. Paper I isolated a virtual KK -class $x_p = T_p - qI \in KK^0(\mathbb{C}^n, \mathbb{C}^n) = M_n(\mathbb{Z})$, arising from the Hecke–Laplacian $\Delta_p = (q+1)I - T_p = 1 - x_p$ of a finite $(q+1)$ -regular graph Y , and left open whether x_p could be realized by a genuine, non-negative C^* -correspondence, so that Δ_p ’s cokernel $\mathbb{Z} \oplus \text{Jac}(Y)$ — the critical group of Y , whose order is a product of Frobenius traces $\prod(p+1 - a_p(f))$ — would be the K_0 -group of an honest Cuntz–Pimsner algebra. This paper proves that no such realization exists, by an elementary and then a deeper obstruction, and identifies where the arithmetic must live instead.

The companion linearization $C = \begin{pmatrix} T_p & -qI \\ I & 0 \end{pmatrix}$, exactly row/column-equivalent to Δ_p over $\mathbb{Z}$, has $\text{Tr}(C) = 0$, $\text{Tr}(C^2) = -n(q-1)$, strictly negative for every $q \geq 2$ and every $(q+1)$ -regular graph. Since $\text{Tr}(B^2) \geq 0$ for any non-negative matrix B , and trace is a similarity invariant, **no matrix of the same size and the same spectrum as C can be non-negative** (Theorem 3.1.2) — an unconditional, four-line fact. Every dilation to a *larger* non-negative matrix, of the kind an auxiliary-vertex or Drinen–Tomforde construction would attempt, is ruled out as well, and unconditionally: shift equivalence preserves the nonzero spectrum with multiplicities — proved here by an elementary two-way injection of generalized eigenspaces (Lemma 3.1.4) — hence preserves $\text{Tr}(\cdot^2)$, closing the realization problem of Paper I, Remark 3.11, in the negative for matrices of every size (Theorem 3.1.5).

We then show that no repair by $\mathbb{Z}/2$ -grading is available: a formal difference of two positive correspondences is not itself a correspondence, and no Cuntz–Pimsner-type functor converts a virtual difference into a single algebra realizing that difference’s index (Section 3.2); if one existed, it would trivialize the realization problem for every virtual K -theory class whatsoever, which is itself evidence against its existence.

The arithmetic is instead formalized where no positivity is required: the Hecke–Dirac operator D_p defines an honest Fredholm module over $C(V_Y)$ with $\text{ind}(D_p) = \mathbb{Z} \oplus \text{Jac}(Y)$ exactly (Theorem 3.3.3), inside $KK^0(C(V_Y), C(V_Y)) \cong M_n(\mathbb{Z})$. We record, rather than disguise, that for a single finite graph this KK -group is nothing but a matrix ring and carries no content beyond ordinary linear algebra; the paper closes by locating the correct venue for a genuinely non-commutative sequel in the projective limit over a congruence tower of graphs, where the coefficient algebra becomes an infinite, totally disconnected profinite space and KK^0 ceases to be finite-dimensional. As a first exact instance of the tower philosophy, we compute four abelian $\mathbb{Z}/\ell^k$ voltage towers over K_4 to nine levels: the torsion obeys the Iwasawa law $\text{ord}_\ell |\text{Jac}(Y_k)| = \mu\ell^k + \lambda k + \nu$ exactly at every computed level, with $(\mu, \lambda) = (3, 1), (0, 3), (2, 3), (0, 1)$, positive μ occurring freely, and the Smith normal forms exhibiting λ as the number of persistently

growing cyclic factors. The canonical combinatorial model of the congruence tower itself — the Iwahori path tower — obeys instead an exact pseudo-Iwasawa law with $\lambda = -1 < 0$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal: the non-normal tower is not an Iwasawa module, and the conjecture for it is restated over the Iwahori–Hecke algebra, with its naive commutative form withdrawn.

Status of the results, stated in advance

As in Paper I, each numbered result is labelled by how it is entitled to be believed. Theorem 3.1.2 is proved here in full, elementarily, using nothing beyond the fact that trace is a similarity invariant and that non-negative matrices have non-negative squared-trace. Theorem 3.1.5, which rules out the auxiliary-vertex and Drinen–Tomforde schemes of any size, is likewise unconditional: the fact it needs — that shift equivalence preserves the nonzero spectrum with multiplicities — is proved here as the elementary Lemma 3.1.4 by a two-way injection of generalized eigenspaces, and no external citation is load-bearing for it. Theorem 3.3.3 is proved in full and requires no citation beyond standard finite-dimensional linear algebra dressed in Fredholm-module language. Section 3.3 is exploratory: it states a research programme, not a theorem, and we say so throughout.

3.1 The trace obstruction

3.1.1 Setup, recalled from Paper I

Y is a finite connected $(q+1)$ -regular graph on n vertices, $T_p \in M_n(\mathbb{Z})$ its adjacency (Hecke) operator, $\Delta_p := (q+1)I_n - T_p$ its Hecke–Laplacian, and $\text{Jac}(Y)$ its critical group, with

$$\text{coker}_{\mathbb{Z}}(\Delta_p) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad |\text{Jac}(Y)| = \kappa(Y) = \frac{1}{n} \prod_f (p+1 - a_p(f)) = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p),$$

the product over the weight-two newforms f whose Hecke eigenvalues occur in $\text{Spec } T_p$, by Kirchhoff’s theorem together with the Eichler–Shimura and Jacquet–Langlands correspondences. Introduce the companion linearization

$$C := \begin{pmatrix} T_p & -qI_n \\ I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{Z}),$$

of the Ihara pencil $1 - uT_p + qu^2$, so that $\det(I_{2n} - uC) = \det(I_n - uT_p + qu^2I_n)$. By Schur elimination against the invertible block I_n in the lower right of $I_{2n} - C = \begin{pmatrix} I_n - T_p & qI_n \\ -I_n & I_n \end{pmatrix}$, one has the exact integral equivalence

$$\begin{pmatrix} I_n & -qI_n \\ 0 & I_n \end{pmatrix} (I_{2n} - C) \begin{pmatrix} I_n & 0 \\ I_n & I_n \end{pmatrix} = \begin{pmatrix} \Delta_p & 0 \\ 0 & I_n \end{pmatrix}, \quad \text{hence} \quad \text{coker}_{\mathbb{Z}}(I_{2n} - C) \cong \mathbb{Z} \oplus \text{Jac}(Y),$$

with no denominators at any step, since both flanking matrices are unimodular. So C is not merely spectrally related to Δ_p ; it is honestly row/column-equivalent to it, and any non-negative realization of C solves the realization problem for Δ_p with no loss.

3.1.2 The elementary, unconditional obstruction

Lemma 3.1.1. *For every $(q + 1)$ -regular simple graph Y on n vertices, $\text{Tr}(C) = 0$ and*

$$\text{Tr}(C^2) = -n(q - 1).$$

Proof. $\text{Tr}(C) = \text{Tr}(T_p) + \text{Tr}(0) = 0$ since Y is simple (no loops). In block form,

$$C^2 = \begin{pmatrix} T_p^2 - qI_n & -qT_p \\ T_p & -qI_n \end{pmatrix},$$

so $\text{Tr}(C^2) = \text{Tr}(T_p^2) - qn - qn = \text{Tr}(T_p^2) - 2qn$. Since T_p is symmetric with entries in $\{0, 1\}$, $\text{Tr}(T_p^2) = \sum_{i,j} (T_p)_{ij}^2 = 2|E_Y| = (q + 1)n$ by regularity. Hence $\text{Tr}(C^2) = (q + 1)n - 2qn = -n(q - 1)$. $\square$

This has been verified directly on the eleven graphs of Table 3.1, in every case with exact agreement; the lemma is general and the table is confirmation, not evidence.

Theorem 3.1.2 (Unconditional obstruction: no equal-size, equal-spectrum realization). *Let $q \geq 2$. No matrix $\tilde{C} \in M_{2n}(\mathbb{Z}_{\geq 0})$ similar to C — in particular, no non-negative matrix with the same characteristic polynomial as C — exists.*

Proof. Trace is invariant under similarity, so $\text{Tr}(\tilde{C}) = \text{Tr}(C) = 0$ and $\text{Tr}(\tilde{C}^2) = \text{Tr}(C^2) = -n(q - 1) < 0$ would be forced. But for any matrix $B \in M_N(\mathbb{Z}_{\geq 0})$, $\text{Tr}(B^2) = \sum_{i,j} B_{ij}B_{ji} \geq 0$, a sum of products of non-negative integers. Contradiction. $\square$

Remark 3.1.3. Theorem 3.1.2 needs nothing beyond Lemma 3.1.1 and the one-line non-negativity of $\text{Tr}(B^2)$ for non-negative B ; it requires no appeal to symbolic dynamics, and it already rules out any construction that keeps C 's size fixed at $2n$ — for instance, any attempt to directly reinterpret the entries of C itself, without adding vertices, as a graph adjacency matrix.

3.1.3 The obstruction to dilation of any size

The constructions actually attempted in the course of this project (auxiliary sink vertices, Drinen–Tomforde-style desingularization, cascading intermediate nodes) all enlarge the matrix, replacing C by some $\tilde{C} \in M_N(\mathbb{Z}_{\geq 0})$, $N > 2n$, related to C not by similarity but by *shift equivalence*: matrices $R \in M_{2n \times N}(\mathbb{Z})$, $S \in M_{N \times 2n}(\mathbb{Z})$ and a lag ℓ with

$$CR = R\tilde{C}, \quad SC = \tilde{C}S, \quad RS = C^\ell, \quad SR = \tilde{C}^\ell.$$

Shift equivalence is the correct and standard notion for comparing presentation matrices of possibly different sizes whose associated shifts of finite type (equivalently, Cuntz–Krieger algebras) are to be considered the same dynamical, and K -theoretic, object: the Bowen–Franks group $\text{coker}_{\mathbb{Z}}(I - \tilde{C})$ is a shift-equivalence invariant, which is exactly why enlarging the matrix could in principle have been a way to keep $\text{coker}(I - \tilde{C}) \cong \mathbb{Z} \oplus \text{Jac}(Y)$ while repairing non-negativity.

Lemma 3.1.4 (Shift equivalence preserves the nonzero spectrum). *Let $A \in M_a(\mathbb{C})$, $B \in M_b(\mathbb{C})$ be lag- ℓ shift equivalent: there exist $R \in M_{a \times b}$, $S \in M_{b \times a}$ with*

$$AR = RB, \quad SA = BS, \quad RS = A^\ell, \quad SR = B^\ell.$$

Then for every $\lambda \neq 0$ the generalized eigenspaces satisfy $\dim V_\lambda(A) = \dim V_\lambda(B)$; hence the nonzero spectra of A and B agree as multisets, and $\text{Tr}(A^k) = \text{Tr}(B^k)$ for every $k \geq 1$, including $k < \ell$.

Proof. Let $v \in V_\lambda(A) = \ker(A - \lambda)^{k_0}$, $\lambda \neq 0$. Then $(B - \lambda)^{k_0}Sv = S(A - \lambda)^{k_0}v = 0$, so S maps $V_\lambda(A)$ into $V_\lambda(B)$. If $Sv = 0$ then $A^\ell v = RSv = 0$; but A restricted to $V_\lambda(A)$ has sole eigenvalue $\lambda \neq 0$, hence is invertible there, forcing $v = 0$. So $S|_{V_\lambda(A)}$ is injective. Symmetrically, R maps $V_\lambda(B)$ injectively into $V_\lambda(A)$. Two finite-dimensional spaces injecting into each other have equal dimension. Algebraic multiplicity equals the dimension of the generalized eigenspace, and $\text{Tr}(M^k) = \sum_\lambda m_\lambda \lambda^k$ with zero eigenvalues contributing nothing for $k \geq 1$; the trace identity follows. $\square$

Theorem 3.1.5 (Unconditional obstruction: no dilation of any size). *Let $q \geq 2$. No matrix $\tilde{C} \in M_N(\mathbb{Z}_{\geq 0})$, of any size N , is shift equivalent to C . Consequently no auxiliary-vertex, desingularization, or other enlargement scheme produces a non-negative integer matrix presenting the same K -theoretic and dynamical data as C .*

Proof. By Lemma 3.1.4, shift equivalence would force $\text{Tr}(\tilde{C}^2) = \text{Tr}(C^2) = -n(q-1) < 0$, contradicting $\text{Tr}(\tilde{C}^2) = \sum_{i,j} \tilde{C}_{ij}\tilde{C}_{ji} \geq 0$. $\square$

Remark 3.1.6 (Relation to the spectral conjecture literature). An earlier draft stated Theorem 3.1.5 conditionally on the resolution of the Boyle–Handelman spectral conjecture by Kim–Ormes–Roush. That was an error of economy, not of substance: the deep content of that literature is the *converse* direction — which trace-positive spectra *are* realizable by non-negative matrices — whereas the direction needed here, invariance of the nonzero spectrum under shift equivalence, is the elementary Lemma 3.1.4. The theorem is unconditional and self-contained; the spectral-conjecture literature remains relevant context for the converse question, and its citations remain unverified by us.

Remark 3.1.7 (What is, and is not, ruled out). Theorem 3.1.5 rules out any *faithful* dilation of C — one preserving its nonzero spectrum, hence its actual Ihara/Hecke dynamical content, which is the only kind of dilation that would let the resulting graph algebra be honestly called a realization of the Hecke correspondence. It does not, and cannot, rule out the bare abstract existence of *some* unrelated non-negative matrix D with $\text{coker}_{\mathbb{Z}}(I - D) \cong \mathbb{Z} \oplus \text{Jac}(Y)$: by Kirchberg–Phillips classification, a Kirchberg algebra realizing any prescribed pair (K_0, K_1) with K_0 finitely generated and K_1 free exists unconditionally, with no relation to T_p required. Such an algebra would answer no question of interest here; the content of this paper is that the *natural* one, built from C itself, does not exist.

3.2 The virtual correspondence barrier

Given Theorems 3.1.2–3.1.5, it is natural to ask whether the obstruction can be sidestepped by weakening what counts as a “correspondence,” for instance by allowing a $\mathbb{Z}/2$ -graded or superstructure in which T_p is placed in even degree and qI_n in odd degree, with the hope that a graded Cuntz–Pimsner-type construction applied to $\mathcal{E}^+ \oplus \mathcal{E}^-$ (with $[\mathcal{E}^+] = T_p$, $[\mathcal{E}^-] = qI_n$, both honest positive correspondences) might produce a single algebra whose K_0 -group is the cokernel of $1 - ([\mathcal{E}^+] - [\mathcal{E}^-]) = \Delta_p$.

Proposition 3.2.1. *No such construction exists. More precisely: for any C^* -algebra A and any two genuine A – A correspondences $\mathcal{E}^+, \mathcal{E}^-$, the Cuntz–Pimsner algebra of the direct sum $\mathcal{E}^+ \oplus \mathcal{E}^-$ — graded or not — has K -theoretic index map $1 - [\mathcal{E}^+] - [\mathcal{E}^-]$, not $1 - [\mathcal{E}^+] + [\mathcal{E}^-]$; a $\mathbb{Z}/2$ -grading records parity for the purpose of assigning classes to KK^0 versus KK^1 , and does not introduce a sign into the index computation of a single Cuntz–Pimsner algebra built from an honestly positive module.*

Discussion, in place of a proof. The Cuntz–Pimsner construction is a functor from the category of genuine Hilbert A – A -bimodules (with a fixed left action by compacts, or by adjointable operators, as appropriate) to C^* -algebras, and its K -theory is computed, via Pimsner’s exact sequence, from the class $[\mathcal{E}]$ of the *single* bimodule fed to it, and the root of the constraint is the axiom $\langle \xi, \xi \rangle \geq 0$ of the Hilbert C^* -module inner product: every channel of the completed module carries non-negative norm, so no summand can act with a negative sign in the completion. There is no step in the construction at which a summand of the bimodule can be assigned a formal minus sign: the bimodule $\mathcal{E}^+ \oplus \mathcal{E}^-$ is a perfectly good, positive, correspondence in its own right, with class $[\mathcal{E}^+ \oplus \mathcal{E}^-] = [\mathcal{E}^+] + [\mathcal{E}^-] = T_p + qI_n$, and its Cuntz–Pimsner algebra has index map $1 - (T_p + qI_n)$, the *opposite* sign adjustment from the one wanted. Introducing a $\mathbb{Z}/2$ -grading changes which Kasparov group (KK^0 or KK^1) a resulting class is assigned to; it does not alter this computation. We are not aware of, and have not been able to construct, any variant Cuntz–Pimsner-type functor that takes a *formal difference* of two correspondences as input and returns an algebra realizing that difference as an index. Were such a functor to exist, it would apply to every virtual class in every $KK^0(A, A)$, converting every formal difference of finitely generated projective modules into a single algebra realizing it as an index map, and would in particular apply to the underlying integer matrix problem addressed by Theorems 3.1.2–3.1.5, trivializing it; the fact that those theorems hold, and are elementary, is itself evidence that no such functor exists. $\square$

Remark 3.2.2 (The obstruction is ontological, not technical). The negative block $-qI_n$ in C is not an artifact of a particular presentation that a cleverer bookkeeping device — grading, desingularization, or otherwise — could absorb. It is a faithful record of the non-backtracking correction subtracted in the Ihara recursion $T \cdot T_1 = T_2 + (q + 1)T_0$: the class $x_p = T_p - qI_n$ is the class of a correspondence *minus* the class of q copies of the identity correspondence, and Theorem 3.1.2 shows this difference is not, for any $q \geq 2$, again the class of a correspondence. A virtual class that is provably not effective cannot be made effective by re-describing it; it can only be relocated to a category that does not demand effectivity. That category is KK -theory, to which we now turn.

3.3 The K-homological bulk, and the congruence tower

3.3.1 The Hecke–Dirac operator and its index

Definition 3.3.1 (Hecke–Dirac operator). Fix an orientation E_Y^+ of the edges of Y and let $\partial: \mathbb{Z}^{E_Y^+} \rightarrow \mathbb{Z}^{V_Y}$, $\partial(f) = t(f) - o(f)$, be the simplicial boundary map, so $\partial^t: \mathbb{Z}^{V_Y} \rightarrow \mathbb{Z}^{E_Y^+}$ is its transpose (coboundary). Define the self-adjoint operator

$$D_p := \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \quad \text{on} \quad \ell^2(V_Y) \oplus \ell^2(E_Y^+), \quad D_p^2 = \begin{pmatrix} \partial\partial^t & 0 \\ 0 & \partial^t\partial \end{pmatrix},$$

i.e. $D_p = d + d^*$ is the discrete Hodge–de Rham operator of the graph, with even part the vertex (Hodge 0-)Laplacian $\partial\partial^t = \Delta_p$ and odd part the edge (Hodge 1-)Laplacian $\partial^t\partial$.

Lemma 3.3.2. $\partial\partial^t = \Delta_p = (q + 1)I - T_p$ on $\ell^2(V_Y)$.

Proof. $(\partial\partial^t)_{uv} = \sum_{f \in E_Y^+} \partial_f(u) \partial_f(v)$; the diagonal term is the degree $q + 1$ (each incident edge contributes ± 1 squared) and the off-diagonal term at $u \neq v$ is -1 for each edge joining u, v (with a sign cancellation independent of orientation choice, since $\partial_f(u)\partial_f(v) = -1$ whenever f joins u, v regardless of which endpoint is called $o(f)$), giving exactly $(q + 1)I - T_p$. $\square$

Theorem 3.3.3 (Fredholm module and index). *($C(V_Y)$, $\ell^2(V_Y) \oplus \ell^2(E_Y^+)$, D_p) is an even Fredholm module over $A := C(V_Y)$, defining a class $[D_p] \in KK^0(A, A) \cong M_n(\mathbb{Z})$, $n = |V_Y|$, and*

$$\text{ind}(D_p) := [\ker D_p] - [\text{coker } D_p] \in K_0(A), \quad \ker D_p \cong \mathbb{Z} \oplus \mathbb{Z}^{\text{rank } H_1(Y)},$$

$$\text{coker}(D_p^2|_{\ell^2(V_Y)}) \cong \mathbb{Z} \oplus \text{Jac}(Y),$$

so that the even part of the index recovers Δ_p 's cokernel exactly: K_0 -theoretically, $[D_p^2|_{\ell^2(V_Y)}]$ has cokernel $\mathbb{Z} \oplus \text{Jac}(Y)$, with no positivity requirement anywhere in the construction — D_p is self-adjoint, not positive, and ∂, ∂^t are integer matrices of arbitrary sign.

Proof. D_p is a finite self-adjoint matrix, hence trivially Fredholm (bounded, and every finite-rank perturbation of 0 is compact), giving a class in $KK^0(A, A)$ since $A = C(V_Y) = \mathbb{C}^n$ acts diagonally on $\ell^2(V_Y)$ and trivially, via a chosen embedding of V_Y -labels, on $\ell^2(E_Y^+)$ compatible with ∂ 's A -bimodule structure. $\ker D_p = \ker \partial^t \oplus \ker \partial$; $\ker \partial^t$ is the space of harmonic 0-cochains, $\cong \mathbb{Z}$ (the constants, Y connected), and $\ker \partial \cong H_1(Y; \mathbb{Z}) \cong \mathbb{Z}^{\text{rank } H_1(Y)}$, the cycle space. The even part follows from Lemma 3.3.2 and Theorem 3.5(i) of Paper I. $\square$

Remark 3.3.4 (The honest scope of this theorem). This is a correct and complete restatement, in Fredholm-module language, of results already proved by direct linear algebra in Paper I. We record it because the language matters for what comes next: D_p 's self-adjointness, not positivity, is all that KK -theory demands, and Theorem 3.1.2 shows this is not a matter of convenience but of necessity — no positive substitute for $D_p^2 - qI$ -type data exists. It is not, however, a new theorem about Y ; every computation in it already appears in Paper I under other names.

3.3.2 Why nothing new happens for a single graph

Remark 3.3.5 (Finite-dimensional triviality). For $A = C(V_Y) = \mathbb{C}^n$, $KK^0(A, A) \cong M_n(\mathbb{Z})$ as a ring under the Kasparov product, and the product of two classes represented by integer matrices is exactly their matrix product. Consequently, any identity one might hope to derive by taking a Kasparov product of $[D_p]$ with the Hecke correspondence class $[\mathcal{E}_p] = T_p$ is nothing more than the corresponding matrix identity, already available by direct computation. We tested the specific proposed identity $[D_p] \otimes [\mathcal{E}_p] \stackrel{?}{=} [D_{p^2}] + q[D_0]$ under both natural readings of $[D_p]$ (as Δ_p and as $T_p - qI$) against the matrices directly: neither matches. What is true, and is already Lemma 2.1 of Paper I, is the unadorned recursion $T_p \cdot T_p = T_2 + (q+1)I$, with no reference to D_p or to any Kasparov product required to state or prove it. **KK -theory for a single finite graph adds no content beyond $M_n(\mathbb{Z})$** ; dressing an identity in Kasparov-product language does not make it a new theorem when the underlying ring is already $M_n(\mathbb{Z})$ with ordinary matrix multiplication.

3.3.3 The congruence tower: where new content could appear

The finite-dimensionality identified in Remark 3.3.5 is specific to working at a single level Y . The genuine non-commutative geometry of the Langlands correspondence, if it is to be found at all along this route, should appear only in the limit over a *tower* of graphs.

Conjecture 3.3.6 (Congruence tower and the profinite KK -limit). *Let $(Y_k)_{k \geq 0}$ be the tower of quotient graphs $Y_k = \Gamma_0(p^k) \backslash X$ for the Bruhat–Tits tree X of $\text{PGL}_2(\mathbb{Q}_p)$, with covering maps $Y_{k+1} \rightarrow Y_k$ inducing transfer maps on vertex algebras $A_k = C(V_{Y_k})$ and hence a directed system*

$$A_0 \hookrightarrow A_1 \hookrightarrow A_2 \hookrightarrow \cdots, \quad A_\infty := \varinjlim_k A_k \cong C(\varprojlim_k V_{Y_k}),$$

with $\varprojlim_k V_{Y_k}$ a profinite (Cantor) set carrying a residual action of $\mathrm{PGL}_2(\mathbb{Z}_p)$. Then:

- (i) the Hecke–Dirac operators $D_{p,k}$ are compatible under the transfer maps up to a controlled error, defining a class $[D_p]_\infty \in KK^0(A_\infty, A_\infty)$;
- (ii) $KK^0(A_\infty, A_\infty)$ is no longer a finite matrix ring, but a genuine inductive limit incorporating the p -adic representation-theoretic structure of $\mathrm{PGL}_2(\mathbb{Z}_p)$ -modules, in which the individual finite invariants $\mathrm{Jac}(Y_k) = \mathrm{Tors} \, \mathrm{coker}(\Delta_{p,k})$ assemble into an inverse system with growth of Iwasawa shape. Here the two kinds of tower must be separated, and the separation is dictated by our own data (Section 3.5), not by preference:
 - (ii-a) Abelian deck towers. For towers carrying a genuine $\mathbb{Z}_p$ -deck action (the voltage towers of Table 3.2), the limit $\varprojlim_k \mathrm{Jac}(Y_k)[p^\infty]$ is conjectured to be a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module with characteristic element the cleared voltage pencil $g(T) = \det D(1+T)$; the finite-level evidence is exact — the growth law and the Newton-polygon identities $\mu_{\mathrm{meas}} = \mu(g)$, $\lambda_{\mathrm{meas}} = \lambda(g) - 1$ of Remarks 3.5.1 and 3.5.3 hold to the integer in all four computed towers — but the limit-module statement itself remains conjectural at the level of this paper, and its attribution in the literature is flagged in the caveats.
 - (ii-b) The congruence tower. The tower $\Gamma_0(p^k) \backslash X$ is non-normal and carries no $\mathbb{Z}_p$ -deck action, and its canonical combinatorial model, the Iwahori path tower of Remark 3.5.2, exhibits the exact pseudo-Iwasawa law $\nu_2 = 6 \cdot 2^k - k - 4$ with $\lambda = -1 < 0$, which is impossible for the characteristic ideal of any finitely generated torsion $\mathbb{Z}_p[[T]]$ -module. The naive $\mathbb{Z}_p[[T]]$ -form of this conjecture, including the tentative identification of the Ihara variable u with $1 + T$ proposed in an earlier draft, is therefore withdrawn as refuted in shape by its own model. What we conjecture in its place is that the correct base object for the congruence tower is the Iwahori–Hecke algebra $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ of the tower, acting through the degeneracy correspondences, with the finite invariants governed by a characteristic ideal over that non-commutative base, and the observed $-k$ term arising as an Euler-characteristic correction — the removal of one Eisenstein rank per level — external to the module invariants proper. We do not know the correct precise formulation, and we prefer to say so rather than to state a formula the data has already falsified;
- (iii) the trace obstruction of Theorem 3.1.5 persists at every finite level k , so the entire tower consists of virtual, non-realizable classes; whether the limiting object $[D_p]_\infty$ is subject to an analogous obstruction, or whether the profinite limit process itself provides the missing room for a genuine realization, is open.

Remark 3.3.7. We state Conjecture 3.3.6 as a research programme, not a theorem, and we have proved none of its parts; part (ii-a) is the best supported, part (ii-b) is deliberately left without a formula. For part (iii) one sharp dichotomy can already be posed, and it is the intended opening theorem of a sequel: if the limit carries a faithful invariant trace τ (type II regime), the obstruction of Theorem 3.1.5 should persist as non-negativity of the τ -weighted second moment of any would-be correspondence, killing realizations exactly as at finite level; if the limit is traceless (type III regime, as the boundary crossed products of Paper I in fact are), the numerical obstruction has no formulation at all, and the question becomes whether absence of an obstruction can be converted into an actual construction. Part (iii) is, in our view, the most important open question raised by this paper: it asks whether the No-Go theorems of Section 3.1, proved level-by-level, survive or dissolve in the projective limit. We do not know the answer, and we think this — rather than any further finite-graph computation — is the correct next object of study.

3.4 Computations

Y	n	q	$\text{Tr}(C^2)$ (computed)	$-n(q-1)$ (formula)
K_4	4	2	-4	-4
K_5	5	3	-10	-10
K_6	6	4	-18	-18
$K_{3,3}$	6	2	-6	-6
Q_3 (cube)	8	2	-8	-8
prism $C_4 \times K_2$	8	2	-8	-8
Wagner V_8	8	2	-8	-8
prism $C_5 \times K_2$	10	2	-10	-10
Möbius V_{10}	10	2	-10	-10
Petersen	10	2	-10	-10
Heawood	14	2	-14	-14

Table 3.1: $\text{Tr}(C^2)$ computed directly from the $2n \times 2n$ matrix C , against the closed form of Lemma 3.1.1, for every graph of Paper I’s Table 1. Exact agreement in all eleven cases, confirming a fact already proved in general; the table is illustrative, not evidential.

3.5 A first computed instance of the tower: abelian $\mathbb{Z}_\ell$ -covers

Conjecture 3.3.6 concerns the congruence tower $\Gamma_0(p^k) \backslash X$, which is neither Galois nor abelian and whose explicit construction requires quaternionic lattice data we do not assemble here. But part (ii) of the conjecture makes a *shape* prediction — Iwasawa-type growth of the torsion — and that shape can be tested rigorously today in the nearest fully constructible situation: abelian $\mathbb{Z}/\ell^k$ -covers of a fixed base graph, built from a voltage assignment $\alpha: E_Y \rightarrow \mathbb{Z}$ on the chords of a spanning tree, with derived graphs Y_k on $n\ell^k$ vertices and compatible covering maps $Y_{k+1} \rightarrow Y_k$. Every Y_k is again $(q+1)$ -regular, so the entire theory of Papers I–II applies level by level; in particular the obstruction density of Theorem 3.1.5 is constant along the tower, $\text{Tr}(C_k^2)/n_k = -(q-1)$ for every k , so no naïve normalization dilutes the No-Go theorem in the limit.

Two independent exact methods were used and cross-checked at every level where both run: the reduced-Laplacian determinant and Smith normal form of the derived graph directly, and the character factorization

$$\kappa(Y_k) = \frac{\kappa(Y_0)}{\ell^k} \prod_{j=1}^{\ell^k-1} \det D(\zeta_{\ell^k}^j), \quad D(t) := (q+1)I_n - A(t),$$

with $A(t)$ the voltage adjacency pencil, the product computed exactly as a resultant of integer polynomials. The two methods agreed exactly in all twelve cross-checked levels.

Remark 3.5.1 (What the table shows, and what it does not). Three points, in decreasing order of confidence. (1) The growth law itself, for abelian ℓ -towers of graphs, is a known theorem of Vallières and of McGown–Vallières [8, 9], with closely related Iwasawa-theoretic structure in Gonet [10]; the bibliographic records are now verified and their titles corroborate the attribution, though the theorems themselves have not been checked against the papers’ texts here. Our computations

voltages on K_4	ℓ	μ	λ	ν	exact law, verified range
(1, 0, 0)	2	3	1	1	$\text{ord}_2 \kappa(Y_k) = 3 \cdot 2^k + k + 1, 2 \leq k \leq 9$
(1, 1, 1)	2	0	3	4	$\text{ord}_2 \kappa(Y_k) = 3k + 4, 0 \leq k \leq 9$
(1, 2, 0)	2	2	3	3	$\text{ord}_2 \kappa(Y_k) = 2^{k+1} + 3k + 3, 2 \leq k \leq 9$
(1, 0, 0)	3	0	1	0	$\text{ord}_3 \kappa(Y_k) = k, 0 \leq k \leq 6$

Table 3.2: Exact ℓ -adic valuations of $|\text{Jac}(Y_k)| = \kappa(Y_k)$ along four $\mathbb{Z}/\ell^k$ voltage towers over K_4 ($q = 2$). In each tower the Iwasawa law $\text{ord}_\ell \kappa(Y_k) = \mu\ell^k + \lambda k + \nu$ holds *exactly* for every computed level in the stated range; the parameters were fitted from two increments and the remaining six to eight levels are genuine verification, not fit.

confirm that theory in four instances; they do not discover it. (2) The Smith normal forms refine the order-growth law structurally: in the $\mu = 0$ tower (1, 1, 1) the 2-primary part of $\text{Jac}(Y_k)$ is, for $1 \leq k \leq 3$,

$$\text{Jac}(Y_k)[2^\infty] \cong \mathbb{Z}/2^k \oplus \mathbb{Z}/2^{k+2} \oplus \mathbb{Z}/2^{k+2},$$

exactly $\lambda = 3$ cyclic factors each growing by one power of ℓ per level, while in the $\mu = 3$ tower the *number* of cyclic factors itself grows (5 factors at $k = 2$, 9 at $k = 3$): structurally, λ counts persistently growing cyclic factors and $\mu > 0$ manifests as exponential proliferation of new ones. We do not know whether this structural statement in this form appears in the graph-Iwasawa literature. (3) Positive μ occurs freely here ($\mu = 3$ and $\mu = 2$ above), in genuine contrast to the cyclotomic number-field setting, where $\mu = 0$ is the Ferrero–Washington theorem; whatever the correct formulation of Conjecture 3.3.6(ii) for the congruence tower turns out to be, it cannot import the $\mu = 0$ expectation from number fields.

Remark 3.5.2 (The Iwahori path tower: exact data against the naive conjecture). The canonical combinatorial model of the congruence tower itself is constructible: take level- k vertices to be the non-backtracking paths of length k in Y_0 (level 1 is the Hashimoto digraph of A^{nb} , level $k \geq 2$ its iterated line digraphs), so that $|V(Y_k)| = n(q+1)q^{k-1}$ reproduces the index $[\Gamma(1) : \Gamma_0(p^k)] = (p+1)p^{k-1}$ exactly. For $Y_0 = K_4$, $q = 2$, the sandpile groups of these Eulerian digraphs were computed exactly for $0 \leq k \leq 6$ (up to 384 vertices). Two exact laws emerge. First, a Levine-type recursion, $|\text{sandpile}(Y_{k+1})| = 2^{n_k-1} |\text{sandpile}(Y_k)|$, verified at five consecutive levels. Second, the valuation obeys a pseudo-Iwasawa law exactly for $2 \leq k \leq 6$:

$$\nu_2(|\text{sandpile}(Y_k)|) = 6 \cdot 2^k - k - 4, \quad \text{i.e.} \quad (\mu, \lambda, \nu) = (6, -1, -4).$$

The invariant $\lambda = -1$ is *negative*, which is impossible for the characteristic ideal of a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module. The canonical combinatorial model of the congruence tower therefore *refutes the naive $\mathbb{Z}_p[[T]]$ -form of part (ii) of Conjecture 3.3.6 as stated above*: whatever governs the congruence tower, it is not an ordinary Iwasawa module — consistently with the fact that this tower carries no $\mathbb{Z}_p$ -deck action, being non-normal. The -1 has a transparent source, the removal of one Eisenstein (trivial-direction) rank per level, an Euler-characteristic correction rather than a module invariant; the corrected form of the conjecture should be stated either over the Iwahori–Hecke algebra of the tower or as the abelian-deck statement, which the data of Table 3.2 and the Newton test below confirm exactly. Structurally, the Smith normal forms at $k \leq 3$ show the same three persistent growing factors $\mathbb{Z}/2^k \oplus (\mathbb{Z}/2^{k+2})^2$ as the abelian (1, 1, 1) tower, plus an exponentially proliferating cloud of bounded factors (8 copies of $\mathbb{Z}/2$ at $k = 2$; 12 copies of $\mathbb{Z}/2$

and 8 of $\mathbb{Z}/4$ at $k = 3$): λ counts persistent factors, μ the proliferation rate, and the negative correction lives outside both.

Remark 3.5.3 (Newton-polygon confirmation of the characteristic element). In the four abelian towers of Table 3.2 the characteristic-element mechanism was tested directly: with $g(T)$ the cleared voltage pencil $\det D(1 + T) \in \mathbb{Z}[T]$, the Newton data give, in all four cases exactly,

$$\mu_{\text{measured}} = \mu(g), \quad \lambda_{\text{measured}} = \lambda(g) - 1,$$

and the uniform offset -1 is fully accounted: $t \mapsto t^{-1}$ symmetry of the pencil forces a double zero $T^2 \mid g$ at the trivial character, contributing $+2k$ to the valuation sum, while the normalization $\kappa(Y_k) = \kappa(Y_0)\ell^{-k} \prod_{\zeta \neq 1} \det D(\zeta)$ removes k ; net $+k$, i.e. $\lambda(g) - 2 + 1$. Within the abelian regime the graph-theoretic p -adic L -function is thus confirmed to the integer.

Remark 3.5.4 (Relation to Conjecture 3.3.6). These towers are a rigorous toy, not the conjecture: the congruence tower is non-normal and its covering combinatorics are those of the full tree, not of an abelian voltage group. What the computation establishes is only this: the Iwasawa shape asserted in part (ii) is exactly right in the nearest constructible abelian analogue, down to the integer, with the characteristic-ideal mechanism (here, the resultant of the voltage pencil against $t^{\ell^k} - 1$) functioning precisely as the graph-theoretic p -adic L -function. Part (iii) — the fate of the trace obstruction in the projective limit — is untouched by any of this, and remains the paper’s central open question.

Acknowledgement of provenance and caveats

This is an unrefereed preprint. Theorem 3.1.2 and Theorem 3.3.3 are proved here in full from elementary linear algebra and require no external citation. Theorem 3.1.5 depends on the claim that shift equivalence forces agreement of the full nonzero spectrum of two integer matrices, which we attribute to the resolution of the Boyle–Handelman spectral conjecture by Kim, Ormes and Roush (both names corrected from an earlier draft’s “Boyer” and “Ozawa”); this attribution has not been checked against the primary literature, since the authors had no network access while writing this paper, and should be verified independently before further citation. Proposition 3.2.1 is an argument, not a theorem in the usual sense: it asserts the non-existence of a construction, supported by an impossibility argument (such a functor would trivialize Theorems 3.1.2–3.1.5) rather than a proof ruling out every conceivable construction. Conjecture 3.3.6 is entirely open; none of its three parts is established here, and part (iii) is explicitly posed as the paper’s main unresolved question. All computations of Section 3.4 are exact and were checked independently of the general proof of Lemma 3.1.1. The tower computations of Section 3.5 are exact integer computations, cross-checked between two independent methods at twelve levels. Bibliography status: every external entry is now verified against its primary source or its Crossref record, with the DOI recorded in place. The three entries that had returned mutually contradictory data in two earlier out-of-band passes (Vallières, McGown–Vallières, Gonet) are resolved, and the resolution is chastening: *all three* earlier variants were wrong, in the Gonet case down to the title and the journal. Residual soft spots are noted inline: the Williams end page (Crossref records the start page only), and the Pimsner chapter year (1996 electronic versus 1997 volume). One distinction is maintained deliberately: what has been verified is the bibliographic *metadata*; the attribution of the abelian growth law to these works rests on their (now corroboratively titled) records and on recollection of their content, not on a reading of their theorems performed within this verification pass. The possibility that the structural refinement of Remark 3.5.1(2) is known to their authors stands as

before. Two author-name errata were applied: “Boyer–Handelman” corrected to Boyle–Handelman, and “Kim–Ozawa–Roush” corrected to Kim–Ormes–Roush; the latter was the present authors’ own error, and both corrected names should themselves be confirmed at the primary-source stage.

Chapter 4

Algorithmic non-commutative class field theory, VI: the shadow algebra — a functorial Kirchberg realization of the critical group

Summary. The central open problem of this series — posed in [Ch. 5, Rem. 1.9] and sharpened in [Ch. 6, §6] — asked for a Kirchberg C^* -algebra, functorial in the Hecke data of a finite $(q+1)$ -regular graph Y , whose K_0 -group realizes the Laplacian-channel invariant $\mathbb{Z} \oplus \text{Jac}(Y)$, with $|\text{Jac}(Y)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ a product of Frobenius point counts. Papers II–III proved two No-Go theorems that appeared to surround the problem: no non-negative matrix of any size is shift equivalent to the Ihara companion C (the trace obstruction $\text{Tr}(C^2) = -n(q-1) < 0$), and the natural tail-groupoid algebra is arithmetically blind. This paper solves the single-level problem, inside the exact gap those theorems leave open: shift equivalence governs spectra, not cokernels. Define the *shadow graph* $Y^\sharp$ on $2n$ vertices, with adjacency

$$B(Y) = \begin{pmatrix} T_p & I_n \\ qI_n & 2I_n \end{pmatrix} :$$

the original graph, one arc from each vertex to its shadow, q return arcs, and a double loop at each shadow. The double loops are unimodular pivots of determinant -1 , and a two-line block factorization gives $U(I - B^t)V = \Delta_p \oplus (-I_n)$ with explicit integer unimodular U, V ; hence the graph algebra $\mathcal{O}_{B(Y)}$ is a unital Kirchberg algebra with

$$K_0(\mathcal{O}_{B(Y)}) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad K_1(\mathcal{O}_{B(Y)}) \cong \mathbb{Z},$$

functorially in Y . Being purely infinite it carries no trace, so the obstruction of [Ch. 3] has no formulation on it; and it does not contradict [Ch. 3], because $B(Y)$ shares with the Ihara pencil only its Bowen–Franks data, not its spectrum — the exact reconciliation is the per-eigenvalue identity $(1 - 2u)(1 - u\lambda) - qu^2|_{u=1} = -(q+1-\lambda)$. What remains open, and what [Ch. 3] proves is impossible, are stated with equal precision: tower-level functoriality is open; a *Hecke-dynamical* realization is impossible, which is the sense in which the present construction is maximal.

Status of the results

Theorems 4.2.1 and 4.3.1 are proved in full; the unimodular factorization at the heart of Theorem 4.2.1 is a two-line block computation reproduced below and machine-verified, together with the Smith-form match against [Ch. 2, Table 1], on K_4 , K_5 , $K_{3,3}$ and Petersen (Table 4.1). The classical inputs — simplicity and pure infiniteness criteria for Cuntz–Krieger/graph algebras and their K -theory — are quoted with a flagged citation. Section 4.4 states exactly what is solved, what is open, and what is impossible; one error in the task specification that prompted this paper is corrected there (Remark 4.4.2).

4.1 The shadow graph

Throughout, Y is a finite connected $(q + 1)$ -regular graph on n vertices, $q \geq 2$, with adjacency (Hecke) matrix T_p and Hecke–Laplacian $\Delta_p = (q + 1)I - T_p$, whose cokernel is $\mathbb{Z} \oplus \text{Jac}(Y)$ with $\ker \Delta_p = \mathbb{Z}$ ([Ch. 2, Thm. 3.5]).

Definition 4.1.1 (Shadow graph). The *shadow graph* $Y^\sharp$ is the directed graph on vertex set $V_Y \sqcup V_Y'$ (each v paired with its shadow v') with arcs:

- (i) $v \rightarrow w$ for every edge vw of Y , in both orientations (the T_p block);
- (ii) one arc $v \rightarrow v'$ for every v ;
- (iii) q parallel arcs $v' \rightarrow v$ for every v ;
- (iv) two loops at every shadow v' .

Its adjacency matrix is $B(Y) = \begin{pmatrix} T_p & I \\ qI & 2I \end{pmatrix}$, and the *shadow algebra* is the graph C^* -algebra $\mathcal{O}_{B(Y)}$.

The design principle is worth stating in one sentence, because it is precisely the loophole left by the No-Go theorems of [Ch. 3]. The obstruction $\text{Tr}(C^2) = -n(q - 1) < 0$ forbids any non-negative matrix from carrying the *spectrum* of the Ihara companion — the $-qI$ block cannot be dilated away. But K -theory sees only the *cokernel*, which is a unimodular-equivalence invariant, and unimodular row operations, unlike similarities, may be implemented by pivots of determinant -1 : a shadow vertex with a double loop contributes the 1×1 block $1 - 2 = -1$, and Schur elimination against it *adds* q back to the diagonal — exactly the operation that is spectrally impossible and cohomologically free.

4.2 The K -theory

Theorem 4.2.1 (Functorial realization). *With $U = \begin{pmatrix} I & -qI \\ 0 & I \end{pmatrix}$ and $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$, both integer unimodular,*

$$U(I_{2n} - B(Y)^t)V = \Delta_p \oplus (-I_n). \quad (4.2.1)$$

Consequently

$$K_0(\mathcal{O}_{B(Y)}) = \text{coker}_{\mathbb{Z}}(I - B(Y)^t) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad K_1(\mathcal{O}_{B(Y)}) = \ker_{\mathbb{Z}}(I - B(Y)^t) \cong \mathbb{Z},$$

and the assignment $Y \mapsto \mathcal{O}_{B(Y)}$ is functorial under graph isomorphisms: any automorphism π of Y satisfies $(\pi \oplus \pi)B(Y)(\pi \oplus \pi)^{-1} = B(Y)$, inducing an action on $\mathcal{O}_{B(Y)}$ compatible with the induced action on $\mathbb{Z} \oplus \text{Jac}(Y)$.

Proof. $B^t = \begin{pmatrix} T_p & qI \\ I & 2I \end{pmatrix}$, so $I - B^t = \begin{pmatrix} I - T_p & -qI \\ -I & -I \end{pmatrix}$. Two lines of block multiplication:

$$U(I - B^t) = \begin{pmatrix} I & -qI \\ 0 & I \end{pmatrix} \begin{pmatrix} I - T_p & -qI \\ -I & -I \end{pmatrix} = \begin{pmatrix} (q+1)I - T_p & 0 \\ -I & -I \end{pmatrix},$$

$$U(I - B^t)V = \begin{pmatrix} \Delta_p & 0 \\ -I & -I \end{pmatrix} \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix} = \begin{pmatrix} \Delta_p & 0 \\ 0 & -I \end{pmatrix}.$$

Cokernel and kernel are unimodular-equivalence invariants, and $\text{coker}(\Delta_p \oplus (-I)) = \text{coker } \Delta_p = \mathbb{Z} \oplus \text{Jac}(Y)$, $\ker = \ker \Delta_p = \mathbb{Z}$ by [Ch. 2, Thm. 3.5(i)]. The K -theory formula for graph C^* -algebras is the classical $K_0 = \text{coker}(I - B^t)$, $K_1 = \ker(I - B^t)$ [11]. Equivariance is the visible block symmetry, verified explicitly for a transposition on K_4 . $\square$

4.3 The Kirchberg properties

Theorem 4.3.1 (Shadow algebra is Kirchberg). *For Y connected and $q \geq 1$, the matrix $B(Y)$ is irreducible (the shadow graph is strongly connected), is not a permutation matrix, and has a vertex — any shadow v' — carrying two distinct loops. Consequently $\mathcal{O}_{B(Y)}$ is separable, nuclear, simple, purely infinite and in the UCT class: a unital Kirchberg algebra. Being purely infinite, it admits no nonzero trace, finite or semifinite; the trace obstruction of [Ch. 3, Thms. 1.5–1.6] therefore has no formulation on $\mathcal{O}_{B(Y)}$.*

Proof. Strong connectivity: within the first layer all vertices are mutually reachable through the arcs of the connected graph Y ; layers communicate through $v \rightarrow v' \rightarrow v$. The double loop at v' gives two distinct return paths at one vertex and period 1. Irreducibility with a non-permutation matrix and the double-loop witness place $B(Y)$ under the classical simplicity and pure infiniteness criteria for Cuntz–Krieger algebras, and graph algebras of finite graphs are separable, nuclear and satisfy the UCT [11]. Pure infiniteness excludes traces. Machine confirmation of the witnesses on all four test graphs: Table 4.1. $\square$

4.4 Non-contradiction, and the exact scope of the solution

Proposition 4.4.1 (Why this does not contradict the No-Go theorems). *$B(Y)$ is not shift equivalent to the Ihara companion C of [Ch. 3]: already $\text{Tr } B(Y) = 2n \neq 0 = \text{Tr } C$, and Lemma [Ch. 3, 1.4] makes traces shift-equivalence invariants. The shadow algebra shares with the Ihara pencil its Bowen–Franks data and nothing spectral: per eigenvalue λ of T_p ,*

$$\det(I - uB(Y)) = \prod_{\lambda} [(1 - 2u)(1 - u\lambda) - qu^2], \quad (1 - 2u)(1 - u\lambda) - qu^2 \Big|_{u=1} = -(q + 1 - \lambda),$$

so the $u = 1$ specialization of the shadow zeta reproduces the Laplacian spectrum up to sign, while for $u \neq 1$ the two polynomial families differ. The construction lives exactly in the gap the No-Go theorems left open: shift equivalence constrains spectra; K -theory is a cokernel invariant; and Remark [Ch. 3, after Thm. 1.6] anticipated precisely this distinction when it separated existence from canonicity.

Proof. The block determinant is computed with the commuting-block formula (three of the four blocks are scalars), and the $u = 1$ evaluation is $(1 - 2)(1 - \lambda) - q = \lambda - 1 - q$. Traces: immediate. $\square$

Remark 4.4.2 (A correction to the task specification). The specification that prompted this paper asked for $K_1 \cong \mathbb{Z}^r$. That is the *boundary*-channel invariant ([Ch. 2, Thm. 3.7]), not the Laplacian-channel one: the target fixed by [Ch. 2, Thm. 3.5], [Ch. 5, Rem. 1.9] and [Ch. 6] is $K_1 = \ker \Delta_p = \mathbb{Z}$, and that is what Theorem 4.2.1 delivers. Realizing $\mathbb{Z}^r$ in K_1 alongside $\mathbb{Z} \oplus \text{Jac}(Y)$ in K_0 would require a non-square-type presentation and a different target; it is not the series' problem.

Remark 4.4.3 (What is solved, what is open, what is impossible). *Solved*: the single-level canonicity problem of [Ch. 5, Rem. 1.9] — a Kirchberg algebra functorial in the Hecke datum T_p , traceless, with $K_0 \cong \mathbb{Z} \oplus \text{Jac}(Y)$, $K_1 \cong \mathbb{Z}$, the torsion order $\frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ carried on the nose. *Open*: tower-level functoriality — whether the tail norms of [Ch. 9, Thm. 3.1] lift to $*$ -homomorphisms or correspondences $\mathcal{O}_{B(Y_{k+1})} \rightarrow \mathcal{O}_{B(Y_k)}$ inducing the norm maps of [IV–V] on K_0 ; nothing here constructs such lifts. *Impossible*: a *Hecke-dynamical* realization, i.e. an algebra whose gauge dynamics is the Ihara flow itself — that is exactly what [Ch. 3, Thm. 1.6] forbids for every non-negative presentation of every size. The shadow algebra is Hecke-functorial, not Hecke-dynamical, and by [Ch. 3] no algebra can be both: the construction is maximal in a theorem-certified sense.

Remark 4.4.4 (Normalization). The construction uses the minimal unimodular pivot: loop weight 2 is the least m with $\det(1 - m) = -1$, and the off-diagonal pair $(1, q)$ is one of the factorizations $\alpha\beta = q$; any such choice yields the same K -theory by the same factorization. We fix $(1, q, 2)$ as the normal form and make no uniqueness claim beyond it.

4.5 The verification battery

Y	n	q	(S) factorization (4.2.1)	$\text{Jac}(Y)$ from $\text{SNF}(I - B^t)$	vs [Ch. 2, Table 1]
K_4	4	2	exact	$(\mathbb{Z}/4)^2$	match
K_5	5	3	exact	$(\mathbb{Z}/5)^3$	match
$K_{3,3}$	6	2	exact	$\mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$	match
Petersen	10	2	exact	$\mathbb{Z}/2 \oplus (\mathbb{Z}/10)^3$	match

Table 4.1: Machine verification (`hecke_kirchberg.py`): the explicit factorization (4.2.1) holds as an exact matrix identity; the invariant factors of $I - B(Y)^t$ reproduce [Ch. 2, Table 1] exactly; strong connectivity, the double-loop witness, $\text{Tr } B = 2n \neq 0$, and the $u=1$ reconciliation identity all verified; automorphism equivariance spot-checked on K_4 .

Acknowledgement of provenance and caveats

Unrefereed preprint. Theorems 4.2.1–4.3.1 and Proposition 4.4.1 are proved; the two-line factorization (4.2.1) can be checked by hand and was machine-verified on four graphs. The classical Cuntz–Krieger inputs (graph-algebra K -theory; simplicity and pure infiniteness criteria) are quoted through [11]. Remark 4.4.3 delimits the result: single-level canonicity solved, tower functoriality open, Hecke-dynamical realization impossible by [Ch. 3].

Chapter 5

Algorithmic non-commutative class field theory, III: the Iwahori–Hecke tower and the type III dissolution of the trace obstruction

Summary. Papers I–II established that the arithmetic of the GL_2 Hecke–Laplacian $\Delta_p = (q+1)I - T_p$ — the critical group $\mathrm{Jac}(Y)$ of order $\frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ — is carried by a virtual KK -class whose companion matrix C satisfies $\mathrm{Tr}(C^2) = -n(q-1) < 0$, so that no non-negative C^* -correspondence of any size realizes it (the unconditional trace obstruction, [Ch. 3, Thm. 1.6]). This paper takes the projective limit over the tower and determines what survives.

Section 5.1 proves that the geometric tail equivalence relation on the Cantor limit $\Omega = \varprojlim_k V_{Y_k}$, with its harmonic measure ν , has Radon–Nikodym cocycle valued in $q^{\mathbb{Z}}$ and asymptotic ratio set $r_\infty = q^{\mathbb{Z}} \cup \{0\}$, computed in full by a density argument; by Krieger’s theorem the associated von Neumann algebra is the hyperfinite factor of type $\mathrm{III}_{1/q}$, the Powers factor $R_{1/q}$, and for Satake-twisted cocycle classes with range $\mu_0^{\mathbb{Z}}$ the same computation gives type $\mathrm{III}_{\mu_0 \wedge \mu_0^{-1}}$.

A type III factor carries no faithful normal semifinite trace, so the inequality $\mathrm{Tr}(\tilde{C}^2) \geq 0$ that powered the finite No-Go theorems has no formulation on the limit: the obstruction dissolves as a *sentence*, not as a defeated inequality. We prove equally that the dissolution buys less than it appears to: any inductive-limit realization by finite-level correspondences still dies level by level, and the weak closure that dissolves the trace also annihilates the K -theory the programme wants to realize — the correct target is a traceless Kirchberg C^* -limit, not a von Neumann algebra, and we say so.

Section 5.2 formalizes the finding of [Ch. 3, Rem. 5.2]: the congruence (Iwahori path) tower obeys the exact pseudo-Iwasawa law $\nu_2 = 6 \cdot 2^k - k - 4$ with $\lambda = -1$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal. The one-step recursion $|\mathrm{sandpile}(Y_{k+1})| = q^{n_k(q-1)-1} |\mathrm{sandpile}(Y_k)|$ — whose exponent corrects the $q = 2$ special form of earlier drafts — is proved here outright, self-containedly, from the AB/BA spectral identity for line digraphs together with the equal-cofactor eigenvalue-product formula for Eulerian Laplacians, making Section 5.2 unconditional; the full law follows in two lines, and $\lambda = -1$ is identified as $-\chi$ for the one-dimensional Eisenstein kernel removed at each level of the degeneracy correspondences: an Euler-characteristic correction over the Iwahori–Hecke base, external to any commutative Iwasawa module. A closing outlook section states, without derivation, the p -adic horizon.

Status of the results

Theorem 5.1.5 and its supporting lemmas are proved here in full up to two classical inputs, quoted and flagged: Krieger's ratio-set criterion for the type of an ergodic amenable relation, and the uniqueness of the hyperfinite type III_λ factor for $0 < \lambda < 1$. Corollary 5.1.7 and Proposition 5.1.8 are complete. Section 5.2 is now unconditional: the line-digraph recursion, previously a five-level computational hypothesis, is proved in full as Theorem 5.2.1, self-containedly and with the exponent corrected from $n_k - 1$ (a $q = 2$ artifact) to $n_k(q - 1) - 1$; the computational confirmations ($q = 2$, five levels; $q = 3$, one instance) now confirm a theorem rather than support a hypothesis. Section 5.3 derives nothing beyond one exactly verified bookkeeping identity, clearly labelled.

5.1 The type of the limit, and what dissolves

5.1.1 The Cantor limit and its harmonic measure

Throughout, $q \geq 2$. Let X be the $(q + 1)$ -regular tree with base vertex x_0 and let

$$\Omega = \{1, \dots, q + 1\} \times \{1, \dots, q\}^{\mathbb{N}}$$

be its geodesic boundary in rooted coordinates: the first symbol chooses among the $q + 1$ edges at x_0 , every later symbol among the q non-backtracking continuations. Under the identifications of [Ch. 2, §1], $\Omega \cong \partial X \cong \varprojlim_k V_{Y_k}$ for the Iwahori path tower, a Cantor set. The *harmonic measure* ν is the product measure with uniform factors: mass $\frac{1}{q+1}$ on the first coordinate, $\frac{1}{q}$ on each subsequent one, so that every cylinder $[w]$ of length $n \geq 1$ has $\nu([w]) = \frac{1}{q+1}q^{-(n-1)}$.

Definition 5.1.1 (Tail relation and degree cocycle). $\mathcal{R}$ is the tail equivalence relation on Ω : $\omega \sim \omega'$ iff $\sigma^m \omega = \sigma^n \omega'$ for some $m, n \geq 0$, where σ deletes the first symbol. The *degree cocycle* of the pair (ω, ω') realized at (m, n) is $c = m - n \in \mathbb{Z}$; it is well defined on $\mathcal{R}$. $\mathcal{R}_0 \subset \mathcal{R}$ is the degree-zero subrelation.

Lemma 5.1.2 (Radon–Nikodym cocycle). ν is quasi-invariant under $\mathcal{R}$, and every partial isomorphism $\gamma \in [[\mathcal{R}]]$ of degree $c(\gamma)$ has

$$\frac{d\nu \circ \gamma}{d\nu} = q^{c(\gamma)} \quad \nu\text{-a.e. on its domain.}$$

In particular the cocycle takes values in the discrete group $q^{\mathbb{Z}} \subset \mathbb{R}_{>0}$, and $\mathcal{R}_0$ preserves ν .

Proof. It suffices to check the generating partial isomorphisms $T_{u,v}: [u] \rightarrow [v]$, $(u\xi) \mapsto (v\xi)$, for finite words u, v of lengths $m, n \geq 1$: these have degree $m - n$ and scale every cylinder inside $[u]$ by $\nu([v])/ \nu([u]) = q^{m-n}$. Every element of $[[\mathcal{R}]]$ is a countable disjoint union of restrictions of such maps. $\square$

Lemma 5.1.3 (Ergodicity). $\mathcal{R}_0$, hence $\mathcal{R}$, is ergodic on (Ω, ν) .

Proof. A Borel set invariant under $\mathcal{R}_0$ is invariant under every change of finitely many initial coordinates (compose $T_{u,v}$ with $|u| = |v|$), hence is a tail event for the product measure ν ; Kolmogorov's zero–one law gives measure 0 or 1. $\square$

5.1.2 The ratio set

Recall Krieger's asymptotic ratio set $r_\infty(\mathcal{R}, \nu) \subseteq [0, \infty)$: $\lambda \in r_\infty$ iff for every $\varepsilon > 0$ and every Borel A with $\nu(A) > 0$ there is $\gamma \in [[\mathcal{R}]]$ with domain $B \subseteq A$ of positive measure, $\gamma(B) \subseteq A$, and $|d\nu \circ \gamma / d\nu - \lambda| < \varepsilon$ on B .

Theorem 5.1.4 (Ratio set of the geometric tail relation). $r_\infty(\mathcal{R}, \nu) = q^{\mathbb{Z}} \cup \{0\}$.

Proof. Upper bound. By Lemma 5.1.2 the derivative of every partial isomorphism is ν -a.e. $q^{\mathbb{Z}}$ -valued, and $q^{\mathbb{Z}}$ is closed and discrete in $\mathbb{R}_{>0}$; hence $r_\infty \cap \mathbb{R}_{>0} \subseteq q^{\mathbb{Z}}$.

Lower bound: $q \in r_\infty$. Let A have positive measure and let $0 < \delta < \frac{1}{3}$. By the martingale (Lebesgue density) theorem for the cylinder filtration, there is a point of A at which the cylinder densities converge to 1; choose nested cylinders $[u] \supset [us]$ (with $|u| \geq 1$, s the next symbol of that point) satisfying

$$\nu(A \cap [u]) > (1 - \delta) \nu([u]), \quad \nu(A \cap [us]) > (1 - \delta) \nu([us]).$$

Let $T = T_{us,u}: [us] \rightarrow [u]$, the degree-(+1) tail map, with constant derivative q by Lemma 5.1.2. The bad set inside the domain is

$$[us] \setminus A \text{ of measure } < \delta \nu([us]), \quad T^{-1}([u] \setminus A) \text{ of measure } = \frac{1}{q} \nu([u] \setminus A) < \delta \nu([us]),$$

so $B := (A \cap [us]) \cap T^{-1}(A \cap [u])$ has $\nu(B) > (1 - 2\delta) \nu([us]) > 0$, and $T|_B$ maps a positive-measure subset of A into A with derivative exactly q . Hence $q \in r_\infty$; applying the same argument to T^{-1} and to compositions gives $q^{\mathbb{Z}} \subseteq r_\infty$.

The value 0. Since $r_\infty \cap \mathbb{R}_{>0} = q^{\mathbb{Z}} \neq \{1\}$, the relation is not of type II, and for an ergodic relation this forces $0 \in r_\infty$ by Krieger's dichotomy; alternatively, iterating $T_{us^k,u}$ realizes derivatives $q^k \rightarrow \infty$ on shrinking sets, whose inverses realize derivatives $\rightarrow 0$. $\square$

5.1.3 The factor and its type

Theorem 5.1.5 (Type of the limit). *Let $M_\infty := W^*(\mathcal{R}, \nu)$ be the Feldman–Moore von Neumann algebra of the geometric tail relation. Then M_∞ is the amenable (hyperfinite) factor of type $\text{III}_{1/q}$, isomorphic to the Powers factor $R_{1/q}$. Moreover, for any quasi-invariant measure class on Ω whose Radon–Nikodym cocycle on $\mathcal{R}$ takes values in a subgroup $\mu_0^{\mathbb{Z}} \subset \mathbb{R}_{>0}$ with $\mu_0 \neq 1$ and for which the density argument of Theorem 5.1.4 applies verbatim — in particular for the Satake-twisted cocycle classes of [Ch. 2, §2] whose weights range in $q^{-\beta\mathbb{Z}}$ — the factor is of type III_λ with $\lambda = \min(\mu_0, \mu_0^{-1})$; for the twisted classes, $\text{III}_{q^{-\beta}}$.*

Proof. $\mathcal{R}$ is a countable Borel equivalence relation with countable classes, generated by the maps $T_{u,v}$, hence amenable as an increasing union of finite subrelations (the level- N truncations of [Ch. 2, Def. 1.6]); ν is quasi-invariant (Lemma 5.1.2) and ergodic (Lemma 5.1.3), so M_∞ is an amenable factor by Feldman–Moore. Krieger's theorem identifies the type of an ergodic amenable relation from its ratio set: $r_\infty \cap \mathbb{R}_{>0} = \lambda^{\mathbb{Z}}$ with $0 < \lambda < 1$ gives type III_λ . Theorem 5.1.4 gives $\lambda = 1/q$. Uniqueness of the amenable type III_λ factor for $0 < \lambda < 1$ (Connes) identifies M_∞ with the Araki–Woods/Powers factor $R_{1/q}$. The twisted statement repeats the proof with q replaced by μ_0 wherever the cocycle value enters; nothing else in the argument refers to the numerical value. $\square$

Remark 5.1.6 (A correction to the loose phrasing of [Ch. 2, Thm. 2.4(iii)]). Paper I stated a family “type $\text{III}_{q-\beta}$ ” for the boundary system. For the *untwisted* geometric cocycle this is misleading as a family: the uniform measure is the unique conformal probability class for every β (all branch weights being equal, the normalized cylinder masses are β -independent), the KMS point on the boundary quotient is unique, and the factor type is $\text{III}_{1/q}$, full stop. A genuine β -family of types arises only when the cocycle itself is deformed — the Satake-twisted classes of Theorem 5.1.5 — and the correct statement is the cocycle-range statement proved above. We record this as a correction of emphasis to [Ch. 2], not of any equation.

5.1.4 Dissolution, and its exact purchase

Corollary 5.1.7 (Dissolution of formulation). *M_∞ admits no faithful normal semifinite trace. Consequently no inequality of the form $\tau(x^*x) \geq 0$ with τ a trace is available on M_∞ , and the finite-level obstruction mechanism of [Ch. 3, Thms. 1.5–1.6] — non-negativity of $\text{Tr}(\cdot^2)$ against $\text{Tr}(C^2) = -n(q-1) < 0$ — has no formulation on the weak closure of the tower. The obstruction is not overcome; the sentence expressing it ceases to exist.*

Proof. Type III means precisely the absence of a nonzero normal semifinite trace; the rest is the observation that the finite-level argument is a statement about a trace. $\square$

Proposition 5.1.8 (Inductive persistence). *Let $(\mathcal{E}_k)_{k \geq 0}$ be genuine C^* -correspondences over the level algebras A_k , compatible with the tower transfers, such that $[\mathcal{E}_k] \in KK^0(A_k, A_k)$ equals the level- k companion class C_k (equivalently, realizes $\Delta_{p,k}$ as its Pimsner index map). Then no such system exists: already each single level is impossible by [Ch. 3, Thm. 1.6], and a fortiori no inductive limit $\mathcal{E}_\infty = \varinjlim \mathcal{E}_k$ realizes $[D_p]_\infty$.*

Proof. Level by level, [Ch. 3, Thm. 1.6]; there is nothing to add, which is the point. $\square$

Remark 5.1.9 (What dissolution buys, and what it destroys). Corollary 5.1.7 and Proposition 5.1.8 bracket the truth from two sides, and a third fact closes the bracket. The von Neumann completion that dissolves the trace also dissolves the prize: K_0 of a type III factor is trivial, so M_∞ retains no shadow of $\text{Jac}(Y_k)$ — the weak closure destroys exactly the invariant the programme exists to realize. The dissolution theorem therefore does *not* say “pass to the limit and construct”; it says that the unique *numerical* obstruction known dies at the measure-theoretic level, while the K -theoretic content survives only at the C^* -level. The sharpened open problem, which we state as the successor to [Ch. 3, Conj. 3.6(iii)], is accordingly: does there exist a *traceless, purely infinite, nuclear* C^* -algebra over $A_\infty = C(\Omega)$ — a Kirchberg-type limit, where no trace exists to obstruct and where K -theory is nontrivial — carrying a genuine correspondence whose index map recovers the tower $(\mathbb{Z} \oplus \text{Jac}(Y_k))_k$? Tracelessness removes the only obstruction we have proved; it does not supply the construction, and we do not possess one.

5.1.5 The C^* -level: traceless with K -theory alive

Proposition 5.1.10 (C^* -dissolution, no K -annihilation). *Let $C_r^*(\mathcal{G}_\infty)$ denote the reduced groupoid C^* -algebra of the rooted tail groupoid of [Ch. 2, Def. 1.6] over Ω . It is a unital Kirchberg algebra: separable, nuclear, simple, purely infinite, and in the UCT class. Being purely infinite, it admits no nonzero trace, finite or semifinite, densely defined or bounded. Its K -theory is*

$$K_0(C_r^*(\mathcal{G}_\infty)) \cong \mathbb{Z}/(q-1), \quad K_1(C_r^*(\mathcal{G}_\infty)) = 0,$$

by stabilization to the Cuntz algebra $\mathcal{O}_q$ ([Ch. 2, Prop. 1.7(iii)]). Consequently, at the C^* -level the trace obstruction of [Ch. 3] has no formulation and K -theory is not annihilated: both properties denied to the von Neumann closure of Corollary 5.1.7 and Remark 5.1.9 hold simultaneously here.

Remark 5.1.11 (What this algebra does not do, stated against temptation). It would be pleasant to add that $K_0(C_r^*(\mathcal{G}_\infty))$ “carries the tower of sandpile groups.” It does not, and asserting it would contradict this series’ own results: the boundary channel is arithmetically blind ([Ch. 2, Thm. 3.7]), its K -theory seeing only $\mathbb{Z}/(q-1)$ locally and Euler-characteristic data globally, and nothing of $\text{Jac}(Y_k)$. Proposition 5.1.10 therefore proves an *existence of habitat*, not an occupancy: the C^* -level is the unique regime where tracelessness and nontrivial K -theory coexist, but the particular algebra naturally supplied by the tail groupoid is the wrong tenant. The sharpened open problem of Remark 5.1.9 stands exactly as posed: construct, functorially from the Hecke data, a Kirchberg algebra over $C(\Omega)$ whose K_0 realizes $(\mathbb{Z} \oplus \text{Jac}(Y_k))_k$; abstract existence for any prescribed UCT K -data is guaranteed by the Kirchberg–Phillips classification (quoted, data-unverified per the series policy), and canonicity, not existence, is the entire problem.

5.2 The Iwahori–Hecke Euler characteristic

The congruence (Iwahori path) tower of [Ch. 3, §5] has levels $Y_k = \text{non-backtracking } k\text{-paths of } Y_0$, with $n_k = n(q+1)q^{k-1}$ vertices for $k \geq 1$, connected by the two degeneracy correspondences (initial and terminal truncation), and it is non-normal: no deck group acts, and [Ch. 3, Rem. 5.2] showed its torsion growth has the exact pseudo-Iwasawa form with $\lambda = -1$, impossible for a $\mathbb{Z}_p[[T]]$ -characteristic ideal. This section derives that law from a single combinatorial input and locates the -1 .

Theorem 5.2.1 (Line-digraph recursion, proved). *Let G be a strongly connected digraph on n vertices in which every vertex has out-degree and in-degree $q \geq 2$ (Eulerian q -regular), $m = qn$ arcs, and let $L(G)$ be its line digraph. Then $L(G)$ is again strongly connected and Eulerian q -regular on m vertices, and*

$$|\text{sandpile}(L(G))| = q^{m-n-1} |\text{sandpile}(G)| = q^{n(q-1)-1} |\text{sandpile}(G)|.$$

In particular, along the Iwahori tower ($k \geq 1$), $|\text{sandpile}(Y_{k+1})| = q^{n_k(q-1)-1} |\text{sandpile}(Y_k)|$; for $q = 2$ this is the five-level-verified 2^{n_k-1} of [Ch. 3, Rem. 5.2], whose exponent is a $q = 2$ coincidence, corrected here.

Proof. Step 1 (spectra). Let $D_{\text{out}}, D_{\text{in}} \in M_{n \times m}(\mathbb{Z})$ be the tail and head incidence matrices, $(D_{\text{out}})_{v,a} = [\text{tail}(a) = v]$, $(D_{\text{in}})_{v,a} = [\text{head}(a) = v]$. Then $A(G) = D_{\text{out}} D_{\text{in}}^t$ (arcs $u \rightarrow v$) and $A(L(G)) = D_{\text{in}}^t D_{\text{out}}$ (consecutive arc pairs). For rectangular $P \in M_{n \times m}$, $Q \in M_{m \times n}$ one has $\det(zI_m - PQ) = z^{m-n} \det(zI_n - PQ)$; applied with $P = D_{\text{out}}$, $Q = D_{\text{in}}^t$,

$$\det(zI_m - A(L(G))) = z^{m-n} \det(zI_n - A(G)) : \quad (5.2.1)$$

the line digraph has the same nonzero spectrum with multiplicity, plus $m - n$ extra zeros. $L(G)$ is Eulerian q -regular because both degrees of an arc equal the corresponding degrees of its endpoints in G , and strongly connected because a path between two arcs is obtained by concatenating a vertex path of G from the head of the first to the tail of the second.

Step 2 (equal cofactors and the eigenvalue product). Let $\Delta = qI - A$ for an Eulerian q -regular strongly connected digraph on N vertices. Row sums of Δ vanish (out-degree q) and column

sums vanish (in-degree q), so the all-ones vector spans both the right and the left kernel; by Perron–Frobenius, q is an algebraically simple eigenvalue of the irreducible non-negative matrix A , so 0 is a simple eigenvalue of Δ . The two-sided kernel $\mathbb{C}\mathbf{1}$ forces all N principal cofactors of Δ to be equal — cofactors agree along a row because the columns of Δ sum to zero, and along a column because the rows do — and the common value is by definition $\kappa := |\text{sandpile}|$, the determinant of the reduced Laplacian at any sink; sink-independence is exactly this equality. Expanding $\det(zI - \Delta) = z \prod_{i=2}^N (z - \mu_i)$ over the nonzero eigenvalues μ_i , the coefficient of z equals $(-1)^{N-1} \prod_{i \geq 2} \mu_i$ and also $(-1)^{N-1} \sum(\text{principal cofactors}) = (-1)^{N-1} N\kappa$, whence

$$\prod_{\mu \in \text{Spec } \Delta \setminus \{0\}} \mu = N\kappa. \quad (5.2.2)$$

Step 3 (assembly). Apply (5.2.2) to G and to $L(G)$. By (5.2.1), the nonzero eigenvalues of $\Delta_{L(G)} = qI_m - A(L(G))$ are those of Δ_G together with $q - 0 = q$ taken with multiplicity $m - n$; the eigenvalue 0 remains simple for $\Delta_{L(G)}$ since q is Perron-simple for $A(L(G))$ as well. Hence

$$m\kappa(L(G)) = q^{m-n} \cdot n\kappa(G) \implies \kappa(L(G)) = \frac{n}{m} q^{m-n} \kappa(G) = q^{m-n-1} \kappa(G),$$

using $m = qn$. Besides the five $q = 2$ tower levels, the formula was confirmed exactly in a $q = 3$ instance (the complete digraph on four vertices: $\kappa = 16$, $\kappa(L) = 34992 = 3^7 \cdot 16$), against which the uncorrected exponent $n - 1$ fails by a factor 3^4 . $\square$

Theorem 5.2.2 (The pseudo-Iwasawa law). *Let the base graph have n vertices and $q = p$. Then, unconditionally by Theorem 5.2.1, for $k \geq 2$,*

$$\text{ord}_p |\text{sandpile}(Y_k)| = \underbrace{\frac{n(q+1)}{q}}_{\mu} p^k - k + \nu_0, \quad \lambda = -1,$$

with ν_0 determined by the first level. For $Y_0 = K_4$, $q = 2$: $\mu = 6$, $\nu_0 = -4$, reproducing $\nu_2 = 6 \cdot 2^k - k - 4$ exactly.

Proof. Summing the recursion of Theorem 5.2.1 from level 1,

$$\text{ord}_p |\text{sandpile}(Y_k)| = \text{ord}_p |\text{sandpile}(Y_1)| + \sum_{j=1}^{k-1} (n_j(q-1) - 1),$$

the multiplier being a pure power of $p = q$, and $\sum_{j=1}^{k-1} n_j(q-1) = n(q+1)(q^{k-1} - 1) = \frac{n(q+1)}{q} p^k - n(q+1)$. With $q = 2$, $n = 4$: $\sum_{j=1}^{k-1} (12 \cdot 2^{j-1} - 1) = 6 \cdot 2^k - 12 - (k-1)$, and $\text{ord}_2 |\text{sandpile}(Y_1)| = 7$ gives $6 \cdot 2^k - k - 4$. The general coefficients read off the same computation. $\square$

Proposition 5.2.3 ($\lambda \geq 0$ for commutative Iwasawa towers). *Let $g \in \mathbb{Z}_p[[T]]$ be nonzero with Weierstrass decomposition $g = p^\mu P(T)U(T)$, P distinguished of degree $\lambda \geq 0$, U a unit. Then for $k \gg 0$, $\text{ord}_p \prod_{\zeta_{p^k}=1, \zeta \neq 1} g(\zeta - 1) = \mu(p^k - 1) + \lambda k + O(1)$ with $\lambda \geq 0$. Hence no tower governed by the characteristic ideal of a finitely generated torsion $\mathbb{Z}_p[[T]]$ -module can exhibit $\lambda = -1$: Theorem 5.2.2 is a rigorous diagnostic that the Iwahori tower is not such a tower.*

Proof. $\text{ord}_p g(\zeta_{p^j} - 1) = \mu + \lambda \text{ord}_p(\zeta_{p^j} - 1) + O(p^{-j}) = \mu + \lambda/\varphi(p^j) + o(1)$ for $j \gg 0$ by the decomposition, and summing over the $\varphi(p^j)$ primitive roots for $1 \leq j \leq k$ gives the formula; λ is a degree, hence non-negative. $\square$

Definition 5.2.4 (Degeneracy pair and its Euler term). Let $s_k, t_k: A_{k+1} \rightrightarrows A_k$ be the initial- and terminal-truncation correspondences of the tower, the graph-level avatars of the two degeneracy maps $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$, generating the action of the Iwahori–Hecke algebra $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ on the tower. The *Eisenstein kernel* at level k is the one-dimensional space of $\mathcal{R}$ -harmonic constants (the Perron direction), simultaneously the kernel of the level- k sandpile Laplacian and the sole obstruction to exactness of the pair (s_k, t_k) on 0-chains.

Theorem 5.2.5 (The -1 is an Euler characteristic). *The transfer of torsion along one level of the Iwahori tower factors, by Theorem 5.2.1, as*

$$\mathrm{ord}_p|\mathrm{sandpile}(Y_{k+1})| - \mathrm{ord}_p|\mathrm{sandpile}(Y_k)| = \underbrace{n_k(q-1)}_{\text{arc growth: } \mathrm{ord}_p q^{m_k - n_k}} - \underbrace{\chi_k}_{=1},$$

where $\chi_k = \dim \ker \Delta_k^{\mathrm{dir}} = 1$ is the rank of the Eisenstein kernel removed at each level. Hence in the growth law of Theorem 5.2.2 the invariants separate cleanly: μ is the arc-proliferation density $n(q+1)/q$ of the correspondences, and $\lambda = -\chi = -1$ is the Euler-characteristic correction of the degeneracy pair — a homological constant of the non-commutative base $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$, not a module invariant over any commutative $\mathbb{Z}_p[[T]]$. In particular the corrected form of [Ch. 3, Conj. 3.6(ii-b)] should assert the following: the characteristic object of the congruence tower is a module over $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ whose growth law carries the intrinsic correction $-\chi \cdot k$, and it reduces to a commutative characteristic ideal exactly on sub-towers with abelian deck action, where the χ -correction vanishes and the Newton identities of [Ch. 3, §5] hold to the integer.

Proof. The recursion’s exponent splits as $(m_k - n_k) - 1 = \mathrm{ord}_p(q^{m_k - n_k}) - 1$, and both terms are now located inside the proof of Theorem 5.2.1 itself. The first is the p -valuation of the extra Perron factors: each of the $m_k - n_k$ new zero eigenvalues of $A(Y_{k+1})$ contributes $q - 0 = q$ to the eigenvalue product (5.2.2). The subtracted 1 is the vertex-count normalization $n_k/m_k = q^{-1}$ in (5.2.2): precisely the one-dimensional Eisenstein kernel $\ker \Delta^{\mathrm{dir}}$, simple at every level by strong connectivity, excluded from the eigenvalue product once per level, independently of k . Summation as in Theorem 5.2.2 turns the per-level $-\chi_k = -1$ into the global $-k$, i.e. $\lambda = -\chi = -1$. $\square$

Remark 5.2.6 (Honest scope). Theorem 5.2.5 is now unconditional, but its final sentence remains a proposal for restating [Ch. 3, Conj. 3.6(ii-b)], not a theorem about $\mathcal{H}(\mathrm{PGL}_2(\mathbb{Q}_p), \mathcal{I})$ -modules: we have defined no characteristic ideal over the Iwahori–Hecke algebra, and constructing that module theory — with a Weierstrass-type preparation in which $-\chi$ appears intrinsically — is the open algebraic problem this section poses. What is unconditional is Proposition 5.2.3: $\lambda = -1$ certifies, rigorously, that whatever that theory is, it is not commutative Iwasawa theory.

5.3 Exploratory outlook

We record, deriving nothing, where the two threads point. The dissolution theorem directs the realization problem toward traceless Kirchberg limits over $C(\Omega)$ (Remark 5.1.9), where the natural analytic invariants are not traces but the KMS functionals of the twisted dynamics of [Ch. 2, §2], with local L -factors as partition functions. The Euler-characteristic correction of Section 5.2 lives at the trivial zero of those L -factors — the same Eisenstein direction removed once per level — and it is natural to ask, in the spirit of the exceptional-zero phenomena of Mazur–Tate–Teitelbaum and of Gross–Stark, whether the per-level derivative data of the tower assembles into a p -adic L -functional on the Iwahori–Hecke base whose trivial-zero corrections reproduce $-\chi k$. We state

this as a direction, not a conjecture: no formulation precise enough to be false is offered here, and the series' methodology is to compute before naming. The first computable instance has in fact been computed. Write $g(T) = T^2 h(T)$ for the cleared voltage pencil (the double trivial zero of [Ch. 3, Rem. 5.4]) and $\varepsilon_j := \text{ord}_\ell \prod_{\zeta \text{ prim. } \ell^j} h(\zeta - 1) - (\mu \varphi(\ell^j) + \lambda_h)$ for the exceptional low-level defects. Then the constant term of the growth law satisfies, exactly in all four abelian towers of [Ch. 3, Table 2],

$$\nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_{j \geq 1} \varepsilon_j,$$

with ε finitely supported: identically zero in three towers, and $(\varepsilon_1, \varepsilon_2) = (-1, +2)$ in the $(1, 2, 0)$ tower. The trivial-zero residue $a_2 = \frac{1}{2}g''(0)$ carries the leading datum precisely when $\lambda_h = 0$, where $\text{ord}_\ell a_2 = \mu$ as in both $(1, 0, 0)$ towers, and not otherwise: an exceptional-zero reading in the Mazur–Tate–Teitelbaum style is available only in the $\lambda_h = 0$ regime, and this bookkeeping identity, verified to the integer, is the whole of what is presently proved.

Acknowledgement of provenance and caveats

Unrefereed preprint. Proved in full: Lemmas 5.1.2–5.1.3, Theorem 5.1.4, Corollary 5.1.7, Propositions 5.1.8 and 5.2.3, Theorem 5.2.1 (self-contained; the only classical input is the algebraic simplicity of the Perron eigenvalue of an irreducible non-negative matrix), Theorem 5.2.2, and Theorem 5.2.5, all now unconditional. Quoted classical inputs: every entry in the bibliography is verified against its primary source or Crossref record — Connes and Powers via JSTOR, Araki–Woods via EMS Press, Krieger via Springer, Feldman–Moore Parts I and II via Crossref and the AMS Digital Library, Levine via Crossref — with no residual soft spots; the DOIs are recorded inline at each entry. The Levine reference [17] is retained for priority context only: Theorem 5.2.1 is proved self-containedly above, and comparing the exponent $q^{n(q-1)-1}$ with the published statement belongs to the primary-source check this series requires. The concluding restatement of [Ch. 3, Conj. 3.6(ii-b)] in Theorem 5.2.5 remains a proposal. Remark 5.1.6 records a correction of emphasis to [Ch. 2, Thm. 2.4(iii)]. Section 5.3 makes no claims.

Part II

Towers: commutative Iwasawa theory and its failure

Chapter 6

Algorithmic non-commutative class field theory, IV: the limit module of abelian towers and the characteristic identity

Summary. For an abelian $\mathbb{Z}/\ell^k$ voltage tower (Y_k) over a finite base graph, Papers II–III established the exact Iwasawa growth law of the sandpile orders and its Newton-polygon mechanism, leaving the limit-module statement of [Ch. 3, Conj. 3.6(ii-a)] open. This paper closes it. Writing $D(t)$ for the voltage Laplacian pencil and $\Lambda = \mathbb{Z}_\ell[[T]]$, we prove:

$$X_\infty := \varprojlim_k (\operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell, \pi_*) \cong \Lambda^n / D(1+T) \Lambda^n,$$

a finitely generated torsion Λ -module with $\operatorname{char}_\Lambda(X_\infty) = (\det D(1+T))$, together with an *exact* control theorem $X_\infty / \omega_k X_\infty \cong \operatorname{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell$ — exact, not merely pseudo-isomorphic, because the tower’s presentations are literally the ω_k -specializations of one Λ -presentation. The proof rests on three structural identities — the derived Laplacian is the block-circulant $D(\sigma)$; the covering pushforward is the group-ring coefficient projection; pushforward is surjective on ℓ -primary torsion with kernel of the exact predicted order — each proved and additionally machine-verified at every level of two towers. Combined with [Ch. 3, Rem. 5.4], the measured invariants $(\mu, \lambda) = (\mu(g), \lambda(g) - 1)$, $g = \det D(1+T)$, are now consequences of a theorem rather than observations. The result is stated with explicit deference to the graph-Iwasawa literature of Vallières, McGown–Vallières and Gonet, whose verified titles indicate substantial overlap; what is new here, at minimum, is the exactness of the control isomorphism at the level of Smith normal forms and the integration into the trilogy’s K -theoretic framework. The non-normal congruence tower ([Ch. 3, Conj. 3.6(ii-b)]) and the canonicity problem ([Ch. 5, Rem. 1.9]) are untouched and remain the open frontier.

Status of the results

Every numbered statement in this paper is proved in full. Lemmas 6.1.1, 6.1.2 and 6.1.3 are elementary and were additionally verified by exact machine computation at every level $k \leq 3$ of the tower with voltages $(1, 1, 1)$, $\ell = 2$, and every level $k \leq 2$ of the tower $(1, 0, 0)$, $\ell = 3$ (Table 6.1). The single external algebraic input is the structure theory of finitely generated Λ -modules, quoted with a flagged citation. The relation of Theorem 6.2.1 to the published graph-Iwasawa literature is discussed, deliberately and prominently, in Section 6.4.

6.1 Setup and the three structural identities

Fix a finite connected base graph Y_0 on n vertices, a prime ℓ , a spanning tree, and a voltage assignment α on the chords, giving the derived $\mathbb{Z}/\ell^k$ -covers Y_k on $n\ell^k$ vertices with deck generator σ and covering maps $\pi: Y_{k+1} \rightarrow Y_k$. Let

$$D(t) = (q+1)I_n - A(t) \in M_n(\mathbb{Z}[t, t^{-1}])$$

be the voltage Laplacian pencil ($A(t)$ the voltage adjacency), Δ_k the ordinary Laplacian of Y_k , $G_k = \mathbb{Z}/\ell^k$, and order V_{Y_k} as $(v, g) \mapsto vN + g$, $N = \ell^k$. Write S_N for the cyclic shift on $\mathbb{Z}_\ell^N$ and $Q_N: \mathbb{Z}_\ell^N \rightarrow \mathbb{Z}_\ell^N$ for reduction of the group coordinate, $e_g \mapsto e_{g \bmod N}$.

Lemma 6.1.1 (Block-circulant identification). *In the ordering above, $\Delta_k = D(S_N)$ exactly: the level- k Laplacian is the image of the pencil under $t \mapsto S_N$. Equivalently, as a $\mathbb{Z}[G_k]$ -module with the deck action, $\text{coker}_{\mathbb{Z}}(\Delta_k) \cong \mathbb{Z}[G_k]^n / D(\sigma)\mathbb{Z}[G_k]^n$.*

Proof. This is the definition of the derived graph read backwards: a base edge (u, v) of voltage a contributes the arcs $(u, g) - (v, g+a)$ for all g , i.e. the block $E_{uv} \otimes S_N^a$ plus its transpose, and the degree term is $(q+1)I_n \otimes I_N$. Summing gives $\Delta_k = D(S_N)$; identifying $\mathbb{Z}_\ell^N \cong \mathbb{Z}_\ell[G_k]$ with $S_N =$ multiplication by σ gives the module form. Verified as an exact matrix identity at all computed levels. $\square$

Lemma 6.1.2 (Pushforward is coefficient projection). *Let $P_k := I_n \otimes Q_{\ell^k}$, the matrix of the covering pushforward $\pi_*: \mathbb{Z}^{V_{Y_{k+1}}} \rightarrow \mathbb{Z}^{V_{Y_k}}$ (sum over fibres). Then*

$$P_k \Delta_{k+1} = \Delta_k P_k, \quad P_k R_{k+1} = R_k P_k,$$

where $R_k = I_n \otimes S_{\ell^k}$ is the deck action; under Lemma 6.1.1 the induced map on cokernels is the group-ring coefficient projection $\mathbb{Z}[G_{k+1}] \rightarrow \mathbb{Z}[G_k]$. Moreover P_k is surjective.

Proof. Both identities reduce to the one-line facts $Q S_{\ell N} = S_N Q$ and $Q S_{\ell N}^a = S_N^a Q$ applied blockwise to $\Delta = D(S)$; surjectivity is immediate since every $e_{(v,h)}$ is hit. Verified as exact matrix identities at all computed levels. $\square$

Lemma 6.1.3 (Norm surjectivity on torsion). *The induced norm $\pi_*: \text{coker}_{\mathbb{Z}}(\Delta_{k+1}) \rightarrow \text{coker}_{\mathbb{Z}}(\Delta_k)$ is surjective; it preserves the degree functional $\deg$ (total mass), hence restricts to a surjection of torsion subgroups, with kernel of order*

$$|\ker(\pi_*|_{\text{Tor}_\ell})| = \ell^{\text{ord}_\ell |\text{Jac}(Y_{k+1})| - \text{ord}_\ell |\text{Jac}(Y_k)|}.$$

Proof. Surjectivity of P_k plus the intertwining gives surjectivity on cokernels. The column sums of Δ_k vanish, so $\deg([y]) = \sum_i y_i$ is well defined and realizes $\text{coker}_{\mathbb{Z}}(\Delta_k) \cong \mathbb{Z} \oplus \text{Tor}$ with $\text{Tor} = \ker(\deg)$ (a finitely generated group surjecting onto $\mathbb{Z}$ with rank-zero kernel has kernel equal to its torsion). Pushforward preserves total mass, so $\deg \circ \pi_* = \deg$; hence for torsion t downstairs any preimage a has $\deg(a) = 0$, i.e. a is torsion: surjectivity on Tor . The kernel order is then forced by counting. Verified, including the exact kernel orders ℓ^3, ℓ^3, ℓ^3 and ℓ^1, ℓ^1 at the computed transitions (Table 6.1). $\square$

6.2 The limit module and the characteristic identity

Theorem 6.2.1 (Limit module). *Let $g(T) := \det D(1+T) \in \mathbb{Z}_\ell[[T]]$ and assume $g \neq 0$ (true in every tower of [Ch. 3, §5]). Then, with transition maps the norms π_* of Lemma 6.1.3,*

$$X_\infty := \varprojlim_k (\text{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell) \cong \Lambda^n / D(1+T) \Lambda^n, \quad \Lambda = \mathbb{Z}_\ell[[T]],$$

under the standard identification $\varprojlim_k \mathbb{Z}_\ell[G_k] \cong \Lambda$, $\sigma \mapsto 1+T$ (the Laurent entries $t^{\pm a}$ of the pencil landing in $\Lambda^\times$ -multiples since $1+T \in \Lambda^\times$). The module X_∞ is finitely generated and torsion over Λ , and

$$\text{char}_\Lambda(X_\infty) = (g(T)) = (\det D(1+T)).$$

Proof. By Lemmas 6.1.1–6.1.2 the tower of cokernels is, level by level and compatibly with the transition maps, the tower of presentations $\mathbb{Z}_\ell[G_k]^n \xrightarrow{D(\sigma)} \mathbb{Z}_\ell[G_k]^n \rightarrow C_k \rightarrow 0$ with transitions the coefficient projections. All modules are finitely generated $\mathbb{Z}_\ell$ -modules, hence compact, and the transition maps are continuous surjections on the presenting copies; inverse limits are exact on such systems (Mittag–Leffler for compact groups), so $\varprojlim C_k = \text{coker}(\varprojlim D(\sigma)) = \Lambda^n/D(1+T)\Lambda^n$. Since $g = \det D(1+T) \neq 0$, the map $D(1+T)$ is injective over the fraction field, so the cokernel is Λ -torsion; it is finitely generated by presentation. For the characteristic ideal, localize at a height-one prime $P \subset \Lambda$: Λ_P is a discrete valuation ring, elementary divisors give $\text{length}_{\Lambda_P}((X_\infty)_P) = v_P(\det D(1+T))$, and taking the product over height-one primes yields $\text{char}_\Lambda(X_\infty) = (\det D(1+T))$ by the structure theory of finitely generated torsion Λ -modules [18]. $\square$

Corollary 6.2.2 (Exact control). *For every $k \geq 0$, with $\omega_k = (1+T)^{\ell^k} - 1$,*

$$X_\infty/\omega_k X_\infty \cong \text{coker}_{\mathbb{Z}}(\Delta_{Y_k}) \otimes \mathbb{Z}_\ell \cong (\mathbb{Z} \oplus \text{Jac}(Y_k)) \otimes \mathbb{Z}_\ell,$$

an isomorphism on the nose — not merely up to bounded kernels and cokernels — because $\Lambda/\omega_k \cong \mathbb{Z}_\ell[G_k]$ carries the level- k presentation exactly.

Corollary 6.2.3 (Invariants, now as a theorem). *The Iwasawa invariants of the tower are $\mu = \mu(g)$ and $\lambda = \lambda(g) - 1$, where $(\mu(g), \lambda(g))$ are read off the Newton polygon of g , the -1 being the trivial-zero and normalization accounting of [Ch. 3, Rem. 5.4]. In particular the four exact laws of [Ch. 3, Table 2] are consequences of Theorem 6.2.1, no longer observations.*

6.3 The verification battery

tower	levels	(B) block	(S) deck	(P)+(C) norm identities	(N) torsion norm
$(1, 1, 1), \ell = 2$	$k \leq 3$	exact	exact	exact	surj.; $ \ker = 2^3, 2^3, 2^3$
$(1, 0, 0), \ell = 3$	$k \leq 2$	exact	exact	exact	surj.; $ \ker = 3^1, 3^1$

Table 6.1: Machine verification of Lemmas 6.1.1–6.1.3: (B) $\Delta_k = D(S_N)$; (S) $R_k \Delta_k = \Delta_k R_k$; (P) $P_k \Delta_{k+1} = \Delta_k P_k$; (C) $P_k R_{k+1} = R_k P_k$; (N) surjectivity of π_* on ℓ -primary torsion with kernel orders exactly $\ell^{\Delta_{\text{ord}}}$. Every check passed as an exact identity; the lemmas are proved above independently, so the table is confirmation, not evidence.

6.4 Relation to the literature, stated prominently

The verified bibliographic records of Vallières [8], McGown–Vallières [9] and, above all, Gonet — whose paper is titled precisely *Iwasawa theory of Jacobians of graphs* [10] — make it very likely that Theorem 6.2.1, or a statement equivalent to it, is present in that literature; the growth law certainly is, as acknowledged since [Ch. 3, §5]. Under the series’ policy, we have verified those papers’ metadata but have not read their theorems, and we therefore claim priority for nothing in Sections 6.1–6.2 beyond what a comparison with their texts will leave standing. What we believe is ours at minimum: the *exactness* of Corollary 6.2.2 in the form stated, its verification at the resolution of Smith normal forms across [Ch. 3, Table 2] and Table 6.1, and the integration of the statement into the trilogy’s K -theoretic framework, where $\text{coker}_{\mathbb{Z}}(\Delta_{Y_k})$ is the K_0 -group of the level- k Laplacian channel ([Ch. 2, Thm. 3.5]), so that Theorem 6.2.1 reads: *the K -theoretic torsion of the Laplacian channel along an abelian tower is one finitely generated torsion Iwasawa module, with characteristic ideal the voltage pencil determinant.*

6.5 What remains open

Nothing here touches the two hard problems. The congruence tower is non-normal, its pseudo-Iwasawa law has $\lambda = -1$, and the correct base object — conjecturally a module theory over the Iwahori–Hecke algebra with an intrinsic Euler-characteristic correction ([Ch. 3, Conj. 3.6(ii-b)], [Ch. 5, §2]) — still lacks a definition of characteristic ideal. And the canonicity problem of [Ch. 5, Rem. 1.9] — a Kirchberg algebra over $C(\Omega)$, functorial in the Hecke data, with $K_0 \cong (\mathbb{Z} \oplus \text{Jac}(Y_k))_k$ — is exactly as open as before; Theorem 6.2.1 sharpens its target, since the prescribed K -theory is now known to be a single Λ -module, but supplies no construction.

Acknowledgement of provenance and caveats

Unrefereed preprint. All numbered statements are proved; the machine battery of Table 6.1 confirms the three lemmas at seven levels across two towers. External inputs: the structure theory of finitely generated torsion Λ -modules and the identification $\varprojlim \mathbb{Z}_\ell[G_k] \cong \Lambda$, both standard, cited to [18], whose bibliographic record is now primary-source verified (only the chapter-level pointer remains a recollection-based reading aid); the graph-Iwasawa literature [8, 9, 10], bibliographic data verified in [II–III], texts unread, priority explicitly ceded in Section 6.4.

Chapter 7

Algorithmic non-commutative class field theory, IX: exceptional zeros, the graph $\mathcal{L}$ -invariant, and the operator-algebraic Mazur–Tate–Teitelbaum formula

Summary. Section 3 of [Ch. 5] recorded, without proof, that the cleared voltage pencil $g(T) = T^2 h(T)$ of an abelian $\mathbb{Z}/\ell^k$ tower over a $(q+1)$ -regular graph has a double zero at the trivial character, and that the constant term of the Iwasawa growth law obeys $\nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_j \varepsilon_j$ in all four towers of [Ch. 3, Table 2]. This paper turns both observations into theorems and identifies the number they conceal. The double zero is forced by the symmetry $D(t^{-1}) = D(t)^t$ of the pencil; its second-order coefficient has the closed form

$$\tfrac{1}{2}\Theta''(0) = -\kappa(Y_0) \|h_\alpha\|^2,$$

where h_α is the harmonic representative of the voltage cochain α — its orthogonal projection onto the cycle space, equivalently onto the kernel of the Hecke–Dirac operator of [Ch. 3] — and $\kappa(Y_0) = |\text{Jac}(Y_0)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ is the algebraic L -value of [Ch. 2]. We therefore define the graph $\mathcal{L}$ -invariant $\mathcal{L}(Y_\bullet) := \frac{1}{2}\Theta''(0)/\kappa(Y_0) = -\|h_\alpha\|^2$, a rational number depending only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$ and not on ℓ ; for a single-chord voltage, $-\frac{1}{2}\Theta''(0)$ counts the spanning trees avoiding the chord. The bookkeeping identity of [Ch. 5] becomes a theorem with an exact description of the finitely supported defects ε_j , and the specification’s proposed alignment is replaced by the correct one: in the exceptional-zero regime $\lambda_h = 0$ all defects vanish and $\nu_0 = -\text{ord}_\ell \mathcal{L}(Y_\bullet)$, while in general $\nu_0 + \text{ord}_\ell \mathcal{L}(Y_\bullet) = (\text{ord}_\ell a_2 - \mu) + \sum_j \varepsilon_j$. On the operator-algebraic side we prove a twisted Bass identity for the character-twisted Satake dynamics of [Ch. 2] and show that at the critical inverse temperature $\beta = 1$ the twisted Bass determinant equals $q^{-n}\Theta(t-1)$: its zero at the trivial character is simple in the temperature direction with derivative proportional to $\kappa(Y_0)$, of order two in the character direction with second derivative proportional to $\mathcal{L}(Y_\bullet)\kappa(Y_0)$, and the partition function of the critical KMS family has a double pole with leading coefficient $n(q-q^{-1})/\mathcal{L}(Y_\bullet)$ — the Mazur–Tate–Teitelbaum and Greenberg–Stevens shapes, with the roles of algebraic L -value and $\mathcal{L}$ -invariant made explicit. The abelian shadow limit of [Ch. 12] carries Θ as the characteristic element of its equivariant K_0 . For the non-normal Iwahori tower there is no character direction: its zero is simple at every level, and the constancy of its residue $n_k \kappa(Y_k) q^{-n_k}$ along the tower is exactly the pseudo-Iwasawa law, with the Euler term $-k$ appearing as the p -adic valuation of the vertex

count. No $\mathcal{L}$ -invariant exists for that tower, and we say so.

Status of the results

Every numbered statement is proved. The proofs use only results of Papers I–VIII, finite-dimensional linear algebra (a Schur complement replaces analytic perturbation theory), the Weierstrass preparation theorem for $\mathbb{Z}_\ell[[T]]$ [23], and the character product formula for abelian covers already used in [Ch. 3, §5] and [Ch. 6]. The twisted Bass identity is the Artin–Ihara L -function formula of Stark–Terras [22]; we give a six-line proof so that nothing is quoted. Every identity was machine-verified exactly on the four abelian towers of [Ch. 3, Table 2] — growth laws recomputed directly from the reduced Laplacians of the derived graphs up to 128 vertices — and on the Iwahori tower at levels $k \leq 3$ (Table 7.1). The Mazur–Tate–Teitelbaum [21] and Greenberg–Stevens [20] formulas are cited for the shape of the analogy only; nothing here depends on them. Section 7.5 records what is open and five corrections to the specification that prompted this paper.

7.1 The exceptional zero of the voltage pencil

Setting as in [Ch. 3, §5] and [Ch. 6]. Y_0 is a finite connected $(q+1)$ -regular graph with n vertices and $m = \frac{(q+1)n}{2}$ edges, $r = m - n + 1$; fix an orientation E^+ of the edges and a voltage $\alpha: E^+ \rightarrow \mathbb{Z}$, extended to oriented edges by $\alpha(\bar{e}) = -\alpha(e)$, so that α is an integer 1-cochain. The voltage adjacency and Laplacian pencils are

$$A(t)_{uv} = \sum_{e: u \rightarrow v} t^{\alpha(e)}, \quad D(t) = (q+1)I_n - A(t) \in M_n(\mathbb{Z}[t, t^{-1}]),$$

and $\Theta(T) := \det D(1+T) \in \mathbb{Z}[T, (1+T)^{-1}] \subset \Lambda := \mathbb{Z}_\ell[[T]]$ is the graph ℓ -adic L -function, the characteristic element of the limit module of [Ch. 6, Thm. 2.1]. Its cleared form $g(T) = (1+T)^d \Theta(T) \in \mathbb{Z}[T]$, d minimal, is the polynomial of [Ch. 3, Rem. 5.4] and [Ch. 5, §3]; since $(1+T)^d = 1 + dT + \dots$ and $\Theta(0) = \Theta'(0) = 0$ (Theorem 7.1.2), the coefficients a_i of T^i in Θ and in g agree for $i \leq 2$. We write $\kappa_0 := \kappa(Y_0) = |\text{Jac}(Y_0)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$, the algebraic L -value of [Ch. 2, Thm. 3.5], and $\Delta_0 = D(1)$ for the Laplacian of Y_0 .

Hodge data. Let $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$, $\partial e = t(e) - o(e)$, be the boundary map, so $\Delta_0 = \partial \partial^t$ [Ch. 3, Lem. 3.2]; the cycle space $\ker \partial = H_1(Y_0; \mathbb{Z})$ and the cut space $\text{im } \partial^t$ are orthogonal complements in $\mathbb{Q}^{E^+}$. With Δ_0^+ the Moore–Penrose inverse (the inverse of Δ_0 on $\mathbf{1}^\perp$, zero on $\mathbf{1}$), the orthogonal projection onto the cycle space is

$$\Pi := I - \partial^t \Delta_0^+ \partial,$$

which is also the projection onto $\ker D_p \cap \ell^2(E^+)$ for the Hecke–Dirac operator D_p of [Ch. 3, Def. 3.1]. Put $h_\alpha := \Pi \alpha$, the harmonic representative of α , and $\|h_\alpha\|^2 = \alpha^t \Pi \alpha = \|\alpha\|^2 - (\partial \alpha)^t \Delta_0^+ (\partial \alpha) \in \mathbb{Q}_{\geq 0}$.

Lemma 7.1.1 (Symmetry and switching). *$D(t^{-1}) = D(t)^t$, hence $f(t) := \det D(t)$ satisfies $f(t^{-1}) = f(t)$. For $\varphi \in \mathbb{Z}^V$, the switched voltage $\alpha + \partial^t \varphi$ has pencil $\text{diag}(t^{\varphi_u}) D(t) \text{diag}(t^{-\varphi_u})$, so f depends only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$.*

Proof. $A(t^{-1})_{uv} = \sum_{e: u \rightarrow v} t^{-\alpha(e)} = \sum_{e: v \rightarrow u} t^{\alpha(e)} = A(t)_{vu}$. Switching multiplies the entry $t^{\alpha(e)}$, $e: u \rightarrow v$, by $t^{\varphi_v - \varphi_u}$. $\square$

Theorem 7.1.2 (Exceptional zero of order two). *(i) $\Theta(0) = 0$ and $\Theta'(0) = 0$.*

$$(ii) \ a_2 = \frac{1}{2} \Theta''(0) = -\kappa(Y_0) \|h_\alpha\|^2.$$

(iii) *The zero has order exactly two if and only if $h_\alpha \neq 0$, i.e. if and only if $[\alpha] \neq 0$ in $H^1(Y_0; \mathbb{Q})$.*

(iv) *If α is the indicator of a single edge e , then $\|h_\alpha\|^2 = 1 - R_{\text{eff}}(e)$ and $-a_2 = \kappa(Y_0)(1 - R_{\text{eff}}(e))$ is the number of spanning trees of Y_0 not containing e .*

Proof. Write $s = T$ and expand $D(1+s) = \Delta_0 + sD_1 + s^2D_2 + O(s^3)$ with $D_1 = D'(1) = -A'(1) =: -W$ and $D_2 = \frac{1}{2}D''(1) = -\frac{1}{2}A''(1)$. By Lemma 7.1.1, $A'(1)^t = -A'(1)$, so W is antisymmetric, and $(W\mathbf{1})_u = \sum_{o(e)=u} \alpha(e) - \sum_{t(e)=u} \alpha(e) = -(\partial\alpha)_u$, i.e. $W\mathbf{1} = -\partial\alpha$. Also $\mathbf{1}^t A''(1)\mathbf{1} = \sum_{e \in E^+} [\alpha(\alpha-1) + \alpha(\alpha+1)] = 2\|\alpha\|^2$, so $\mathbf{1}^t D_2 \mathbf{1} = -\|\alpha\|^2$.

Decompose $\mathbb{Q}^n = \mathbb{Q}\mathbf{1} \oplus \mathbf{1}^\perp$ orthogonally, with unit vector $\mathbf{1}/\sqrt{n}$. In this decomposition $\Delta_0 = 0 \oplus \Delta'$ with Δ' invertible and $\det \Delta' = \prod_{\mu \neq 0} \mu = n\kappa_0$ [Ch. 5, eq. (2.2)]. Writing $E := sD_1 + s^2D_2 + O(s^3)$ in block form $\begin{pmatrix} e_{00} & e_{0*} \\ e_{*0} & E_{**} \end{pmatrix}$, the Schur complement gives

$$f(1+s) = \det(\Delta' + E_{**}) \cdot (e_{00} - e_{0*}(\Delta' + E_{**})^{-1}e_{*0}).$$

Here $e_{00} = \frac{1}{n}\mathbf{1}^t E \mathbf{1} = \frac{s}{n}\mathbf{1}^t D_1 \mathbf{1} + \frac{s^2}{n}\mathbf{1}^t D_2 \mathbf{1} + O(s^3) = -\frac{s^2}{n}\|\alpha\|^2 + O(s^3)$, because $\mathbf{1}^t W \mathbf{1} = 0$ by antisymmetry; and $e_{0*}, e_{*0} = O(s)$ with $e_{*0} = \frac{s}{\sqrt{n}}D_1 \mathbf{1}|_{\mathbf{1}^\perp} + O(s^2)$, $e_{0*} = \frac{s}{\sqrt{n}}(D_1^t \mathbf{1})^t|_{\mathbf{1}^\perp} + O(s^2)$; note $D_1 \mathbf{1} = -W\mathbf{1} = \partial\alpha \in \mathbf{1}^\perp$ and $D_1^t \mathbf{1} = W\mathbf{1} = -\partial\alpha$. Since $(\Delta' + E_{**})^{-1} = \Delta_0^+ + O(s)$ on $\mathbf{1}^\perp$,

$$e_{0*}(\Delta' + E_{**})^{-1}e_{*0} = \frac{s^2}{n}(-\partial\alpha)^t \Delta_0^+ (\partial\alpha) + O(s^3).$$

Hence $f(1+s) = (n\kappa_0 + O(s)) \cdot \frac{s^2}{n}(-\|\alpha\|^2 + (\partial\alpha)^t \Delta_0^+ (\partial\alpha)) + O(s^3) = -\kappa_0 \|\Pi\alpha\|^2 s^2 + O(s^3)$, which is (i) and (ii). (iii): $\Pi\alpha = 0$ iff α lies in the cut space $\text{im } \partial^t \otimes \mathbb{Q}$ iff $[\alpha] = 0$ in $H^1(Y_0; \mathbb{Q})$. (iv): for $\alpha = e$, $\|\Pi e\|^2 = 1 - e^t \partial^t \Delta_0^+ \partial e = 1 - R_{\text{eff}}(e)$, and $\kappa_0 R_{\text{eff}}(e)$ is the number of spanning trees containing e (Kirchhoff). On K_4 : $R_{\text{eff}} = \frac{1}{2}$, $16 \cdot \frac{1}{2} = 8$ trees avoid the chord, and $a_2 = -8$ in both $(1, 0, 0)$ towers (Table 7.1). $\square$

Definition 7.1.3 (Graph $\mathcal{L}$ -invariant). $\mathcal{L}(Y_\bullet) := \frac{\Theta''(0)}{2\kappa(Y_0)} = -\|h_\alpha\|^2 = -\|\Pi_{\ker D_p} \alpha\|^2 \in \mathbb{Q}_{\leq 0}$, nonzero iff $[\alpha] \neq 0$ rationally. It depends only on $[\alpha] \in H^1(Y_0; \mathbb{Z})$ and not on ℓ : the same rational number serves every prime. On K_4 the four towers of [Ch. 3, Table 2] have $\mathcal{L} = -\frac{1}{2}, -1, -\frac{3}{2}, -\frac{1}{2}$ for $(1, 0, 0), (1, 1, 1), (1, 2, 0), (1, 0, 0)$ respectively.

Theorem 7.1.4 (The graph Mazur–Tate–Teitelbaum formula). $\frac{1}{2}\Theta''(0) = \mathcal{L}(Y_\bullet) \cdot \kappa(Y_0) = \mathcal{L}(Y_\bullet) \cdot \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$: the leading coefficient of the graph ℓ -adic L -function at its exceptional zero is the $\mathcal{L}$ -invariant times the algebraic L -value.

Proof. Theorem 7.1.2(ii), Definition 7.1.3, and [Ch. 2, Thm. 3.5] for the product formula for $\kappa(Y_0)$. $\square$

Remark 7.1.5 (The analogy, stated as an analogy). For an elliptic curve with split multiplicative reduction at p , the p -adic L -function has an exceptional zero at $s = 1$ and [21] conjectured, [20] proved, $L'_p(E, 1) = \mathcal{L}(E) L(E, 1)/\Omega_E$ with $\mathcal{L}(E) = \log_p q_E / \text{ord}_p q_E$ the $\mathcal{L}$ -invariant of the Tate period. Theorem 7.1.4 has that shape: a zero forced by a symmetry, an algebraic part that is a product of Frobenius point counts, and a transcendental-looking factor that is in fact a period-type quantity — here the Hodge energy of the class $[\alpha]$, i.e. its length in the harmonic metric of Y_0 . The zero is of order two rather than one because the graph pencil is self-dual under $t \mapsto t^{-1}$ (Lemma 7.1.1), which kills the first derivative; the analogue in the classical setting would be a functional equation forcing an even order. We claim the shape and nothing more.

7.2 The ℓ -adic bookkeeping

Fix a prime ℓ with the tower Y_k connected for all k (equivalently $\alpha \neq 0$ in $H^1(Y_0; \mathbb{F}_\ell)$, true in the four towers). Write $h(T) := g(T)/T^2 \in \mathbb{Z}[T]$, so $h(0) = a_2$, and let $h = \ell^\mu P(T)U(T)$ be its Weierstrass decomposition in Λ : P distinguished of degree λ_h , U a unit [23]. Thus $\mu = \min_i \text{ord}_\ell(\text{coefficients of } h)$ and λ_h is the index of the first coefficient attaining it; [Ch. 3, Rem. 5.4] and [Ch. 6, Cor. 2.3] give the Iwasawa invariants of the tower as μ and $\lambda = \lambda(g) - 1 = \lambda_h + 1$. For $j \geq 1$ put

$$N_j := \text{ord}_\ell \prod_{\zeta \text{ prim. } \ell^j} h(\zeta - 1) = \text{ord}_\ell \text{Res}_T(\Phi_{\ell^j}(1+T), h(T)), \quad \varepsilon_j := N_j - (\mu \varphi(\ell^j) + \lambda_h),$$

the exceptional low-level defects of [Ch. 5, §3].

Lemma 7.2.1 (Character product). *For every $k \geq 0$, $\kappa(Y_k) = \kappa(Y_0) \ell^{-k} \prod_{\zeta^{\ell^k}=1, \zeta \neq 1} \det D(\zeta)$ [Ch. 3, §5], and consequently*

$$\text{ord}_\ell \kappa(Y_k) = \text{ord}_\ell \kappa_0 - k + \sum_{j=1}^k (2 + N_j).$$

Proof. $\det D(\zeta) = \zeta^{-d} g(\zeta - 1)$ and $\prod_{\zeta \neq 1} \zeta^{-d}$ is a root of unity; $g(\zeta - 1) = (\zeta - 1)^2 h(\zeta - 1)$ and $\prod_{\zeta \text{ prim. } \ell^j} (\zeta - 1) = \pm \Phi_{\ell^j}(1) = \pm \ell$, of valuation 1; group the nontrivial roots by exact order ℓ^j , $1 \leq j \leq k$. $\square$

Theorem 7.2.2 (The bookkeeping identity of [Ch. 5, §3]). *Let $\rho_1, \dots, \rho_{\lambda_h}$ be the roots of P (all of positive valuation) and let $j_0 \geq 1$ be least with $1/\varphi(\ell^{j_0}) < \min_i \text{ord}_\ell \rho_i$ ($j_0 = 1$ if $\lambda_h = 0$). Then:*

(i) $\varepsilon_j = 0$ for all $j \geq j_0$; the defects are finitely supported;

(ii) for all $k \geq j_0 - 1$,

$$\text{ord}_\ell \kappa(Y_k) = \mu \ell^k + (\lambda_h + 1)k + \nu_0, \quad \nu_0 = \text{ord}_\ell \kappa(Y_0) - \mu + \sum_{j \geq 1} \varepsilon_j;$$

(iii) in particular $\lambda = \lambda_h + 1 = \lambda(g) - 1$, recovering [Ch. 3, Rem. 5.4] and [Ch. 6, Cor. 2.3], and the constant term is the identity recorded in [Ch. 5, §3].

Proof. (i) For ζ primitive of order ℓ^j , $\text{ord}_\ell(\zeta - 1) = 1/\varphi(\ell^j)$; if this is smaller than every $\text{ord}_\ell \rho_i$ then $\text{ord}_\ell(\zeta - 1 - \rho_i) = 1/\varphi(\ell^j)$ for all i , so $\text{ord}_\ell h(\zeta - 1) = \mu + \lambda_h/\varphi(\ell^j)$, and summing over the $\varphi(\ell^j)$ primitive roots gives $N_j = \mu\varphi(\ell^j) + \lambda_h$. (ii) By Lemma 7.2.1, $\text{ord}_\ell \kappa(Y_k) = \text{ord}_\ell \kappa_0 - k + 2k + \sum_{j \leq k} (\mu\varphi(\ell^j) + \lambda_h + \varepsilon_j) = \text{ord}_\ell \kappa_0 + k + \mu(\ell^k - 1) + \lambda_h k + \sum_{j \leq k} \varepsilon_j$, and for $k \geq j_0 - 1$ the last sum is the full sum. (iii) is read off. Machine confirmation (Table 7.1): the law holds with the predicted $(\mu, \lambda, \nu_0) = (3, 1, 1), (0, 3, 4), (2, 3, 3), (0, 1, 0)$ — the four rows of [Ch. 3, Table 2] — at every computed level $k \geq 2$, and in fact at every $k \geq 0$ except in the $(1, 2, 0)$ tower, where $(\varepsilon_1, \varepsilon_2) = (-1, +2)$ and the law starts at $k = 2$, exactly as [Ch. 5, §3] reported. $\square$

Corollary 7.2.3 (The exceptional-zero regime). *The following are equivalent: $\lambda_h = 0$; $\text{ord}_\ell a_2 = \mu$; $\text{ord}_\ell \mathcal{L}(Y_\bullet) = \mu - \text{ord}_\ell \kappa(Y_0)$. In this regime $\varepsilon_j = 0$ for every $j \geq 1$, the law of Theorem 7.2.2(ii) holds for every $k \geq 0$, and*

$$\nu_0 = -\text{ord}_\ell \mathcal{L}(Y_\bullet) :$$

the constant term of the Iwasawa law is minus the ℓ -adic valuation of the $\mathcal{L}$ -invariant. This is the case in both $(1, 0, 0)$ towers: $\nu_0 = 1 = -\text{ord}_2(-\frac{1}{2})$ and $\nu_0 = 0 = -\text{ord}_3(-\frac{1}{2})$.

Proof. $\mu \leq \text{ord}_\ell a_2$ always, with equality iff the constant coefficient attains the minimum iff $\lambda_h = 0$; the third condition is the second rewritten. If $\lambda_h = 0$ then $P = 1$, $\text{ord}_\ell h(\zeta - 1) = \mu$ for every ζ , so $N_j = \mu\varphi(\ell^j)$ and $\varepsilon_j = 0$, $j_0 = 1$; then $\nu_0 = \text{ord}_\ell \kappa_0 - \mu = \text{ord}_\ell \kappa_0 - \text{ord}_\ell a_2 = -\text{ord}_\ell \mathcal{L}$. $\square$

Theorem 7.2.4 (The $\mathcal{L}$ -defect identity). *In general,*

$$\nu_0 + \text{ord}_\ell \mathcal{L}(Y_\bullet) = (\text{ord}_\ell a_2 - \mu) + \sum_{j \geq 1} \varepsilon_j =: \delta(Y_\bullet, \ell),$$

with $\text{ord}_\ell a_2 - \mu \geq 0$ and equality iff $\lambda_h = 0$. On the four towers $\delta = 0, 4, 2, 0$.

Proof. Substitute $\text{ord}_\ell \mathcal{L} = \text{ord}_\ell a_2 - \text{ord}_\ell \kappa_0$ into Theorem 7.2.2(ii). $\square$

Remark 7.2.5 (A correction to the specification). The specification asked to prove that $\text{ord}_\ell \mathcal{L}(Y_\bullet)$ “aligns exactly” with $\nu_0 - \text{ord}_\ell \kappa(Y_0) + \mu$. By Theorem 7.2.2 the latter quantity is $\sum_j \varepsilon_j$, which is $0, 0, 1, 0$ on the four towers, whereas $\text{ord}_\ell \mathcal{L} = -1, 0, -1, 0$: there is no such identity. The statements that hold are Corollary 7.2.3 and Theorem 7.2.4. The specification’s other claim, that $\text{ord}_\ell a_2 = \mu$ “controls the main term in the $\lambda_h = 0$ regime”, is the definition of that regime; its content is the vanishing of all defects and the formula $\nu_0 = -\text{ord}_\ell \mathcal{L}$.

Proposition 7.2.6 (The exceptional zero inside the Iwasawa module). *Let $X_\infty \cong \Lambda^n/D(1+T)\Lambda^n$ be the limit module of [Ch. 6, Thm. 2.1], with $\text{char}_\Lambda(X_\infty) = (\Theta)$. Then $X_\infty/TX_\infty \cong (\mathbb{Z} \oplus \text{Jac}(Y_0)) \otimes \mathbb{Z}_\ell$ has $\mathbb{Z}_\ell$ -rank one, while $\text{ord}_T \Theta = 2$; at the augmentation prime the localization is*

$$X_\infty \otimes_\Lambda \Lambda_{(T)} \cong \Lambda_{(T)}/T^2\Lambda_{(T)},$$

cyclic of length two. The exceptional zero is the statement that the Eisenstein eigenspace of the Iwasawa module is not semisimple.

Proof. The first isomorphism is the exact control theorem [Ch. 6, Cor. 2.2] at $k = 0$. $\Lambda_{(T)}$ is a discrete valuation ring with uniformizer T and residue field $\mathbb{Q}_\ell$, so $D(1+T)$ has elementary divisors over it; reducing modulo T gives $D(1) = \Delta_0$ of rank $n - 1$ over $\mathbb{Q}_\ell$, so exactly one elementary divisor is divisible by T , and since the T -adic order of the determinant is 2 (Theorem 7.1.2) that divisor is T^2 times a unit. $\square$

7.3 The operator-algebraic formulation

Setting from [Ch. 2]. $\mathcal{G}_\Gamma$ is the Hecke groupoid, $C^*(\mathcal{G}_\Gamma) = \mathcal{O}_{A^{\text{nb}}}$ the Cuntz–Krieger algebra of the non-backtracking matrix $A^{\text{nb}} = T^t S - J$ [Ch. 2, Def. 1.7, Prop. 1.9(ii)], where $S, T \in M_{n \times 2m}(\mathbb{Z})$ are the start and terminal incidence matrices of oriented edges and J is edge reversal. The Satake dynamics $\sigma^{(\pi)}$ [Ch. 2, Def. 2.1] has partition function $Z_p(\beta; \pi) = L_p(\beta, \pi_p)$ [Ch. 2, Thm. 2.2]; for the Eisenstein parameter $(\alpha_1, \alpha_2) = (1, q)$ the KMS_β simplex is nonempty exactly at the critical inverse temperature $\beta = 1$, where it is a single Perron state [Ch. 2, Thm. 2.4(ii),(iv)]; off the Perron locus the Satake data survive only as KMS functionals [Ch. 2, Thm. 2.5, Rem. 2.6]. Throughout $u = q^{-\beta}$, so $\beta = 1$ is $u = q^{-1}$.

Definition 7.3.1 (Character twist). The voltage α is a $\mathbb{Z}$ -valued continuous 1-cocycle c_α on $\mathcal{G}_\Gamma$ — the sum of α along the edges of a Hecke move, well defined on $(\omega, i - j, \omega')$ because the tails agree — commuting with the degree cocycle c . It induces a $\mathbb{T}$ -action γ^α on $C^*(\mathcal{G}_\Gamma)$ commuting with $\sigma^{(\pi)}$, and for $t \in \mathbb{C}^\times$ the *twisted transfer operator* and *twisted Hecke operators*

$$B_t := B\Lambda_t, \quad \Lambda_t := \text{diag}(t^{\alpha(e)})_e, \quad T_k(t)_{uv} := \sum_{\text{nb paths } u \rightarrow v \text{ of length } k} t^{\alpha(\text{path})},$$

with $B = A^{\text{nb}}$ on $\mathbb{C}^{2m}$; for $|t| = 1$ these are the character-twisted Ruelle–Hecke transfer operator of [Ch. 2, Def. 2.1] at the Eisenstein potential and the twisted Hecke correspondences. Note $T_1(t) = A(t)$ and $\Lambda_t J \Lambda_t = J$, since $\alpha(\bar{e}) = -\alpha(e)$.

Lemma 7.3.2 (Twisted Hecke recursion). *$A(t)T_k(t) = T_{k+1}(t) + qT_{k-1}(t)$ for $k \geq 2$, $A(t)^2 = T_2(t) + (q+1)I$, hence*

$$\sum_{k \geq 0} T_k(t)u^k = (1 - u^2)(I - uA(t) + qu^2I)^{-1}$$

as an identity of matrix-valued rational functions of (u, t) .

Proof. $A(t)T_k(t)$ sums over an edge $e: u \rightarrow w$ followed by a non-backtracking path $w \rightarrow v$ of length k , with weight $t^{\alpha(e)}t^{\alpha(\text{path})}$. The terms in which the path begins with $\bar{e}$ have weight $t^{\alpha(e)+\alpha(\bar{e})}t^{\alpha(\text{rest})} = t^{\alpha(\text{rest})}$, the rest being a non-backtracking path $u \rightarrow v$ of length $k - 1$ whose first edge is not e ; for each such rest there are q admissible e . This is the proof of [Ch. 2, Lem. 1.5] with the multiplicative weight carried along; the generating function follows by the same algebra. $\square$

Theorem 7.3.3 (Twisted Bass identity). *With $P(u, t) := \det(I_n - uA(t) + qu^2I_n)$,*

$$\det(I_{2m} - uB\Lambda_t) = (1 - u^2)^{r-1} P(u, t).$$

Proof. Recall $SA_\Gamma T^t = A(t)$ (entry (u, v) sums $t^{\alpha(e)}$ over $e: u \rightarrow v$), $SJ = T$, $TT^t = (q+1)I$, and $\Lambda_t J \Lambda_t = J$. The operator $J\Lambda_t$ is an involution $((J\Lambda_t)^2 = J\Lambda_t J\Lambda_t = JJ = I)$ with trace 0 (no edge is its own reverse),

so its eigenvalues are ± 1 with multiplicity m each and $\det(I + uJ\Lambda_t) = (1 - u^2)^m$, with $(I + uJ\Lambda_t)^{-1} = (1 - u^2)^{-1}(I - uJ\Lambda_t)$. Then $I - uB\Lambda_t = (I + uJ\Lambda_t) - uT^tS\Lambda_t$, and by $\det(I - XY) = \det(I - YX)$,

$$\begin{aligned}\det(I - uB\Lambda_t) &= (1 - u^2)^m \det\left(I_n - \frac{u}{1-u^2} S\Lambda_t(I - uJ\Lambda_t)T^t\right) \\ &= (1 - u^2)^{m-n} \det\left((1 - u^2)I_n - u(A(t) - uSJT^t)\right) \\ &= (1 - u^2)^{m-n} \det(I - uA(t) + qu^2I),\end{aligned}$$

using $S\Lambda_tJ\Lambda_tT^t = SJT^t = TT^t = (q+1)I$ and $m - n = r - 1$. For $t = 1$ this is Bass's identity [Ch. 2, Prop. 3.4]; for $|t| = 1$ it is the Artin-Ihara L -function formula of [22]. Verified as a polynomial identity in (u, t) on all four voltage assignments (Table 7.1). $\square$

Theorem 7.3.4 (Exceptional zero at the critical temperature). *Let $P(u, t)$ be as above and $Z(\beta; t) := \text{tr} \sum_k T_k(t) q^{-k\beta}$ the twisted partition function of the Eisenstein Satake dynamics.*

- (i) *Functional equation: $P(\frac{1}{qu}, t) = (qu^2)^{-n} P(u, t)$.*
- (ii) *At the critical inverse temperature $\beta = 1$, $P(q^{-1}, t) = q^{-n} \Theta(t - 1)$: the twisted Bass determinant of the Satake dynamics at $\beta = 1$ is the graph ℓ -adic L -function.*
- (iii) *At $(u, t) = (q^{-1}, 1)$ the zero is simple in the temperature direction and of order two in the character direction:*

$$\begin{aligned}\partial_u P(q^{-1}, 1) &= -n(q - 1) q^{1-n} \kappa(Y_0), \\ \partial_t P(q^{-1}, 1) &= 0, \quad \frac{1}{2} \partial_t^2 P(q^{-1}, 1) = q^{-n} \mathcal{L}(Y_\bullet) \kappa(Y_0).\end{aligned}$$

The ratio $\frac{1}{2} \partial_t^2 P / \partial_u P = -\mathcal{L}(Y_\bullet) / (nq(q - 1)) = \|h_\alpha\|^2 / (nq(q - 1))$ is independent of $\kappa(Y_0)$.

- (iv) *The unique KMS_1 state of [Ch. 2, Thm. 2.4] is the $t = 1$ member of the family; for $t \neq 1$ on the unit circle with $t^{\alpha(C)} \neq 1$ for some cycle C there is no KMS_1 state, only the functional of [Ch. 2, Rem. 2.6], and the critical partition function has a double pole at the trivial character with*

$$Z(1; t) = q(1 - q^{-2}) \text{tr} D(t)^{-1} = \frac{n(q - q^{-1})}{\mathcal{L}(Y_\bullet)} \frac{1}{(t - 1)^2} + O((t - 1)^{-1}).$$

Proof. (i) $I - \frac{1}{qu}A + \frac{q}{q^2u^2}I = (qu^2)^{-1}(qu^2I - uA + I)$. (ii) At $u = q^{-1}$, $I - q^{-1}A(t) + q^{-1}I = q^{-1}D(t)$. (iii) $\partial_t P(q^{-1}, t) = q^{-n} f'(t)$ and $\frac{1}{2} \partial_t^2 P(q^{-1}, 1) = q^{-n} a_2$, so Theorem 7.1.2 gives the t -derivatives. For the u -derivative, $\partial_u \det M(u) = \text{tr}(\text{adj} M(u) M'(u))$ with $M(q^{-1}) = q^{-1} \Delta_0$, $\text{adj}(q^{-1} \Delta_0) = q^{1-n} \text{adj} \Delta_0 = q^{1-n} \kappa_0 \mathbf{1} \mathbf{1}^t$ (all cofactors of a Laplacian equal κ_0), and $M'(q^{-1}) = -T_p + 2I$, so $\partial_u P = q^{1-n} \kappa_0 \mathbf{1}^t (2I - T_p) \mathbf{1} = q^{1-n} \kappa_0 n(2 - (q+1))$. (iv) Existence of a KMS_1 state requires $\rho(q^{-1}B\Lambda_t) = 1$ [Ch. 2, Thm. 2.4(i)]; for $|t| = 1$ the twisted Laplacian $D(t)$ is Hermitian positive semidefinite with $D(t)v = 0$ forcing $|v|$ constant and $t^{\alpha(C)} = 1$ on every cycle, so under the hypothesis $D(t)$ is positive definite, $P(q^{-1}, t) \neq 0$, and the pressure equation fails (Perron obstruction, as in [Ch. 2, Thm. 2.5]). By Lemma 7.3.2, $Z(1; t) = (1 - q^{-2}) \text{tr}(I - q^{-1}A(t) + q^{-1}I)^{-1} = q(1 - q^{-2}) \text{tr} D(t)^{-1}$. The Perron eigenvalue branch of $D(t)$ through 0 at $t = 1$ is $\lambda(1 + s) = -\frac{\|h_\alpha\|^2}{n} s^2 + O(s^3)$ (this is the Schur-complement computation in the proof of Theorem 7.1.2: $f(1 + s) / \det \Delta'$), and the other $n - 1$ eigenvalues are bounded away from zero, so $\text{tr} D(t)^{-1} = -\frac{n}{\|h_\alpha\|^2} (t - 1)^{-2} + O((t - 1)^{-1}) = \frac{n}{\mathcal{L}(Y_\bullet)} (t - 1)^{-2} + O((t - 1)^{-1})$. On K_4 the leading coefficients are $-12, -6, -4, -12$ (Table 7.1). $\square$

Remark 7.3.5 (Which derivative). The specification asked for “the second derivative of the KMS functional at $\beta = 1$ ”. Theorem 7.3.4 locates the derivatives precisely: at the critical point the *temperature* derivative is simple and carries the algebraic L -value $\kappa(Y_0)$; the *character* derivative is of second order and carries $\mathcal{L}(Y_\bullet) \kappa(Y_0)$; their ratio is the $\mathcal{L}$ -invariant up to the explicit normalization $nq(q - 1)$, with $\kappa(Y_0)$ cancelling — which is the Greenberg–Stevens shape [20], a ratio of a derivative in the “weight” direction to one in the cyclotomic direction. The specification’s request for a cyclic-cohomology formulation is not met here: the

natural candidate pairs the derivation generating γ^α with the Fredholm module of [Ch. 3, Thm. 3.3], and we have not established that pairing. We record it as open (Remark 7.5.1) rather than dress the second derivative in cohomological language it has not earned.

Theorem 7.3.6 (The abelian shadow limit carries Θ). *Let (Y_k) be a connected abelian $\mathbb{Z}/\ell^k$ voltage tower, $\mathcal{O}_k := \mathcal{O}_{B(Y_k)}$ the shadow algebras of [Ch. 4] ($c = q$ at every level, all levels being $(q+1)$ -regular), and $\pi: Y_{k+1} \rightarrow Y_k$ the covering maps. Then π extends to an in-covering $\pi^\sharp$ of shadow graphs, [Ch. 12, Lem. 2.1] yields unital injective $*$ -homomorphisms $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, and $\mathcal{O}_\infty^{\text{ab}} := \varinjlim (\mathcal{O}_k, \varphi_k)$ is a unital Kirchberg algebra with*

$$K_1(\mathcal{O}_\infty^{\text{ab}}) \cong \mathbb{Z}, \quad \text{Tor } K_0(\mathcal{O}_\infty^{\text{ab}}) \cong \text{Hom}_{\text{cont}} \left(\varprojlim_k (\text{Jac}(Y_k), \pi_*), \mathbb{Q}/\mathbb{Z} \right), \quad K_0/\text{Tor} \cong \mathbb{Z}[1/\ell].$$

The deck transformations induce a continuous action of $\mathbb{Z}_\ell = \varprojlim \mathbb{Z}/\ell^k$ on $\mathcal{O}_\infty^{\text{ab}}$ commuting with the gauge action, and the ℓ -primary torsion of K_0 is, as a discrete Λ -module, the Pontryagin dual of the compact Λ -module $\text{Tor } X_\infty$ of [Ch. 6, Thm. 2.1], whose characteristic ideal is (Θ) . The ideal (Θ) is stable under the involution $1+T \mapsto (1+T)^{-1}$ (Lemma 7.1.1), so the same Θ — with its exceptional zero and its $\mathcal{L}$ -invariant — is the characteristic element on both sides of the duality.

Proof. A covering map is bijective on in-arcs and on out-arcs at every vertex, so $\pi^\sharp (v \mapsto \pi(v), v' \mapsto \pi(v)')$, shadow arcs label-wise) is an in-covering, and [Ch. 12, Lem. 2.1, Thm. 2.3] apply verbatim with τ replaced by π : $K_0(\varphi_k)$ is the pullback π^* on coker Δ_k (the Laplacians are symmetric, so the transpose convention of [Ch. 12, Conv. 1.1] is invisible), which is the Pontryagin adjoint of the norm π_* of [Ch. 6, Lem. 1.2–1.3] by [Ch. 12, Lem. 1.5]; $\pi^* \mathbf{1} = \mathbf{1}$ gives unitality, $\deg \pi^* = \ell \deg$ gives the free part, and $K_1(\varphi_k) = \text{id}$ as in [Ch. 12, Thm. 2.3(iii)]. The limit is Kirchberg by [Ch. 12, Thm. 3.2] and its K -theory follows as in [Ch. 12, Thm. 3.3]. Each deck generator σ_k is an automorphism of $Y_k^\sharp$ covering σ_{k-1} , hence induces an automorphism of $\mathcal{O}_k$ with $\sigma_{k+1} \circ \varphi_k = \varphi_k \circ \sigma_k$ (both send p_v to $\sum_{\pi(w)=\sigma_k v} p_w$), so $\mathbb{Z}_\ell$ acts on the limit, continuously because on each $\mathcal{O}_k$ the action factors through $\mathbb{Z}/\ell^k$. On $K_0(\mathcal{O}_k) = \text{coker } \Delta_k \otimes \mathbb{Z}_\ell$ the deck action is multiplication by $\sigma \in \mathbb{Z}_\ell[G_k]$ [Ch. 6, Lem. 1.1], so the ℓ -part of $\text{Tor } K_0(\mathcal{O}_\infty^{\text{ab}}) = \varinjlim \text{Jac}(Y_k)^\vee$ is the Pontryagin dual of $\varinjlim \text{Jac}(Y_k) \otimes \mathbb{Z}_\ell = \text{Tor } X_\infty$ with the contragredient Λ -action, and $\text{char}_\Lambda(X_\infty) = (\Theta)$ is [Ch. 6, Thm. 2.1]. The involution stability is $f(t^{-1}) = f(t)$. $\square$

Remark 7.3.7 (Two algebras, one zero). The shadow algebras are Hecke-functorial and not Hecke-dynamical [Ch. 4, Rem. 4.3]: their gauge dynamics reach the Laplacian at $u = 1$, i.e. $\beta = 0$, where pure infiniteness forbids traces and hence KMS_0 states. So the exceptional zero has two operator-algebraic homes, and they do not overlap: the *dynamical* side — partition functions, the critical temperature $\beta = 1$, the double pole of Theorem 7.3.4(iv) — lives on $C^*(\mathcal{G}_\Gamma)$ of [Ch. 2]; the *K-theoretic* side — Θ as characteristic element, Proposition 7.2.6 — lives on the equivariant limit $\mathcal{O}_\infty^{\text{ab}}$ of Theorem 7.3.6. The specification's request to read the derivative off “the gauge action of $\mathcal{O}_\infty$ ” conflates the two; by [Ch. 3, Thm. 1.6] no single algebra can carry both.

7.4 The Iwahori tower: a simple zero and no $\mathcal{L}$ -invariant

Theorem 7.4.1 (Residue constancy along the Iwahori tower). *For the Iwahori path tower $(Y_k)_{k \geq 1}$ of [Ch. 3, §5] over a $(q+1)$ -regular base with $q = p$, let $\zeta_k(u)^{-1} := \det(I - uA_k)$. Then:*

- (i) $\zeta_k(u)^{-1}$ is independent of $k \geq 1$ and equals $\det(I - uA^{\text{nb}}) = (1 - u^2)^{r-1} P(u, 1)$;
- (ii) $u = q^{-1}$ is a simple zero of $\zeta_k(u)^{-1}$ at every level, with

$$\frac{d}{du} \det(I - uA_k) \Big|_{u=q^{-1}} = -q^{2-n_k} n_k \kappa(Y_k);$$

- (iii) consequently $n_k \kappa(Y_k) q^{-n_k}$ is independent of $k \geq 1$, which is the recursion $\kappa(Y_{k+1}) = q^{n_k(q-1)-1} \kappa(Y_k)$ of [Ch. 5, Thm. 2.1], and comparing with level 0,

$$\kappa(Y_1) = \frac{(1 - q^{-2})^{r-1} (q-1) n q^{n_1-1-n}}{n_1} \kappa(Y_0);$$

- (iv) the pseudo-Iwasawa law [Ch. 5, Thm. 2.2] reads $\text{ord}_p \kappa(Y_k) = n_k - \text{ord}_p n_k + \text{const} = \mu p^k - k + \nu_0$: the Euler term $-k$ is the p -adic valuation of the vertex count $n_k = n(q+1)q^{k-1}$, i.e. of the Eisenstein normalization in the residue.

Proof. (i) $A_1 = A^{\text{nb}}$, and $\det(zI - A_{k+1}) = z^{n_{k+1}-n_k} \det(zI - A_k)$ [Ch. 5, eq. (2.1)] gives $\det(I - uA_{k+1}) = \det(I - uA_k)$; the last equality is Bass [Ch. 2, Prop. 3.4]. (ii) With $M(u) = I - uA_k$, $M(q^{-1}) = q^{-1}\Delta_k$ has the simple eigenvalue 0 (Perron), so the zero is simple; $\text{adj}(q^{-1}\Delta_k) = q^{1-n_k} \kappa(Y_k) \mathbf{1}\mathbf{1}^t$ since all cofactors of the Eulerian Laplacian equal $\kappa(Y_k)$ [Ch. 5, proof of Thm. 2.1, Step 2], and $\text{tr}(\mathbf{1}\mathbf{1}^t(-A_k)) = -\mathbf{1}^t A_k \mathbf{1} = -qn_k$. (iii) Equate the derivatives at consecutive levels, using $n_{k+1} = qn_k$; for the level-0 formula differentiate $(1 - u^2)^{r-1}P(u, 1)$ at q^{-1} using Theorem 7.3.4(iii). (iv) $\text{ord}_p(n_k \kappa_k q^{-n_k})$ constant gives $\text{ord}_p \kappa_k = n_k - \text{ord}_p n_k + \text{const}$, and $\text{ord}_p n_k = \text{ord}_p(n(q+1)) + k - 1$. On K_4 : $n_k \kappa_k q^{-n_k} = \frac{9}{8}$ at $k = 1, 2, 3$, the derivative is $-\frac{9}{2}$ at every level, and $\kappa(Y_1) = \frac{(3/4)^2 \cdot 4 \cdot 2^7}{12} \cdot 16 = 384$ (Table 7.1). $\square$

Remark 7.4.2 (No $\mathcal{L}$ -invariant for the non-normal tower). The specification asked for $\Theta''(0)$, the $\mathcal{L}$ -invariant and the defects ε_j “on the Iwahori path tower as well”. There is no Θ there: the tower carries no deck action, hence no character variable, hence no pencil, and the Hecke variable u is the only direction available. In that direction the zero at the critical temperature is simple at every level (Theorem 7.4.1(ii)), with residue $\kappa(Y_k)$ rather than a second-order coefficient. What replaces the $\mathcal{L}$ -invariant is the tower constant $n_k \kappa(Y_k) q^{-n_k}$, and what replaces the bookkeeping identity is the observation that its constancy is the growth law. This gives the Euler term $-k$ its third reading, after the Eisenstein rank removed once per level [Ch. 5, Thm. 2.4] and the Eisenstein index through the connecting map of the snake [Ch. 10, Cor. 2.2]: it is the p -adic valuation of the vertex count, the normalization $\prod_{\mu \neq 0} \mu = n_k \kappa_k$ of [Ch. 5, eq. (2.2)] read p -adically. The three readings are one fact seen from three sides, and none of them is an $\mathcal{L}$ -invariant.

7.5 Scope, residue, corrections

Remark 7.5.1 (Solved, open, impossible). *Solved*: the direction of [Ch. 5, §3], made precise. The double zero is a theorem with the closed form $\frac{1}{2}\Theta''(0) = -\kappa(Y_0)\|h_\alpha\|^2$; the $\mathcal{L}$ -invariant is defined, computed, and shown to be a Hodge energy and ℓ -independent; the bookkeeping identity holds with an exact description of the defects, and the constant term is $-\text{ord}_\ell \mathcal{L}$ in the exceptional-zero regime; the twisted Bass identity and the critical-temperature theorem give the Mazur–Tate–Teitelbaum and Greenberg–Stevens shapes on $C^*(\mathcal{G}_T)$; the abelian shadow limit carries Θ as equivariant characteristic element. *Open*: (a) a cyclic-cohomology formulation of Theorem 7.3.4(iii) (Remark 7.3.5); (b) a converse to Theorem 7.2.4 — whether $\delta(Y_\bullet, \ell) = 0$ forces $\lambda_h = 0$; (c) a structural formula for the defects ε_j at $j < j_0$ in terms of the Newton polygon of P at the levels ℓ^j , beyond the resultant definition; (d) everything about the non-normal tower that [Ch. 9, Prob. 5.1] and [Ch. 10, Rem. 3.1] left open, which this paper does not touch. *Impossible or false as stated*: an $\mathcal{L}$ -invariant for the Iwahori tower in the sense of Definition 7.1.3 (Remark 7.4.2); the alignment formula of the specification (Remark 7.2.5); a single algebra carrying both the dynamical and the K -theoretic side (Remark 7.3.7, by [Ch. 3, Thm. 1.6]).

Remark 7.5.2 (Five corrections to the task specification). (1) $\Theta(T) = \det D(1+T)$ lies in $\mathbb{Z}[T, (1+T)^{-1}] \subset \Lambda$, not in $\mathbb{Z}_\ell[T]$; the cleared g is the polynomial, and a_2 is the same for both. (2) “ $\text{ord}_\ell a_2 = \mu$ in the $\lambda_h = 0$ regime” is the definition of the regime; the theorem is Corollary 7.2.3. (3) The alignment “ $\text{ord}_\ell \mathcal{L} \leftrightarrow \nu_0 - \text{ord}_\ell \kappa_0 + \mu$ ” is false; Theorem 7.2.4 is the identity that holds. (4) The KMS second derivative is in the character direction, at $\beta = 1$, on $C^*(\mathcal{G}_T)$ and not on the shadow limit; the temperature derivative carries $\kappa(Y_0)$ (Theorem 7.3.4, Remarks 7.3.5–7.3.7); no cyclic-cohomology statement is claimed. (5) The Iwahori tower has no Θ , no Θ'' , and no ε_j ; Theorem 7.4.1 is what it has. Separately, [Ch. 5, Rem. 1.11] and [Ch. 12] cite the tail-groupoid K -theory as [Ch. 2, Prop. 1.7(iii)]; in the compiled numbering of [Ch. 2] it is Proposition 1.9(iii), and [Ch. 12] has been corrected accordingly.

7.6 The verification battery

Table 7.1 lists the exact checks of `exceptional_zero.py`; the letters match the docstring of the script and the references in the text. The four abelian towers are those of [Ch. 3, Table 2]; the growth laws were

recomputed directly from the reduced Laplacians of the derived graphs ($k \leq 5$ for $\ell = 2$, up to 128 vertices; $k \leq 3$ for $\ell = 3$) and independently from the character product of Lemma 7.2.1, and the two agree at every level.

check	towers / levels	result
(Z) $\Theta(0) = \Theta'(0) = 0$; $a_2 = -8, -16, -24, -8$; $g = -8T^2, -T^4 - 16T^3 - 16T^2, -4T^4 - 24T^3 - 24T^2$	4 towers	exact
(H) $a_2 = -\kappa_0 \ h_\alpha\ ^2$; $\ h_\alpha\ ^2 = \frac{1}{2}, 1, \frac{3}{2}, \frac{1}{2}$; $\mathcal{L} = -\frac{1}{2}, -1, -\frac{3}{2}, -\frac{1}{2}$	4 towers	exact
(H') single chord: $R_{\text{eff}} = \frac{1}{2}$, 8 spanning trees avoid the chord $= -a_2$	(1, 0, 0)	exact
(K) double pole of $Z(1; t)$: leading coefficient $= n(q - q^{-1})/\mathcal{L} = -12, -6, -4, -12$	4 towers	exact
(M) Newton data $(\mu, \lambda_h) = (3, 0), (0, 2), (2, 2), (0, 0)$; regime iff $\text{ord}_\ell a_2 = \mu$	4 towers	exact
(E) $\varepsilon_j, j \leq 5$: all zero except $(\varepsilon_1, \varepsilon_2) = (-1, 2)$ for $(1, 2, 0)$	4 towers	exact
(G) $\text{ord}_\ell \kappa(Y_k)$ direct = character formula = law; (μ, λ, ν_0) as [Ch. 3, Table 2]	$k \leq 5 / k \leq 3$	exact
(L) $\nu_0 + \text{ord}_\ell \mathcal{L} = (\text{ord}_\ell a_2 - \mu) + \sum \varepsilon_j = 0, 4, 2, 0$; regime $\Rightarrow \nu_0 = -\text{ord}_\ell \mathcal{L}$	4 towers	exact
(B) $\det(I - uB\Lambda_t) = (1 - u^2)^{r-1} P(u, t)$ as a polynomial identity in (u, t)	4 towers	exact
(F) functional equation; $P(q^{-1}, t) = q^{-n} \Theta(t - 1)$; $\partial_u P = -8$; $\frac{1}{2} \partial_t^2 P = q^{-n} a_2$	4 towers	exact
(S) $\text{rk } D(1) = 3 = n - 1, \text{ord}_T \Theta = 2$	4 towers	exact
(I) Iwahori: level-independent $\det(I - uA_k)$; $n_k \kappa_k q^{-nk} = \frac{9}{8}$; slope $-\frac{9}{2}$ at $u = \frac{1}{2}$; $\kappa(Y_1) = 384$	$k \leq 3$	exact

Table 7.1: Exact verification (`exceptional_zero.py`) on K_4 ($q = 2, n = 4, r = 3, \kappa_0 = 16$). Tower order: $(1, 0, 0), \ell = 2$; $(1, 1, 1), \ell = 2$; $(1, 2, 0), \ell = 2$; $(1, 0, 0), \ell = 3$. For the fixed towers each line is itself a proof; the general statements are proved in the text.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from Papers I–VIII, finite linear algebra, Weierstrass preparation [23] and the character product formula; the twisted Bass identity is proved rather than quoted from [22]. The Mazur–Tate–Teitelbaum and Greenberg–Stevens references [21, 20] fix the shape of an analogy (Remarks 7.1.5, 7.3.5) and are load-bearing for nothing. Remark 7.5.2 records the corrections to the specification that prompted the paper and one citation correction propagated from [Ch. 5] to [Ch. 12]. Bibliographic data of [21, 20, 22, 19] were checked against publisher or indexing pages (titles, venues, years, volumes, pages); [23] is the entry verified in [Ch. 6].

Chapter 8

Algorithmic non-commutative class field theory, XIV: the graph $\mathcal{L}$ -invariant as spectral action, Bloch effective mass and Albanese length, and what the cyclic Chern character sees

Summary. [Ch. 7] identified the graph $\mathcal{L}$ -invariant of an abelian voltage tower as $\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2$, the negative harmonic energy of the voltage cochain, and left open “a cyclic-cohomology formulation”. This paper settles the question in both directions. Negatively: for the finite-dimensional spectral triple $(C(V) \oplus C(E^+), \ell^2(V) \oplus \ell^2(E^+), D_p)$ of [Ch. 3] the odd cyclic cohomology vanishes, every Fredholm index is zero, and the pairing proposed in the task specification is linear in α (it equals $4 \sum_e \alpha(e)$); a non-integer, quadratic quantity is not an index and does not pair with HC^1 . Positively: $\mathcal{L}(Y_\bullet)$ is the second variation of the spectral action of D_p along the flat $U(1)$ -connection $\varphi\alpha$, in the infrared,

$$\mathcal{L}(Y_\bullet) = \frac{n}{2} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \operatorname{Tr} e^{-t\Delta_\varphi} \Big|_{\varphi=0},$$

equivalently minus $n/2$ times the effective mass $\lambda_0''(0)$ of the bottom Bloch band of the $\mathbb{Z}$ -cover $\tilde{Y}_\alpha$ whose finite quotients are the tower; equivalently $-m\sigma^2$, where σ^2 is the diffusion constant of the voltage along simple random walk; equivalently minus the squared length of the class $[\alpha] \in H^1(Y_0; \mathbb{Z})$ in the Jacobian torus of Y_0 , the dual of the Albanese torus of Kotani–Sunada. The ℓ -adic L -function $\Theta(T) = \det D(1+T)$ is the Floquet determinant of $\tilde{Y}_\alpha$, and its exceptional zero of order two is the parabolic bottom of the band; the Kirchhoff identity $\det G = \kappa(Y_0)$ for the cycle Gram matrix gives $\kappa(Y_0)\mathcal{L}(Y_\bullet) \in \mathbb{Z}$. All identities are exact and machine-verified on K_4 and K_5 .

Status of the results

Every numbered statement is proved. The external inputs are the vanishing of odd cyclic cohomology of finite-dimensional commutative semisimple algebras [29, Ch. 2], which is also verified by direct linear

algebra, the Markov-chain central limit theorem for additive functionals of a finite reversible chain, and Kirchhoff's theorem in its cycle form. The Kotani–Sunada Albanese torus [27, 28] and the spectral action [24, 26, 25] supply names and context, not steps. Section 8.9 says what was asked, what is proved, and what is impossible.

8.1 Setting

Y_0 is a finite connected $(q+1)$ -regular graph with n vertices and $m = \frac{(q+1)n}{2}$ edges, E^+ an orientation, $r = m - n + 1$, $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$, $\partial e = t(e) - o(e)$, $\Delta_0 = \partial\partial^t = (q+1)I - A$, $\kappa_0 = \kappa(Y_0)$. The Hecke–Dirac operator of [Ch. 3, Def. 3.1] is

$$D_p = \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \text{ on } \ell^2(V) \oplus \ell^2(E^+), \quad D_p^2 = \Delta_0 \oplus \partial^t\partial,$$

and $\mathcal{A} := C(V) \oplus C(E^+)$ acts diagonally, so $(\mathcal{A}, \ell^2(V) \oplus \ell^2(E^+), D_p)$ is an even finite-dimensional spectral triple with grading $\gamma = 1 \oplus (-1)$. A voltage $\alpha \in \mathbb{Z}^{E^+}$ has harmonic representative $h_\alpha = \Pi\alpha$, $\Pi = I - \partial^t\Delta_0^+\partial$ the projection onto the cycle space $\ker \partial = \ker D_p \cap \ell^2(E^+)$, and [Ch. 7, Def. 1.3]

$$\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2 = -(\|\alpha\|^2 - (\partial\alpha)^t\Delta_0^+(\partial\alpha)) = \frac{\Theta''(0)}{2\kappa_0}, \quad \Theta(T) = \det D(1+T),$$

with $D(t) = (q+1)I - A(t)$, $A(t)_{uv} = \sum_{e: u \rightarrow v} t^{\alpha(e)}$, the voltage pencil. On K_4 the values are $-\frac{1}{2}, -1, -\frac{3}{2}$ for $(1, 0, 0), (1, 1, 1), (1, 2, 0)$ [Ch. 7, Table 1]; on K_5 ($q = 3$, $\kappa_0 = 125$) they are $-\frac{3}{5}$ for a single edge and $-\frac{7}{5}$ for a triangle (Table 8.1).

8.2 What cyclic cohomology sees, and what it cannot

Proposition 8.2.1 (Three impossibilities). *(i) $HC^1(\mathcal{A}) = 0$; more precisely the space $Z_\lambda^1(\mathcal{A})$ of cyclic 1-cocycles is zero, so there is no class $\text{Ch}^1(D_p) \in HC^1(\mathcal{A})$, and $HH_1(\mathcal{A}) = 0$, so there is no class $[\delta_\alpha] \in HH_1(\mathcal{A})$ either.*

(ii) Every index pairing of the even Fredholm module $(\mathcal{A}, \mathcal{H}, D_p)$ with $K_0(\mathcal{A})$ or of any compression PuP with a unitary u is an integer, and every Fredholm index of an operator between finite-dimensional spaces of equal dimension is zero. $\mathcal{L}(Y_\bullet) \notin \mathbb{Z}$ in general $(-\frac{1}{2}, -\frac{3}{2}, -\frac{3}{5}, -\frac{7}{5})$, so it is not an index.

(iii) With α acting by multiplication on $\ell^2(E^+)$,

$$\lim_{t \rightarrow 0^+} \text{Tr}_s([D_p, \alpha] D_p e^{-tD_p^2}) = \text{Tr}_s([D_p, \alpha] D_p) = 2 \text{Tr}(\alpha \partial^t \partial) = 4 \sum_{e \in E^+} \alpha(e),$$

linear in α : the formula proposed in the specification does not compute a quadratic invariant.

Proof. (i) $\mathcal{A} \cong \mathbb{C}^N$, $N = n + m$, with minimal idempotents e_i . A cyclic 1-cocycle τ satisfies $\tau(a, b) = -\tau(b, a)$ and $\tau(ab, c) - \tau(a, bc) + \tau(ca, b) = 0$. Taking $(a, b, c) = (e_i, e_i, e_k)$ with $k \neq i$ gives $\tau(e_i, e_k) = 0$, and cyclicity gives $\tau(e_i, e_i) = 0$; so $\tau = 0$. Since $\mathcal{A}$ is commutative, $b: C^0 \rightarrow C^1$ is zero and $HC^1 = Z_\lambda^1 = 0$. This is the case $n = 0$ of $HC^{2n+1}(\mathbb{C}^N) = 0$ [29, 2.1.12]; the linear system is solved by machine for $N = 10, 15$ (Table 8.1, line (C)). $HH_1(\mathbb{C}^N) = \Omega_{\mathbb{C}^N/\mathbb{C}}^1 = 0$ because $\mathbb{C}^N$ is separable. (ii) is linear algebra. (iii)

$[D_p, \alpha] D_p = \begin{pmatrix} \partial\alpha\partial^t & 0 \\ 0 & -\alpha\partial^t\partial \end{pmatrix}$, whose supertrace is $\text{Tr}(\partial\alpha\partial^t) + \text{Tr}(\alpha\partial^t\partial) = 2 \text{Tr}(\alpha\partial^t\partial)$, and $(\partial^t\partial)_{ee} = 2$. Continuity in t gives the limit. Verified on all six assignments: 4, 12, 12, 4, 12. $\square$

Remark 8.2.2 (What the even Chern character does see). $HC^{\text{even}}(\mathcal{A}) = HC^0(\mathcal{A})$ is the space of traces, $\mathbb{C}^N$, and the Chern character of the even Fredholm module $(\mathcal{A}, \mathcal{H}, D_p)$ pairs with $K_0(\mathcal{A}) = \mathbb{Z}^N$ to give the index of D_p localized at each idempotent — the content of [Ch. 3, Thm. 3.3]: $\ker D_p \cong \mathbb{Z} \oplus \mathbb{Z}^r$. This is topology (b_0 and b_1) and does not depend on α . The $\mathcal{L}$ -invariant depends on the metric (the unit-weight inner product on edges) and on α quadratically; it lives in the spectral, not the cohomological, side of the triple. Section 8.5 says where.

8.3 The heat-kernel formulas

Theorem 8.3.1 (Infrared, not ultraviolet). *For every voltage α ,*

$$\begin{aligned}\|h_\alpha\|^2 &= \lim_{t \rightarrow \infty} \langle \alpha, e^{-tD_p^2} \alpha \rangle = \|\alpha\|^2 - \int_0^\infty \langle \partial \alpha, e^{-s\Delta_0} \partial \alpha \rangle ds, \\ \langle \alpha, e^{-tD_p^2} \alpha \rangle &= \|h_\alpha\|^2 + \sum_{\lambda > 0} e^{-t\lambda} \|P_\lambda \alpha\|^2,\end{aligned}$$

the sum over the nonzero eigenvalues λ of $\partial^t \partial$ (equivalently of Δ_0) with spectral projections P_λ . The ultraviolet limit $t \rightarrow 0^+$ gives $\|\alpha\|^2$, not $\|h_\alpha\|^2$.

Proof. Π is the projection onto $\ker \partial^t \partial$, so $e^{-t\partial^t \partial} \rightarrow \Pi$ strongly and the spectral expansion follows. For the integral, $\Delta_0^+ = \int_0^\infty (e^{-s\Delta_0} - P_{\ker \Delta_0}) ds$ and $\partial \alpha \perp \mathbf{1} = \ker \Delta_0$. On K_4 : $\langle \alpha, e^{-tD_p^2} \alpha \rangle = \frac{1}{2} + \frac{1}{2}e^{-4t}$, $1 + 2e^{-4t}$, $\frac{3}{2} + \frac{7}{2}e^{-4t}$; on K_5 : $\frac{3}{5} + \frac{2}{5}e^{-5t}$, $\frac{7}{5} + \frac{8}{5}e^{-5t}$ (Table 8.1, line (H), symbolic in t). $\square$

8.4 The $\mathbb{Z}$ -cover, its Bloch bands, and the effective mass

Definition 8.4.1 (The $\mathbb{Z}$ -cover of the tower). $\tilde{Y}_\alpha$ is the derived graph of (Y_0, α) with voltage group $\mathbb{Z}$: vertices $V \times \mathbb{Z}$, an edge $(o(e), j) \rightarrow (t(e), j + \alpha(e))$ for each $e \in E^+$. The level- k graph Y_k of the tower [Ch. 6, §1] is $\tilde{Y}_\alpha / \ell^k \mathbb{Z}$. Its Laplacian $\tilde{\Delta}$ is $\mathbb{Z}$ -periodic, and the Floquet transform gives $\ell^2(\tilde{Y}_\alpha) \cong \int_{[0, 2\pi)}^\oplus \mathbb{C}^n \frac{d\varphi}{2\pi}$, $\tilde{\Delta} \cong \int^\oplus \Delta_\varphi$, $\Delta_\varphi := D(e^{i\varphi})$, the Hermitian twisted Laplacian. Its eigenvalues $\lambda_0(\varphi) \leq \dots \leq \lambda_{n-1}(\varphi)$ are the *Bloch bands*. The block-character decomposition [Ch. 6, Lem. 1.1] says

$$\text{spec } \Delta(Y_k) = \bigcup_{\varphi \in 2\pi\ell^{-k}\mathbb{Z}/2\pi\mathbb{Z}} \text{spec } \Delta_\varphi, \quad \Theta(e^{i\varphi} - 1) = \det \Delta_\varphi,$$

so the tower spectra sample the bands at the ℓ^k -th roots of unity and the ℓ -adic L -function is the Floquet determinant of the $\mathbb{Z}$ -cover.

Theorem 8.4.2 (Effective mass). λ_0 is even in φ , simple near $\varphi = 0$, and

$$\lambda_0(\varphi) = \frac{\|h_\alpha\|^2}{n} \varphi^2 + O(\varphi^4), \quad \text{i.e.} \quad \mathcal{L}(Y_\bullet) = -\frac{n}{2} \lambda_0''(0).$$

The exceptional zero of order two of Θ [Ch. 7, Thm. 1.2] is the parabolic bottom of the band: $\det \Delta_\varphi = \lambda_0(\varphi) \prod_{j \geq 1} \lambda_j(\varphi) = n\kappa_0 \cdot \frac{\|h_\alpha\|^2}{n} \varphi^2 + O(\varphi^3)$, and $\varphi^2 = -T^2 + O(T^3)$ for $T = e^{i\varphi} - 1$ gives $a_2 = -\kappa_0 \|h_\alpha\|^2$ again.

Proof. $D(t^{-1}) = D(t)^t$ [Ch. 7, Lem. 1.1] gives $\text{spec } \Delta_{-\varphi} = \text{spec } \Delta_\varphi$, so λ_0 is even and the odd Taylor terms vanish. At $\varphi = 0$ the kernel of Δ_0 is $\mathbb{C}\mathbf{1}$, simple, with gap $\lambda_1(0) > 0$; so λ_0 is analytic near 0 and Rayleigh–Schrödinger applies. Write $\Delta_\varphi = \Delta_0 + \varphi D_1 + \varphi^2 D_2 + O(\varphi^3)$ with $(D_1)_{uv} = -\sum_{e: u \rightarrow v} i\alpha(e)$, $(D_2)_{uv} = \sum_{e: u \rightarrow v} \alpha(e)^2/2$ (both sums over both orientations). Then $(D_1 \mathbf{1})_u = i(\partial \alpha)_u$, $\langle \mathbf{1}, D_1 \mathbf{1} \rangle = 0$, $\langle \mathbf{1}, D_2 \mathbf{1} \rangle = \|\alpha\|^2$, and

$$\lambda_0(\varphi) = \frac{\varphi^2}{n} \left(\langle \mathbf{1}, D_2 \mathbf{1} \rangle - \langle D_1 \mathbf{1}, \Delta_0^+ D_1 \mathbf{1} \rangle \right) + O(\varphi^4) = \frac{\varphi^2}{n} (\|\alpha\|^2 - (\partial \alpha)^t \Delta_0^+ \partial \alpha) + O(\varphi^4).$$

This is the Schur-complement branch $\lambda(1+s) = -\frac{\|h_\alpha\|^2}{n} s^2$ of [Ch. 7, proof of Thm. 3.4] with $s = e^{i\varphi} - 1$. The coefficient is computed exactly and checked to 10^{-12} numerically on all six assignments $(\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{3}{25}, \frac{7}{25})$; Table 8.1, line (B)). $\square$

8.5 The spectral action

$\Delta_\varphi = \partial_\varphi \partial_\varphi^*$ with $\partial_\varphi = \partial U_\varphi$, $U_\varphi = \text{diag}(e^{i\varphi\alpha(e)})$ on $\ell^2(E^+)$, is the Hodge Laplacian of the flat $U(1)$ -connection with holonomy $e^{i\varphi\alpha(C)}$ around each cycle C ; the twisted Dirac operator is $D_p^\varphi = \begin{pmatrix} 0 & \partial U_\varphi \\ U_\varphi^* \partial^t & 0 \end{pmatrix}$, $(D_p^\varphi)^2 = \Delta_\varphi \oplus U_\varphi^* \partial^t \partial U_\varphi$, and the *spectral action* [24] of the fluctuated triple is $\text{Tr } f((D_p^\varphi)^2)$.

Theorem 8.5.1 ($\mathcal{L}$ as second variation of the spectral action). *For the heat cutoff $f(x) = e^{-tx}$,*

$$\mathcal{L}(Y_\bullet) = \frac{n}{2} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi} \Big|_{\varphi=0} = \frac{n}{4} \lim_{t \rightarrow \infty} \frac{1}{t} \partial_\varphi^2 \text{Tr } e^{-t(D_p^\varphi)^2} \Big|_{\varphi=0},$$

with error $O(te^{-t\lambda_1(0)})$. For an arbitrary smooth cutoff f ,

$$\partial_\varphi^2 \text{Tr } f(\Delta_\varphi) \Big|_0 = f'(0) \lambda_0''(0) + \sum_{j \geq 1} \partial_\varphi^2 f(\lambda_j(\varphi)) \Big|_0,$$

and $\mathcal{L}(Y_\bullet) = -\frac{n}{2}$ times the coefficient of $f'(0)$ in the ground-band term.

Proof. $\text{Tr } e^{-t\Delta_\varphi} = e^{-t\lambda_0(\varphi)} + \sum_{j \geq 1} e^{-t\lambda_j(\varphi)}$, the first term analytic near 0 by Theorem 8.4.2, the remainder smooth in φ (a symmetric function of the other eigenvalues of an analytic Hermitian family) with φ -derivatives bounded by $Ct^2 e^{-t\lambda_1(0)}$ near 0. Since $\lambda_0'(0) = 0$, $\partial_\varphi^2 e^{-t\lambda_0} \Big|_0 = -t\lambda_0''(0) = -2t\|h_\alpha\|^2/n$. For the Dirac version, $(D_p^\varphi)^2 = \partial_\varphi \partial_\varphi^* \oplus \partial_\varphi^* \partial_\varphi$, and the two blocks have the same nonzero spectrum, so $\text{Tr } e^{-t(D_p^\varphi)^2} = 2 \text{Tr } e^{-t\Delta_\varphi} + (m - n)$. The general- f statement is the chain rule. Numerically on all six assignments the $t = 10$ value agrees with $\mathcal{L}$ to 10^{-14} (Table 8.1, line (S)). $\square$

Remark 8.5.2 (Reading). The Chern character of [26, 25] is the ultraviolet ($t \rightarrow 0$), cohomological, integer-valued side of the spectral triple; Proposition 8.2.1 shows that side is blind to α . The $\mathcal{L}$ -invariant is on the infrared ($t \rightarrow \infty$), spectral, rational side: the curvature of the ground state energy along the moduli of flat connections. On a Riemannian manifold the analogous quantity is the Hessian of the bottom of the spectrum of the twisted Laplacian at the trivial character, which is the Albanese metric [27]; the graph statement is Theorem 8.7.1 below. The Greenberg–Stevens shape of [Ch. 7, Rem. 3.5] — $\mathcal{L}$ as the ratio of a second character derivative to a first temperature derivative — is here the statement that the band is parabolic while the Perron eigenvalue moves linearly with β .

8.6 Diffusion

Let (X_i) be simple random walk on Y_0 started from the uniform distribution, and let

$$S_N = \sum_{i < N} \alpha(X_i, X_{i+1}), \quad \alpha(y, x) = -\alpha(x, y),$$

be the accumulated voltage; on the $\mathbb{Z}$ -cover, S_N is the fibre coordinate of the lifted walk.

Theorem 8.6.1 ($\mathcal{L}$ as diffusion constant). $\mathbb{E}S_N = 0$, $\text{Var}(S_N) = N\sigma^2 + O(1)$ and $S_N/\sqrt{N} \Rightarrow \mathcal{N}(0, \sigma^2)$ with

$$\sigma^2 = \frac{\|h_\alpha\|^2}{m} = -\frac{\mathcal{L}(Y_\bullet)}{m}.$$

Proof. The chain is reversible with uniform stationary law π and transition matrix $P = A/(q+1)$. Let $F(x) = \sum_y P(x, y)\alpha(x, y) = -(\partial\alpha)(x)/(q+1)$ be the drift; $\pi(F) = 0$. The Poisson equation $(I - P)g = F$, i.e. $\Delta_0 g = -\partial\alpha$, has the solution $g = -\Delta_0^+ \partial\alpha$, and $M_i = \alpha(X_i, X_{i+1}) + g(X_{i+1}) - g(X_i)$ are stationary martingale increments with $S_N = \sum M_i + g(X_0) - g(X_N)$. The martingale CLT gives the limit law with $\sigma^2 = \mathbb{E}_\pi M_0^2 = \sum_{x,y} \pi(x)P(x, y)(\alpha + \partial^t g)(x, y)^2$. Now $\alpha + \partial^t g = \alpha - \partial^t \Delta_0^+ \partial\alpha = h_\alpha$, and summing over both orientations of each edge, $\sigma^2 = \frac{2}{n(q+1)} \|h_\alpha\|^2 = \|h_\alpha\|^2/m$. The $O(1)$ term is $2\mathbb{E}[g(X_0)^2] - 2\mathbb{E}[g(X_0)g(X_N)] + \dots$, convergent geometrically. The Poisson-equation value of σ^2 is computed exactly, and $\text{Var}(S_N)$ is computed exactly for $N \leq 24$ from the moment-generating matrix over $\mathbb{Q}[s]/(s^3)$: on K_4 , $\sigma^2 = \frac{1}{12}, \frac{1}{6}, \frac{1}{4}$ and $\text{Var}(S_N) - N\sigma^2 \rightarrow \frac{1}{16}, \frac{1}{4}, \frac{7}{16}$; on K_5 , $\sigma^2 = \frac{3}{50}, \frac{7}{50}$ (Table 8.1, line (V)). $\square$

This is the one-dimensional case of the Kotani–Sunada central limit theorem for periodic random walks on crystal lattices [27], where the covariance is the Albanese metric.

8.7 Albanese length and integrality

Let $c_1, \dots, c_r$ be a $\mathbb{Z}$ -basis of the cycle lattice $H_1(Y_0; \mathbb{Z}) = \ker \partial \cap \mathbb{Z}^{E^+}$, $C = [c_1 \cdots c_r]$, $G = C^t C$ its Gram matrix, and $a_c = C^t \alpha$ the vector of periods $\langle \alpha, c_j \rangle$, which depends only on the class $[\alpha] \in H^1(Y_0; \mathbb{Z}) = \text{Hom}(H_1(Y_0; \mathbb{Z}), \mathbb{Z})$.

Theorem 8.7.1 ($\mathcal{L}$ as Albanese length).

$$\|h_\alpha\|^2 = a_c^t G^{-1} a_c = \|[\alpha]\|_{H^1}^2,$$

the squared norm of $[\alpha]$ in $H^1(Y_0; \mathbb{R})$ for the inner product dual to the cycle inner product on $H_1(Y_0; \mathbb{R}) \subset \mathbb{R}^{E^+}$ — the flat metric of the Jacobian torus $H^1(Y_0; \mathbb{R})/H^1(Y_0; \mathbb{Z})$, dual to the Albanese torus $\text{Alb}(Y_0) = H_1(Y_0; \mathbb{R})/H_1(Y_0; \mathbb{Z})$ of [28] in the unit-weight normalization. Hence $\mathcal{L}(Y_\bullet) = -\|[\alpha]\|_{H^1}^2$ depends only on $[\alpha]$, is invariant under switching $\alpha \mapsto \alpha + \partial^t \varphi$, and

$$\det G = \kappa(Y_0), \quad \kappa(Y_0) \mathcal{L}(Y_\bullet) \in \mathbb{Z},$$

which is the integrality $a_2 \in \mathbb{Z}$ of [Ch. 7, Thm. 1.2(ii)] seen from the cycle side.

Proof. h_α is the orthogonal projection of α onto the column space of C , so $h_\alpha = C(C^t C)^{-1} C^t \alpha$ and $\|h_\alpha\|^2 = \alpha^t C G^{-1} C^t \alpha$. The harmonic form h_α is the unique cycle with periods a_c , so $\|h_\alpha\|$ is the dual norm of the functional $[\alpha]$ on $(H_1(Y_0; \mathbb{R}), \langle \cdot, \cdot \rangle)$. Kirchhoff's theorem in its cycle form (the Gram determinant of a fundamental cycle basis equals the number of spanning trees) gives $\det G = \kappa_0$; a $\mathbb{Z}$ -basis change has determinant ± 1 , so $\det G = \kappa_0$ for every $\mathbb{Z}$ -basis, and $G^{-1} \in \kappa_0^{-1} M_r(\mathbb{Z})$. Verified: $\det G = 16, 125$; $\kappa_0 \mathcal{L} = -8, -16, -24, -75, -175$ (Table 8.1, lines (A),(T)). $\square$

Corollary 8.7.2 (Five faces of one number). *For an abelian voltage tower over Y_0 the following are equal: $\mathcal{L}(Y_\bullet) = \Theta''(0)/2\kappa_0$; $-\|h_\alpha\|^2$; $-\frac{n}{2} \lambda_0''(0)$; $\frac{n}{2} \lim_{t \rightarrow \infty} t^{-1} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi}|_0$; $-m\sigma^2$; $-\|[\alpha]\|_{H^1}^2$.*

8.8 The verification battery

check	cases	result
(C) $\dim Z_\lambda^1(\mathcal{A}) = \dim HC^1(\mathcal{A}) = 0$, $\mathcal{A} = \mathbb{C}^{10}, \mathbb{C}^{15}$	K_4, K_5	exact
(N) $\text{Tr}_s([D_p, \alpha] D_p) = 4 \sum \alpha(e) = 4, 12, 12, 4, 12$	all six	exact
(H) $\langle \alpha, e^{-tD_p^2} \alpha \rangle = \ h_\alpha\ ^2 + ce^{-t\lambda_1}$, limit and Duhamel integral $= \ h_\alpha\ ^2$	all six	exact, symbolic t
(B) $\lambda_0(\varphi) = \frac{\ h_\alpha\ ^2}{n} \varphi^2 + O(\varphi^4)$: exact coefficient $\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{3}{25}, \frac{7}{25}$; Richardson numerics	all six	exact + 40 digits
(S) $\frac{n}{2} t^{-1} \partial_\varphi^2 \text{Tr } e^{-t\Delta_\varphi} _0$ at $t = 5, 10, 20, 40$: $\mathcal{L}$ to $10^{-8}, 10^{-14}, 10^{-14}, 10^{-13}$	all six	numeric
(V) $\sigma^2 = \ h_\alpha\ ^2/m = \frac{1}{12}, \frac{1}{6}, \frac{1}{4}, \frac{3}{50}, \frac{7}{50}$; $\text{Var}(S_N)$ exact, $N \leq 24$	all six	exact
(A) $a_c^t G^{-1} a_c = \ h_\alpha\ ^2$; switching invariance; $\det G = \kappa_0$; $\kappa_0 \mathcal{L} \in \mathbb{Z}$	all six	exact
(T) [Ch. 7, Thm. 1.2(ii)] on K_5 : $a_2 = -75, -175 = -\kappa_0 \ h_\alpha\ ^2$	K_5	exact

Table 8.1: Verification (`cyclic_index.py`). “All six” = K_4 with $(1, 0, 0)$, $(1, 1, 1)$, $(1, 2, 0)$ [the $(1, 0, 0)$, $\ell = 3$ tower has the same voltage and $\mathcal{L}$] and K_5 with a single-edge and a triangle voltage.

8.9 Scope, residue, corrections

Remark 8.9.1 (Solved, open, impossible). *Solved:* open item (a) of [Ch. 7, Rem. 5.1], “a cyclic-cohomology formulation of Theorem 3.4(iii)”, in the only form it admits: the $\mathcal{L}$ -invariant is the infrared second variation of the spectral action of the Hecke–Dirac triple along the flat-connection direction α (Theorem 8.5.1), the effective mass of the $\mathbb{Z}$ -cover (Theorem 8.4.2), its diffusion constant (Theorem 8.6.1), and the Albanese length of $[\alpha]$ (Theorem 8.7.1); the exceptional zero is the parabolic band bottom, and $\kappa_0 \mathcal{L} \in \mathbb{Z}$ is Kirchhoff. *Open:* (a) whether a nonzero odd cyclic class on a genuinely non-commutative algebra of the tower, such as the crossed product $C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$ or the Cuntz–Krieger algebra of [Ch. 12], pairs to $\|h_\alpha\|^2$ or to its integer multiple $\kappa_0 \|h_\alpha\|^2$; the integrality of Theorem 8.7.1 makes the latter the only candidate for an index. (b) The rank-two analogue for the A_2 towers of Chs. 16–18, where $b_1 = 0$ [Ch. 18] and there is no α : the Bloch picture survives (the Hecke pencil of [Ch. 16] is a Floquet determinant over a 2-torus when a $\mathbb{Z}^2$ -cover exists) but the $\mathcal{L}$ -invariant does not. *Impossible as stated:* Proposition 8.2.1, all three items.

Remark 8.9.2 (Corrections to the task specification). (1) There is no “ $\text{Ch}^1(D_p) \in HC^1(\mathcal{A})$ ” and no “ $[\delta_\alpha] \in H_1(\mathcal{A})$ ”: both groups vanish for $\mathcal{A} = C(V) \oplus C(E^+)$, and every derivation of $\mathcal{A}$ is zero. (2) The proposed pairing $\lim_{t \rightarrow 0^+} \text{Tr}_s([D_p, \alpha] D_p e^{-t D_p^2})$ equals $4 \sum_e \alpha(e)$; it is linear in α and cannot equal $-\|\Pi\alpha\|^2$. (3) The $\mathcal{L}$ -invariant is not an index: it is rational and non-integral, and finite-dimensional Fredholm indices vanish. (4) The correct limit is $t \rightarrow \infty$, not $t \rightarrow 0^+$ (Theorem 8.3.1); the $t \rightarrow 0^+$ side of the JLO cocycle is the topological Chern character, which does not see α . (5) The theorem that does hold is a second-variation (curvature) statement for the spectral action, Theorem 8.5.1, and its geometric name is the Albanese metric, not “quantum topological index”. (6) The $(1, 0, 0)$, $\ell = 3$ tower is not a separate case: $\mathcal{L}$ depends on α only, and ℓ enters through [Ch. 7, §2] alone. The battery adds K_5 ($q = 3$) as asked.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from Papers II, IV and IX and finite linear algebra, plus the Markov-chain CLT and Kirchhoff’s theorem in cycle form. The vanishing of $HC^1(\mathbb{C}^N)$ is both quoted [29] and verified directly. Bibliographic data of [27, 28, 26, 24, 29, 25] were checked against publisher or indexing pages; [27, 28] are cited for the definition of the Albanese torus and the crystal-lattice CLT, and no statement of this paper depends on them.

Chapter 9

Algorithmic non-commutative class field theory, V: the degeneracy structure of the Iwahori tower and the Euler correction

Summary. The congruence-model (Iwahori path) tower $Y_0 \leftarrow Y_1 \leftarrow \cdots$ of [Ch. 3, §5] is non-normal, carries no deck action, and obeys the exact pseudo-Iwasawa law $\nu_p = \mu p^k - k + \nu_0$ with $\lambda = -1$, impossible over any commutative $\mathbb{Z}_p[[T]]$ base ([Ch. 3, Rem. 5.2], [Ch. 5, §2]). This paper equips that tower with its canonical algebraic structure and identifies, without fabricating a theory that does not yet exist, exactly what any future non-commutative characteristic formalism must specialize to.

The two degeneracy maps — initial and terminal truncation of non-backtracking paths, the graph avatars of $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$ — are packaged into an incidence calculus with six exact identities, among them $\tau_*\eta^* = A_k$ (the level- k Hecke operator realized through level $k+1$; at the bottom, $\tau_*\eta^* = T_p$ of [Ch. 2]) and $\tau_*\tau^* = \eta_*\eta^* = qI$. Theorem 9.2.1 computes the degeneracy complex: $\ker(\tau^* - \eta^*) = \mathbb{Z}\mathbf{1}$ and $\operatorname{coker}(\tau_* - \eta_*) \cong \mathbb{Z}$ torsion-free, so the Euler rank $\chi = 1$ at every level — the $-\chi$ of [Ch. 5, Thm. 2.4] is the H^0 of this complex. Theorem 9.3.1 shows the tail degeneracy alone intertwines the sandpile Laplacians and the Hecke operators, is surjective on p -primary torsion with kernel of the exact order $q^{n_k(q-1)-1}$ forced by [Ch. 5, Thm. 2.1], and hence assembles the tower into one pro-module $X_\infty = \varprojlim (\operatorname{coker} \Delta_k \otimes \mathbb{Z}_p, \tau_*)$ carrying a limit Hecke operator; the head degeneracy satisfies the mirror identities for the in-Laplacian tower and the exact asymmetry $\eta_*\Delta_{k+1} = q(\eta_* - \tau_*)$, so the structure is genuinely two-sided. Proposition 9.3.4 records a second, independent rigorous certificate that no commutative base suffices: $A_k A_k^t \neq A_k^t A_k$ at every computed level. We then state, as axioms rather than as a theorem, the specialization data any characteristic theory over the Iwahori–Hecke base must reproduce, and we decline — for the same reason as in [Ch. 5] — to define an Akashi-type series whose hypotheses our pro-ring does not satisfy.

Status of the results

Theorems 9.2.1 and 9.3.1 and Propositions 9.1.2, 9.3.2 are proved for a general Eulerian q -regular strongly connected level ≥ 1 (the base level $k = 0$ is the boundary case, treated in Remark 9.3.3); every identity was additionally machine-verified exactly on the K_4 tower at all levels $k \leq 3$ (Table 9.1). Proposition 9.3.4 is a finite exact computation on the fixed tower, hence a proof for it; no general-graph claim is made. Section 9.4 proves nothing and says so: it is a specification, and Problem 9.4.1 is open.

9.1 The degeneracy pair and its incidence calculus

Levels: Y_0 a finite connected $(q+1)$ -regular graph; for $k \geq 1$, V_{Y_k} = the non-backtracking k -paths of Y_0 , with an arc $p \rightarrow p'$ when p' is the shift of a common $(k+1)$ -path; $Y_{k+1} = L(Y_k)$ for $k \geq 1$, each Y_k Eulerian q -regular, $n_k = n(q+1)q^{k-1}$ ([Ch. 3, §5], [Ch. 5, §2]).

Definition 9.1.1 (Degeneracy maps). For $k \geq 1$ let $\tau, \eta: Y_{k+1} \rightarrow Y_k$ be terminal and initial truncation of paths, $\tau(v_0 \cdots v_{k+1}) = (v_0 \cdots v_k)$, $\eta(v_0 \cdots v_{k+1}) = (v_1 \cdots v_{k+1})$; under $V_{Y_{k+1}} = \text{Arcs}(Y_k)$ these are the tail and head maps. Write $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ for the pushforwards (sum over fibres) and $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ for the pullbacks; thus $\tau_* = D_{\text{out}}$, $\eta_* = D_{\text{in}}$, the incidence matrices of [Ch. 5, Thm. 2.1].

Proposition 9.1.2 (Incidence calculus). *For every $k \geq 1$:*

$$\tau_* \eta^* = A_k, \quad \eta_* \tau^* = A_k^t, \quad \eta^* \tau_* = A_{k+1}, \quad \tau^* \eta_* = A_{k+1}^t, \quad \tau_* \tau^* = \eta_* \eta^* = q I_{n_k}.$$

At the boundary level, with $\tau, \eta: Y_1 \rightarrow Y_0$ the endpoint maps on oriented edges, $\tau_ \eta^* = T_p$, the Hecke operator of [Ch. 2].*

Proof. Entrywise bookkeeping of tails and heads; the middle two are [Ch. 5, Thm. 2.1, Step 1], the outer two their transposes, the last two the regularity of degrees. The boundary identity is $ST^t = T_p$ in the notation of [Ch. 2, Thm. 3.7]. All six verified as exact matrix identities at $k = 1, 2$ (Table 9.1). $\square$

The pair (τ_*, η_*) is the graph avatar of the modular degeneracy pair $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$, and Proposition 9.1.2 is the statement that the level- k Hecke operator is the correspondence $\tau_* \circ \eta^*$ computed through level $k+1$ — the tower knows its own Hecke theory.

9.2 The degeneracy complex and the Euler rank

Theorem 9.2.1 (Euler rank of the degeneracy complex). *For every $k \geq 1$, with $\partial_k := \tau_* - \eta_*: \mathbb{Z}^{V_{Y_{k+1}}} \rightarrow \mathbb{Z}^{V_{Y_k}}$ and $\partial_k^* := \tau^* - \eta^*$:*

- (i) $\ker \partial_k^* = \mathbb{Z} \cdot \mathbf{1}$: the only pulled-back-equal functions are the constants — the Perron–Eisenstein line, of rank $\chi_k = 1$;
- (ii) $\text{coker } \partial_k \cong \mathbb{Z}$ via the degree functional, and it is torsion-free: the image of ∂_k is exactly the augmentation-zero sublattice.

Hence the Euler rank of the two-term degeneracy complex is $\chi = 1$ at every level, and the correction $-\chi \cdot k = -k$ of [Ch. 5, Thm. 2.4] is the accumulated H^0 of this complex, once per level.

Proof. (i) $(\partial_k^* x)_{(a \rightarrow b)} = x_a - x_b$; vanishing forces x constant along arcs, hence constant by strong connectivity. (ii) $\partial_k(e_{(a \rightarrow b)}) = e_a - e_b$; over a connected digraph the differences $e_a - e_b$ along arcs generate, integrally, the full kernel of the degree map (telescope along paths; any z with $\sum z_i = 0$ is an integer combination of consecutive differences), so $\text{im } \partial_k = \ker(\text{deg})$ and $\text{coker} \cong \mathbb{Z}$ with no torsion. Machine confirmation: the Smith form of ∂_k has exactly $n_k - 1$ unit invariants at $k = 1, 2$ (Table 9.1). $\square$

9.3 The tail norm and the pro-module

Theorem 9.3.1 (Tail norm). *For every $k \geq 1$, with $\Delta_k = qI - A_k$ the out-Laplacian:*

- (i) $\tau_* \Delta_{k+1} = \Delta_k \tau_*$ and $\tau_* A_{k+1} = A_k \tau_*$;
- (ii) τ_* is surjective and preserves the degree functional; hence the induced map on sandpile cokernels is surjective, restricts to a surjection of p -primary torsion, and its kernel there has order exactly $q^{n_k(q-1)-1}$, the increment proved in [Ch. 5, Thm. 2.1];

(iii) consequently $X_\infty := \varprojlim_k (\text{coker}_{\mathbb{Z}}(\Delta_k) \otimes \mathbb{Z}_p, \tau_*)$ is a well-defined pro-module carrying the limit Hecke operator $A_\infty := \varprojlim A_k$, and its torsion-size sequence is the pseudo-Iwasawa law $\text{ord}_p = \mu p^k - k + \nu_0$, $\mu = n(q+1)/q$, of [Ch. 5, Thm. 2.2] — now the size function of a single pro-object rather than a sequence of unrelated groups.

Proof. (i) $\tau_* A_{k+1} = D_{\text{out}} D_{\text{in}}^t D_{\text{out}} = A_k D_{\text{out}} = A_k \tau_*$ by associativity and Proposition 9.1.2; subtract from $q\tau_*$. (ii) Every level- k vertex extends, so τ_* hits every basis vector; $\deg \circ \tau_* = \deg$ since fibres are summed; torsion-surjectivity then follows exactly as in [Ch. 6, Lem. 1.3] (the kernel of $\deg$ is the torsion), and the kernel order is forced by counting against [Ch. 5, Thm. 2.1]. (iii) Inverse limits over surjections of compact modules; the Hecke compatibility is (i). Explicit machine confirmation of (ii) at the transition $2 \rightarrow 1$: surjective on the 2-part with kernel order exactly $2^{11} = 2^{n_1-1}$, the orders 7, 18 matching [Ch. 3, Rem. 5.2] (Table 9.1). $\square$

Proposition 9.3.2 (Head mirror, and the exact asymmetry). *For every $k \geq 1$: η_* intertwines the in-Laplacians, $\eta_*(qI - A_{k+1}^t) = (qI - A_k^t)\eta_*$, so the mirrored construction yields the in-sandpile pro-module X_∞^{in} ; on the out-tower, however,*

$$\eta_* \Delta_{k+1} = q(\eta_* - \tau_*) = -q\partial_k,$$

so η_* does not descend to the out-cokernels: the failure is exactly q times the degeneracy boundary of Theorem 9.2.1. The tower is thus canonically two-sided — a (τ, η) -bimodule system, not a single inverse system — and the obstruction to collapsing the two sides is again the Euler complex.

Proof. $\eta_* A_{k+1}^t = D_{\text{in}} D_{\text{out}}^t D_{\text{in}} = A_k^t D_{\text{in}}$; and $\eta_* \Delta_{k+1} = qD_{\text{in}} - D_{\text{in}} D_{\text{in}}^t D_{\text{out}} = q(D_{\text{in}} - D_{\text{out}})$ by $D_{\text{in}} D_{\text{in}}^t = qI$. Verified at $k = 1, 2$. $\square$

Remark 9.3.3 (The boundary level). At $k = 0$ the base is undirected and $(q+1)$ -regular, and the clean intertwining fails by exactly the degeneracy boundary: $\tau_* \Delta_1 - \Delta_0 \tau_* = -(\tau_* - \eta_*) = -\partial_0$, verified as an exact identity. The Eisenstein special layer of the whole series — the $(q+1)$ -versus- q mismatch of [Ch. 2, Rem. 1.8], the shared Perron root of [Ch. 3], the excluded trivial eigenvalue of [Ch. 5] — reappears here as the unique level where the norm formalism acquires a correction term, and the correction is once more ∂ .

Proposition 9.3.4 (Second non-commutativity certificate). *On the K_4 tower, $A_k A_k^t \neq A_k^t A_k$ for $k = 1, 2, 3$ (exact computation, hence a proof for this tower). Together with the growth-theoretic certificate $\lambda = -1 < 0$ of [Ch. 5, Prop. 2.3], this gives two independent rigorous witnesses that the level operator algebras $\mathbb{Z}\langle A_k, A_k^t \rangle$, and a fortiori any base ring over which the tower's characteristic theory could live, are non-commutative: the first says the growth cannot come from $\mathbb{Z}_p[[T]]$, the second that the operators themselves do not commute with their adjoints.*

9.4 What a characteristic theory must specialize to

We now state — as a specification, not a theorem — the data pinned down by Papers II–V that any characteristic formalism Char for (τ, η) -towers over the (pro-)Iwahori–Hecke base must reproduce:

- (A1) a multiplicative rank-density term recovering $\mu = n(q+1)/q$, the arc proliferation of [Ch. 5, Thm. 2.2];
- (A2) an additive Euler term $-\chi \cdot k$ with $\chi = \text{rk ker}(\tau^* - \eta^*)$, the H^0 of the degeneracy complex (Theorem 9.2.1) — intrinsic to the base, external to any commutative characteristic ideal;
- (A3) on sub-towers carrying an abelian deck action, reduction to the commutative characteristic ideal $(\det D(1+T))$ of [Ch. 6, Thm. 2.1], with the Newton-polygon evaluation of [Ch. 3, Rem. 5.4].

Problem 9.4.1. *Construct Char satisfying (A1)–(A3) on the pro-module $(X_\infty, A_\infty, \tau, \eta)$ of Theorem 9.3.1.*

We explicitly decline to define an Akashi-type series here. The non-commutative Iwasawa formalism in which Akashi series live is built over Iwasawa algebras of compact p -adic Lie groups [30]; our pro-ring is the inverse limit of the finite-dimensional, non-normal level algebras of Proposition 9.3.4, generated by degeneracy correspondences rather than by a group, and none of the hypotheses of that formalism are available. Writing down an “Akashi series” whose only property is to output the already-proved growth law would be circular dressing, and this series does not do that ([Ch. 5, Rem. 2.6]). What Papers II–V have done instead is to make Problem 9.4.1 well-posed: the category, the operators, the two invariants any answer must produce, and the commutative shadow it must reduce to, are now all theorems.

9.5 The verification battery

check	levels	result
(F) six incidence identities, incl. $\tau_*\eta^* = A_k$, $\tau_*\tau^* = \eta_*\eta^* = qI$	$k = 1, 2$	exact
(T) $\tau_*\Delta_{k+1} = \Delta_k\tau_*$, $\tau_*A_{k+1} = A_k\tau_*$	$k = 1, 2$	exact
(H) in-mirror for η_* ; asymmetry $\eta_*\Delta_{k+1} = q(\eta_* - \tau_*)$	$k = 1, 2$	exact
(E) $\ker(\tau^* - \eta^*) = \mathbb{Z}\mathbf{1}$; SNF($\tau_* - \eta_*$): $n_k - 1$ units	$k = 1, 2$	exact
(N) tail norm on 2-part: surjective, $ \ker = 2^{11} = 2^{n_1-1}$	$2 \rightarrow 1$	exact
(C) $A_k A_k^t \neq A_k^t A_k$	$k = 1, 2, 3$	exact
(Z) $\tau_*\Delta_1 - \Delta_0\tau_* = -\partial_0$; $\tau_*\eta^* = T_p$ at the bottom	$0 \rightarrow 1$	exact

Table 9.1: Exact integer verification (scripts `iw_stage1-3.py`) on the K_4 tower, $q = 2$, levels ≤ 3 (up to 48 vertices). Each general lemma is proved in the text; for the fixed tower each line is itself a proof.

Acknowledgement of provenance and caveats

Unrefereed preprint. All numbered statements are proved, with Proposition 9.3.4 proved for the fixed K_4 tower by exact computation and asserted for no other graph. Section 9.4 is a specification; Problem 9.4.1 is open, and the paper’s deliberate refusal to define an Akashi-type invariant is explained there. The single external citation [30] is contextual and load-bearing for nothing; its bibliographic data are now primary-source verified.

Chapter 10

Algorithmic non-commutative class field theory, VII: the snake factorization of the transfer and the evaluated characteristic form

Summary. Problem 5.1 of [Ch. 9] asked for a characteristic formalism on the degeneracy tower satisfying three axioms: an arc-density term recovering $\mu = n(q+1)/q$ (A1), an additive Euler term $-\chi k$ with χ the rank of the degeneracy H^0 (A2), and reduction to the commutative characteristic ideal of [Ch. 6] on abelian sub-towers (A3). This paper solves the problem in evaluated form, by a single application of the snake lemma, and the mechanism turns out to be simpler than anything the series anticipated: because the level- $(k+1)$ Hecke operator factors through the tail norm, $A_{k+1} = \eta^* \tau_*$ [Ch. 9, Prop. 1.2], it *vanishes identically* on $\ker \tau_*$, so the Laplacian restricted to the norm kernel is exactly qI and the “arc determinant” is exactly $q^{n_k(q-1)}$ — no nilpotency estimate, no unit ambiguity. The snake lemma applied to $0 \rightarrow \ker \nu \rightarrow \mathbb{Z}_p^{N_{k+1}} \rightarrow \mathbb{Z}_p^{N_k} \rightarrow 0$ with Laplacian verticals then yields, for every exactly intertwining, surjective, degree-preserving norm ν ,

$$\mathrm{ord}_p \mathrm{Tor}_{k+1} - \mathrm{ord}_p \mathrm{Tor}_k = \mathrm{ord}_p |\mathrm{coker}(\Delta_{k+1}|_{\ker \nu})| - \mathrm{ord}_p \mathrm{ind}_1(\nu),$$

where $\mathrm{ind}_1(\nu)$ is the factor by which ν scales the Eisenstein line. For $\nu = \tau_*$ this gives $n_k(q-1)-1$: axiom (A1) with $\mathrm{Det}_{\mathrm{arc}} = q^{n_k(q-1)}$ and axiom (A2) with $\mathrm{Det}_{\mathrm{Euler}} = \mathrm{ind}_1(\tau_*) = q$, one factor per level, $\chi = 1$. For $\nu = \pi_*$ on an abelian voltage tower the same snake gives $\mathrm{ord}_\ell \prod_{\text{new } \zeta} \det D(\zeta) - 1$: axiom (A3), the character mechanism of [Ch. 3, §5] and [Ch. 6], verified to the integer $(\det(\Delta|_{\ker \pi}) = 48 \text{ and } 784 \text{ against the character products})$. One formula, two specializations, all three axioms now theorems. What is not delivered — a lift of this evaluated form to a single class in K_1 of a localized completed degeneracy algebra — is stated as the precise residue of Problem 5.1, and, consistently with [Ch. 5] and [Ch. 9], we decline to axiomatize a localization theory that does not exist.

Status of the results

All numbered statements are proved; the proofs use only results of Papers I–VI and the snake lemma. Every identity was additionally machine-verified exactly (Table 10.1): Iwahori transitions $1 \rightarrow 2$, $2 \rightarrow 3$ over K_4 , and abelian transitions $0 \rightarrow 1$, $1 \rightarrow 2$ of the voltage tower $(1, 1, 1)$, $\ell = 2$. Remark 10.3.1 states what remains open of [Ch. 9, Prob. 5.1]; Remark 10.3.2 records two corrections to the task specification that prompted this paper. There are no external citations.

10.1 The kernel of the norm, and the vanishing lemma

Setting as in [Ch. 9]: Y_k the Iwahori path tower over a $(q+1)$ -regular base ($n_k = n(q+1)q^{k-1}$ for $k \geq 1$), τ_* the tail norm, A_k the level Hecke (adjacency) operator, $\Delta_k = qI - A_k$ for $k \geq 1$, sandpile torsion Tor_k .

Lemma 10.1.1 (Vanishing on the norm kernel). *For $k \geq 1$ let $K_k := \ker_{\mathbb{Z}_p}(\tau_*) \subseteq \mathbb{Z}_p^{n_{k+1}}$. Then:*

- (i) K_k is a saturated direct summand of rank $n_{k+1} - n_k = n_k(q-1)$ (kernels of integer matrices are saturated);
- (ii) A_{k+1} vanishes on K_k : since $A_{k+1} = \eta^* \tau_*$ [Ch. 9, Prop. 1.2], $A_{k+1}x = \eta^*(\tau_*x) = 0$ for $x \in K_k$; in particular K_k is invariant and

$$\Delta_{k+1}|_{K_k} = q \text{id}_{K_k}, \quad |\text{coker}(\Delta_{k+1}|_{K_k})| = q^{n_k(q-1)} \text{ exactly.}$$

Proof. (i) is the saturation principle of [Ch. 3] (if $ka \in \ker$, $k \neq 0$, then $a \in \ker$); the rank is surjectivity of τ_* . (ii) is the displayed one-line computation; then $\Delta_{k+1} = qI - A_{k+1}$ restricts to $q \text{id}$, whose cokernel on a rank- $n_k(q-1)$ lattice has order exactly $q^{n_k(q-1)}$. Verified: the machine search for the nilpotency index of $\Delta_{k+1}|_{K_k}$ returned $s = 1$ at both computed transitions, and $\det(\Delta_{k+1}|_{K_k}) = 2^{12}, 2^{24}$ on the nose. $\square$

Lemma 10.1.1(ii) is the structural surprise of this paper: the series expected, at best, a nilpotency estimate through the characteristic-polynomial ratio of [Ch. 5, eq. (2.1)]; instead the incidence calculus of [Ch. 9] kills the operator outright, because the Hecke operator of the upper level literally factors through the norm to the lower one.

10.2 The snake factorization

Theorem 10.2.1 (Snake factorization of the transfer). *Let $\nu: \mathbb{Z}_p^{N_{k+1}} \rightarrow \mathbb{Z}_p^{N_k}$ be surjective and degree-preserving, with $\nu\Delta_{k+1} = \Delta_k\nu$, and suppose the Eisenstein lines satisfy $\nu(\mathbf{1}) = c\mathbf{1}$ for an integer $c =: \text{ind}_1(\nu) \neq 0$. Then $K := \ker \nu$ is a saturated invariant summand, the snake lemma applied to*

$$0 \longrightarrow K \longrightarrow \mathbb{Z}_p^{N_{k+1}} \xrightarrow{\nu} \mathbb{Z}_p^{N_k} \longrightarrow 0$$

with verticals $\Delta_{k+1}|_K, \Delta_{k+1}, \Delta_k$ yields the exact sequence

$$0 \rightarrow \mathbb{Z}_p \xrightarrow{\cdot c} \mathbb{Z}_p \xrightarrow{\delta} \text{coker}(\Delta_{k+1}|_K) \rightarrow \text{coker} \Delta_{k+1} \rightarrow \text{coker} \Delta_k \rightarrow 0,$$

and hence, whenever $\text{coker}(\Delta_{k+1}|_K)$ is finite,

$$\text{ord}_p \text{Tor}_{k+1} - \text{ord}_p \text{Tor}_k = \text{ord}_p |\text{coker}(\Delta_{k+1}|_K)| - \text{ord}_p c. \quad (10.2.1)$$

Proof. Invariance of K is the intertwining; saturation as in Lemma 10.1.1(i). The kernels of the verticals are: $\ker(\Delta_{k+1}|_K) = K \cap \mathbb{Z}_p \mathbf{1} = 0$ (the constants are not in $\ker \nu$ since $\nu \mathbf{1} = c\mathbf{1} \neq 0$), $\ker \Delta_{k+1} = \mathbb{Z}_p \mathbf{1}$, $\ker \Delta_k = \mathbb{Z}_p \mathbf{1}$, and the induced map between them is $\mathbf{1} \mapsto c\mathbf{1}$. The snake lemma gives the displayed sequence; the free ranks of the two cokernels match through ν (degree preservation, as in [Ch. 6, Lem. 1.3]), so taking orders of the finite parts yields (10.2.1). $\square$

Corollary 10.2.2 (Iwahori specialization: axioms (A1) and (A2)). *For $\nu = \tau_*$, $k \geq 1$: $c = q$ ($[V]$, $\tau_* \mathbf{1} = q\mathbf{1}$) and Lemma 10.1.1 give*

$$\text{ord}_p \text{Tor}_{k+1} - \text{ord}_p \text{Tor}_k = n_k(q-1) - 1,$$

recovering [Ch. 5, Thm. 2.1] homologically. Summation yields the pseudo-Iwasawa law with $\mu = n(q+1)/q$ carried entirely by $\text{Det}_{\text{arc}} := \det(\Delta_{k+1}|_{\ker \tau_}) = q^{n_k(q-1)}$ (axiom A1), and the Euler term $-\chi k$ carried entirely by $\text{Det}_{\text{Euler}} := \text{ind}_1(\tau_*) = q$, one factor per level, with $\chi = 1 = \text{rk} \ker(\tau^* - \eta^*)$ by [Ch. 9, Thm. 2.1] (axiom A2): the -1 is, finally and precisely, the Eisenstein line passing through the connecting map of the snake.*

Corollary 10.2.3 (Abelian specialization: axiom (A3)). *For an abelian $\mathbb{Z}/\ell^k$ voltage tower with covering pushforward $\nu = \pi_*$ (exactly intertwining by [Ch. 6, Lem. 1.2]): $c = \ell$, and under the block-character decomposition of [Ch. 6, Lem. 1.1] the kernel $\ker \pi_*$ is the sum of the new character blocks, on which Δ acts with determinant $\prod_{\text{new } \zeta} \det D(\zeta)$; hence (10.2.1) reads*

$$\text{ord}_\ell \text{Tor}_{k+1} - \text{ord}_\ell \text{Tor}_k = \text{ord}_\ell \prod_{\text{new } \zeta} \det D(\zeta) - 1,$$

which is precisely the character/Newton mechanism of [Ch. 3, §5] and [Ch. 6, Cor. 2.3]. Verified exactly: $\det(\Delta|_{\ker \pi}) = 48 = \det D(-1)$ at $0 \rightarrow 1$ and $784 = |\det D(i) \det D(-i)|$ at $1 \rightarrow 2$, increments $4 - 1 = 3$ in both, matching [Ch. 3, Table 2].

One snake, two kernels: on the Iwahori side the kernel is killed by the Hecke operator and contributes the pure arc power $q^{n_k(q-1)}$; on the abelian side the kernel is the new-character space and contributes the L -value block $\prod_{\text{new}} \det D(\zeta)$. The Eisenstein index c is q or ℓ respectively — one factor per level in either world, which is the whole content of $\lambda = -\chi$. The three axioms of [Ch. 9, §5] are now theorems with these referents.

10.3 Scope, residue, corrections

Remark 10.3.1 (The residue of Problem 5.1). Theorem 10.2.1 solves [Ch. 9, Prob. 5.1] *in evaluated form*: the characteristic form is the level-wise pair $\text{Char}_k := (\det(\Delta_{k+1}|_{\ker \nu}), \text{ind}_1(\nu))$, and (A1)–(A3) hold for it as theorems. What is not constructed is a single element of K_1 of a localized completion of the degeneracy algebra — a Dieudonné-type determinant — whose evaluations reproduce Char_k . Defining such an object would require an Ore localization theory for the non-normal pro-ring of [Ch. 9, Prop. 4.1], which does not exist; consistently with [Ch. 5, Rem. 2.6] and [Ch. 9, §5], we decline to axiomatize it, and we record the lift as the precise open residue. Everything quantitative that a lift could evaluate to is already proved here.

Remark 10.3.2 (Two corrections to the task specification). The specification prompting this paper asserted an exact sequence $0 \rightarrow X_{k+1} \xrightarrow{\partial} X_k \rightarrow 0$ with $\partial = \tau_* - \eta_*$; that map is far from injective ($\text{rk } \ker \partial = n_{k+1} - n_k + 1$ by [Ch. 9, Thm. 2.1(ii)]), and the exact sequence that carries the theory is the norm sequence of Theorem 10.2.1. It also placed the Eisenstein class in $K_0(\mathcal{H}_k)$; the Eisenstein projector $\mathbf{1}^t/N$ is not integral and does not lie in the algebra, and the class that exists is the module-level line $\mathbb{Z}_p \mathbf{1} = \ker \Delta$, entering the theory through the connecting map — which is where Corollary 10.2.2 puts it.

Remark 10.3.3 (Boundary level). At $k = 0$ the tail map fails to intertwine by exactly the degeneracy boundary ([Ch. 9, Rem. 3.3]), $\ker S$ is not invariant, and Theorem 10.2.1 does not apply; the Eisenstein index there is $q + 1$, not q ($S\mathbf{1} = (q + 1)\mathbf{1}$, verified). The transfer $0 \rightarrow 1$ enters the law as the initial value, as everywhere in the series.

10.4 The verification battery

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from Papers I–VI and the snake lemma; the battery of Table 10.1 confirms each ingredient exactly. No external sources are cited and none are load-bearing. Remark 10.3.1 states the open residue of [Ch. 9, Prob. 5.1]; Remark 10.3.2 corrects the specification that prompted the paper.

transition	$\text{rk ker } \nu$	invariant	$A_{k+1} _{\text{ker}}$	$\det(\Delta _{\text{ker } \nu})$	$\text{ind}_1(\nu)$
Iwahori $1 \rightarrow 2$	12	yes	$= 0$ ($s=1$)	2^{12} exactly	2
Iwahori $2 \rightarrow 3$	24	yes	$= 0$ ($s=1$)	2^{24} exactly	2
abelian $0 \rightarrow 1$	4	yes	—	$48 = \det D(-1)$	2
abelian $1 \rightarrow 2$	8	yes	—	$784 = \det D(i) \det D(-i) $	2
boundary $0 \rightarrow 1$	—	no	—	—	$3 = q + 1$

Table 10.1: Exact verification (`iwahori_char.py`). Increments (10.2.1): $12 - 1 = 11$, $24 - 1 = 23$ (Iwahori, matching [Ch. 3, Rem. 5.2]); $4 - 1 = 3$, $4 - 1 = 3$ (abelian 2-adic orders, matching [Ch. 3, Table 2]). The vanishing $A_{k+1}|_{\text{ker } \tau_*} = 0$ was found by the machine ($s = 1$) before its one-line proof was noticed.

Chapter 11

Algorithmic non-commutative class field theory, XIII: universal localization of the degeneracy algebra, the K_1 -characteristic class, and the Eisenstein boundary

Summary. Problem 5.1 of [Ch. 9] and its residue [Ch. 10, Rem. 3.1] asked for a single class in K_1 of a localized degeneracy algebra whose evaluations reproduce the snake factorization of [Ch. 10]: the arc term $q^{n_k(q-1)}$ and the Eisenstein index q at every level of the Iwahori tower. Ore localization is unavailable because the degeneracy algebra is non-commutative and non-normal ([Ch. 9, Prop. 4.1]). This paper constructs the class with Cohn’s universal localization, and finds that the construction reorganizes the whole picture. The degeneracy order $\mathcal{H}$ is the $\mathbb{Z}_p$ -algebra generated by the four degeneracy maps $\tau_*, \tau^*, \eta_*, \eta^*$ acting on the augmentation-zero lattices $\mathcal{M}_k^0 = \ker(\deg) \subset \mathbb{Z}_p^{n_k}$; the tail Laplacian factors inside it as $\Delta_k = \tau_*(\tau^* - \eta^*)$, tail norm times degeneracy coboundary. On $\mathcal{M}_k^0$ the Laplacian is injective with finite cokernel, so $\Delta = \bigoplus_k \Delta_k$ is Σ -inverting for $\Sigma = \{\Delta\}$; the universal localization $\mathcal{H} \rightarrow \mathcal{H}_\Sigma$ exists, the low-degree localization sequence $K_1(\mathcal{H}) \rightarrow K_1(\mathcal{H}_\Sigma) \xrightarrow{\partial} K_0(\mathcal{H}, \Sigma) \rightarrow K_0(\mathcal{H})$ holds, and $[\Theta_{\mathcal{H}}] := [\Delta] \in K_1(\mathcal{H}_\Sigma)$ is the characteristic class. Its content is in two theorems. The level evaluations $\text{ev}_k: K_1(\mathcal{H}_\Sigma) \rightarrow \mathbb{Q}_p^\times$ give $\text{ev}_k[\Theta_{\mathcal{H}}] = \pm \text{pdet}(\Delta_k) = \pm n_k \kappa(Y_k)$, and on K_4 these have 2-adic order exactly $n_k - 3$ at $k = 1, 2, 3$: the K_1 class carries the pure arc growth μp^k and nothing else. The boundary class $\partial[\Theta_{\mathcal{H}}]$ at level k is the cokernel of Δ_k on $\mathcal{M}_k^0$, which sits in a non-split extension $0 \rightarrow \mathbb{Z}/n_k \rightarrow \text{coker } \Delta_k^0 \rightarrow \text{Jac}(Y_k) \rightarrow 0$: the Euler term $-k = -\text{ord}_p n_k + \text{const}$ of [Ch. 5] is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary. Axioms (A1) and (A2) of [Ch. 9] are thus separated inside a single K -theoretic object, and (A3) holds on abelian towers, where $\text{pdet}(\Delta_k) = n \kappa(Y_0) \prod_{\chi \neq 1} \det D(\chi)$ is the characteristic element of [Ch. 6] evaluated at all nontrivial characters (exact on the $(1, 1, 1)$ tower). What the specification called the “non-commutative main conjecture” $\partial[\Theta] = [X]$ is the definition of ∂ ; the genuine main conjecture — an analytic p -adic L -function of the tower, defined independently, equal to $[\Theta_{\mathcal{H}}]$ — is stated and left open.

Status of the results

Every numbered statement is proved. The external inputs are the existence and universal property of Cohn's universal localization [25] and the low-degree localization exact sequence for universal localizations [33, Ch. 4]; the higher sequence of Neeman–Ranicki [32] requires stable flatness, which we do not verify and do not use. Everything else is finite linear algebra over $\mathbb{Z}_p$, machine-verified exactly on the K_4 Iwahori tower at $k \leq 3$ and on the $(1, 1, 1)$ abelian tower (Table 11.1). Section 11.5 says what the main conjecture is and is not.

11.1 The degeneracy order and the augmentation-zero lattices

Setting as in [Ch. 9], [Ch. 10]: Y_k the Iwahori path tower over a $(q+1)$ -regular base, $q = p$, $n_k = n(q+1)q^{k-1}$, A_k the level Hecke operator, $\Delta_k = qI - A_k$, $\text{Tor}_k = \text{Jac}(Y_k) = \text{Tor coker } \Delta_k$; $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ the tail and head pushforwards, τ^*, η^* their transposes, with [Ch. 9, Prop. 1.2]

$$\tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI_{n_k}, \quad (11.1.1)$$

and $\deg: \mathbb{Z}^{n_k} \rightarrow \mathbb{Z}$ the coordinate sum.

Lemma 11.1.1 (Degree compatibility and the Laplacian factorization). *$\deg \circ \tau_* = \deg \circ \eta_* = \deg$ and $\deg \circ \tau^* = \deg \circ \eta^* = q \deg$; hence all four degeneracy maps preserve the augmentation-zero lattices $\mathcal{M}_k^0 := \ker(\deg) \subset \mathbb{Z}_p^{n_k}$. Moreover*

$$\Delta_k = \tau_* (\tau^* - \eta^*) = \tau_* \partial_k^*,$$

the tail norm composed with the degeneracy coboundary ∂_k^ of [Ch. 9, Thm. 2.1]; in particular Δ_k lies in the algebra generated by the degeneracy maps and maps $\mathbb{Z}_p^{n_k}$ into $\mathcal{M}_k^0$.*

Proof. Pushforwards sum over fibres and pullbacks have q elements per fibre. By (11.1.1), $\tau_* \tau^* - \tau_* \eta^* = qI - A_k$. Column sums of Δ_k vanish (in-degree q). Verified exactly at $k = 1, 2, 3$ (Table 11.1, lines (I), (B)). $\square$

Definition 11.1.2 (Degeneracy order). For $K \geq 1$ let $\mathcal{M}_{\leq K}^0 := \bigoplus_{k \leq K+1} \mathcal{M}_k^0$, and let $\mathcal{H}_K \subset \text{End}_{\mathbb{Z}_p}(\mathcal{M}_{\leq K}^0)$ be the $\mathbb{Z}_p$ -subalgebra generated by the degeneracy maps between consecutive levels $\leq K+1$, restricted to the augmentation-zero lattices by Lemma 11.1.1. It is an order in a semisimple $\mathbb{Q}_p$ -algebra (a $\mathbb{Z}_p$ -subalgebra of a matrix algebra, finitely generated as a module), non-commutative and non-normal since $A_k A_k^t \neq A_k^t A_k$ inside it [Ch. 9, Prop. 4.1]. Put

$$\Delta_{(K)} := \bigoplus_{k \leq K} \Delta_k^0 \oplus q \cdot \text{id}_{\mathcal{M}_{K+1}^0} \in \mathcal{H}_K, \quad \Delta_k^0 := \Delta_k|_{\mathcal{M}_k^0},$$

where the last summand $q \text{id} = \tau_* \tau^*$ keeps the element inside $\mathcal{H}_K$. We write $\mathcal{H}$ for $\mathcal{H}_K$ with K fixed and large, and $\Sigma := \{\Delta_{(K)}\}$.

Proposition 11.1.3 (Σ -inverting). Δ_k^0 is injective with finite cokernel and invertible over $\mathbb{Q}_p$, with

$$\det \Delta_k^0 = \pm \text{pdet}(\Delta_k) = \pm n_k \kappa(Y_k),$$

the product of the nonzero eigenvalues of Δ_k . Hence $\Delta_{(K)} \otimes \mathbb{Q}_p$ is invertible while $\Delta_{(K)}$ is not invertible in $\mathcal{H}$, its determinant not being a $\mathbb{Z}_p$ -unit: localization is needed and is non-trivial.

Proof. $\ker \Delta_k = \mathbb{Z}_p \mathbf{1}$ and $\deg \mathbf{1} = n_k \neq 0$, so Δ_k is injective on $\ker(\deg)$; Δ_k maps $\mathbb{Z}_p^{n_k}$ into $\ker(\deg)$, so Δ_k^0 is an endomorphism of the rank- $(n_k - 1)$ lattice $\mathcal{M}_k^0$ with finite cokernel. Its determinant is the product of the eigenvalues of Δ_k on the complement of the kernel, i.e. the pseudo-determinant, which equals $n_k \kappa(Y_k)$ by [Ch. 5, eq. (2.2)]. On K_4 : $4608 = 12 \cdot 384$, $18874368 = 24 \cdot 786432$, $316659348799488 = 48 \cdot 6597069766656$, with 2-adic orders 9, 21, 45. $\square$

11.2 Universal localization and the characteristic class

Definition 11.2.1 (Universal localization and the relative group). By Cohn [25] there is a ring homomorphism $\lambda: \mathcal{H} \rightarrow \mathcal{H}_\Sigma$, universal among ring homomorphisms under which $\Delta_{(K)}$ becomes invertible. Let $K_0(\mathcal{H}, \Sigma)$ be the relative group of triples (P, Q, f) , P, Q finitely generated projective $\mathcal{H}$ -modules and $f: P \rightarrow Q$ a map with $\lambda(f)$ invertible, modulo the usual relations [33, Ch. 4]. Then

$$K_1(\mathcal{H}) \longrightarrow K_1(\mathcal{H}_\Sigma) \xrightarrow{\partial} K_0(\mathcal{H}, \Sigma) \longrightarrow K_0(\mathcal{H}) \quad (11.2.1)$$

is exact [33, Ch. 4], and ∂ sends the class of $\lambda(f)$, for $f: P \rightarrow Q$ with $\lambda(f)$ invertible, to $[(P, Q, f)]$. Forgetting to modules, $[(P, Q, f)] \mapsto [\text{coker } f] - [\ker f]$ when these are finite.

Definition 11.2.2 (Characteristic class). $[\Theta_{\mathcal{H}}] := [\lambda(\Delta_{(K)})] \in K_1(\mathcal{H}_\Sigma)$. It is well defined by Proposition 11.1.3, and $\mathcal{H}_\Sigma \neq 0$ because λ extends to the representation $\mathcal{H} \rightarrow \text{End}_{\mathbb{Q}_p}(\mathcal{M}_{\leq K}^0 \otimes \mathbb{Q}_p)$ in which $\Delta_{(K)}$ is invertible.

Theorem 11.2.3 (The tautological identity, and what it says). $\partial[\Theta_{\mathcal{H}}] = [(\mathcal{M}_{\leq K}^0, \mathcal{M}_{\leq K}^0, \Delta_{(K)})] \in K_0(\mathcal{H}, \Sigma)$, whose image under the forgetful map is

$$\sum_{k \leq K} [\text{coker } \Delta_k^0] + [\mathcal{M}_{K+1}^0 / q \mathcal{M}_{K+1}^0].$$

This is the definition of ∂ ; it is the statement the specification called the “non-commutative main conjecture”, and it has no content beyond Definition 11.2.1. The content is in Theorems 11.3.1 and 11.3.2.

11.3 Evaluation and the Eisenstein boundary

For each $k \leq K$ the projection $\mathcal{H} \rightarrow \text{End}_{\mathbb{Z}_p}(\mathcal{M}_k^0)$ followed by $\otimes \mathbb{Q}_p$ is a ring homomorphism $\mathcal{H} \rightarrow M_{n_k-1}(\mathbb{Q}_p)$ under which $\Delta_{(K)}$ becomes invertible; by universality it factors through $\mathcal{H}_\Sigma$, and composing with the determinant gives the *level evaluation*

$$\text{ev}_k: K_1(\mathcal{H}_\Sigma) \longrightarrow K_1(M_{n_k-1}(\mathbb{Q}_p)) = \mathbb{Q}_p^\times.$$

Theorem 11.3.1 (Level evaluations: pure arc growth). $\text{ev}_k[\Theta_{\mathcal{H}}] = \det \Delta_k^0 = \pm n_k \kappa(Y_k)$, and for all $k \geq 1$

$$\frac{\text{ev}_{k+1}[\Theta_{\mathcal{H}}]}{\text{ev}_k[\Theta_{\mathcal{H}}]} = \pm q^{n_k(q-1)}, \quad \text{ord}_p \text{ev}_k[\Theta_{\mathcal{H}}] = n_k + c = \frac{n(q+1)}{q} p^k + c,$$

with a constant c depending only on the base. The arc term $\text{Det}_{\text{arc}} = q^{n_k(q-1)}$ of [Ch. 10, Cor. 2.2] is the ratio of consecutive evaluations of the single class $[\Theta_{\mathcal{H}}]$; the Euler term does not appear. On K_4 : $c = -3$ and $\text{ord}_2 \text{ev}_k = 9, 21, 45$ at $k = 1, 2, 3$.

Proof. The first equality is Proposition 11.1.3. The ratio: $\text{pdet}(\Delta_{k+1}) / \text{pdet}(\Delta_k) = q^{n_{k+1} - n_k} = q^{n_k(q-1)}$ because the nonzero spectrum of A_{k+1} is that of A_k plus $n_{k+1} - n_k$ zeros [Ch. 5, eq. (2.1)], each contributing the eigenvalue q of Δ_{k+1} . Equivalently this is the residue constancy $n_k \kappa_k q^{-n_k} = \text{const}$ of [Ch. 7, Thm. 4.1(iii)]. Summing, $\text{ord}_p \text{ev}_k = n_k + c$ with $n_k = n(q+1)q^{k-1}$. $\square$

Theorem 11.3.2 (The Eisenstein boundary). For every $k \geq 1$ there is an exact sequence of finite $\mathbb{Z}_p$ -modules

$$0 \longrightarrow \mathbb{Z}/n_k \longrightarrow \text{coker } \Delta_k^0 \longrightarrow \text{Jac}(Y_k) \longrightarrow 0,$$

so that in the Grothendieck group of finite $\mathbb{Z}_p$ -modules the level- k component of $\partial[\Theta_{\mathcal{H}}]$ is

$$[\text{coker } \Delta_k^0] = [\text{Jac}(Y_k)] + [\mathbb{Z}/n_k], \quad \text{ord}_p |\text{Jac}(Y_k)| = \text{ord}_p \text{ev}_k[\Theta_{\mathcal{H}}] - \text{ord}_p n_k = \frac{n(q+1)}{q} p^k - k + \nu_0.$$

The Euler term $-k$ of [Ch. 5, Thm. 2.2] is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary, of p -adic order $\text{ord}_p n_k = k + \text{ord}_p(n(q+1)) - 1$. On K_4 the extension is non-split at $k = 1, 2, 3$: $\text{coker } \Delta_k^0 = \mathbb{Z}/8 \oplus (\mathbb{Z}/24)^2$, $(\mathbb{Z}/2)^9 \oplus \mathbb{Z}/16 \oplus (\mathbb{Z}/48)^2$, $(\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^9 \oplus \mathbb{Z}/32 \oplus (\mathbb{Z}/96)^2$, against $\text{Jac}(Y_k) \oplus \mathbb{Z}/n_k$, whose invariant factors differ (at $k = 1$: $\mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus (\mathbb{Z}/24)^2$).

Proof. $\Delta_k(\mathbb{Z}_p^{n_k}) \subseteq \ker(\deg)$, so $\text{Jac}(Y_k) = \text{Tor coker } \Delta_k = \ker(\deg)/\Delta_k(\mathbb{Z}_p^{n_k})$ by [Ch. 10, Lem. 1.1] and [Ch. 6, Lem. 1.3]. The lattice $\Delta_k(\mathbb{Z}_p^{n_k})$ contains $\Delta_k(\ker \deg)$ with quotient $\Delta_k(\mathbb{Z}_p^{n_k})/\Delta_k(\ker \deg) \cong \mathbb{Z}_p^{n_k}/(\ker \deg + \mathbb{Z}_p \mathbf{1}) \cong \mathbb{Z}/\deg(\mathbf{1}) = \mathbb{Z}/n_k$, since Δ_k is injective modulo $\mathbb{Z}_p \mathbf{1}$. The exact sequence follows, and orders multiply: $|\text{coker } \Delta_k^0| = n_k |\text{Jac}(Y_k)|$, consistent with Proposition 11.1.3. The p -adic law is Theorem 11.3.1 minus $\text{ord}_p n_k$; on K_4 , $n_k - 3 - (k + 1) = 6 \cdot 2^k - k - 4$, the law of [Ch. 5, Thm. 2.2]. Non-splitting: compare invariant factors (Table 11.1, line (B)). $\square$

Remark 11.3.3 (Axioms (A1), (A2), (A3) in K -theory). [Ch. 9, §5] asked for a characteristic formalism producing an arc-density term μ (A1), an Euler term $-\chi k$ intrinsic to the base (A2), and the commutative characteristic ideal on abelian sub-towers (A3); [Ch. 10] delivered all three in evaluated form and left the lift to a single K_1 class as the residue. Theorems 11.3.1–11.3.2 are that lift: the class $[\Theta_{\mathcal{H}}]$ evaluates to the pure arc law (A1); its boundary contains the Eisenstein subclass $[\mathbb{Z}/n_k]$ whose order is the vertex count, and $-\text{ord}_p n_k$ is the Euler term (A2) — the fourth reading of $-\chi k$ after [Ch. 5], [Ch. 10] and [Ch. 7], and the first in which it is a class rather than a number; (A3) is Theorem 11.3.4. The three axioms are properties of one object, not three formulas.

Theorem 11.3.4 (Abelian reduction). *For an abelian $\mathbb{Z}/\ell^k$ voltage tower of [Ch. 6] with pencil $D(t)$ and $\Delta_k = D(S_{\ell^k})$ [Ch. 6, Lem. 1.1], the same construction (the group ring $\mathbb{Z}_{\ell}[G_k]$ playing the role of $\mathcal{H}$) gives*

$$\text{ev}_k[\Theta] = \text{pdet}(\Delta_k) = n_k \kappa(Y_k) = n \kappa(Y_0) \prod_{\chi \neq 1} \det D(\chi) = n \kappa(Y_0) \prod_{\zeta^{\ell^k}=1, \zeta \neq 1} \zeta^{-d} g(\zeta - 1),$$

the characteristic element $g = \det D(1 + T)$ of [Ch. 6, Thm. 2.1] evaluated at all nontrivial characters, times the Eisenstein constant $n \kappa(Y_0)$; the Eisenstein subclass is $\mathbb{Z}/n_k = \mathbb{Z}/n \ell^k$ and $\text{ord}_{\ell} n_k = k + \text{ord}_{\ell} n$ is the “ $-k$ ” of [Ch. 3, Rem. 5.4]. Exact on the $(1, 1, 1)$ tower, $\ell = 2$, $k \leq 3$: 3072, 2408448, 362577395712 (Table 11.1, line (A3)).

Proof. The block-character decomposition [Ch. 6, Lem. 1.1] splits the nonzero spectrum of Δ_k into the spectra of $D(\chi)$, $\chi \neq 1$, and the nonzero spectrum of $D(1) = \Delta_0$, whose product is $n \kappa(Y_0)$ [Ch. 5, eq. (2.2)]; $\det D(\zeta) = \zeta^{-d} g(\zeta - 1)$ as in [Ch. 7, Lem. 2.1]. $\square$

11.4 The verification battery

check	levels	result
(I) (11.1.1); $A_k A_k^t \neq A_k^t A_k$; degree compatibility of $\tau_*, \eta_*, \tau^*, \eta^*$	$k = 1, 2, 3$	exact
(S) Δ_k^0 injective, invertible over $\mathbb{Q}$, det not a $\mathbb{Z}_2$ -unit (2-adic orders 9, 21, 45)	$k = 1, 2, 3$	exact
(E) $\det \Delta_k^0 = n_k \kappa_k = \text{pdet}$; ratios $2^{12}, 2^{24}$; $\text{ord}_2 = n_k - 3$	$k = 1, 2, 3$	exact (pdet numeric)
(B) $ \text{coker } \Delta_k^0 / \text{Jac}(Y_k) = n_k$; structures; extension non-split; $\Delta_k = \tau_*(\tau^* - \eta^*)$	$k = 1, 2, 3$	exact
(A3) $(1, 1, 1)$, $\ell = 2$: $\text{pdet}(\Delta_k) = n \kappa_0 \prod_{\chi \neq 1} \det D(\chi)$	$k = 1, 2, 3$	exact

Table 11.1: Verification (`cohn_localization.py`) on the K_4 Iwahori tower and the $(1, 1, 1)$ abelian tower.

11.5 Scope, residue, corrections

Remark 11.5.1 (What a main conjecture would be). In the theory of [30] a main conjecture asserts that an *analytic* object — a p -adic L -function characterized by interpolation of special values, constructed independently of the module — equals the algebraic characteristic element, an element of K_1 of the localized Iwasawa algebra whose boundary is the class of the Selmer module. The algebraic half is what this paper constructs: $[\Theta_{\mathcal{H}}] \in K_1(\mathcal{H}_{\Sigma})$ with $\partial[\Theta_{\mathcal{H}}]$ the class of the tower’s cokernels, the analogue of “ $\partial \xi = [X]$ ”. The

analytic half does not exist yet: there is no independently defined element of $K_1(\mathcal{H}_\Sigma)$ interpolating the L -values of the tower — the candidates are the pseudo-determinants $n_k \kappa_k = \prod_j (q+1 - \lambda_j)$, products of local L -values at the Eisenstein point as in [Ch. 2, Thm. 3.5], but a construction of a single element from them, rather than from Δ , is exactly what is missing. The non-commutative main conjecture for Hecke towers is therefore: *construct $\mathcal{L}_\mathcal{H} \in K_1(\mathcal{H}_\Sigma)$ from the L -values alone and prove $\mathcal{L}_\mathcal{H} = [\Theta_\mathcal{H}]$ modulo the image of $K_1(\mathcal{H})$* . Its evaluated form is [Ch. 10, Thm. 2.1]; its K_1 form is open.

Remark 11.5.2 (Solved, open, impossible). *Solved*: the residue of [Ch. 9, Prob. 5.1] and [Ch. 10, Rem. 3.1] in the form asked — a single class in K_1 of a localized degeneracy algebra whose level evaluations and boundary reproduce the snake factorization, with (A1) and (A2) separated as evaluation and Eisenstein subclass and (A3) as the abelian reduction. *Open*: (a) the analytic side (Remark 11.5.1); (b) stable flatness of $\mathcal{H} \rightarrow \mathcal{H}_\Sigma$ and hence the higher localization sequence [32]; (c) the structure of $\mathcal{H}_\Sigma$ itself — whether it is an order in a semisimple algebra or has torsion, and the image of $K_1(\mathcal{H})$ in $K_1(\mathcal{H}_\Sigma)$, which measures how far $[\Theta_\mathcal{H}]$ is from unique; (d) the rank-two version over the degeneracy algebra of [Ch. 18], where no abelian sector exists. *Impossible or empty as stated*: “ $\partial[\Theta] = [X]$ ” as a theorem with content (Theorem 11.2.3).

Remark 11.5.3 (Corrections to the task specification). (1) The identity $\partial([\Theta_\mathcal{H}]) = [X_\infty]$ is the definition of the boundary map in the relative sequence, not a theorem; the paper proves what does have content (Theorems 11.3.1–11.3.4). (2) The class must be formed on the augmentation-zero lattices: on $\mathbb{Z}_p^{n_k}$ the Laplacian has the Eisenstein kernel and is not Σ -invertible; the specification’s “ $\Delta_\infty \in M_n(\widehat{\mathcal{H}}_\infty)$ ” has no $n \times n$ presentation because n_k grows (contrast [Ch. 6], where $\mathbb{Z}_\ell[G_k]^n$ is free of constant rank). (3) The Euler term is not produced by “the connecting map” inside K_1 ; it is the Eisenstein subclass $[\mathbb{Z}/n_k]$ of the boundary, and it is invisible to the K_1 -evaluations, which see the arc term only. (4) “ Σ = the set of morphisms inducing surjections with finite cokernel on X_∞ ” is replaced by the single element $\Delta_{(K)}$; universal localization at one element already gives the class. (5) The higher K -theory sequence needs stable flatness [32]; only the low-degree sequence [33] is used, and only its first map matters for the construction.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from Papers III–XII and finite linear algebra, given the existence of universal localization [25] and the low-degree exact sequence [33, Ch. 4], both quoted at chapter level (the precise theorem numbers were not consulted). Bibliographic data of [25, 33, 32] were checked against publisher or indexing pages; [30] is the entry verified in [Ch. 9].

Part III

Operator algebras of the tower

Chapter 12

Algorithmic non-commutative class field theory, VIII: the tower-level Hecke–Kirchberg limit and the non-commutative Iwasawa K -module

Summary. The single open problem left by [Ch. 4] (Remark 4.3 there) asked whether the tail norms of [Ch. 9, Thm. 3.1] lift from the sandpile groups to the shadow algebras $\mathcal{O}_{B(Y_k)}$ of the Iwahori tower, and whether the lifts assemble into one Kirchberg algebra whose K -theory carries the pro-module X_∞ of [Ch. 9]. We answer both questions, and the answers differ from the expected shape in three places, each forced by a theorem. First, at the q -regular levels $k \geq 1$ the shadow must carry $q - 1$ return arcs, not q : the level-0 normalization of [Ch. 4] is the $(q+1)$ -regular special case, and with q return arcs the shadow K_0 is a finite group unrelated to $\text{Jac}(Y_k)$ and K_1 vanishes. Second, the tail map $\tau: Y_{k+1} \rightarrow Y_k$ extends to a map of shadow graphs that is bijective on *in*-arcs at every vertex; such in-coverings induce unital injective $*$ -homomorphisms of graph algebras, but in the direction $\mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, and on K_0 they induce the *pullback* τ^* . The pushforward τ_* acts by the block matrix $\text{diag}(\tau_*, \tau_*)$ on the Bowen–Franks group $\text{coker}(I - B_k)$, where it is exactly the tail norm of [Ch. 9]; the two actions are adjoint under the canonical perfect pairing between the torsion of $\text{coker } M$ and of $\text{coker } M^t$. Third, X_∞ is an uncountable profinite group and K -theory commutes with inductive, not projective, limits, so no separable C^* -algebra has $K_0 \supseteq X_\infty$; what the inductive limit $\mathcal{O}_\infty = \varinjlim \mathcal{O}_k$ realizes is the Pontryagin dual. Precisely, $\mathcal{O}_\infty$ is a unital Kirchberg algebra with

$$K_1(\mathcal{O}_\infty) \cong \mathbb{Z}, \quad 0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0(\mathcal{O}_\infty) \rightarrow \mathbb{Z}[1/q] \rightarrow 0, \quad \mathfrak{X}_\infty := \varprojlim_k (\text{Jac}(Y_k), \tau_*),$$

so that $\mathfrak{X}_\infty \cong \text{Hom}(\text{Tor } K_0(\mathcal{O}_\infty), \mathbb{Q}/\mathbb{Z})$ recovers the pro-module together with its level filtration and the tail norms; by the UCT it also appears in odd K -homology. Along the way the machine found, and we then proved, a structural fact the series had not anticipated: for $q = p$ prime the kernel of the tail norm on $\text{Jac}(Y_{k+1})$ is exactly the p -torsion subgroup, an elementary abelian group of rank $n_k(p - 1) - 1$, so the p -part of the Iwasawa pro-module is a torsion-free compact $\mathbb{Z}_p$ -module of infinite rank, its dual is divisible, and the K_0 sequence above splits. The plain tower algebras $C^*(Y_k)$, by contrast, are all isomorphic to the boundary algebra of [Ch. 2] and see nothing of the tower.

Status of the results

Every numbered statement is proved. The C^* -algebraic inputs are: the Cuntz–Krieger K -theory formula and the gauge-invariant uniqueness theorem for graph algebras [36]; the closure of nuclearity, simplicity, pure infiniteness and the UCT class under inductive limits [34, 37, 38]; the Pontryagin duality between profinite and discrete torsion abelian groups; and Kirchberg–Phillips classification [35], which is quoted once, in Remark 12.2.6, for an existence statement we do not use. The in-covering lemma (Lemma 12.2.1) and the dual-graph isomorphism (Proposition 12.4.1) are proved in full. Every finite identity was machine-verified exactly on the K_4 tower at levels $k \leq 3$ (shadow matrices of size 24, 48, 96; Table 12.2). Section 12.5 states what is solved, what remains open, and what is impossible, and records four corrections to the specification that prompted this paper.

12.1 The level shadow and its normalization

Setting as in [Ch. 9] and [Ch. 10]. Y_0 is a finite connected $(q+1)$ -regular graph on n vertices, $q \geq 2$; for $k \geq 1$, Y_k is the Iwahori path graph, a strongly connected q -regular digraph on $n_k = n(q+1)q^{k-1}$ vertices with $Y_{k+1} = L(Y_k)$ the line digraph, so that $V_{Y_{k+1}} = \text{Arcs}(Y_k)$; A_k is the level Hecke (adjacency) operator, $\Delta_k = qI - A_k$ the out-Laplacian, $\text{coker}_{\mathbb{Z}} \Delta_k = \mathbb{Z} \oplus \text{Jac}(Y_k)$ with $\ker \Delta_k = \mathbb{Z}\mathbf{1}$. The degeneracy maps $\tau, \eta: Y_{k+1} \rightarrow Y_k$ are the tail (source) and head (range) maps of arcs; $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ are their pushforwards, $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ the pullbacks, and

$$\tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI_{n_k}, \quad \tau_* \Delta_{k+1} = \Delta_k \tau_*, \quad \eta_* \Delta_{k+1}^t = \Delta_k^t \eta_* \quad (12.1.1)$$

by [Ch. 9, Prop. 1.2, Thm. 3.1, Prop. 3.2]. Transposing the last two identities,

$$\Delta_{k+1}^t \tau^* = \tau^* \Delta_k^t, \quad \Delta_{k+1} \eta^* = \eta^* \Delta_k, \quad (12.1.2)$$

so τ^* descends to $\text{coker} \Delta_k^t \rightarrow \text{coker} \Delta_{k+1}^t$ and η^* to $\text{coker} \Delta_k \rightarrow \text{coker} \Delta_{k+1}$, both *up* the tower. We write $\text{Tor}_k := \text{Jac}(Y_k) = \text{Tor coker} \Delta_k$ and $\text{Tor}_k^t := \text{Tor coker} \Delta_k^t$.

Convention 12.1.1 (Graph algebras). For a finite directed graph E without sinks, $C^*(E)$ is the universal C^* -algebra generated by mutually orthogonal projections p_v ($v \in E^0$) and partial isometries s_e ($e \in E^1$) subject to

$$(\text{CK1}) \quad s_e^* s_e = p_{r(e)}, \quad (\text{CK2}) \quad p_v = \sum_{s(e)=v} s_e s_e^*,$$

with vertex matrix $A_E(v, w) = \#\{e : s(e) = v, r(e) = w\}$. Then $K_0(C^*(E)) = \text{coker}(1 - A_E^t)$, generated by the classes $[p_v] \mapsto e_v$, and $K_1(C^*(E)) = \ker(1 - A_E^t)$ [36, Thm. 7.16]; this is the convention of [Ch. 4], where $K_0(\mathcal{O}_{B(Y)}) = \text{coker}(I - B^t)$. The other standard convention reverses all arcs. Because T_p is symmetric the choice was invisible in [Ch. 4]; at the directed levels it is not, and we make it once and for all here. Under the reversed convention every statement of this paper holds with $(\tau, \Delta_k, X_\infty)$ replaced by $(\eta, \Delta_k^t, X_\infty^{\text{in}})$ of [Ch. 9, Prop. 3.2] (Remark 12.2.5).

Definition 12.1.2 (Level shadow). For a strongly connected d -regular digraph Y with adjacency A (every vertex of out-degree $d \geq 2$), the *shadow graph* $Y^\sharp$ is the digraph on $V_Y \sqcup V_Y'$ with the arcs of Y , one arc $v \rightarrow v'$, $c := d - 1$ labelled return arcs $v' \rightarrow v$, and two labelled loops at v' ; its vertex matrix is

$$B(Y) = \begin{pmatrix} A & I \\ cI & 2I \end{pmatrix}, \quad c = d - 1,$$

and the *shadow algebra* is $\mathcal{O}_{B(Y)} := C^*(Y^\sharp)$. For $Y = Y_0$ one has $d = q+1$, $c = q$: this is [Ch. 4, Def. 1.1]. For the levels $k \geq 1$ one has $d = q$, $c = q - 1$, and we write $B_k := B(Y_k)$, $\mathcal{O}_k := \mathcal{O}_{B_k}$, $Y_k^\sharp$ for the level- k shadow graph.

Theorem 12.1.3 (Level K -theory). *Let $k \geq 1$, $c = q - 1$, $U = \begin{pmatrix} I & -cI \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$. Then*

$$U(I - B_k^t)V = \Delta_k^t \oplus (-I_{n_k}), \quad V^t(I - B_k)U^t = \Delta_k \oplus (-I_{n_k}), \quad (12.1.3)$$

and consequently

$$K_0(\mathcal{O}_k) = \text{coker}(I - B_k^t) \cong \text{coker } \Delta_k^t \cong \mathbb{Z} \oplus \text{Jac}(Y_k), \quad K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z} \cdot (\mathbf{1}, -\mathbf{1}),$$

the isomorphism $K_0(\mathcal{O}_k) \rightarrow \text{coker } \Delta_k^t$ being $[x] \mapsto [\text{first block of } Ux]$. The Bowen–Franks group $\text{coker}(I - B_k) \cong \text{coker } \Delta_k = \mathbb{Z} \oplus \text{Jac}(Y_k)$ via V^t , and $\ker(I - B_k) = \mathbb{Z} \cdot (\mathbf{1}, -c\mathbf{1})$. The algebra $\mathcal{O}_k$ is a unital Kirchberg algebra (separable, nuclear, simple, purely infinite, UCT), hence traceless. Its unit class is $[1_{\mathcal{O}_k}] = (2 - q) \sum_v [p_v]$; in particular $[1_{\mathcal{O}_k}] = 0$ when $q = 2$.

Proof. $I - B_k^t = \begin{pmatrix} I - A_k^t & -cI \\ -I & I \end{pmatrix}$, and the two block products of [Ch. 4, Thm. 2.1] give $((1 + c)I - A_k^t) \oplus (-I)$; with $c = q - 1$ the first block is Δ_k^t . The second identity is the transpose of the first. Cokernels and kernels are invariant under unimodular equivalence, and $\text{coker } \Delta_k^t$ has the same invariant factors as $\text{coker } \Delta_k$, whence the abstract isomorphism with $\mathbb{Z} \oplus \text{Jac}(Y_k)$ (the canonical relation between the two torsion groups is Lemma 12.1.5). For the kernels: $(x, y) \in \ker(I - B_k^t)$ forces $y = -x$ and $A_k^t x = qx$, and the Perron eigenspace of the irreducible A_k^t (row sums = in-degrees = q) is $\mathbb{Z}\mathbf{1}$, saturated because $\mathbf{1}$ is primitive; $\ker(I - B_k)$ similarly. Kirchberg: $Y_k^\#$ is strongly connected (Y_k is, and layers communicate through $v \rightarrow v' \rightarrow v$; here $c \geq 1$ is used), B_k is not a permutation matrix, and every shadow vertex carries two loops, so the criteria quoted in [Ch. 4, Thm. 3.1] apply verbatim. Unit class: (CK2) at v' gives $[p_{v'}] = c[p_v] + 2[p_{v'}]$, i.e. $[p_{v'}] = -c[p_v]$, and summing over v gives $[1] = (1 - c) \sum_v [p_v] = (2 - q) \sum_v [p_v]$. $\square$

Remark 12.1.4 (A correction to the specification: $q - 1$, not q). The specification prompting this paper prescribed $B(Y_k) = \begin{pmatrix} A_k & I \\ qI & 2I \end{pmatrix}$ at every level. The block computation of Theorem 12.1.3 shows that with c return arcs the Laplacian block is $(1 + c)I - A_k^t$, so the shadow K_0 is the sandpile group only when $1 + c$ equals the out-degree: $c = q$ at the $(q+1)$ -regular base, $c = q - 1$ at the q -regular levels. With $c = q$ at a level $k \geq 1$ the block is $(q+1)I - A_k^t$, nonsingular because every eigenvalue of A_k has modulus $\leq q$; then $K_1 = 0$ and K_0 is a finite group with no relation to $\text{Jac}(Y_k)$. On the K_4 tower it is $\mathbb{Z}/2 \oplus \mathbb{Z}/14 \oplus (\mathbb{Z}/112)^2$ at $k = 1$ and $(\mathbb{Z}/3)^8 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}/42 \oplus (\mathbb{Z}/336)^2$ at $k = 2$ (Table 12.2, line (W)), against $\text{Jac}(Y_1) = \mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$ and $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$. The $(q+1)$ -versus- q mismatch is the Eisenstein special layer of the whole series ([Ch. 2, Rem. 1.8], [Ch. 9, Rem. 3.3], [Ch. 10, Rem. 3.3]); here it fixes the number of return arcs.

Lemma 12.1.5 (Bowen–Franks duality). *Let $M \in M_n(\mathbb{Z})$. For $[x] \in \text{Tor coker } M$ and $[y] \in \text{Tor coker } M^t$ put*

$$\langle [x], [y] \rangle_M := y^t z \bmod \mathbb{Z}, \quad \text{where } z \in \mathbb{Q}^n, \quad Mz = x.$$

- (i) *The pairing is well defined, bilinear and perfect: $\text{Tor coker } M^t \cong \text{Hom}(\text{Tor coker } M, \mathbb{Q}/\mathbb{Z})$ canonically.*
- (ii) *If $F \in M_{n \times n'}(\mathbb{Z})$ satisfies $FM' = MF$, so that F descends to $\text{coker } M' \rightarrow \text{coker } M$ and F^t to $\text{coker } M^t \rightarrow \text{coker } M'^t$, then $\langle Fx, y \rangle_M = \langle x, F^t y \rangle_{M'}$: the two induced maps are adjoint.*

Proof. (i) If $[x]$ is torsion, $mx = Mz'$ for some $m \neq 0$ and $z = z'/m$ exists; z is unique up to $\ker_{\mathbb{Q}} M$. If $[y]$ is torsion, $my = M^t u$, so for $w \in \ker M$ one has $y^t w = \frac{1}{m} u^t M w = 0$: the ambiguity is invisible. Replacing x by $x + Ma$ changes $y^t z$ by $y^t a \in \mathbb{Z}$; replacing y by $y + M^t b$ changes it by $b^t M z = b^t x \in \mathbb{Z}$. For perfectness write $M = P^{-1} D Q^{-1}$ with D diagonal and P, Q unimodular; then $x \mapsto Px$ identifies $\text{coker } M$ with $\text{coker } D$, $y \mapsto Q^t y$ identifies $\text{coker } M^t$ with $\text{coker } D^t = \text{coker } D$, and if $Mz = x$ then $D(Q^{-1}z) = Px$ and $(Q^t y)^t (Q^{-1}z) = y^t z$, so the pairing is transported to $\langle e_i, e_j \rangle_D = \delta_{ij}/d_i$ on $\bigoplus \mathbb{Z}/d_i$, which is perfect. (ii) If $M'z' = x$ then $M(Fz') = FM'z' = Fx$, so $\langle Fx, y \rangle_M = y^t Fz' = (F^t y)^t z' = \langle x, F^t y \rangle_{M'}$. That F^t descends is $F^t M^t = (MF)^t = (FM')^t = M'^t F^t$. $\square$

Applied to $M = \Delta_k$, $M' = \Delta_{k+1}$, $F = \tau_*$ (using $\tau_* \Delta_{k+1} = \Delta_k \tau_*$), the lemma says: $\text{Tor}_k^t \cong \text{Jac}(Y_k)^\vee := \text{Hom}(\text{Jac}(Y_k), \mathbb{Q}/\mathbb{Z})$ canonically, and under this identification the pullback $\tau^*: \text{Tor}_k^t \rightarrow \text{Tor}_{k+1}^t$ is the Pontryagin dual of the tail norm $\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$.

12.2 The shadow transfer

Lemma 12.2.1 (In-coverings induce homomorphisms). *Let $p: F \rightarrow E$ be a morphism of finite directed graphs without sinks (maps on vertices and arcs with $s \circ p = p \circ s$, $r \circ p = p \circ r$), surjective on vertices, and an in-covering: for every $w \in F^0$ the restriction $p: r^{-1}(w) \rightarrow r^{-1}(p(w))$ is a bijection. Then*

$$P_v := \sum_{p(w)=v} p_w, \quad S_e := \sum_{p(f)=e} s_f$$

is a Cuntz–Krieger E -family in $C^(F)$, and the induced map $p^\sharp: C^*(E) \rightarrow C^*(F)$ is a unital, injective, gauge-equivariant $*$ -homomorphism. Its K_0 -map is the pullback $p^*: \text{coker}(1 - A_E^t) \rightarrow \text{coker}(1 - A_F^t)$, $e_v \mapsto \sum_{p(w)=v} e_w$, which satisfies $p^* A_E^t = A_F^t p^*$; its K_1 -map is the restriction of p^* to $\ker(1 - A_E^t)$.*

Proof. Fix $e \in E^1$ with $r(e) = u$. By the in-covering property, r restricts to a bijection from $p^{-1}(e)$ onto $p^{-1}(u)$: each w over u receives exactly one lift of e . Hence for distinct lifts $f \neq f'$ of e one has $r(f) \neq r(f')$ and $s_f s_{f'}^* = s_f p_{r(f)} p_{r(f')}^* s_{f'}^* = 0$, while always $s_f^* s_{f'} = 0$. Therefore

$$S_e^* S_e = \sum_{p(f)=e} p_{r(f)} = \sum_{p(w)=u} p_w = P_{r(e)}, \quad S_e S_e^* = \sum_{p(f)=e} s_f s_f^*,$$

which is (CK1). For (CK2), using (CK2) in $C^*(F)$ and $s(p(f)) = p(s(f))$,

$$P_v = \sum_{p(w)=v} \sum_{s(f)=w} s_f s_f^* = \sum_{s(p(f))=v} s_f s_f^* = \sum_{s(e)=v} \sum_{p(f)=e} s_f s_f^* = \sum_{s(e)=v} S_e S_e^*.$$

The P_v are mutually orthogonal projections, nonzero by surjectivity, with $\sum_v P_v = \sum_w p_w = 1$. The universal property gives $p^\sharp$; it intertwines the gauge actions since each S_e is a sum of degree-one generators, so the gauge-invariant uniqueness theorem [36, Thm. 2.2] makes it injective. On K_0 , $[P_v] = \sum_{p(w)=v} [p_w]$ is the pullback. Intertwining: $(p^* A_E^t)(w, v) = \#\{\text{arcs } v \rightarrow p(w)\}$ and $(A_F^t p^*)(w, v) = \#\{\text{arcs into } w \text{ from vertices over } v\}$, equal by the bijection on in-arcs. For K_1 let $\mathcal{F}_E = \bigcup_m \mathcal{F}_E^m$ be the gauge-fixed AF core, with $\mathcal{F}_E^m = \text{span}\{s_\mu s_\nu^* : |\mu| = |\nu| = m, r(\mu) = r(\nu)\}$, $K_0(\mathcal{F}_E^m) = \mathbb{Z}^{E^0}$ (the class of $s_\mu s_\mu^*$ being $e_{r(\mu)}$), the inclusions inducing A_E^t , and $K_1(C^*(E))$ identified through the Pimsner–Voiculescu sequence of the dual action with the kernel of $1 - A_E^t$ on $\varinjlim (\mathbb{Z}^{E^0}, A_E^t)$, into which the stage-0 kernel maps isomorphically because A_E^t fixes it [36, Ch. 7]. In-coverings lift paths uniquely backwards from a prescribed terminal vertex: for μ with $r(\mu) = x$ and each w over x there is a unique lift μ_w ending at w , and $p^\sharp(s_\mu) = \sum_w s_{\mu_w}$, so $p^\sharp(s_\mu s_\mu^*) = \sum_w s_{\mu_w} s_{\mu_w}^*$ (cross terms vanish as above), of class $\sum_{p(w)=x} e_w = p^* e_x$ in $K_0(\mathcal{F}_F^m)$. Thus $p^\sharp$ maps $\mathcal{F}_E^m$ into $\mathcal{F}_F^m$ and acts by p^* at every stage, compatibly with A^t by the intertwining, and naturality of the Pimsner–Voiculescu sequence gives $K_1(p^\sharp) = p^*|_{\ker(1 - A_E^t)}$. $\square$

Proposition 12.2.2 (The tail map is an in-covering of shadows). *For $k \geq 1$ define $\tau^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ by $w \mapsto \tau(w)$, $w' \mapsto \tau(w)'$ on vertices, and on arcs: an arc (e, f) of $Y_{k+1} = L(Y_k)$ (arcs e, f of Y_k with $r(e) = s(f)$) maps to the arc e of Y_k ; the arc $w \rightarrow w'$ maps to $\tau(w) \rightarrow \tau(w)'$; return arcs and loops map to the return arcs and loops of the same label at $\tau(w)'$. Then $\tau^\sharp$ is a surjective graph morphism, bijective on in-arcs at every vertex, and not bijective on out-arcs.*

Proof. Morphism: (e, f) runs from e to f in $L(Y_k)$, and its image e runs from $s(e) = \tau(e)$ to $r(e) = s(f) = \tau(f)$; the shadow arcs are compatible by construction. Surjective since τ is. In-arcs at $f \in V_{Y_{k+1}}$: the q arcs (e, f) with $r(e) = s(f)$, mapping to the q arcs e into $s(f) = \tau(f)$ — all of them, each once — and the c return arcs $f' \rightarrow f$, mapping label-wise onto the return arcs $\tau(f)' \rightarrow \tau(f)$. In-arcs at f' : the arc $f \rightarrow f'$ and the two loops, mapping onto the arc $\tau(f) \rightarrow \tau(f)'$ and the two loops. Out-arcs at f : the q arcs (f, g) with $s(g) = r(f)$ all map to the single arc f of Y_k , so $\tau^\sharp$ is q -to-one there and misses the other out-arcs of $\tau(f)$. $\square$

Theorem 12.2.3 (Shadow transfer). *For $k \geq 1$ let $\varphi_k := (\tau^\sharp)^\natural: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$,*

$$\varphi_k(p_v) = \sum_{\tau(w)=v} p_w, \quad \varphi_k(p_{v'}) = \sum_{\tau(w)=v} p_{w'}, \quad \varphi_k(s_a) = \sum_{\tau^\sharp(b)=a} s_b.$$

Then φ_k is a canonical unital injective gauge-equivariant $$ -homomorphism. Put*

$$\mathcal{T}^k := \text{diag}(\tau^*, \tau^*) \in M_{2n_{k+1} \times 2n_k}(\mathbb{Z}).$$

Then:

- (i) $\mathcal{T}^k(I - B_k^t) = (I - B_{k+1}^t)\mathcal{T}^k$, $\mathcal{T}^k \mathbf{1}_{2n_k} = \mathbf{1}_{2n_{k+1}}$, $\mathcal{T}^k(\mathbf{1}, -\mathbf{1}) = (\mathbf{1}, -\mathbf{1})$;
- (ii) $K_0(\varphi_k)$ is the map induced by $\mathcal{T}^k$, which under the isomorphisms of Theorem 12.1.3 is the pullback $\tau^*: \text{coker } \Delta_k^t \rightarrow \text{coker } \Delta_{k+1}^t$; on torsion it is the Pontryagin dual $\text{Jac}(Y_k)^\vee \rightarrow \text{Jac}(Y_{k+1})^\vee$ of the tail norm τ_* (Lemma 12.1.5), hence injective; on the torsion-free quotient $\text{coker } \Delta^t / \text{Tor} \cong \mathbb{Z}$ (via the degree) it is multiplication by q ; and $K_0(\varphi_k)$ is injective;
- (iii) $K_1(\varphi_k): \mathbb{Z} \rightarrow \mathbb{Z}$ is the identity.

Proof. Lemma 12.2.1 applies by Proposition 12.2.2. (i) The intertwining reduces blockwise to $\tau^* A_k^t = A_{k+1}^t \tau^*$, the transpose of $\tau_* A_{k+1} = A_k \tau_*$ (12.1.1); $\tau^* \mathbf{1} = \mathbf{1}$ because every vertex of Y_{k+1} has exactly one tail. (ii) Since $\mathcal{T}^k U_k = U_{k+1} \mathcal{T}^k$ (block scalars commute with $\text{diag}(\tau^*, \tau^*)$), the first-block isomorphisms of Theorem 12.1.3 carry the map induced by $\mathcal{T}^k$ to the map induced by τ^* on $\text{coker } \Delta^t$. The dual of a surjection is injective, and τ_* is surjective on Jac [Ch. 9, Thm. 3.1(ii)]. Degree: $\deg \circ \Delta^t = (\Delta \mathbf{1})^t = 0$ so $\deg$ descends, and $\deg(\tau^* y) = q \deg y$ because fibres of τ have q elements. If $K_0(\varphi_k)[x] = 0$ then $q \deg x = 0$, so $[x]$ is torsion, so $[x] = 0$. (iii) is Lemma 12.2.1 with (i). $\square$

Theorem 12.2.4 (The transfer matrix and the tail norm). *For $k \geq 1$ let $\mathcal{T}_k := \text{diag}(\tau_*, \tau_*) \in M_{2n_k \times 2n_{k+1}}(\mathbb{Z})$. Then:*

- (i) $\mathcal{T}_k(I - B_{k+1}) = (I - B_k)\mathcal{T}_k$, so $\mathcal{T}_k$ induces $(\mathcal{T}_k)_*: \text{coker}(I - B_{k+1}) \rightarrow \text{coker}(I - B_k)$ on the Bowen–Franks groups, and under V^t of (12.1.3) this map is exactly the tail norm $\tau_*: \text{coker } \Delta_{k+1} \rightarrow \text{coker } \Delta_k$ of [Ch. 9, Thm. 3.1]: surjective, surjective on Jac , with kernel of order $q^{n_k(q-1)-1}$ on $\text{Jac}(Y_{k+1})$;
- (ii) on $\ker(I - B_{k+1}) = \mathbb{Z}(\mathbf{1}, -c\mathbf{1})$ the matrix $\mathcal{T}_k$ acts by multiplication by $q = \text{ind}_1(\tau_*)$, the Eisenstein index of [Ch. 10, Cor. 2.2];
- (iii) $(\mathcal{T}_k)_*$ on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ and $K_0(\varphi_k)$ on $\text{Jac}(Y_k)^\vee \rightarrow \text{Jac}(Y_{k+1})^\vee$ are Pontryagin adjoint;
- (iv) $\mathcal{T}_k$ does not descend to K_0 : $\mathcal{T}_k(I - B_{k+1}^t) - (I - B_k^t)\mathcal{T}_k$ has the single nonzero block $\tau_* A_{k+1}^t - A_k^t \tau_* = \eta_*(qI - \tau^* \tau_*) \neq 0$, whose (v, w) entry is $q[\eta(w) = v] - A_k(\tau(w), v)$.

Proof. (i) Blockwise the intertwining is $\tau_* A_{k+1} = A_k \tau_*$; since $V_k^t \mathcal{T}_k = \mathcal{T}_k V_{k+1}^t$ the induced map on $\text{coker}(\Delta \oplus (-I)) = \text{coker } \Delta$ is the first block τ_* ; the properties listed are [Ch. 9, Thm. 3.1(ii)]. (ii) $\tau_* \mathbf{1} = q\mathbf{1}$. (iii) is Lemma 12.1.5(ii) with $F = \tau_*$. (iv) By (12.1.1), $\tau_* A_{k+1}^t = (A_{k+1} \tau^*)^t = (\eta^* \tau_* \tau^*)^t = q\eta_*$ and $A_k^t \tau_* = (\tau^* A_k)^t = (\tau^* \tau_* \eta^*)^t = \eta_* \tau^* \tau_*$; and $(\eta_* \tau^* \tau_*)(v, w) = \#\{w' : \eta(w') = v, \tau(w') = \tau(w)\} = A_k(\tau(w), v)$, which is 0 or 1 for q different v , whereas $q[\eta(w) = v]$ is q at one v . $\square$

Remark 12.2.5 (The mirror). Under Convention 12.1.1 exactly one degeneracy map lifts to the C^* -level: the one that is a covering on the (CK1) side, τ . The head map $\eta^\sharp$ is an out-covering (bijective on out-arcs, the same proof with s and r exchanged), and the sum-over-lifts formula fails for it because the cross terms $s_f s_{f'}$, with $r(f) = r(f')$ do not vanish. Dually, on K_0 the pushforward that descends is η_* ($\eta_* A_{k+1}^t = A_k^t \eta_*$ by (12.1.1)), not τ_* (Theorem 12.2.4(iv)). Reversing all arcs exchanges $\tau \leftrightarrow \eta$, $\Delta \leftrightarrow \Delta^t$, and the pro-module X_∞ of [Ch. 9, Thm. 3.1] with the in-module X_∞^{in} of [Ch. 9, Prop. 3.2]. This is the C^* -algebraic face of the two-sidedness proved in [Ch. 9]: the tower is a (τ, η) -bimodule system, and each side is realized by one of the two conventions.

Remark 12.2.6 (Downward homomorphisms). The specification asked for $*$ -homomorphisms $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ inducing τ_* . Under Convention 12.1.1 no map $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ can induce τ_* on K_0 , because τ_* does not descend (Theorem 12.2.4(iv)); the descending pushforward is η_* , and it sends $[1_{\mathcal{O}_{k+1}}] \mapsto q[1_{\mathcal{O}_k}]$ -type classes ($\eta_* \mathbf{1} = q\mathbf{1}$), so any $*$ -homomorphism inducing it is non-unital. By Kirchberg–Phillips classification [35] a $*$ -homomorphism $\psi_k: \mathcal{O}_{k+1} \rightarrow P\mathcal{O}_kP$ with $K_0(\psi_k) = \eta_*$ and $K_1(\psi_k) = q$ exists for a projection P with $[P] = q[1]$ (quoted, not used); it depends on a choice of KK -lift, ambiguous by $\text{Ext}(K_*(\mathcal{O}_{k+1}), K_{*+1}(\mathcal{O}_k))$, and is not canonical. The canonical object is φ_k ; the tail norm is its K_0 -adjoint, realized on the nose by the integer matrix $\mathcal{T}_k$.

Proposition 12.2.7 (Prime-to- q stability). *For $k \geq 1$, $\tau_*\eta^* = A_k$ acts on $\text{Jac}(Y_k)$ as multiplication by q , and $\eta^*\tau_* = A_{k+1}$ acts on $\text{Jac}(Y_{k+1})$ as multiplication by q . Hence for every prime $\ell \nmid q$ the tail norm $\tau_*: \text{Jac}(Y_{k+1})[\ell^\infty] \rightarrow \text{Jac}(Y_k)[\ell^\infty]$ is an isomorphism with inverse $q^{-1}\eta^*$, the prime-to- q torsion is the same at every level $k \geq 1$, and $\ker(\tau_*|_{\text{Jac}(Y_{k+1})}) \subseteq \text{Jac}(Y_{k+1})[q]$.*

Proof. η^* descends by (12.1.2); on coker Δ_k one has $A_k \equiv qI$ since $\Delta_k = qI - A_k \equiv 0$; the two composites are (12.1.1). On the K_4 tower the odd part is $\mathbb{Z}/3$ at $k = 1, 2, 3$ (Table 12.2). $\square$

Theorem 12.2.8 (The kernel of the tail norm is the p -torsion). *Let $q = p$ be prime and $k \geq 1$. Then:*

- (i) $\text{rk}_{\mathbb{F}_p} A_{k+1} = n_k$; hence $\dim_{\mathbb{F}_p}(\text{coker } \Delta_{k+1} \otimes \mathbb{F}_p) = n_k(p-1)$ and $\text{Jac}(Y_{k+1})[p] \cong (\mathbb{Z}/p)^{n_k(p-1)-1}$; exactly $n_k(p-1) - 1 = n_{k+1} - n_k - 1$ invariant factors of Δ_{k+1} are divisible by p ;
- (ii) $\ker(\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[p]$;
- (iii) the pro-module $\mathfrak{X}_\infty := \varprojlim_k (\text{Jac}(Y_k), \tau_*)$ has no p -torsion. Put $X_\infty^{(p)} := \varprojlim_k (\text{Jac}(Y_k)[p^\infty], \tau_*)$, the torsion part of the module X_∞ of [Ch. 9, Thm. 3.1]. Then $X_\infty^{(p)}$ is a torsion-free compact $\mathbb{Z}_p$ -module of infinite rank, topologically isomorphic to $\mathbb{Z}_p^{\mathbb{N}}$; its Pontryagin dual is the divisible group $(\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})}$; and $\mathfrak{X}_\infty \cong X_\infty^{(p)} \oplus \text{Jac}(Y_1)[p']$, where $\text{Jac}(Y_1)[p']$ is the finite prime-to- p part.

Proof. (i) $A_{k+1} = \eta^*\tau_*$; η^* has full column rank over every field (its columns are indicator vectors of the disjoint nonempty fibres of η) and τ_* is surjective over $\mathbb{Z}$, hence of full row rank n_k over $\mathbb{F}_p$; so $\text{rk}_{\mathbb{F}_p} A_{k+1} = \text{rk}_{\mathbb{F}_p} \tau_* = n_k$. Since $p \mid q$, $\Delta_{k+1} \equiv -A_{k+1} \pmod{p}$, so the $\mathbb{F}_p$ -dimension of $\text{coker } \Delta_{k+1} \otimes \mathbb{F}_p$ is $n_{k+1} - n_k = n_k(p-1)$; one dimension is the free summand $\mathbb{Z}$, the rest is the p -rank of $\text{Jac}(Y_{k+1})$. (ii) The kernel is contained in $\text{Jac}(Y_{k+1})[p]$ by Proposition 12.2.7; its order is $p^{n_k(p-1)-1}$ by [Ch. 9, Thm. 3.1(ii)] (the increment of [Ch. 5, Thm. 2.1], re-proved homologically in [Ch. 10, Cor. 2.2]); and $|\text{Jac}(Y_{k+1})[p]| = p^{n_k(p-1)-1}$ by (i). (iii) Let $x = (x_k)_k \in \mathfrak{X}_\infty$ with $px = 0$. For $k \geq 2$, $x_k \in \text{Jac}(Y_k)[p] = \ker \tau_*$ by (ii), so $x_{k-1} = \tau_* x_k = 0$; hence $x = 0$. The p -part is a pro- p abelian group, i.e. a compact $\mathbb{Z}_p$ -module, torsion-free by what was just shown; its Pontryagin dual D is a discrete p -torsion group on which multiplication by p is surjective (dual to injectivity), i.e. divisible, hence $D \cong (\mathbb{Q}_p/\mathbb{Z}_p)^{(I)}$ and $X_\infty^{(p)} \cong \mathbb{Z}_p^I$. Since $X_\infty^{(p)} \twoheadrightarrow \text{Jac}(Y_k)[p^\infty]$ for every k (surjective transitions) and the p -ranks $n_{k-1}(p-1) - 1$ are unbounded, I is infinite; it is countable because a countable inverse limit of finite groups is second countable. The splitting off of the prime-to- p part is Proposition 12.2.7. Machine confirmation on K_4 ($p = 2$): $\text{rk}_{\mathbb{F}_2} A_2 = 12$, $\text{rk}_{\mathbb{F}_2} A_3 = 24$; the 2-ranks of $\text{Jac}(Y_2)$, $\text{Jac}(Y_3)$ are 11, 23; τ_* annihilates $\text{Jac}(Y_{k+1})[2]$; and $2^{11}, 2^{23}$ are the kernel orders (Table 12.2, line (Q)). $\square$

The machine found (ii) before we did: at both computed transitions the number of even invariant factors coincided with $\text{ord}_2 |\ker \tau_*|$, and the one-line inclusion of Proposition 12.2.7 turned the coincidence into a theorem. Its meaning for the Iwasawa-theoretic reading of Chs. 5–9 is discussed in Remark 12.4.2. Note that (ii) has no meaning at the boundary transition $0 \rightarrow 1$: there τ_* does not descend to the cokernels ([Ch. 9, Rem. 3.3]) and $A_1 \neq \eta^*\tau_*$ ([Ch. 2, Def. 1.7]: $A_1 = T^t S - J$), so the identity $\text{rk}_{\mathbb{F}_p} A_{k+1} = n_k$ driving (i) has no analogue. On K_4 the count of (i) happens to hold at level 1 as well (2-rank $3 = n_0(q-1) - 1$), but no proof is offered and none is claimed.

12.3 The limit algebra

Definition 12.3.1. $\mathcal{O}_\infty(Y_\bullet) := \varinjlim_k (\mathcal{O}_k, \varphi_k)$, the C^* -inductive limit of the level shadow algebras along the shadow transfers of Theorem 12.2.3, starting at $k = 1$. We abbreviate it $\mathcal{O}_\infty$.

Theorem 12.3.2 (The limit is Kirchberg). *$\mathcal{O}_\infty$ is a unital Kirchberg algebra: separable, nuclear, simple, purely infinite, in the UCT class; it admits no trace, finite or semifinite; and it carries the gauge action γ of $\mathbb{T}$ induced from the levels.*

Proof. Separability and nuclearity pass to inductive limits. Simplicity: if J is a closed ideal with $J \cap \varphi(\mathcal{O}_k) \neq 0$ for some k then, $\mathcal{O}_k$ being simple and φ injective, $J \ni 1$; otherwise the quotient map is isometric on the dense union $\bigcup_k \varphi(\mathcal{O}_k)$, hence on $\mathcal{O}_\infty$, and $J = 0$. Pure infiniteness passes to inductive limits [34, §4], [37, §4.1], and the bootstrap class is closed under inductive limits [38]. Purely infinite simple algebras have no traces. The φ_k are gauge-equivariant (Lemma 12.2.1). $\square$

Theorem 12.3.3 (K -theory of the limit). *Write $\mathcal{K}_\infty := \varinjlim_k (\text{coker } \Delta_k^t, \tau^*)$ and $\mathfrak{X}_\infty := \varprojlim_k (\text{Jac}(Y_k), \tau_*)$, the full torsion pro-module (its p -part for $p \mid q$ is the torsion of X_∞ of [Ch. 9, Thm. 3.1]). Then:*

- (i) $K_1(\mathcal{O}_\infty) \cong \mathbb{Z}$, generated by the common image of the classes $(\mathbf{1}, -\mathbf{1})_k$;
- (ii) $K_0(\mathcal{O}_\infty) \cong \mathcal{K}_\infty$, and each $K_0(\varphi_k)$ is injective, so the levels embed as an exhaustive filtration $K_0(\mathcal{O}_1) \subset K_0(\mathcal{O}_2) \subset \cdots \subset K_0(\mathcal{O}_\infty)$ whose k -th torsion term is $\text{Jac}(Y_k)^\vee$, of order $|\text{Jac}(Y_k)|$;
- (iii) $\text{Tor } K_0(\mathcal{O}_\infty) \cong \varinjlim_k (\text{Jac}(Y_k)^\vee, \tau_*^\vee) \cong \text{Hom}_{\text{cont}}(\mathfrak{X}_\infty, \mathbb{Q}/\mathbb{Z}) =: \mathfrak{X}_\infty^\vee$ and $K_0(\mathcal{O}_\infty)/\text{Tor} \cong \mathbb{Z}[1/q]$; thus

$$0 \longrightarrow \mathfrak{X}_\infty^\vee \longrightarrow K_0(\mathcal{O}_\infty) \longrightarrow \mathbb{Z}[1/q] \longrightarrow 0$$

is exact, and $\mathfrak{X}_\infty \cong \text{Hom}(\text{Tor } K_0(\mathcal{O}_\infty), \mathbb{Q}/\mathbb{Z})$ recovers the pro-module, with $\text{Jac}(Y_k) = \text{Hom}(\text{Tor } K_0(\mathcal{O}_k), \mathbb{Q}/\mathbb{Z})$ at each level and τ_* the dual of $K_0(\varphi_k)$;

- (iv) if $q = p$ is prime the sequence splits: $K_0(\mathcal{O}_\infty) \cong \mathbb{Z}[1/p] \oplus (\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})} \oplus \text{Jac}(Y_1)[p']$ as an abstract group;
- (v) $[1_{\mathcal{O}_\infty}] = (2 - q)u$ where u is the common image of $\sum_v [p_v]$, a class of infinite order; for $q = 2$ the unit class is 0.

Proof. K -theory commutes with C^* -inductive limits, so $K_*(\mathcal{O}_\infty) = \varinjlim K_*(\mathcal{O}_k)$ along $K_*(\varphi_k)$, computed in Theorem 12.2.3. (i) The maps are the identity. (ii) Injectivity is Theorem 12.2.3(ii); the torsion of $K_0(\mathcal{O}_k)$ is $\text{Tor}_k^t \cong \text{Jac}(Y_k)^\vee$ by Lemma 12.1.5. (iii) Direct limits are exact, so $\text{Tor } \varinjlim = \varinjlim \text{Tor}$ and $\varinjlim (G_k)/\text{Tor} = \varinjlim (G_k/\text{Tor}) = \varinjlim (\mathbb{Z}, \times q) = \mathbb{Z}[1/q]$ (Theorem 12.2.3(ii)). Pontryagin duality exchanges inverse limits of compact groups with direct limits of their discrete duals, and τ^* is dual to τ_* by Lemma 12.1.5(ii). The recovery statement is double duality for the discrete torsion group $\mathfrak{X}_\infty^\vee$. (iv) By Theorem 12.2.8(iii), $\mathfrak{X}_\infty^\vee = D \oplus F$ with D divisible (hence an injective $\mathbb{Z}$ -module) and F finite of order prime to p . $\text{Ext}(\mathbb{Z}[1/p], D) = 0$; and writing $\mathbb{Z}[1/p] = \varinjlim (\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \cdots)$, $\text{Ext}(\mathbb{Z}[1/p], F)$ is the $\varprojlim^1$ of the tower $(F \xleftarrow{p} F \xleftarrow{p} \cdots)$, which vanishes because multiplication by p is an automorphism of F . So the extension splits. (v) Theorem 12.1.3 and $\tau^* \sum_v e_v = \sum_w e_w$; $\deg u = n_k \neq 0$ at level k . $\square$

Corollary 12.3.4 (The pro-module in K -homology). *By the UCT [38] for $\mathcal{O}_\infty$ there are canonical exact sequences*

$$\begin{aligned} 0 \rightarrow \text{Ext}(K_0(\mathcal{O}_\infty), \mathbb{Z}) \rightarrow K^1(\mathcal{O}_\infty) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z} \rightarrow 0, \\ 0 \rightarrow \left(\prod_{p \mid q} \mathbb{Z}_p \right) / \mathbb{Z} \rightarrow \text{Ext}(K_0(\mathcal{O}_\infty), \mathbb{Z}) \rightarrow \mathfrak{X}_\infty \rightarrow 0. \end{aligned}$$

The pro-module itself is therefore a canonical quotient of the Ext-part of the odd K -homology of $\mathcal{O}_\infty$, the kernel being the q -adic Eisenstein term $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z})$.

Proof. Apply $\text{Hom}(-, \mathbb{Z})$ to the sequence of Theorem 12.3.3(iii): $\text{Hom}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) = 0$ and $\text{Hom}(\mathbb{Z}[1/q], \mathbb{Z}) = 0$, so $0 \rightarrow \text{Ext}(\mathbb{Z}[1/q], \mathbb{Z}) \rightarrow \text{Ext}(K_0, \mathbb{Z}) \rightarrow \text{Ext}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) \rightarrow 0$. For a torsion group T , $\text{Ext}(T, \mathbb{Z}) \cong \text{Hom}(T, \mathbb{Q}/\mathbb{Z})$ (apply $\text{Hom}(T, -)$ to $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$), so $\text{Ext}(\mathfrak{X}_\infty^\vee, \mathbb{Z}) = \mathfrak{X}_\infty$ by duality. And from $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[1/q] \rightarrow \bigoplus_{p|q} \mathbb{Q}_p/\mathbb{Z}_p \rightarrow 0$ one gets $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z}) \cong (\prod_{p|q} \mathbb{Z}_p)/\mathbb{Z}$, $\mathbb{Z}$ embedded diagonally. $\square$

Theorem 12.3.5 (No-Go: the pro-module is not a K -group). $\mathfrak{X}_\infty$ (and already its p -part for any $p \mid q$) is an uncountable group, while K_0 and K_1 of a separable C^* -algebra are countable. Hence no separable C^* -algebra — in particular no Kirchberg algebra — has K_0 containing X_∞ or $\mathbb{Z} \oplus X_\infty$, and the formula $K_0(\mathcal{O}_\infty) \cong \mathbb{Z} \oplus X_\infty$ of the specification is false for every candidate $\mathcal{O}_\infty$. The correct statement is the dual one, Theorem 12.3.3(iii).

Proof. $\mathfrak{X}_\infty \rightarrow \text{Jac}(Y_k)$ for all k with $|\text{Jac}(Y_k)| \rightarrow \infty$ (the pseudo-Iwasawa law [Ch. 5, Thm. 2.2]), so $\mathfrak{X}_\infty$ is an infinite profinite group. A countable compact Hausdorff space has an isolated point (Baire); a topological group with an isolated point is discrete; an infinite discrete group is not compact. Hence $\mathfrak{X}_\infty$ is uncountable. Countability of K_* for separable algebras is standard. That K -theory is continuous for inductive and not for projective limits is the reason the inverse system $(\text{coker } \Delta_k, \tau_*)$ cannot be the K_0 of any limit algebra; the inductive system dual to it is. $\square$

12.4 Spectral blindness, broken at the tower level

Proposition 12.4.1 (The plain tower algebras are constant). For $k \geq 1$ the tail map $\tau: Y_{k+1} = L(Y_k) \rightarrow Y_k$ is an in-covering of plain graphs, and $\tau^\natural: C^*(Y_k) \rightarrow C^*(Y_{k+1})$ is an isomorphism, with inverse $\Psi: p_e \mapsto s_e s_e^*$, $s_{(e,f)} \mapsto s_e s_f s_f^*$. Consequently $C^*(Y_k) \cong C^*(Y_1) = \mathcal{O}_{A_1}$ for all $k \geq 1$, $\varinjlim_k C^*(Y_k) \cong \mathcal{O}_{A_1}$, and at every level and in the limit

$$K_0 \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1), \quad K_1 \cong \mathbb{Z}^r, \quad r = \frac{(q-1)n}{2} + 1,$$

the boundary-channel groups of [Ch. 2, Thm. 3.7]; the rooted tail groupoid algebra of [Ch. 2, Prop. 1.9(iii)] and [Ch. 5, §1] has $K_0 \cong \mathbb{Z}/(q-1)$, $K_1 = 0$. Neither sees $\text{Jac}(Y_k)$ for any k .

Proof. In-arcs at $f \in L(Y_k)^0$ are the (e, f) with $r(e) = s(f)$, mapping bijectively to the in-arcs e of $\tau(f) = s(f)$: Lemma 12.2.1 gives $\tau^\natural$. The family $\{s_e s_e^*, s_e s_f s_f^*\}$ is a Cuntz–Krieger $L(Y_k)$ -family in $C^*(Y_k)$: $(s_e s_f s_f^*)^*(s_e s_f s_f^*) = s_f s_f^* p_{r(e)} s_f s_f^* = s_f s_f^*$ since $p_{s(f)} s_f = s_f$, and $\sum_{s(f)=r(e)} s_e s_f s_f^* s_f s_f^* s_e^* = s_e p_{r(e)} s_e^* = s_e s_e^*$; so Ψ exists. $\Psi \tau^\natural(p_v) = \sum_{s(e)=v} s_e s_e^* = p_v$, $\Psi \tau^\natural(s_e) = \sum_{s(f)=r(e)} s_e s_f s_f^* = s_e$; $\tau^\natural \Psi(p_e) = \sum_{f, f'} s_{(e,f)} s_{(e,f')}^* = \sum_f s_{(e,f)} s_{(e,f)}^* = p_e$ (cross terms vanish, $r(e, f) = f \neq f'$), and $\tau^\natural \Psi(s_{(e,f)}) = (\sum_{f_1} s_{(e, f_1)}) p_f = s_{(e,f)}$. The K -groups are Morita invariants of $C^*(Y_1) = \mathcal{O}_{A_1^{\text{nb}}}$, Morita equivalent to $C(\partial X) \rtimes \Gamma$ [Ch. 2, Prop. 1.9(ii)]. On K_4 : $\text{coker}(I - A_k^t) = \mathbb{Z}^3 \oplus \mathbb{Z}/2$ at $k = 1, 2, 3$, $r = 3$ (Table 12.2, line (B)). $\square$

algebra	K_0	K_1	tower content
rooted tail groupoid $C_r^*(\mathcal{G}_\infty)$ [Chs. 2, 5]	$\mathbb{Z}/(q-1)$	0	none
plain tower limit $\varinjlim C^*(Y_k) \cong \mathcal{O}_{A_1}$	$\mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$	$\mathbb{Z}^r$	none: constant in k
shadow limit $\mathcal{O}_\infty(Y_\bullet)$	$0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0 \rightarrow \mathbb{Z}[1/q] \rightarrow 0$	$\mathbb{Z}$	all of it, dually

Table 12.1: Spectral blindness and its breaking. The shadow limit carries the Pontryagin dual of the full torsion pro-module, filtered by levels; $\text{Jac}(Y_k)$ is recovered at level k as $\text{Hom}(\text{Tor } K_0(\mathcal{O}_k), \mathbb{Q}/\mathbb{Z})$ and the tail norm as the dual of $K_0(\varphi_k)$.

Remark 12.4.2 (Where μ , $-k$ and the Euler line live). The pseudo-Iwasawa law $\text{ord}_p |\text{Jac}(Y_k)| = \mu p^k - k + \nu_0$, $\mu = n(q+1)/q$ [Ch. 5, Thm. 2.2], is the growth law of the torsion of the level filtration of $K_0(\mathcal{O}_\infty)$ (Theorem 12.3.3(ii)): 7, 18, 41 on K_4 . The Eisenstein index $\text{ind}_1(\tau_*) = q$ of [Ch. 10] appears three times in the limit: as the action of $\mathcal{T}_k$ on $\ker(I - B)$ (Theorem 12.2.4(ii)), as the scaling of the free part of K_0 that produces $\mathbb{Z}[1/q]$ (Theorem 12.2.3(ii)), and as the q -adic term $\text{Ext}(\mathbb{Z}[1/q], \mathbb{Z})$ separating K -homology from the pro-module (Corollary 12.3.4). Theorem 12.2.8 sharpens the arithmetic reading: for $q = p$ the p -rank of $\text{Jac}(Y_{k+1})$ is $(n_{k+1} - n_k) - 1$, the arc increment minus the Eisenstein line, exactly the two terms of the snake formula [Ch. 10, eq. (2.1)] now read as a rank count rather than an order count; and the p -part of the pro-module is $\mathbb{Z}_p^{\mathbb{N}}$ — torsion-free of infinite $\mathbb{Z}_p$ -rank. In classical Iwasawa theory the exponential term μp^k in the growth of the finite layers is produced by the p -power-torsion part of a finitely generated $\mathbb{Z}_p[[T]]$ -module, which is infinite whenever $\mu > 0$; here the same growth is produced by a module with no p -torsion at all. This is a third certificate, alongside [Ch. 5, Prop. 2.3] ($\lambda = -1$) and [Ch. 9, Prop. 4.1] ($A_k A_k^t \neq A_k^t A_k$), that the classical mechanism is not the one at work. We state it as a certificate, not as a theory: Problem 5.1 of [Ch. 9] and its residue [Ch. 10, Rem. 3.1] are untouched by this paper.

12.5 Scope, residue, corrections

Remark 12.5.1 (Solved, open, impossible). *Solved*: the tower-level functoriality of [Ch. 4, Rem. 4.3]. The tail norms lift canonically to the shadow algebras — as unital injective $*$ -homomorphisms $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$ whose K_0 -maps are the Pontryagin adjoints of τ_* , with τ_* itself realized by the integer transfer matrix $\mathcal{T}_k$ on the Bowen–Franks side — and the lifts assemble into a unital Kirchberg algebra $\mathcal{O}_\infty$ with $K_1 \cong \mathbb{Z}$ and $\text{Tor } K_0 \cong \mathfrak{X}_\infty^\vee$, from which the pro-module, its level filtration and the tail norms are recovered by duality. *Open*: (a) a canonical C^* -correspondence $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ induced by the out-covering $\eta^\sharp$ with K_0 -map η_* ; existence is abstract (Remark 12.2.6), canonicity is not established; (b) whether the sequence of Theorem 12.3.3(iii) splits for composite q ; (c) the Hecke-module structure of $\mathfrak{X}_\infty$ as a module over the degeneracy pro-ring of [Ch. 9, Prop. 4.1], and with it [Ch. 9, Prob. 5.1]. *Impossible*: a separable C^* -algebra with $K_0 \supseteq X_\infty$ (Theorem 12.3.5); and, as before, a Hecke-dynamical realization ([Ch. 3, Thm. 1.6], [Ch. 4, Rem. 4.3]).

Remark 12.5.2 (Four corrections to the task specification). (1) The level shadow needs $q - 1$ return arcs, not q (Remark 12.1.4). (2) The matrix $\mathcal{T}_k = \text{diag}(\tau_*, \tau_*)$ does not induce a map $\text{coker}(I - B_{k+1}^t) \rightarrow \text{coker}(I - B_k^t)$ (Theorem 12.2.4(iv)); it induces τ_* on $\text{coker}(I - B_{k+1}) \rightarrow \text{coker}(I - B_k)$, the Bowen–Franks groups, which are the Pontryagin duals of the K_0 -groups on torsion. The K_0 -map of the canonical $*$ -homomorphism is the pullback τ^* , in the opposite direction. (3) The target $K_0(\mathcal{O}_\infty) \cong \mathbb{Z} \oplus X_\infty$ is impossible (Theorem 12.3.5); the theorem is $\text{Tor } K_0(\mathcal{O}_\infty) \cong \mathfrak{X}_\infty^\vee$, $K_0/\text{Tor} \cong \mathbb{Z}[1/q]$, $K_1 \cong \mathbb{Z}$. (4) The blind algebra with $K_0 \cong \mathbb{Z}/(q - 1)$ is the rooted tail groupoid algebra of [Ch. 2, Prop. 1.9(iii)] and [Ch. 5, §1]; [Ch. 2, Thm. 3.7] is the boundary channel, with $K_0 \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$; both are blind, and the plain tower algebras are the latter at every level (Proposition 12.4.1). We also decline the specification’s historical superlative; what is proved is stated in Remark 12.5.1.

Remark 12.5.3 (Boundary level). The transition $0 \rightarrow 1$ is not covered: $\tau: Y_1 \rightarrow Y_0$ is not an in-covering (in-degrees q against $q+1$), the shadow normalizations differ ($c = q$ at level 0, $c = q - 1$ at level 1), the intertwining fails by the degeneracy boundary [Ch. 9, Rem. 3.3], and $A_1 \neq \eta^* \tau_*$. As everywhere in the series the base enters as the initial value ν_0 , and the limit $\mathcal{O}_\infty$ starts at $k = 1$. The single-level algebra of [Ch. 4] at level 0 stands beside the tower, not at its foot.

12.6 The verification battery

Table 12.2 lists the exact checks performed by `tower_kirchberg.py`; the letters match the docstring of the script and the references in the text. The levels $k \leq 3$ of the K_4 tower are the same ones used in [Ch. 9] and [Ch. 10], so every number here can be compared line by line with the batteries of those papers.

check	levels	result
(S) both factorizations (12.1.3), $c = q - 1$	$k \leq 3$	exact
(K) $\text{SNF}(I - B_k^t)$: one zero; torsion = $\text{Jac}(Y_k)$; $\text{ord}_2 = 7, 18, 41$; odd part $\mathbb{Z}/3$	$k \leq 3$	exact
(W) $c = q$: no zero invariant, finite $K_0 \neq \mathbb{Z} \oplus \text{Jac}$	$k \leq 3$	Rem. 12.1.4
(C) $\tau^\sharp$: morphism, surjective, bijective on in-arcs, not on out-arcs	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(U) $\mathcal{T}^k(I - B_k^t) = (I - B_{k+1}^t)\mathcal{T}^k$; $\mathcal{T}^k \mathbf{1} = \mathbf{1}$; $\mathcal{T}^k(\mathbf{1}, -\mathbf{1}) = (\mathbf{1}, -\mathbf{1})$	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(D) $\mathcal{T}_k(I - B_{k+1}) = (I - B_k)\mathcal{T}_k$; transposed version fails; $\mathcal{T}_k(\mathbf{1}, -c\mathbf{1}) = q(\mathbf{1}, -c\mathbf{1})$	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(N) $(\mathcal{T}_k)_*$ on torsion: surjective, $ \ker = 2^{11}, 2^{23}$; identical to τ_* on Jac	$2 \rightarrow 1, 3 \rightarrow 2$	exact
(Q) $\text{rk}_{\mathbb{F}_2} A_{k+1} = n_k$; 2-rank of $\text{Jac} = n_k - 1$; τ_* kills $\text{Jac}[2]$; $ \text{Jac}[2] = \ker \tau_* $	$2 \rightarrow 1, 3 \rightarrow 2$	exact
(P) $\langle \tau_* x, y \rangle_k = \langle x, \tau^* y \rangle_{k+1}$ in $\mathbb{Q}/\mathbb{Z}$ on random torsion classes	$1 \rightarrow 2, 2 \rightarrow 3$	exact
(G) strong connectivity, double loop at a shadow vertex	$k \leq 3$	exact
(B) $\text{SNF}(I - A_k^t) = \mathbb{Z}^3 \oplus \mathbb{Z}/2$ at every level	$k \leq 3$	exact
(L) $\deg \tau^* y = q \deg y$, $\deg \tau_* x = \deg x$	$1 \rightarrow 2, 2 \rightarrow 3$	exact

Table 12.2: Exact integer verification (`tower_kirchberg.py`) on the K_4 tower, $q = 2$, shadow matrices of size 24, 48, 96. Invariant factors found: $\text{Jac}(Y_1) = \mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$, $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$, $\text{Jac}(Y_3) = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^8 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/32 \oplus \mathbb{Z}/96$. For the fixed tower each line is itself a proof; the general statements are proved in the text.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved; the C^* -algebraic inputs are quoted with flagged citations in the Status paragraph, and Kirchberg–Phillips classification is quoted only for an existence statement that nothing depends on (Remark 12.2.6). Theorem 12.2.8 was found by machine and then proved; it is stated for $q = p$ prime, and for composite q only the inclusion $\ker \tau_* \subseteq \text{Jac}[q]$ is claimed. Remark 12.5.2 records the corrections to the specification that prompted the paper. Bibliographic data of [36, 35, 37, 38] were checked against publisher or indexing pages (titles, venues, years, volumes); page ranges are as commonly cited.

Chapter 13

Algorithmic non-commutative class field theory, XVI: the out-covering correspondence of the head map, its K_0 -index η_* , and the KK -category of the Iwahori tower

Summary. [Ch. 12] realized the tail norm of the Iwahori tower on the shadow Kirchberg algebras $\mathcal{O}_k$: the tail map τ is an in-covering, so it induces a canonical unital $*$ -homomorphism $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$, whose K_0 -map is the pullback τ^* up the tower. It left open [Ch. 12, Rem. 2.5, 2.6, 5.1(a)] the mirror question: the head map η is an out-covering, the pushforward η_* is the one that descends on the K_0 -side ($\eta_* A_{k+1}^t = A_k^t \eta_*$), yet no $*$ -homomorphism $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ can induce it, because $\eta_*[1] = q[1]$ forces non-unitality and the Kirchberg–Phillips lift is non-canonical. This paper settles it: the canonical object is not a homomorphism but a C^* -correspondence. We build from the out-covering $\eta^\sharp$ a right Hilbert $\mathcal{O}_k$ -module $\mathcal{X}_\eta = \bigoplus_w p_{\eta^\sharp(w)} \mathcal{O}_k$, finitely generated and projective, and a left action $\phi: \mathcal{O}_{k+1} \rightarrow \mathcal{K}(\mathcal{X}_\eta)$ by compact operators; the out-covering bijection on out-arcs is exactly what makes ϕ satisfy (CK2), where the sum-over-lifts homomorphism of [Ch. 12] had failed. The pair $(\mathcal{X}_\eta, \phi)$ is an injective, full, proper correspondence, so it defines a canonical class $[\mathcal{X}_\eta] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$, and its K_0 -index map is precisely the head pushforward η_* ; as a right module $[\mathcal{X}_\eta] = q[1_{\mathcal{O}_k}]$. The correspondence is not invertible in KK : its index η_* has kernel the 2-torsion of $\text{Jac}(Y_{k+1})$, of order $q^{n_k(q-1)-1}$. The two families of morphisms organize the levels into a small KK -category with objects $\{\mathcal{O}_k\}$: alongside the upward homomorphisms $[\varphi_k]$ (K_0 -map τ^*) sit the downward correspondences $[\mathcal{X}_\eta]$ (K_0 -map η_*), and their Kasparov products are the Hecke operators — $[\varphi_k] \otimes [\mathcal{X}_\eta]$ acts on $K_0(\mathcal{O}_k)$ as A_k^t and $[\mathcal{X}_\eta] \otimes [\varphi_k]$ on $K_0(\mathcal{O}_{k+1})$ as A_{k+1}^t . This is the C^* -bivariant form of the degeneracy calculus $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ of [Ch. 9], and it completes the two-sided operator-algebraic picture of the tower. Everything is verified exactly on the K_4 tower at levels $k \leq 3$.

Status of the results

Every numbered statement is proved. The C^* -algebraic inputs are the standard theory of C^* -correspondences and their Kasparov classes [40, 39], the graph-algebra K -theory and gauge-invariant uniqueness of [36], and Kaminker–Putnam Spanier–Whitehead duality [41] (cited once, for context, in Remark 13.3.3); the

construction of $\mathcal{X}_\eta$ and the computation of its index are done in full. The in-covering homomorphism φ_k and the matrix identities of the tower are quoted from [Ch. 12] and re-verified. Every finite identity was machine-verified exactly on the K_4 tower, $q = 2$, at the transitions $k + 1 \rightarrow k$ for $k \leq 3$ (shadow graphs on 24, 48, 96 vertices; Table 13.1). Section 13.6 records what is solved, what stays open, and where the specification is corrected.

13.1 Setting: the two degeneracy maps and the mirror problem

We keep the conventions of [Ch. 12]. Y_0 is a finite connected $(q+1)$ -regular graph, $q \geq 2$; for $k \geq 1$, Y_k is the Iwahori path graph, a strongly connected q -regular digraph on $n_k = n(q+1)q^{k-1}$ vertices with $Y_{k+1} = L(Y_k)$ its line digraph, so $V_{Y_{k+1}} = \text{Arcs}(Y_k)$. The tail and head maps $\tau, \eta: Y_{k+1} \rightarrow Y_k$ send an arc to its source and range; $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ are their pushforwards, $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ the pullbacks, with [Ch. 9, Prop. 1.2]

$$\tau_* \eta^* = A_k, \quad \eta^* \tau_* = A_{k+1}, \quad \tau_* \tau^* = \eta_* \eta^* = qI, \quad \tau_* \Delta_{k+1} = \Delta_k \tau_*, \quad \eta_* \Delta_{k+1}^t = \Delta_k^t \eta_*, \quad (13.1.1)$$

$\Delta_k = qI - A_k$. The shadow graph $Y_k^\sharp$ has vertex set $V_{Y_k} \sqcup V'_{Y_k}$, the arcs of Y_k , one arc $v \rightarrow v'$, $c := q - 1$ return arcs $v' \rightarrow v$, and two loops at each v' ; its vertex matrix is $B_k = \begin{pmatrix} A_k & I \\ cI & 2I \end{pmatrix}$ and $\mathcal{O}_k := C^*(Y_k^\sharp)$ is a unital Kirchberg algebra with, in the convention $K_0(C^*(E)) = \text{coker}(1 - A_E^t)$ of [36, Thm. 7.16],

$$K_0(\mathcal{O}_k) = \text{coker}(I - B_k^t) \cong \mathbb{Z} \oplus \text{Jac}(Y_k), \quad K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z}, \quad (13.1.2)$$

by the factorization $U(I - B_k^t)V = \Delta_k^t \oplus (-I)$ of [Ch. 12, Thm. 1.4] with $U = \begin{pmatrix} I & -cI \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$.

[Ch. 12] proved that τ extends to a graph morphism $\tau^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ that is an *in*-covering — bijective on the in-arcs at every vertex — and that in-coverings induce, by $p_v \mapsto \sum_{\tau^\sharp(w)=v} p_w$, $s_a \mapsto \sum_{\tau^\sharp(b)=a} s_b$, a canonical unital injective $*$ -homomorphism $\varphi_k: \mathcal{O}_k \rightarrow \mathcal{O}_{k+1}$ [Ch. 12, Lem. 2.2, Thm. 2.5], whose K_0 -map is the pullback τ^* (up the tower) and whose K_1 -map is the identity.

The head map is the mirror. Its shadow extension $\eta^\sharp: Y_{k+1}^\sharp \rightarrow Y_k^\sharp$ is an *out*-covering, bijective on out-arcs and q -to-one on in-arcs (Table 13.1, line (O)), and the sum-over-lifts formula *fails* to be a homomorphism: for two arcs $f \neq f'$ of $Y_{k+1}^\sharp$ over the same arc with the same range, the cross term $s_f s_{f'}$, does not vanish, so $(\sum_b s_b)$ is not a Cuntz–Krieger partial isometry [Ch. 12, Rem. 2.4]. Yet on K_0 it is the head pushforward η_* , not τ_* , that descends:

$$\eta_* A_{k+1}^t = A_k^t \eta_*, \quad H_k := \begin{pmatrix} \eta_* & 0 \\ 0 & \eta_* \end{pmatrix} \text{ satisfies } H_k(I - B_{k+1}^t) = (I - B_k^t)H_k, \quad (13.1.3)$$

the first being the transpose of $\tau_* A_{k+1} = A_k \tau_*$ from (13.1.1); whereas $\text{diag}(\tau_*, \tau_*)$ does *not* intertwine $I - B^t$ [Ch. 12, Thm. 2.5(iv)]. So the arrow the tower asks for — a morphism $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ realizing η_* downward — exists on K_0 but cannot be a homomorphism, since additionally $H_k \mathbf{1} = q\mathbf{1}$ makes $\eta_*[1_{\mathcal{O}_{k+1}}] = q[1_{\mathcal{O}_k}]$ (a unit class cannot map to q times a unit class). The resolution is that the morphism is a C^* -correspondence.

13.2 The out-covering correspondence

We recall that a C^* -correspondence ${}_A \mathcal{X}_B$ is a right Hilbert B -module $\mathcal{X}$ with a $*$ -homomorphism $\phi: A \rightarrow \mathcal{L}(\mathcal{X})$; if $\mathcal{X}$ is finitely generated projective over a unital B then $\mathcal{L}(\mathcal{X}) = \mathcal{K}(\mathcal{X})$, and $(\mathcal{X}, \phi, 0)$ is an even Kasparov cycle defining $[\mathcal{X}] \in KK(A, B)$ whose product with $[e] \in K_0(A)$, $e = e^2 = e^* \in M_\infty(A)$, is $[\phi(e)\mathcal{X}] \in K_0(B)$ [40, 39].

Throughout this section $p := \eta^\sharp: F \rightarrow E$, $F = Y_{k+1}^\sharp$, $E = Y_k^\sharp$, an out-covering: $s^{-1}(w) \rightarrow s^{-1}(p(w))$ is a bijection for every $w \in F^0$. Write $\{p_v, s_e\}$ for the generators of $\mathcal{O}_k = C^*(E)$ and $\{q_w, t_f\}$ for those of $\mathcal{O}_{k+1} = C^*(F)$.

Definition 13.2.1 (The head module). Let

$$\mathcal{X}_\eta := \bigoplus_{w \in F^0} p_{p(w)} \mathcal{O}_k,$$

the right Hilbert $\mathcal{O}_k$ -module of finitely supported families $\xi = (\xi_w)_{w \in F^0}$ with $\xi_w \in p_{p(w)}\mathcal{O}_k$, right action $(\xi \cdot a)_w = \xi_w a$, and $\mathcal{O}_k$ -valued inner product $\langle \xi, \zeta \rangle = \sum_{w \in F^0} \xi_w^* \zeta_w$. Let $e_w \in \mathcal{X}_\eta$ be the generator with w -component $p_{p(w)}$ and all others 0.

Lemma 13.2.2 (The module is finitely generated projective and full). *$\mathcal{X}_\eta$ is a right Hilbert $\mathcal{O}_k$ -module with $\langle e_w, e_{w'} \rangle = \delta_{ww'} p_{p(w)}$, finite orthogonal generating set $\{e_w\}_{w \in F^0}$, and reconstruction $\xi = \sum_w e_w \langle e_w, \xi \rangle$; hence $\mathcal{X}_\eta$ is finitely generated projective, $\mathcal{L}(\mathcal{X}_\eta) = \mathcal{K}(\mathcal{X}_\eta)$, and*

$$[\mathcal{X}_\eta] = \sum_{w \in F^0} [p_{p(w)}] = q [1_{\mathcal{O}_k}] \in K_0(\mathcal{O}_k),$$

because every fibre of $\eta^\sharp$ has exactly q elements. $\mathcal{X}_\eta$ is full: $\overline{\langle \mathcal{X}_\eta, \mathcal{X}_\eta \rangle} = \mathcal{O}_k$.

Proof. $\mathcal{X}_\eta$ is an orthogonal direct sum of the corners $p_{p(w)}\mathcal{O}_k$, each a Hilbert $\mathcal{O}_k$ -module with $\langle a, b \rangle = a^*b$; the displayed inner products and reconstruction are immediate, so $\{e_w\}$ is a finite frame and $\mathcal{X}_\eta \cong \bigoplus_w p_{p(w)}\mathcal{O}_k$ is finitely generated projective with $[\mathcal{X}_\eta] = \sum_w [p_{p(w)}]$. The fibre of $\eta^\sharp$ over a Y -vertex v is $\{\text{arcs } f \text{ of } Y_k \text{ with } r(f) = v\}$, of size the in-degree q ; over v' it is the q shadow lifts likewise, so each vertex of E is hit q times and $\sum_w [p_{p(w)}] = q \sum_v [p_v] = q[1]$. Fullness: $\langle e_w, e_w \rangle = p_{p(w)}$ and $\{p_v\}$ generates $\mathcal{O}_k$ as an ideal (it contains the unit). $\square$

Theorem 13.2.3 (The left action). *There is a unique $*$ -homomorphism $\phi: \mathcal{O}_{k+1} \rightarrow \mathcal{K}(\mathcal{X}_\eta)$ with*

$$\phi(q_w) = \Theta_{e_w, e_w} \quad (\text{projection onto the } w\text{-summand}), \quad \phi(t_f) e_w = \delta_{w, r(f)} e_{s(f)} \cdot s_{p(f)}.$$

It is injective. The verification of the Cuntz–Krieger relation (CK2) for the operators $\phi(t_f)$ uses precisely that $\eta^\sharp$ is an out-covering.

Proof. Define $V_f := \phi(t_f)$ by the stated formula: V_f maps the $r(f)$ -summand $p_{p(r(f))}\mathcal{O}_k$ into the $s(f)$ -summand by left multiplication by $s_{p(f)}$, and kills the other summands. Since p is a graph morphism, $p(f)$ runs $p(s(f)) \rightarrow p(r(f))$, so $s_{p(f)} = p_{p(s(f))} s_{p(f)} p_{p(r(f))}$ and V_f does land in $p_{p(s(f))}\mathcal{O}_k$; each V_f is adjointable with $V_f^* \xi = \delta$ -shift back by $s_{p(f)}^*$, hence $V_f \in \mathcal{K}(\mathcal{X}_\eta)$ (finite rank in the frame $\{e_w\}$). Now

$$V_f^* V_f \xi = s_{p(f)}^* s_{p(f)} \xi_{r(f)} = p_{r(p(f))} \xi_{r(f)} = p_{p(r(f))} \xi_{r(f)} = \xi_{r(f)} \quad (\xi \in \mathcal{X}_\eta),$$

using (CK1) in $\mathcal{O}_k$ and $r(p(f)) = p(r(f))$; so $V_f^* V_f = \Theta_{e_{r(f)}, e_{r(f)}} = \phi(q_{r(f)})$, which is (CK1) for ϕ . For (CK2), fix $w \in F^0$ and compute $\sum_{s(f)=w} V_f V_f^*$. Each such $V_f V_f^*$ acts on the $w = s(f)$ -summand by $\xi \mapsto s_{p(f)} s_{p(f)}^* \xi$, so

$$\sum_{s(f)=w} V_f V_f^* \Big|_{w\text{-summand}} = \sum_{s(f)=w} s_{p(f)} s_{p(f)}^* = \sum_{s(e)=p(w)} s_e s_e^* = p_{p(w)},$$

where the middle equality is the out-covering property: $f \mapsto p(f)$ is a *bijection* from $\{f : s(f) = w\}$ onto $\{e : s(e) = p(w)\}$, so the sum reindexes exactly, and the last equality is (CK2) in $\mathcal{O}_k$. The right side is the identity of the w -summand, i.e. $\phi(q_w)$; and $V_f V_f^*$ kills every other summand. Hence $\sum_{s(f)=w} \phi(t_f) \phi(t_f)^* = \phi(q_w)$, which is (CK2) for ϕ . The universal property of $C^*(F)$ gives the $*$ -homomorphism ϕ . Injectivity: $\phi(q_w) \neq 0$ for every w (its range $p_{p(w)}\mathcal{O}_k \neq 0$), and $\mathcal{O}_{k+1}$ is simple, so $\ker \phi = 0$.

The out-covering hypothesis enters at exactly one point — the reindexing $\{p(f) : s(f) = w\} = \{e : s(e) = p(w)\}$ in (CK2) — and nowhere else. For the in-covering $\tau^\sharp$ this reindexing holds on *in*-arcs, which is why $\tau^\sharp$ gave a homomorphism into $\mathcal{O}_{k+1}$ rather than a correspondence; for $\eta^\sharp$ it is the out-arcs that biject, and the natural home is $\mathcal{K}(\mathcal{X}_\eta)$. The cross terms $s_{p(f)} s_{p(f')}^*$, $s_{p(f')} s_{p(f)}^*$ that obstructed the sum-over-lifts homomorphism of [Ch. 12] are here mutually orthogonal (distinct out-arcs $p(f) \neq p(f')$ of $p(w)$), so they add rather than interfere. $\square$

Definition 13.2.4 (The head correspondence and its Kasparov class). $\mathcal{E}_\eta := (\mathcal{X}_\eta, \phi)$ is an injective, full, proper C^* -correspondence from $\mathcal{O}_{k+1}$ to $\mathcal{O}_k$; its Kasparov class is

$$[\mathcal{E}_\eta] := [(\mathcal{X}_\eta, \phi, 0)] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k).$$

13.3 The index is the head pushforward

Theorem 13.3.1 (K_0 -index of the head correspondence). *The map $\otimes_{\mathcal{O}_{k+1}}[\mathcal{E}_\eta]: K_0(\mathcal{O}_{k+1}) \rightarrow K_0(\mathcal{O}_k)$ is the head pushforward: under (13.1.2) it is induced by $H_k = \text{diag}(\eta_*, \eta_*)$, i.e. the descent of $\eta_*: \text{coker } \Delta_{k+1}^t \rightarrow \text{coker } \Delta_k^t$ (13.1.3). On generators $[q_w] \mapsto [p_{\eta^\sharp(w)}]$; the K_1 -map is multiplication by q ; and $[\mathcal{E}_\eta] \otimes [1_{\mathcal{O}_k}]^\vee$ recovers $[\mathcal{X}_\eta] = q[1_{\mathcal{O}_k}]$.*

Proof. For a vertex projection $q_w \in \mathcal{O}_{k+1}$, $\phi(q_w)\mathcal{X}_\eta$ is the w -summand $p_{p(w)}\mathcal{O}_k$, of class $[p_{p(w)}] = [p_{\eta^\sharp(w)}]$ in $K_0(\mathcal{O}_k)$; since the classes $[q_w]$ generate $K_0(\mathcal{O}_{k+1})$ and the product is additive, the index map sends $[q_w] \mapsto [p_{\eta^\sharp(w)}]$, which on the generators $e_w \mapsto e_{\eta^\sharp(w)}$ of $\text{coker}(I - B^t)$ is the matrix $H_k = \text{diag}(\eta_*, \eta_*)$. That H_k descends to $\text{coker}(I - B_{k+1}^t) \rightarrow \text{coker}(I - B_k^t)$ is (13.1.3); conjugating by U of [Ch. 12, Thm. 1.4] (which commutes with diag blocks) turns it into η_* on $\text{coker } \Delta^t$. The K_1 -map: $K_1(\mathcal{O}_k) = \ker(I - B_k^t) = \mathbb{Z}(\mathbf{1}, -\mathbf{1})$ and $H_k(\mathbf{1}, -\mathbf{1}) = q(\mathbf{1}, -\mathbf{1})$ (Table 13.1, line (K)). The last statement is Lemma 13.2.2. $\square$

Proposition 13.3.2 (The correspondence is not invertible). *$[\mathcal{E}_\eta]$ is not a KK -equivalence. Its index η_* on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ is surjective with kernel the p -torsion subgroup $\text{Jac}(Y_{k+1})[p]$ when $q = p$ is prime, of order $p^{n_k(p-1)-1}$; equivalently $\mathcal{X}_\eta$ is not a Morita equivalence bimodule ($\phi(\mathcal{O}_{k+1}) \subsetneq \mathcal{K}(\mathcal{X}_\eta)$).*

Proof. By reversal symmetry (Remark 13.3.3) η_* on $\text{Tor coker } \Delta_{k+1}^t \rightarrow \text{Tor coker } \Delta_k^t$ is the mirror of the tail norm τ_* on $\text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)$ of [Ch. 12, Thm. 2.8]: surjective with kernel the p -torsion of order $p^{n_k(p-1)-1}$ (Table 13.1, line (N)). A KK -equivalence induces isomorphisms on K_* ; η_* is not injective, so $[\mathcal{E}_\eta]$ is not invertible. Were $\mathcal{X}_\eta$ an imprimitivity bimodule, ϕ would be onto $\mathcal{K}(\mathcal{X}_\eta)$ and the index an isomorphism. $\square$

Remark 13.3.3 (Reversal symmetry). Let ρ_k be the path-reversal permutation of V_{Y_k} (reverse each non-backtracking path). Then $\rho_k A_k \rho_k^{-1} = A_k^t$ and $\rho_k \tau_* = \eta_* \rho_{k+1}$ (Table 13.1, line (R)): reversing all arrows exchanges $\tau \leftrightarrow \eta$, $\Delta \leftrightarrow \Delta^t$, $\mathcal{O}_B \leftrightarrow \mathcal{O}_{B^t}$, and carries the in-covering theory of [Ch. 12] for (τ, Δ) to the out-covering theory of this paper for (η, Δ^t) . At the algebra level reversal is Kaminker–Putnam Spanier–Whitehead duality [41], $\mathcal{O}_B$ and $\mathcal{O}_{B^t}$ dual, which exchanges K_0 with K^1 ; we use only the elementary permutation identity, not the duality, and cite [41] for context.

13.4 Kasparov products and the Hecke operators

Write $[\varphi_k] \in KK(\mathcal{O}_k, \mathcal{O}_{k+1})$ for the class of the tail homomorphism (K_0 -map τ^* , upward) and $[\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ for the head correspondence (K_0 -map η_* , downward).

Theorem 13.4.1 (Round trips are Hecke operators). *The Kasparov products*

$$[\varphi_k] \otimes_{\mathcal{O}_{k+1}} [\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_k, \mathcal{O}_k), \quad [\mathcal{E}_\eta^{(k)}] \otimes_{\mathcal{O}_k} [\varphi_k] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_{k+1})$$

induce on K_0 the maps

$$\eta_* \tau^* = A_k^t \text{ on } K_0(\mathcal{O}_k), \quad \tau^* \eta_* = A_{k+1}^t \text{ on } K_0(\mathcal{O}_{k+1}),$$

i.e. the level Hecke operators. These are the C^ -bivariant form of the degeneracy identities $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ of [Ch. 9, Prop. 1.2].*

Proof. The K_0 -functor takes Kasparov products to composites, so $\otimes([\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]) = (\otimes[\mathcal{E}_\eta^{(k)}]) \circ (\otimes[\varphi_k]) = H_k \circ T^k$ with $T^k = \text{diag}(\tau^*, \tau^*) = K_0(\varphi_k)$ (Theorem 13.3.1, [Ch. 12, Thm. 2.5]). Blockwise $H_k T^k = \text{diag}(\eta_* \tau^*, \eta_* \tau^*)$, and $\eta_* \tau^* = (\tau_* \eta^*)^t = A_k^t$ by transposing (13.1.1); similarly $T^k H_k = \text{diag}(\tau^* \eta_*, \tau^* \eta_*)$ with $\tau^* \eta_* = (\eta^* \tau_*)^t = A_{k+1}^t$ (Table 13.1, line (H)). Both descend to $\text{coker}(I - B^t)$, since each factor does. $\square$

Remark 13.4.2 (The opposite side and the full calculus). The other two degeneracy maps, τ_* and η^* , descend not on K_0 but on the Bowen–Franks group $\text{coker}(I - B) \cong \text{coker } \Delta$, the K_0 of the opposite algebra $\mathcal{O}_k^{\text{op}} \cong C^*(\overline{Y_k^\sharp})$: there $\text{diag}(\tau_*, \tau_*)$ and $\text{diag}(\eta^*, \eta^*)$ intertwine $I - B$, with composites A_k and A_{k+1}

(Table 13.1, line (B)). Thus all four degeneracy maps of [Ch. 9] are realized, two by KK -classes on the K_0 -side (τ^* up, η_* down) and two by their reversal-dual partners on the Bowen–Franks side; the “ qI ” identities $\tau_*\tau^* = \eta_*\eta^* = qI$ pair one map from each side and are the Pontryagin adjointness of [Ch. 12, Lem. 2.1]. The correspondence $\mathcal{E}_\eta$ supplies the one arrow that was missing: the canonical downward morphism inducing η_* .

Definition 13.4.3 (The KK -category of the tower). Let $\mathcal{T}$ be the small category enriched in KK (a subcategory of the Kasparov category KK) with objects $\{\mathcal{O}_k : k \geq 1\}$ and morphisms generated under Kasparov product and $\mathbb{Z}$ -linear combination by $[\varphi_k] \in KK(\mathcal{O}_k, \mathcal{O}_{k+1})$ and $[\mathcal{E}_\eta^{(k)}] \in KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ for all k .

Corollary 13.4.4 (Structure of $\mathcal{T}$). *In $\mathcal{T}$ the endomorphism $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]$ of $\mathcal{O}_k$ acts on $K_0(\mathcal{O}_k)$ as the Hecke operator A_k^t , and $[\mathcal{E}_\eta^{(k)}] \otimes [\varphi_k]$ on $\mathcal{O}_{k+1}$ as A_{k+1}^t ; the diagonal composite $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}] \otimes \cdots$ over a τ -then- η round trip realizes, on K_0 , the polynomial algebra generated by the Hecke operators of the tower. No morphism of $\mathcal{T}$ is invertible except the identities: $[\varphi_k]$ has index τ^* (injective, not onto: cokernel the new p -torsion) and $[\mathcal{E}_\eta^{(k)}]$ has index η_* (onto, not injective), so $\mathcal{T}$ is a genuinely one-way-plus-one-way category, the operator-algebraic shadow of the two-sidedness of the pro-module X_∞ of [Ch. 9].*

Proof. Theorems 13.3.1, 13.4.1; injectivity/surjectivity from [Ch. 12, Thm. 2.5] and Proposition 13.3.2. $\square$

13.5 The verification battery

check	transitions	result
(O) $\eta^\sharp$ morphism, bijective on out-arcs, q -to-one on in-arcs; $\tau^\sharp$ the mirror in-covering	$k \leq 3$	exact
(R) $\rho_k A_k \rho_k^{-1} = A_k^t$, $\rho_k \tau_* = \eta_* \rho_{k+1}$	$k \leq 3$	exact
(K) $H_k(I - B_{k+1}^t) = (I - B_k^t)H_k$; $\text{diag}(\tau_*, \tau_*)$ does not; $H_k \mathbf{1} = q\mathbf{1}$; $H_k(\mathbf{1}, -\mathbf{1}) = q(\mathbf{1}, -\mathbf{1})$	$k \leq 3$	exact
(H) $H_k T^k = \text{diag}(A_k^t)$, $T^k H_k = \text{diag}(A_{k+1}^t)$; witness $\text{diag}(A^t - q) = (I - B^t)VN$	$k \leq 3$	exact
(B) opposite side: $\text{diag}(\tau_*, \tau_*)$, $\text{diag}(\eta^*, \eta^*)$ descend on $\text{coker}(I - B)$; composites A_k, A_{k+1}	$k \leq 3$	exact
(N) η_* on $\text{Tor coker } \Delta^t$: surjective, $ \ker = 2^{11}, 2^{23}, 2^{47} = 2^{n_k(q-1)-1}$; $\text{rank}_{\mathbb{F}_2} A_{k+1} = n_k$	$k \leq 3$	exact
(X) $\text{Ext}(K_0(\mathcal{O}_{k+1}), K_1(\mathcal{O}_k)) = \text{Jac}(Y_{k+1})$ (lift ambiguity); $\text{Ext}(K_0(\mathcal{O}_k), K_1(\mathcal{O}_k)) = \text{Jac}(Y_k)$	$k \leq 3$	exact

Table 13.1: Exact integer verification (`head_correspondence.py`) on the K_4 tower, $q = 2$, at the transitions $k + 1 \rightarrow k$, $k \leq 3$; shadow matrices of size 24, 48, 96, 192. Jac data as in [Ch. 12, Table 1].

13.6 Scope, residue, corrections

Remark 13.6.1 (Solved, open, impossible). *Solved:* the open item [Ch. 12, Rem. 5.1(a)] — a canonical C^* -correspondence $\mathcal{O}_{k+1} \rightsquigarrow \mathcal{O}_k$ induced by the out-covering $\eta^\sharp$ with K_0 -map η_* . It is $\mathcal{E}_\eta = (\mathcal{X}_\eta, \phi)$ of Definition 13.2.4: a finitely generated projective, full, injective proper correspondence whose class in $KK(\mathcal{O}_{k+1}, \mathcal{O}_k)$ induces η_* (Theorem 13.3.1), completing the two-sided realization of the degeneracy calculus (Theorem 13.4.1, Remark 13.4.2) and organizing the levels into the KK -category $\mathcal{T}$ (Definition 13.4.3, Corollary 13.4.4). *Open:* (a) the Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{X}_\eta}$ of the head correspondence, and whether the inductive system it generates gives a second model of the limit $\mathcal{O}_\infty$ of [Ch. 12] dual to the first; (b) an explicit unbounded (spectral-triple) representative of $[\mathcal{E}_\eta]$ pairing with the $\mathcal{L}$ -invariant KK -classes of [Ch. 8]; (c) the Hecke-module structure of $K_0(\mathcal{O}_\infty)$ under the operators $[\varphi_k] \otimes [\mathcal{E}_\eta^{(k)}]$ of Corollary 13.4.4, and with it [Ch. 9, Prob. 5.1]. *Impossible:* a unital $*$ -homomorphism $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ inducing η_* (Section 13.1: $\eta_*[1] = q[1]$); and, as throughout the series, a KK -equivalence between adjacent levels (Proposition 13.3.2).

Remark 13.6.2 (Corrections to the task specification). (1) The correspondence’s right module is not “freely generated by the graph’s generators”; it is the finitely generated projective module $\mathcal{X}_\eta = \bigoplus_w p_{\eta^\sharp(w)} \mathcal{O}_k$, a sum of corners, of K -theory class $q[1]$. (2) Positive-definiteness and self-adjointness of the inner product are automatic (the inner product is $\sum_w \xi_w^* \zeta_w$, a sum of positives) and use no Cuntz–Krieger relation; what (CK2) — together with the out-covering bijection on *out*-arcs — establishes is that the *left action* ϕ is a $*$ -homomorphism (Theorem 13.2.3). This is the one place the geometry is used, and it is on out-arcs, not on the (CK1) side as the specification suggested. (3) The left action lands in $\mathcal{K}(\mathcal{X}_\eta)$, not in $\mathcal{O}_k$; that is exactly why the object is a correspondence and not a homomorphism, and why it exists where the sum-over-lifts homomorphism of [Ch. 12] could not. (4) The head pushforward η_* , not τ_* , is the map the downward correspondence induces; τ_* descends on the opposite (Bowen–Franks) side, not on K_0 (Remark 13.4.2). (5) “All six degeneracy identities in KK ” is made precise as the Kasparov products of Theorem 13.4.1 and the opposite-side identities of Remark 13.4.2: the two Hecke operators are the two round-trip products, and $\tau_* \tau^* = \eta_* \eta^* = qI$ are cross-side adjointness identities, not products of two K_0 -side classes.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved; the correspondence $\mathcal{X}_\eta$, its left action, and its index are constructed and computed in full, and the tower matrix identities are quoted from [Ch. 12] and re-verified exactly on K_4 . The bivariant inputs — C^* -correspondences define KK -classes, and K_0 turns Kasparov products into composites — are standard [40, 39]; [41] is cited for context only. Bibliographic data of [40, 39, 36, 41] were checked against publisher or indexing pages.

Chapter 14

Algorithmic non-commutative class field theory, XVII: the tail norm at composite q , and the splitting of the limit K -module

Summary. Paper VIII proved that for $q = p$ prime the kernel of the tail norm on the Iwahori tower is the full p -torsion of the sandpile group, and deduced from the resulting torsion-freeness that the sequence $0 \rightarrow \mathfrak{X}^\vee \rightarrow K_0(\mathcal{O}_\infty) \rightarrow \mathbb{Z}[1/q] \rightarrow 0$ splits. Both statements were left open for composite q , on the ground that $\mathbb{Z}/q\mathbb{Z}$ has zero divisors. That ground is illusory. The only ingredient of the prime proof that used primality is the rank identity $\text{rank}_{\mathbb{F}_p} A_{k+1} = n_k$, and A_{k+1} is the adjacency matrix of a line digraph, hence factors as $H^T T$ through the head and tail incidence matrices; the rows of H and of T are non-empty and disjointly supported, so the rank is n_k *in every characteristic*. With that, the two facts already known for all q — $|\ker \tau_*| = q^{n_k(q-1)-1}$ and $\ker \tau_* \subseteq \text{Jac}(Y_{k+1})[q]$ — close the problem by counting: the two orders are forced to agree, so

$$\ker(\tau_* : \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[q] \cong (\mathbb{Z}/q)^{n_k(q-1)-1}$$

for every $q \geq 2$. Equality in the counting has a second consequence, which is new even for prime powers and is the sharpest statement here: *every invariant factor of $\text{Jac}(Y_{k+1})$ divisible by a prime $p \mid q$ is divisible by $p^{v_p(q)}$* . So at $q = 4$ the group has no element of order exactly 2, and the kernel of the tail norm strictly contains the 2-torsion — a structural phenomenon with no analogue in the prime case. Duality then gives the splitting for every q in three lines and without identifying $\mathfrak{X}$: $\mathfrak{X}[q] = 0$, so $\mathfrak{X}^\vee$ is q -divisible, so $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}^\vee) = \varprojlim^1(\mathfrak{X}^\vee, \times q) = 0$. Six checks on eight towers ($q = 2, 3, 4, 5, 6$, complete and incomplete bases, to level three), all exact; the $q = 4$ towers could have falsified the divisibility statement and did not.

14.1 The rank is characteristic-free

Throughout, Y_0 is a $(q+1)$ -regular graph on n vertices, Y_1 its non-backtracking directed-edge graph — Eulerian q -regular on $n_1 = n(q+1)$ vertices — and $Y_{k+1} = L(Y_k)$ the line digraph, so $n_{k+1} = qn_k$, for $k \geq 1$. The degeneracy maps τ (tail) and η (head) send the vertex $i \rightarrow j$ of Y_{k+1} to i and to j respectively; τ_* is the pushforward and η^* the pullback. We write

$$r_k := n_k(q-1) - 1,$$

$\Delta_k = qI - A_k$ for the out-Laplacian of Y_k , and $\text{Jac}(Y_k)$ for the torsion of $\text{coker } \Delta_k^\top$, so that $\text{coker } \Delta_k^\top \cong \mathbb{Z} \oplus \text{Jac}(Y_k)$. Three facts from the earlier papers hold for *every* $q \geq 2$ and will be used as stated:

- (i) $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ and $\tau_* \tau^* = \eta_* \eta^* = qI$ [Ch. 9, Prop. 1.2];
- (ii) τ_* is surjective on Jacobians with $|\ker \tau_*| = q^{r_k}$ [Ch. 9, Thm. 3.1(ii)], a restatement of the line-digraph recursion $\kappa(Y_{k+1}) = q^{r_k} \kappa(Y_k)$ [Ch. 5, Thm. 2.1];
- (iii) A_{k+1} acts on $\text{Jac}(Y_{k+1})$ as multiplication by q , and consequently $\ker \tau_* \subseteq \text{Jac}(Y_{k+1})[q]$ [Ch. 12, Prop. 2.7].

Only (iii) needs a word: $\Delta_{k+1} = qI - A_{k+1}$ annihilates the cokernel, so $A_{k+1} = q$ there, and by (i) $\tau_* x = 0$ forces $A_{k+1}x = \eta^* \tau_* x = 0$, i.e. $qx = 0$.

What was available only for $q = p$ prime is the rank of A_{k+1} modulo p [Ch. 12, Thm. 2.8]. It is in fact insensitive to the characteristic.

Lemma 14.1.1 (Rank of a line-digraph adjacency matrix). *Let G be a digraph on n vertices and m arcs in which every vertex is the head of at least one arc and the tail of at least one arc, and let $A_{L(G)}$ be the adjacency matrix of its line digraph. Then $\text{rank}_F A_{L(G)} = n$ for every field F .*

Proof. Let $H, T \in F^{n \times m}$ be the head and tail incidence matrices, $H_{v,e} = [\text{head}(e) = v]$ and $T_{v,f} = [\text{tail}(f) = v]$. For arcs e, f ,

$$(H^\top T)_{e,f} = \sum_v [\text{head}(e) = v][\text{tail}(f) = v] = [\text{head}(e) = \text{tail}(f)] = (A_{L(G)})_{e,f},$$

so $A_{L(G)} = H^\top T$. Every arc has exactly one head, so the rows of H have pairwise disjoint supports; each is non-zero by hypothesis; hence they are linearly independent over any field and $\text{rank } H = n$, and likewise $\text{rank } T = n$. Thus $T: F^m \rightarrow F^n$ is surjective and $H^\top: F^n \rightarrow F^m$ injective, so $\text{rank}(H^\top T) = \dim H^\top(F^n) = n$. $\square$

Corollary 14.1.2 (The p -rank of the Jacobian). *For $k \geq 1$ and every prime $p \mid q$, the p -rank of $\text{Jac}(Y_{k+1})$ equals r_k .*

Proof. $Y_{k+1} = L(Y_k)$ is q -regular, so $\Delta_{k+1} \equiv -A_{k+1} \pmod{p}$ and $\text{rank}_{\mathbb{F}_p} \Delta_{k+1} = n_k$ by Lemma 14.1.1. Tensoring $\text{coker } \Delta_{k+1}^\top \cong \mathbb{Z} \oplus \text{Jac}(Y_{k+1})$ with $\mathbb{F}_p$ gives

$$n_{k+1} - \text{rank}_{\mathbb{F}_p} \Delta_{k+1} = 1 + \dim_{\mathbb{F}_p} (\text{Jac}(Y_{k+1}) \otimes \mathbb{F}_p),$$

and the left side is $qn_k - n_k$. As $\text{Jac}(Y_{k+1}) \otimes \mathbb{F}_p \cong \text{Jac}(Y_{k+1})[p]$ for a finite abelian group, the p -rank is $n_k(q-1) - 1 = r_k$. $\square$

14.2 The tail norm at composite q

Theorem 14.2.1 (The kernel of the tail norm, for every q). *Let $q \geq 2$ be arbitrary and $k \geq 1$. Then*

$$\ker(\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[q] \cong (\mathbb{Z}/q)^{r_k},$$

and moreover every invariant factor of $\text{Jac}(Y_{k+1})$ divisible by a prime $p \mid q$ is divisible by $p^{v_p(q)}$.

Proof. Write $e_p = v_p(q)$ and let the p -primary part of $\text{Jac}(Y_{k+1})$ be $\bigoplus_{j=1}^{s_p} \mathbb{Z}/p^{a_{p,j}}$ with $a_{p,j} \geq 1$; by Corollary 14.1.2, $s_p = r_k$ for every $p \mid q$. By (ii) and (iii) of Section 14.1,

$$q^{r_k} = |\ker \tau_*| \leq |\text{Jac}(Y_{k+1})[q]| = \prod_{p \mid q} p^{\sum_j \min(e_p, a_{p,j})} \leq \prod_{p \mid q} p^{e_p r_k} = q^{r_k}.$$

Both inequalities are therefore equalities. The first forces $\ker \tau_* = \text{Jac}(Y_{k+1})[q]$, two finite groups of the same order one contained in the other. The second forces $\min(e_p, a_{p,j}) = e_p$, that is $a_{p,j} \geq e_p$, for every $p \mid q$ and every j ; the p -primary part of $\text{Jac}(Y_{k+1})[q]$ is then $(\mathbb{Z}/p^{e_p})^{r_k}$, and taking the product over $p \mid q$ gives $(\mathbb{Z}/q)^{r_k}$. $\square$

Remark 14.2.2 (What was and was not primality). For $q = p$ this is [Ch. 12, Thm. 2.8]. The present proof differs from that one only in Corollary 14.1.2, and the corollary differs only in proving the rank identity by a factorisation valid in every characteristic rather than by an argument in characteristic p . Nothing else in the chain ever meets a zero divisor of $\mathbb{Z}/q\mathbb{Z}$: the counting is carried out one prime at a time, and q enters only through its valuations.

Remark 14.2.3 (Prime powers behave differently from primes). For q squarefree, $\text{Jac}(Y_{k+1})[q] = \bigoplus_{p|q} \text{Jac}(Y_{k+1})[p]$ and Theorem 14.2.1 says only that the kernel of the tail norm is the torsion killed by each prime factor — the prime picture, assembled by the Chinese remainder theorem. For q not squarefree the theorem says something with no prime analogue. At $q = 4$ it asserts that $\text{Jac}(Y_{k+1})$ has *no element of order exactly 2*: every invariant factor divisible by 2 is divisible by 4. Consequently $\text{Jac}(Y_{k+1})[2] \subsetneq \text{Jac}(Y_{k+1})[4] = \ker \tau_*$, of index 2^{r_k} , and the kernel of the tail norm strictly contains the p -torsion for each $p \mid q$. This is a falsifiable statement about a concrete integer matrix; check (E) tests it on three $q = 4$ towers, including level three of the K_6 tower, where $\text{Jac}(Y_3)$ has 2-rank 359 and the smallest 2-adic valuation among its invariant factors is 2.

Corollary 14.2.4 (No q -torsion in the pro-module). $\mathfrak{X} := \varprojlim (\text{Jac}(Y_k), \tau_*)$ satisfies $\mathfrak{X}[q] = 0$.

Proof. Let $x = (x_k)_{k \geq 1} \in \mathfrak{X}$ with $qx = 0$. Then $x_k \in \text{Jac}(Y_k)[q]$ for every k , and for $k \geq 2$ that group is $\ker(\tau_*: \text{Jac}(Y_k) \rightarrow \text{Jac}(Y_{k-1}))$ by Theorem 14.2.1; hence $x_{k-1} = \tau_* x_k = 0$ for every $k \geq 2$, i.e. $x = 0$. $\square$

14.3 The splitting, for every q

Theorem 14.3.1 (Splitting of the limit K -module). *For every $q \geq 2$ the short exact sequence of [Ch. 12, Thm. 3.3(iii)],*

$$0 \longrightarrow \mathfrak{X}^\vee \longrightarrow K_0(\mathcal{O}_\infty) \longrightarrow \mathbb{Z}[1/q] \longrightarrow 0,$$

splits, and $K_0(\mathcal{O}_\infty) \cong \mathbb{Z}[1/q] \oplus \mathfrak{X}^\vee$.

Proof. The extension class lies in $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}^\vee)$. Write $\mathbb{Z}[1/q] = \varinjlim (\mathbb{Z} \xrightarrow{q} \mathbb{Z} \xrightarrow{q} \cdots)$. For any abelian group B the Milnor sequence of this colimit reads

$$0 \rightarrow \varprojlim^1 \text{Hom}(\mathbb{Z}, B) \rightarrow \text{Ext}^1(\mathbb{Z}[1/q], B) \rightarrow \varprojlim \text{Ext}^1(\mathbb{Z}, B) \rightarrow 0,$$

and $\text{Ext}^1(\mathbb{Z}, B) = 0$, so $\text{Ext}^1(\mathbb{Z}[1/q], B) \cong \varprojlim^1 (B \xleftarrow{q} B \xleftarrow{q} \cdots)$. If B is q -divisible every transition map is surjective and this $\varprojlim^1$ vanishes. Now $\mathfrak{X}$ is profinite and multiplication by q on it is injective by Corollary 14.2.4; Pontryagin duality is exact, so multiplication by q on $\mathfrak{X}^\vee$ is surjective, i.e. $\mathfrak{X}^\vee$ is q -divisible. Hence $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}^\vee) = 0$. $\square$

Remark 14.3.2 (A shorter route to the prime case). For $q = p$, [Ch. 12, Thm. 3.3] obtained the splitting by first identifying $\mathfrak{X}^{(p)} \cong \mathbb{Z}_p^{\mathbb{N}}$ and then observing that its dual $(\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})}$ is divisible. The proof above needs neither identification: it uses only that multiplication by q is injective on $\mathfrak{X}$, which is Corollary 14.2.4. The structure of $\mathfrak{X}$ is not an input to the splitting, and for composite q we do not determine it.

14.4 Verification

The script `composite_tail_norm.py` runs the six checks of Table 14.1 in about five seconds; its output is archived as `composite_tail_norm_output.txt`. Eight towers are used: base graphs K_4 and Petersen ($q = 2$), K_5 and $K_{4,4}$ ($q = 3$), K_6 and $K_{5,5}$ ($q = 4$), K_7 ($q = 5$) and K_8 ($q = 6$), carried to level three where the size permits, the largest being Y_3 of the K_6 tower on 480 vertices. All arithmetic is exact: Bareiss determinants for the orders, and p -adic elimination (no coefficient explosion) for the valuations of the invariant factors.

Check (E) is the one that could have failed. For each $q = 4$ tower it reports the multiset of 2-adic valuations of the invariant factors of $\text{Jac}(Y_k)$: on K_6 at level two these are $2^{84}, 4, 5^4$ and at level three

key	check	mode
(I)	the degeneracy identities of §14.1(i) hold verbatim at composite q	exact
(N)	$\kappa(Y_{k+1}) = q^{r_k} \kappa(Y_k)$, so $ \ker \tau_* = q^{r_k}$	exact
(R)	Lemma 14.1.1: $\text{rank } A_{k+1} = n_k$ over $\mathbb{F}_p$ for $p \mid q$ and $p \nmid q$ alike	exact
(P)	Corollary 14.1.2: the p -rank of $\text{Jac}(Y_{k+1})$ is r_k for $p \mid q$	exact
(E)	Theorem 14.2.1: every invariant factor divisible by p is divisible by $p^{v_p(q)}$	exact
(T)	the published prime- q values of [Ch. 12] are recovered	exact

Table 14.1: The verification battery, 6/6.

the 359 valuations are all at least 2; on $K_{5,5}$ at level two the 149 valuations are all at least 2. A single invariant factor of valuation 1 anywhere would have contradicted Theorem 14.2.1. Check (T) reproduces $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$ and $\text{Jac}(Y_3) = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^8 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/32 \oplus \mathbb{Z}/96$ of [Ch. 12, Table 1], with $\text{ord}_2 \kappa = 18$ and 41.

14.5 Status, scope, corrections

Proved. Lemma 14.1.1, Corollary 14.1.2, Theorem 14.2.1, Corollary 14.2.4 and Theorem 14.3.1, for arbitrary $q \geq 2$ and $k \geq 1$. Together they close [Ch. 12, Rem. 5.1(b)] in both of its parts.

Quoted, for every q and used as stated. The degeneracy calculus [Ch. 9, Prop. 1.2]; the tail norm and its kernel order [Ch. 9, Thm. 3.1], resting on the line-digraph recursion [Ch. 5, Thm. 2.1]; the containment $\ker \tau_* \subseteq \text{Jac}[q]$ [Ch. 12, Prop. 2.7]; the exact sequence [Ch. 12, Thm. 3.3(iii)]; the Milnor $\varprojlim^1$ sequence and the exactness of Pontryagin duality, both textbook.

Remark 14.5.1 (Solved, open, impossible). *Solved:* the composite- q half of [Ch. 12, Rem. 5.1(b)], in the strong form $\ker \tau_* \cong (\mathbb{Z}/q)^{r_k}$, together with the splitting for every q . *Open:* (a) the structure of $\mathfrak{X}$ itself for composite q — the splitting no longer needs it, and we have not determined whether $\mathfrak{X}^{(p)}$ is again a free $\mathbb{Z}_p$ -module when $p^2 \mid q$; (b) whether the divisibility statement of Theorem 14.2.1 — no invariant factor of valuation below $v_p(q)$ — has a direct combinatorial proof, independent of the counting, and what it says about the spanning-tree lattice of a line digraph; (c) the remaining items of [Ch. 12, Rem. 5.1], which this note does not touch. *Impossible:* nothing new is shown impossible here. In particular the obstruction the task specification anticipated — that zero divisors in $\mathbb{Z}/q\mathbb{Z}$ would make the kernel smaller than $\text{Jac}[q]$ and force an order formula in terms of $\gcd(q, \cdot)$ — does not exist.

Remark 14.5.2 (Corrections to the task specification). Four, all in the same direction: the composite case is easier than expected, not harder. (1) The specification asked for the Smith normal form of A_{k+1} over $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$ and for a fallback order formula in terms of $\gcd(q, \text{Jac})$ should the kernel fail to be $\text{Jac}[q]$. Neither is needed: the kernel *is* $\text{Jac}[q]$, and the only rank computation required is over a field, where Lemma 14.1.1 makes it characteristic-free. (2) “Modulo q localisation has zero divisors” is true and irrelevant; no step localises at q . (3) The specification proposed expanding $\mathbb{Z}[1/q]$ as a direct limit of localisations over the primes $p \mid q$; the single tower $\mathbb{Z} \xrightarrow{q} \mathbb{Z} \xrightarrow{q} \cdots$ is what gives the $\varprojlim^1$ description, and splitting into primes is unnecessary. (4) The specification asked for the divisible subgroup structure of $\mathfrak{X}^\vee$. What is needed and true is only that $\mathfrak{X}^\vee$ is q -divisible, dual to $\mathfrak{X}[q] = 0$; no finer structure enters, and Remark 14.3.2 notes that this also shortens the published prime-case proof.

Provenance of this chapter

Unrefereed preprint. Every numbered statement is proved in full above from the three quoted facts of Section 14.1, each of which is stated in the earlier papers for all q ; the computations verify those inputs independently (checks (I) and (N)), the new lemma (check (R)), and the falsifiable consequence of the theorem (check (E)). The divisibility statement of Theorem 14.2.1 was found by the counting and then tested, not the other way round; the $q = 4$ towers were computed specifically because they could refute it.

Chapter 15

Algorithmic non-commutative class field theory, XIX: the odd cyclic class that does not exist, and what the exceptional zero sees instead

Summary. Paper XIV showed that the finite-dimensional commutative spectral triple of Paper II has $HC^1 = 0$, so no odd cyclic cocycle there can pair to the graph $\mathcal{L}$ -invariant $-\|h_\alpha\|^2$, and asked whether an infinite-dimensional non-commutative algebra would help. Three answers, in increasing order of finality. First, the algebra usually proposed, $\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$, is Morita-trivial: the $\mathbb{Z}$ -cover has vertex set $V \times \mathbb{Z}$ with the deck group acting freely on the second factor, so $\mathcal{A}_\alpha \cong \bigoplus_{v \in V} \mathcal{K}$, with $K_1 = 0$ and no odd K -theory to pair with at all. Second, the algebra where Bloch theory does live is the $\mathbb{Z}$ -equivariant one, $M_n(C(S^1))$, with $K_1 = \mathbb{Z}$; but its natural class, the voltage pencil $D(t) = (q+1)I - A(t)$, is *not invertible*, because $D(1)$ is the ordinary Laplacian. On a circle where the pencil is invertible the odd pairing is the winding number of $\det D$, and that number is the *order* of the exceptional zero — equal to 2 for every voltage class — whereas $\kappa_0 \mathcal{L}$ is its leading *coefficient*, $\frac{1}{2}\Theta''(0) = -\kappa_0 \|h_\alpha\|^2$. Index theory sees orders, not coefficients. Third, and this is a theorem rather than a diagnosis: an index pairing is additive in each slot, so if $\alpha \mapsto \tau_\alpha$ and $\alpha \mapsto [u_\alpha]$ are both linear and K_1 has rank one — as it does for $M_n(C(S^1))$ and for the shadow algebra alike — then $\langle \tau_\alpha, [u_\alpha] \rangle$ is a product of two linear forms in α . It cannot be $\kappa_0 \|h_\alpha\|^2$, which is positive definite on $H^1(Y; \mathbb{R})$, as soon as $b_1(Y) \geq 2$. So the impossibility of Paper XIV is not an artefact of finite dimensions. Five checks, on the K_4 and K_5 voltage classes.

15.1 The proposed algebra is the compacts

Let Y be a finite connected $(q+1)$ -regular graph on n vertices, α an integer 1-cochain with $[\alpha] \neq 0$, and $\tilde{Y}_\alpha$ the associated $\mathbb{Z}$ -cover, with deck group $\mathbb{Z}$. Write κ_0 for the number of spanning trees of Y , h_α for the harmonic representative of $[\alpha]$, and $\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2$ [Ch. 8, Def. 1.3].

Proposition 15.1.1. *The deck group acts freely on the vertex set of $\tilde{Y}_\alpha$ with exactly n orbits, and*

$$\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z} \cong \bigoplus_{v \in V} \mathcal{K}(\ell^2(\mathbb{Z})).$$

Consequently $K_0(\mathcal{A}_\alpha) \cong \mathbb{Z}^n$, $K_1(\mathcal{A}_\alpha) = 0$, and every odd cyclic cocycle pairs trivially.

Proof. A connected $\mathbb{Z}$ -cover of a connected graph has vertex set $V \times \mathbb{Z}$ with the deck group translating the second coordinate; the action is free and the orbits are the fibres, n of them (check (A)). Hence $C_0(\tilde{Y}_\alpha) \cong c_0(V) \otimes c_0(\mathbb{Z})$ equivariantly, with $\mathbb{Z}$ acting only on the second factor, and $c_0(V) \otimes (c_0(\mathbb{Z}) \rtimes \mathbb{Z}) \cong c_0(V) \otimes \mathcal{K}(\ell^2(\mathbb{Z}))$, the crossed product of c_0 of a free transitive action being the compacts. The K -theory and the vanishing of HC^{odd} for a direct sum of copies of $\mathcal{K}$ are standard. $\square$

So the premise that $\mathcal{A}_\alpha$ “has nontrivial K_1 and HC^1 ” is false: it is stably $\mathbb{C}^n$, and its only cyclic cohomology is the even part, spanned by the n traces. The non-commutativity of a crossed product by a *free* action is illusory.

15.2 The Floquet algebra, and what its pairing sees

The algebra that does carry Bloch theory is the commutant, not the crossed product. After Floquet transform the $\mathbb{Z}$ -equivariant bounded operators on $\ell^2(\tilde{Y}_\alpha)$ are $M_n(C(S^1))$, whose K_1 is $\mathbb{Z}$, generated by the winding of a determinant. The natural element of it is the voltage pencil $D(t) = (q+1)I - A(t)$ of [Ch. 6, Lem. 1.1], whose Floquet determinant is Θ .

Proposition 15.2.1. *D is not invertible in $M_n(C(S^1))$: $\det D(1) = 0$, since $D(1)$ is the ordinary Laplacian of Y . The zero has order exactly two, and $\frac{1}{2}\Theta''(0) = -\kappa_0\|h_\alpha\|^2$ [Ch. 7, Thm. 1.2]. Hence the class $[D] \in K_1$ does not exist.*

Corollary 15.2.2 (What the pairing does see). *Let ε be small enough that the circle $|t-1| = \varepsilon$ meets no other zero of $\det D$. There the pencil is invertible and defines a class in K_1 , and its pairing with the generator of $HC^1(C^\infty(S^1))$ is the winding number of $\det D$. That number is the order of the exceptional zero, namely 2, for every voltage class; it does not depend on α , and so cannot be $\kappa_0\|h_\alpha\|^2$.*

Check (W) computes this winding number for the three K_4 classes and two K_5 classes and finds 2 in all five, against values of $\kappa_0\|h_\alpha\|^2$ equal to 8, 16, 24, 75 and 175 (Table 15.1). The contrast is the whole point: an index pairing is a topological degree, and a degree reads the *order* of vanishing; $\mathcal{L}$ is the *coefficient* at the same point.

	b_1	κ_0	$\ h_\alpha\ ^2$	$\kappa_0\ h_\alpha\ ^2$	winding of $\det D$
$K_4, \alpha = (1, 0, 0)$	3	16	1/2	8	2
$K_4, \alpha = (1, 1, 1)$	3	16	1	16	2
$K_4, \alpha = (1, 2, 0)$	3	16	3/2	24	2
K_5 , single edge	6	125	3/5	75	2
K_5 , triangle	6	125	7/5	175	2

Table 15.1: The cleared pencil has $[T^0] = [T^1] = 0$ and $[T^2] = -\kappa_0\|h_\alpha\|^2$ in every case (check (P)); the winding number is 2 in every case (check (W)).

Remark 15.2.3 (Where the integrality comes from). The task specification asked why the index should “absorb” $\kappa_0 = \det G$. It does not, and no index is involved: by [Ch. 8, Thm. 7.1], $\|h_\alpha\|^2 = a_c^\top G^{-1} a_c$ with G the cycle Gram matrix and $a_c = C^\top \alpha$ integral, so

$$\kappa_0\|h_\alpha\|^2 = a_c^\top (\det G \cdot G^{-1}) a_c = a_c^\top \text{adj}(G) a_c \in \mathbb{Z},$$

the adjugate of an integer matrix being an integer matrix. Integrality is Cramer’s rule, not an index theorem (check (I)).

15.3 The obstruction

Theorem 15.3.1 (No linear construction, in any algebra with K_1 of rank one). *Let A be a C^* -algebra with $\text{rank}_{\mathbb{Z}} K_1(A) = 1$. Suppose given group homomorphisms*

$$H^1(Y; \mathbb{Z}) \rightarrow HC^1(A), \quad \alpha \mapsto \tau_\alpha, \quad H^1(Y; \mathbb{Z}) \rightarrow K_1(A), \quad \alpha \mapsto [u_\alpha].$$

If $b_1(Y) \geq 2$ then $\langle \tau_\alpha, [u_\alpha] \rangle \neq \kappa_0 \|h_\alpha\|^2$ for some α .

Proof. Torsion classes pair trivially with a $\mathbb{C}$ -valued cocycle, so after tensoring with $\mathbb{Q}$ we may write $[u_\alpha] = \ell(\alpha)g$ for a fixed generator g of $K_1(A) \otimes \mathbb{Q} \cong \mathbb{Q}$ and a linear form ℓ on $H^1(Y; \mathbb{Q})$. Additivity of the pairing in the second slot gives $\langle \tau_\alpha, [u_\alpha] \rangle = \ell(\alpha) \mu(\alpha)$ with $\mu(\alpha) = \langle \tau_\alpha, g \rangle$, linear by additivity in the first slot. So the pairing is a product of two linear forms. If $b_1 \geq 2$ then $\ker \ell \neq 0$, so the product vanishes at some α with $[\alpha] \neq 0$; but $\kappa_0 \|h_\alpha\|^2 > 0$ there, since $h_\alpha = 0$ only for $[\alpha] = 0$. $\square$

Corollary 15.3.2. *The theorem applies to $M_n(C(S^1))$, where $K_1 \cong \mathbb{Z}$, and to the shadow algebra $\mathcal{O}_{B(Y)}$ of [Ch. 4], where $K_1 \cong \mathbb{Z}$ by [Ch. 4, Thm. 2.1]. Both candidate algebras of the task specification are therefore excluded, for every base graph with $b_1 \geq 2$ — in particular for K_4 ($b_1 = 3$) and K_5 ($b_1 = 6$), the graphs on which the $\mathcal{L}$ -invariant was computed.*

Check (Q) verifies the hypothesis that makes the argument bite: $\kappa_0 \|h_\alpha\|^2$ is the quadratic form $\text{adj}(G)$, which is positive definite of rank exactly b_1 — 3 for K_4 and 6 for K_5 — so it is not a product of two linear forms.

Remark 15.3.3 (What would be needed). Theorem 15.3.1 is not a proof that no odd cyclic class anywhere computes $\mathcal{L}$; it rules out the linear constructions in rank-one K_1 . Two escapes remain, and we state them as the honest residue of [Ch. 8, Rem. 9.1(a)]. *Either* one finds an algebra whose K_1 has rank at least b_1 , with a natural $H^1(Y; \mathbb{Z}) \rightarrow K_1$ — the pairing may then be a genuine quadratic form; *or* one abandons linearity, in which case the assignment $\alpha \mapsto \tau_\alpha$ is not a cohomological construction but part of the input, and the exercise is circular.

The first escape is a real question and it has an obvious candidate, which we record because the rank matches exactly. The fundamental group of a connected graph is free of rank b_1 , and by Pimsner–Voiculescu

$$K_0(C_r^*(F_m)) \cong \mathbb{Z}, \quad K_1(C_r^*(F_m)) \cong \mathbb{Z}^m,$$

so $C_r^*(\pi_1(Y)) = C_r^*(F_{b_1})$ has rank $K_1 = b_1$ — precisely the threshold at which Theorem 15.3.1 stops applying, and the smallest rank at which a positive definite form on $H^1(Y; \mathbb{R})$ could be a pairing at all. Whether the natural map $H^1(Y; \mathbb{Z}) \cong \mathbb{Z}^{b_1} \rightarrow K_1(C_r^*(F_{b_1})) \cong \mathbb{Z}^{b_1}$ and a family τ_α can be chosen so that the pairing is $\kappa_0 \|h_\alpha\|^2$ we do not know; we note only that $\kappa_0 \|h_\alpha\|^2 = a_c^\top \text{adj}(G) a_c$ is already a quadratic form on that lattice, so the shape is right and only the construction is missing.

15.4 Verification

key	check	mode
(A)	Proposition 15.1.1: free deck action, n orbits	exact
(P)	the cleared pencil: $[T^0] = [T^1] = 0$, $[T^2] = -\kappa_0 \ h_\alpha\ ^2$	exact
(W)	Corollary 15.2.2: the winding number is 2 for every α	numeric
(I)	Remark 15.2.3: $\kappa_0 \ h_\alpha\ ^2 = a_c^\top \text{adj}(G) a_c \in \mathbb{Z}$	exact
(Q)	$\text{adj}(G)$ is positive definite of rank $b_1 \geq 2$	exact

Table 15.2: The verification battery (`cyclic_obstruction.py`), 5/5, about one second.

15.5 Status, scope, corrections

Proved. Proposition 15.1.1, Proposition 15.2.1 (given [Ch. 7, Thm. 1.2]), Corollary 15.2.2, Theorem 15.3.1 and Corollary 15.3.2.

Not proved. That no odd cyclic class in any algebra computes $\mathcal{L}$. What is proved is that the two algebras named, and every algebra with K_1 of rank one, fail for a structural reason, and that the natural class in the Floquet algebra fails for a different and more interesting one — it does not exist, and the object obstructing it is the exceptional zero whose coefficient is the invariant sought.

Remark 15.5.1 (Solved, open, impossible). *Solved:* the two candidate algebras of [Ch. 8, Rem. 9.1(a)] are ruled out, the first because it is Morita-trivial and the second by Theorem 15.3.1. *Open:* an algebra with K_1 of rank $\geq b_1$ and a natural map from $H^1(Y; \mathbb{Z})$ (Remark 15.3.3); and, unchanged from [Ch. 8], the question whether $\mathcal{L}$ has any index-theoretic description at all, as opposed to the spectral-action and Albanese descriptions it is already known to have [Ch. 8, Cor. 7.2]. *Impossible:* an odd pairing linear in α against a rank-one K_1 , whenever $b_1 \geq 2$. Together with [Ch. 8, Prop. 2.1] this closes the naive routes in both the finite-dimensional and the infinite-dimensional directions.

Remark 15.5.2 (Corrections to the task specification). Three. (1) “ $\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$ has nontrivial K_1 and HC^1 ” is false; the action is free, so the crossed product is a sum of copies of the compacts (Proposition 15.1.1). The specification’s own hint is the obstruction to following it. (2) The proposed mechanism — pair a cocycle built from the integral period vector a_c with the unit shift $u \in \mathbb{Z}$ and read off $\|h_\alpha\|^2$ from a heat-kernel limit — is linear in α in the K_1 slot and cannot produce a positive definite quadratic form (Theorem 15.3.1); this is the same objection [Ch. 8, Prop. 2.1(iii)] raised against the finite-dimensional proposal, and it survives the passage to infinite dimensions unchanged. (3) There is nothing for an index to “absorb”: $\kappa_0 \|h_\alpha\|^2 = a_c^\top \text{adj}(G) a_c$ is an integer by Cramer’s rule (Remark 15.2.3), and no index theorem is needed to see it.

Provenance of this chapter

Unrefereed preprint. Every numbered statement is proved above. The computations verify the exceptional-zero coefficient and the winding number independently on five voltage classes, and confirm the positive definiteness that makes Theorem 15.3.1 bite. The winding number was computed before the theorem was formulated, in order to find out what the available pairing actually measures.

Part IV

Buildings: rank two, and rank d

Chapter 16

Algorithmic non-commutative class field theory, X: $\tilde{A}_2$ complexes, higher simplicial Laplacians, and the fate of the trace obstruction in rank two

Summary. The series has so far lived on quotients of the Bruhat–Tits tree of PGL_2 . This paper moves one rank up, to finite $\tilde{A}_2$ complexes — quotients of two-dimensional buildings of order q , locally the building of PGL_3 over a local field with residue field $\mathbb{F}_q$ — and asks which of the rank-one theorems survive. Three things are established. First, the Hodge Laplacians $\Delta_0, \Delta_1, \Delta_2$ of the typed chain complex give higher critical groups $\mathrm{Jac}_i(Y) = \mathrm{Tor} \, \mathrm{coker} \, \Delta_i$, computed exactly on thin tori and on five thick complexes of order 2 with 21, 24, 42, 84 and 168 vertices, built here from a triangle presentation over $\mathrm{PG}(2, 2)$ compatible with a Singer cycle and verified to have Heawood-graph links; the Hodge vertex Laplacian is no longer a Hecke operator, and the arithmetic bridge of [Ch. 2, Thm. 3.5] is carried instead by the *Hecke–Laplacian* $\Delta^H = (q^3 - 1)I + A_1 - qA_2$, the value at $u = 1$ of the cubic pencil $P(u) = \det(I - uA_1 + qu^2A_2 - q^3u^3I)$, whose critical group has order $\frac{1}{n} \prod_j |P_j(1)|$, a product of inverse local GL_3 L -values. Second, the companion $C^{(3)}$ of the cubic pencil is unimodularly equivalent to $\Delta^H \oplus I \oplus I$ exactly as in [Ch. 3], but the trace obstruction of [Ch. 3] does *not* transfer: the $\mathbb{Z}/3$ type grading forces $\mathrm{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$, and a block computation gives $\mathrm{Tr}(C^{(3)})^3 = n(q-1)^2(q+1) \geq 0$. There is no cubic trace obstruction; on the five thick complexes, all of which turn out to be Ramanujan, every net trace of the reduced (type-preserving) pencil is non-negative and its Perron condition holds, so by the theorem of Kim–Ormes–Roush the nonzero spectrum of $C^{(3)}$ is that of a non-negative integer matrix, irreducible of period three — the rank-two pencil is spectrally realizable, the opposite of the rank-one situation, and the realization problem for the PGL_3 Hecke critical group is open and unobstructed. Third, the shadow construction of [Ch. 4] extends to arbitrary integer matrices with positive diagonal by a signed-shadow lemma in which positive off-diagonal entries are produced by one-directional auxiliary vertices with double loops; it yields unital Kirchberg algebras $\mathcal{O}(Y, \Delta_1)$ and $\mathcal{O}(Y, \Delta^H)$ with $K_0 \cong \mathbb{Z}^{b_1} \oplus \mathrm{Jac}_1(Y)$ and $K_0 \cong \mathbb{Z} \oplus \mathrm{Jac}^H(Y)$ respectively, functorially in the complex. What the specification predicted and what is proved differ in two places, and the paper says where.

Status of the results

Every numbered statement is proved. The inputs are finite linear algebra, the classical Cuntz–Krieger facts quoted in [Chs. 4, 12], the Cartwright–Mantero–Steger–Zappa theorem that a triangle presentation gives a group acting simply transitively on the vertices of an $\tilde{A}_2$ building [52], and, for one corollary that nothing else depends on, the Kim–Ormes–Roush theorem on the nonzero spectra of primitive integer matrices [55]. The Kang–Li zeta function of PGL_3 complexes [54] is context for the cubic pencil, not an input. Every finite identity was machine-verified exactly (Table 16.1); the five thick complexes are constructed by the script, and their local structure (Heawood links, regularity, flag condition) is checked rather than assumed — with the result that the flag condition holds only for the largest of them, a distinction the theorems respect. Section 16.7 records what is open and the corrections to the specification.

16.1 $\tilde{A}_2$ complexes of order q

Definition 16.1.1 ($\tilde{A}_2$ complex). An $\tilde{A}_2$ complex of order $q \geq 1$ is a finite connected pure two-dimensional simplicial complex Y with vertex set V ($n := |V|$), edge set E and triangle set F , together with a type function $\tau: V \rightarrow \mathbb{Z}/3$, such that

- (i) every triangle has one vertex of each type;
- (ii) every vertex has exactly $q^2 + q + 1$ neighbours of type $\tau(v) + 1$ and $q^2 + q + 1$ of type $\tau(v) + 2$;
- (iii) every edge lies in exactly $q + 1$ triangles;
- (iv) (flag condition) every closed walk $x \rightarrow y \rightarrow z \rightarrow x$ with $\tau(y) = \tau(x) + 1$, $\tau(z) = \tau(x) + 2$ is the boundary of a triangle.

Every edge is oriented from type t to type $t + 1$ and every triangle (a, b, c) is listed with types $(t, t + 1, t + 2)$, so that $\partial_2(a, b, c) = (a, b) + (b, c) + (c, a)$ and $\partial_1(a, b) = b - a$ are canonical; $A_1 \in M_n(\{0, 1\})$ is the adjacency $x \rightarrow y$ for edges, $A_2 := A_1^t$, and we assume $A_1 A_2 = A_2 A_1$ (true for building quotients, where A_1, A_2 generate the commutative spherical Hecke algebra; verified on every complex used).

Quotients of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a type-rotating, freely acting group with injectivity radius ≥ 2 are $\tilde{A}_2$ complexes in this sense, with vertex links the incidence graph of $\mathrm{PG}(2, q)$; the flag condition holds because buildings are flag complexes. Thin complexes ($q = 1$) are triangulated flat tori (Construction 16.1.3).

Lemma 16.1.2 (Counts). $|E| = n(q^2 + q + 1)$, $|F| = \frac{1}{3}n(q^2 + q + 1)(q + 1)$, $\mathrm{Tr} A_1^3 = 3|F|$, $\mathrm{Tr}(A_1 A_2) = |E|$, and $\chi(Y) = n(1 + \frac{1}{3}(q^2 + q + 1)(q - 2))$. For $q = 2$: $|E| = |F| = 7n$ and $\chi(Y) = n$.

Proof. (ii) gives $|E|$; each vertex lies in $(q^2 + q + 1)(q + 1)$ triangles by (ii), (iii), each counted three times; $\mathrm{Tr} A_1^3$ counts closed type-increasing 3-walks, three per triangle by (iv); $\mathrm{Tr}(A_1 A_1^t) = \sum_{x,y} A_1(x, y)^2 = |E|$. $\square$

Construction 16.1.3 (Thin tori). On $\mathbb{Z}^2$ let $d_1 = (1, 0)$, $d_2 = (0, 1)$, $d_3 = (-1, -1)$, edges $v \rightarrow v + d_i$, type $\tau(a, b) = a + b \bmod 3$, triangles $\{v, v + d_i, v + d_i + d_j\}$, $i \neq j$. This is the $\tilde{A}_2$ Coxeter complex (one apartment); its quotient by $m\mathbb{Z}^2$, $3 \mid m$, $m \geq 3$, is an $\tilde{A}_2$ complex of order 1 with m^2 vertices, $b_1 = 2$, $b_2 = 1$, hexagonal links.

Construction 16.1.4 (Thick complexes of order 2). Label the points of $\mathrm{PG}(2, 2)$ by $\mathbb{Z}/7$ with lines $\{i, i + 1, i + 3\}$ and let $\lambda(x) = \{x + 1, x + 2, x + 4\}$, the Singer cycle. A triangle presentation compatible with λ [52] is a set T of 21 triples with (A) $(x, y, z) \in T \Rightarrow (y, z, x) \in T$, (B) $(x, y, \cdot) \in T$ iff $y \in \lambda(x)$, (C) the third entry is determined by the first two. Exhaustive search over all bijections without absolute points finds, for λ the Singer cycle, the presentation with cyclic orbits

$$T/\mathbb{Z}_3 = \{(0, 1, 3), (0, 2, 6), (0, 4, 5), (1, 2, 4), (1, 5, 6), (2, 3, 5), (3, 4, 6)\}$$

(axioms re-verified by the script). By [52] the group $\Gamma = \langle a_0, \dots, a_6 \mid a_x a_y a_z = 1, (x, y, z) \in T \rangle$ acts simply transitively and type-rotatingly on the vertices of an $\tilde{A}_2$ building of order 2; its type-preserving subgroup Γ^0 has index 3, and $X_3 := \Gamma^0 \backslash B$ is the Δ -complex with three vertices, 21 edges and 21 triangles. We compute $H_1(X_3; \mathbb{Z}) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14$ and build the abelian voltage covers of X_3 with deck group A for the quotients $A = \mathbb{Z}/7, (\mathbb{Z}/2)^3, \mathbb{Z}/14$ and $H_1(X_3)$ itself, exactly as the derived graphs of [Ch. 3, §5] in one dimension higher: vertices (i, g) , edges lifted by voltages in A , triangles lifting because their boundaries are null-homologous. The results $Y_{21}, Y_{24}, Y_{42}, Y_{84}, Y_{168}$ are simplicial, every vertex link is the Heawood graph (bipartite, 3-regular, 14 vertices, girth 6), and conditions (i)–(iii) hold, with $|E| = |F| = 7n$ (Table 16.1, line (P)). The flag condition (iv) holds for Y_{168} and fails for the four smaller covers: there the deck group is too small, some product $a_x a_y a_{z'}$ with $(x, y, z') \notin T$ has trivial voltage, and $\text{Tr } A_1^3$ exceeds $3|F|$ by $28n, 21n, 12n, 4n$ respectively. So (iv) is not implied by the link condition, and the five complexes separate the two hypotheses cleanly: Y_{168} is an $\tilde{A}_2$ complex in the full sense of Definition 16.1.1, the others satisfy (i)–(iii) only. The count $168 = |\text{PGL}_3(\mathbb{F}_2)|$ for the maximal abelian cover suggests that Y_{168} is a congruence quotient; we neither need nor claim this. Whether the building is that of $\text{PGL}_3(\mathbb{F}_2((t)))$, of $\text{PGL}_3(\mathbb{Q}_2)$, or exotic is not decided here: every statement below depends only on the local structure of Definition 16.1.1, and the statements that use (iv) are marked.

16.2 Hodge Laplacians and higher critical groups

Definition 16.2.1. $\Delta_0 := \partial_1 \partial_1^t$ on C_0 , $\Delta_1 := \partial_1^t \partial_1 + \partial_2 \partial_2^t$ on C_1 , $\Delta_2 := \partial_2^t \partial_2$ on C_2 , and $\text{Jac}_i(Y) := \text{Tor coker}_{\mathbb{Z}} \Delta_i$, the i -th critical group. Following Duval–Klivans–Martin [53] we also record $K(Y) := \ker \partial_1 / \text{im}(\partial_2 \partial_2^t)$.

Proposition 16.2.2 (Structure). $\partial_1 \partial_2 = 0$; $\text{coker } \Delta_i \cong \mathbb{Z}^{b_i} \oplus \text{Jac}_i(Y)$ with b_i the Betti numbers of Y ; $b_0 = 1$; $\Delta_0 = 2(q^2 + q + 1)I - (A_1 + A_2)$ is the Laplacian of the 1-skeleton, so $\text{Jac}_0(Y)$ is its sandpile group with $|\text{Jac}_0(Y)|$ the number of spanning trees of the 1-skeleton; Δ_1 has diagonal entries $q + 3$, entry 0 at pairs of edges spanning a triangle, and entries ± 1 at pairs sharing a vertex and no triangle.

Proof. $\partial_1 \partial_2(a, b, c) = (b - a) + (c - b) + (a - c) = 0$. Each Δ_i is symmetric with $\ker \Delta_i = \ker \partial_i \cap \ker \partial_{i+1}^t$ of rank b_i (Hodge), so its cokernel has free rank b_i . The diagonal of Δ_1 is $2 + (q + 1)$ by (iii); for edges $e = (a, b)$, $f = (b, c)$ in a triangle the down-term $\langle \partial e, \partial f \rangle = -1$ cancels the up-term $+1$, and without a triangle only the down-term survives. $\square$

16.3 The Hecke–Laplacian and the GL_3 L -values

Definition 16.3.1 (Cubic pencil and Hecke–Laplacian). $P(u) := I - uA_1 + qu^2A_2 - q^3u^3I \in M_n(\mathbb{Z}[u])$ and

$$\Delta^H := -P(1) = (q^3 - 1)I + A_1 - qA_2, \quad \text{Jac}^H(Y) := \text{Tor coker}_{\mathbb{Z}} \Delta^H.$$

For a joint eigenpair (a_j, b_j) of (A_1, A_2) write $P_j(u) = 1 - a_j u + q b_j u^2 - q^3 u^3 = \prod_{i=1}^3 (1 - \lambda_{j,i} u)$; the trivial pair $a = b = q^2 + q + 1$ gives $(1 - u)(1 - qu)(1 - q^2 u)$.

The pencil is the vertex factor of the Kang–Li zeta function of a PGL_3 complex [54], and $P_j(u)^{-1}$ is the local L -factor of the unramified GL_3 Hecke eigenform with Satake parameters $\lambda_{j,i}/q$; we use only the polynomial.

Theorem 16.3.2 (Hecke critical group and L -values). (i) $\Delta^H \mathbf{1} = 0$ and $\mathbf{1}^t \Delta^H = 0$.

(ii) If $P_j(1) \neq 0$ for every nontrivial j (no nontrivial eigenform has a Satake parameter $\lambda_{j,i} = 1$; automatic when $|\lambda_{j,i}| = q$), then $\text{rk } \Delta^H = n - 1$, all n principal cofactors of Δ^H are equal, and

$$\text{coker}_{\mathbb{Z}} \Delta^H \cong \mathbb{Z} \oplus \text{Jac}^H(Y), \quad |\text{Jac}^H(Y)| = \frac{1}{n} \prod_{j \text{ nontrivial}} |P_j(1)|.$$

(iii) $\Delta^H \neq \Delta_0$ for $q \geq 1$, and Δ_0 is not a specialization of the pencil: no u gives $P(u) \in \mathbb{Q}^\times \Delta_0$. For PGL_2 the two operators coincide $((q+1)I - T_p = \Delta_0 = -P(1)$ with $P(u) = I - uT_p + qu^2$).

Proof. (i) Row sums: $(q^3 - 1) + (q^2 + q + 1) - q(q^2 + q + 1) = 0$; column sums likewise since $\mathbf{1}^t A_1 = \mathbf{1}^t A_2 = (q^2 + q + 1)\mathbf{1}^t$. (ii) The eigenvalues of Δ^H on the joint eigenspaces are $-P_j(1)$, zero exactly for the trivial pair under the hypothesis, so the kernel is $\mathbb{Z}\mathbf{1}$ on both sides; the equal-cofactor argument of [Ch. 5, proof of Thm. 2.1, Step 2] applies verbatim to a matrix with two-sided kernel $\mathbf{1}$ and gives $\prod_{\mu \neq 0} \mu = n\kappa$ with κ the common cofactor = |Tor coker|. (iii) $P(u) \in \mathbb{Q}\Delta_0$ would need $-u : qu^2 = -1 : -1$, i.e. $u = -1/q$, where $P(-1/q) = 2I + (A_1 + A_2)/q$ has the wrong constant term unless $2q = 2(q^2 + q + 1)$, impossible. $\square$

Theorem 16.3.3 (Companion reduction). *Let $C^{(3)} = \begin{pmatrix} A_1 & -qA_2 & q^3I \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix}$, so that $\det(I - uC^{(3)}) = \det P(u)$.*

With $U = \begin{pmatrix} I & q^3I - qA_2 & q^3I \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix}$ and $V = \begin{pmatrix} I & 0 & 0 \\ I & I & 0 \\ I & I & I \end{pmatrix}$, both unimodular,

$$U(I - C^{(3)})V = P(1) \oplus I_n \oplus I_n = (-\Delta^H) \oplus I_n \oplus I_n, \quad \text{hence} \quad \mathrm{coker}_{\mathbb{Z}}(I - C^{(3)}) \cong \mathbb{Z} \oplus \mathrm{Jac}^H(Y).$$

Proof. $I - C^{(3)} = \begin{pmatrix} I - A_1 & qA_2 & -q^3I \\ -I & I & 0 \\ 0 & -I & I \end{pmatrix}$; the first block row of $U(I - C^{(3)})$ is $(I - A_1, qA_2, -q^3I) + (q^3I - qA_2)(-I, I, 0) + q^3(0, -I, I) = (P(1), 0, 0)$, and the column operations V clear the subdiagonal $-I$'s. This is [Ch. 3, §1.1] one degree higher. $\square$

16.4 The fate of the trace obstruction

Theorem 16.4.1 (Grading). $\mathrm{Tr}(A_1^\alpha A_2^\beta) = 0$ unless $\alpha \equiv \beta \pmod{3}$. Consequently $\mathrm{Tr}(C^{(3)})^k = 0$ for every k with $3 \nmid k$, and the spectrum of $C^{(3)}$ is invariant under multiplication by cube roots of unity.

Proof. A closed walk contributing to the trace changes type by $\alpha + 2\beta \equiv \alpha - \beta$. $\mathrm{Tr}(C^{(3)})^k = \sum_j p_k(a_j, b_j)$ with p_k the k -th power sum of the roots of $\lambda^3 - a\lambda^2 + qb\lambda - q^3$, a polynomial in monomials $a^\alpha (qb)^\beta (q^3)^\gamma$ with $\alpha + 2\beta + 3\gamma = k$, so $\alpha \equiv \beta$ forces $3 \mid k$. Equivalently $D_\omega = \mathrm{diag}(\omega^{\tau(v)}) \oplus \cdots$ conjugates $\omega C^{(3)}$ to $C^{(3)}$. $\square$

Theorem 16.4.2 (No cubic trace obstruction). *For every $\tilde{A}_2$ complex of order q ,*

$$\mathrm{Tr} C^{(3)} = 0, \quad \mathrm{Tr}(C^{(3)})^2 = 0, \quad \mathrm{Tr}(C^{(3)})^3 = 3|F| - 3q|E| + 3nq^3 = n(q-1)^2(q+1) \geq 0,$$

with equality only in the thin case $q = 1$. Hence the mechanism of [Ch. 3, Thms. 1.5–1.6] — a negative trace of a power of the companion, incompatible with any non-negative matrix of any size — has no analogue in rank two at orders ≤ 5 : the first net trace that is not forced to be non-negative by these identities is tr_6 .

Proof. $p_1 = a$, $p_2 = a^2 - 2qb$, $p_3 = a^3 - 3qab + 3q^3$ (Newton). $\mathrm{Tr} A_1 = 0$ (no loops), $\mathrm{Tr} A_1^2 = 0$ and $\mathrm{Tr} A_2 = 0$ by Theorem 16.4.1, $\mathrm{Tr} A_1^3 = 3|F|$ and $\mathrm{Tr}(A_1 A_2) = |E|$ by Lemma 16.1.2; substitute the counts: $(q^2 + q + 1)(q + 1) - 3q(q^2 + q + 1) + 3q^3 = q^3 - q^2 - q + 1 = (q - 1)^2(q + 1)$. The net trace $\mathrm{tr}_m = \sum_{d|m} \mu(m/d) \mathrm{Tr}(C^{(3)})^d$ vanishes for $m \leq 5$, $m \neq 3$, and equals $\mathrm{Tr}(C^{(3)})^3 \geq 0$ for $m = 3$. Without the flag condition the identity reads $\mathrm{Tr}(C^{(3)})^3 = \mathrm{Tr} A_1^3 - 3q|E| + 3nq^3$, still non-negative since $\mathrm{Tr} A_1^3 \geq 3|F|$: on Y_{168} the value is $504 = n(q - 1)^2(q + 1)$ exactly; on Y_{21} it is $651 = 63 + 588$, the excess being the spurious closed 3-walks (Table 16.1). $\square$

Remark 16.4.3 (Why rank one is different). In [Ch. 3] the companion of $1 - uT_p + qu^2$ has $\mathrm{Tr} C^2 = \mathrm{Tr} T_p^2 - 2qn = n(q + 1) - 2qn < 0$: the $\mathbb{Z}/2$ type structure of a graph allows a walk to return in two steps (edge reversal), so the symmetric adjacency contributes $\mathrm{Tr} T_p^2 = 2|E| > 0$ but not enough to beat the $-2qn$ of the negative block. In rank two the type grading is $\mathbb{Z}/3$: a type-increasing walk cannot return before three steps, the negative block $-qA_2$ is invisible to $\mathrm{Tr}(C^{(3)})^k$ for $k < 3$, and at $k = 3$ the chambers contribute $3|F| = n(q^2 + q + 1)(q + 1)$, which exceeds the $3q|E| - 3nq^3 = n(q^2 + q + 1) \cdot 3q - 3nq^3$ removed by the negative block. The obstruction of [Ch. 3] is a rank-one accident of parity.

Definition 16.4.4 (Reduced pencil). By Theorem 16.4.1, $\det P(u) \in \mathbb{Z}[u^3]$; write $\det P(u) = Q(u^3)$ with $Q \in \mathbb{Z}[v]$ of degree n . Its roots are λ^3 , one for each ω -orbit $\{\lambda, \omega\lambda, \omega^2\lambda\}$ of eigenvalues of $C^{(3)}$: the trivial orbits give $1, q^3, q^6$, and the power sums of the multiset Λ_Q of roots are $p_d(\Lambda_Q) = \frac{1}{3} \text{Tr}(C^{(3)})^{3d}$. Λ_Q is the spectrum of the type-preserving (three-step) dynamics.

Theorem 16.4.5 (Spectral realizability of the thick examples). *For each of $Y_{21}, Y_{24}, Y_{42}, Y_{84}, Y_{168}$:*

- (i) $C^{(3)}$ has exactly nine eigenvalues of modulus $\geq q$: the orbits of $1, q, q^2$ under ω ; every other eigenvalue has modulus exactly q (numeric, to 10^{-6}): the complexes are Ramanujan in the sense of [54]. In particular $C^{(3)}$ itself never satisfies the Perron condition of a primitive matrix — $q^2, q^2\omega, q^2\omega^2$ are all dominant — which is forced by the grading and is the signature of an irreducible matrix of period 3.
- (ii) For Λ_Q : q^6 is simple and strictly dominant; the net traces

$$\text{tr}_m(\Lambda_Q) = \sum_{d|m} \mu(m/d) \frac{1}{3} \text{Tr}(C^{(3)})^{3d}$$

are non-negative for $m \leq 10$ (exact integers) and provably positive for $m \geq m_0 \in \{2, 3\}$ (Lemma 16.4.6); hence by [55] there is a primitive non-negative integer matrix B_0 with nonzero spectrum Λ_Q .

- (iii) Consequently the block-cyclic non-negative integer matrix $\begin{pmatrix} 0 & B_0 & 0 \\ 0 & 0 & I \\ I & 0 & 0 \end{pmatrix}$ has the nonzero spectrum of $C^{(3)}$: the PGL_3 pencil of these complexes is spectrally realizable by a non-negative integer matrix, irreducible of period 3.

Proof. (i) is computation (Table 16.1, line (R)); the grading statement is Theorem 16.4.1. (ii) The Boyle–Handelman conditions for a primitive integer matrix are: Λ_Q is the root multiset of a monic integer polynomial (Q), a simple strictly dominant positive root (q^6 , by (i)), and $\text{tr}_m(\Lambda_Q) \geq 0$ for all m ; [55] proves them sufficient. (iii) If $M = \begin{pmatrix} 0 & X & 0 \\ 0 & 0 & Y \\ Z & 0 & 0 \end{pmatrix}$ then $M^3 = \text{diag}(XYZ, YZX, ZXY)$, so the nonzero spectrum of M consists of all cube roots of the nonzero spectrum of $XYZ = B_0$, i.e. of the ω -orbits of $\Lambda_Q^{1/3}$, which is the nonzero spectrum of $C^{(3)}$. $\square$

Lemma 16.4.6 (Tail bound). *Let $M < q^2$ bound the moduli of the genuinely nontrivial eigenvalues and $N = n - 1$. Then $\text{tr}_m(\Lambda_Q) \geq q^{6m} + q^{3m} + 1 - NM^{3m} - \sum_{d \leq m/2} (q^{6d} + q^{3d} + 1 + NM^{3d})$, positive for all $m \geq m_0$ with m_0 computable from (n, q, M) ; for the five complexes $m_0 \leq 3$.*

Proof. $|p_d(\Lambda_Q) - (1 + q^{3d} + q^{6d})| \leq NM^{3d}$; the proper divisors of m are $\leq m/2$; the bound increases with m once q^{6m} dominates, which for $M = q$ and $n \leq 168$ happens by $m = 3$. $\square$

Remark 16.4.7 (What remains of the realization problem). Theorem 16.4.5 is about the nonzero spectrum. The realization problem proper — a non-negative integer matrix B shift equivalent over $\mathbb{Z}$ to $C^{(3)}$, so that $\text{coker}(I - B) \cong \mathbb{Z} \oplus \text{Jac}^H(Y)$ with the Hecke dynamics — asks for more than spectral data, and we do not solve it. What is established is that the one tool which closed it in rank one, Theorem [Ch. 3, 1.6], is unavailable in rank two, and that the necessary conditions known to us are all satisfied. The problem is open and, as far as the trace method can see, unobstructed. Whether the double-loop mechanism of Section 16.5 can be upgraded from a K -theoretic to a dynamical realization is the natural next question, and [Ch. 3, Thm. 1.6] no longer forbids it here.

16.5 Signed shadows and the higher Kirchberg algebras

Lemma 16.5.1 (Signed shadow lemma). *Let $M \in M_N(\mathbb{Z})$ with $M_{ee} \geq 1$ for all e , and let $\mathcal{P} = \{(e, f) : e \neq f, M_{ef} > 0\}$. Define a non-negative integer matrix B on the vertex set $\{e\} \sqcup \{w_{ef} : (e, f) \in \mathcal{P}\} \sqcup \{e'\}$ by the arcs*

- $|M_{ef}|$ arcs $f \rightarrow e$ for each $e \neq f$ with $M_{ef} < 0$;
- for $(e, f) \in \mathcal{P}$: M_{ef} arcs $f \rightarrow w_{ef}$, one arc $w_{ef} \rightarrow e$, two loops at w_{ef} ;

- for each e : one arc $e \rightarrow e'$, $M_{ee} - 1$ arcs $e' \rightarrow e$, two loops at e' .

Then, with $X := (I - B^t)_{\text{main}, \text{aux}}$, $Y := (I - B^t)_{\text{aux}, \text{main}}$ and $U = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ Y & I \end{pmatrix}$,

$$U(I - B^t)V = M \oplus (-I_{|\mathcal{P}|+N}).$$

If the graph on $\{e\}$ with edges $\{e, f\}$ for $M_{ef} \neq 0$ is connected, B is irreducible with double-loop vertices, and $\mathcal{O}_B := C^*(\text{digraph of } B)$ is a unital Kirchberg algebra with $K_0(\mathcal{O}_B) \cong \text{coker}_{\mathbb{Z}} M$ and $K_1(\mathcal{O}_B) \cong \ker_{\mathbb{Z}} M$.

Proof. The auxiliary block of $I - B^t$ is $-I$ (each auxiliary vertex carries exactly two loops and no other arcs among auxiliaries), so the Schur complement of that block is $R + XY$ where R is the main block: diagonal 1, entry M_{ef} at negative positions, 0 elsewhere. The auxiliary w_{ef} contributes $X_{e,w}Y_{w,f} = (-1)(-M_{ef}) = M_{ef}$ at (e, f) and nothing else; the shadow e' contributes $X_{e,e'}Y_{e',e} = (-M_{ee} - 1)(-1) = M_{ee} - 1$ at (e, e) . Hence $R + XY = M$, and the displayed identity is the block factorization $\begin{pmatrix} R & X \\ Y & -I \end{pmatrix} = U^{-1}((R + XY) \oplus (-I))V^{-1}$. Irreducibility: every arc $f \rightarrow e$ or $f \rightarrow w_{ef} \rightarrow e$ is matched by the reverse connection through M_{fe} or through the shared vertex structure of a connected support graph, and every e' communicates with e in both directions; the double loops give the witness of [Ch. 4, Thm. 3.1]; K -theory is the Cuntz–Krieger formula transported by U, V . For $M = \Delta_p$ of a graph ($M_{ef} \leq 0$ off the diagonal) $\mathcal{P} = \emptyset$ and the construction is [Ch. 4, Def. 1.1] with $c = M_{ee} - 1$. $\square$

Theorem 16.5.2 (Higher shadow algebras). *For an $\tilde{A}_2$ complex Y let $\mathcal{O}(Y, \Delta_1) := \mathcal{O}_{B(\Delta_1)}$ and $\mathcal{O}(Y, \Delta^H) := \mathcal{O}_{B(\Delta^H)}$ be the algebras of Lemma 16.5.1 for $M = \Delta_1$ (diagonal $q+3 \geq 1$) and $M = \Delta^H$ (diagonal $q^3 - 1 \geq 1$ for $q \geq 2$). Both are unital Kirchberg algebras, hence traceless, and*

$$K_0(\mathcal{O}(Y, \Delta_1)) \cong \mathbb{Z}^{b_1} \oplus \text{Jac}_1(Y), \quad K_1 \cong \mathbb{Z}^{b_1}; \quad K_0(\mathcal{O}(Y, \Delta^H)) \cong \mathbb{Z} \oplus \text{Jac}^H(Y), \quad K_1 \cong \mathbb{Z},$$

the second under the hypothesis of Theorem 16.3.2(ii). Type- and orientation-preserving automorphisms of Y act compatibly. Changing the edge orientations conjugates Δ_1 by a diagonal ± 1 matrix and changes $B(\Delta_1)$, but not the K -theory.

Proof. Lemma 16.5.1 with Proposition 16.2.2 and Theorem 16.3.2; the support graph of Δ_1 is connected because two edges of a connected complex are joined by a chain of edges sharing vertices, and pairs spanning a triangle (where the entry is 0) are joined through a third edge; the support of Δ^H is the 1-skeleton. $\square$

Remark 16.5.3 (Digraph, not cellular algebra). The specification asked for a “two-dimensional shadow complex with higher self-loop faces” and a “higher graph or cellular C^* -algebra”. No such theory is needed: the invariant to be realized is a cokernel, cokernels are unimodular-equivalence invariants, and Lemma 16.5.1 realizes any integer matrix with positive diagonal as the Bowen–Franks presentation of an ordinary graph algebra. The two dimensions of Y enter through Δ_1 and Δ^H , not through the algebra.

16.6 The verification battery

All computations are exact except the spectra. Smith forms are computed p -adically: the determinant of the (reduced) matrix by Bareiss elimination, its prime factors by trial division, and the p -primary invariants by elimination over $\mathbb{Z}/p^k$ with valuation-minimal pivots; for singular symmetric M the torsion found is certified complete by the lattice identity

$$|\text{Tor } \text{coker}_{\mathbb{Z}} M| \cdot \text{disc}(\ker_{\mathbb{Z}} M) = \text{pdet}(M), \quad (16.6.1)$$

where disc is the Gram determinant of a saturated basis of the kernel lattice and pdet the product of the nonzero eigenvalues; the two sides agree to 10^{-6} in every case (this also verifies (16.6.1) itself, a matrix-tree theorem for symmetric integer matrices, on twelve instances).

Data

Y_{168} : $n = 168$, $|E| = |F| = 1176$, $\chi = 168$, $(b_0, b_1, b_2) = (1, 0, 167)$, flag condition holds;

$$\begin{aligned} \text{Jac}_0 &= (\mathbb{Z}/2)^{31} \oplus (\mathbb{Z}/16)^6 \oplus (\mathbb{Z}/592)^6 \oplus (\mathbb{Z}/1184)^{18} \oplus \mathbb{Z}/4736 \\ &\oplus (\mathbb{Z}/33152)^3 \oplus (\mathbb{Z}/430976)^{12} \oplus \mathbb{Z}/3016832 \oplus \mathbb{Z}/9050496, \end{aligned}$$

$$\text{Jac}^H = (\mathbb{Z}/7)^{18} \oplus (\mathbb{Z}/299593)^{13} \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/14680057 \oplus (\mathbb{Z}/117440456)^5 \oplus \mathbb{Z}/352321368,$$

where $299593 = 7 \cdot 127 \cdot 337$ and $2097151 = 2^{21} - 1 = 7^2 \cdot 127 \cdot 337$; $|\text{Jac}^H| \approx 6.18 \cdot 10^{136}$ agrees with $\frac{1}{n} \prod_j |P_j(1)|$ to 10^{-14} ; $\text{Tr}(C^{(3)})^3 = 504 = n(q-1)^2(q+1)$.

Y_{21} : $(b_0, b_1, b_2) = (1, 0, 20)$. $\text{Jac}_0 = \mathbb{Z}/2 \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294$; $\text{Jac}_1 = (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/14)^{10} \oplus (\mathbb{Z}/98)^{26} \oplus \mathbb{Z}/2058 \oplus \mathbb{Z}/18522$ ($|\text{Jac}_1| \approx 6.7 \cdot 10^{73}$); $\text{Jac}_2 = (\mathbb{Z}/2)^{13} \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294$; $K(Y) = (\mathbb{Z}/7)^6 \oplus \mathbb{Z}/14 \oplus (\mathbb{Z}/98)^{10} \oplus \mathbb{Z}/294$; $\text{Jac}^H = (\mathbb{Z}/7)^{17} \oplus \mathbb{Z}/147$, $|\text{Jac}^H| = 3 \cdot 7^{19}$; $\text{Tr}(C^{(3)})^3 = 651 = 63 + 588$.

Y_{24} : $(1, 0, 23)$. $\text{Jac}_0 = (\mathbb{Z}/13)^7 \oplus \mathbb{Z}/26 \oplus (\mathbb{Z}/208)^4 \oplus \mathbb{Z}/1456 \oplus \mathbb{Z}/4368$; $\text{Jac}_1 = (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/51376)^2 \oplus (\mathbb{Z}/102752)^6 \oplus (\mathbb{Z}/2877056)^4 \oplus \mathbb{Z}/60418176 \oplus \mathbb{Z}/543763584$; $\text{Jac}_2 = \mathbb{Z}/2 \oplus (\mathbb{Z}/26)^6 \oplus (\mathbb{Z}/52)^2 \oplus (\mathbb{Z}/208)^2 \oplus (\mathbb{Z}/1456)^3 \oplus \mathbb{Z}/4368$; $K(Y) = (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/38)^7 \oplus \mathbb{Z}/152 \oplus (\mathbb{Z}/304)^2 \oplus (\mathbb{Z}/2128)^3 \oplus \mathbb{Z}/6384$; $\text{Jac}^H = (\mathbb{Z}/7)^2 \oplus \mathbb{Z}/49 \oplus (\mathbb{Z}/392)^5 \oplus \mathbb{Z}/1176$.

Y_{42} : $(1, 0, 41)$. $\text{Jac}_0 = (\mathbb{Z}/4)^2 \oplus (\mathbb{Z}/28)^{10} \oplus \mathbb{Z}/56 \oplus (\mathbb{Z}/8288)^4 \oplus \mathbb{Z}/754208 \oplus \mathbb{Z}/2262624$;

$$\begin{aligned} \text{Jac}_1 &= (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^4 \oplus (\mathbb{Z}/28)^{10} \oplus (\mathbb{Z}/196)^{14} \oplus (\mathbb{Z}/784)^6 \oplus (\mathbb{Z}/1568)^2 \\ &\oplus (\mathbb{Z}/21865186112)^4 \oplus \mathbb{Z}/N_1 \oplus \mathbb{Z}/N_2, \end{aligned}$$

$N_1 = 1474391364718272$, $N_2 = 13269522282464448 = 9N_1$; $\text{Jac}_2 = (\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^6 \oplus (\mathbb{Z}/28)^{10} \oplus \mathbb{Z}/56 \oplus (\mathbb{Z}/8288)^4 \oplus \mathbb{Z}/754208 \oplus \mathbb{Z}/2262624$; $K(Y) = \mathbb{Z}/7 \oplus (\mathbb{Z}/14)^6 \oplus (\mathbb{Z}/98)^5 \oplus \mathbb{Z}/1078 \oplus (\mathbb{Z}/2156)^3 \oplus \mathbb{Z}/40964 \oplus \mathbb{Z}/245784$; $\text{Jac}^H = (\mathbb{Z}/7)^{18} \oplus \mathbb{Z}/299593 \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/176160684$.

Y_{84} (vertex level): $\text{Jac}_0 = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^7 \oplus (\mathbb{Z}/16)^2 \oplus \mathbb{Z}/1184 \oplus (\mathbb{Z}/8288)^5 \oplus (\mathbb{Z}/16576)^6 \oplus (\mathbb{Z}/215488)^4 \oplus \mathbb{Z}/1508416 \oplus \mathbb{Z}/4525248$.

Thin torus $m = 6$ ($n = 36$, $|E| = 108$, $|F| = 72$, $\chi = 0$, $(1, 2, 1)$, flag holds): $\text{Jac}_0 = \mathbb{Z}/2 \oplus (\mathbb{Z}/8)^2 \oplus (\mathbb{Z}/16)^6 \oplus \mathbb{Z}/80 \oplus (\mathbb{Z}/240)^3 \oplus \mathbb{Z}/720 \oplus \mathbb{Z}/2160$; $\text{coker } \Delta_1 = \mathbb{Z}^2 \oplus (\mathbb{Z}/2)^{12} \oplus \mathbb{Z}/8 \oplus (\mathbb{Z}/16)^3 \oplus (\mathbb{Z}/640)^3 \oplus (\mathbb{Z}/1920)^5 \oplus (\mathbb{Z}/5760)^2 \oplus \mathbb{Z}/17280 \oplus \mathbb{Z}/51840$; $K(Y) = \mathbb{Z}^2 \oplus \mathbb{Z}/72$; $\text{Tr}(C^{(3)})^3 = 0$. The torus $m = 3$ violates the flag condition ($\text{Tr } A_1^3 = 81 > 54$) and has $\text{Tr}(C^{(3)})^3 = 27$.

In every thick example the odd parts of Jac_0 and Jac_2 nearly coincide while the 2-parts differ, $K(Y)$ is much smaller than Jac_1 , and Jac^H shares no visible structure with any Hodge group; we record this without interpretation.

check	complexes	result
(P) presentation axioms; simplicial; Heawood/hexagon links; (ii),(iii); flag (iv)	all	exact; (iv) only for Y_{168} , T_6
(D) $\partial_1 \partial_2 = 0$; Betti numbers; χ ; $b_2 = \chi - 1 + b_1$	all	exact
(H) Smith forms of $\Delta_0, \Delta_1, \Delta_2$; $K(Y)$; completeness via (16.6.1)	thin, Y_{21}, Y_{24}, Y_{42}	exact, complete
(L) Δ^H : two-sided kernel; equal cofactors; $ \text{Jac}^H = \frac{1}{n} \prod P_j(1) $	thick	exact; numeric to 10^{-14}
(C) $U(I - C^{(3)})V = P(1) \oplus I \oplus I$, $P(1) = -\Delta^H$	all	exact
(T) $\text{Tr}(C^{(3)})^k$, $k \leq 6$, directly and by Newton; $= 0$ for $3 \nmid k \leq 30$; $\text{Tr}(C^{(3)})^3$ formula	all	exact
(R) nine trivial eigenvalues; nontrivial $ \lambda = q$ (Ramanujan); Λ_Q net traces $m \leq 10$; m_0	thick	numeric / exact
(S) signed shadows of Δ^H and Δ_1 : identity, $B \geq 0$, irreducible, double loops; Smith check	all	exact

Table 16.1: Exact verification (`pgl3_building.py`). Sizes: shadow of Δ^H has $3n + |E|$ vertices (1512 for Y_{168}); shadow of Δ_1 has 2058 vertices for Y_{21} and 2352 for Y_{24} . Net traces of Λ_Q for Y_{168} : 168, 3920, 257544, 16751728, $\dots$; $m_0 = 3$.

16.7 Scope, residue, corrections

Remark 16.7.1 (Solved, open, impossible). *Solved*: the Hodge and Hecke Laplacians of $\tilde{A}_2$ complexes are set up and their critical groups computed exactly on thin and thick examples; the Hecke critical group

is identified with the product of inverse GL_3 L -values and the companion is reduced to it unimodularly; the trace question is settled, negatively for the obstruction and positively for spectral realizability on the computed complexes; the signed-shadow lemma gives Kirchberg realizations of Δ_1 and Δ^H . *Open*: (a) a non-negative matrix shift equivalent over $\mathbb{Z}$ to $C^{(3)}$ (Remark 16.4.7); (b) a general theorem that all net traces of the reduced pencil are non-negative for every $\tilde{A}_2$ complex of order $q \geq 2$ (we have it for five complexes, and in general for the Ramanujan case once $m \geq m_0$); (c) a relation between $\mathrm{Jac}_1(Y)$, $K(Y)$ and the Hecke data — none is visible in the computed examples, and we conjecture none exists: the edge critical group is combinatorial, the vertex Hecke group arithmetic; (d) towers of $\tilde{A}_2$ complexes (the natural continuation of Chs. 5–7). *Impossible or false as stated*: a cubic trace obstruction (Theorem 16.4.2).

Remark 16.7.2 (Corrections to the task specification). (1) The predicted “higher cubic trace obstruction” does not exist: $\mathrm{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$ and $\mathrm{Tr}(C^{(3)})^3 = n(q-1)^2(q+1) \geq 0$; the paper proves the contrary and, for the computed complexes, spectral realizability. (2) The “duality bridge between the cellular sandpile group and the GL_3 local L -factors” runs through the Hecke–Laplacian Δ^H and its critical group $\mathrm{Jac}^H(Y)$, not through the Hodge groups $\mathrm{Jac}_0, \mathrm{Jac}_1$, which in rank two are different objects (Theorem 16.3.2(iii)). (3) The test complexes are $\tilde{A}_2$ complexes of order 2 built from a Cartwright–Steger-type triangle presentation, i.e. quotients of an $\tilde{A}_2$ building of order 2; the specification’s “ $\mathrm{PGL}_3(\mathbb{Q}_p)$ ” is not literally available (arithmetic lattices in $\mathrm{PGL}_3(\mathbb{Q}_p)$ were not constructed), and nothing proved depends on the field. (4) No “two-dimensional shadow complex” is needed (Remark 16.5.3). (5) The companion in the specification is correct, and its reduction to Δ^H (Theorem 16.3.3) is the rank-two form of [Ch. 3, §1.1].

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from finite linear algebra and the results of Papers I–IX; the classical inputs are [52] (existence of the building for a triangle presentation), the Cuntz–Krieger facts quoted in [Ch. 4], and [55] for Theorem 16.4.5 only. Spectral statements marked “numeric” in Table 16.1 are floating-point computations with the stated tolerances; all group-theoretic and trace statements are exact. The identification of the thick complexes with quotients of the Bruhat–Tits building of PGL_3 over a specific local field is not claimed. Bibliographic data of [52, 54, 55, 53] were checked against publisher or indexing pages.

Chapter 17

Algorithmic non-commutative class field theory, XI: the PGL_3 positive Hecke-dynamical system: shift equivalence over $\mathbb{Z}$ and the geodesic edge flow

Summary. Paper X left one problem at the top of its list: a non-negative integer matrix shift equivalent over $\mathbb{Z}$ to the companion $C^{(3)}$ of the PGL_3 cubic pencil, which would be a positive Hecke-dynamical system carrying the Hecke critical group $\mathbb{Z} \oplus \mathrm{Jac}^H(Y)$ in its K -theory. This paper settles what can be settled about that problem and builds the positive system that does exist. First, the $\mathbb{Z}/3$ grading $\deg(i, v) = i - \tau(v)$ makes $C^{(3)}$ a period-three cyclic matrix; shift equivalence of $C^{(3)}$ to a non-negative matrix reduces, with an explicit lag formula, to shift equivalence of the return map $P_0 = C^{(3)3}|_{\deg 0}$ (Perron root q^6 , $\det P_0 = q^{3n}$) to a primitive matrix. Second, the required statement is identified precisely: since the nonzero spectrum of P_0 satisfies the Perron and net-trace conditions ([Ch. 16], Kim–Ormes–Roush), its shift equivalence over $\mathbb{Z}$ to a primitive matrix is an instance of the *weak Generalized Spectral Conjecture* of Boyle–Handelman, which is open over $\mathbb{Z}$; the specification’s request for explicit R, S, ℓ is a request to prove a case of an open conjecture, and we do not claim to. Third, the natural explicit route — a positive presentation $(tI - Y)(tI - P_0)$ with a nilpotent cofactor Y — is proved impossible on Y_{21} and Y_{24} by exact rational Farkas certificates: P_0 has a positive right but not left Perron vector, no positive power, and a negative-entry digraph with cycles. Fourth, and positively: the *geodesic edge flow* L_E on typed edges $((x, y) \rightarrow (y, z))$ with z not adjacent to x is a non-negative integer matrix with row sums q^2 , irreducible, and for every joint eigenvector f of (A_1, A_2) with eigenvalues (a, b) the three edge functions $f \circ \text{tail}$, $f \circ \text{head}$, m_f (sum over the third vertices of the chambers on the edge) span an L_E -invariant space on which the characteristic polynomial is exactly $\lambda^3 - a\lambda^2 + qb\lambda - q^3$. Hence an explicit integer intertwiner Ψ with $L_E\Psi = \Psi C'$, C' lag-one shift equivalent to $C^{(3)}$, and the whole nontrivial PGL_3 Hecke spectrum inside the spectrum of a non-negative matrix; the trivial pair contributes only q^2 ($\mathrm{rk} \Psi = 3n - 6$), and the complementary spectrum has modulus $q^{1/2}$ (numeric on Y_{168}). The Cuntz–Krieger algebra $\mathcal{O}_{L_E}$ is a Kirchberg algebra whose gauge dynamics is the gallery flow and whose K_0 is finite: the positive system sees the cuspidal Hecke spectrum but not the Hecke critical group, the rank-two form of the bulk–boundary dichotomy of [Ch. 2].

Status of the results

Every numbered statement is proved except where marked “numeric” (floating-point spectra, Table 17.1). Theorem 17.2.3 rests on exact rational Farkas certificates verified by integer arithmetic. The classical inputs are the module characterization of shift equivalence over $\mathbb{Z}$ [58, §2], the Spectral Conjecture over $\mathbb{Z}$ [55], and the statements (not proofs) of the Generalized Spectral Conjecture [56, 57, 58]. Kang–Li [54] is context for the edge flow. The complexes are those of [Ch. 16]; the edge-flow theorem uses the flag condition [Ch. 16, Def. 1.1(iv)], which holds for Y_{168} and fails for the smaller covers, so its numerical checks are on Y_{168} . Section 17.5 records what is open and the corrections to the specification; Proposition 17.3.8 answers its item on KMS states.

17.1 The grading, the return map, and the cyclic reduction

Throughout, Y is an $\tilde{A}_2$ complex of order q in the sense of [Ch. 16, Def. 1.1] with n vertices, type function τ , type-raising adjacency A_1 , $A_2 = A_1^t$, and companion $C^{(3)} = \begin{pmatrix} A_1 & -qA_2 & q^3I \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix}$ on $\mathbb{Z}^{3n} = \bigoplus_{i=0}^2 \mathbb{Z}^V$, $\det(I - uC^{(3)}) = \det P(u)$ with $P(u) = I - uA_1 + qu^2A_2 - q^3u^3I$ [Ch. 16, Thm. 3.3].

Lemma 17.1.1 (Grading). *Give the coordinate (i, v) of $\mathbb{Z}^{3n}$ the degree $\deg(i, v) := i - \tau(v) \in \mathbb{Z}/3$. Then $C^{(3)}$ maps the degree- d summand $\mathbb{Z}_d^{3n}$ into $\mathbb{Z}_{d+1}^{3n}$: in the graded decomposition $C^{(3)}$ is the cyclic block matrix with blocks $X_d: \mathbb{Z}_d^{3n} \rightarrow \mathbb{Z}_{d+1}^{3n}$, each $\mathbb{Z}_d^{3n}$ of rank n , and $C^{(3)3}$ is block diagonal with return maps $P_d := X_{d+2}X_{d+1}X_d \in M_n(\mathbb{Z})$.*

Proof. A_1 sends $(0, y)$ to $(0, x)$ with $\tau(x) = \tau(y) - 1$, raising the degree by one; the identity blocks send $(0, v) \mapsto (1, v)$ and $(1, v) \mapsto (2, v)$, raising i by one; $-qA_2$ sends $(1, v)$ to $(0, x)$ with $\tau(x) = \tau(v) + 1$: degree $1 - \tau(v) \mapsto -\tau(v) - 1 \equiv 1 - \tau(v) + 1$; q^3I sends $(2, v) \mapsto (0, v)$: $2 - \tau(v) \mapsto -\tau(v) \equiv 3 - \tau(v)$. Each $\mathbb{Z}_d^{3n}$ has rank n because the n vertices split equally among the three types [Ch. 16, Lem. 1.2]. Verified as an exact block identity on four complexes (Table 17.1, line (G)). $\square$

Proposition 17.1.2 (The return map). *The nonzero spectrum of P_0 is Λ_Q of [Ch. 16, Def. 4.4]: one cube λ^3 for each ω -orbit of eigenvalues of $C^{(3)}$, with q^6 simple and dominant; $\det P_0 = q^{3n}$; P_0 has a positive right Perron vector, namely the restriction of $(q^4\mathbf{1}, q^2\mathbf{1}, \mathbf{1})$ to $\mathbb{Z}_0^{3n}$. On $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ (numeric): the left Perron vector of P_0 is not positive, no power P_0^k , $k \leq 30$, is positive, and the digraph of negative entries of P_0 contains cycles.*

Proof. $C^{(3)3} = \bigoplus_d P_d$ and the three P_d are conjugate over $\mathbb{Z}$ ($P_{d+1}X_d = X_dP_d$ with X_d invertible over $\mathbb{Q}$), so each has the spectrum of $C^{(3)3}$ with multiplicities divided by three: $\{\lambda^3\}$, one per orbit. $\det C^{(3)} = (q^3)^n$ (block expansion), so $\det P_0^3 = \det C^{(3)3} = q^{9n}$ and $\det P_0 = q^{3n}$ (positive, as q^6 dominates). For $\lambda = q^2$ the vector $(\lambda^2\mathbf{1}, \lambda\mathbf{1}, \mathbf{1})$ is a $C^{(3)}$ -eigenvector because $\lambda^2(q^2 + q + 1) - q\lambda(q^2 + q + 1) + q^3 = \lambda^3$; it is positive and graded-homogeneous pieces of it are eigenvectors of the P_d . The remaining statements are computations. $\square$

Lemma 17.1.3 (Cyclic lifting of shift equivalence). *Let $B_0 \in M_N(\mathbb{Z})$ and suppose (R_0, S_0, ℓ) is a shift equivalence over $\mathbb{Z}$ between P_0 and B_0 : $P_0R_0 = R_0B_0$, $S_0P_0 = B_0S_0$, $R_0S_0 = P_0^\ell$, $S_0R_0 = B_0^\ell$. Put $B := \text{cyc}(I, I, B_0)$, the cyclic block matrix with blocks $I: 0 \rightarrow 1$, $I: 1 \rightarrow 2$, $B_0: 2 \rightarrow 0$ on $\mathbb{Z}^N \oplus \mathbb{Z}^N \oplus \mathbb{Z}^N$, and*

$$R := \text{diag}(R_0, X_0R_0, X_1X_0R_0), \quad S := \text{diag}(S_2X_1X_0, S_2X_1, S_2), \quad S_2 := S_0X_2.$$

Then $C^{(3)}R = RB$, $SC^{(3)} = BS$, $RS = C^{(3)3(\ell+1)}$, $SR = B^{3(\ell+1)}$: a shift equivalence over $\mathbb{Z}$ of lag $3(\ell+1)$. If $B_0 \geq 0$ then $B \geq 0$, and B is irreducible of period 3 when B_0 is primitive. Conversely a shift equivalence between $C^{(3)}$ and a graded cyclic B restricts to one between the return maps.

Proof. Direct verification: $X_0R_0 = (X_0R_0) \cdot I$, $X_1(X_0R_0) = (X_1X_0R_0) \cdot I$, $X_2(X_1X_0R_0) = P_0R_0 = R_0B_0$ give $C^{(3)}R = RB$; $S_2X_1X_0 \cdot X_2 = S_0P_0X_2 = B_0S_0X_2 = B_0S_2$ and the two identities $S_{d+1}X_d = S_d$ give $SC^{(3)} = BS$; on the degree- d block RS equals $R_dS_d = (X_{d-1} \cdots X_0R_0)(S_0X_2 \cdots X_d) = X_{d-1} \cdots X_0P_0^\ell X_2 \cdots X_d = P_d^{\ell+1}$, and SR equals $S_0P_0R_0 = B_0^{\ell+1}$ on each block, while $B^3 = \text{diag}(B_0, B_0, B_0)$. Checked with the trivial

equivalence $B_0 = P_0$ on four complexes (Table 17.1, line (L)). The converse is the restriction of the four equations to graded pieces. $\square$

17.2 Shift equivalence over $\mathbb{Z}$: criterion, cofactors, obstructions

Proposition 17.2.1 (Module criterion). *For square integer matrices A, B (any sizes), $A \sim_{\mathbb{Z}} B$ if and only if the $\mathbb{Z}[t, t^{-1}]$ -modules $M_A := \mathbb{Z}[t^{\pm}]^n / (tI - A)\mathbb{Z}[t^{\pm}]^n$ and M_B are isomorphic [58, §2]. For nonsingular A , $M_A \cong \mathbb{Z}^n[A^{-1}] = \bigcup_k A^{-k}\mathbb{Z}^n$ with t acting as A . A non-negative B with $B \sim_{\mathbb{Z}} P_0$ is therefore the same thing as a finite set of generators $g_1, \dots, g_N$ of M_{P_0} over $\mathbb{Z}[t^{\pm}]$ with $t g_j = \sum_i B_{ij} g_i$, $B_{ij} \in \mathbb{Z}_{\geq 0}$: a positive presentation of the Hecke module.*

Proposition 17.2.2 (Cofactor construction). *Let $U(t) \in GL_n(\mathbb{Z}[t^{\pm}])$ and suppose $F(t) := U(t)(tI - P_0) = t^{K+1}I - \sum_{k=0}^K C_k t^k$ with all $C_k \in M_n(\mathbb{Z}_{\geq 0})$. Then the companion $B_F = \begin{pmatrix} C_K & C_{K-1} & \cdots & C_0 \\ I & 0 & \cdots & 0 \\ & \ddots & \ddots & \\ 0 & \cdots & I & 0 \end{pmatrix} \geq 0$ satisfies $B_F \sim_{\mathbb{Z}} P_0$. In particular, if $Y \in M_n(\mathbb{Z})$ is nilpotent with $P_0 + Y \geq 0$ and $-Y P_0 \geq 0$, then $U = tI - Y$ works and $B = \begin{pmatrix} P_0 + Y & -Y P_0 \\ 0 & 0 \end{pmatrix} \geq 0$ is shift equivalent over $\mathbb{Z}$ to P_0 , hence $\text{cyc}(I, I, B) \sim_{\mathbb{Z}} C^{(3)}$ by Lemma 17.1.3.*

Proof. $\text{coker}(tI - B_F) \cong \text{coker } F$ by the companion reduction (as in [Ch. 16, Thm. 3.3]), and $\text{coker } F = \text{coker}(U(tI - P_0)) \cong \text{coker}(tI - P_0)$ because U is invertible over $\mathbb{Z}[t^{\pm}]$; Proposition 17.2.1. For $U = tI - Y$: $\det(tI - Y) = t^n$ is a unit in $\mathbb{Z}[t^{\pm}]$ when Y is nilpotent, and $(tI - Y)(tI - P_0) = t^2 I - (P_0 + Y)t + Y P_0$. $\square$

Theorem 17.2.3 (No nilpotent cofactor on Y_{21}, Y_{24}). *For $Y = Y_{21}$ and $Y = Y_{24}$ there is no matrix $Y \in M_n(\mathbb{Q})$ with $P_0 + Y \geq 0$, $Y P_0 \leq 0$ and $\text{Tr } Y = 0$; a fortiori no nilpotent integer cofactor exists, and Proposition 17.2.2 with $K = 1$ cannot produce a positive presentation.*

Proof. Write $Y = -P_0 + Z$, $Z \geq 0$; the system becomes $Z P_0 \leq P_0^2$ entrywise and $\text{Tr } Z = \text{Tr } P_0$. By Farkas' lemma it is infeasible iff there exist $\Lambda \in M_n(\mathbb{Q}_{\geq 0})$ and $\mu \in \mathbb{Q}$ with $\Lambda P_0^t + \mu I \geq 0$ entrywise and $\langle \Lambda, P_0^2 \rangle + \mu \text{Tr } P_0 < 0$. The script finds such certificates with denominators 2 (Y_{21} , dual value -4340) and 128 (Y_{24} , dual value $-53185/8$) and verifies both inequalities in exact rational arithmetic. (For Y_{42} the LP relaxation is infeasible in floating point but a rational certificate was not extracted; we claim nothing there.) $\square$

Remark 17.2.4 (The precise status of the realization problem). Let Λ be the nonzero spectrum of P_0 . By [Ch. 16, Thm. 4.5] and [55], Λ is the nonzero spectrum of a primitive integer matrix. The statement the specification asks for $-P_0 \sim_{\mathbb{Z}} B_0$ with B_0 primitive, equivalently $C^{(3)} \sim_{\mathbb{Z}} B$ with $B \geq 0$ (Lemma 17.1.3) — is exactly the *weak form of the Generalized Spectral Conjecture* [56, p. 253], [57, p. 124] for the ring $\mathbb{Z}$ and the matrix P_0 : that the three spectral conditions suffice for shift equivalence over $\mathbb{Z}$ to a primitive matrix. Boyle–Schmieding [58] show that over dense subrings of $\mathbb{R}$ the weak form implies the strong form (strong shift equivalence); $\mathbb{Z}$ is not dense and the weak form over $\mathbb{Z}$ is, to our knowledge, open. The requested explicit (B, R, S, ℓ) would therefore prove a case of an open conjecture. This paper does not. What it proves is that the problem is equivalent to the primitive-matrix problem for the $n \times n$ return map (Lemma 17.1.3), that the first explicit route is closed (Theorem 17.2.3), and that a positive system carrying the Hecke spectrum — though not the Hecke module — exists and is canonical (Section 17.3).

17.3 The geodesic edge flow

Let E be the set of typed edges $e = (x \rightarrow y)$, $\tau(y) = \tau(x) + 1$, and for each chamber $\{x, y, z\}$ on e call z its third vertex. Let $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ be the tail, head and third-vertex incidence matrices: $S_{v,e} = [v = \text{tail}(e)]$, $T_{v,e} = [v = \text{head}(e)]$, $M_{v,e} = [\{v\} \cup e \text{ is a chamber}]$; M has column sums $q + 1$. For a vertex function f put $f \circ t := S^t f$, $f \circ h := T^t f$, $m_f := M^t f$ (the sum of f over the third vertices of the chambers on the edge).

Definition 17.3.1 (Geodesic edge flow). $L_E \in M_{|E|}(\{0, 1\})$, $L_E((x \rightarrow y), (y \rightarrow z)) = 1$ iff $\tau(z) = \tau(y) + 1$ and z is not adjacent to x ; all other entries 0. A step of L_E extends a type-raising edge to a type-raising path of length two that does not lie in a chamber: the rank-two analogue of the non-backtracking condition of the Hashimoto matrix A^{nb} of [Ch. 2].

Lemma 17.3.2 (Local counts). *Let Y satisfy [Ch. 16, Def. 1.1(i)–(iv)] and have vertex links isomorphic to the incidence graph of $\text{PG}(2, q)$. For a typed edge $e = (x \rightarrow y)$:*

- (a) y has $q^2 + q + 1$ type-raising neighbours z ; exactly $q + 1$ of them are adjacent to x , and those are exactly the third vertices of the chambers on e (flag condition); hence the row sum of L_E at e is q^2 ;
- (b) $\sum_{z: y \rightarrow z, z \not\sim x} f(z) = (A_1 f)(y) - m_f(e)$;
- (c) $\sum_{z: y \rightarrow z, z \not\sim x} m_f(y \rightarrow z) = q((A_2 f)(y) - f(x))$.

Proof. (a), (b) are the flag condition. (c) Expand $m_f(y \rightarrow z) = \sum_w f(w)$ over third vertices w of chambers $\{y, z, w\}$; such w are type- $\tau(x)$ neighbours of y , i.e. the in-neighbours $w \rightarrow y$ counted by A_2 . For fixed w the number of z with $y \rightarrow z$, $\{y, z, w\}$ a chamber and $z \not\sim x$ is: 0 if $w = x$ (every chamber on $(x \rightarrow y)$ has $z \sim x$); and if $w \neq x$, in the link of y — the incidence graph of $\text{PG}(2, q)$, with x, w two distinct points and the z 's the $q + 1$ lines through w — the lines through w that also pass through x number exactly one, so q of them have $z \not\sim x$. Summing, $q \sum_{w \rightarrow y, w \neq x} f(w) = q((A_2 f)(y) - f(x))$. $\square$

Theorem 17.3.3 (The edge flow carries the Hecke pencil). *Under the hypotheses of Lemma 17.3.2, with $\Psi := \begin{bmatrix} S^t & T^t & M^t \end{bmatrix} \in M_{|E| \times 3n}(\mathbb{Z})$ and*

$$C' := \begin{pmatrix} 0 & 0 & -qI \\ q^2I & A_1 & qA_2 \\ 0 & -I & 0 \end{pmatrix} \in M_{3n}(\mathbb{Z}) : \quad L_E \Psi = \Psi C'.$$

Consequently, for every joint eigenvector f of (A_1, A_2) with eigenvalues (a, b) , the space $V_f := \text{span}\{f \circ t, f \circ h, m_f\}$ is L_E -invariant and the matrix of L_E on the ordered basis $(f \circ t, f \circ h, m_f)$ is $\begin{pmatrix} 0 & 0 & -q \\ q^2 & a & qb \\ 0 & -1 & 0 \end{pmatrix}$, with characteristic polynomial $\lambda^3 - a\lambda^2 + qb\lambda - q^3$, the cubic pencil of the pair (a, b) . Every root of the cubic pencil attached to a joint eigenpair with $\dim V_f = 3$ is an eigenvalue of the non-negative matrix L_E .

Proof. For $g = \alpha f \circ t + \beta f \circ h + \gamma m_f$ and $e = (x \rightarrow y)$,

$$\begin{aligned} (L_E g)(e) &= \sum_{z: y \rightarrow z, z \not\sim x} (\alpha f(y) + \beta f(z) + \gamma m_f(y \rightarrow z)) \\ &= \alpha q^2 f(y) + \beta((A_1 f)(y) - m_f(e)) + \gamma q((A_2 f)(y) - f(x)) \end{aligned}$$

by Lemma 17.3.2, i.e. $L_E g = (-q\gamma) f \circ t + (q^2\alpha f + \beta A_1 f + q\gamma A_2 f) \circ h + (-\beta) m_f$. Read for arbitrary (not necessarily eigen-) triples (f_0, f_1, f_2) this is $L_E \Psi(f_0, f_1, f_2)^t = \Psi(-qf_2, q^2f_0 + A_1f_1 + qA_2f_2, -f_1)^t = \Psi C'(f_0, f_1, f_2)^t$. For an eigenvector, $A_1 f = af$, $A_2 f = bf$ give the 3×3 matrix; expanding its characteristic determinant along the first row gives $\lambda((\lambda - a)\lambda + qb) - q^3$. Verified as an exact integer identity $L_E \Psi = \Psi C'$ on Y_{168} (Table 17.1, line (I)). $\square$

Proposition 17.3.4 (C' is the companion). *With $D := \begin{pmatrix} 0 & 0 & qI \\ I & 0 & 0 \\ 0 & -I & 0 \end{pmatrix}$, both D and $S := C^{(3)} D^{-1}$ are integer matrices, and $C' D = D C^{(3)}$, $S C' = C^{(3)} S$, $D S = C'$, $S D = C^{(3)}$: C' and $C^{(3)}$ are shift equivalent over $\mathbb{Z}$ with lag one (indeed elementary strong shift equivalent over $\mathbb{Z}$), so $M_{C'} \cong M_{C^{(3)}}$ and C' has the spectrum of the pencil.*

Proof. $D^{-1} = \begin{pmatrix} 0 & I & 0 \\ q^{-1}I & 0 & 0 \end{pmatrix}$, and the third block column of $C^{(3)}$ is $(q^3I, 0, 0)^t$, so $C^{(3)} D^{-1}$ has first block column $(q^2I, 0, 0)^t$ and is integral. The four identities are then formal: $C' = D C^{(3)} D^{-1}$ by direct multiplication (the new coordinates are $f_0 = qg_2$, $f_1 = g_0$, $f_2 = -g_1$), $D S = D C^{(3)} D^{-1} = C'$, $S D = C^{(3)}$. Verified exactly on Y_{168} . $\square$

Corollary 17.3.5 (What the edge flow sees and what it does not). *Under the hypotheses of Theorem 17.3.3:*

- (i) $\ker \Psi \supseteq \{(f, -f, 0), ((q+1)f, 0, -f) : f = \omega^{k\tau} \mathbf{1}, k = 0, 1, 2\}$, a rank-6 sublattice: for the trivial pair and its two ω -conjugates the three edge functions are proportional, V_f is one-dimensional and carries only the eigenvalue $q^2 \omega^k$; hence $\text{rk } \Psi \leq 3n - 6$, with equality on Y_{168} (numeric rank 498).
- (ii) On Y_{168} the spectrum of L_E contains every pencil root except the six $\{1, \omega, \omega^2, q, q\omega, q\omega^2\}$; the remaining $|E| - (3n - 6) = 4n + 6$ eigenvalues all have modulus $q^{1/2}$ (numeric, to 10^{-4}): with $P(u) = \det(I - uC^{(3)})$,

$$\det(I - uL_E) \cdot (1 - u^3)(1 - q^3 u^3) = \det P(u) \cdot G(u),$$

$$G \in \mathbb{Z}[u], \quad \deg G = 4n + 6, \quad \text{all roots of } G \text{ on } |u| = q^{-1/2}.$$

- (iii) L_E is irreducible with row sums q^2 (its Perron eigenvalue), and 1 is not an eigenvalue of L_E (numeric, distance 0.58 to the nearest eigenvalue on Y_{168}).

Proof. (i) For $f = \omega^{k\tau} \mathbf{1}$: $f \circ t = \omega^{k\tau(x)}$, $f \circ h = \omega^{k(\tau(x)+1)}$, $m_f = (q+1)\omega^{k(\tau(x)+2)}$ on $(x \rightarrow y)$, proportional; the two displayed vectors are integral for $k = 0$, and over $\mathbb{Z}[\omega]$ for $k = 1, 2$, so the rational kernel has rank at least 6. (ii),(iii) are computations; the divisibility in (ii) is exact because $\det P(u)$ contains $(1 - u^3)(1 - q^3 u^3)(1 - q^6 u^3)$ and L_E has the eigenvalues $q^2 \omega^k$. $\square$

Remark 17.3.6 (Kang–Li context). Kang–Li [54] express the zeta function of a PGL_3 complex as a rational function in which the vertex factor $\det P(u)$ and a chamber (parahoric) factor appear; the operator L_E is the type-(1,0) gallery transfer operator whose determinant computes the zeta function counting closed type-raising geodesic galleries, and the modulus- $q^{1/2}$ remainder of Corollary 17.3.5(ii) is the parahoric spectrum, which for a Ramanujan complex lies on $|u| = q^{-1/2}$. We use nothing from [54]; the identity (ii) is established here numerically and its general form is left as it stands. The chamber flow L_F on oriented chambers $((a, b, c) \rightarrow (b, c, d), d \neq a)$, row sums q , has spectrum of moduli $\{1, q^{1/4}, q^{1/2}, q\}$ on Y_{168} and contains no nontrivial pencil root: the Hecke spectrum lives on edges, not on chambers.

Theorem 17.3.7 (The positive Hecke-dynamical algebra). *Let Y satisfy the hypotheses of Theorem 17.3.3 and let $\mathcal{O}_{L_E}$ be the Cuntz–Krieger algebra of L_E with its gauge action γ . Then $\mathcal{O}_{L_E}$ is a unital Kirchberg algebra whose gauge dynamics is the geodesic gallery flow; its partition function is $\det(I - uL_E)^{-1}$, whose poles include every nontrivial Satake datum $\lambda_{j,i}$ of the complex at $u = \lambda_{j,i}^{-1}$; its K -theory is $K_1(\mathcal{O}_{L_E}) = \ker(I - L_E^t) = 0$ and $K_0(\mathcal{O}_{L_E}) = \text{coker}(I - L_E^t)$ finite of order $|\det(I - L_E)|$ (on Y_{168} , $\log |\det(I - L_E)| = 603.249$, numeric). The integer intertwiner Ψ induces a $\mathbb{Z}[t^{\pm}]$ -module map $M_{C^{(3)}} \cong M_{C'} \rightarrow M_{L_E}$ of the Hecke module into the edge-flow module. $\mathcal{O}_{L_E}$ is not shift equivalent to $C^{(3)}$ (different nonzero spectra), and its K_0 is not the Hecke critical group $\mathbb{Z} \oplus \text{Jac}^H(Y)$.*

Proof. L_E is irreducible with row sums $q^2 \geq 4$ and is not a permutation matrix, so it satisfies condition (I) of Cuntz–Krieger and $\mathcal{O}_{L_E}$ is simple and purely infinite, separable, nuclear and in the UCT class, by the criteria quoted in [Ch. 4, Thm. 3.1]; purely infinite algebras are traceless. K -theory is the Cuntz–Krieger formula; $1 \notin \text{spec } L_E$ gives the kernel statement and finiteness. The module map is Theorem 17.3.3 and Proposition 17.3.4. The last sentence is Corollary 17.3.5(ii) and [Ch. 16, Thm. 3.3]. $\square$

Proposition 17.3.8 (KMS states of the edge flow). *Let $\sigma_t(s_e) = e^{it}s_e$ be the gauge dynamics of $\mathcal{O}_{L_E}$. A KMS_β state exists if and only if $\beta = \beta_c := 2 \log q = \log \rho(L_E)$, and it is then unique: the state attached to the Perron eigenvector of L_E (the Parry measure of the gallery shift), which is the constant vector since the row sums are q^2 . The partition function $Z(\beta) = \sum_k \text{Tr}(L_E^k) e^{-k\beta} = -u \frac{d}{du} \log \det(I - uL_E) \Big|_{u=e^{-\beta}}$ has poles at $u = \lambda^{-1}$ for every eigenvalue λ of L_E , hence at the inverses of all nontrivial Satake data by Theorem 17.3.3; the KMS state sees only $\lambda = q^2$. The Perron obstruction of [Ch. 2, Thm. 2.5] therefore holds verbatim in rank two: the cuspidal GL_3 L -factors $P_j(u)^{-1}$ are carried by the partition function and by the KMS functionals obtained from the nontrivial eigenvectors of L_E (Theorem 17.3.3 gives them explicitly, $\Psi(f \circ t, f \circ h, m_f)$ -combinations), not by any KMS state.*

Proof. For an irreducible non-negative matrix with the gauge dynamics, KMS_β states of the Cuntz–Krieger algebra correspond to non-negative eigenvectors of L_E^t for the eigenvalue e^β (as in [Ch. 2, Thm. 2.4]); by Perron–Frobenius the only such eigenvalue is $\rho(L_E) = q^2$ (row sums), with a unique normalized positive eigenvector. The rest is Theorem 17.3.3 and the identity $\text{Tr}(L_E^k) = \sum_\lambda \lambda^k$. $\square$

Remark 17.3.9 (Bulk and boundary, in rank two). [Ch. 2, Thm. 3.7] found that the natural positive system of a graph — the boundary algebra $\mathcal{O}_{\text{Anb}}$ — has K -theory blind to the Hecke data, while the Hecke critical group $\mathbb{Z} \oplus \text{Jac}(Y)$ lives in the Laplacian channel; [Ch. 4] realized the latter by a Hecke-functorial, not Hecke-dynamical, Kirchberg algebra, and [Ch. 3] showed no single algebra can be both. In rank two the same dichotomy appears with one difference: the positive system $\mathcal{O}_{L_E}$ carries the whole nontrivial Hecke *spectrum* (Theorem 17.3.3), and its K_0 is a finite group unrelated to $\text{Jac}^H(Y)$, while $\mathcal{O}(Y, \Delta^H)$ of [Ch. 16] carries $\mathbb{Z} \oplus \text{Jac}^H(Y)$ without dynamics. An algebra carrying both would be $\mathcal{O}_B$ for a non-negative $B \sim_{\mathbb{Z}} C^{(3)}$; [Ch. 3, Thm. 1.6] no longer forbids it (there is no trace obstruction, [Ch. 16, Thm. 4.2]), and its existence is the open instance of the weak Generalized Spectral Conjecture of Remark 17.2.4. That is the exact residue of the problem.

17.4 The verification battery

check	complexes	result
(G) $\deg = i - \tau$ makes $C^{(3)}$ degree one; $P_d = C^{(3)3} _d$; $\text{spec } P_0 = \Lambda_Q$; $\det P_0 = q^{3n}$	$Y_{21}, Y_{24}, Y_{42}, Y_{168}$	exact (spectrum numeric)
(Y) P_0 : negative entries 147, 184, 476, 5936; right Perron > 0 ; left Perron $\not> 0$; no $P_0^k > 0$, $k \leq 30$; negative digraph cyclic	same	exact / numeric
(N) LP relaxation $Y \geq -P_0$, $YP_0 \leq 0$, $\text{Tr } Y = 0$ infeasible; exact Farkas certificates (denominators 2, 128)	Y_{21}, Y_{24} (Y_{42} float)	exact
(L) cyclic lift of the trivial shift equivalence: four equations, lag 3	four complexes	exact
(E) L_E : 1176 edges, row sums $4 = q^2$, irreducible; 498 of 504 pencil roots present; missing $\{1, \omega, \omega^2, 2, 2\omega, 2\omega^2\}$; remainder $678 = 4n + 6$ of modulus $\sqrt{2}$	Y_{168}	numeric
(I) $L_E \Psi = \Psi C'$; $\text{rk } \Psi = 498 = 3n - 6$; $D, C^{(3)}D^{-1}$ integral; lag-one equations	Y_{168}	exact
(T) $\text{Tr } L_E^k$ vs $\text{Tr}(C^{(3)})^k$, $k \leq 10$: unequal (parahoric spectrum)	Y_{168}	exact
(F) L_F : 3528 oriented chambers, row sums 2, irreducible; spectrum moduli $\{1, 2^{1/4}, 2^{1/2}, 2\}$; 3 pencil roots	Y_{168}	numeric

Table 17.1: Verification (`pgl3_dynamical.py`). Y_{168} is the only computed complex satisfying the flag condition; the edge-flow theorem is checked there. $\log |\det(I - L_E)| = 603.249$ on Y_{168} .

17.5 Scope, residue, corrections

Remark 17.5.1 (Solved, open, impossible). *Solved*: the graded structure of $C^{(3)}$ and the reduction of its realization problem to the primitive-matrix problem for the $n \times n$ return map, with an explicit lag formula; the module criterion and the cofactor construction; the impossibility of the nilpotent-cofactor route on Y_{21}, Y_{24} (exact certificates); the geodesic edge flow theorem with its explicit intertwiner, giving a non-negative integer matrix whose spectrum contains the entire nontrivial PGL_3 Hecke spectrum, and the Kirchberg algebra $\mathcal{O}_{L_E}$ it generates. *Open*: (a) $C^{(3)} \sim_{\mathbb{Z}} B \geq 0$, an instance of the weak Generalized Spectral Conjecture over $\mathbb{Z}$ (Remark 17.2.4); (b) a proof of the determinant identity of Corollary 17.3.5(ii) for all $\tilde{A}_2$ complexes, and the identification of $G(u)$ with Kang–Li’s chamber factor; (c) the structure of the finite group $K_0(\mathcal{O}_{L_E})$; (d) whether positive presentations of M_{P_0} of higher degree $K \geq 2$ exist (Proposition 17.2.2) — the Farkas method extends but the search space grows. *Impossible or false as stated*: the “century deadlock since Bost–Connes” that positive correspondences cannot carry higher automorphic Hecke spectra — Theorem 17.3.3 carries it, as Hashimoto’s matrix already did in rank one; what positive systems have so far failed to carry is the Hecke *module* (the K -theory), and whether they can is the open conjecture.

Remark 17.5.2 (Corrections to the task specification). (1) The specification asked for explicit (B, R, S, ℓ) with $B \geq 0$ shift equivalent over $\mathbb{Z}$ to $C^{(3)}$, as a corollary of [Ch. 16]’s “Boyle–Handelman spectral conditions”. Those conditions give the nonzero spectrum (Spectral Conjecture, a theorem over $\mathbb{Z}$ by [55]); shift equivalence is the weak Generalized Spectral Conjecture, open. No such (B, R, S, ℓ) is supplied, and none could be without proving a case of that conjecture. (2) “Method one” — a block-cyclic B whose three-step product cancels $-qA_2$ — is exactly Lemma 17.1.3: correct as a reduction, and it converts the problem into the

primitive-matrix problem for P_0 rather than solving it. (3) “Method two” (state splitting) applies to non-negative matrices; it cannot start from $C^{(3)}$, and the algebraic substitute (cofactors, Proposition 17.2.2) is obstructed at degree two (Theorem 17.2.3). (4) The genuine positive GL_3 dynamics is the geodesic edge flow L_E , whose existence and spectral content are proved here; it is the rank-two Hashimoto matrix, not a realization of $C^{(3)}$. (5) The claim that KMS states at the critical temperature “reproduce the GL_3 local L -factors” is false in the same way it was false in rank one: the unique KMS state sees the Perron eigenvalue only, the L -factors live in the partition function (Proposition 17.3.8); the “curse” of [Ch. 3] is not broken but located — it concerns the Hecke module, not the Hecke spectrum. (6) Of the specification’s four checks, $B \geq 0$, irreducibility and the spectral containment hold for L_E ; the four shift-equivalence identities hold for the intertwiner pair $(D, C^{(3)}D^{-1})$ between $C^{(3)}$ and C' (lag one) and for Ψ as a one-sided intertwiner into L_E . The trace identities $\mathrm{Tr} L_E^k = \mathrm{Tr}(C^{(3)})^k$ fail, and so does the Smith-form identity $\mathrm{coker}(I - L_E) \cong \mathbb{Z} \oplus \mathrm{Jac}^H$; both must fail, because L_E carries the additional parahoric spectrum (Table 17.1, line (T)).

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved except the items marked numeric in Table 17.1; Theorem 17.2.3 is certified by exact rational arithmetic. The statement of the weak Generalized Spectral Conjecture and its status are taken from the introduction of [58] (consulted), which attributes the weak form to [56, p. 253] and [57, p. 124]; those two pages were not consulted directly. The identification of the thick complexes with quotients of the building of PGL_3 over a specific field is not claimed. Bibliographic data of [56, 57, 58, 55, 54] were checked against publisher or arXiv pages.

Chapter 18

Algorithmic non-commutative class field theory, XII: $\tilde{A}_2$ towers — no abelian sector, the covering norm theorem, and the two digraph towers

Summary. The specification asked for a two-dimensional congruence tower of $\tilde{A}_2$ complexes, a recursion for the Hecke critical groups along it, and a “rank-two Iwasawa law” $\mu_2 p^{2k} + \mu_1 p^k - \chi_2 k + \nu_0$ with $\chi_2 = 2$. Three things are established, and they change the framework. First, thick $\tilde{A}_2$ complexes have $b_1 = 0$ — the vanishing of first L^2 -cohomology for groups acting on thick $\tilde{A}_2$ buildings (Garland, Ballmann–Świątkowski) — and $H_1(Y; \mathbb{Z})$ is finite, computed exactly on our complexes $((\mathbb{Z}/2)^6, (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14, (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2)$: there are no $\mathbb{Z}_\ell$ -voltage towers in rank two, the commutative Iwasawa theory of [Ch. 6] has no rank-two sector, and every tower is a finite abelian chain or a congruence tower. Second, along the finite abelian chains $X_3 \leftarrow Y_{21} \leftarrow Y_{84} \leftarrow Y_{168}$ (and through Y_{24}, Y_{42}) the covering pushforwards intertwine A_1, A_2 and the Hecke–Laplacian, are surjective on the Hecke critical groups, and their kernel orders obey the rank-two character mechanism $|\text{Jac}^H(Y_{k+1})|/|\text{Jac}^H(Y_k)| = \frac{n_k}{n_{k+1}} \prod_{\text{new } j} |P_j(1)|$, a product of twisted GL_3 local L -values, verified to 10^{-6} on five coverings together with its Hodge analogue. Third, the two natural digraph towers of an $\tilde{A}_2$ complex — the geodesic edge tower E_k (q^2 -regular, $n_k = n(q^2 + q + 1)q^{2(k-1)}$, the $(k, 0)$ -coset counts of PGL_3) and the chamber tower F_k (q -regular) — obey the line-digraph recursion of [Ch. 5] with degree q^2 and q , giving the exact laws $\text{ord}_p \kappa(E_k) = \frac{2n(q^2+q+1)}{q^2} p^{2k} - 2k + \nu$ and $\text{ord}_p \kappa(F_k) = \frac{3|F|}{q} p^k - k + \nu'$: the “higher Euler term” $\chi_2 = 2$ is $\text{ord}_p(q^2)$, the valuation of the regularity degree, one exponential term per tower and no mixed law. The sandpile of the geodesic edge digraph factors as an Eisenstein term, the product $\prod_j q^6 P_j(q^{-2})$ of GL_3 local L -values at $u = q^{-2}$, and the parahoric factor, by the edge-flow theorem of [Ch. 17]. A two-dimensional degeneracy calculus for the incidence matrices (S, T, M) and the three face maps d_i is proved, and it yields a combinatorial proof that A_1 and A_2 commute. The growth law of Jac^H along a genuine infinite congruence tower is left open, with the reason stated: no computable second level exists, and no abelian shortcut exists in principle.

Status of the results

Every numbered statement is proved except where marked “numeric”. The classical inputs are the vanishing theorem for the first L^2 -cohomology of groups acting on thick $\tilde{A}_2$ buildings [60, 59] (quoted; our complexes’

Betti numbers are also computed exactly), the line-digraph recursion [Ch. 5, Thm. 2.1], the edge-flow theorem [Ch. 17, Thm. 3.3], and the norm-surjectivity argument of [Ch. 6, Lem. 1.3]. All Smith forms are p -adic with the prime sets certified as in [Ch. 16, §6] or by determinants of maximal minors (Section 18.1). Section 18.6 records what is open and the corrections to the specification.

18.1 No abelian sector in rank two

Theorem 18.1.1 (No abelian towers). *Let $Y = \Gamma \backslash B$ be a finite quotient of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a group acting freely. Then $H^1(\Gamma; \mathbb{Q}) = H^1(Y; \mathbb{Q}) = 0$, i.e. $b_1(Y) = 0$, and $H_1(Y; \mathbb{Z})$ is finite. Consequently Y has no connected $\mathbb{Z}_\ell$ -voltage covers for any prime ℓ : every abelian cover of Y is a finite cover with group a quotient of the finite group $H_1(Y; \mathbb{Z})$, and the limit modules X_∞ of [Ch. 6] have no rank-two analogue.*

Proof. Γ acts properly and cocompactly on B ; by Garland’s vanishing theorem as extended to all $q \geq 2$ by Ballmann–Świątkowski [60, 59], $H^1(\Gamma; \mathbb{R}) = 0$ (indeed Γ has property (T)). Since Y is aspherical with $\pi_1(Y) = \Gamma$, $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z}) = \Gamma^{\text{ab}}$ is finite. A voltage cover with group $\mathbb{Z}/\ell^k$ is classified by a class in $H^1(Y; \mathbb{Z}/\ell^k) = \text{Hom}(H_1(Y; \mathbb{Z}), \mathbb{Z}/\ell^k)$, whose image has bounded order; a $\mathbb{Z}_\ell$ -tower would need unbounded ones. On Y_{21}, Y_{24}, Y_{42} the groups $H_1(Y; \mathbb{Z})$ are $(\mathbb{Z}/2)^6$, $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$, $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ (exact: p -adic Smith forms of the coordinate matrix of ∂_2 in a basis of $\ker \partial_1$, with the prime set certified by the gcd of maximal minors, 64, 448, 256); on Y_{84}, Y_{168} , $b_1 = 0$ by exact ranks (Table 18.1, line (B)). $\square$

Remark 18.1.2 (What this changes). In rank one every finite graph has $b_1 = r \geq 2$ and carries $\mathbb{Z}_\ell$ -voltage towers for every ℓ ; Papers II, IV, VII and IX built the commutative Iwasawa theory on them, and the Iwahori tower of [Ch. 5], [Ch. 9], [Ch. 10], [Ch. 12] was the non-normal exception. In rank two the exception is the rule: the only infinite towers of $\tilde{A}_2$ complexes are congruence-type towers with non-abelian (indeed property-(T)) deck structure, and the only abelian structure available is the finite group $H_1(Y; \mathbb{Z})$. The specification’s “rank-two Iwasawa theory” is therefore, from the start, the non-commutative theory that [Ch. 9] and [Ch. 10] asked for in rank one; nothing commutative can be borrowed.

18.2 The finite abelian chains and the covering norm theorem

The complexes of [Ch. 16] are abelian covers of the three-vertex type quotient X_3 of the Cartwright–Steger-type group with deck group $H_1(X_3; \mathbb{Z}) = \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14$; every quotient of that group gives a cover, and quotient maps give covering maps π between covers. We use the chains

$$X_3 \leftarrow Y_{21} \leftarrow Y_{84} \leftarrow Y_{168}, \quad Y_{21} \leftarrow Y_{42} \leftarrow Y_{168}, \quad Y_{24} \leftarrow Y_{168},$$

of degrees 7, 4, 2; 2, 4; 7. Write $P_\pi \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ for the vertex pushforward (sum over fibres), $A_i^{(k)}$ for the Hecke operators of Y_k , and

$$\Delta_k^H = (q^3 - 1)I + A_1^{(k)} - qA_2^{(k)}.$$

Theorem 18.2.1 (Covering norm theorem). *For a covering $\pi: Y_{k+1} \rightarrow Y_k$ of $\tilde{A}_2$ complexes:*

- (i) $P_\pi A_i^{(k+1)} = A_i^{(k)} P_\pi$ for $i = 1, 2$, and hence $P_\pi \Delta_{k+1}^H = \Delta_k^H P_\pi$;
- (ii) the induced norm π_* from $\text{coker } \Delta_{k+1}^H$ to $\text{coker } \Delta_k^H$ is surjective, preserves the degree functional, and restricts to a surjection of $\text{Jac}^H(Y_{k+1})$ onto $\text{Jac}^H(Y_k)$ whose kernel has order $|\text{Jac}^H(Y_{k+1})|/|\text{Jac}^H(Y_k)|$;
- (iii) (character mechanism) that order equals $\frac{n_k}{n_{k+1}} \prod_{j \text{ new}} |P_j(1)|$, the product over the joint eigenpairs of $(A_1^{(k+1)}, A_2^{(k+1)})$ not lifted from Y_k , i.e. over the nontrivial characters χ of the deck group and the joint eigenpairs of the χ -twisted operators $(A_1(\chi), A_2(\chi))$ on Y_k : a product of twisted GL_3 local L -values at $u = 1$. The same holds for the Hodge critical groups with $|P_j(1)|$ replaced by the Laplacian eigenvalues $2(q^2 + q + 1) - 2 \text{Re } a_j$.

Proof. (i) A covering is bijective on in- and out-arcs at every vertex, so fibre sums commute with the type-raising and type-lowering adjacencies; Δ^H is a polynomial in them. (ii) The argument of [Ch. 6, Lem. 1.3] applies verbatim: P_π is surjective, $\mathbf{1}^t \Delta^H = 0$ makes $\deg$ well defined, $\deg \circ \pi_* = \deg$, and Tor coker $\Delta^H = \ker \deg$ by [Ch. 16, Thm. 3.2(ii)]; surjectivity on torsion follows, and the kernel order is forced by counting. (iii) The deck group acts on Y_{k+1} commuting with A_1, A_2 , so the joint spectrum of $(A_1^{(k+1)}, A_2^{(k+1)})$ is the union over characters χ of the joint spectra of the twisted operators on Y_k ; the trivial character gives the spectrum of Y_k . Apply [Ch. 16, Thm. 3.2(ii)] at both levels and divide. Verified numerically to 10^{-6} on the five coverings, together with the exact integrality of the ratios and the exact intertwining identities (Table 18.1, line (A)). $\square$

Data

$|\text{Jac}^H|$ along the chains, with the 2-adic valuations: Y_{21} : $3 \cdot 7^{19}$ (2^0); Y_{42} : 2^2 -odd; Y_{24} : 2^{18} -odd; Y_{84} : 2^7 -odd; Y_{168} : 2^{18} -odd. The groups: $\text{Jac}^H(Y_{84}) = (\mathbb{Z}/7)^{18} \oplus (\mathbb{Z}/299593)^5 \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/29360114 \oplus \mathbb{Z}/117440456 \oplus \mathbb{Z}/352321368$ (new here); the others are in [Ch. 16]. Kernel orders of the norms have 2-adic valuations 11 ($Y_{168} \rightarrow Y_{84}$), 7 ($Y_{84} \rightarrow Y_{21}$), 0 ($Y_{168} \rightarrow Y_{24}$), 16 ($Y_{168} \rightarrow Y_{42}$), 2 ($Y_{42} \rightarrow Y_{21}$). There is no monotone p -adic law along these finite chains, and none is to be expected: a finite abelian chain is not a tower.

18.3 Two-dimensional degeneracy calculus

Let $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ be the tail, head and third-vertex incidence matrices of [Ch. 17, §3], and $d_0, d_1, d_2 \in M_{|E| \times |F|}(\mathbb{Z})$ the three face maps sending a chamber (a, b, c) (types 0, 1, 2) to its typed edges $(a \rightarrow b)$, $(b \rightarrow c)$, $(c \rightarrow a)$.

Theorem 18.3.1 (Incidence calculus in rank two). *For an $\tilde{A}_2$ complex satisfying [Ch. 16, Def. 1.1(i)–(iv)] with $\text{PG}(2, q)$ links:*

$$\begin{aligned} ST^t &= A_1, & TS^t &= A_2, & SS^t &= TT^t = (q^2 + q + 1)I, \\ SM^t &= (q + 1)A_2, & TM^t &= (q + 1)A_1, \\ MM^t &= (q^2 + q + 1)(q + 1)I + A_{(1,1)}, \\ A_{(1,1)} &:= A_1 A_2 - (q^2 + q + 1)I = A_2 A_1 - (q^2 + q + 1)I, \\ \partial_1 &= T - S, & \Delta_0 &= (T - S)(T - S)^t = 2(q^2 + q + 1)I - A_1 - A_2, \\ \partial_2 &= d_0 + d_1 + d_2, \\ \sum_i d_i d_i^t &= (q + 1)I_E, & d_i^t d_j &= 0 \ (i \neq j), \\ K &:= \sum_i d_i d_{i+1}^t, & L_E &= T^t S - K. \end{aligned}$$

In particular $A_1 A_2 = A_2 A_1$: the two Hecke operators commute, by the symmetry of MM^t .

Proof. The first line is [Ch. 17, proof of Thm. 3.3] and Lemma [Ch. 17, 3.2(a)]: $(SM^t)(x, w)$ counts edges $x \rightarrow y$ whose chamber contains w , i.e. chambers on the edge $\{x, w\}$ with w a type-raising-twice neighbour, $q + 1$ of them; similarly TM^t . For MM^t : the diagonal counts chambers through v , $(q^2 + q + 1)(q + 1)$; for $v \neq w$, $(MM^t)(v, w)$ counts edges $x \rightarrow y$ with $\{x, y, v\}$ and $\{x, y, w\}$ chambers, so x is a common out-neighbour of v, w and y a common in-neighbour; given the common out-neighbour x , in the link of x the vertices v, w are two distinct points and y must be a line through both — exactly one — so $(MM^t)(v, w) = \#\{x: v \rightarrow x, w \rightarrow x\} = (A_1 A_2)(v, w)$ for $v \neq w$, and the diagonal of $A_1 A_2$ is $q^2 + q + 1$. Since MM^t is symmetric, $A_1 A_2 = (A_1 A_2)^t = A_2 A_1$. The Laplacian line is $\partial_1 = T - S$ expanded with the first line. The face maps: every chamber has one edge of each type, so $\sum_i d_i d_i^t$ is the diagonal matrix of chambers per edge, $q + 1$; $d_i^t d_j = 0$ for $i \neq j$ because an edge has one type; $K(e, e')$ counts chambers in which e, e' are consecutive, which is the chamber-continuation term subtracted in [Ch. 17, Def. 3.1]. Verified exactly on Y_{168} (Table 18.1, line (D)). $\square$

Remark 18.3.2 (The three degeneracy operators). In rank one, [Ch. 9] had two degeneracy maps τ, η with $\tau_*\eta^* = A_k$. In rank two the three face maps $d_i: C_2 \rightarrow C_1$ together with the two vertex maps S, T generate both Hecke operators, their commutation, the Hodge Laplacian Δ_0 , the chamber continuation K and the edge flow L_E : the whole vertex- and edge-level Hecke theory of the complex is a polynomial identity in (S, T, M, d_0, d_1, d_2) , and the identity $A_1 A_2 = A_2 A_1$ — normally quoted from the commutativity of the spherical Hecke algebra — is here a count of lines through two points.

18.4 The two digraph towers

Definition 18.4.1 (Edge and chamber towers). For an $\tilde{A}_2$ complex Y with the hypotheses of Theorem 18.3.1 let $E_1 := (E, L_E)$ be the geodesic edge digraph and $F_1 := (F^\pm, L_F)$ the chamber digraph of [Ch. 17, Rem. 3.6] (oriented chambers $(a, b, c) \rightarrow (b, c, d)$, $d \neq a$), and for $k \geq 1$ let $E_{k+1} := L(E_k)$, $F_{k+1} := L(F_k)$ be their line digraphs. Vertices of E_k are geodesic type-raising galleries of k edges; their number is $n_k(E) = n(q^2 + q + 1)q^{2(k-1)}$, the number of vertices at coweight distance $(k, 0)$ from a vertex of the building, i.e. the index of the PGL_3 congruence subgroup fixing a $(k, 0)$ -flag: E_k is the rank-two Iwahori path tower in the $(k, 0)$ direction.

Theorem 18.4.2 (Recursions and growth laws). E_1 is Eulerian q^2 -regular and strongly connected, F_1 is Eulerian q -regular and strongly connected (exact on Y_{168} ; in general by [Ch. 17, Lem. 3.2(a)] and its chamber analogue). Hence, by [Ch. 5, Thm. 2.1] applied with $d = q^2$ and $d = q$, for all $k \geq 1$

$$\kappa(E_{k+1}) = (q^2)^{n_k(E)(q^2-1)-1} \kappa(E_k), \quad \kappa(F_{k+1}) = q^{n_k(F)(q-1)-1} \kappa(F_k),$$

and for $q = p$ the exact growth laws

$$\mathrm{ord}_p \kappa(E_k) = \frac{2n(q^2 + q + 1)}{q^2} p^{2k} - 2k + \nu_E, \quad \mathrm{ord}_p \kappa(F_k) = \frac{3|F|}{q} p^k - k + \nu_F,$$

with constants fixed by $\kappa(E_1), \kappa(F_1)$. The nonzero spectra of A_{E_k} and of A_{F_k} are independent of k (spectral blindness), so the zeta function of E_k is $\det(I - uL_E)^{-1}$ at every level and contains the whole nontrivial GL_3 Hecke spectrum [Ch. 17, Thm. 3.3].

Proof. Regularity: each typed edge has q^2 continuations and q^2 predecessors ([Ch. 17, Lem. 3.2(a)] and its reverse); each oriented chamber has q successors and q predecessors (the $q + 1$ chambers on an edge minus the one you came from). Line digraphs of Eulerian d -regular strongly connected digraphs are again such, on d times as many vertices, with $\det(zI - A_{L(G)}) = z^{m-n} \det(zI - A_G)$ [Ch. 5, eq. (2.1)]; the recursion is [Ch. 5, Thm. 2.1]; summing, $\sum_{j < k} 2(n_j(E)(q^2 - 1) - 1) = 2n_1(E)(q^{2(k-1)} - 1) - 2(k - 1)$ gives the edge law and likewise the chamber law. Sanity check of [Ch. 5, Thm. 2.1] with $d = 4$ on the complete digraph K_5 : $\kappa = 125$, $\kappa(L) = 4^{14} \cdot 125$ (Table 18.1, line (L)). $\square$

Remark 18.4.3 (The higher Euler term). The specification predicted $\chi_2 = 2$. It is right, and for a transparent reason: in [Ch. 7, Thm. 4.1(iv)] the Euler term of the rank-one Iwahori tower was identified as $-\mathrm{ord}_p n_k$, the valuation of the vertex count in the residue normalization $\prod_{\mu \neq 0} \mu = n_k \kappa_k$; here $n_k(E) = n_1 q^{2(k-1)}$ has $\mathrm{ord}_p n_k = 2k + \mathrm{const}$, so the Euler term is $-2k$, and for the chamber tower $-k$. In general the Euler term of a line-digraph tower of degree d is $-\mathrm{ord}_p(d) \cdot k$: one Eisenstein direction removed per level, weighted by the p -adic size of the regularity degree. There is no mixed law $\mu_2 p^{2k} + \mu_1 p^k$: each tower has one exponential term, with μ the arc density n_1/d times $\mathrm{ord}_p d$.

Proposition 18.4.4 (The sandpile of the geodesic edge digraph). With the notation of [Ch. 17, Cor. 3.5],

$$\kappa(E_1) = \frac{3q^4}{n_1} \cdot \prod_{j \text{ nontrivial}} q^6 P_j(q^{-2}) \cdot \prod_{\lambda \text{ parahoric}} (q^2 - \lambda),$$

where $q^6 P_j(q^{-2}) = \prod_i (q^2 - \lambda_{j,i})$ is, up to the factor q^6 , the inverse local L -value of the j -th eigenform at $u = q^{-2}$, and the parahoric factor runs over the $4n + 6$ eigenvalues of L_E of modulus $q^{1/2}$. On Y_{168} (numeric): $\log \kappa(E_1) = 1623.9567$, decomposed as $\log(3q^4/n_1) = -3.1987$, pencil part 684.7469, parahoric part 942.4085, which add up to 1623.9567.

Proof. E_1 is Eulerian and strongly connected, so [Ch. 5, eq. (2.2)] gives $n_1\kappa(E_1) = \prod_{\mu \neq 0} \mu$ over the nonzero eigenvalues of $q^2I - L_E$, i.e. over all eigenvalues $\lambda \neq q^2$ of L_E . By [Ch. 17, Cor. 3.5] these are: the two rotations $q^2\omega, q^2\omega^2$, contributing $(q^2 - q^2\omega)(q^2 - q^2\omega^2) = 3q^4$; the three roots of each nontrivial pencil P_j ; and the parahoric eigenvalues. Exact $\kappa(E_1)$ (a 1176×1176 determinant of size e^{1624}) is not computed. $\square$

18.5 The verification battery

check	complexes	result
(B) $b_1 = 0$; $H_1(Y; \mathbb{Z}) = (\mathbb{Z}/2)^6, (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14, (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$; prime sets certified by minor gcds 64, 448, 256	Y_{21}, Y_{24}, Y_{42} (b_1 also Y_{84}, Y_{168})	exact
(A) $P_\tau A_i = A_i P_\tau$, $P_\tau \Delta^H = \Delta^H P_\tau$; Jac^H at five levels; ratios integral; character mechanism to 10^{-6} ; Hodge analogue	five coverings	exact / numeric
(D) the fourteen identities of Theorem 18.3.1, including $A_1 A_2 = A_2 A_1$	Y_{168}	exact
(E) E_1 : q^2 -regular in and out, strongly connected, $n_1 = 1176$; spectral accounting of $\kappa(E_1)$; constants $\mu_2 = 588, \chi_2 = 2$	Y_{168}	exact / numeric
(F) F_1 : q -regular in and out, 3528 vertices; constants $\mu_1 = 1764, \chi = 1$	Y_{168}	exact
(L) [Ch. 5, Thm. 2.1] with $d = 4$ on K_5 ; $\kappa(L) = 4^{14}\kappa$	—	exact

Table 18.1: Verification (`pgl3_tower.py`).

18.6 Scope, residue, corrections

Remark 18.6.1 (Solved, open, impossible). *Solved*: the non-existence of abelian towers in rank two and its consequence for the series’ framework; the covering norm theorem with the rank-two character mechanism on the finite abelian chains; the two-dimensional incidence calculus with the combinatorial commutation of A_1, A_2 ; the edge and chamber digraph towers with their exact recursions and growth laws and the identification of χ_2 ; the L -value structure of the sandpile of the geodesic edge digraph. *Open*: (a) the growth law of Jac^H along a genuine infinite congruence tower $\Gamma \supset \Gamma_1 \supset \dots$ of $\tilde{A}_2$ complexes — no second level is computable here (the maximal abelian cover of Y_{168} would have $168 \cdot |H_1(Y_{168})|$ vertices), and Theorem 18.1.1 shows no abelian shortcut exists; (b) an exact value of $\kappa(E_1)$ on Y_{168} and the p -adic constant ν_E ; (c) a closed form for the parahoric factor of Proposition 18.4.4, which should be Kang–Li’s chamber factor at $u = q^{-2}$; (d) the pro-module of an infinite congruence tower — which, by Theorem 18.1.1, is a module over a non-commutative completed Hecke algebra from the start, the object [Ch. 9, Prob. 5.1] asked for in rank one. *Impossible or false as stated*: a $\mathbb{Z}_p$ -tower of $\tilde{A}_2$ complexes; a two-term law $\mu_2 p^{2k} + \mu_1 p^k$ for a single line-digraph tower.

Remark 18.6.2 (Corrections to the task specification). (1) “ $X_\infty^{(2)} = \varprojlim (\text{Jac}^H(Y_k) \otimes \mathbb{Z}_p)$ ” over an abelian tower does not exist: there are no abelian towers (Theorem 18.1.1); over the finite chains the inverse limit is just the top group. (2) The proposed recursion $|\text{Jac}^H(Y_{k+1})| = q^{f(n_k, E_k, F_k)} |\text{Jac}^H(Y_k)|$ is not what happens along coverings of $\tilde{A}_2$ complexes: the ratio is a product of twisted L -values (Theorem 18.2.1(iii)), not a pure power of q (its 2-adic valuations along our chains are 11, 7, 0, 16, 2); a pure-power recursion is what the *digraph* towers obey (Theorem 18.4.2), for the ordinary sandpile groups, not for Jac^H . (3) The law $\mu_2 p^{2k} + \mu_1 p^k - \chi_2 k + \nu_0$ splits into two separate one-term laws, one per tower; $\chi_2 = 2$ is $\text{ord}_p(q^2)$ (Remark 18.4.3). (4) The “2D line complex” $\mathcal{L}_2(Y_0)$ is a digraph, not a two-dimensional complex; the levels E_k carry no A_2 , and the Hecke–Laplacian Δ_k^H is not defined on them — the Hecke data live at level one, in $\det(I - uL_E)$, and are constant along the tower.

Acknowledgement of provenance and caveats

Unrefereed preprint. Every numbered statement is proved from Papers I–XI and finite combinatorics, except the vanishing of b_1 for general thick quotients, which is quoted from [60, 59] and independently computed on our five complexes. Spectral statements marked numeric are floating point. Bibliographic data of [59] were checked against the publisher page and those of [60] against the JSTOR record.

Chapter 19

Algorithmic non-commutative class field theory, XV: the parahoric factorization of the geodesic edge flow, the unitarity of the remainder, and the chamber companion

Summary. [Ch. 17, Cor. 3.5] found numerically, on one $\tilde{A}_2$ complex, that the characteristic polynomial of the geodesic edge flow L_E factors as $\det(I - uL_E)(1 - u^3)(1 - q^3u^3) = \det P(u)G(u)$, with P the cubic Hecke pencil and G a polynomial of degree $4n + 6$ all of whose roots lie on $|u| = q^{-1/2}$, and left the proof and the identification of G open. This paper proves the identity for every $\tilde{A}_2$ complex of order q , identifies G , and proves the Riemann hypothesis for it unconditionally. The mechanism is a two-sided intertwining: the image of $\Psi = [S^t T^t M^t]$ is invariant under L_E and L_E^t , so its orthogonal complement $W = \ker S \cap \ker T \cap \ker M$ reduces L_E , and the local identities $L_E L_E^t = qI + (q^2 - q)T^t T$, $L_E^t L_E = qI + (q^2 - q)S^t S$ — two points of a projective plane lie on one line — make $R := L_E|_W$ satisfy $RR^t = R^t R = qI$. Hence $R/\sqrt{q}$ is orthogonal, $G(u) = \det(I - uR)$, and every root of G has $|u| = q^{-1/2}$ for every complex, Ramanujan or not; the Ramanujan property of the complex lives entirely in the vertex pencil. The same calculus on pointed chambers gives the Kang–Li operator L_B a companion B on $\ell^2(E)^3$, with $L_B \Phi = \Phi B$ and $\det(I + uB) = \det((1 + qu^3)I + uK + u^2 K^t) = (1 - u^3)^n \det(I - uL_E) \det(I - u^2 L_E) / \det P(u)$: the Kang–Li zeta identity in companion form, with the parahoric block $G(u)G(u^2)$ explaining the eigenvalues of modulus $q^{1/4}$ of the chamber operator as square roots of the unitary remainder. A correction to [Ch. 17, Def. 3.1] is recorded (the geodesic condition is “not in a common chamber”). Verified exactly on $Y_{21}, Y_{24}, Y_{42}, T_6$ and structurally on Y_{168} .

Status of the results

Every numbered statement is proved; the proofs use only the link structure of an $\tilde{A}_2$ complex of order q ([Ch. 16, Def. 1.1(i)–(iii)], vertex links the incidence graph of $\text{PG}(2, q)$), linear algebra over $\mathbb{Q}$, and the results of [Ch. 17], [Ch. 18] that are re-derived where used. Kang–Li [54] and Kang–Li–Wang [61] are cited for context and for the one statement we do not reprove (the relation between L_B and its companion for $q \neq 2$, Corollary 19.4.3); Lubotzky–Samuels–Vishne [62] for the notion of Ramanujan complex. The flag condition [Ch. 16, Def. 1.1(iv)] is not assumed anywhere.

19.1 Setting, and a correction

Y is a finite $\tilde{A}_2$ complex of order q : a connected pure 2-dimensional simplicial complex with a $\mathbb{Z}/3$ vertex typing, n vertices, typed edges E ($e = (x \rightarrow y)$, $\tau(y) = \tau(x) + 1$, $|E| = n(q^2 + q + 1) =: nN$), chambers F (one vertex of each type, $|F| = (q+1)|E|/3$), such that the link of every vertex y is the incidence graph of $\text{PG}(2, q)$: in-neighbours of y are the points, out-neighbours the lines, and $\{x, y, z\}$ is a chamber iff the point x lies on the line z . $A_1, A_2 = A_1^t$ are the Hecke operators, $P(u) := I - A_1 u + q A_2 u^2 - q^3 u^3$ the pencil, $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ the tail, head and third-vertex incidences, $K := \sum_i d_i d_{i+1}^t$ the chamber continuation ($K(e, e') = \#\{\text{chambers in which } e' \text{ follows } e\}$), all as in [Ch. 17, §3] and [Ch. 18, Thm. 3.1].

Definition 19.1.1 (Geodesic edge flow, chamber form). $L_E \in M_{|E|}(\{0, 1\})$, $L_E((x \rightarrow y), (y \rightarrow z)) = 1$ iff $(y \rightarrow z) \in E$ and $\{x, y, z\} \notin F$; equivalently $L_E = T^t S - K$ [Ch. 18, Thm. 3.1].

Remark 19.1.2 (Correction to [Ch. 17, Def. 3.1]). [Ch. 17, Def. 3.1] used “ z not adjacent to x ”. The two conditions agree exactly when Y is a flag complex ([Ch. 16, Def. 1.1(iv)]), which holds for Y_{168} , the only complex on which [Ch. 17] evaluated L_E . On Y_{21} , whose 1-skeleton is $K_{7,7,7}$, every z is adjacent to x and the adjacency version is the zero matrix, while Definition 19.1.1 gives row sums q^2 ; on Y_{24} and Y_{42} the two versions differ as well (Table 19.1, line (D)). Definition 19.1.1 is the projection of the geodesic operator of the building, the operator L_E of [54], and every statement of [Ch. 17, §3] and [Ch. 18] holds for it with the proofs unchanged; the quoted numerics of [Ch. 17] concern Y_{168} and are unaffected.

Lemma 19.1.3 (Local counts). For $e = (x \rightarrow y)$, $e' = (x' \rightarrow y)$ with the same head, and $f \in \ell^2(V)$, $m_f := M^t f$:

- (a) e has q^2 geodesic continuations and q^2 geodesic predecessors;
- (b) $\sum_{z: e \rightarrow (y \rightarrow z)} f(z) = (A_1 f)(y) - m_f(e)$, $\sum_z m_f(y \rightarrow z) = q((A_2 f)(y) - f(x))$;
- (c) $\#\{\text{common geodesic continuations of } e, e'\} = q^2$ if $x = x'$ and $q^2 - q$ if $x \neq x'$; dually for two edges with the same tail;
- (d) $\sum_{x: (x \rightarrow y) \rightarrow e''} f(x) = (A_2 f)(y) - m_f(e'')$ and $\sum_x m_f(x \rightarrow y) = q((A_1 f)(y) - f(z))$ for $e'' = (y \rightarrow z)$.

Proof. In the link of y the continuations of e are the lines z not through the point x : $N - (q+1) = q^2$. (b) is [Ch. 17, Lem. 3.2(b),(c)] with “chamber” in place of “adjacent”, the proof verbatim. (c): lines through neither of two distinct points: $N - 2(q+1) + 1 = q^2 - q$. (d) is (b) for the opposite complex (all edges reversed, types negated), whose L_E is L_E^t . $\square$

19.2 Two-sided intertwining and the unitary remainder

Let $\Psi := [S^t \ T^t \ M^t]: \ell^2(V)^3 \rightarrow \ell^2(E)$ and $W := (\text{im } \Psi)^\perp = \ker S \cap \ker T \cap \ker M$.

Theorem 19.2.1 (Two-sided intertwining). With $I = I_n$ and block matrices over $\ell^2(V)^3$,

$$\begin{aligned} L_E \Psi &= \Psi C', & C' &= \begin{pmatrix} 0 & 0 & -qI \\ q^2 I & A_1 & qA_2 \\ 0 & -I & 0 \end{pmatrix}; & L_E^t \Psi &= \Psi C'', & C'' &= \begin{pmatrix} A_2 & q^2 I & qA_1 \\ 0 & 0 & -qI \\ -I & 0 & 0 \end{pmatrix}; \\ K \Psi &= \Psi \hat{C}, & \hat{C} &= \begin{pmatrix} 0 & 0 & qI \\ (q+1)I & 0 & A_2 \\ 0 & I & 0 \end{pmatrix}; & K^t \Psi &= \Psi \hat{C}', & \hat{C}' &= \begin{pmatrix} 0 & (q+1)I & A_1 \\ 0 & 0 & qI \\ I & 0 & 0 \end{pmatrix}; \\ L_E L_E^t &= qI_E + (q^2 - q)T^t T, & L_E^t L_E &= qI_E + (q^2 - q)S^t S. \end{aligned}$$

Consequently $\text{im } \Psi$ and W are invariant under L_E, L_E^t, K, K^t , and $\ell^2(E) = \text{im } \Psi \oplus W$ is an orthogonal decomposition reducing all four.

Proof. The first identity is [Ch. 17, Thm. 3.3], whose proof uses only Lemma 19.1.3(a),(b). For L_E^t : $(L_E^t S^t f)(y \rightarrow z) = \sum_x f(x)$ over geodesic predecessors, which is $(A_2 f)(y) - m_f(y \rightarrow z)$ by (d), i.e. $L_E^t S^t = S^t A_2 - M^t$; $(L_E^t T^t f)(y \rightarrow z) = q^2 f(y)$ by (a), i.e. $L_E^t T^t = q^2 S^t$; $(L_E^t M^t f)(y \rightarrow z) = \sum_x m_f(x \rightarrow y) = q((A_1 f)(y) - f(z))$ by (d), i.e. $L_E^t M^t = q S^t A_1 - q T^t$. Reading the three columns gives C'' . For $K = T^t S - L_E$: $T^t S \Psi = T^t [S S^t S T^t S M^t] = T^t [N I A_1 (q+1) A_2]$ by [Ch. 18, Thm. 3.1], and subtracting $L_E \Psi = \Psi C'$ gives $K S^t = (q+1) T^t$, $K T^t = M^t$, $K M^t = T^t A_2 + q S^t$, i.e. $\widehat{C}$; $K^t = S^t T - L_E^t$ likewise gives $\widehat{C}'$. The two product identities are Lemma 19.1.3(c): $(L_E L_E^t)(e, e')$ counts common continuations, zero unless the heads agree, q^2 on the diagonal and $q^2 - q$ off it, and $(T^t T)(e, e') = [\text{head}(e) = \text{head}(e')]$. Invariance of $\text{im } \Psi$ under an operator X and of $W = (\text{im } \Psi)^\perp$ under X^t are the same statement; since all four operators and their transposes preserve $\text{im } \Psi$, both summands are invariant under all four. Verified as exact integer identities on all five complexes (Table 19.1, line (D)). $\square$

Theorem 19.2.2 (The remainder is $\sqrt{q}$ times an orthogonal matrix). *Let $R := L_E|_W$. Then $R^t = L_E^t|_W$ and*

$$R R^t = R^t R = q I_W.$$

Hence R is invertible, every eigenvalue of R has modulus $\sqrt{q}$, and $G(u) := \det(I - uR) \in \mathbb{Z}[u]$ has all its roots on the circle $|u| = q^{-1/2}$, with $G(u) = (-u)^d \det(R) G(1/(qu))$, $d = \dim W$, $\det R = \pm q^{d/2}$.

Proof. W is invariant under L_E and L_E^t (Theorem 19.2.1), so the restriction of L_E^t to W is the adjoint of R . For $v \in W$, $Tv = 0$ and $Sv = 0$, so the product identities give $L_E L_E^t v = qv$ and $L_E^t L_E v = qv$. Thus $R/\sqrt{q}$ is orthogonal, its eigenvalues have modulus one, and $\det(I - uR) = \det(-uR) \det(I - (uR)^{-1}) = (-u)^d \det R \cdot \det(I - \frac{1}{qu} R^t)$. $G \in \mathbb{Z}[u]$ because $W \cap \mathbb{Z}^E$ is a saturated L_E -invariant sublattice of full rank in W . On the five complexes the eigenvalues of R have modulus $\sqrt{q}$ to 10^{-12} (Table 19.1, line (U)). $\square$

Remark 19.2.3 (What this says about the Riemann hypothesis). Kang–Li–Wang [61, Thm. 1.5.1] prove that Y is Ramanujan iff the nontrivial zeros of $\det P(u)$ have $|u| = q^{-1}$, iff the nontrivial zeros of $\det(I - uL_E)$ have $|u| \in \{q^{-1}, q^{-1/2}\}$. Theorem 19.2.2 shows that the second clause of the edge criterion is automatic: the zeros of $\det(I - uL_E)$ off the pencil are always on $|u| = q^{-1/2}$. The Riemann hypothesis for the zeta function of an $\tilde{A}_2$ complex is therefore exactly the Ramanujan property of the *vertex* spectrum, and the “parahoric” zeros are on the critical line for every complex, Ramanujan or not — for the same reason that the Hashimoto remainder of a graph is ± 1 : it is a local, not a global, statement.

19.3 The factorization theorem

Theorem 19.3.1 (Parahoric factorization). *For every $\tilde{A}_2$ complex of order q , with $K_Y(u) := \det(I - uC'|_{\ker \Psi})$,*

$$\begin{aligned} \det(I - uL_E) \cdot K_Y(u) &= \det P(u) \cdot G(u), \\ G(u) &= \det(I - uL_E|_W) \in \mathbb{Z}[u], \quad \deg G = \dim W = |E| - \text{rk } \Psi, \end{aligned}$$

K_Y divides $\det P$, and all roots of G lie on $|u| = q^{-1/2}$.

Proof. By Theorem 19.2.1, $L_E = L_E|_{\text{im } \Psi} \oplus R$, so $\det(I - uL_E) = \det(I - uL_E|_{\text{im } \Psi}) G(u)$. Ψ induces an isomorphism $\ell^2(V)^3 / \ker \Psi \rightarrow \text{im } \Psi$ intertwining the map induced by C' with $L_E|_{\text{im } \Psi}$, and $\ker \Psi$ is C' -invariant ($\Psi C'v = L_E \Psi v = 0$); hence $\det(I - uC') = K_Y(u) \det(I - uL_E|_{\text{im } \Psi})$. Finally $\det(I - uC') = \det P(u)$: the blocks of C' commute, and the formal determinant of $I - uC'$ expanded along the first block row is $(I - uA_1 + qu^2 A_2) + qu \cdot (-q^2 u^2) = P(u)$ [Ch. 17, Prop. 3.4]. Multiply out. Degree and roots: Theorem 19.2.2, R invertible. $\square$

Proposition 19.3.2 (The kernel of Ψ). *The Gram matrix*

$$\Psi^t \Psi = \begin{pmatrix} NI & A_1 & (q+1)A_2 \\ A_2 & NI & (q+1)A_1 \\ (q+1)A_1 & (q+1)A_2 & qNI + A_1 A_2 \end{pmatrix}$$

lies in M_3 of the commutative algebra $\mathbb{Q}[A_1, A_2]$; for a joint eigenvector with $A_1 f = \lambda f$, $A_2 f = \bar{\lambda} f$ it is the Hermitian matrix

$$H(\lambda) = \begin{pmatrix} N & \lambda & (q+1)\bar{\lambda} \\ \bar{\lambda} & N & (q+1)\lambda \\ (q+1)\lambda & (q+1)\bar{\lambda} & qN + |\lambda|^2 \end{pmatrix},$$

$$\det H(\lambda) = (qN + |\lambda|^2)(N^2 - |\lambda|^2) - 2N(q+1)^2|\lambda|^2 + 2(q+1)^2 \operatorname{Re} \lambda^3,$$

and $\dim \ker \Psi = \sum_{\lambda} \dim \ker H(\lambda)$ over the eigenvalues of A_1 with multiplicity. For the three trivial eigenvalues $\lambda = N\omega^j$, H has rank one, so $\dim \ker \Psi \geq 6$ [Ch. 17, Cor. 3.5(i)]. Say Y is non-degenerate (ND) if $\det H(\lambda) \neq 0$ for every nontrivial eigenvalue. Under (ND): $\ker \Psi$ is the 6-dimensional space of type-constant triples (f, g, h) with $f_i + g_{i+1} + (q+1)h_{i+2} = 0$,

$$K_Y(u) = (1 - u^3)(1 - q^3 u^3), \quad \deg G = 4n + 6,$$

and Theorem 19.3.1 is the identity of [Ch. 17, Cor. 3.5(ii)]. $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ satisfy (ND); the thin torus T_6 ($q = 1$) does not (Table 19.1, line (K)).

Proof. The block entries are [Ch. 18, Thm. 3.1]; A_1 is normal, so $\ell^2(V)$ is an orthogonal sum of joint eigenlines and $\Psi^t \Psi$ is the direct sum of the $H(\lambda)$. Expanding $\det H$ with $x = \lambda$, $y = \bar{\lambda}$, $r = xy$, $c = q+1$, $D = qN + r$: $N(ND - c^2 r) - x(yD - c^2 x^2) + cy(cy^2 - cNx) = D(N^2 - r) - 2Nc^2 r + c^2(x^3 + y^3)$. For $\lambda = N$ the rows are (N, N, cN) , (N, N, cN) , $(cN, cN, (c-1)N + N^2)$, and $(c-1)N + N^2 - c^2 N = N(q + N - (q+1)^2) = 0$, so the rank is one; the ω -rotations are conjugate. Under (ND) the kernel is the 6-dimensional type-constant space described (it is C' -invariant and contained in the 9-dimensional space V_0 of type-constant triples). C' on V_0 has characteristic polynomial $\prod_j P_{N\omega^j}(u) = (1 - u^3)(1 - q^3 u^3)(1 - q^6 u^3)$, and on the quotient $V_0 / \ker \Psi \cong \Psi(V_0)$, the type-constant edge functions, L_E acts by q^2 times the type rotation, with characteristic polynomial $1 - q^6 u^3$; dividing gives K_Y . Then $\deg G = |E| - (3n - 6) = 4n + 6$. $\square$

19.4 The chamber companion and the Kang–Li identity

A *pointed chamber* is a pair (e, z) with $e = (x \rightarrow y) \in E$ and $\{x, y, z\} \in F$; write it $(x, y; z)$. There are $|P| = 3|F| = (q+1)|E|$ of them. The Kang–Li operator [54] is

$$L_B((x, y; z), (y, z; w)) = 1 \quad \text{iff} \quad w \neq x,$$

row sums q ; it is the chamber flow L_F of [Ch. 17, Rem. 3.6] and [Ch. 18, Def. 4.1]. Let $\sigma, \rho, \tau: P \rightarrow E$ send $(x, y; z)$ to $(x \rightarrow y)$, $(y \rightarrow z)$, $(z \rightarrow x)$, and $\Phi := [\sigma^* \rho^* \tau^*]: \ell^2(E)^3 \rightarrow \ell^2(P)$, $\Phi(f, g, h)(x, y; z) = f(x, y) + g(y, z) + h(z, x)$.

Theorem 19.4.1 (Chamber companion).

$$L_B \Phi = \Phi B, \quad B = \begin{pmatrix} 0 & 0 & -I_E \\ qI_E & K & K^t \\ 0 & -I_E & 0 \end{pmatrix}, \quad \det(I + uB) = \det((1 + qu^3)I_E + uK + u^2 K^t).$$

Proof. All sums run over the q vertices $w \neq x$ with $\{y, z, w\} \in F$; with $K((y, z), (z, w)) = [\{y, z, w\} \in F]$,

$$\begin{aligned} (L_B \sigma^* f)(x, y; z) &= \sum_w f(y, z) = q(\rho^* f)(x, y; z), \\ (L_B \rho^* g)(x, y; z) &= \sum_w g(z, w) = (Kg)(y, z) - g(z, x) = (\rho^* Kg - \tau^* g)(x, y; z), \\ (L_B \tau^* h)(x, y; z) &= \sum_w h(w, y) = (K^t h)(y, z) - h(x, y) = (\rho^* K^t h - \sigma^* h)(x, y; z). \end{aligned}$$

Reading the three columns gives B . For the determinant, permute the block rows and columns of $I + uB$ to

$$\begin{pmatrix} I & -uI & 0 \\ 0 & I & -uI \\ quI & uK^t & I + uK \end{pmatrix};$$

the upper-left 2×2 block is unipotent with inverse $\begin{pmatrix} I & uI \\ 0 & I \end{pmatrix}$, and the Schur complement is

$$I + uK - (quI \quad uK^t) \begin{pmatrix} I & uI \\ 0 & I \end{pmatrix} \begin{pmatrix} 0 \\ -uI \end{pmatrix} = I + uK + qu^3I + u^2K^t.$$

No commutation of K, K^t is used. Exact on all five complexes (Table 19.1, line (B)). $\square$

Theorem 19.4.2 (Kang–Li identity in companion form). *For an $\tilde{A}_2$ complex of order q satisfying (ND),*

$$\det((1 + qu^3)I_E + uK + u^2K^t) = (1 - u^3)^n \frac{\det(I - uL_E) \det(I - u^2L_E)}{\det P(u)},$$

and the restriction of B to W^3 has characteristic polynomial $G(u)G(u^2)$.

Proof. $\ell^2(E)^3 = (\text{im } \Psi)^3 \oplus W^3$ orthogonally, both summands B -invariant (Theorem 19.2.1). On W , $K = T^tS - L_E$ acts as $-R$ and K^t as $-R^t$, so $B|_{W^3} = \begin{pmatrix} 0 & 0 & -I \\ qI & -R & -R^t \\ 0 & -I & 0 \end{pmatrix}$ and, by the elimination of Theorem 19.4.1, $\det(I + uB|_{W^3}) = \det((1 + qu^3)I - uR - u^2R^t) = \det((I - uR)(I - u^2R^t)) = G(u)G(u^2)$, using $RR^t = qI$ (Theorem 19.2.2) and $\det(I - u^2R^t) = G(u^2)$. On $(\text{im } \Psi)^3$, $\Psi^{\oplus 3}$ intertwines B with $\hat{B} = \begin{pmatrix} 0 & 0 & -I \\ qI & \hat{C} & \hat{C}' \\ 0 & -I & 0 \end{pmatrix}$ on $\ell^2(V)^9$ modulo $(\ker \Psi)^3$, so $\det(I + uB|_{(\text{im } \Psi)^3}) = \det(I + u\hat{B}) / \det(I + u\hat{B}|_{(\ker \Psi)^3})$. The same elimination gives $\det(I + u\hat{B}) = \det((1 + qu^3)I + u\hat{C} + u^2\hat{C}')$, a 3×3 block matrix over $\mathbb{Q}[A_1, A_2]$; on a joint eigenline $(a, b) = (\lambda, \bar{\lambda})$ it is

$$D_{a,b}(u) = \det \begin{pmatrix} 1 + qu^3 & (q+1)u^2 & qu + au^2 \\ (q+1)u & 1 + qu^3 & bu + qu^2 \\ u^2 & u & 1 + qu^3 \end{pmatrix} = (1 - u^3)(1 - bu^2 + qau^4 - q^3u^6)$$

(a polynomial identity in a, b, q, u , verified by expansion), so $\det(I + u\hat{B}) = (1 - u^3)^n \det(I - u^2A_2 + qu^4A_1 - q^3u^6) = (1 - u^3)^n \det P(u^2)$, the last step by transposition. Under (ND), $(\ker \Psi)^3 \subset V_0^3$ with V_0 the type-constant space; $\det(I + u\hat{B}|_{V_0^3}) = \prod_j D_{N\omega^j, N\omega^{-j}}(u) = (1 - u^3)^3(1 - u^6)(1 - q^3u^6)(1 - q^6u^6)$, and on $\Psi(V_0)^3$ — type-constant edge functions, on which K and K^t act as $(q+1)$ times the type rotation and its inverse — the factor is $\prod_j (1 + qu^3 + (q+1)u\omega^j + (q+1)u^2\omega^{2j}) = \prod_j (1 + u\omega^j + u^2\omega^{2j})(1 + qu\omega^j) = (1 - u^3)^2(1 + q^3u^3)$; the quotient is $\det(I + u\hat{B}|_{(\ker \Psi)^3}) = (1 - u^3)(1 - q^3u^3)(1 - u^6)(1 - q^3u^6) = K_Y(u)K_Y(u^2)$. Assembling,

$$\det(I + uB) = (1 - u^3)^n \frac{\det P(u^2) G(u) G(u^2)}{K_Y(u) K_Y(u^2)},$$

and Theorem 19.3.1 at u and at u^2 turns the right side into the statement. Certified modulo two 31-bit primes at $3|E| + 1$ points on $Y_{21}, Y_{24}, Y_{42}, T_6$ (a polynomial identity of degree $\leq 3|E|$) and at random points on Y_{168} (Table 19.1, line (B)). $\square$

Corollary 19.4.3 (Kang–Li’s zeta identity). $\det(I + uL_B) K_\Phi(u) = \det(I + uB) G_\Phi(u)$ with K_Φ, G_Φ the characteristic polynomials of B on $\ker \Phi$ and of L_B on $\ell^2(P)/\text{im } \Phi$. For $q = 2$, where $|P| = 3|E|$, $K_\Phi = G_\Phi$ on the four thick complexes, and hence, with $\chi(Y) = n$,

$$\frac{(1 - u^3)^{\chi(Y)}}{\det P(u)} = \frac{\det(I + uL_B)}{\det(I - uL_E) \det(I - u^2L_E^t)},$$

which is [54, Thm. C], [61, eq. (1.6)], for the zeta function $Z(Y, u) = \det(I - uL_E)^{-1}$. For general q , [54] gives $G_\Phi/K_\Phi = (1 - u^3)^{\chi(Y)-n}$; verified on T_6 , where $\chi = 0$ and $n = 36$. The eigenvalues of L_B off the pencil are the roots of $G(u)G(u^2)$: moduli $q^{1/2}$ and $q^{1/4}$, the latter being square roots of the unitary remainder — the $q^{-1/4}$ zeros of [61, Thm. 1.5.1(4)].

Proof. Φ intertwines, $\ker \Phi$ is B -invariant and $\text{im } \Phi$ is L_B -invariant, so both characteristic polynomials factor through $\ell^2(E)^3/\ker \Phi \cong \text{im } \Phi$. The rest is Theorem 19.4.2, $\det(I - u^2L_E) = \det(I - u^2L_E^t)$, and the computations of Table 19.1; the equality $K_\Phi = G_\Phi$ and the general ratio are not proved here and are the one point where [54] is used. $\square$

19.5 The verification battery

check	complexes	result
(D) $L_E L_E^t = qI + (q^2 - q)T^t T$, $L_E^t L_E = qI + (q^2 - q)S^t S$; $L_E \Psi = \Psi C'$, $L_E^t \Psi = \Psi C''$, $K \Psi = \Psi \widehat{C}$, $K^t \Psi = \Psi \widehat{C}'$ adjacency version of [Ch. 17, Def. 3.1] equals the chamber version	all five Y_{168}, T_6 yes; Y_{21}, Y_{24}, Y_{42} no	exact exact
(K) $\text{rk } \Psi = 3n - 6$ (mod two primes + explicit rational kernel); $K_Y = (1 - u^3)(1 - q^3 u^3)$; $\dim W = 4n + 6$ T_6 : $\text{rk } \Psi = 93$, $\dim \ker \Psi = 15$, (ND) fails, $\dim W = 15$, K_Y of degree 15	$Y_{21}, Y_{24}, Y_{42}, Y_{168}$ T_6	exact exact
(U) $RR^t = R^t R = qI$ on W ; $ \text{spec } R = \sqrt{q}$ to 10^{-12}	all five	numeric
(E) $\det(I - uL_E)KY = \det P \cdot G$ in $\mathbb{Z}[u]$; $\deg G = \dim W$; functional equation; roots on $ u = q^{-1/2}$ Y_{168} : six missing pencil roots $\{1, \omega, \omega^2, q, q\omega, q\omega^2\}$, 678 remainder eigenvalues of modulus $\sqrt{q}$	$Y_{21}, Y_{24}, Y_{42}, T_6$ Y_{168}	exact (roots numeric) numeric
(B) $L_B \Phi = \Phi B$; Theorem 19.4.2 at $3 E + 1$ points mod two primes; $\det(I + uL_B) = \det(I + uB)$ ($q = 2$); ratio $(1 - u^3)^{X-n}$ on T_6	all five (Y_{168} : random points)	exact / modular
(V) nontrivial pencil roots of modulus q (vertex Ramanujan property)	all five	numeric

Table 19.1: Verification (`pgl3_parahoric.py`). Complexes: $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ ($q = 2$) and T_6 ($q = 1$) of [Ch. 16].

19.6 Scope, residue, corrections

Remark 19.6.1 (Solved, open, impossible). *Solved*: [Ch. 17, Rem. 5.1(b)] in full — the determinant identity for all $\tilde{A}_2$ complexes (Theorem 19.3.1, with the exact form of the trivial factor under (ND), Proposition 19.3.2) and the identification of G with Kang–Li’s chamber factor (Theorem 19.4.2: $G(u)G(u^2)$ is the parahoric block of the chamber companion); the Riemann hypothesis for the parahoric factor, unconditionally (Theorem 19.2.2); the Kang–Li identity itself in companion form, with a proof by the degeneracy calculus of [Ch. 18] (Theorem 19.4.2). *Open*: (a) a direct proof that $G_\Phi/K_\Phi = (1 - u^3)^{X-n}$ (Corollary 19.4.3), which would make the Kang–Li identity independent of [54]; (b) an exact (rather than numeric) certificate of the vertex Ramanujan property of $Y_{21}, \dots, Y_{168}$, i.e. of the Riemann hypothesis for their zeta functions; (c) the exact polynomial identity on Y_{168} , verified here only structurally and at random points modulo primes; (d) the $\tilde{A}_2 \rightarrow \tilde{A}_d$ generalization, where the parahoric remainders should again be $\sqrt{q^k}$ times orthogonal matrices. *Impossible or false as stated*: see Remark 19.6.2.

Remark 19.6.2 (Corrections to the task specification). (1) The identity $\det(I - uL_E)(1 - u^3)(1 - q^3 u^3) = \det P(u) \det(I - uL_E^{\text{par}})$ with L_E^{par} an operator on the $3|F|$ pointed chambers is false as stated: the right factor would have degree up to $3|F| = (q+1)|E|$, not $4n + 6$. The remainder G is the characteristic polynomial of L_E on $W \subset \ell^2(E)$, and it enters the chamber operator as the block $G(u)G(u^2)$. (2) “ $L_E^{\text{par}} \cong \text{im}(\Psi)^\perp$ ” is false: $\dim W = 4n + 6$ while L_B acts on $(q+1)|E|$ coordinates; the correct relation is $L_B \Phi = \Phi B$ with B on $\ell^2(E)^3$. (3) The “Riemann hypothesis for $\tilde{A}_2$ complexes” for the parahoric poles is true but needs no Ramanujan hypothesis: it holds for every complex (Theorem 19.2.2); the Ramanujan property is exactly the vertex condition (Remark 19.2.3). (4) The formula requires (ND); on T_6 the trivial factor has degree 15 and $\deg G = 15 \neq 4n + 6$. (5) The “block triangularization” is in fact a block diagonalization: $\text{im } \Psi$ is invariant under L_E^t too. (6) Our own [Ch. 17, Def. 3.1] is corrected (Remark 19.1.2). (7) The representation-theoretic reading of the remainder as “the subdominant Iwahori–Hecke spectrum on a parahoric” is [61]’s; here the remainder is characterized combinatorially and its unitarity proved, which is more than the representation theory gives without a temperedness input.

Acknowledgement of provenance and caveats

Unrefereed preprint. All numbered statements are proved from the link structure of an $\tilde{A}_2$ complex and linear algebra; the one external input is the relation between L_B and its companion for $q \neq 2$ in Corollary 19.4.3, quoted from [54]. The modular certificates of Theorem 19.4.2 are identities in $\mathbb{F}_p[u]$ for two primes, not in $\mathbb{Z}[u]$; the theorem itself is proved. Bibliographic data of [54, 61, 62] were checked against publisher pages.

Chapter 20

Algorithmic non-commutative class field theory, XVIII: the unitary remainder in rank d

Summary. Paper XV proved that for an $\tilde{A}_2$ complex of order q the geodesic edge flow satisfies $L_E L_E^\top = qI + (q^2 - q)T^\top T$, so that its remainder is $\sqrt{q}$ times an orthogonal matrix and the parahoric zeros of the edge zeta lie on $|u| = q^{-1/2}$ whether or not the complex is Ramanujan; whether this survives in higher rank was left open. It does, with the exponent $(d-1)/2$, and the proof is one line of projective geometry. In an $\tilde{A}_d$ complex the types force the tails of the edges into a vertex y to be the *hyperplanes* of the link $\text{PG}(d, q)$ and the successors to be its *points*, and $\{x, y, z\}$ is a simplex exactly when the point z lies in the hyperplane x . Since two distinct hyperplanes of $\text{PG}(d, q)$ always meet in codimension two — for every d — inclusion–exclusion gives a count that does not depend on the pair, and

$$L_E L_E^\top = q^{d-1}I + q^{d-1}(q-1)T^\top T, \quad L_E^\top L_E = q^{d-1}I + q^{d-1}(q-1)S^\top S,$$

which is Paper XV’s identity at $d = 2$. Two unconditional consequences follow with no hypothesis on the complex: *exactly* $|E| - n$ singular values of L_E equal $q^{(d-1)/2}$, and L_E restricts to a bijective $q^{(d-1)/2}$ -isometry from $\ker S$ onto $\ker T$ (the mixed identities $TL_E = q^d S$ and $SL_E^\top = q^d T$). On any subspace of $\ker S \cap \ker T$ invariant under L_E and $L_E^\top$ the remainder is therefore $q^{(d-1)/2}$ times an orthogonal matrix and its zeros lie on $|u| = q^{-(d-1)/2}$. The typing is not a convention one may vary: run the same construction on lines against planes in $\text{PG}(3, q)$ and the count takes two values, because two lines meet or are skew. Four checks, exhaustive in $\text{PG}(d, q)$ for $d \leq 5$ and reproducing Paper XV on Y_{21}, Y_{24}, Y_{42} . What is *not* done: the invariant subspace itself for $d \geq 3$, and the worked $\tilde{A}_3$ example, for want of a thick $\tilde{A}_3$ complex.

20.1 The link, and what the types force

Let X be an $\tilde{A}_d$ complex of order q in the sense of [Ch. 16, Def. 1.1] generalised to rank d : a pure d -dimensional simplicial complex with vertices typed by $\mathbb{Z}/(d+1)$, one vertex of each type in every chamber, and the link of every vertex isomorphic to the flag complex of $\text{PG}(d, q)$. Write n for the number of vertices, E for the set of type-increasing directed edges (type(y) = type(x) + 1 for $x \rightarrow y$), and $S, T \in \{0, 1\}^{n \times E}$ for the tail and head incidence matrices. The geodesic edge flow is

$$L_E((x \rightarrow y), (y \rightarrow z)) = 1 \iff \{x, y, z\} \text{ is not a simplex of } X,$$

the rank- d form of [Ch. 19, Def. 1.1]; for $d = 2$ a 2-simplex is a chamber and this is [Ch. 19, Def. 1.1] verbatim, including the correction recorded there.

Lemma 20.1.1 (What the typing forces). *Fix a vertex y and identify its link with the flag complex of $\text{PG}(d, q)$, a neighbour of type $\text{type}(y) + i$ corresponding to a subspace of vector dimension i ($1 \leq i \leq d$). Then the tails of the edges into y are exactly the hyperplanes of $\text{PG}(d, q)$, the heads of the edges out of y are exactly its points, and for such x and z the set $\{x, y, z\}$ is a simplex of X if and only if the point z lies in the hyperplane x .*

Proof. A tail x has $\text{type}(x) = \text{type}(y) - 1 \equiv \text{type}(y) + d$, so $i = d$ and x is a subspace of vector dimension d , a hyperplane; a successor z has $\text{type}(z) = \text{type}(y) + 1$, so $i = 1$ and z is a point. Two vertices of the link span a simplex with y precisely when they are incident in the link, which for a point and a hyperplane means containment. $\square$

That one paragraph is the whole difference between rank two and rank d , and it is what makes the rank-two count survive: of all the pairings of types that one might have had to handle, the typing selects the single pairing whose intersection behaviour is uniform.

20.2 The identity

Write $\theta_j = \theta_j(q) = (q^{j+1} - 1)/(q - 1)$ for the number of points of $\text{PG}(j, q)$.

Theorem 20.2.1 (Unitarity identity in rank d). *For every $\tilde{A}_d$ complex of order q , $d \geq 2$,*

$$L_E L_E^\top = q^{d-1} I + q^{d-1}(q-1) T^\top T, \quad L_E^\top L_E = q^{d-1} I + q^{d-1}(q-1) S^\top S.$$

Proof. $(L_E L_E^\top)(e, e')$ counts the common successors of e and e' , which is zero unless e and e' have the same head y ; so the matrix is block diagonal by head, and $T^\top T$ is the indicator of “same head”. Fix y and use Lemma 20.1.1: the block is $BB^\top$ with $B[x][z] = 1$ iff the point z is not in the hyperplane x . On the diagonal, the number of points off a hyperplane is $\theta_d - \theta_{d-1} = q^d$. Off the diagonal, two *distinct* hyperplanes of $\text{PG}(d, q)$ meet in a subspace of codimension two, which carries θ_{d-2} points, so by inclusion–exclusion the number of points on neither is

$$\theta_d - 2\theta_{d-1} + \theta_{d-2} = (\theta_d - \theta_{d-1}) - (\theta_{d-1} - \theta_{d-2}) = q^d - q^{d-1} = q^{d-1}(q-1),$$

independent of the pair and of d . Hence each block is $q^d I + q^{d-1}(q-1)(J - I) = q^{d-1} I + q^{d-1}(q-1)J$, which is the first identity. The second is the same computation in the dual plane: two distinct points of $\text{PG}(d, q)$ span a line, the hyperplanes containing both are the θ_{d-2} hyperplanes containing that line, and the count of hyperplanes missing both points is again $q^{d-1}(q-1)$. $\square$

Remark 20.2.2 (Reduction to Paper XV). At $d = 2$ the constants are $q^{d-1} = q$ and $q^{d-1}(q-1) = q^2 - q$, so Theorem 20.2.1 is [Ch. 19, Thm. 2.1] and its proof is [Ch. 19, Lem. 1.3] with the counting done once instead of case by case. Check (A) verifies the two identities, with these constants, on the complexes Y_{21} , Y_{24} and Y_{42} of [Ch. 16, Constr. 1.4].

Proposition 20.2.3 (The mixed identities). *$TL_E = q^d S$ and $SL_E^\top = q^d T$.*

Proof. $(TL_E)(v, f)$ counts the edges e with head v and $e \rightarrow f$; since $e \rightarrow f$ forces $\text{head}(e) = \text{tail}(f)$, the entry vanishes unless $v = \text{tail}(f)$, and then it counts the hyperplanes of the link of v not containing the point $\text{head}(f)$, of which there are $\theta_d - \theta_{d-1} = q^d$. The second identity is dual. $\square$

Corollary 20.2.4 (Unconditional consequences). *For every $\tilde{A}_d$ complex of order q :*

- (i) *exactly $|E| - n$ singular values of L_E equal $q^{(d-1)/2}$;*
- (ii) *L_E maps $\ker S$ into $\ker T$, bijectively, and $\|L_E w\|^2 = q^{d-1} \|w\|^2$ there; symmetrically $L_E^\top$ maps $\ker T$ onto $\ker S$ as a $q^{(d-1)/2}$ -isometry.*

Proof. (i) $L_E L_E^\top - q^{d-1} I = q^{d-1} (q-1) T^\top T$ has rank $\text{rank } T = n$, every vertex being the head of some edge; so q^{d-1} is an eigenvalue of $L_E L_E^\top$ of multiplicity exactly $|E| - n$. (ii) For $w \in \ker S$, $T L_E w = q^d S w = 0$ by Proposition 20.2.3, and $\|L_E w\|^2 = \langle L_E^\top L_E w, w \rangle = q^{d-1} \|w\|^2$ by Theorem 20.2.1; both spaces have dimension $|E| - n$, so the isometry is onto. $\square$

Corollary 20.2.5 (The critical circle in rank d). *Let $W \subseteq \ker S \cap \ker T$ be invariant under L_E and $L_E^\top$ and put $R = L_E|_W$. Then $RR^\top = R^\top R = q^{d-1} I_W$, so $q^{-(d-1)/2} R$ is orthogonal and every root of $G(u) = \det(I - uR)$ lies on $|u| = q^{-(d-1)/2}$. No hypothesis on the complex — in particular no Ramanujan hypothesis — is used.*

Proof. Immediate from Theorem 20.2.1, since $T^\top T$ and $S^\top S$ vanish on W ; an orthogonal matrix has all eigenvalues of modulus one. $\square$

Remark 20.2.6 (The typing is not a convention). The uniformity in Theorem 20.2.1 is a property of the hyperplane/point pairing, not of $\text{PG}(d, q)$ in general. Run the same construction on lines against planes in $\text{PG}(3, q)$: two distinct lines either meet or are skew, so the planes containing both number 1 or 0, and the off-diagonal of $BB^\top$ takes two values differing by one — 9 and 10 for $q = 2$, 32 and 33 for $q = 3$, 144 and 145 for $q = 5$ (check (C)). Had the types of an $\tilde{A}_d$ complex put intermediate dimensions against each other, the correction term would have been a sum over the classes of the Grassmann association scheme instead of a multiple of $T^\top T$, and the remainder would have carried several moduli rather than one. Lemma 20.1.1 is what rules this out.

20.3 Verification

The script `ad_unitarity.py` runs the four checks of Table 20.1 in about three seconds; its output is archived as `ad_unitarity_output.txt`.

key	check	mode
(H)	$BB^\top = B^\top B = q^{d-1} I + q^{d-1} (q-1) J$ in $\text{PG}(d, q)$, 14 pairs (d, q)	exact
(C)	lines against planes in $\text{PG}(3, q)$: the off-diagonal takes two values	exact
(A)	the identities of §20.2 on Y_{21}, Y_{24}, Y_{42} , with $d = 2$	exact
(S)	exactly $ E - n$ singular values of L_E equal $q^{(d-1)/2}$	numeric

Table 20.1: The verification battery, 4/4.

Check (H) is exhaustive: for $d = 2, 3, 4, 5$ and $q = 2, 3, 5, 7, 11$ with $\theta_d \leq 1300$ it forms the full point/hyperplane non-incidence matrix of $\text{PG}(d, q)$ and both of its Gram matrices, and finds the diagonal constantly q^d and the off-diagonal constantly $q^{d-1}(q-1)$ — 14 instances, including $\text{PG}(5, 3)$ on 364 points and $\text{PG}(4, 5)$ on 781. Check (A) builds the thick $\tilde{A}_d$ complexes of [Ch. 16] and verifies Theorem 20.2.1, Proposition 20.2.3 and Corollary 20.2.4(ii) on each; check (S) finds 126, 144 and 252 singular values equal to $\sqrt{2}$ out of 147, 168 and 294, which are exactly $|E| - n$.

20.4 Status, scope, corrections

Proved. Lemma 20.1.1, Theorem 20.2.1, Proposition 20.2.3 and Corollary 20.2.4 for every $\tilde{A}_d$ complex of order q , $d \geq 2$; Corollary 20.2.5 for every such complex admitting an invariant $W \subseteq \ker S \cap \ker T$.

Not proved, and this is the gap. The existence of that W . For $d = 2$ it is [Ch. 19, Thm. 2.1], obtained from the intertwining $L_E \Psi = \Psi C'$ with $\Psi = [S^\top T^\top M^\top]$ and the cubic Hecke pencil. Corollary 20.2.4 is what survives unconditionally: L_E is already $q^{(d-1)/2}$ times an isometry from $\ker S$ onto $\ker T$, and only the passage from that pair of spaces to a single invariant one is missing.

Remark 20.4.1 (The candidate intertwiner in rank d). A chamber of an $\tilde{A}_d$ complex has $d + 1$ vertices, so an edge $e = (x \rightarrow y)$ lying in a chamber determines $d - 1$ further vertices, one of each type $\text{type}(y) + 1, \dots, \text{type}(y) + d - 1$. Let $M_i(v, e)$ be the number of chambers containing e whose vertex of type $\text{type}(y) + i$ is v , and put

$$\Psi_d := \begin{bmatrix} S^\top & T^\top & M_1^\top & \cdots & M_{d-1}^\top \end{bmatrix} \in \mathbb{Z}^{|E| \times (d+1)n},$$

so that $\Psi_2 = [S^\top \ T^\top \ M^\top]$ is the intertwiner of [Ch. 19, Thm. 2.1] and the block count $d + 1$ matches the degree of the Hecke pencil. We record this as the natural candidate and do not claim it works: what must be shown is $L_E \Psi_d = \Psi_d C^{(d+1)}$ and $L_E^\top \Psi_d = \Psi_d C^{(d+1)^\top}$ for $(d + 1) \times (d + 1)$ block matrices, after which $W = (\text{im } \Psi_d)^\perp$ would satisfy the hypothesis of Corollary 20.2.5 and complete the argument. Both identities are entrywise *local* — each entry is a count inside the star of a single vertex — so they can in principle be settled in $\text{PG}(d, q)$ without a global complex, as Theorem 20.2.1 was.

Added in proof. That has now been carried out, in Paper XX. The local identity is a single flag count: in $\text{PG}(d, q)$ the number of complete flags F with $F_{i+1} = V$ and F_1 inside a hyperplane H takes exactly two values, according as $V \subseteq H$ or not, and their difference $K_i = f(d - i)f(i + 1)q^i/[i + 1]_q$ depends on neither V nor H ; with $M_0 = cT$ and $M_d = cS$ this gives $L_E M_i^\top = K_i(T^\top A_{i+1} - c_{i+1}^{-1} M_{i+1}^\top)$ and, by the self-duality of $\text{PG}(d, q)$, the mirror identity for $L_E^\top$. So Ψ_d does work, Corollary 20.2.5 is unconditional for every $d \geq 2$, and at $d = 2$ the characteristic polynomial of $C^{(3)}$ is the cubic Hecke pencil $x^3 - A_1 x^2 + q A_2 x - q^3$.

Remark 20.4.2 (Solved, open, impossible). *Solved:* the counting half of [Ch. 19, Rem. 6.1(d)] — the exact decomposition of $L_E L_E^\top$ in rank d — with the exponent $(d - 1)/2$ as anticipated. *Open:* (a) *closed in Paper XX* — the invariant subspace $W_d = (\text{im } \Psi_d)^\perp$ exists for every d , so Corollary 20.2.5 is unconditional; what remains open there is the exact rank of Ψ_d , hence $\dim W_d$; (b) a thick $\tilde{A}_3$ complex small enough to compute with, and with it the degree-four block companion and a worked example; (c) the exact certificate of the vertex Ramanujan property, the other half of [Ch. 19, Rem. 6.1(d)], which this note does not touch. *Impossible:* nothing new. In particular the mechanism does *not* degenerate in higher rank, which was the outcome one might have expected from Remark 20.2.6.

Remark 20.4.3 (Corrections to the task specification). Three. (1) The specification proposed building the identity from “the typical counting formula for the intersection of k -dimensional with $(d - k)$ -dimensional subspaces of $\text{PG}(d, q)$ ”. No such formula is needed and none would serve: the typing of an $\tilde{A}_d$ complex puts hyperplanes against points (Lemma 20.1.1), and the only geometric input is that two distinct hyperplanes meet in codimension two. With intermediate dimensions the count is not constant (Remark 20.2.6), so an identity of the stated shape would be false. (2) The specification asked for $W_d = \bigcap_{i=0}^d \ker S_i$. Only two operators appear in Theorem 20.2.1, and $\ker S \cap \ker T$ already carries $RR^\top = q^{d-1}I$; whether a smaller intersection is needed is a question about *invariance*, not about the constant. (3) The predicted $R_d R_d^\top = q^{d-1}I$ and the circle $|u| = q^{-(d-1)/2}$ are correct, and are Corollary 20.2.5. The third task — the explicit $\tilde{A}_3$ block companion and a verified example — is not done: no thick $\tilde{A}_3$ complex is at hand. The rank-two test complexes came from an exhaustive search for triangle presentations over $\text{PG}(2, 2)$ [Ch. 16, Constr. 1.4]; there is no comparably small search in rank three, and the known constructions of finite $\tilde{A}_d$ quotients go through arithmetic lattices in PGL_{d+1} over a local field. That is a separate problem and we do not pretend to have solved it.

Provenance of this chapter

Unrefereed preprint. Theorem 20.2.1 and everything derived from it are proved in full; the computations verify the geometric input exhaustively in $\text{PG}(d, q)$ for $d \leq 5$ and confirm that the rank- d statement specialises at $d = 2$ to the published identity on three complexes constructed independently. The one-line proof was found by asking which pair of types the $\mathbb{Z}/(d + 1)$ grading actually selects, and the negative half of Remark 20.2.6 was computed before the positive half was proved, to check that the question had teeth.

Chapter 21

Algorithmic non-commutative class field theory, XX: the invariant subspace in rank d , and one flag count that produces it

Summary. Paper XVIII proved that on an $\tilde{A}_d$ complex of order q the geodesic edge flow satisfies $L_E L_E^T = q^{d-1} I + q^{d-1}(q-1)T^T T$ and its dual, so that on any subspace $W \subseteq \ker S \cap \ker T$ invariant under L_E and L_E^T the restriction is $q^{(d-1)/2}$ times an orthogonal matrix. It left the existence of such a W open, recording only a candidate intertwiner built from the incidence operators $M_i(v, e) = \#\{\text{chambers } C \supseteq e : C_{\tau(e)+i} = v\}$. This note supplies the subspace. The entire content is a single flag count: in $\text{PG}(d, q)$, the number of complete flags F with $F_{i+1} = V$ and F_1 inside a hyperplane H takes exactly *two* values, according as $V \subseteq H$ or not, and their difference $K_i = f(d-i)f(i+1)q^i/[i+1]_q$ depends on neither V nor H . Because the tail of an edge is a hyperplane of the link of its head and the geodesic successors are the points off that hyperplane (Paper XVIII, Lem. 2.2), this dichotomy is exactly an intertwining; and after normalising M_i to the 0/1 indicator N_i the constant collapses to a bare power, $K_i/c_i = q^i$, leaving

$$L_E N_i^T = q^i (N_0^T A_{i+1} - N_{i+1}^T), \quad N_0 = T, \quad N_d = S,$$

with the mirror identity for L_E^T from the self-duality of $\text{PG}(d, q)$. Hence $\Psi_d = [N_0^T \cdots N_d^T]$ satisfies $L_E \Psi_d = \Psi_d C$ and $L_E^T \Psi_d = \Psi_d C'$, and $W_d = (\text{im } \Psi_d)^\perp$ is a two-sided invariant subspace of $\ker S \cap \ker T$. Paper XVIII's Corollary 2.7 therefore holds with no hypothesis: every eigenvalue of $L_E|_{W_d}$ has modulus exactly $q^{(d-1)/2}$. The companion matrix is explicit in every rank and its characteristic polynomial is $\sum_j (-1)^j q^{j(j-1)/2} A_j x^{d+1-j}$, the local Hecke polynomial of PGL_{d+1} — the cubic pencil of Kang and Li at $d=2$, and at $d=3$ the quartic $x^4 - A_1 x^3 + q A_2 x^2 - q^3 A_3 x + q^6$, obtained with no global $\tilde{A}_3$ complex. The relations $N_i^T 1_{t+i} = \binom{d}{i}_q N_0^T 1_t$ give $\text{corank } \Psi_d \geq d(d+1)$, with equality at $d=2$. The upshot is a decomposition of the spectrum of L_E into a Hecke part on $\text{im } \Psi_d$, where the Ramanujan property is exactly the question, and a complement on W_d that is on the critical circle unconditionally. Eight checks, verified on $\text{PG}(d, q)$ for fourteen triples (d, q, i) , symbolically in q for $d \leq 7$, and on the three thick $\tilde{A}_2$ complexes of Paper X.

21.1 The gap, and what closes it

Let X be a thick $\tilde{A}_d$ complex of order q : a finite quotient of the Bruhat–Tits building of PGL_{d+1} over a local field with residue field of size q , with vertices typed by $\mathbb{Z}/(d+1)$. Write E for the set of directed edges $x \rightarrow y$ with $\tau(y) = \tau(x) + 1$, S and T for the tail and head incidence matrices, and L_E for the geodesic edge flow,

$$(L_E)_{e,f} = 1 \iff e = (x, y), f = (y, z) \text{ and } \{x, y, z\} \text{ is not a simplex of } X.$$

Paper XVIII proved

$$L_E L_E^\top = q^{d-1} I + q^{d-1} (q-1) T^\top T, \quad L_E^\top L_E = q^{d-1} I + q^{d-1} (q-1) S^\top S, \quad (21.1.1)$$

for every $d \geq 2$, and deduced that on any subspace $W \subseteq \ker S \cap \ker T$ stable under both L_E and $L_E^\top$ the restriction $R = L_E|_W$ satisfies $RR^\top = R^\top R = q^{d-1} I_W$, so that all its eigenvalues have modulus $q^{(d-1)/2}$ — with no Ramanujan hypothesis. What it could not do was produce a single such W . Its Remark 4.1 recorded a candidate and observed that the identities it would need are “entrywise local”, so that they could in principle be settled inside $\mathrm{PG}(d, q)$ without a global complex. That is what is carried out here, and the local computation turns out to be one line.

Definition 21.1.1. For $0 \leq i \leq d$ and an edge $e = (x, y) \in E$ put

$$N_i(v, e) = \begin{cases} 1, & \{x, y, v\} \text{ lies in a chamber and } \tau(v) = \tau(y) + i, \\ 0, & \text{otherwise,} \end{cases}$$

and $\Psi_d = [N_0^\top \ N_1^\top \ \cdots \ N_d^\top] \in \mathbb{Z}^{E \times (d+1)n}$. Write c_i for the number of chambers containing such a triple, so that the unnormalised operator of Paper XVIII, Rem. 4.1 is $M_i = c_i N_i$.

Two blocks are old: $N_0 = T$ and $N_d = S$, because a vertex of relative type 0 in a chamber through $e = (x, y)$ is y and one of relative type d is x . So $\mathrm{im} \Psi_d \supseteq \mathrm{im} S^\top + \mathrm{im} T^\top$ and therefore

$$W_d := (\mathrm{im} \Psi_d)^\perp \subseteq (\mathrm{im} S^\top)^\perp \cap (\mathrm{im} T^\top)^\perp = \ker S \cap \ker T. \quad (21.1.2)$$

So the only thing to prove is that $\mathrm{im} \Psi_d$ absorbs L_E and $L_E^\top$.

21.2 One flag count

Write $[j]_q = (q^j - 1)/(q - 1)$ for the number of points of a j -dimensional $\mathbb{F}_q$ space and $f(m) = \prod_{j=1}^m [j]_q$ for the number of complete flags of an m -dimensional one. A *complete flag* of $\mathrm{PG}(d, q)$ is a chain $F_1 \subset F_2 \subset \cdots \subset F_d$ of subspaces of an ambient $(d+1)$ -dimensional space, $\dim F_j = j$.

Theorem 21.2.1. Fix $0 \leq i \leq d-1$, a hyperplane H and a subspace V with $\dim V = i+1$. Then

$$N(V, H) := \#\{\text{complete flags } F : F_{i+1} = V \text{ and } F_1 \subseteq H\} = \begin{cases} f(i+1) f(d-i), & V \subseteq H, \\ f(i+1) f(d-i) - K_i, & V \not\subseteq H, \end{cases}$$

where

$$K_i = \frac{f(d-i) f(i+1) q^i}{[i+1]_q}$$

is a positive integer depending only on d, i and q . In particular N takes exactly two values, and the jump is independent of both V and H .

Proof. A complete flag with $F_{i+1} = V$ is a complete flag of V together with a complete flag of the quotient by V , and the two choices are independent; the quotient has dimension $(d+1) - (i+1) = d-i$, so the number of flags above V is $f(d-i)$ regardless of H .

d	q	i	$N(V \subseteq H)$	$N(V \not\subseteq H)$	K_i
2	2	0	3	0	3
2	2	1	3	1	2
2	3	0	4	0	4
2	3	1	4	1	3
3	2	1	9	3	6
3	2	2	21	9	12
3	3	1	16	4	12
3	3	2	52	16	36
4	2	1	63	21	42
4	2	2	63	27	36
4	2	3	315	147	168

Table 21.1: The two values of $N(V, H)$ and their difference, over all pairs (V, H) in $\text{PG}(d, q)$. The rows with $i = 0$ are omitted for $d \geq 3$ (there V is a point, $N = 0$ when $V \not\subseteq H$, and $K_0 = f(d)$). Checks (F) and (K).

Below V the constraint is $F_1 \subseteq H \cap V$. If $V \subseteq H$ this is vacuous and the count is $f(i+1)$. Otherwise $H \cap V$ is a hyperplane of V , of dimension i . Every point of V lies in the same number $f(i+1)/[i+1]_q$ of complete flags of V , by transitivity of $\text{GL}(V)$ on points; hence the count is $[i]_q f(i+1)/[i+1]_q$, and

$$f(i+1) - \frac{[i]_q f(i+1)}{[i+1]_q} = f(i+1) \frac{[i+1]_q - [i]_q}{[i+1]_q} = \frac{f(i+1) q^i}{[i+1]_q}.$$

Multiplying by $f(d-i)$ gives K_i . It is an integer because it counts flags. $\square$

Theorem 21.2.1 is verified exhaustively — over *every* pair (V, H) — for $(d, q) \in \{(2, 2), (2, 3), (3, 2), (3, 3), (4, 2)\}$ and every i , fourteen triples in all; checks (F) and (K). Table 21.1 lists the values.

21.3 The intertwining

Lemma 21.3.1 (Paper XVIII, Lem. 2.2). *Let y be a vertex of X and identify its link with the flag complex of $\text{PG}(d, q)$ so that a neighbour of relative type $+j$ corresponds to a subspace of dimension j . Then the tail of an edge with head y is a hyperplane, the geodesic successors are exactly the points not on it, and the chambers of X through y are the complete flags.*

Theorem 21.3.2. *Let $A_j(y, v) = 1$ when v is a neighbour of y of relative type $+j$ and 0 otherwise, and put $N_{d+1} := 0$, $A_{d+1} := I$. Then for every $\tilde{A}_d$ complex of order q and every $0 \leq i \leq d$,*

$$L_E N_i^\top = q^i (N_0^\top A_{i+1} - N_{i+1}^\top), \quad (21.3.1)$$

and dually, with $N_{-1} := 0$ and $A_0 := I$, $L_E^\top N_i^\top = q^{d-i} (N_d^\top A_i - N_{i-1}^\top)$. Consequently $L_E \Psi_d = \Psi_d C$ and $L_E^\top \Psi_d = \Psi_d C'$ for $(d+1) \times (d+1)$ block matrices with entries in the algebra generated by $A_1, \dots, A_d$.

The collapse of the constant from K_i to the bare power q^i is what makes the rank- d statement usable, and it is forced: a chamber through $\{x, y, v\}$ is a flag through the chain $V \subset H$ of dimensions $i \subset d$, so $c_i = f(i)f(d-i)$, and with $f(i+1) = [i+1]_q f(i)$,

$$\frac{K_i}{c_i} = \frac{f(d-i)f(i+1)q^i}{[i+1]_q} \cdot \frac{1}{f(i)f(d-i)} = q^i. \quad (21.3.2)$$

Proof. Fix $e = (x, y)$ and a vertex v of type $\tau(y) + 1 + i$. Then

$$c_i(L_E N_i^\top)(e, v) = \sum_{z: (y, z) \text{ geodesic after } e} M_i(v, (y, z)) = \#\{(z, C) : z \notin x, y, z \in C, C_{\tau(y)+1+i} = v\},$$

the condition $z \notin x$ being Lemma 21.3.1. Work in the link of y . There x is a hyperplane H , each z is a point p , v is a subspace V of dimension $i + 1$, and a chamber C through y and z with $C_{\tau(y)+1+i} = v$ is a complete flag F with $F_1 = p$ and $F_{i+1} = V$. Since z is recovered from C as F_1 , the pair (z, C) is just the flag, and

$$c_i(L_E N_i^\top)(e, v) = \#\{F : F_{i+1} = V\} - \#\{F : F_{i+1} = V, F_1 \subseteq H\} = f(i+1)f(d-i) - N(V, H).$$

By Theorem 21.2.1 this is 0 when $V \subseteq H$ and K_i otherwise. Now $V \subseteq H$ says precisely that v and x lie in a common chamber with y , i.e. that $\{x, y, v\}$ is a simplex, which is exactly what N_{i+1} records; and the value K_i is attained only when v is a neighbour of y of relative type $i + 1$ at all, which is $A_{i+1}(y, v)$. Assembling and dividing by c_i , which is q^i by (21.3.2),

$$(L_E N_i^\top)(e, v) = q^i (A_{i+1}(y, v) - N_{i+1}(v, e)),$$

and $A_{i+1}(y, v) = (N_0^\top A_{i+1})(e, v)$ since $N_0 = T$ and y is the head of e . At $i = d$ the first term is $q^d N_0^\top$ and the second vanishes, recovering $L_E S^\top = q^d T^\top$, which is Paper XVIII, Prop. 2.4 transposed.

For the dual, $(L_E^\top N_i^\top)(f, v)$ with $f = (y, z)$ sums $N_i(v, (x, y))$ over the x with $\{x, y, z\}$ not a simplex. In the link of y this counts complete flags F with $F_i = V$ and $p \not\subseteq F_d$, where now V has dimension i and p is the point z . The duality of $\text{PG}(d, q)$, which exchanges subspaces of dimension j with those of dimension $d + 1 - j$ and reverses inclusion, carries this count to the count of Theorem 21.2.1 with i replaced by $d - i$. The same argument then gives the mirror identity, with $N_d = S$ in place of $N_0 = T$ and N_{i-1} in place of N_{i+1} .

Finally, each right-hand side of (21.3.1) is a $\mathbb{Z}$ -linear combination of the blocks $N_0^\top$ and $N_{i+1}^\top$, the first multiplied on the right by an operator on the vertex space. That is exactly the statement $L_E \Psi_d = \Psi_d C$ with C of the asserted shape. $\square$

Corollary 21.3.3. $W_d = (\text{im } \Psi_d)^\perp$ is contained in $\ker S \cap \ker T$ and is invariant under L_E and $L_E^\top$. Hence, by Paper XVIII, Cor. 2.7, $R = L_E|_{W_d}$ satisfies $RR^\top = R^\top R = q^{d-1}I$, every eigenvalue of R has modulus $q^{(d-1)/2}$, and every root of $\det(I - uR)$ lies on $|u| = q^{-(d-1)/2}$. No hypothesis on X is used — in particular, no Ramanujan hypothesis.

Proof. The inclusion is (21.1.2). If $L_E \Psi_d = \Psi_d C$ then $L_E(\text{im } \Psi_d) \subseteq \text{im } \Psi_d$, so $L_E^\top$ preserves $(\text{im } \Psi_d)^\perp$; and $L_E^\top \Psi_d = \Psi_d C'$ gives the other half. Now apply (21.1.1), whose correction terms $T^\top T$ and $S^\top S$ vanish on W_d . $\square$

21.4 The companion matrix, and the Hecke polynomial in rank d

Reading (21.3.1) columnwise gives C outright: the block in row 0, column i is $q^i A_{i+1}$ and the block in row $i + 1$, column i is $-q^i I$, all others zero.

Theorem 21.4.1. $L_E \Psi_d = \Psi_d C$ with

$$C = \begin{pmatrix} A_1 & qA_2 & q^2A_3 & \cdots & q^{d-1}A_d & q^dI \\ -I & 0 & 0 & & & 0 \\ 0 & -qI & 0 & & & 0 \\ & & \ddots & & & \vdots \\ 0 & 0 & & -q^{d-1}I & 0 & 0 \end{pmatrix},$$

and, the blocks $A_1, \dots, A_d$ being commuting operators on the vertex space,

$$\det(xI - C) = \sum_{j=0}^{d+1} (-1)^j q^{j(j-1)/2} A_j x^{d+1-j}, \quad A_0 = A_{d+1} = I,$$

the local Hecke polynomial of PGL_{d+1} . Equivalently, on $\mathrm{im} \Psi_d$,

$$L_E^{d+1} - L_E^d A_1 + q L_E^{d-1} A_2 - q^3 L_E^{d-2} A_3 + \cdots + (-1)^{d+1} q^{d(d+1)/2} = 0.$$

Proof. The shape of C is (21.3.1). For the determinant, eliminate: (21.3.1) gives $N_{i+1}^\top = N_0^\top A_{i+1} - q^{-i} L_E N_i^\top$, so by induction $N_j^\top = \sum_{t=0}^j (-1)^t q^{-(tj-t(t+1)/2)} L_E^t N_0^\top A_{j-t}$. Applying L_E to the case $j = d$ and using $L_E N_d^\top = q^d N_0^\top$ gives a relation of degree $d+1$ in L_E annihilating $N_0^\top$; clearing the negative powers of q turns it into the displayed polynomial, whose coefficient of $L_E^{d+1-j} A_j$ is $(-1)^j q^{j(j-1)/2}$. $\square$

At $d = 2$ this is $x^3 - A_1 x^2 + q A_2 x - q^3$, the cubic pencil of Kang and Li [68]. At $d = 3$, which no published work reaches, it is

$$C^{(4)} = \begin{pmatrix} A_1 & q A_2 & q^2 A_3 & q^3 I \\ -I & 0 & 0 & 0 \\ 0 & -q I & 0 & 0 \\ 0 & 0 & -q^2 I & 0 \end{pmatrix}, \quad \det(xI - C^{(4)}) = x^4 - A_1 x^3 + q A_2 x^2 - q^3 A_3 x + q^6,$$

the quartic local Hecke polynomial of PGL_4 — obtained here with no global $\tilde{A}_3$ complex, purely from the flag count. The determinant identity is verified symbolically for $d = 2, 3, 4, 5$, and (21.3.2) for $d \leq 6$; checks (H) and (A).

Theorem 21.4.2. *For every $t \in \mathbb{Z}/(d+1)$ and every i , let $1_t \in \mathbb{Z}^n$ be the indicator of the vertices of type t , and let $b_i = \binom{d}{i}_q = f(d)/(f(i)f(d-i))$ be the number of vertices of relative type i forming a simplex with a given edge. Then*

$$N_i^\top 1_{t+i} = b_i N_0^\top 1_t,$$

so the vectors $w = (w_0, \dots, w_d)$ with $w_i = \sum_s a_{i,s} 1_s$ lie in $\ker \Psi_d$ exactly when $\sum_{i=0}^d b_i a_{i,t+i} = 0$ for each t . That is $(d+1)^2$ unknowns subject to $d+1$ independent equations, so

$$\mathrm{corank} \Psi_d \geq d(d+1), \quad \dim W_d \leq |E| - (d+1)n + \mathrm{corank} \Psi_d.$$

Proof. $(N_i^\top 1_s)(e) = \sum_{\tau(v)=s} N_i(v, e)$ counts the vertices of relative type i lying in a chamber with e , when $s = \tau(\mathrm{head} e) + i$, and is 0 otherwise. In the link of the head those vertices are the i -dimensional subspaces of the hyperplane x , so there are $b_i = \binom{d}{i}_q$ of them, independently of e ; hence $N_i^\top 1_s = b_i N_0^\top 1_{s-i}$. The $d+1$ vectors $N_0^\top 1_t$ are the indicators of the type classes of E and so are independent, which makes the $d+1$ equations independent. $\square$

On all three complexes of Section 21.5 the corank is exactly $d(d+1) = 6$ and these relations span the kernel, so $\dim W_d = |E| - (d+1)n + d(d+1)$ there; check (N). Whether the inequality is an equality for $d \geq 3$ we do not know. We note that $d(d+1)$, $2(d+1)$ and $(d+1)!$ all equal 6 at $d = 2$ and diverge at $d = 3$, where they are 12, 8 and 24; the value derived above is $d(d+1)$.

21.5 Rank two, and the Hecke pencil

At $d = 2$ we have $N_0 = T$, $N_2 = S$ and $N_1 = M$, the chamber indicator; the three cases of (21.3.1) read

$$L_E T^\top = T^\top A_1 - M^\top, \quad L_E M^\top = q(T^\top A_2 - S^\top), \quad L_E S^\top = q^2 T^\top, \quad (21.5.1)$$

with the mirror obtained by exchanging $S \leftrightarrow T$ and $A_1 \leftrightarrow A_2$. Only the first and third were available before: the first is the definition of the geodesic condition rearranged, the third is Paper XVIII, and the middle one — the one that closes the cycle — is new. With $\Psi_2 = [T^\top \ M^\top \ S^\top]$,

$$C = \begin{pmatrix} A_1 & q A_2 & q^2 I \\ -I & 0 & 0 \\ 0 & -q I & 0 \end{pmatrix}, \quad \det(xI - C) = x^3 - A_1 x^2 + q A_2 x - q^3.$$

X	n	$ E $	q	$\text{rank } \Psi_2$	$\dim W_2$	eigenvalues of $L_E _{W_2} $
Y_{21}	21	147	2	57	90	$1.414214 = \sqrt{2}$
Y_{24}	24	168	2	66	102	$1.414214 = \sqrt{2}$
Y_{42}	42	294	2	120	174	$1.414214 = \sqrt{2}$

Table 21.2: The invariant subspace on the three thick $\tilde{A}_2$ complexes of Paper X. In every case Ψ_2 has corank 6 in $\mathbb{Z}^{3n}$ — the two relations per type saying that the columns of S , T and M sum to the indicator of a type class — and *every* eigenvalue of $L_E|_{W_2}$ has modulus $\sqrt{q}$ to machine precision. Checks (P) and (U).

That cubic is the local Hecke polynomial of PGL_3 , and the determinant identity built on it is Kang and Li’s [68]. The three identities of (21.5.1), their mirrors, $L_E\Psi_2 = \Psi_2C$ exactly, and the pencil relation $C^3 = A_1C^2 - qA_2C + q^3$ are verified on the thick $\tilde{A}_2$ complexes Y_{21} , Y_{24} and Y_{42} of Paper X; checks (I) and (P). Table 21.2 records the subspace they produce.

Remark 21.5.1 (What the decomposition says). Corollary 21.3.3 splits the spectrum of L_E into a part carried by $\text{im } \Psi_d$ and a part carried by W_d . On $\text{im } \Psi_d$ the flow is governed by C , whose entries are Hecke operators, so that part of the spectrum is a function of the Satake parameters and the Ramanujan property is exactly the question of where it sits. On W_d the flow is $q^{(d-1)/2}$ times an orthogonal matrix, unconditionally. At $d = 2$ this explains the shape of Kang and Li’s criterion [68, Thm. 1.0.2]: they prove that X is Ramanujan if and only if the non-trivial zeros of $\det(I - uL_E)$ have $|u| \in \{q^{-1}, q^{-1/2}\}$. The $q^{-1/2}$ family is precisely the W_d part, and Corollary 21.3.3 says it is there for every complex, Ramanujan or not; the Ramanujan content is entirely in the q^{-1} family, which is the Hecke part.

Remark 21.5.2 (Solved, open, impossible). *Solved*: the open problem (a) of Paper XVIII, Rem. 4.2 — the invariant subspace, and with it Corollary 2.7 of that paper unconditionally, for every $d \geq 2$. *Open*: (i) whether the bound $\text{corank } \Psi_d \geq d(d+1)$ of Theorem 21.4.2 is an equality for $d \geq 3$, hence $\dim W_d$ exactly; at $d = 2$ it is an equality on all three complexes tested, and we have not proved it in general. (ii) Whether $\text{im } \Psi_d$ is the *full* Hecke-isotypic part, i.e. whether W_d is the largest two-sided invariant subspace inside $\ker S \cap \ker T$. (iii) A rank- d analogue of the determinant identity itself; see the provenance note. *Impossible by the same argument*: nothing here bounds the spectrum on $\text{im } \Psi_d$. The Hecke part is where Ramanujan lives, and Corollary 21.3.3 is silent about it by construction — that is the price of being unconditional.

21.6 Verification

The script `ad_invariant_subspace.py` runs eight checks in a few seconds; its output is archived as `ad_invariant_subspace_output.txt`.

Check (F) enumerates every complete flag of $\text{PG}(d, q)$ and every pair (V, H) — no sampling — for the five pairs (d, q) with at most 20 000 flags. Check (U) is the only numerical item; it is a consequence of the exact checks, and is reported because the eigenvalues are what the statement is about.

21.7 Scope and provenance

Scope. Theorem 21.2.1 and Theorem 21.3.2 are proved for every $d \geq 2$ and every $\tilde{A}_d$ complex of order q ; so is Corollary 21.3.3. The global identities are *verified* only at $d = 2$, because we have no $\tilde{A}_d$ complex for $d \geq 3$ to verify them on; the proof of Theorem 21.3.2, however, uses nothing beyond Lemma 21.3.1 and the flag count, and both hold in every rank. The corank of Ψ_d is not determined.

Provenance. Identity (21.1.1) and Lemma 21.3.1 are Paper XVIII. The $d = 2$ determinant identity $Z(X, u)^{-1} = \det(I - A_1u + qA_2u^2 - q^3u^3) \cdot \det(I + L_Bu) \cdot (1 - u^3)^{-x}$ and the four-way Ramanujan criterion

key	claim	cases	mode
(F)	$N(V, H)$ takes exactly two values, jump K_i	14 triples (d, q, i)	exact
(K)	the closed forms f, φ, K_i match the enumeration	14 triples	exact
(A)	$K_i/c_i = q^i$, symbolically in q	$2 \leq d \leq 7$, all i	exact
(H)	$\det(xI - C)$ is the PGL_{d+1} Hecke polynomial	$d = 2, 3, 4, 5$	exact
(I)	the six rank-2 identities of (21.5.1) and their mirrors	3 complexes	exact
(P)	$\mathrm{im} \Psi_2$ absorbs L_E and $L_E^\top$; C obeys the pencil	3 complexes	exact
(U)	every eigenvalue of $L_E _{W_2}$ has modulus $q^{1/2}$	3 complexes	numeric
(N)	$\mathrm{corank} \Psi_2 = 6 = d(d+1)$, and the type relations span	3 complexes	exact

Table 21.3: The battery. “Exact” is exact integer arithmetic; (U) is a floating-point eigenvalue computation, agreeing with $\sqrt{q}$ to 10^{-12} .

quoted in Remark 21.5.1 are Kang and Li [68]; a representation-theoretic reproof is Kang, Li and Wang [69], and an earlier approximation is Deitmar and Hoffman [66]. The rank-two case for Sp_4 is Fang, Li and Wang [67]. For general rank the state of the art is an alternating-product Ihara identity, uniform in n , of Kang and Yu [71]; it carries no Riemann-hypothesis statement and no single-operator determinant formula, and Kang and McCallum [70] had recorded the higher-rank case as a conjecture. A targeted search found no published statement of an *operator-level* intertwining $L_E \Psi = \Psi C$ — at any rank, including $d = 2$ — and none of an unconditional modulus statement for a complementary part of $\mathrm{spec}(L_E)$. Those two, and the flag count that produces them, are what is new here. The spectral definition of a Ramanujan complex of type $\tilde{A}_d$ is Lubotzky, Samuels and Vishne [72].

Part V

Applications outside number theory

Chapter 22

Effective mass from spanning trees, and the absence of Bloch phases on rank-two hyperbolic lattices

Summary. Every $\mathbb{Z}^d$ -periodic tight-binding model with a finite unit cell is an abelian cover of a finite graph, the winding numbers of the hoppings playing the role of a cocycle. We show that for such a lattice the inverse effective-mass tensor at the bottom of the band is, exactly, the Albanese metric of the unit cell: $(M^{-1})_{ab} = (2t/n)\langle h_a, h_b \rangle$, where h_a is the harmonic projection of the a -th winding cochain and n is the number of sites per cell. Three consequences follow. (i) When the flux threads a single link e , the effective mass has a closed combinatorial form, $M^* = n\kappa_0/[2t\tau(Y_0 - e)]$, where κ_0 is the number of spanning trees of the unit cell and $\tau(Y_0 - e)$ the number of those avoiding e . (ii) The tensor $\kappa_0(n/2t)M^{-1}$ has integer entries: *effective masses on such lattices are quantized*, in units fixed by a spanning-tree count. (iii) The diffusion tensor of the classical walk on the same lattice obeys $\sigma^2 = M^{-1}/[t(q+1)]$ with $q+1$ the coordination number — an Einstein relation with no other non-universal factor. We then prove a no-go theorem in the opposite direction. For a lattice whose unit cell is a free quotient of a thick A_2 building — the rank-two analogue of the trees and $\{p, q\}$ tessellations used in circuit-QED hyperbolic lattices — the first Betti number vanishes, so no $\mathbb{Z}^d$ cover exists at all: there is no Brillouin zone, no crystal momentum, and no band structure, and by property (T) no continuous family of unitary characters deforms the trivial one either. The threadable fluxes form a *finite* group, computed exactly as $(\mathbb{Z}/2)^6$, $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$ and $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ on complexes with 21, 24 and 42 sites; on the first of these only π -fluxes exist. Hyperbolic band theory does not survive the passage from rank one to rank two. All statements are verified by machine: 90 checks, exact except where labelled.

22.1 Introduction

Band theory is a statement about $\mathbb{Z}^d$ symmetry. A tight-binding model on a lattice with a finite unit cell and a $\mathbb{Z}^d$ translation group decomposes over a Brillouin zone $\mathbb{T}^d$, and the low-energy physics is controlled by the curvature of the lowest band at its minimum — the effective mass. Two recent developments make it worth asking how far this picture is combinatorial rather than geometric.

The first is the experimental realization of *hyperbolic lattices* in circuit quantum electrodynamics [73], where a coplanar-waveguide network is laid out as a $\{p, q\}$ tessellation of the hyperbolic plane, and the subsequent construction of a hyperbolic band theory [74, 75] in which the role of the Brillouin zone is played by a higher-dimensional Jacobian and by a character variety of the (non-abelian) translation group. The platform has since broadened to electrical circuits [76], integrated photonics [77], and scalable

superconducting architectures with the unit cell on a genus-three surface [78]. The unit cell of such a lattice is a finite graph, and what is threaded through it is flux.

The second is the observation, standard in the mathematics of crystal lattices since Kotani and Sunada [27, 28], that a $\mathbb{Z}^d$ -periodic graph carries a canonical flat metric on H^1 of its unit cell — the Albanese metric — which controls the long-time heat kernel and the central limit theorem of the random walk, and that this metric is bound up with the *complexity* of the unit cell, i.e. its spanning-tree count [79]; explicit Kirchhoff expressions for it were given by Baker and Faber [80]. What does not seem to have been said is that this metric *is* the inverse effective-mass tensor of the tight-binding band bottom, exactly and without approximation, and that reading it so turns those Kirchhoff expressions into a constraint on a measurable quantity. That is the content of Secs. 22.3–22.4, whose sharpest consequence is a quantization statement (Corollary 22.4.2): the entries of M^{-1} , measured in the natural unit $2t/(n\kappa_0)$, are integers.

The paper then turns the question around. Trees and $\{p, q\}$ tessellations are rank-one objects: the tree is the Bruhat–Tits building of GL_2 over a local field, and its finite quotients are ordinary finite graphs, which always have $b_1 \geq 2$ and therefore always admit $\mathbb{Z}^d$ covers. The natural rank-two generalization is a two-dimensional building of type $\tilde{A}_2$ — a simplicial complex of sites, links and plaquettes in which every vertex link is the incidence graph of a projective plane — whose finite quotients are the Ramanujan complexes [62]. These are the higher-rank hyperbolic lattices. We show in Sec. 22.6 that for them band theory does not merely become harder, it ceases to exist: $b_1 = 0$ by Garland’s vanishing theorem [60] as extended by Ballmann and Świątkowski [59], so there is no $\mathbb{Z}^d$ cover, no crystal momentum, and no continuous flux parameter. The Aharonov–Bohm phases that a rank-two hyperbolic lattice admits form a finite abelian group, which we compute exactly on three complexes.

The two halves meet at a single graph. The Heawood graph — the incidence graph of the Fano plane — is a perfectly good rank-one unit cell: it has $b_1 = 8$, $50421 = 3 \cdot 7^5$ spanning trees, and a single-link flux gives it the effective mass $M^* = 147/(8t)$ (Table 22.1). It is also exactly the vertex link of a thick $\tilde{A}_2$ building of order two. Assembling Heawood links into a rank-two complex destroys b_1 completely (Table 22.3). The obstruction is not local.

Relation to previous work

Theorem 22.3.1 lies close to known mathematics. Once the Albanese metric is identified with the covariance of the crystal-lattice central limit theorem [27] and with a Kirchhoff ratio [79, 80], Eq. (22.3.2) follows from a second-order perturbation argument, which we give in full for completeness rather than for novelty. What we claim as new is: the tensor form of Theorem 22.3.1 together with the explicit statement that this object is the physical effective-mass tensor; the quantization Corollary 22.4.2 and its reading as a rigid constraint on measurable band curvature; the deletion–contraction closed form (22.4.2); the loop-flux Corollary 22.4.3; and the whole of Sec. 22.6. The closest prior result in the physics literature that we are aware of is that of Korotyaev and Saburova [81], who bound and estimate effective masses at band edges of periodic discrete and metric graphs from geometric parameters, without spanning trees or the Albanese metric. An “effective-mass approximation” of a different kind — the long-distance reduction of the hyperbolic tight-binding Hamiltonian to a Laplace–Beltrami operator — appears in Ref. [74].

For Sec. 22.6 we have found no prior work: we are not aware of any physics literature that places a tight-binding model on an affine building of rank two or on a Ramanujan complex, let alone one that asks whether it has a band structure. A literature search cannot establish a negative, and we state this only as the limit of our knowledge.

22.2 Lattices from abelian covers

Let Y_0 be a finite connected $(q+1)$ -regular graph with n vertices and $m = (q+1)n/2$ edges, E^+ a choice of orientation, and $r = m - n + 1 = b_1(Y_0)$ its cycle rank. Write $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$ for the boundary map, $\partial e = t(e) - o(e)$, and $\Delta_0 = \partial\partial^t = (q+1)I - A$ for the graph Laplacian. Let κ_0 denote the number of spanning trees of Y_0 .

Definition 22.2.1 (The lattice). A *winding cochain* is a d -tuple $\alpha = (\alpha_1, \dots, \alpha_d)$ with $\alpha_a \in \mathbb{Z}^{E^+}$. The associated lattice $\tilde{Y}_\alpha$ has sites $V \times \mathbb{Z}^d$ and a link from $(o(e), \mathbf{j})$ to $(t(e), \mathbf{j} + \alpha(e))$ for each $e \in E^+$ and each $\mathbf{j} \in \mathbb{Z}^d$. We assume throughout that the period map $H_1(Y_0; \mathbb{Z}) \rightarrow \mathbb{Z}^d$, $c \mapsto (\langle \alpha_a, c \rangle)_a$, is surjective, which is exactly the condition that $\tilde{Y}_\alpha$ be connected; in particular $d \leq r$.

This is not a special construction: *every* $\mathbb{Z}^d$ -periodic tight-binding lattice with n sites per cell and uniform hopping is of this form, with Y_0 the unit cell and $\alpha(e)$ recording how many cells the hopping e crosses. Changing which representative of a site one calls “in cell $\mathbf{0}$ ” replaces α by $\alpha + \partial^t \varphi$; the class $[\alpha] \in H^1(Y_0; \mathbb{Z})^d$ is the gauge-invariant datum.

The Hamiltonian is $H = t \tilde{\Delta}$, with $\tilde{\Delta}$ the Laplacian of $\tilde{Y}_\alpha$: uniform hopping amplitude $-t$ on every link and on-site energy $(q+1)t$. Floquet transform in the $\mathbb{Z}^d$ direction gives $\ell^2(\tilde{Y}_\alpha) \cong \int_{\mathbb{T}^d}^{\oplus} \mathbb{C}^n d^d k / (2\pi)^d$ and $\tilde{\Delta} \cong \int^{\oplus} \Delta_{\mathbf{k}}$ with the Peierls-substituted Bloch Laplacian

$$(\Delta_{\mathbf{k}})_{uv} = (q+1)\delta_{uv} - \sum_{e: u \rightarrow v} e^{i\mathbf{k} \cdot \alpha(e)} - \sum_{e: v \rightarrow u} e^{-i\mathbf{k} \cdot \alpha(e)}, \quad (22.2.1)$$

a Hermitian positive-semidefinite $n \times n$ matrix, and bands $E_j(\mathbf{k}) = t\lambda_j(\mathbf{k})$, $\lambda_0 \leq \dots \leq \lambda_{n-1}$. We set $\hbar = 1$ and the lattice constant to unity, so that

$$(M^{-1})_{ab} := \left. \frac{\partial^2 E_0}{\partial k_a \partial k_b} \right|_{\mathbf{k}=0} = t \left. \frac{\partial^2 \lambda_0}{\partial k_a \partial k_b} \right|_{\mathbf{k}=0}. \quad (22.2.2)$$

Since Δ_0 is a Laplacian of a connected graph its kernel is the constant line, simple, with gap $\lambda_1(0) > 0$; the band bottom sits at $\mathbf{k} = 0$ and λ_0 is analytic near it.

22.3 The effective-mass tensor is the Albanese metric

Let $\Pi = I - \partial^t \Delta_0^+ \partial$ be the orthogonal projection of $\mathbb{R}^{E^+}$ onto the cycle space $\ker \partial = H_1(Y_0; \mathbb{R})$, and let $h_a := \Pi \alpha_a$ be the harmonic representative of the a -th winding cochain. Define the *Albanese matrix*

$$H_{ab} := \langle h_a, h_b \rangle = \alpha_a^t \Pi \alpha_b. \quad (22.3.1)$$

Theorem 22.3.1 (Effective mass). *Under the hypothesis of Definition 22.2.1, λ_0 is even in $\mathbf{k}$, simple and analytic near the origin, and*

$$\lambda_0(\mathbf{k}) = \frac{1}{n} \mathbf{k}^t H \mathbf{k} + O(|\mathbf{k}|^4), \quad \text{so} \quad M^{-1} = \frac{2t}{n} H. \quad (22.3.2)$$

H is positive definite, and it is the Gram matrix of the classes $[\alpha_a]$ in the flat metric of the Jacobian torus $H^1(Y_0; \mathbb{R})/H^1(Y_0; \mathbb{Z})$, dual to the Albanese torus of Kotani–Sunada [28]: if C is the matrix of a $\mathbb{Z}$ -basis of $H_1(Y_0; \mathbb{Z})$, $G = C^t C$ its Gram matrix and $\mathbf{a}_a = C^t \alpha_a$ the vector of periods, then

$$H_{ab} = \mathbf{a}_a^t G^{-1} \mathbf{a}_b. \quad (22.3.3)$$

Equivalently, in terms of the Floquet determinant $\Theta(\mathbf{k}) = \det \Delta_{\mathbf{k}}$,

$$\left. \frac{\partial^2 \Theta}{\partial k_a \partial k_b} \right|_{\mathbf{k}=0} = 2 \kappa_0 H_{ab}. \quad (22.3.4)$$

Proof. $\Delta_{-\mathbf{k}} = \overline{\Delta_{\mathbf{k}}} = \Delta_{\mathbf{k}}^t$, so the spectrum is even in $\mathbf{k}$ and all odd Taylor coefficients of λ_0 vanish. Expanding (22.2.1) about $\mathbf{k} = 0$ gives $\Delta_{\mathbf{k}} = \Delta_0 + \sum_a k_a D_a^{(1)} + \frac{1}{2} \sum_{ab} k_a k_b D_{ab}^{(2)} + O(|\mathbf{k}|^3)$ with $(D_a^{(1)})_{uv} = -i \sum_{e: u \rightarrow v} \alpha_a(e) + i \sum_{e: v \rightarrow u} \alpha_a(e)$ and $(D_{ab}^{(2)})_{uv} = \sum_{e: u \rightarrow v} \alpha_a(e) \alpha_b(e) + (u \leftrightarrow v)$. On the constant vector $\mathbf{1}$ one has $D_a^{(1)} \mathbf{1} = i \partial \alpha_a$, $\langle \mathbf{1}, D_a^{(1)} \mathbf{1} \rangle = 0$ and $\langle \mathbf{1}, D_{ab}^{(2)} \mathbf{1} \rangle = 2 \langle \alpha_a, \alpha_b \rangle$. Second-order Rayleigh–Schrödinger perturbation theory at the simple eigenvalue 0 therefore gives

$$\lambda_0(\mathbf{k}) = \frac{1}{n} \sum_{ab} k_a k_b \left(\langle \alpha_a, \alpha_b \rangle - (\partial \alpha_a)^t \Delta_0^+ (\partial \alpha_b) \right) + O(|\mathbf{k}|^4),$$

and the bracket is $\alpha_a^t(I - \partial^t \Delta_0^+ \partial) \alpha_b = H_{ab}$. For (22.3.3), h_a is the orthogonal projection of α_a onto the column space of C , so $h_a = C(C^t C)^{-1} C^t \alpha_a$ and $\langle h_a, h_b \rangle = \alpha_a^t C G^{-1} C^t \alpha_b$. Positive definiteness is the surjectivity hypothesis: $\mathbf{v}^t H \mathbf{v} = 0$ forces $\sum_a v_a [\alpha_a] = 0$ in $H^1(Y_0; \mathbb{Q})$, impossible for $\mathbf{v} \neq 0$ when the periods span. Finally $\Theta(\mathbf{k}) = \lambda_0(\mathbf{k}) \prod_{j \geq 1} \lambda_j(\mathbf{k})$, and by the matrix-tree theorem $\prod_{j \geq 1} \lambda_j(0) = n \kappa_0$, which with (22.3.2) gives (22.3.4). $\square$

For $d = 1$ Eq. (22.3.2) is the effective-mass identity of Ref. [Ch. 8], where $-n\lambda_0''(0)/2$ was identified with an $\mathcal{L}$ -invariant and with a diffusion constant. The content of (22.3.3) for $d = 1$ — that the Albanese metric evaluates the harmonic energy — is Kotani–Sunada [27, 79]; what is added here is the $d > 1$ tensor, the identification of the whole object with M^{-1} , and the consequences drawn in Secs. 22.4–22.5.

Remark 22.3.2. Equation (22.3.4) is the practically useful form: $\Theta(\mathbf{k})$ is a Laurent polynomial in $e^{ik_1}, \dots, e^{ik_d}$ with integer coefficients, computable exactly, and its Hessian at the origin delivers $\kappa_0 H$ without any eigenvalue computation. This is how the battery of Sec. 22.7 verifies Theorem 22.3.1 in exact arithmetic.

22.4 Spanning trees and the quantization of effective mass

Theorem 22.4.1 (Single-link flux). *Let $d = 1$ and let the winding cochain be the indicator of a single link e , so that the lattice is the unit cell repeated along one direction with exactly one hopping crossing the cell boundary. Then*

$$H = 1 - R_{\text{eff}}(e) = \frac{\tau(Y_0 - e)}{\kappa_0}, \quad (22.4.1)$$

where $R_{\text{eff}}(e)$ is the effective resistance across e when every link carries unit resistance, $\kappa_0 = \tau(Y_0)$ is the number of spanning trees of the unit cell, and $\tau(Y_0 - e)$ is the number of spanning trees avoiding e . Consequently

$$M^* = \frac{n}{2t} \cdot \frac{\kappa_0}{\tau(Y_0 - e)}. \quad (22.4.2)$$

Proof. $H = \Pi_{ee} = 1 - (\partial^t \Delta_0^+ \partial)_{ee} = 1 - R_{\text{eff}}(e)$ by the standard resistance formula. Kirchhoff's theorem gives $\kappa_0 R_{\text{eff}}(e) = \tau(Y_0/e)$, the number of spanning trees containing e , and deletion-contraction $\kappa_0 = \tau(Y_0 - e) + \tau(Y_0/e)$ finishes it. Equation (22.4.2) is (22.3.2). $\square$

Corollary 22.4.2 (Quantization of effective mass). *For any winding cochain, $\kappa_0 H$ is a matrix of integers. Hence every entry of the inverse effective-mass tensor is an integer multiple of $2t/(n\kappa_0)$,*

$$\frac{n\kappa_0}{2t} (M^{-1})_{ab} \in \mathbb{Z}, \quad \det \left[\frac{n\kappa_0}{2t} M^{-1} \right] \in \mathbb{Z}, \quad (22.4.3)$$

and the admissible effective masses of a lattice with unit cell Y_0 form a discrete set determined by the single integer κ_0 .

Proof. By Kirchhoff's theorem in its cycle form, the Gram matrix G of any $\mathbb{Z}$ -basis of $H_1(Y_0; \mathbb{Z})$ has $\det G = \kappa_0$; since $\text{adj } G$ is an integer matrix, $G^{-1} \in \kappa_0^{-1} M_r(\mathbb{Z})$. The period vectors $\mathbf{a}_a = C^t \alpha_a$ are integral, so $\kappa_0 H_{ab} = \mathbf{a}_a^t (\kappa_0 G^{-1}) \mathbf{a}_b \in \mathbb{Z}$ by (22.3.3). $\square$

This is the most directly falsifiable statement in the paper. The band curvature at the Γ point of such a lattice cannot be an arbitrary real number: measured in units of $2t/n$, it is a rational with denominator dividing κ_0 . Table 22.1 lists the values for six unit cells.

The algebra behind Corollary 22.4.2 is elementary, and in the equivalent form “the Albanese lattice of Y_0 has covolume κ_0 ” it is classical [79, 80]. What is at stake here is only what is being quantized: not a lattice covolume but the curvature of a band, which an experiment can measure.

Corollary 22.4.3 (Loop flux). *If the winding cochain is the indicator of an oriented cycle $C \subset Y_0$ of length ℓ , then $h = \alpha$, $H = \ell$ exactly, and*

$$M^* = \frac{n}{2t\ell}. \quad (22.4.4)$$

The effective mass depends only on the length of the flux loop and on the number of sites per cell — not on the rest of the unit cell.

Y_0	n	m	q	κ_0	$R_{\text{eff}}(e)$	$\tau(Y_0 - e)$	tM^*
K_4	4	6	2	16	1/2	8	4
$K_{3,3}$	6	9	2	81	5/9	36	27/4
Q_3	8	12	2	384	7/12	160	48/5
K_5	5	10	3	125	2/5	75	25/6
Petersen	10	15	2	2000	3/5	800	25/2
Heawood	14	21	2	50421	13/21	19208	147/8

Table 22.1: Single-link flux. κ_0 is the number of spanning trees of the unit cell, $\tau(Y_0 - e)$ the number avoiding the flux link, and $tM^* = n\kappa_0/[2\tau(Y_0 - e)]$ by Eq. (22.4.2). For the Heawood graph $\kappa_0 = 3 \cdot 7^5$ and $\tau(Y_0 - e) = 8 \cdot 7^4$. All entries exact.

Y_0	κ_0	$\kappa_0 H$	$\det(\kappa_0 H)$
K_4	16	$\begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix}$	48
Q_3	384	$\begin{pmatrix} 160 & -32 \\ -32 & 160 \end{pmatrix}$	24576
K_5	125	$\begin{pmatrix} 75 & -25 \\ -25 & 75 \end{pmatrix}$	5000
Heawood	50421	$\begin{pmatrix} 19208 & -9604 \\ -9604 & 19208 \end{pmatrix}$	276710448

Table 22.2: The effective-mass tensor for $d = 2$, two single-link fluxes in independent directions. $\kappa_0 H = (n\kappa_0/2t)M^{-1}$ is an integer matrix (Corollary 22.4.2); the off-diagonal entries are the mass anisotropy. All entries exact.

Proof. A cycle lies in $\ker \partial$, so $\Pi\alpha = \alpha$ and $H = \|\alpha\|^2 = \ell$. $\square$

Corollary 22.4.3 is the sharp opposite of Theorem 22.4.1: a single link is maximally sensitive to the ambient cell (through R_{eff}), a closed loop is completely insensitive to it. Both are verified in Sec. 22.7 on K_5 , where the cyclically oriented triangle gives $H = 3$ exactly while the same three links with a non-cyclic orientation give $H = 7/5$.

22.5 An Einstein relation

The same matrix H governs the classical transport problem on $\tilde{Y}_\alpha$. Let (X_i) be the simple random walk on Y_0 and $\mathbf{S}_N = \sum_{i < N} \alpha(X_i, X_{i+1}) \in \mathbb{Z}^d$ the accumulated winding — the cell coordinate of the lifted walk on $\tilde{Y}_\alpha$.

Theorem 22.5.1 (Diffusion tensor). *$\mathbf{S}_N/\sqrt{N}$ converges in law to a centred Gaussian with covariance $\sigma_{ab}^2 = H_{ab}/m$, and therefore*

$$\sigma^2 = \frac{1}{t(q+1)} M^{-1}. \quad (22.5.1)$$

Proof. The walk is reversible with uniform stationary law and transition matrix $A/(q+1)$; the drift is $-\partial\alpha_a/(q+1)$ and the Poisson equation $\Delta_0 g_a = -\partial\alpha_a$ has solution $g_a = -\Delta_0^+ \partial\alpha_a$. Then $\alpha_a + \partial^t g_a = \Pi\alpha_a = h_a$ are the martingale increments, and the martingale central limit theorem gives $\sigma_{ab}^2 = \frac{2}{n(q+1)} \langle h_a, h_b \rangle = H_{ab}/m$. Substituting $H = (n/2t)M^{-1}$ and $m = n(q+1)/2$ gives (22.5.1). $\square$

Equation (22.5.1) says that the quantum band curvature and the classical diffusion tensor of a lattice of this type differ by the coordination number alone. In particular the mass anisotropy and the diffusion anisotropy are the *same* tensor up to scale, and both are quantized by Corollary 22.4.2. The $d = 1$ case is the crystal-lattice central limit theorem of Kotani and Sunada [27], in which the covariance is the Albanese metric; Theorem 22.3.1 is what identifies that covariance with a band curvature.

Y	sites	links	plaquettes	b_1	$H_1(Y; \mathbb{Z})$
Y_{21}	21	147	147	0	$(\mathbb{Z}/2)^6$
Y_{24}	24	168	168	0	$(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$
Y_{42}	42	294	294	0	$(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$

Table 22.3: Three finite quotients of a thick $\tilde{A}_2$ building of order $q = 2$, built as abelian covers of the three-site type quotient of a triangle presentation over $\text{PG}(2, 2)$. Every vertex link is a Heawood graph. $b_1 = 0$ in every case, so none of them is the unit cell of any periodic lattice; the entire flux group is the finite group in the last column. On Y_{21} the only admissible fluxes are 0 and π . All entries exact (p -adic Smith normal forms; see Sec. 22.7).

22.6 Rank two: no Brillouin zone

Everything above needs $b_1(Y_0) \geq 1$. For an ordinary finite graph this is automatic and harmless: a connected $(q+1)$ -regular graph with $q \geq 2$ has $r = m - n + 1 \geq 2$, so $\mathbb{Z}^d$ covers exist for every $d \leq r$. The hyperbolic lattices of Refs. [73, 74] are of this type — their unit cells are finite graphs, and the difficulty there is that the deck group is non-abelian, not that abelian directions are missing.

We now consider the rank-two analogue. A thick $\tilde{A}_2$ building of order q is the two-dimensional simplicial complex on which PGL_3 of a local field with residue field of size q acts; it is the rank-two counterpart of the $(q+1)$ -regular tree, and every vertex link in it is the incidence graph of the projective plane over $\mathbb{F}_q$ (for $q = 2$, the Heawood graph). Its finite quotients are lattices of sites, links and plaquettes; the Ramanujan complexes of Lubotzky, Samuels and Vishne [62] are among them, and they are built from groups acting simply transitively on the vertices in the sense of Cartwright, Mantero, Steger and Zappa [52]. These are the natural candidates for a higher-rank hyperbolic lattice.

Theorem 22.6.1 (No band structure in rank two). *Let $Y = \Gamma \backslash B$ be a finite quotient of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a group Γ acting freely. Then*

- (i) $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z})$ is finite;
- (ii) Y admits no connected $\mathbb{Z}^d$ cover for any $d \geq 1$. There is no $\mathbb{Z}^d$ -periodic lattice with unit cell Y , hence no Brillouin zone, no crystal momentum, and no band structure;
- (iii) the abelian covers of Y are exactly those with deck group a quotient of the finite group $H_1(Y; \mathbb{Z})$, so the threadable flux configurations form the finite group $\text{Hom}(H_1(Y; \mathbb{Z}), U(1)) \cong H_1(Y; \mathbb{Z})$;
- (iv) Γ has Kazhdan's property (T); the trivial representation is isolated in its unitary dual, so no continuous family of unitary characters — abelian or not — deforms the trivial one.

Proof. Γ acts properly and cocompactly on B , so by Garland's vanishing theorem [60] in the form extended to all $q \geq 2$ by Ballmann and Świątkowski [59], $H^1(\Gamma; \mathbb{R}) = 0$ and Γ has property (T). Y is aspherical with $\pi_1(Y) = \Gamma$, so $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z}) = \Gamma^{\text{ab}}$ is finite. A cover with deck group $\mathbb{Z}^d$ requires a surjection $\Gamma \rightarrow \mathbb{Z}^d$, hence a surjection $\Gamma^{\text{ab}} \rightarrow \mathbb{Z}^d$, impossible for $d \geq 1$ from a finite group; this is (ii), and (iii) is the classification of abelian covers by $H^1(Y; \cdot) = \text{Hom}(H_1(Y; \mathbb{Z}), \cdot)$. Part (iv) is property (T). $\square$

Table 22.3 makes the statement concrete. On Y_{21} the flux group is $(\mathbb{Z}/2)^6$: there are exactly 64 distinct flux configurations, each link pattern carrying flux 0 or π , and nothing in between. There is no knob to turn continuously, and no momentum to plot a band against.

Remark 22.6.2 (What this does *not* say). Theorem 22.6.1 is a statement about translational symmetry, not about spectra. The Laplacian and the Hecke operators of Y have perfectly good spectra — indeed the Ramanujan complexes of Ref. [62] have optimal spectral gaps, which is precisely what makes them attractive as synthetic lattices. What fails is the specific device of labelling eigenstates by a crystal momentum in a torus. Any transport theory on a rank-two hyperbolic lattice must be built without a Brillouin zone, and by Theorem 22.6.1(iv) not even an infinitesimal one is available near the trivial character. The effective mass of Sec. 22.3 is simply not defined there: it is the curvature of a band that does not exist.

key	check	#	mode
(S)	$H = 1 - R_{\text{eff}}(e)$ and $\kappa_0 H = \tau(Y_0 - e)$, Eq. (22.4.1)	6/6	exact
(M)	$\partial_{ab}^2 \Theta _0 = 2\kappa_0 H$, Eq. (22.3.4)	15/15	exact
(M)	$\lambda_0(\mathbf{k}) = \mathbf{k}^t H \mathbf{k} / n + O(\mathbf{k} ^4)$	15/15	numeric
(Q)	$\kappa_0 H$ integral, Eq. (22.4.3)	15/15	exact
(E)	$\sigma^2 = H/m$ by Green–Kubo, Eq. (22.5.1)	15/15	exact
(G)	gap $\lambda_1(0) > 0$ and $\lambda_0(0) = 0$	15/15	numeric
(X)	H reproduces Refs. [Chs. 7, 8]	4/4	exact
(C)	loop flux gives $H = \ell$, Corollary 22.4.3	1/1	exact
(B)	$b_1 = 0$ and $H_1(Y; \mathbb{Z})$, Table 22.3	4/4	exact
total		90/90	

Table 22.4: The verification battery (`bloch_mass.py`). Unit cells: K_4 , $K_{3,3}$, Q_3 , K_5 , Petersen, Heawood, with single-link fluxes ($d = 1$), two-link fluxes ($d = 2$), and on K_4 and K_5 the multi-chord and triangle patterns of Refs. [Chs. 7, 8].

Remark 22.6.3 (The obstruction is global). The Heawood graph appears on both sides of this paper. As a unit cell in its own right it is unremarkable: $b_1 = 8$, $\kappa_0 = 3 \cdot 7^5$, and a single-link flux gives $tM^* = 147/8$ (Table 22.1). As the *link* of a rank-two complex it is part of a structure with $b_1 = 0$ (Table 22.3). No local inspection of a rank-two hyperbolic lattice reveals the obstruction; it is a property (T) phenomenon, visible only at the level of the fundamental group.

22.7 Numerical verification

All statements were checked by machine. The script `bloch_mass.py` accompanying this paper runs 90 checks in 13 s and reports every one as passing; its output is archived as `bloch_mass_output.txt`. Exact arithmetic is used throughout except where a row is labelled *numeric*.

A word on the two *numeric* rows. The band identity (22.3.2) is checked by evaluating λ_0 at $|\mathbf{k}| = 10^{-6}$ and 5×10^{-7} in 40-digit arithmetic and taking one Richardson step, which removes the $O(|\mathbf{k}|^2)$ truncation and leaves a residual below 10^{-24} ; the observed relative deviations from $\mathbf{k}^t H \mathbf{k} / n$ range from 2×10^{-27} to 2×10^{-26} . The exact row (M) above it, which verifies the Floquet-determinant form (22.3.4) by interpolating Θ as a Laurent polynomial with rational coefficients and differentiating it symbolically, is the mathematically complete check; the numeric row is a redundant confirmation that the object being differentiated really is the bottom band.

The rank-two rows use the complex builder and the p -adic Smith-normal-form routine of Ref. [Ch. 16]: the complexes of Table 22.3 are abelian covers of the three-site type quotient of a triangle presentation over $\text{PG}(2, 2)$ compatible with a Singer cycle, the presentation axioms are re-verified at start-up, and every vertex link is confirmed to be a Heawood graph. The Betti numbers are exact ranks; the torsion subgroups are exact p -adic Smith forms with the prime set certified by the greatest common divisor of the maximal minors.

22.8 Discussion

22.8.1 What is testable

Corollary 22.4.2 is a constraint on a measurable quantity. For a synthetic lattice assembled from a known unit cell — a coplanar-waveguide network, a photonic-crystal supercell, a cold-atom superlattice — the number κ_0 of spanning trees of the cell is a finite combinatorial datum, and the band curvature at Γ must be

an integer multiple of $2t/(n\kappa_0)$. Departures measure either the failure of the uniform-hopping idealization or the presence of longer-range terms; the statement is rigid enough to be diagnostic.

Theorem 22.4.1 says more: for a single flux link, the mass is set by the ratio of two spanning-tree counts, so it can be engineered by choosing where the flux link sits, with no computation beyond counting trees. Corollary 22.4.3 gives the complementary design rule — route the flux around a closed loop of length ℓ and the mass becomes $n/2t\ell$, insensitive to everything else in the cell.

Equation (22.5.1) is a consistency check available without any quantum measurement: the classical diffusion tensor on the same network fixes the effective-mass tensor up to the coordination number.

22.8.2 What is open

(a) The non-abelian case. Hyperbolic band theory [74, 75] replaces $\mathbb{T}^d$ by a character variety of a non-abelian surface group. Theorem 22.3.1 is a statement about the abelian directions only; whether a “non-abelian effective mass” exists at a higher-genus analogue of the Γ point, and whether it is still a spanning-tree quantity, we do not know.

(b) The quartic term. Band non-parabolicity is the next Taylor coefficient of λ_0 . It is a rational function of the same combinatorial data, but we have no closed form for it and no quantization statement.

(c) Rank two without a Brillouin zone. Theorem 22.6.1 removes the tool but not the problem: finite quotients of $\tilde{A}_2$ buildings are excellent expanders with explicit spectra, and the question of what replaces band theory for transport on them — given that property (T) blocks even an infinitesimal character deformation — seems to us the most interesting thing this paper leaves behind.

22.8.3 What is impossible

A $\mathbb{Z}^d$ -periodic lattice whose unit cell is a free quotient of a thick $\tilde{A}_2$ building, for any $d \geq 1$ and any $q \geq 2$ (Theorem 22.6.1). Any construction claiming a Brillouin zone for such a lattice is in error, and the error will be that the quotient is not free, or the building is not thick, or the complex is not a quotient of a building at all. This is a theorem and does not depend on the state of the literature; the remark in Sec. 22.1, that no one appears to have asked the question, does.

Provenance and caveats

This is an unrefereed preprint. Every numbered statement is proved from finite linear algebra together with three classical inputs that are quoted, not reproved: Kirchhoff’s matrix–tree theorem in its cycle form, the martingale central limit theorem for additive functionals of a finite reversible chain, and the Garland–Ballmann–Świątkowski vanishing theorem [60, 59]. Kotani–Sunada [27, 28, 79] and Baker–Faber [80] supply the Albanese metric, its relation to the spanning-tree count, and the crystal-lattice central limit theorem; Secs. 22.3–22.5 are a physical reading of that material and not a replacement for it.

Attribution, statement by statement. The $d = 1$ case of Theorem 22.3.1 is Ref. [Ch. 8], Theorem 4.2; Eq. (22.4.1) for $d = 1$ is Ref. [Ch. 7], Theorem 1.2(iv); Theorem 22.6.1(i) is Ref. [Ch. 18], Theorem 1.1, which in turn quotes Refs. [60, 59]. The identification of the Albanese metric with a Kirchhoff ratio, and hence the mathematical substance of Corollary 22.4.2, is in Refs. [79, 80]. What the present paper adds is the $d > 1$ tensor form, the identification of that tensor with M^{-1} , the deletion–contraction form (22.4.2), Corollary 22.4.3, the Einstein relation in the form (22.5.1), and Sec. 22.6 — the reading of the rank-two vanishing theorem as a no-go theorem for band structure, which is the part we believe to be without precedent. We have not found prior physics work on tight-binding models over rank-two affine buildings; we cannot prove that none exists.

Acknowledgments

The complex builder and the exact p -adic Smith normal form used in Sec. 22.7 are those of Ref. [Ch. 16].

Chapter 23

What the boundary algebra of p -adic AdS/CFT does not see

Summary. In p -adic AdS/CFT the bulk is the Bruhat–Tits tree of PGL_2 over a local field and the boundary is $\mathbb{P}^1(F)$. The boundary carries two standard operator algebras, the crossed products $C(\mathbb{P}^1(F)) \rtimes \Gamma$ and $L^\infty(\mathbb{P}^1(F)) \rtimes \Gamma$, and both have been completely determined: the first is a Cuntz–Krieger algebra with $K_0 = \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$ depending only on the rank of Γ , and the second is the hyperfinite factor of type III_λ with $\lambda = q^{-2}$ or q^{-1} according as the quotient graph is bipartite or not (Robertson, 2005). Neither result has ever been cited in the holography literature, and no p -adic AdS/CFT paper has used an operator-algebraic invariant at all. We import them, and add one arithmetic layer. The bulk Hecke–Laplacian channel has $K_0 = \mathbb{Z} \oplus \mathrm{Jac}(Y)$, the critical group of the quotient graph, whose order is a product of Frobenius traces $\frac{1}{n} \prod_f (p+1 - a_p(f))$. We exhibit two cubic graphs on eight vertices whose boundary algebras agree in K -theory *and* in von Neumann type, and whose bulk critical groups have coprime orders $256 = 2^8$ and $363 = 3 \cdot 11^2$: the boundary data do not determine the bulk arithmetic, not even which primes divide it. This is strictly stronger than Robertson’s own counterexample, which the type does separate. We do not conclude that holography fails: Cornelissen and Marcolli proved that the boundary quantum statistical mechanical system, with its Busemann time evolution, reconstructs the graph completely, so what we exhibit is a precise gap in a hierarchy whose top is a reconstruction theorem. Finally we observe that $\lambda \neq 1$ is itself a diagnostic. Continuum holography is forced to type III_1 by theorems (mixing; half-sided modular inclusion); p -adic holography is type III_λ with $\lambda < 1$, so its modular flow is periodic and its boundary algebra is not a Haag–Kastler algebra. All computations are exact and machine-verified.

23.1 Introduction

p -adic AdS/CFT [97, 99] replaces anti-de Sitter space by the Bruhat–Tits tree X of $\mathrm{PGL}_2(F)$, F a non-archimedean local field with residue field of size q , and the conformal boundary by $\mathbb{P}^1(F)$, the space of ends of X . A discrete torsion-free cocompact lattice $\Gamma \leq \mathrm{PGL}_2(F)$ — free of finite rank r — plays the role of a quotient identification, exactly as a Schottky group does for BTZ [100], and $Y = \Gamma \backslash X$ is a finite $(q+1)$ -regular graph on n vertices. The field has matured on correlators, tensor networks, Wilson lines and finite temperature [98, 105, 100, 106], but it has remained kinematic in one specific respect: no invariant from operator algebras has ever been computed in it.

That is unfortunate, because the relevant invariants were computed twenty years ago, in a literature the holography community does not read. Robertson [108] determined both boundary algebras of exactly this situation, and Cornelissen and Marcolli [90] determined precisely how much of the bulk each one remembers.

The purpose of this note is to import those results, to add one layer they do not contain, and to draw one physical consequence.

What is imported. For Γ a free uniform tree lattice of rank r , Robertson proved [108, Thm. 1] that $C(\partial X) \rtimes \Gamma$ is a Cuntz–Krieger algebra with

$$K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)\mathbb{Z}, \quad K_1 \cong \mathbb{Z}^r, \quad (23.1.1)$$

[1] generating the torsion summand — so by Kirchberg–Phillips the algebra *depends only on* r . (The same K -groups were obtained independently in [89].) He proved further [108, Thm. 2, Cor. 1] that

$$L^\infty(\partial X) \rtimes \Gamma \cong \text{hyperfinite III}_\lambda, \quad \lambda = \begin{cases} q^{-2} & \Gamma \subset \text{PSL}_2(F) \text{ (} Y \text{ bipartite)}, \\ q^{-1} & \text{otherwise,} \end{cases} \quad (23.1.2)$$

the ratio set being generated by the Busemann cocycle; the rank-one case goes back to Ramagge and Robertson [107] and the general hyperbolic-group statement to [101, 85]. Cornelissen and Marcolli [90] then showed that the C^* -level invariants “reconstruct only topological information” — nine natural equivalences all reduce to equality of the first Betti number — while the boundary *quantum statistical mechanical* system, $C(\partial X) \rtimes \Gamma$ equipped with the Busemann time evolution, reconstructs Y up to isomorphism.

What is added. The bulk of this geometry has an invariant that none of these papers considers. The Hecke–Laplacian $\Delta_p = (q+1)I - T_p$ of Y has

$$\text{coker}_{\mathbb{Z}} \Delta_p \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad |\text{Jac}(Y)| = \frac{1}{n} \prod_f (p+1 - a_p(f)), \quad (23.1.3)$$

where $\text{Jac}(Y)$ is the critical (sandpile) group [84, 83] and the product runs over the weight-two newforms attached to Γ , so that $|\text{Jac}(Y)|$ is a product of Frobenius traces [Ch. 2]. The critical group has appeared in physics — as the global symmetry group of fracton models on graphs [95, 96] — but never in holography. Our contribution is Theorem 23.3.1: the boundary data of (23.1.1) and (23.1.2) *together* fail to determine (23.1.3), and fail badly enough that two bulks in the same boundary class can have coprime arithmetic.

What we do not claim. Not that holography fails. [90] is a reconstruction theorem: keep the time evolution and the bulk comes back. What Theorem 23.3.1 locates is a gap between two rungs of a ladder whose top rung is a positive result. Nor do we claim (23.1.1) or (23.1.2); both are [108].

23.2 Setting

$Y = \Gamma \backslash X$ is a finite connected $(q+1)$ -regular graph, $n = |V_Y|$, $m = (q+1)n/2$, and

$$r = \text{rank } \Gamma = m - n + 1, \quad r - 1 = m - n = \frac{(q-1)n}{2}.$$

The *boundary channel* is the crossed product by the Γ -action on ∂X ; concretely $C(\partial X) \rtimes \Gamma \cong \mathcal{O}_{A^{\text{nb}}}$ with A^{nb} the non-backtracking matrix on the $2m$ oriented edges of Y [108], and $K_0 = \text{coker}(I - (A^{\text{nb}})^t)$, $K_1 = \ker(I - (A^{\text{nb}})^t)$ by [91]. The *bulk channel* is Δ_p on the n vertices, with (23.1.3).

Remark 23.2.1 (Two relations, two types). Equation (23.1.2) concerns the orbit relation of Γ on ∂X . It must be distinguished from the tail equivalence relation on the Cantor limit $\varprojlim_k V_{Y_k}$ of a tower of covers, whose ratio set is all of $q^{\mathbb{Z}} \cup \{0\}$ and which therefore gives type $\text{III}_{1/q}$ unconditionally [Ch. 5]. The orbit relation of a type-preserving lattice realises only $q^{2\mathbb{Z}}$, because the Busemann cocycle of such an element has even value; this is the source of the dichotomy in (23.1.2) and it is easy to conflate the two.

Proposition 23.2.2. *The boundary K -theory (23.1.1) is a function of (q, n) alone, and the von Neumann type (23.1.2) is a function of q and the bipartiteness of Y alone. In particular all $(q+1)$ -regular graphs on n vertices with the same bipartiteness have the same boundary data.*

Proof. $r - 1 = (q - 1)n/2$ and $r = (q - 1)n/2 + 1$ depend only on (q, n) ; (23.1.2) is as stated. $\square$

23.3 The bulk is not determined

Theorem 23.3.1. *There are two cubic graphs Y_A, Y_B on eight vertices with*

- *the same boundary K -theory, $K_0 = \mathbb{Z}^5 \oplus \mathbb{Z}/4$ and $K_1 = \mathbb{Z}^5$;*
- *the same von Neumann type, both being non-bipartite and hence $\text{III}_{1/q}$;*
- *coprime bulk critical groups,*

$$\text{Jac}(Y_A) = \mathbb{Z}/4 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16, \quad |\text{Jac}(Y_A)| = 256 = 2^8,$$

$$\text{Jac}(Y_B) = \mathbb{Z}/33 \oplus \mathbb{Z}/11, \quad |\text{Jac}(Y_B)| = 363 = 3 \cdot 11^2.$$

Explicitly, with vertices $0, \dots, 7$,

$$E(Y_A) = \{03, 06, 07, 12, 15, 16, 24, 25, 34, 37, 45, 67\},$$

$$E(Y_B) = \{02, 03, 07, 14, 16, 17, 24, 25, 34, 35, 56, 67\}.$$

Consequently the boundary K -theory and the boundary von Neumann type, taken together, do not determine $|\text{Jac}(Y)|$, nor the set of primes dividing it, nor — by (23.1.3) — the Frobenius traces $a_p(f)$ of the associated automorphic forms.

Proof. Direct computation of the three invariants in exact integer arithmetic; this is check (P) of Section 23.5. Both graphs contain an odd cycle, hence are non-bipartite. $\square$

Remark 23.3.2 (Comparison with [108]). Robertson gives his own counterexample to boundary rigidity [108, Ex. 1.5]: the theta graph and the dumbbell graph, both with $\pi_1 = F_2$, non-isomorphic, with the same $C(\partial X) \rtimes \Gamma$, realised concretely inside $\text{PGL}_2(\mathbb{Q}_2)$. His example is separated by the von Neumann type, $\text{III}_{1/4}$ against $\text{III}_{1/2}$ — indeed that is his point, that the measure-theoretic invariant is finer. Theorem 23.3.1 defeats both invariants at once, and does so in a way that is arithmetically meaningful rather than merely graph-theoretic.

Table 23.1 shows the two channels on a list of standard quotients, and Table 23.2 shows how much bulk arithmetic a single boundary class holds.

Y	q	n	r	$K_0(\text{boundary})$	vN type	$ \text{Jac}(Y) $
K_4	2	4	3	$\mathbb{Z}^3 \oplus \mathbb{Z}/2$	$\text{III}_{1/q}$	16
$K_{3,3}$	2	6	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/3$	III_{1/q^2}	81
prism $C_3 \times K_2$	2	6	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/3$	$\text{III}_{1/q}$	75
prism $C_4 \times K_2$	2	8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$	III_{1/q^2}	384
Möbius M_8	2	8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$	$\text{III}_{1/q}$	392
Möbius M_{10}	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	III_{1/q^2}	1815
prism $C_5 \times K_2$	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	1805
Petersen	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	2000
Heawood	2	14	8	$\mathbb{Z}^8 \oplus \mathbb{Z}/7$	III_{1/q^2}	50421
K_5	3	5	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	125
K_6	4	6	10	$\mathbb{Z}^{10} \oplus \mathbb{Z}/9$	$\text{III}_{1/q}$	1296

Table 23.1: The two channels. $K_1 = \mathbb{Z}^r$ throughout. Within a block of equal (q, n) the boundary K -theory is constant; the type refines it by bipartiteness; $|\text{Jac}(Y)|$ is constrained by neither. Compare prism $C_5 \times K_2$ and Petersen: same (q, n) , same type, $|\text{Jac}| = 1805$ against 2000. All entries exact.

n	r	$K_0(\text{boundary})$	distinct	$ \text{Jac} $	found	range
8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$			5	256 – 392
10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$			16	960 – 2000
12	7	$\mathbb{Z}^7 \oplus \mathbb{Z}/6$			38	2808 – 9747
14	8	$\mathbb{Z}^8 \oplus \mathbb{Z}/7$			54	12100 – 46998

Table 23.2: Bulk critical groups inside a single boundary K -theory class, from 60 random cubic quotients at each n . Sampled, not exhaustive: the true counts are at least these.

23.4 The type as a physical diagnostic

The recent algebraic program in holography [103, 104, 110, 87, 88] turns on the fact that the large- N algebra of a holographic CFT, and the algebra of a black hole exterior, are of type III_1 , and that crossed products by the modular flow convert them to semifinite type II, which is what makes generalized entropy finite. Type III_1 is not an accident there: it is forced. If the correlation functions decay — if the algebra is generated by mixing operators in a KMS state — the algebra must be III_1 [93]; and the half-sided modular inclusion structure that [103, 104] use forces III_1 by Wiesbrock’s theorem [109]. Local algebras of relativistic quantum field theory are the hyperfinite III_1 factor [86].

Equation (23.1.2) says that p -adic holography is not in that class. Its boundary algebra is III_λ with $\lambda = q^{-1}$ or q^{-2} , a Powers factor. The physical content of $\lambda < 1$ is that the modular flow is *periodic*: the flow of weights is the translation flow on $\mathbb{R}/T\mathbb{Z}$ with

$$T = \frac{2\pi}{\log \lambda^{-1}} = \frac{2\pi}{\kappa \log q}, \quad \kappa \in \{1, 2\},$$

so that the modular Hamiltonian has spectrum in $(\log q^\kappa)\mathbb{Z}$ rather than a continuum. A III_λ algebra has a nontrivial discrete invariant — Connes’ S invariant is $\lambda^\mathbb{Z} \cup \{0\}$ — where a III_1 algebra has none.

We draw one conclusion and refrain from a second. The conclusion: the type is a sharp, computable obstruction to any claim that a p -adic holographic boundary theory is a local quantum field theory in the Haag–Kastler sense, and it identifies precisely what would have to change — the residue field would have to be made continuous, $q \rightarrow 1$ — for the III_1 of [103, 110] to be recovered. What we refrain from: we do not propose a p -adic analogue of the crossed-product construction of [110]. For III_λ with $\lambda < 1$ the crossed product by the modular flow is a type II_∞ algebra as well, but the relevant flow is periodic, and whether the resulting trace has anything to do with an entropy is a question we have not investigated.

23.5 Verification

The script `boundary_blind.py` accompanying this note runs the checks of Table 23.3 in 0.3 s, all passing; its output is archived as `boundary_blind_output.txt`. Arithmetic is exact integer throughout except the two rows marked *sampled*.

key	check	mode
(K)	(23.1.1) holds on 11 standard quotients, including $q = 3, 4$	exact
(B)	$r - 1 = (q - 1)n/2$: the boundary K -theory is a function of (q, n)	exact
(J)	witness families at $(q, n) = (2, 6), (2, 8), (2, 10)$: same boundary, different Jac	exact
(P)	Theorem 23.3.1: same K -theory, same type, coprime $ \text{Jac} $	exact
(S)	Table 23.2: the spread inside one boundary class	sampled
(R)	the Busemann cocycle is $q^\mathbb{Z}$ -valued (normalisation check only)	exact

Table 23.3: The verification battery (`boundary_blind.py`), 7/7 passing.

Row (R) deserves a caveat we state rather than bury: it verifies only that the cylinder measures $\nu(\Omega_v) = q^{1-d(O,v)}$ give a $q^{\mathbb{Z}}$ -valued Radon–Nikodym cocycle. That the *essential range* of that cocycle is attained — which is what Krieger’s theorem [102] needs — is quoted from [107, 108], not verified here; no finite computation could verify it.

23.6 Scope

Quoted, not proved here. Equations (23.1.1) and (23.1.2) and the rigidity observation [108]; the reconstruction theorem [90]; Krieger’s classification [102]; the Cuntz machine [91]; the forcing of III_1 in the continuum [93, 109, 86].

New here. Theorem 23.3.1 and Tables 23.1–23.2: the arithmetic layer, and the observation that K -theory and type together are still blind to it. The import of [108, 90] into p -adic holography, where no operator-algebraic invariant has previously been computed. The reading of $\lambda \neq 1$ in Section 23.4 as a diagnostic separating p -adic from continuum holography.

Open. (a) Whether the Cornelissen–Marcolli reconstruction [90], which recovers Y and hence $\text{Jac}(Y)$, can be made to recover the arithmetic *directly* — an explicit formula for $|\text{Jac}(Y)|$ from the boundary KMS data would be a genuine holographic dictionary entry for Frobenius traces. (b) The higher-rank case: for $\tilde{A}_n$ buildings the boundary type is $\text{III}_{1/q}$ or III_{1/q^2} according to the parity of n [92], and holographic constructions on buildings exist [94, 105], but none is arithmetic; the bulk invariant there is the rank-two critical group of [Ch. 16], and we have not computed the corresponding witnesses. (c) Whether the periodicity of the modular flow has an entropic meaning.

Provenance of this chapter

Unrefereed preprint. The three exact computations behind Theorem 23.3.1 — the Cuntz–Krieger K -groups, the bipartiteness, and the Smith normal form of the Hecke–Laplacian — are performed by the accompanying script and are reproducible in seconds. Everything else in Sections 23.1–23.4 is quoted, with the attribution set out in Section 23.6. We searched the physics literature for any prior use of an operator-algebraic invariant in p -adic AdS/CFT, and for any occurrence of type III_λ with $\lambda \neq 1$ in holography, and found none; a search cannot establish a negative, and we state this as the limit of our knowledge.

Chapter 24

The sign of λ : an algebraic certificate for how a graph tower is generated

Summary. For a tower of finite graphs $Y_0 \leftarrow Y_1 \leftarrow \cdots$ the ℓ -adic size of the sandpile group often obeys a three-parameter law $\text{ord}_\ell |\text{Jac}(Y_k)| = \mu \ell^k + \lambda k + \nu$. In the established theory of abelian ℓ -towers this law is a theorem and μ, λ are non-negative by construction, being the Iwasawa invariants of a torsion $\mathbb{Z}_\ell[[T]]$ -module. We observe that the tower of iterated line digraphs — the standard design iteration for de Bruijn and Kautz interconnection networks — obeys the same law with $\lambda = -1$, on every base graph we tested, with $\mu = n(q+1)/q$ exactly. Since $\lambda < 0$ cannot be the λ -invariant of any torsion $\mathbb{Z}_\ell[[T]]$ -module, a measured $\lambda < 0$ certifies that the family is not an abelian ℓ -tower. We also find that the certificate detects *normality*, not commutativity: dihedral (non-abelian but Galois) towers give $\lambda = 3 > 0$, and only the non-normal line-digraph tower goes negative. We are careful about what is new. The abelian law is due to Vallières and collaborators; the line-digraph tower is classical in network design (Fiol–Alegre–Yebra, 1983) and its spanning-tree arithmetic is Knuth’s, in the sandpile form of Levine; indeed we show that $\lambda = -1$ telescopes directly from that classical recursion, and verify the telescoping on five bases. What appears to be new is organising the iteration as a tower, extracting (μ, λ, ν) from it, obtaining a negative λ at all, and reading the -1 as an Euler characteristic — the one-dimensional Eisenstein kernel removed at each level. Since the spanning-tree count is a standard network reliability measure, the law is a scaling law for a reliability measure whose linear coefficient records the design method. All 27 towers are computed in exact integer arithmetic.

24.1 Introduction

Given a family of networks that grows by refinement — a sequence $Y_0 \leftarrow Y_1 \leftarrow \cdots$ in which each term covers or refines the last — one may ask how the family was generated. Existing answers in network science are statistical (likelihood, cross-validation, graphlet summaries) or geometric: the renormalisation exponents of Song, Havlin and Makse [127, 128] infer a growth mechanism from how box-covering statistics scale. This note offers an *algebraic* answer, in a narrow but exactly computable setting.

The invariant is the sandpile (critical) group $\text{Jac}(Y)$ [84], whose order is the number of spanning trees — itself a classical measure of network reliability [113, 115, 111]. Along a tower, $\text{ord}_\ell |\text{Jac}(Y_k)|$ frequently obeys

$$\text{ord}_\ell |\text{Jac}(Y_k)| = \mu \ell^k + \lambda k + \nu \quad (k \text{ large}). \quad (24.1.1)$$

What is established. For abelian ℓ -towers, (24.1.1) is a theorem. Vallières [129] proved it for a family of towers over bouquets; McGown and Vallières [126] proved it over an arbitrary connected multigraph;

Gonet [121] obtained the Sylow structure from the $\Lambda = \mathbb{Z}_\ell[[T]]$ -module decomposition, and Kleine and Müller [123] and others have since extended the picture to $\mathbb{Z}_\ell^d$ -towers, ramified towers, weighted graphs and p -adic Lie voltage groups. In every one of these constructions the tower is a *voltage* (derived-graph) tower, hence Galois by construction, and $\mu, \lambda \geq 0$ because they are the Iwasawa invariants of a finitely generated torsion Λ -module [18]. The non-negativity is stated flatly throughout that literature.

Separately, the tower of iterated line digraphs is classical. Fiol, Alegre and Yebra [118, 119] introduced line-digraph iteration as *the* design method for dense interconnection networks, which is how de Bruijn and Kautz networks are built; and the spanning-tree arithmetic of one iteration is Knuth's [124], refined by Levine [125] to the group statement $\text{Jac}(G) \cong \text{Jac}(LG)/(q\text{-torsion})$, with the de Bruijn and Kautz cases worked out in [112, 114].

What is here. These two bodies of work have not met. We put the line-digraph iteration into the form (24.1.1) and find

$$\mu = \frac{n(q+1)}{q}, \quad \lambda = -1, \quad \nu = \text{const}, \quad (24.1.2)$$

on every base graph tested (Section 24.3). Since $\lambda < 0$ is impossible for a torsion Λ -module, this is a certificate (Section 24.4). We show in Section 24.5 that (24.1.2) telescopes from the classical recursion, so the arithmetic is not new — only its reading is — and we give the Euler-characteristic interpretation of the -1 . Section 24.6 draws the network consequence.

What we do not claim. Not (24.1.1) in the abelian case [129, 126, 121]; not the line-digraph tower as an object [118, 119]; not its spanning-tree counts [124, 125, 112, 114]. A referee will observe that (24.1.2) follows from [124, 125], and will be right; we say so in Section 24.5 and verify it.

24.2 Setting

A *tower* is a sequence of finite connected graphs with maps $Y_{k+1} \rightarrow Y_k$. We consider four generating mechanisms.

- (a) *Abelian*: Y_k is the derived graph of a voltage assignment in $\mathbb{Z}/\ell^k$, so the tower is Galois with group $\mathbb{Z}/\ell^k$.
- (b) *Rank two*: voltages in $(\mathbb{Z}/\ell)^k$.
- (c) *Non-abelian but normal*: voltages in the dihedral group D_{ℓ^k} .
- (d) *Non-normal*: $Y_{k+1} = L(Y_k)$, the line digraph, starting from the non-backtracking digraph of a base graph. Level- k vertices are the non-backtracking k -paths; this is the Fiol–Alegre–Yebra iteration [118].

For (a)–(c) $|\text{Jac}(Y_k)|$ is the number of spanning trees; for (d), which is a digraph tower, it is the number of spanning arborescences. Both are computed exactly as integer determinants.

24.3 Measurements

Table 24.1 reports (μ, λ, ν) fitted from the last three levels of each tower — fitting from the tail is deliberate, since the defects of (24.1.1) are finitely supported [Ch. 7] and the first levels may be transient — together with the number of levels on which the fit holds.

The contrast in Table 24.1 is the point, and it is not the contrast we expected. Negative λ does not track non-commutativity: the dihedral towers are non-abelian and give $\lambda = 3 > 0$. It tracks *normality*. Every tower in the graph-Iwasawa literature is a voltage tower and hence Galois; the line-digraph tower is not, and it is the only one that goes negative.

mechanism	instances	λ observed	law holds?
(a) abelian $\mathbb{Z}/2^k$	15 towers, 6 bases	1, 3, 5, 9 (all > 0)	yes
(b) rank two $(\mathbb{Z}/2)^k$	3 towers	—	no (as required)
(c) dihedral D_{2^k}	4 towers	3	yes
(d) line digraph	5 bases	−1 (every base)	yes

Table 24.1: The four mechanisms. For (b) the one-variable law (24.1.1) fails on all three towers, as it must: a $\mathbb{Z}_\ell^2$ -extension needs the two-variable theory of [116, 117, 123]. All entries exact.

base Y_0	n	q	μ measured	$n(q+1)/q$	$\text{ord}_q \text{Jac}(Y_k) $, $k = 1, 2, \dots$
K_4	4	2	6	6	7, 18, 41, 88, 183, 374
$K_{3,3}$	6	2	9	9	5, 22, 57, 128, 271
Q_3 (cube)	8	2	12	12	14, 37, 84, 179
Petersen	10	2	15	15	13, 42, 101, 220
$C_4 \times K_2$	8	2	12	12	14, 37, 84, 179

Table 24.2: The non-normal (line-digraph) towers. In every case $\lambda = -1$ and μ equals $n(q+1)/q$ exactly, and the fitted law reproduces every computed level. All entries exact.

24.4 The certificate

Theorem 24.4.1. *Suppose a tower satisfies (24.1.1) with $\lambda < 0$. Then the tower is not an abelian ℓ -tower: there is no voltage assignment in $\mathbb{Z}_\ell$ whose derived graphs are the Y_k .*

Proof. For an abelian ℓ -tower the inverse limit $X_\infty = \varprojlim_k \text{Jac}(Y_k) \otimes \mathbb{Z}_\ell$ is a finitely generated torsion $\Lambda = \mathbb{Z}_\ell[[T]]$ -module [Ch. 6], [121, 126], and its Iwasawa invariants satisfy $\mu = \sum_i k_i \geq 0$ and $\lambda = \sum_j m_j \deg g_j \geq 0$ by the structure theorem [18, Thm. 13.12]; these are the coefficients in (24.1.1). $\square$

Remark 24.4.2 (What the certificate does not say). Theorem 24.4.1 rules out the characteristic ideal of a *single* torsion Λ -module. It does not rule out a virtual object — a formal difference of two Λ -modules, or the cohomology of a complex — and indeed the interpretation in Section 24.5 is exactly of that kind. It is therefore wrong to say that $\lambda = -1$ contradicts Iwasawa theory; the correct statement is that it is the invariant of a difference, not of a module. Two further limitations: the certificate is one-directional (Table 24.1(c) shows non-abelian towers with $\lambda > 0$), and it presupposes that (24.1.1) holds at all, which for rank-two towers it does not.

Remark 24.4.3 (A parity coincidence). Kataoka [122] shows that every pair (λ, μ) with λ odd is realised by an unramified $\mathbb{Z}_\ell$ -covering of a bouquet. Our $\lambda = -1$ is odd. We have no reason to think this is more than a coincidence, but it is the kind of thing a reader notices, so we record it.

24.5 Where the -1 comes from

Proposition 24.5.1. *Let Y_1 be a q -regular Eulerian digraph on n_1 vertices and $Y_{k+1} = L(Y_k)$. The classical line-digraph tree formula [124], in the form used in [Ch. 5] and equivalent to Levine’s $\text{Jac}(G) \cong \text{Jac}(LG)/(q\text{-torsion})$ [125], gives*

$$\text{ord}_q |\text{Jac}(Y_{k+1})| - \text{ord}_q |\text{Jac}(Y_k)| = n_k(q-1) - 1.$$

With $n_k = n(q+1)q^{k-1}$ this telescopes to (24.1.2).

Proof. Summing the increment from level 1 to level k gives $n(q+1)(q^{k-1} - 1) - (k-1) + \text{ord}_q |\text{Jac}(Y_1)|$, which is $\frac{n(q+1)}{q}q^k - k + \text{const.}$ $\square$

So $\lambda = -1$ is a consequence of a 1967 formula. We verify the telescoping numerically on all five bases (check (K) of Section 24.7): the measured sequences of Table 24.2 agree term by term with the telescoped prediction.

The Euler characteristic. The -1 in the increment is not a normalisation artefact. The one-step increment is $n_k(q-1) - \chi_k$ with $\chi_k = \dim \ker \Delta_k^{\text{dir}} = 1$: at every level a one-dimensional kernel — the constant, or Eisenstein, direction — is removed [Ch. 5]. Summing a constant Euler characteristic over k levels is what produces a term linear in k with negative coefficient, and it is why no single Λ -module can have this as its characteristic ideal: the object is a difference, one term of which is the Eisenstein line. Check (E) verifies the increment identity level by level.

24.6 The network reading

The two mechanisms of Table 24.1 are not artificial. Torus, circulant and chordal-ring interconnects are voltage covers of a small base, hence abelian towers; de Bruijn and Kautz interconnects are built by line-digraph iteration [118, 119, 120], hence non-normal towers. The quantity being measured, $|\text{Jac}(Y_k)|$, is the spanning-tree count, a standard reliability measure [113, 115, 111]. Equation (24.1.1) is therefore a three-parameter scaling law for a reliability measure along a design family, in which

- μ is the bulk growth rate — for the line-digraph family, $n(q+1)/q$, i.e. proportional to the level-one size;
- λ records the design method: positive for covering constructions, -1 for line-digraph iteration;
- ν is an offset with no evident structural meaning.

We are not aware of prior use of an algebraic growth exponent for this purpose; the closest precedent is the use of renormalisation exponents to infer a growth mechanism [128]. Whether the law survives the perturbations a real network family suffers — irregularity, missing edges, coarse-grainings that are not covers — we have not tested, and Section 24.8 lists it first among the open questions.

24.7 Verification

The script `tower_lambda.py` accompanying this note builds and measures 27 towers and runs the checks of Table 24.3 in about 7 minutes, all passing; its output is archived as `tower_lambda_output.txt`. All spanning-tree and arborescence counts are exact integer determinants.

key	check	mode
(A)	(24.1.1) holds with $\lambda > 0$ on 15 abelian towers over 6 bases	exact
(L)	$\lambda = -1$ and $\mu = n(q+1)/q$ on 5 line-digraph towers (Table 24.2)	exact
(E)	the increment is $n_k(q-1) - \chi$ with $\chi = 1$, level by level	exact
(K)	Prop. 24.5.1: the measured sequence equals the telescoped recursion	exact
(R)	the one-variable law fails on all 3 rank-two $(\mathbb{Z}/2)^k$ towers	exact
(D)	dihedral towers: $\lambda = 3 > 0$, so the certificate detects normality	exact
(C)	the certificate assembled over the whole battery	exact

Table 24.3: The verification battery (`tower_lambda.py`), 7/7 passing.

24.8 Scope

Quoted. The abelian law [129, 126, 121, 123]; non-negativity of Iwasawa invariants [18]; the line-digraph iteration [118, 119]; its tree arithmetic [124, 125, 112, 114]; spanning trees as a reliability measure [113, 115, 111].

New here, as far as we can determine. Putting the line-digraph iteration in the form (24.1.1); the value $\lambda = -1$, there being no negative λ anywhere in the graph-Iwasawa literature; the Euler-characteristic reading of it; the certificate (Theorem 24.4.1) with its caveats (Remark 24.4.2); and the observation that the certificate separates normal from non-normal rather than abelian from non-abelian.

Open. (a) Robustness. Every tower here is exact; real network families are not, and we do not know how (24.1.1) degrades under edge noise or under refinements that are not covers. This is the question that decides whether any of this is usable. (b) Whether $\lambda < 0$ occurs for any tower other than the line-digraph family — we found none, but tested only four mechanisms. (c) Whether the structural refinement of λ as a count of persistently growing cyclic factors is already contained in [121] or in Kataoka’s work on Fitting ideals; we were not able to settle this. (d) The rank-two non-normal case, where [Chs. 16–18] shows no abelian sector exists at all.

Provenance of this chapter

Unrefereed preprint. Theorem 24.4.1 is a restatement of the structure theorem for Λ -modules and is proved above modulo that theorem. Proposition 24.5.1 is a summation of a classical formula. Everything numerical is produced by the accompanying script in exact integer arithmetic. We searched for a negative λ , for a non-normal graph tower with a growth law, and for an Euler-characteristic reading of such a law, and found none; a search cannot establish a negative, and our coverage of engineering proceedings in particular is incomplete.

Chapter 25

Power-sum lower bounds on hidden eigenvalues in positive realizations

Summary. A positive realization of a transfer function of order n is a nonnegative matrix of some size $N \geq n$ carrying the n poles among its eigenvalues; the $s = N - n$ remaining eigenvalues are the *hidden* eigenvalues, whose characterization Benvenuti and Farina list as an open problem. We give a lower bound on s from the power sums of the poles: if p_k denotes the k -th power sum and ρ the spectral radius, then $s \geq \max_{k \geq 1} \lceil [-p_k]_+ / \rho^k \rceil$. At $k = 1$ this reproduces the quantity ζ of Benvenuti’s lower bound; for $k \geq 2$ it is, as far as we can determine, new, and it is strictly stronger. On the family $\mathcal{M}_m(\theta) = \{1\} \cup m \cdot \{\pm i\theta\}$ the published lower bound is vacuous ($N \geq n$) while ours gives $N \geq 2m(1 + \theta^2)$; when $\theta^2 > 1 - 1/2m$ this meets Benvenuti’s own upper bound $4m$ and therefore determines the minimal size exactly, where previously only $n \leq N \leq 4m$ was known. The bound is attained: on $\{1\} \cup m \cdot \{\pm i\}$ it gives $N \geq 4m$, and a direct sum of m copies of the 4-cycle permutation matrix realizes $N = 4m$. We also record two limitations. The method has a hard ceiling — it can never prove more than $N \leq 2n$ — so it is complementary to, not competitive with, the digraph bounds that give unbounded ratios; and the ratio 2 itself is not new territory, being exactly what the Benvenuti–Farina family of 1999 achieves. Finally we explain why no such bound can exist in the Boyle–Handelman/Kim–Ormes–Roush setting: there one appends only zeros, and zeros change no power sum. All claims are machine-verified in exact rational arithmetic.

25.1 Introduction

Let $H(z)$ be a rational transfer function of order n with pole multiset $\Lambda = \{\lambda_1, \dots, \lambda_n\}$ and spectral radius $\rho = \max_i |\lambda_i|$. A *positive realization* of H is a triple (A, b, c) with $A \in \mathbb{R}_{\geq 0}^{N \times N}$, $b, c \geq 0$ and $H(z) = c(zI - A)^{-1}b$. Existence is settled [135, 130, 140], and a clean criterion is available in terms of an invariant polyhedral cone [134]. What is not settled is *size*. The poles of H are a subset of the eigenvalues of any positive realization, and the remaining $s = N - n$ eigenvalues — the *hidden eigenvalues* — are the reason N may exceed n . Benvenuti and Farina put it as follows [134, §IX]: “An important issue related to minimality of positive systems is the study of ‘hidden’ eigenvalues, i.e. of the eigenvalues which are not poles and that possibly one has to add in order to obtain a minimal positive realization. A full characterization of this property may lead to a deeper and valuable insight into the problem.” The minimality problem remains open; a recent statement of its status is [141].

This note contributes one inequality. It is elementary, and its only merit is that it appears not to have been written down beyond its first instance.

The bound. Write $p_k = p_k(\Lambda) = \sum_i \lambda_i^k$ for the k -th power sum of the poles and $[x]_+ = \max(x, 0)$. Theorem 25.2.1 states

$$s \geq \left\lceil \max_{k \geq 1} \frac{[-p_k]_+}{\rho^k} \right\rceil. \quad (25.1.1)$$

What is already known. The case $k = 1$ of (25.1.1) is published. Benvenuti [131, Thm. 4.7] proves $N \geq N_D + \sum_{|s| < 1} m(s) + \zeta$ with $\zeta = \lceil -m_{\mathcal{R}_1} - \sum_{|s| < 1} s m(s) \rceil$ when that quantity is positive, and the proof is exactly the $k = 1$ argument: “since the trace of a nonnegative matrix is nonnegative ... since each one of these additional eigenvalues has to be not greater than 1, then their number is not lesser than ζ ” [131, pp. 379–380]. The companion paper [132] transfers the same quantity to the realization dimension. We have found no power sum beyond the first anywhere in this literature: not in [131, 132], not in the digraph bound of Nagy and Matolcsi [139], not in [134, 141]. Section 25.3 shows that $k = 2$ already does strictly better.

What we do not claim. Two disclaimers, stated here rather than buried.

- (a) The ratio $N/n = 2$ is *not* new territory. The family $H(z, m) = 1/((z-1)(z^{2^m}+1))$ of Benvenuti and Farina [133] has $n = 2^m + 1$ and minimal positive realization of dimension $N = 2^{m+1} = 2n - 2$, so ratio 2 is exactly what they already achieve; and Nagy and Matolcsi [139] obtain *unbounded* ratios, with $n = 4$ fixed and $N \geq m + 1$ for arbitrary m .
- (b) Worse, ratio 2 is a hard ceiling for the present method. Since $p_k \geq -\sum_i |\lambda_i|^k \geq -n\rho^k$, the right-hand side of (25.1.1) never exceeds n , so (25.1.1) can never prove more than $N \leq 2n$ (Proposition 25.2.3). The bound is therefore complementary to the combinatorial bounds of [139], not a competitor to them.

What we do claim is (i) the inequality for $k \geq 2$, (ii) that it strictly improves the published lower bound, and (iii) that on an explicit family it closes the published gap completely.

25.2 The bound

Theorem 25.2.1. Let $A \in \mathbb{R}_{\geq 0}^{N \times N}$ and let Λ be a sub-multiset of $\text{spec } A$ of size n containing an eigenvalue of modulus $\rho(A)$; write $\rho = \rho(A)$ and $s = N - n$. Then for every $k \geq 1$

$$s \geq \frac{-p_k(\Lambda)}{\rho^k}, \quad \text{hence} \quad s \geq \left\lceil \max_{k \geq 1} \frac{[-p_k(\Lambda)]_+}{\rho^k} \right\rceil.$$

Proof. Let $\mathcal{M}$ be the complementary multiset, $|\mathcal{M}| = s$, so that $\text{spec } A = \Lambda \uplus \mathcal{M}$. Every eigenvalue of A has modulus at most $\rho(A) = \rho$, so $|\sum_{\mu \in \mathcal{M}} \mu^k| \leq \sum_{\mu \in \mathcal{M}} |\mu|^k \leq s\rho^k$. Since A is nonnegative, $\text{Tr}(A^k) \geq 0$: it is a sum of products of nonnegative entries. Therefore

$$0 \leq \text{Tr}(A^k) = p_k(\Lambda) + \sum_{\mu \in \mathcal{M}} \mu^k \leq p_k(\Lambda) + s\rho^k,$$

which rearranges to the claim; s is an integer, so the ceiling may be taken. $\square$

The hypothesis that Λ contains a dominant eigenvalue is what makes ρ computable from the poles alone; in the realization setting it holds because $\rho(H)$ is a pole [134, Lem. 7].

Proposition 25.2.2 ($k = 1$ is Benvenuti’s ζ). At $k = 1$, (25.1.1) reads $s \geq \lceil -p_1(\Lambda)/\rho \rceil$, which after normalising $\rho = 1$ and splitting Λ into its dominant and non-dominant parts is the quantity ζ of [131, Thm. 4.7].

Proof. Normalise $\rho = 1$. The dominant part of Λ contributes $m_{\mathcal{R}_1}$ to p_1 , since the h -th roots of unity sum to zero for $h > 1$ [131, proof of Thm. 4.7], and the non-dominant part contributes $\sum_{|s| < 1} s m(s)$. Hence $-p_1 = -m_{\mathcal{R}_1} - \sum_{|s| < 1} s m(s)$, and the ceiling of its positive part is ζ . $\square$

Proposition 25.2.3 (Ceiling of the method). *The right-hand side of (25.1.1) is at most n ; consequently Theorem 25.2.1 can never certify $N > 2n$.*

Proof. $-p_k \leq \sum_i |\lambda_i|^k \leq n\rho^k$. □

25.3 Strict improvement, and an exact determination

Fix $0 < \theta < 1$ and $m \geq 1$, and let

$$\mathcal{M}_m(\theta) = \{1\} \cup m \text{ copies of } \{i\theta, -i\theta\}, \quad n = 2m + 1, \quad \rho = 1.$$

Only 1 is dominant, so in the notation of [131] we have $r = 1$, $N_D = 1$ and $m_{\mathcal{R}_1} = 1$.

Theorem 25.3.1. *For $\mathcal{M}_m(\theta)$:*

- (i) *the lower bound of [131, Thm. 4.7] gives $N \geq 2m + 1 = n$, i.e. it is vacuous, because $\zeta = 0$;*
- (ii) *Theorem 25.2.1 at $k = 2$ gives $s \geq \lceil 2m\theta^2 - 1 \rceil$, hence $N \geq 2m + 1 + \lceil 2m\theta^2 - 1 \rceil$, which exceeds n whenever $\theta^2 > 1/2m$;*
- (iii) *the upper bound of [131, Thm. 4.4] gives $N \leq 4m$;*
- (iv) *if $\theta^2 > 1 - \frac{1}{2m}$ then the bounds in (ii) and (iii) coincide and*

$$N_{\min} = 4m.$$

Proof. (i) $p_1(\mathcal{M}_m(\theta)) = 1$ and the non-dominant numbers cancel in conjugate pairs, so $m_{\mathcal{R}_1} + \sum_{|s| < 1} s m(s) = 1 \geq 0$ and $\zeta = 0$; the bound reads $N \geq N_D + 2m + 0 = 2m + 1$. (ii) $p_2 = 1 + m((i\theta)^2 + (-i\theta)^2) = 1 - 2m\theta^2$, so $[-p_2]_+/\rho^2 = 2m\theta^2 - 1$ when positive. (iii) is [131, Thm. 4.4] with $\kappa(i\theta) = 4$, valid since $i\theta \in \Pi_4 \setminus \Pi_3$ for $\theta > 1/\sqrt{3}$. (iv) The bound in (ii) equals $4m$ as soon as $\lceil 2m\theta^2 - 1 \rceil \geq 2m - 1$, i.e. $2m\theta^2 - 1 > 2m - 2$, i.e. $\theta^2 > 1 - 1/2m$. □

Table 25.1 shows the three quantities. The criterion $\theta^2 > 1 - 1/2m$ of Theorem 25.3.1(iv) predicts exactly which rows close: it holds for $\theta = 9/10$ only at $m = 1$ ($0.81 > 0.5$) and fails at $m = 4$ ($0.81 < 0.875$), where the bound indeed falls one short of the upper bound; it holds throughout for $\theta = 99/100$ from $m = 2$ on and for $\theta = 999/1000$, and there the gap closes and N is determined.

θ	m	n	[131, Thm. 4.7] lower	Theorem 25.2.1 lower	[131, Thm. 4.4] upper
9/10	1	3	3	4	4
9/10	4	9	9	15	16
9/10	16	33	33	58	64
99/100	2	5	5	8	8
99/100	4	9	9	16	16
99/100	8	17	17	32	32
99/100	16	33	33	64	64
999/1000	4	9	9	16	16
999/1000	16	33	33	64	64

Table 25.1: Lower and upper bounds on the minimal size N for $\mathcal{M}_m(\theta)$. Bold: the new lower bound meets the published upper bound, so N is determined exactly. All entries exact.

25.4 Sharpness

Let $\mathcal{M}_m = \{1\} \cup m \cdot \{i, -i\}$, all of modulus one, $n = 2m + 1$.

Theorem 25.4.1. $p_2(\mathcal{M}_m) = 1 - 2m$, so Theorem 25.2.1 gives $s \geq 2m - 1$ and $N \geq 4m$. The bound is attained: if C_4 is the 4-cycle permutation matrix, with characteristic polynomial $x^4 - 1$ and spectrum $\{1, i, -1, -i\}$, then $C_4^{\oplus m}$ is nonnegative of size $4m$ and contains $\mathcal{M}_m$ in its spectrum. Hence $N_{\min} = 4m = 2n - 2$.

Proof. The power sum is immediate. For the construction, $\text{spec } C_4^{\oplus m}$ is m copies of $\{1, i, -1, -i\}$, which contains one 1 and m copies of each of $\pm i$. $\square$

Remark 25.4.2. Two caveats. First, for $m \geq 2$ the multiset $\mathcal{M}_m$ violates the standing hypothesis of [131, Thms. 3.2, 4.1] that the spectral radius occur with dominant multiplicity: here $m(1) = 1$ while the dominant multiplicity is m . The formula of [131, Thm. 4.1] nevertheless evaluates to $N_D = 4m$ on $\mathcal{M}_m$, so the conclusion $N_{\min} = 4m$ is also reachable by that route, by an argument through roots of unity rather than through traces; we note this so that Theorem 25.4.1 is not mistaken for a new determination. It is a sharpness check on (25.1.1), not a new value of N . Second, $4m = 2n - 2$ is the Benvenuti–Farina ratio of [133], as recalled in Section 25.1.

Remark 25.4.3 (Tutorial Example 8). For the poles $\{1, \pm 0.9i\}$ of [134, Ex. 8], $p_2 = -31/50$ and (25.1.1) gives $s \geq 1$, i.e. $N \geq 4$. This is attained by the nonnegative circulant with first row $(0, \frac{19}{20}, 0, \frac{1}{20})$, whose characteristic polynomial is $(x - 1)(x + 1)(x^2 + \frac{81}{100})$: the hidden eigenvalue is -1 . The example in [134] is 4-dimensional, so the bound is tight there as well.

25.5 Why there is no analogue for the Boyle–Handelman problem

It is worth saying why the same traces behave completely differently one category over. In the inverse eigenvalue problem solved by Boyle and Handelman over $\mathbb{R}$ [56] and by Kim, Ormes and Roush over $\mathbb{Z}$ [136], one asks whether Λ is the *nonzero* spectrum of a nonnegative matrix; the matrix may be enlarged, but only by appending *zeros*. Zeros contribute nothing to any power sum, so the trace conditions $\text{tr}_k \geq 0$ must already hold for Λ itself: they are a genuine obstruction, not a counting device, and no lower bound of the shape (25.1.1) on the size of the dilation can exist. Equivalently: in that category the analogue of Theorem 25.2.1 is the statement $p_k(\Lambda) \geq 0$, with no s in it.

This is the exact point at which results from symbolic dynamics fail to transfer to the positive realization problem. A negative p_2 closes the fixed-nonzero-spectrum problem outright — shift equivalence preserves the nonzero spectrum with multiplicities [137, §7.4], so no dilation of any size helps, an argument used in this form in [Ch. 3] — while in the present setting the same negative p_2 merely counts hidden eigenvalues, and indeed [134, Ex. 8] exhibits a transfer function with $p_2 = -0.62$ and a 4-dimensional positive realization. The two categories differ exactly by the freedom to add nonzero eigenvalues.

25.6 Verification

The script `hidden_eigs.py` accompanying this note runs the checks of Table 25.2 in 23s, all passing; its output is archived as `hidden_eigs_output.txt`. Arithmetic is exact rational throughout; the only floating-point quantity anywhere is the spectral radius in check (T), and that row is labelled accordingly.

In check (T) the spectral radius is replaced by a rational *lower* bound, which makes the tested inequality strictly stronger than the theorem; no violation occurred. The bound was attained in none of those 120 random instances, which is the expected behaviour: like the $k = 1$ bound it is weak on generic spectra and bites only when the poles are strongly oscillatory.

key	check	mode
(T)	Theorem 25.2.1 holds on 120 random nonnegative integer matrices, Λ a rational factor	exact + numeric ρ
(K)	at $k = 1$ the bound equals ζ of [131, Thm. 4.7]	exact
(I)	$k = 2$ strictly exceeds [131, Thm. 4.7] in 15/15 instances (Table 25.1)	exact
(S)	Theorem 25.4.1: bound = $4m$ attained by $C_4^{\oplus m}$, $m \leq 32$	exact
(E)	Remark 25.4.3: bound = 1, attained by an explicit rational circulant	exact
(C)	Proposition 25.2.3: the bound never exceeds n , on 300 random pole sets	exact
(Z)	appending zeros leaves every power sum unchanged (Section 25.5)	exact

Table 25.2: The verification battery (`hidden_eigs.py`), 7/7 passing.

25.7 Scope

Established here. Theorem 25.2.1 for all k ; its coincidence with [131, Thm. 4.7] at $k = 1$; strict improvement and an exact determination of $N_{\min}$ on $\mathcal{M}_m(\theta)$ for $\theta^2 > 1 - 1/2m$; sharpness on $\mathcal{M}_m$ and on [134, Ex. 8]; the ceiling $N \leq 2n$; the structural reason the bound has no analogue in [56, 136].

Open. (a) Whether power sums can be combined with the cone criterion [134] or with the digraph bound [139] to pass the ratio-2 ceiling. (b) Whether the Johnson–Loewy–London inequalities [138], which mix power sums of different orders, give a bound not implied by (25.1.1); we have not investigated this and it is the obvious next step. (c) Whether the bound is ever tight outside the families of Sections 25.3–25.4; it was not tight in any of our 120 random instances.

Not claimed. Ratio 2, which is [133]; unbounded ratios, which are [139]; and the $k = 1$ case, which is [131].

Provenance of this chapter

Unrefereed preprint. Theorem 25.2.1 is elementary and its proof is complete above. The bounds of [131] quoted for comparison in Table 25.1 are evaluated by us from the published statements of Theorems 4.4 and 4.7 of that paper and are reproduced in the script; we have not reproved them. The symbolic-dynamics facts in Section 25.5 are quoted from [137, 56, 136]. We searched the literature for a power-sum bound with $k \geq 2$ and found none; a search cannot establish a negative, and we state this as the limit of our knowledge.

Chapter 26

Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs

Summary. The number of arc-rooted circular sequences sharing a given k -mer spectrum is, by the BEST theorem, the product of a *local* factor $\prod_v (d^+(v) - 1)!$, which depends only on the out-degree multiset of the de Bruijn graph, and a *global* factor, which is the order of the graph's critical (sandpile) group. We make four points about the global factor. First, it is not determined by the degree data: we exhibit two sequences of length 60 whose de Bruijn graphs at $k = 4$ have identical vertex counts, identical out-degree multisets and hence identical local factors, but admit 4 and 256 reconstructions — a factor of 64. This is a strict sharpening of the observation of Kingsford, Schatz and Pop that repeat counting is insufficient. Second, the group structure reads off repeat parameters: a word of length L occurring r times in an otherwise k -unique sequence contributes $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$. Third — and this is new in this version — the group detects how repeats *interleave*: for two repeats of length L occurring twice each, the disjoint and nested arrangements $A A B B$ and $A B B A$ have isomorphic compacted graphs and $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2}$, the direct sum of the two single-repeat contributions, while the interleaved arrangement $A B A B$ has $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}$, one extra $\mathbb{Z}/2$ that survives to $k = L$ after both individual contributions have died. The degree data are identical in all three cases. Fourth, the invariant is cheap where it matters: contracting unitigs leaves the critical group unchanged, so it is computed on the compacted graph. We demonstrate this on a real genome: for bacteriophage ϕ X174 (5386 bp) the $k = 10$ graph has 5339 vertices but 47 branch points, and its critical group, of order 4.4×10^{12} , is computed in ten milliseconds; the spectrum collapses through 132, 2, 1 at $k = 11, 12, 13$, one past the longest exact repeat. All claims are verified by machine: 30 checks, exact integer arithmetic except where labelled.

26.1 Introduction

Given a collection of k -mers with multiplicities — an idealised sequencing experiment — how many genomes are consistent with it? The k -mers and their overlaps form a de Bruijn multigraph G_k in which the genome is an Eulerian circuit, so the answer is the number of Eulerian circuits, which the BEST theorem [142, 152] evaluates as

$$\text{ec}(G_k) = |\mathcal{K}(G_k)| \cdot \prod_v (d^+(v) - 1)!, \quad (26.1.1)$$

$\mathcal{K}(G_k)$ being the group of spanning arborescences — the *critical* or *sandpile* group. Kingsford, Schatz and Pop [146] computed this for real prokaryotic genomes as a measure of assembly complexity; Loukides, Pissis

and co-authors [148] gave fast algorithms for assessing it.

Equation (26.1.1) splits the ambiguity into two factors of different character. The product of factorials is a function of the out-degree multiset alone — the *branch statistics*, read off the graph by inspection. The other factor is not. This paper is about what the other factor knows.

What is already known. Two literatures meet here, and we claim nothing in either. On the combinatorial side the critical groups of *complete* de Bruijn graphs are completely understood: Knuth [147] gave the line-digraph recursion, Levine [7] the group structure $\mathcal{K}(\text{DB}(2, n)) \cong \bigoplus_{j=1}^{n-1} (\mathbb{Z}/2^j)^{2^{n-1-j}}$, and Chan, Hollmann and Pasechnik [145] the generalised de Bruijn and Kautz cases. On the assembly side, counting reconstructions via BEST is established [146, 148], as are uniqueness and safety criteria [150] and feasibility thresholds [143, 149]. What appears not to have happened is that the two ever met: we have found no work applying the critical group of a graph to assembly, and none that looks at the *structure* of $\mathcal{K}(G_k)$ for a genome de Bruijn graph, which is a subgraph of a complete one and not covered by [7, 145].

What is new here.

- (i) **Theorem 26.6.1:** the global factor is not a function of the degree data. Two explicit sequences have identical branch statistics and reconstruction counts differing by $64\times$.
- (ii) **Theorem 26.4.2:** for a single repeat of length L and multiplicity r , $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$.
- (iii) **Theorem 26.5.1** (new in this version): for two repeats the group is additive in the disjoint and nested arrangements and picks up exactly one extra $\mathbb{Z}/2$ in the interleaved arrangement. Since interleaved repeats are the notoriously hard case for assembly, this is the first place the algebra and the practice visibly agree. Proposition 26.5.3 then shows that for three repeats the correction is the sandpile group of a small digraph and not a chord-diagram rank — it can be $\mathbb{Z}/3$.
- (iv) **Lemma 26.4.1** makes the invariant computable on the compacted graph, and Section 26.8 computes it on a real genome.

Relation to [146]. Kingsford, Schatz and Pop already argue that repeat counting is insufficient, with the example that three repeats in two copies each (8 reconstructions) is easier than one repeat in six copies (720). Their two situations have *different* out-degree multisets, $(2, 2, 2)$ against (6) , so the local factor already distinguishes them. Theorem 26.6.1 fixes the out-degree multiset and still finds a 64-fold spread, and Theorem 26.5.1 says what the residual information is: not how many repeats or how many copies, but how they are wired to one another.

26.2 Setting

Let S be a circular sequence of length N over an alphabet of size σ . For $k \geq 1$ the *de Bruijn multigraph* $G_k = G_k(S)$ has as vertices the distinct k -mers of S and one arc for each of the N occurrences of a $(k+1)$ -mer, from its length- k prefix to its length- k suffix. S traverses every arc once, so G_k is connected and balanced, and S is an Eulerian circuit; every Eulerian circuit is a circular sequence with the same k -mer spectrum, counted with multiplicities. We write $n_k = |V(G_k)|$, $d^+(v)$ for out-degree (the number of occurrences of v), and $\text{Loc}(G) := \prod_v (d^+(v) - 1)!$.

Remark 26.2.1 (Counting convention). Throughout, reconstruction counts are *arc-rooted*: circuits are counted as beginning with a fixed arc, which is the convention under which (26.1.1) holds [142, 152]. Circular sequences up to cyclic shift are counted by dividing by N when S is aperiodic, and by Burnside's lemma otherwise; none of the algebra below depends on which convention is chosen, since the correction is a function of N alone.

The *critical group* is $\mathcal{K}(G) := \text{coker}(\Delta^\top)$, Δ the out-Laplacian $D^+ - A$ with the row and column of a fixed root deleted; for a connected balanced digraph the isomorphism class does not depend on the root, and $|\mathcal{K}(G)|$ is the number of spanning arborescences. We compute it as a Smith normal form.

Remark 26.2.2 (The tower). $G_{k+1}(S)$ is a spanning subgraph of the line digraph $L(G_k(S))$, with equality for the complete de Bruijn graph; the prefix and suffix maps are the tail and head maps of G_k and are the degeneracy maps of the non-normal tower of [Ch. 9]. The line-digraph recursion of [Ch. 5] specialises to Knuth's formula and reproduces $|\mathcal{K}(\text{DB}(\sigma, k))| = \sigma^{\sigma^k - k - 1}$ (check (T)); check (L) reproduces Levine's decomposition. These are consistency checks on our conventions, not results.

26.3 The local and global factors

Proposition 26.3.1 (Decomposition). *The number of arc-rooted circular sequences with the same k -mer spectrum as S equals $|\mathcal{K}(G_k)| \cdot \text{Loc}(G_k)$. The factor $\text{Loc}(G_k)$ depends only on the multiset $\{d^+(v)\}$; the factor $|\mathcal{K}(G_k)|$ does not.*

The first sentence is (26.1.1). The second is Theorem 26.6.1.

26.4 Contraction, and a single repeat

Lemma 26.4.1 (Unitig contraction). *Let v be a vertex of a connected balanced multidigraph G with $d^-(v) = d^+(v) = 1$, whose incoming and outgoing arcs are distinct, say $u \rightarrow v$ and $v \rightarrow w$. Let G' be obtained by deleting v and replacing those two arcs by a single arc $u \rightarrow w$. Then $\mathcal{K}(G') \cong \mathcal{K}(G)$. Self-loops may be deleted freely: a loop at v adds 1 to both D_{vv}^+ and A_{vv} and so does not appear in Δ .*

Proof. Root away from v and u . The row of v in Δ is $e_v - e_w$ and the column of v has a single off-diagonal entry, in the row of u ; adding the row of v to the row of u clears it and turns the row of u into its row in $\Delta(G')$, after which the row and column of v form a unit block. Both operations are unimodular. The statement about loops is immediate from $\Delta = D^+ - A$ and is check (P). $\square$

Iterating the lemma yields the *compact* (unitig) de Bruijn graph, in which every maximal non-branching path is one arc and only branch vertices survive.

Theorem 26.4.2 (A repeat of length L and multiplicity r). *Let W have length L and let S be circular with every k -mer occurring once except the $L - k + 1$ k -mers of W , each occurring exactly $r \geq 2$ times. Then for $1 \leq k \leq L$*

$$\mathcal{K}(G_k(S)) \cong (\mathbb{Z}/r)^{L-k}.$$

Proof. Put $s = L - k + 1$ and let $v_1, \dots, v_s$ be the k -mers of W in order. Every other vertex has in- and out-degree 1; contract them. What remains is the spine $v_i \rightarrow v_{i+1}$ with multiplicity r and r arcs $v_s \rightarrow v_1$. Rooted at v_1 , on $v_2, \dots, v_s$ the out-Laplacian is $r(I - N)$ with N the nilpotent shift; $I - N$ is unimodular, so the cokernel is $(\mathbb{Z}/r)^{s-1}$. $\square$

26.5 Two repeats: the group sees interleaving

Fix two distinct words A, B of length L and a circular sequence in which each occurs exactly twice, non-overlapping, with every other k -mer unique; put $s = L - k + 1$. The four occurrences can be arranged in three ways around the circle:

$$\text{disjoint } A A B B, \quad \text{nested } A B B A, \quad \text{interleaved } A B A B.$$

The k -mer spectra of the three sequences have identical multiplicity profiles, so n_k , the out-degree multiset, and hence $\text{Loc}(G_k)$ agree. This is Theorem 26.6.1's setting one level up, and the question is whether $\mathcal{K}$ can tell them apart.

Theorem 26.5.1 (Two repeats). *With $a_1, \dots, a_s$ and $b_1, \dots, b_s$ the k -mers of A and B :*

(i) the disjoint and nested arrangements have isomorphic compacted graphs, and

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2} = (\mathbb{Z}/2)^{s-1} \oplus (\mathbb{Z}/2)^{s-1},$$

the direct sum of the two single-repeat groups of Theorem 26.4.2;

(ii) the interleaved arrangement has

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}.$$

In particular the arrangement is detected by $\mathcal{K}$ and not by the degree data, and at $k = L$ ($s = 1$) the interleaved arrangement retains a $\mathbb{Z}/2$ after both single-repeat contributions have become trivial.

Proof. Contract by Lemma 26.4.1. The surviving vertices are the $2s$ shared k -mers, each of in- and out-degree 2. The spine arcs $a_i \rightarrow a_{i+1}$ and $b_i \rightarrow b_{i+1}$ have multiplicity 2. The four spacer paths contract to four connector arcs from $\{a_s, b_s\}$ to $\{a_1, b_1\}$, one per transition between consecutive occurrences around the circle. Reading the cyclic order: disjoint gives $a_s \rightarrow a_1, a_s \rightarrow b_1, b_s \rightarrow b_1, b_s \rightarrow a_1$; nested gives $a_s \rightarrow b_1, b_s \rightarrow b_1, b_s \rightarrow a_1, a_s \rightarrow a_1$ — the same multiset, so the two compacted graphs coincide; interleaved gives $a_s \rightarrow b_1$ twice and $b_s \rightarrow a_1$ twice.

(ii) Root at a_1 . On $a_2, \dots, a_s, b_1, \dots, b_s$ in this order the reduced out-Laplacian is $2(I - N)$, N the shift along that path (both arcs of b_s go to the deleted root). $I - N$ is unimodular, so $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-1}$.

(i) Root at a_1 ; write x_v for the generator of the cokernel presentation at v . The rows of a_s and b_s read $2x_{a_s} - x_{b_1}$ and $2x_{b_s} - x_{b_1}$, so $x_{b_1} = 2x_{a_s}$ and x_{b_1} can be eliminated. On the remaining $2s - 2$ generators $a_2, \dots, a_s, b_2, \dots, b_s$ every relation is now even — the spine relations $2x_{a_i} - 2x_{a_{i+1}}$ and $2x_{b_i} - 2x_{b_{i+1}}$, the relation $2x_{b_s} - 2x_{a_s}$, and the former row of b_1 , which becomes $4x_{a_s} - 2x_{b_2}$ — so the relation matrix is $2M'$ with

$$M' = \begin{pmatrix} \text{rows } e_{a_i} - e_{a_{i+1}}, & 2 \leq i \leq s-1 \\ & e_{b_s} - e_{a_s} \\ & 2e_{a_s} - e_{b_2} \\ \text{rows } e_{b_i} - e_{b_{i+1}}, & 2 \leq i \leq s-1 \end{pmatrix}.$$

Expanding $\det M'$ along the columns $a_2, \dots, a_{s-1}$ in turn (each has a single entry, $+1$) leaves the $s \times s$ block on columns $(a_s, b_2, \dots, b_s)$ with rows $2e_{a_s} - e_{b_2}, e_{b_i} - e_{b_{i+1}}$ ($2 \leq i \leq s-1$), $e_{b_s} - e_{a_s}$. Expanding along the first row with the cofactor signs written out, $\det M' = 2(-1)^{1+1}M_{11} + (-1)(-1)^{1+2}M_{12} = 2M_{11} + M_{12}$. The minor M_{11} (the remaining rows on columns $b_2, \dots, b_s$) is upper bidiagonal with unit diagonal, so $M_{11} = 1$; the minor M_{12} (the same rows on columns $a_s, b_3, \dots, b_s$) has a single entry -1 in its first column, in its last row, and expanding along that column leaves a lower bidiagonal block with diagonal -1 , the signs combining to $M_{12} = -1$. Hence $\det M' = 2 - 1 = 1$ in this column order, $|\det M'| = 1$ in any order, M' is unimodular and $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-2}$. $\square$

Remark 26.5.2 (Reading). For a single repeat the group is trivial at $k = L$: the repeated L -mer is a branch vertex but its two exits lead to different places and there is only one way to thread it. For two interleaved repeats at $k = L$ there is a genuine choice — leave A for the first B or the second — and $\mathbb{Z}/2$ records it. Nested and disjoint repeats offer no such choice: whichever way one threads A , the two B 's are traversed consecutively. This is the algebraic form of the folklore that interleaved repeats are what makes assembly hard. Additivity of $\mathcal{K}$ over repeats, asked for in version 1, therefore holds exactly when the repeats do not interleave; the general correction term is indexed by the interleaving pattern, and Theorem 26.5.1 is its two-repeat case.

Check (I) verifies the theorem three ways: on clean random sequence instances (the prediction is exact in every one), on the explicit compacted graphs (six branch vertices; connector multiplicities $\{1, 1, 1, 1\}$ for disjoint and nested and $\{2, 2\}$ for interleaved), and by symbolic Smith normal form of the matrices in the proof for $s \leq 7$.

Three repeats, and what the correction is not

With three repeats the natural guess — suggested to us by a reader — is that the interleaving correction is a function of the chord diagram of the six occurrences through the rank over $\mathbb{F}_2$ of its intersection matrix. It is not.

Proposition 26.5.3 (Three repeats). *Let A, B, C be distinct words of length L , each occurring exactly twice, with every other k -mer unique and $s = L - k + 1$. Let D_w be the digraph on the three symbols with one arc from the symbol of each occurrence to the symbol of the next occurrence around the circle, loops deleted. Then $\mathcal{K}(G_L) \cong \mathcal{K}(D_w)$, and according to which pairs of the three chords cross:*

<i>crossing pairs</i>	$\mathbb{F}_2$ -rank of intersection matrix	$\mathcal{K}(G_k), s = 1$	$\mathcal{K}(G_k), s = 2$
<i>none</i>	0	0	$(\mathbb{Z}/2)^3$
<i>one pair</i>	2	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^4$
<i>a path of two</i>	2	$\mathbb{Z}/3$	$(\mathbb{Z}/2)^3 \oplus \mathbb{Z}/3$
<i>all three pairs</i>	2	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^5$

In every case $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{3(s-1)} \oplus \mathcal{K}(D_w)$ for $s = 1, 2$.

Proof. At $s = 1$ the compacted graph is D_w by construction (Lemma 26.4.1 deletes the loops), which gives $\mathcal{K}(G_L) = \mathcal{K}(D_w)$; the four values are the sandpile groups of the four 2-regular digraphs on three vertices that occur. For instance the path pattern $ABACBC$ gives the complete symmetric digraph on three vertices, whose sandpile group is $\mathbb{Z}/3$. The $s = 2$ column and the splitting are verified exactly on all 90 circular words with two of each letter (check (M)); we have not proved the splitting for general s . $\square$

Remark 26.5.4. Three of the four patterns have intersection rank 2 and give three different groups, one of them of odd order: multiplicity-two repeats can interleave into 3-torsion. So the correction is not a chord-diagram rank, nor even a 2-group. The right object is the sandpile group of the connector digraph D_w , which for two repeats is trivial or $\mathbb{Z}/2$ (Theorem 26.5.1) and for three is the table above.

26.6 Branch statistics are insufficient

Theorem 26.6.1 (The global factor is not degree-determined). *There exist circular sequences $S_{\text{lo}}, S_{\text{hi}}$ of length 60 over $\{A, C, G, T\}$ such that at $k = 4$ their de Bruijn graphs have the same number of vertices $n_4 = 51$ and the same out-degree multiset (seven branch vertices of out-degree 2 and one of out-degree 3), hence the same local factor $\text{Loc} = 2$, but*

$$\mathcal{K}(G_4(S_{\text{lo}})) = \mathbb{Z}/2, \quad \mathcal{K}(G_4(S_{\text{hi}})) = (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/8,$$

so that the numbers of reconstructions are 4 and 256. Explicit witnesses:

$S_{\text{lo}} :$ **GAACGAGAGAAAAACACGTACCAGCGCGGGGGCGGGACGGCAGGTGACGCTCTGGGAAG**
 $S_{\text{hi}} :$ **GCACAGGGTCCAGTCTGCGAACGCGCTACAGTCTGGGTCAAGCGGTCCTCTGTGCATCG**

Proof. Exact computation; check (N). $\square$

Corollary 26.6.2. *No function of the out-degree multiset of G_k determines the number of reconstructions.*

In a randomised search over 1500 sequences of length 60 at $k \in \{4, 5\}$, 67 of the 98 degree-classes containing more than one sequence contained more than one critical group (check (N), sampled).

26.7 Computation

Computing $\mathcal{K}(G_k)$ directly is one Smith normal form of an $(n_k - 1) \times (n_k - 1)$ integer matrix, $O(n_k^3)$ with the usual bit-length control [Ch. 2], which for a genome is impractical. Lemma 26.4.1 removes the difficulty: the critical group of G_k equals that of its compacted graph, whose vertices are the branch points. The compaction is linear in N — one pass to find the branch vertices, one walk per outgoing arc — and assemblers already build this graph. The cost is therefore governed by repeat content, not genome length; the next section shows what this means in practice.

26.8 A real genome

Bacteriophage φ X174 (GenBank NC_001422.1, 5386 bp, circular) was the first genome ever sequenced. Its longest exact circular repeat has length $R = 12$. Table 26.1 gives the critical-group spectrum around that scale. The compaction is dramatic: at $k = 10$ the graph has 5339 vertices and 47 branch points, and the Smith normal form takes ten milliseconds. The group order at $k = 10$, 4.4×10^{12} , is the global part of the number of 10-mer reconstructions of the phage; it falls to 132 at $k = 11$, to 2 at $k = 12$, and the graph becomes a single cycle at $k = 13 = R + 1$.

k	n_k	branch vertices	$ \mathcal{K}(G_k) $	invariant factors
≤ 9	≤ 5233	≥ 146	—	above the exact-SNF cap used here
10	5339	47	4 408 682 819 584	$2^8, \dots, 4305354316$
11	5376	10	132	2, 66
12	5384	2	2	2
13	5386	0	1	— (single cycle)

Table 26.1: φ X174, NC_001422.1. Branch vertices are the vertices of the compacted graph. All entries exact (check (F)); the sequence file is distributed with the script and no network access is needed.

Two remarks. The levels $k \leq 9$ have compacted graphs of 146 to 1384 vertices; they are within reach of a sparse or p -adic Smith normal form but not of the dense routine used here, and we have not pursued them. And $k^* = R + 1$ for this genome. A single isolated repeat collapses at $k = R$ (Theorem 26.4.2), so a surviving $\mathbb{Z}/2$ at $k = 12$ is *consistent with* Theorem 26.5.1 rather than implied by it; the companion paper [Ch. 27] settles the question by exhibiting the two reconstructions: the longest 12-mer repeat is interleaved with a second repeated 12-mer, and the residual bit is exactly the one of Theorem 26.5.1 at $s = 1$.

26.9 What is open, and what is false

False. The collapse order is not one more than the longest repeat. With $R(S)$ the longest repeated substring and $k^*(S)$ the least k with $\mathcal{K}(G_k)$ trivial, over 40 random sequences of length 100 we find $k^* = R$ in 26 cases and $k^* = R + 1$ in 13 (check (C)). Exhaustive search over short binary sequences gives minimal witnesses: AACAC has $R = 3$, $k^* = 3$; AACACC has $R = 2$, $k^* = 3$; and homopolymer runs such as AAAA have $k^* < R$, because a repeat that always extends the same way branches nowhere. Theorem 26.4.2 explains the first family and Theorem 26.5.1 the second, but we have no closed form for k^* in general.

Open. (a) *Additivity beyond two repeats.* Theorem 26.5.1 and Proposition 26.5.3 identify the correction at $s = 1$ with the sandpile group of the connector digraph D_w of the circular word of occurrences, and show it is not a function of the chord diagram's intersection rank. We conjecture $\mathcal{K}(G_k) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w)$ for m repeats of multiplicities r_i (with D_w then carrying multiplicities); it is proved for $m \leq 2$ and verified for $m = 3$, $r_i = 2$, $s \leq 2$. (b) *Higher multiplicity.* We have not computed the interleaving correction for repeats of multiplicity $r > 2$. (c) *Double-strandedness.* Real assembly graphs are bidirected. This is a different combinatorial object, and we warn against the tempting decomposition of $\mathcal{K}$ into ± 1 -eigenspaces of the reverse-complement involution: an involution splits an abelian group only after 2 is inverted, and the 2-torsion — which is where all of our examples live, and where inverted repeats would show up — is exactly what such a splitting discards. The tools that survive are the signed and $\mathbb{Z}/2$ -voltage Laplacians [153, 151], with one caveat: reverse complementation reverses arc direction, so the passage to the bidirected graph is not a covering of digraphs and the covering theory of [151] does not apply as it stands. (d) *Errors and coverage.* We assume an exact k -mer spectrum with true multiplicities.

26.10 Verification

The script `dbg_sandpile.py` runs 30 checks in about two minutes, all passing; its output is archived as `dbg_sandpile_output.txt`. Arithmetic is exact except in the two rows marked *sampled*.

key	check	#	mode
(T)	$\text{DB}(\sigma, k+1) = L(\text{DB}(\sigma, k)); \mathcal{K} = \sigma^{\sigma^k - k - 1}$	8/8	exact
(L)	$\mathcal{K}(\text{DB}(2, n))$ reproduces Levine [7], $n \leq 5$	4/4	exact
(E)	BEST (26.1.1) against brute-force arc-rooted enumeration	6/6	exact
(R)	Theorem 26.4.2, 54 parameter settings	1/1	exact
(I)	Theorem 26.5.1: sequences, compacted graphs, symbolic SNF	3/3	exact
(M)	Proposition 26.5.3: all 90 circular words of three repeats, $s = 1, 2$	1/1	exact
(N)	Theorem 26.6.1 witnesses; degree-class splitting rate	2/2	exact / sampled
(C)	collapse order: statistics; minimal binary witnesses	2/2	sampled / exact
(P)	Lemma 26.4.1: contraction and self-loops	2/2	exact
(F)	Table 26.1: $\varphi X174$	1/1	exact
total		30/30	

Table 26.2: The verification battery (`dbg_sandpile.py`).

Provenance and changes in version 2

Unrefereed preprint. The classical inputs are the BEST theorem [142, 152], Knuth’s recursion [147] and the matrix–tree theorem for digraphs; the critical groups of complete de Bruijn graphs [7, 144, 145] are used only as checks. Proposition 26.3.1 is a reading of (26.1.1); the use of BEST to count genome reconstructions is due to [146]. What we claim as new is Theorems 26.4.2, 26.5.1 and 26.6.1 with Corollary 26.6.2, and Lemma 26.4.1 with its use in Sections 26.7–26.8.

Version 2 adds Theorem 26.5.1 (resolving the two-repeat case of the additivity question left open in version 1), Proposition 26.5.3 (three repeats, refuting the chord-rank guess), the real-genome computation of Section 26.8, the counting convention of Remark 26.2.1, the self-loop clause of Lemma 26.4.1, the minimal collapse witnesses, and the warning about eigenspace decompositions in Open (c). These changes respond to a detailed reading of version 1, for which we are grateful; the referee’s four corrections were all right and are all incorporated.

Chapter 27

The sandpile group as a coordinate system on genome reconstructions

Summary. The reconstructions of a circular genome from its k -mer spectrum are the Eulerian circuits of its de Bruijn multigraph G , and by the BEST theorem they correspond bijectively to pairs (spanning arborescence, choice of exit order at every branch vertex). The sandpile group $\mathcal{K}(G)$ acts simply transitively on the arborescences by rotor-routing (Holroyd, Levine, Mészáros, Peres, Propp and Wilson), whose proof of BEST is itself a rotor walk. We put the two facts to a use we have not found in the literature: $\mathcal{K}(G)$ is a *torsor of coordinates on the global part of the reconstruction space*, one chip placed on a k -mer is an explicit rearrangement carrying one candidate genome to another with the same spectrum, and the difference of two candidates is a group element. We exhibit the moves, verify the torsor structure and the bijection on assembly graphs by exhaustive enumeration, and show that the classical uniform sampler of Eulerian circuits — known since Altschul and Erickson (1985) and implemented in uShuffle — is the torsor in Smith-normal-form coordinates, which is what makes the distance between two samples computable. The centrepiece is a real genome: for bacteriophage ϕ X174 the 12-mer spectrum leaves exactly one global choice, $\mathcal{K} = \mathbb{Z}/2$; the two reconstructions are exhibited, they are $A X B Y A Z B U$ and $A X B U A Z B Y$ for two interleaved repeated 12-mers, and a single chip exchanges the 433 and 181 bp segments. This is the $\mathbb{Z}/2$ of the interleaving theorem of the companion paper, realised in the first genome ever sequenced. Seven checks, exact except the two sampling rows.

27.1 Introduction

A companion paper [Ch. 26] showed that the number of reconstructions of a circular sequence from its k -mer spectrum factors, by the BEST theorem, into a local part $\prod_v (d^+(v) - 1)!$ determined by branch statistics and a global part $|\mathcal{K}(G_k)|$, the order of the sandpile group, which branch statistics do not determine; and that the group structure of $\mathcal{K}(G_k)$ records repeat length, multiplicity and interleaving. This note is about the group itself rather than its order: what does the group *do* to the set of reconstructions?

What is known. Three literatures bear on the question, and we claim nothing in any of them.

- *Counting and sampling.* The BEST theorem [142, 152] counts Eulerian circuits of a directed Eulerian graph as $(\text{arborescences}) \times \prod (d - 1)!$, and since arborescences can be sampled uniformly in polynomial time [158, 166, 167], so can Eulerian circuits; Creed and Cryan state this explicitly [24]. (For *undirected* graphs counting is $\#P$ -complete [157]; the directed case, which is ours, is easy.) Bioinformatics has used exactly this since 1985 to shuffle sequences preserving k -let counts [154, 162, 161], and to count reconstructions from k -mer spectra [155, 146].

- *Rotor-routing.* Holroyd et al. [160] prove that the sandpile group of a digraph with a sink acts freely and transitively on the spanning arborescences toward the sink (their §3), and in their §4 give a bijective proof of BEST by a rotor walk: an Eulerian tour is exactly one full turn of every rotor (Cor. 4.10). The rotor-router model itself is the “Eulerian walkers” model of Priezzhev et al. [165].
- *Sandpile groups of de Bruijn graphs.* Levine [7] computed $\mathcal{K}(\text{DB}_n)$ for the complete binary de Bruijn graph; Bidkhori–Kishore and Chan–Hollmann–Pasechnik extended this [156, 145]. None concern genome graphs.

What is here. We compose these into a statement about assembly (Theorem 27.3.1): the set of reconstructions, modulo the local exit choices, is a torsor for $\mathcal{K}(G)$. Consequences: a chip placed on a k -mer is an explicit rearrangement between two candidate genomes (Section 27.4); the difference of two candidates is a group element, computable as a chip word; and the classical uniform sampler is the torsor written in Smith-normal-form coordinates (Section 27.5). Section 27.6 applies this to φX174 , where the 12-mer spectrum leaves one binary choice and we show what that choice is.

What we do not claim. Not the torsor theorem, which is [160, §3]; not the bijection between tours and (arborescence, rotor order), which is [160, Cor. 4.10] and which our check (B) merely re-implements; not uniform sampling, which is [158, 166, 24] and, in bioinformatics, [154, 162, 161]. The contribution is the reading of these as a coordinate system on assemblies, the identification of chip-firing with rearrangement, and the φX174 computation.

27.2 Setting

S is a circular sequence, $G = G_k(S)$ its de Bruijn multigraph (vertices the distinct k -mers, one arc per $(k+1)$ -mer occurrence), which is connected and balanced with S an Eulerian circuit; $\text{Rec}(G)$ denotes the set of arc-rooted Eulerian circuits, i.e. the reconstructions [Ch. 26, Rem. 1]. Fix a sink $s \in V$ and let $\text{Arb}(G)$ be the set of spanning arborescences oriented toward s ; for balanced G the count $|\text{Arb}(G)|$ and the group $\mathcal{K}(G) = \text{coker } \Delta^\top$ do not depend on s [160, Lem. 4.12]. Here $\Delta = D^+ - A$ is the out-Laplacian with the row and column of s deleted; its determinant counts arborescences *toward* s because each non-sink vertex chooses one of its out-arcs, and the transpose enters because firing v sends one chip along each out-arc, $\sigma \mapsto \sigma - \Delta^\top e_v$, so the firing lattice is the column span of $\Delta^\top$. For balanced G , $\Delta^\top = D^+ - A^\top$ is the in-Laplacian, so the choice is between two presentations of one group. A *rotor configuration* is a choice of one out-arc at every non-sink vertex; the acyclic ones are exactly $\text{Arb}(G)$. A chip placed at v is routed to the sink by the rule “advance the rotor at the current vertex by one, then move along it” [160, §3]; the map $\text{Arb}(G) \rightarrow \text{Arb}(G)$ this induces is the action of the class of e_v in $\mathcal{K}(G)$. The same action can be realised algebraically, by adding e_v to a recurrent chip configuration and toppling back to the recurrent representative, since the rotor-router group and the sandpile group coincide [160, Lem. 3.19]; we use the dynamical description because it delivers the new arborescence directly.

By the contraction lemma of [Ch. 26, Lem. 4], both $\mathcal{K}(G)$ and, arc for arc, $\text{Arb}(G)$ are unchanged when every vertex of in- and out-degree 1 is smoothed away, so all of the following may be computed on the compacted (unitig) graph; the contracted arcs carry the concatenated labels and a circuit on the compacted graph is spelled back into a full-length sequence.

27.3 The torsor

Theorem 27.3.1 (Coordinates on reconstructions). *Let G be the de Bruijn multigraph of a circular sequence. Then*

- (i) $\mathcal{K}(G)$ acts freely and transitively on $\text{Arb}(G)$ by rotor-routing [160, Lem. 3.12, 3.13, 3.17];
- (ii) the map $(T, \rho) \mapsto$ “walk from the sink taking, at each vertex, the next unused out-arc in the cyclic order ρ_v with the tree arc of T last” is a bijection from pairs (arborescence, exit orders) onto $\text{Rec}(G)$ [160, Cor. 4.10];

- (iii) consequently $\text{Rec}(G)/\sim$, the reconstructions modulo the local exit choices, is a $\mathcal{K}(G)$ -torsor: after fixing one reconstruction, every other one is labelled by a unique group element, and $|\text{Rec}(G)| = |\mathcal{K}(G)| \cdot \prod_v (d^+(v) - 1)!$.

The theorem is a composition of two results of [160]; what we add is (iii) as a statement about genomes, and the two verifications of Section 27.7: check (A) enumerates $\text{Arb}(G)$ for six assembly graphs, builds the permutation representation of the chip generators, and confirms that the generated group has order $|\mathcal{K}(G)|$, a single orbit, and no fixed points; check (B) implements the walk of (ii) and confirms it is a bijection onto the brute-force enumeration of Eulerian circuits (288, 576, 144, 8 circuits on four graphs).

Remark 27.3.2 (Why the quotient). The local choices at a branch vertex of out-degree d — which of the $d - 1$ non-tree exits to take first — are what the local factor $\prod (d - 1)!$ counts, and they are independent of one another and of the arborescence. They are the ambiguity that branch statistics *do* see. The torsor lives on what remains, which [Ch. 26] showed is invisible to branch statistics; Theorem 27.3.1 says it nevertheless has the structure of a finite abelian group, given explicitly by the Smith normal form of the out-Laplacian.

27.4 A chip is a rearrangement

Fix exit orders once and for all (say, in arc order). Then each arborescence determines one reconstruction, and adding a chip at a k -mer v moves the arborescence and hence the reconstruction. Check (M) does this on $S = \text{ACACCAAAACCA}$, $k = 2$, where $\mathcal{K}(G) = \mathbb{Z}/2 \oplus \mathbb{Z}/6$:

base reconstruction	AAAACCAACCAC	
+ one chip at CA	ACCAAAACCAAC	(6 positions changed)
+ one chip at CC	ACCAAAACCAAC	(6 positions changed)
+ one chip at AA	AAAACCAACCAC	(unchanged: $[e_{AA}] = 0$ in $\mathcal{K}$)

Each row has the 2-mer spectrum of S . The chip at AA is trivial in the group and moves nothing, which is the correct behaviour: not every k -mer carries a nonzero coordinate.

Proposition 27.4.1 (Distance). *For two reconstructions with the same exit orders there is a unique $g \in \mathcal{K}(G)$ carrying one arborescence to the other. Writing g as a sum of chip classes $[e_v]$ with non-negative coefficients — always possible in a finite group — gives a sequence of physically realisable moves converting the first genome into the second. The minimal number of chips $\delta(T, T')$ is a directed distance on $\text{Rec}(G)/\sim$: a quasi-metric, since $\delta(T, T')$ and $\delta(T', T)$ need not agree. Admitting chips of both signs, i.e. the symmetric generating set $\{\pm[e_v]\}$, gives the Cayley word metric of $\mathcal{K}(G)$.*

This is the sense in which the group is a coordinate system rather than merely a count. Check (W) computes the word for two independently sampled reconstructions of a 27-element torsor: three chips.

One thing a chip is *not* is a single transposition. The rotor walk re-threads every branch vertex it passes on its way to the sink, so one chip can change the tree arc at several junctions at once. On the compacted φX174 graph at $k = 11$, which has ten branch vertices, the nine generators change 6, 5, 3, 2, 4, 2, 3, 1, 0 tree arcs respectively (check (J)); only one of them is a single re-threading. The directed distance of Proposition 27.4.1 is therefore not the transposition or reversal distance of comparative genomics, and we see no immediate inequality between them.

27.5 Sampling, and what the group adds

Uniform sampling of $\text{Rec}(G)$ is classical: sample an arborescence uniformly by a loop-erased random walk [167, 166] or by unranking [158], then choose exit orders uniformly [24]; this is what [154, 162] do and what uShuffle implements [161]. Theorem 27.3.1 gives the same distribution in a different form: with $\mathcal{K}(G) \cong \bigoplus \mathbb{Z}/d_i$ from the Smith normal form $U\Delta^\top V = D$, draw r_i uniformly mod d_i , take $U^{-1}r$ as a chip configuration, and route it from a fixed base arborescence. Check (U) draws 2700 samples this way on a torsor of order 27 and finds all 27 arborescences with $\chi^2 = 31.2$ on 26 degrees of freedom ($p = 0.22$); Wilson's

sampler on the same graph gives $\chi^2 = 22.3$ ($p = 0.67$, check (W)). The two are the same distribution. What the coordinate form adds is not speed — Wilson’s algorithm is faster — but the ability to say *where* a sample is. A shuffler such as uShuffle returns independent draws and nothing else: two of its outputs stand in no computable relation, and the sample space has no structure beyond its cardinality. In the torsor two samples differ by a computable group element, a sample can be moved by a prescribed chip, the whole space can be enumerated by stepping through Smith coordinates, and a sample can be conditioned on a prescribed value of some coordinates. What it does *not* give is a linear way to impose local constraints, as the next remark records.

Remark 27.5.1 (Local constraints are not cosets). A long read that spans a branch vertex v fixes which out-arc of v is the tree arc, selecting $S_e = \{T \in \text{Arb}(G) : T(v) = e\}$. If several such constraints could be combined by linear algebra in $\mathcal{K}(G)$, each S_e , transported to the group through the torsor, would have to be a coset of a subgroup. It is not: on the graph of ACACAAAACCA at $k = 2$ ($|\mathcal{K}| = 12$) two of the seven non-empty constraint sets are not cosets, on CCAACAACCAAC at $k = 3$ three of nine, on AABABBAABABB two of eight; only the single-repeat graph AABBAABBAABB ($\mathcal{K} = (\mathbb{Z}/3)^2$) has every constraint a coset (check (N)). The group structure is global; the local structure of arborescences is not linear in it, and a “congruence solver” for long-read disambiguation would have to be a search over the torsor, not a system of congruences.

27.6 φ X174: one global choice

The companion paper computed the sandpile spectrum of bacteriophage φ X174 (NC_001422.1, 5386 bp) and found $\mathcal{K}(G_{12}) = \mathbb{Z}/2$ and $\mathcal{K}(G_{11}) = \mathbb{Z}/2 \oplus \mathbb{Z}/66$ [Ch. 26, Tab. 1]. Theorem 27.3.1 says that the 12-mer spectrum of the phage leaves exactly *two* global reconstructions. Check (F) exhibits them.

Proposition 27.6.1 (φ X174 at $k = 12$). *The compacted 12-mer graph of φ X174 has two branch vertices, the repeated 12-mers*

$$A = \text{CTTCTGCCGTTT}, \quad B = \text{CGTCAAGGACTG},$$

at positions 457 and 3205 (A) and 264 and 2760 (B) of NC_001422.1. Reading the reference from the second A, the genome is

$$\begin{aligned} &A X B U A Z B Y, \\ &X = 3217..263 \text{ (2433 nt, wrapping)}, \quad U = 276..456 \text{ (181 nt)}, \\ &Z = 469..2759 \text{ (2291 nt)}, \quad Y = 2772..3204 \text{ (433 nt)}, \end{aligned}$$

an interleaved pair. Its two reconstructions, both of length 5386 and both with the phage’s 12-mer spectrum, are the reference A X B U A Z B Y and A X B Y A Z B U, the segments following the two copies of B exchanged; one chip at a branch vertex carries one to the other.

The coordinates account for the whole genome: $2433 + 181 + 2291 + 433 + 4 \cdot 12 = 5386$, the four unique segments and the four occurrences of the two 12-mers; the exchange permutes segments, so both reconstructions have the same length and, segment by segment, the same content.

This is precisely the situation of the interleaving theorem [Ch. 26, Thm. 6] at $s = 1$: a single repeat of multiplicity two leaves no circular ambiguity (its two threadings are rotations of one another), so a surviving $\mathbb{Z}/2$ at $k = L$ must come from two interleaved repeats, and the one bit it records is which of the two B -exits follows which A . The first genome ever sequenced has exactly one such pair of 12-mers, and the group finds it. At $k = 11$ the 132 classes of $\mathbb{Z}/2 \oplus \mathbb{Z}/66$ are the corresponding statement one level down; we have not analysed their structure.

The biology of the one bit is worth stating, because it is not what one might hope. By the feature table of NC_001422.1, the exchanged segment U (276..456) spans the junction of genes C (133..393) and D (390..848), and Y (2772..3204) spans the junction of the spike genes G (2395..2922) and H (2931..3917); the four repeat occurrences lie in C , D , G and H respectively. Exchanging a C/D junction for a G/H junction breaks the reading frames on both sides, so the alternative reconstruction is not a viable isomer of the phage. It is the misassembly that a 12-mer assembler is at risk of producing, and the group locates it exactly: one bit, two named 12-mers, two named segments.

27.7 Verification

The script `rotor_assembly.py` runs the eight checks of Table 27.1 in about five seconds, all passing; its output is archived as `rotor_assembly_output.txt`, and the φ X174 sequence is distributed with it.

key	check	mode
(A)	Theorem 27.3.1(i) on six assembly graphs, $ \mathcal{K} \leq 27$: free and transitive	exact
(B)	Theorem 27.3.1(ii): the walk is a bijection onto the enumerated circuits	exact
(M)	Section 27.4: chips are spectrum-preserving rearrangements	exact
(U)	Smith-coordinate sampler uniform, $\chi^2 = 31.2$ on 26 df	sampled
(W)	Wilson's sampler agrees; the chip word between two samples	sampled
(F)	Proposition 27.6.1	exact
(J)	junctions re-threaded per chip on φ X174 at $k = 11$: 0 to 6	exact
(N)	Remark 27.5.1: constraint sets are not cosets on 3 of 4 graphs	exact

Table 27.1: The verification battery (`rotor_assembly.py`), 8/8.

27.8 Scope

Quoted. The torsor and the rotor proof of BEST [160]; uniform sampling [158, 166, 167, 24] and its bioinformatic use [154, 162, 161]; counting [155, 146]; the spectrum of φ X174 and the interleaving theorem [Ch. 26].

New here, as far as we can determine. Theorem 27.3.1(iii) as a statement about assemblies; chip-firing as rearrangement and the word metric of Proposition 27.4.1; the coordinate form of the sampler; and Proposition 27.6.1.

Open. (a) *Double strands.* Real assembly graphs are bidirected [163, 164]. The reverse-complement map sends an arc $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$, reversing direction, so the passage from the k -mer digraph to the bidirected graph is not a covering of digraphs and the covering theory of critical groups [151] does not apply as it stands; a sandpile theory for bidirected de Bruijn graphs would have to start from the signed or bidirected Laplacian [153], and we know of none. (b) The 132 classes of φ X174 at $k = 11$, and more generally what the invariant factors of a real genome's group mean position by position. (c) Whether the directed distance of Proposition 27.4.1 relates to any edit distance between assemblies; since a single chip can re-thread many junctions (Section 27.4), it is not the transposition distance, and any relation would have to pass through the structure of rotor walks on the compacted graph.

Provenance of this chapter

Unrefereed preprint. Theorem 27.3.1 is a composition of published results and is verified, not reproved; Proposition 27.4.1 is immediate from it; Proposition 27.6.1 is an exact computation reproducible in seconds. We searched for prior use of the sandpile torsor on genome graphs and for any critical-group treatment of bidirected de Bruijn graphs, and found neither; a search cannot establish a negative.

Chapter 28

The sandpile group of a de Bruijn graph splits along its repeats

Summary. Two companion papers showed that the sandpile group $\mathcal{K}(G_k)$ of a genome’s de Bruijn graph records repeat length, multiplicity and interleaving — a single repeat of length L and multiplicity r contributes $(\mathbb{Z}/r)^{L-k}$, two interleaved repeats add a $\mathbb{Z}/2$, three can add a $\mathbb{Z}/3$ — and showed that the correction is not a rank of the chord diagram. Here we prove the general statement. *Bundle-chain lemma:* in any balanced multidigraph, a chain $v_1 \rightarrow \cdots \rightarrow v_s$ in which every vertex sends all r of its arcs to the next can be contracted to a point, and the sandpile group splits off a direct summand $(\mathbb{Z}/r)^{s-1}$ exactly. The proof is a unimodular change of variables and takes ten lines; the $r = 1$ case is the familiar unitig contraction. Contracting every bundle chain leaves a *skeleton* with one vertex per repeat family and one arc per adjacency of repeat occurrences around the genome — a line-digraph dual of the repeat graph of Pevzner, Tang and Tesler — and

$$\mathcal{K}(G_k) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}).$$

The skeleton is canonical, and the repeat graph itself is the case in which every chain has length two. The first summand is what longer k -mers remove; the second, the sandpile group of the repeat-adjacency skeleton, is what no k removes. For repeats of multiplicity two the second summand is the critical group of the one-vertex map whose edges are the repeats: it is $\text{coker}(A + I)$ for Bouchet’s signed interlace matrix A (Merino–Moffatt–Noble), its order is the interlace polynomial at 1 (Arratia–Bollobás–Sorkin), and — as their example shows and we recompute — it is not a function of the unsigned chord-crossing graph, although it happens to be for up to five repeats. For φX174 the decomposition is verified against direct computation at $k = 10, 11, 12$ and extends the computable spectrum to $k = 9$, where the compacted graph (146 branch vertices) exceeded our exact Smith-form range but its skeleton does not: $\mathcal{K}(G_9) = (\mathbb{Z}/2)^{42} \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \cdots \oplus \mathbb{Z}/2707831823$, of which $(\mathbb{Z}/2)^{36}$ is length-driven and the rest is the arrangement of repeats. Seven checks, all exact.

28.1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ is the de Bruijn multigraph whose vertices are the distinct k -mers and whose arcs are the occurrences of $(k+1)$ -mers; S is an Eulerian circuit, and by the BEST theorem [142, 152] the number of reconstructions of S from its k -mer spectrum is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$ with $\mathcal{K}(G_k)$ the sandpile group. The first companion paper [Ch. 26] showed that $\mathcal{K}(G_k)$ is not determined by the branch statistics and computed it in three model situations: a single repeat of length L and multiplicity

r gives $(\mathbb{Z}/r)^{L-k}$; two repeats of multiplicity two give $(\mathbb{Z}/2)^{2s-2}$ if they are disjoint or nested and $(\mathbb{Z}/2)^{2s-1}$ if interleaved, $s = L - k + 1$; three repeats give a correction that is $\mathbb{Z}/2$, $\mathbb{Z}/3$ or $(\mathbb{Z}/2)^2$ according to their crossing pattern; since the three non-trivial patterns have intersection matrices of the same rank over $\mathbb{F}_2$, the correction is not a function of that rank (Remark 28.4.6 sharpens this). The second paper [Ch. 27] used the group as a coordinate system on reconstructions and found that for φ X174 at $k = 12$ the single surviving $\mathbb{Z}/2$ is a pair of interleaved 12-mers.

Those three model computations are special cases of one structural fact, which this note proves. The fact is local and holds in any balanced multidigraph; its consequence for genomes is that the sandpile group splits as a direct sum of a *length-driven* part, one cyclic factor per shared k -mer of each repeat, and an *arrangement* part, the sandpile group of a repeat-adjacency skeleton.

What is known. Contracting a vertex of in- and out-degree one leaves the sandpile group unchanged [Ch. 26, Lem. 4]; this is the $r = 1$ case of Lemma 28.3.1 and is the reason assemblers' compacted (unitig) graphs [170, 163] carry the same group. For *undirected* graphs with multiple edges the sandpile groups of thick trees [169] and thick cycles [168] are known, and the thick-tree answer, $\bigoplus \mathbb{Z}/m_e$ over the edge multiplicities, looks like our formula; but the undirected analogue of Lemma 28.3.1 is false (Remark 28.3.3), so the lemma is not a corollary of those results. The nearest directed relative is Levine's theorem that $\mathcal{K}(G)$ is a quotient of $\mathcal{K}(LG)$ for the line digraph [7, 177]. The Eulerian-path formulation of assembly is due to Pevzner, Tang and Waterman [179], and the *repeat graph*, in which repeat families are edges of multiplicity > 1 and vertices are repeat boundaries, to Pevzner, Tang and Tesler [178]; it is the central object of the long-read assembler Flye [171]. Our skeleton has one *vertex* per repeat family and one arc per adjacency of occurrences, so it stands to the repeat graph roughly as a line digraph to its graph; we keep the two apart by name. For two-copy repeats the skeleton is the directed medial graph of a one-vertex map, and its sandpile group is the critical group of that map in the sense of Merino, Moffatt and Noble [174], which is where the question of what determines it has already been answered (Remark 28.4.6). We have not found the $r > 1$ splitting of Lemma 28.3.1 in the sandpile literature, nor the sandpile group of the repeat graph considered as such; a search cannot establish a negative.

28.2 Setting

All digraphs are finite multidigraphs, balanced ($d^-(v) = d^+(v)$ at every vertex) and connected. The out-Laplacian is $\Delta = D^+ - A$; the sandpile group is $\mathcal{K}(G) = \text{coker}(\Delta_s^T)$, the row and column of a sink s deleted; for balanced G the isomorphism class does not depend on s [160, Lem. 4.12], and $|\mathcal{K}(G)|$ is the number of spanning arborescences toward s . We compare finite abelian groups by their elementary divisors.

Definition 28.2.1 (Bundle chain). A *bundle chain* of multiplicity r and length $s \geq 2$ is a sequence of distinct vertices $v_1, \dots, v_s$ such that for each $t < s$ the vertex v_t has out-degree r and all r of its out-arcs go to v_{t+1} , and v_s has out-degree r . Since G is balanced, every in-arc of v_{t+1} ($t < s$) comes from v_t . The *contraction* G/C replaces the chain by a single vertex carrying the in-arcs of v_1 and the out-arcs of v_s ; it is again balanced, and an arc $v_s \rightarrow v_1$ becomes a loop, which does not affect Δ .

28.3 The lemma

Lemma 28.3.1 (Bundle-chain contraction). *Let $C = (v_1, \dots, v_s)$ be a bundle chain of multiplicity r in a balanced multidigraph G with at least one vertex outside C . Then*

$$\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1} \oplus \mathcal{K}(G/C).$$

Proof. Root at a vertex outside C and write x_v for the generator of the cokernel presentation at each non-root vertex, the relations being the rows of Δ . Change generators on the chain:

$$z := x_{v_1}, \quad y_t := x_{v_t} - x_{v_{t+1}} \quad (1 \leq t < s),$$

so that $x_{v_t} = z - \sum_{u < t} y_u$ and in particular $x_{v_s} = z - \sum_{u < s} y_u$; this is a unimodular (triangular) change of basis. The rows of Δ fall into three kinds.

- The row of v_t for $t < s$ is $r x_{v_t} - r x_{v_{t+1}} = r y_t$.
- The row of v_s is $r x_{v_s} - \sum_w c_{v_s w} x_w = r z - r \sum_{u < s} y_u - \sum_w c_{v_s w} x_w$. Subtracting the rows $r y_u$ — a unimodular row operation — leaves $r z - \sum_w c_{v_s w} x_w$, which is the row of the contracted vertex in $\Delta(G/C)$.
- The row of a vertex $u \notin C$ involves x_{v_t} only through arcs from u into the chain, and all of those land on v_1 ; so it reads $d^+(u)x_u - \sum_w c_{uw} x_w$ with $x_{v_1} = z$, which is its row in $\Delta(G/C)$.

The relation lattice is therefore generated by $\{r y_t\}_{t < s}$ together with the rows of $\Delta(G/C)$ in the variables z and x_u , and the generators separate accordingly. Hence $\mathcal{K}(G) \cong \bigoplus_{t < s} \mathbb{Z}/r \oplus \mathcal{K}(G/C)$. $\square$

If G consists of the chain alone — the only vertices are $v_1, \dots, v_s$ and the r arcs of v_s return to v_1 — the same computation rooted at v_1 gives $\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1}$; this is the single-repeat theorem of [Ch. 26, Thm. 5].

Remark 28.3.2. For $r = 1$ the summand is trivial and the lemma says that smoothing a vertex of in- and out-degree one preserves the group, which is [Ch. 26, Lem. 4]. The content of the lemma for $r > 1$ is that a vertex which *cannot* be smoothed — it has r parallel exits — can nevertheless be removed at the exact cost of a $\mathbb{Z}/r$.

Remark 28.3.3 (The direction is essential). The undirected statement one might guess — a path of doubled edges contributes $(\mathbb{Z}/2)^{\text{length}-1}$ — is false. The undirected 4-cycle with every edge doubled has reduced Laplacian $2M$ with M the reduced Laplacian of the simple 4-cycle, whose Smith form is $\text{diag}(1, 1, 4)$; so its sandpile group is $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/8$, whereas the directed doubled 4-cycle is a single bundle chain and has $(\mathbb{Z}/2)^3$. What makes the directed lemma work is that every arc *into* the chain lands on its head v_1 and every arc *out* of it leaves from v_s with the full multiplicity r ; an undirected doubled edge carries flow both ways and the change of variables in the proof no longer separates.

28.4 The splitting theorem

Contracting bundle chains one after another terminates, since each contraction reduces the number of vertices, and the result contains no bundle chain of length ≥ 2 . We call it the *skeleton* $\text{Sk}(G)$. Its sandpile group is well defined by the lemma; the skeleton itself is too.

Lemma 28.4.1 (The skeleton is canonical). *Call $v \rightarrow w$ a chain step if $v \neq w$, all out-arcs of v go to w , and $d^+(w) = d^+(v)$. In a balanced multidigraph the chain steps form a partial injection on the vertex set, so the maximal bundle chains are its maximal paths: they are pairwise vertex-disjoint and determined by G . Contracting them in any order gives the same digraph up to isomorphism, which is chain-free.*

Proof. If $v \rightarrow w$ is a chain step then w receives $d^+(v) = d^+(w) = d^-(w)$ arcs from v , hence all of its in-arcs, so no $v' \neq v$ steps to w ; and v has a single target. Thus the step relation is a partial injection and its maximal paths partition the vertices it touches (a path may close into a cycle only if G is a single closed chain, when the skeleton is one vertex with r loops). Contracting a maximal chain C to a vertex c changes no degree outside C , gives c the in-arcs of v_1 and the out-arcs of v_s , and so creates no new chain step: $u \rightarrow c$ would require $u \rightarrow v_1$ to have been a step and $c \rightarrow w$ would require $v_s \rightarrow w$ to have been one, both excluded by maximality. Nor does it destroy a step outside C . Hence the contractions of distinct maximal chains commute, and the result is G with every maximal chain contracted. $\square$

Check (E) verifies this on 100 random balanced digraphs with planted chains: the partition into maximal chains, the multiset of (r_c, ℓ_c) and the skeleton's group are invariant under random relabelling, and the skeleton contains no chain step.

Theorem 28.4.2 (Splitting). *For any balanced connected multidigraph G , with $\{(r_c, \ell_c)\}$ the multiplicities and lengths of the chains contracted on the way to a skeleton,*

$$\mathcal{K}(G) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}(G)), \quad |\mathcal{K}(G)| = \prod_c r_c^{\ell_c-1} \cdot |\mathcal{K}(\text{Sk}(G))|.$$

Proof. Induction on the number of contractions, using Lemma 28.3.1 at each step. $\square$

Now let S be a sequence with repeats $W_1, \dots, W_m$ of lengths L_i and multiplicities r_i , and suppose the *clean* hypothesis of [Ch. 26]: the k -mers inside W_i occur exactly r_i times each and every other k -mer once. Write w for the circular word of repeat occurrences and D_w for the *connector digraph*: one vertex per repeat, one arc $i \rightarrow j$ for each occurrence of W_i immediately followed around the circle by an occurrence of W_j (loops deleted). D_w is balanced with out-degree r_i at vertex i .

Corollary 28.4.3 (Clean sequences). *Under the clean hypothesis, with $s_i = L_i - k + 1$,*

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{i=1}^m (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w).$$

Proof. After unitig contraction the graph consists of the m spines — bundle chains of multiplicity r_i and length s_i — joined by the connector arcs; contracting the spines gives D_w . Apply Theorem 28.4.2. $\square$

Corollary 28.4.4 (The published cases). *For $m = 1$, D_w is a single vertex and $\mathcal{K} = (\mathbb{Z}/r)^{s-1}$ [Ch. 26, Thm. 5]. For $m = 2$, $r_1 = r_2 = 2$: D_w is trivial for $AABB$ and $ABBA$ and is the doubled 2-cycle, with $\mathcal{K}(D_w) = \mathbb{Z}/2$, for $ABAB$ [Ch. 26, Thm. 6]. For $m = 3$ the four crossing patterns give $\mathcal{K}(D_w) = 0, \mathbb{Z}/2, \mathbb{Z}/3, (\mathbb{Z}/2)^2$ [Ch. 26, Prop. 8]; the $\mathbb{Z}/3$ is the sandpile group of the complete symmetric digraph on three vertices, which is D_w for $ABACBC$.*

Remark 28.4.5 (The skeleton and the repeat graph). The duality with the repeat graph of [178, 171] can be made exact. Under the clean hypothesis, collapse each spine of the compacted graph to a single arc: the result is the multidigraph RG_w with two vertices h_i, t_i per family, r_i parallel arcs $h_i \rightarrow t_i$, and one arc $t_i \rightarrow h_j$ per adjacency ij in w — repeats as multi-arcs between their boundary vertices, unique segments as single arcs, which is the repeat graph. Each pair (h_i, t_i) is a bundle chain of length two, and contracting it identifies the multi-arc with a vertex: D_w is the transition digraph of the repeat arcs of RG_w (the line digraph, restricted to repeat arcs, with unique segments as the adjacencies). By Lemma 28.3.1,

$$\mathcal{K}(\text{RG}_w) \cong \bigoplus_{i=1}^m \mathbb{Z}/r_i \oplus \mathcal{K}(D_w),$$

so the sandpile group of the repeat graph is the arrangement part plus one $\mathbb{Z}/r_i$ per family — the length-driven part at $s_i = 2$ (check (E), 200 random words). The two objects carry the same information; we work with D_w because it has no length-driven residue.

Remark 28.4.6 (Which invariant of the chord diagram). For $r_i = 2$ the $2m$ occurrences form a chord diagram with m chords — a double-occurrence word, or a one-vertex map (bouquet) with the repeats as edges — and D_w is a 2-in 2-out digraph. On what invariant of the diagram does $\mathcal{K}(D_w)$ depend? Not on the rank of the unsigned intersection matrix, over $\mathbb{F}_2$ or over $\mathbb{Z}$, and not on its integer Smith form: for $m = 3$ the three non-trivial patterns all have $\mathbb{F}_2$ -rank 2, and $AABCB C$ (one crossing) and $ABACBC$ (path) share the Smith form $(1, 1, 0)$ while their groups are $\mathbb{Z}/2$ and $\mathbb{Z}/3$; for $m = 5$ the rank-2 diagrams carry nine different groups, among them $\mathbb{Z}/7$ and $\mathbb{Z}/9$ (check (D)). The order $|\mathcal{K}(D_w)|$ is the number of Eulerian circuits of D_w , and this is a function of the intersection graph H alone (the *interlace graph* of [172], whose Definition 1 is exactly chord crossing): it is the interlace polynomial $q(H; 1)$ [172, Thm. 9, Cor. 13], with roots in Martin's and Las Vergnas' circuit-partition polynomials [173]. Exhaustively for $m \leq 5$ the group is also constant on the 4, 11 and 34 isomorphism classes of H — but this is an accident of small m . Merino, Moffatt and Noble [174, Rem. 6.7] give two 6-chord bouquets, 123456132465 and 123654132564, with the same interlace graph (K_6 minus two disjoint edges, so $|\mathcal{K}| = 18$ for both) and critical groups $\mathbb{Z}/3 \oplus \mathbb{Z}/6$ and $\mathbb{Z}/18$; check (D) recomputes both from D_w and exhibits two further such pairs. What does determine the group is the *signed* interlace matrix A of Bouchet ($A_{ef} = \pm 1$ according as the occurrences run $efef$ or $fefe$, 0 if not interlaced): by [174, Thm. 7.3], whose directed medial graph of a bouquet is our D_w ,

$$\mathcal{K}(D_w) \cong \text{coker}(A + I),$$

the critical group of the map; its order $\det(A + I)$ as an Euler-circuit count is older [175, 176]. Check (D) confirms the isomorphism on every double-occurrence word with $m \leq 5$ and on 1200 random words with $6 \leq m \leq 9$. So the arrangement part of an assembly's ambiguity, for two-copy repeats, is the critical group of the bouquet whose edges are the repeats, computable from a signed $m \times m$ matrix rather than from the de Bruijn graph; and the reason it is not a function of the chord-crossing graph is the reason [174] record: the signs matter. For $r_i > 2$ the occurrences form a hypergraph rather than a chord diagram and we know of no analogue.

Remark 28.4.7 (What k can and cannot remove). Increasing k shortens every spine: s_i decreases by one per unit of k , and the length-driven summand $\bigoplus (\mathbb{Z}/r_i)^{s_i-1}$ shrinks accordingly, vanishing at $k = \max L_i$. The skeleton summand $\mathcal{K}(D_w)$ does not depend on k at all, as long as $k \leq \min_i L_i$. So the global ambiguity of an assembly splits into a part that longer k -mers resolve and a part that only information beyond the repeats — long reads, pairs, a reference — can resolve, and the second part is exactly the sandpile group of the repeat-adjacency skeleton. By BEST applied to D_w , $|\mathcal{K}(D_w)| \cdot \prod_i (r_i - 1)!$ is the number of Eulerian circuits of D_w , i.e. of circular words with the same adjacency counts as w : the number of ways to thread the repeats, ignoring their lengths. That this is $|\mathcal{K}(D_w)|$ up to the local factor is the repeat-level form of the decomposition of [Ch. 26].

28.5 $\varphi X174$

Table 28.1 applies the theorem to the first sequenced genome (NC_001422.1, 5386 bp). At each k the compacted graph is bundle-contracted; the chains, the skeleton, and the two summands are listed, and where the compacted graph is small enough the group is also computed directly for comparison. Two things happen. The direct and decomposed groups agree wherever both exist ($k = 10, 11, 12$). And at $k = 9$, where the compacted graph has 146 branch vertices and was beyond the dense exact Smith form used in [Ch. 26], the skeleton has 110 and is not: the spectrum extends by one level, with the length-driven part $(\mathbb{Z}/2)^{36}$ split off exactly and the arrangement part read off the skeleton.

k	branch	chains	skeleton	$\mathcal{K}(\text{Sk})$ (arrangement part)	length-driven part
12	2	1	1	0	$\mathbb{Z}/2$
11	10	2	8	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	$(\mathbb{Z}/2)^2$
10	47	8	37	$\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$	$(\mathbb{Z}/2)^{10}$
9	146	28	110	$(\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/523 \oplus \mathbb{Z}/3779$ $\oplus \mathbb{Z}/5297 \oplus \mathbb{Z}/6561 \oplus \mathbb{Z}/7669 \oplus \mathbb{Z}/2707831823$	$(\mathbb{Z}/2)^{36}$
8	401	65	312	(skeleton beyond the exact-SNF range used)	

Table 28.1: $\varphi X174$ under bundle-chain contraction: number of branch vertices of the compacted graph, number of bundle chains contracted, number of skeleton vertices, and the two summands. $\mathcal{K}(G_k)$ is their direct sum; at $k = 10, 11, 12$ this agrees with the group computed directly on the compacted graph (check (F)). All chains have multiplicity 2. All entries exact.

The reading at $k = 11$ is the cleanest: $\mathcal{K}(G_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/11$, and the theorem says which is which — the two $\mathbb{Z}/2$'s are two repeat families each with one extra shared 11-mer beyond the minimum, and $\mathbb{Z}/3 \oplus \mathbb{Z}/11$, of order 33, is the arrangement ambiguity of the eight-vertex skeleton, which no choice of $k \leq 11$ can reduce.

28.6 Verification

The script `repeat_splitting.py` runs the seven checks of Table 28.2 in about thirty seconds, all passing; its output is archived as `repeat_splitting_output.txt`. All arithmetic is exact; groups are compared by elementary divisors.

key	check	mode
(A)	Corollary 28.4.3 on 120 clean random sequences, $m \leq 4$, $r_i \leq 4$	exact
(B)	Lemma 28.3.1 on 60 random balanced digraphs with a planted chain, $r \leq 6$, $s \leq 4$	exact
(C)	Corollary 28.4.4: the nine published values are the formula on D_w	exact
(D)	Remark 28.4.6: all double-occurrence words, $m \leq 5$; rank, Smith form, graph	exact
(E)	Lemma 28.4.1 on 100 relabelled digraphs; Remark 28.4.5 on 200 words	exact
(F)	Table 28.1: decomposition vs. direct computation; extension to $k = 9$	exact

Table 28.2: The verification battery, 7/7 (check (E) has two rows).

28.7 Scope

Proved. Lemma 28.3.1, Theorem 28.4.2 and its corollaries, for arbitrary balanced multidigraphs and arbitrary multiplicities; the clean hypothesis enters only in identifying the skeleton with D_w .

Quoted. Sink-independence [160]; BEST [142, 152]; the Eulerian formulation and the repeat graph [179, 178, 171]; the thick-graph and line-digraph results that are the lemma’s nearest relatives [169, 168, 7, 177]; the special cases and the φ X174 spectrum [Chs. 26, 27].

Open. (a) A structure theorem for $\mathcal{K}(D_w)$ itself — for which circular words is it trivial, and what bounds its exponent? For $r_i = 2$ this is the critical group of a bouquet (Remark 28.4.6), and [174] note that for 3-connected regular delta-matroids the group is well defined on the delta-matroid; whether $\mathcal{K}(D_w)$ is determined by the interlace graph whenever that graph is prime (no split) we have not examined. For $r_i > 2$ nothing of the kind is available. (b) Real genomes are not clean: the skeleton of φ X174 at $k = 9$ has 110 vertices and carries factors as large as 2707831823. All six factors above 100 in Table 28.1 (523, 3779, 5297, 7669, 1076338579, 2707831823) are prime, which is what the sandpile group of a generic digraph looks like (typically cyclic, or nearly so); we see no reason to expect a topological reading of a large prime cyclic factor and claim none. (c) Double strands: reverse complementation reverses arc direction, so a bidirected version of Lemma 28.3.1 must start from the signed Laplacian; we have not attempted it.

Provenance of this chapter

Unrefereed preprint. Lemmas 28.3.1 and 28.4.1 are proved in full above and are the only new mathematics; Remark 28.4.6 is a translation of published results [172, 174] into the assembly setting, verified computationally — an earlier draft of this note conjectured that the group was an invariant of the unsigned interlace graph, which [174, Rem. 6.7] had already shown to be false; everything else is either quoted or an exact computation reproducible in seconds. We searched for the $r > 1$ splitting in the sandpile literature and for any use of the sandpile group of a repeat graph or repeat-adjacency skeleton and found neither; the undirected thick-graph results [169, 168] are the closest, and Remark 28.3.3 shows they do not imply the lemma.

Chapter 29

Multi-copy repeats: where the two-copy theory of assembly ambiguity stops

Summary. The sandpile group of a genome’s de Bruijn graph splits into a length-driven part and the group $\mathcal{K}(D_w)$ of the transition digraph of the circular word of repeat occurrences [Ch. 28]. For repeats of multiplicity two, $\mathcal{K}(D_w)$ is the critical group of a one-vertex map: $\text{coker}(A + I)$ for Bouchet’s signed interlace matrix, by a theorem of Merino, Moffatt and Noble — a presentation whose entries are *pairwise* data about the repeats. This note asks what survives for repeats with three or more copies. Two families of any multiplicities r, s give $\mathcal{K}(D_w) = \mathbb{Z}/c$, where c is the number of transitions from one family to the other, $1 \leq c \leq \min(r, s)$; this contains the interleaving theorem for two-copy repeats. Beyond two families nothing pairwise survives: for three-copy repeats, two words with identical pairwise occurrence patterns already have groups $\mathbb{Z}/3$ and $\mathbb{Z}/4$ at $m = 3$, so no matrix with pairwise entries can present $\mathcal{K}(D_w)$; detaching a three-copy vertex into two-copy vertices does not control the group; and the literature has no interlacement formula for r -in r -out digraphs with $r \geq 3$. A census of the groups realised by three- and four-copy arrangements is given (9, 17 and 28 groups for $(3, 3, 3)$, $(4, 4, 4)$ and $(3, 3, 3, 3)$, up to $(\mathbb{Z}/3)^3$). On φ X174 the skeleton is 2-in 2-out at $k = 11, 10$, where the signed interlace matrix reproduces the group exactly, and acquires vertices of out-degree 3 to 6 from $k = 9$ down: the two-copy presentation stops exactly where the arrangement group becomes large. Four checks, all exact.

29.1 Introduction

Let S be a circular sequence with repeat families $W_1, \dots, W_m$ of multiplicities r_i , clean in the sense of [Ch. 26], and let w be the circular word of their occurrences. The bundle-chain splitting theorem [Ch. 28] gives

$$\mathcal{K}(G_k(S)) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w),$$

where D_w has one vertex per family and one arc per adjacency of occurrences. The first summand is what longer k -mers remove. The second, the *arrangement group*, is what no k removes, and for $r_i = 2$ it has a striking description: D_w is the directed medial graph of a bouquet whose edges are the repeats, and by [174, Thm. 7.3] $\mathcal{K}(D_w) \cong \text{coker}(A + I)$ with A the skew-symmetric $\{0, \pm 1\}$ matrix recording, for each pair of repeats, whether and in which order their occurrences interlace. The entries of A are pairwise data; its determinant is the interlace polynomial at 1 [172], the number of Euler circuits of D_w [175, 176].

Real genomes have repeats with more than two copies. In φ X174 the skeleton of [Ch. 28] is 2-in 2-out at $k = 11$ and $k = 10$ but has vertices of out-degree 3 and 4 at $k = 9$ and up to 6 at $k = 7$ (Table 29.2). This note asks whether the two-copy description extends. The answer is: only in the smallest case.

What is known. For 2-in 2-out digraphs the determinant formula $\# \text{Euler circuits} = \det(I + A)$ is due to Macris and Pulé [175], with elementary proofs by Lauri and Jonsson [176]; the group statement is [174]. Jonsson [176, p. 192] refers to an unpublished report generalising the intersection matrix to arbitrary Eulerian digraphs, where the determinant counts arborescences rather than circuits; we could not locate it. Lauri's own attempt at extensions [180] is described by its reviewer as inconclusive. For rotor-routing the picture is different: Tóthmérész [181] constructs a canonical simply transitive action of the sandpile group on compatible Euler tours of any Eulerian ribbon digraph, whatever the degrees, so the torsor of [Ch. 27] generalises even though, as we show, the matrix presentation does not.

29.2 Setting

D_w is the transition digraph of the circular word w : vertices the letters, one arc $e \rightarrow f$ per occurrence of e immediately followed by f around the circle; loops are deleted (they do not affect the Laplacian). It is balanced with $d^\pm(e) = r_e$. Its sandpile group is $\mathcal{K}(D_w) = \text{coker}(\Delta_s^T)$ with $\Delta = D^+ - T$, T_{ef} the number of transitions $e \rightarrow f$; the isomorphism class does not depend on the sink s [160, Lem. 4.12]. Groups are compared by elementary divisors. The *pairwise pattern* of two letters e, f is the circular binary word obtained by deleting all other letters from w , taken up to rotation.

29.3 Two families

Theorem 29.3.1 (Two families). *Let w be a circular word over $\{A, B\}$ with r copies of A and s of B , and let c be the number of transitions $A \rightarrow B$ (equivalently the number of maximal blocks of A 's). Then $1 \leq c \leq \min(r, s)$, every such c occurs, and*

$$\mathcal{K}(D_w) \cong \mathbb{Z}/c.$$

Proof. D_w has two vertices; the number of arcs $A \rightarrow B$ equals the number of arcs $B \rightarrow A$ (each block of A 's is entered once and left once), namely c , and the remaining arcs are loops. Reduced at B , the Laplacian is the 1×1 matrix (c) . The word $A^{r-c+1}B^{s-c+1}(AB)^{c-1}$ realises c , and $c \leq \min(r, s)$ since each block contains at least one letter. $\square$

For $r = s = 2$ this is the interleaving theorem [Ch. 26, Thm. 6]: $c = 1$ for $A A B B$ and $A B B A$ (disjoint or nested, no arrangement ambiguity) and $c = 2$ for $A B A B$. For two families the arrangement ambiguity is thus a single cyclic factor whose order is a crossing number, and no sign is needed. Check (A) verifies the theorem over all 119 circular words with $r, s \leq 5$.

29.4 Beyond two families nothing pairwise survives

Proposition 29.4.1 (Pairwise patterns do not determine the group). *The circular words*

$$w_1 = A A B A C C B B C, \quad w_2 = A A B A B C B C C,$$

each letter three times, have the same pairwise pattern for every pair of letters, and

$$\mathcal{K}(D_{w_1}) = \mathbb{Z}/3, \quad \mathcal{K}(D_{w_2}) = \mathbb{Z}/4.$$

Consequently no square matrix whose (e, f) entry is a function of the pairwise pattern of e and f can present $\mathcal{K}(D_w)$ for three-copy repeats, even for $m = 3$.

Proof. Restricting to $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$ gives, for both words, the cyclic patterns 001101, 000111 and 001101 (with the first letter of the pair written as 1). In w_1 all six off-diagonal transition counts are 1, so the Laplacian reduced at C is $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, of determinant 3 and content 1: $\mathbb{Z}/3$. In w_2 the transitions are $A \rightarrow B$ twice, $A \rightarrow C$ never, $B \rightarrow A$ once, $B \rightarrow C$ twice, giving $\begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$, of determinant 4 and content 1: $\mathbb{Z}/4$. $\square$

Remark 29.4.2 (Why this is sharper than the two-copy failure). For two-copy repeats the pairwise pattern of e, f is a single bit (interlaced or not), and the group is likewise not a function of these bits alone — but the first failure is at $m = 6$ [174, Rem. 6.7], and it is repaired by a sign: reading from a base point, record whether the occurrences run $efef$ or $fe fe$. The sign depends on the base point, and the cokernel does not, because changing base point acts by pivots [174, Lem. 4.8]. For three or more copies the analogous base-pointed pairwise data — the linear restriction of w to each pair — determine the whole word for every m (they fix the relative order of any two occurrences), so “the group is a function of them” is vacuous, and the only genuine pairwise invariants are the rotation classes of Proposition 29.4.1, which fail already at $m = 3$. There is no room between the two for a sign.

Proposition 29.4.3 (No reduction to two copies). *Let G be a balanced digraph with a vertex v of in- and out-degree 3, and let G' be obtained by detaching v into v' (two of the in-arcs, one of the out-arcs) and v'' (the remaining in-arc, the other two out-arcs) joined by a new arc $v' \rightarrow v''$, so that G' is 2-in 2-out at both. Then $\mathcal{K}(G)$ is in general neither a quotient nor a subgroup of $\mathcal{K}(G')$: there is a G with $\mathcal{K}(G) = \mathbb{Z}/4 \oplus \mathbb{Z}/5$ whose nine detachments have groups $\mathbb{Z}/23$, $\mathbb{Z}/25$, $\mathbb{Z}/31$, $\mathbb{Z}/32$, $\mathbb{Z}/2 \oplus \mathbb{Z}/13$, $\mathbb{Z}/3 \oplus \mathbb{Z}/8$, $\mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/4$ and $\mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/5$.*

Over 150 random balanced digraphs and all nine detachments each, $\mathcal{K}(G)$ is a quotient of $\mathcal{K}(G')$ in 140 of 1350 cases (check (C)). Euler circuits of G' are those of G compatible with the chosen transition at v , a subset of no fixed proportion, and the group follows the count. So the arrangement group of a multi-copy skeleton cannot be computed by splitting it into a two-copy one and applying [174].

Remark 29.4.4 (What does determine it). Trivially, the transition matrix T : the arrangement group is the sandpile group of the transition digraph, which is what [Ch. 28] says. The content of the two-copy theory is that T can be traded for interlacement — a different, topological parametrisation of the same group. Propositions 29.4.1 and 29.4.3 say that for three or more copies no such trade is available through pairwise data or through two-copy surrogates. A Gram-matrix route (the form $x^T(I - S)y$ of the position cycle restricted to consecutive-occurrence differences) was also tried and is not the Merino–Moffatt–Noble matrix even for two copies; it is recorded in the ancillary README so that it is not retried.

29.5 Census

Table 29.1 lists, for each multiplicity vector, the number of circular words (one per rotation class), the number of distinct transition digraphs up to relabelling, the number of distinct groups, the largest order, and the non-cyclic groups that occur. The last column counts the classes of pairwise patterns and, in parentheses, how many of them carry more than one group. For two families the groups are exactly $\mathbb{Z}/c$, $c \leq \min(r, s)$, as Theorem 29.3.1 predicts, and every pairwise class is pure. From three families on the arrangement group is far richer than for two-copy repeats — three two-copy repeats give only 0, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$ [Ch. 26, Prop. 8], three three-copy repeats give nine groups up to order 9 — and most pairwise classes are mixed.

29.6 φ X174

Table 29.2 gives the out-degree profile of the bundle-contracted skeleton of φ X174 (NC_001422.1) at $k = 12, \dots, 7$. At $k = 11$ and $k = 10$ every skeleton vertex has in- and out-degree 2, and the signed interlace matrix of an Euler circuit found by Hierholzer’s algorithm gives $\text{coker}(A + I) = \mathbb{Z}/3 \oplus \mathbb{Z}/11$ and $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$, the groups computed in [Ch. 28] from the Laplacian (check (D)) — the theorem of [174] on a real genome. From $k = 9$ down the skeleton has vertices of out-degree 3 (5 of them at $k = 9$, 42 at $k = 8$, 147 at $k = 7$), 4, 5 and 6: the two-copy presentation stops applying at the same k at which the skeleton grows from 37 to 110 vertices and the group acquires its large prime factors.

mult.	words	D_w	groups	$\max \mathcal{K} $	non-cyclic groups	pairwise (mixed)
(3, 3)	4	3	3	3	—	3 (0)
(4, 4)	10	4	4	4	—	7 (0)
(3, 3, 3)	188	12	9	9	$(\mathbb{Z}/2)^2, (\mathbb{Z}/3)^2$	15 (6)
(3, 3, 2)	70	15	7	6	$(\mathbb{Z}/2)^2$	10 (4)
(2, 2, 3)	30	10	5	4	$(\mathbb{Z}/2)^2$	6 (2)
(4, 4, 4)	2896	26	17	16	$(\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3, (\mathbb{Z}/3)^2, (\mathbb{Z}/4)^2, \mathbb{Z}/2 \oplus \mathbb{Z}/4$	174 (86)
(3, 3, 3, 3)	30804	96	28	27	those, and $(\mathbb{Z}/2)^3, (\mathbb{Z}/3)^3, \mathbb{Z}/2 \oplus (\mathbb{Z}/3)^2$	180 (132)

Table 29.1: Arrangement groups of multi-copy repeats, exhaustive over circular words (check (B)). D_w : distinct transition digraphs. All entries exact.

k	skeleton	arcs	out-degree multiset
12	1	0	(single vertex)
11	8	16	2^8
10	37	74	2^{37}
9	110	227	$2^{104} 3^5 4^1$
8	312	679	$2^{264} 3^{42} 4^5 5^1$
7	826	1945	$2^{619} 3^{147} 4^{41} 5^{12} 6^7$

Table 29.2: φ X174: size and out-degree profile of the skeleton ($d^n = n$ vertices of out-degree d). At $k = 11, 10$ the skeleton is 2-in 2-out and $\text{coker}(A + I)$ equals $\mathcal{K}(\text{Sk})$; below, multi-copy vertices appear. Exact.

29.7 Verification

The script `multicopy.py` runs the four checks of Table 29.3 in about forty seconds, all passing; its output is archived as `multicopy_output.txt`. All arithmetic is exact; groups are compared by elementary divisors.

key	check	mode
(A)	Theorem 29.3.1 on all 119 two-letter circular words, $r, s \leq 5$	exact
(B)	Table 29.1; the witness of Proposition 29.4.1	exact
(C)	Proposition 29.4.3: 150 digraphs, 1350 detachments; the witness	exact
(D)	Table 29.2; $\text{coker}(A + I) = \mathcal{K}(\text{Sk})$ at $k = 11, 10$	exact

Table 29.3: The verification battery, 4/4.

29.8 Scope

Proved. Theorem 29.3.1; Propositions 29.4.1 and 29.4.3 by explicit witness.

Quoted. The splitting theorem [Ch. 28]; the two-copy presentation [174]; the determinant formula [175, 176] and interlace polynomial [172]; sink-independence [160]; the rotor torsor for Eulerian ribbon digraphs [181].

Open. (a) Whether any presentation of $\mathcal{K}(D_w)$ for $r \geq 3$ exists whose entries are “local” in a weaker sense than pairwise — e.g. functions of the patterns of triples — and whether Jonsson’s unpublished generalised intersection matrix, which counts arborescences, has a cokernel equal to $\mathcal{K}$. Theorem 29.3.1 shows that at least for two families the group is a crossing number, so a multi-chord crossing calculus is not ruled out; Proposition 29.4.1 shows it cannot be pairwise. (b) The census suggests questions we have not pursued: which groups occur for (r, r, r) as r grows, and whether the largest order is always attained by an

elementary abelian group. (c) The φ X174 skeleton at $k = 8$ (312 vertices) is beyond the dense exact Smith form used here; a modular Smith form would extend Table 29.2 of [Ch. 28].

Provenance of this chapter

Unrefereed preprint. Theorem 29.3.1 is elementary and, as far as we can find, unstated in this form; the propositions are negative results established by explicit computation and are the reason this note exists: the natural extension of [174] to multi-copy repeats was the obvious next step after [Ch. 28], and it does not exist in any pairwise form. We searched for an interlacement or determinant formula for Euler circuits of r -in r -out digraphs with $r \geq 3$ and found none; a search cannot establish a negative.

Chapter 30

The bundle-chain decomposition of a bacterial genome: *Escherichia coli* K-12 at five k -mer lengths

Summary. The companion notes computed the sandpile group of a de Bruijn graph on a 5386 bp phage. This one computes it on a 4 641 652 bp bacterium, *Escherichia coli* str. K-12 substr. MG1655 (NC_000913.3), at $k = 31, 51, 75, 100, 150$. The computation is possible because of one observation that costs a line: in the convention used throughout this series every vertex of G_k satisfies $d^+(v) = d^-(v) = \text{mult}(v)$, so a vertex survives unitig contraction *if and only if its k -mer is repeated*. The 4.6-million-vertex graph is therefore never built; the compacted graph is read off the repeated k -mers directly, and the whole pipeline runs in about 13 seconds. Three findings. *One:* the irreducible part of the ambiguity of a 4.6 Mbp genome is a digraph on 115 vertices at $k = 150$ — the chain $4\,641\,652 \rightarrow 23\,933 \rightarrow 115$ — and its group is computed exactly. *Two:* the splitting is extremely lopsided; at $k = 150$ the length-driven summand has order 10^{10439} and the arrangement summand $10^{49.7}$, so 99.5% of the exponent is removable by longer k -mers and 0.5% is removable by no k at all, while the irreducible part still grows by 10^{344} as k falls to 31. *Three:* the seven ribosomal operons do *not* form a multiplicity-seven repeat family, at any k . Five lie on one strand and two on the other, and the single-strand theory of this series sees a repeat only between co-oriented copies; the two minus-strand operons give the longest clean spine in the genome (4030 bp at $k = 150$) and the five plus-strand ones a separate, fragmented family. This is a quantitative measure of what a bidirected theory would have to supply. Six checks, all exact, including a rerun of the whole fast pipeline against the published phage decomposition.

30.1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ is the de Bruijn multigraph whose vertices are the distinct k -mers and whose arcs are the occurrences of $(k+1)$ -mers; S is an Eulerian circuit, and by the BEST theorem the number of reconstructions of S from its k -mer spectrum is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$ with $\mathcal{K}(G_k)$ the sandpile group [Ch. 26]. The bundle-chain splitting theorem [Ch. 28] writes

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}),$$

a *length-driven* summand that longer k -mers remove and an *arrangement* summand, the sandpile group of the repeat-adjacency skeleton, that no k removes. Everything so far has been computed on φ X174, 5386 bp.

The question this note answers is what the decomposition looks like on a genome three orders of magnitude larger.

What is known. Assemblers have compacted de Bruijn graphs to unitigs for twenty years [179, 163, 170], and the repeat graph is the central object of the long-read assembler Flye [178, 171]; the repeat content of *E. coli* K-12 — seven *rrn* operons, some fifty IS elements in a dozen families, several hundred REP/BIME palindromic elements — has been catalogued since the genome was published [182]. What is new here is not the repeat catalogue but the algebra attached to it: the exact invariant factors of the arrangement group, and the exact split between the part of the ambiguity that a longer read length removes and the part it cannot.

30.2 Why a 4.6 Mbp genome is tractable

Lemma 30.2.1 (Branch vertices are the repeated k -mers). *In $G_k(S)$, $d^+(v) = d^-(v) = \text{mult}(v)$, the number of occurrences of the k -mer v in the circular sequence. Consequently a vertex has in- and out-degree one — i.e. is smoothed away by the unitig contraction of [Ch. 26, Lem. 4] — if and only if its k -mer occurs exactly once.*

Proof. Each position i contributes exactly one arc, from $S[i..i+k)$ to $S[i+1..i+k+1)$. The arcs leaving v are indexed by the positions at which v occurs, and the arcs entering v by the positions at which v occurs shifted by one; both sets have $\text{mult}(v)$ elements. $\square$

This is immediate, and it is the whole algorithm. The compacted graph is obtained without constructing G_k : let $R = (p_1 < \dots < p_m)$ be the positions at which a repeated k -mer starts. Then the compacted multidigraph has one vertex per distinct repeated k -mer and one arc $S[p_t..p_t+k) \rightarrow S[p_{t+1}..p_{t+1}+k)$ for each t cyclically, because the walk leaving a branch vertex runs through unique k -mers until the next repeat position. For *E. coli* this replaces 4 641 652 vertices by between 23 933 and 34 272 (Table 30.1), and the decomposition of [Ch. 28] then runs in one pass, since the chain-step relation is a partial injection [Ch. 28, Lem. 6].

Finding the repeated k -mers is the only step that touches the whole genome. We hash each of the N positions to 64 bits, take multiplicities with a sort, and then — this is what keeps the result exact — recover the actual k -mer *strings* at the positions whose hash repeats and group by string. A hash collision can therefore only add a candidate that is discarded; it cannot manufacture a repeat. No collision occurred at any k (check (H)).

30.3 Results

k	distinct k -mers	branch	arcs	chains	skeleton	$\log_{10} \mathcal{K}(\text{Sk}) $	$\log_{10} \mathcal{K}(G_k) $
150	4 594 348	23 933	71 237	107	115	49.71	10 488.89
100	4 588 709	26 631	79 574	166	175	71.11	11 690.57
75	4 584 636	28 579	85 595	243	261	99.80	12 563.08
51	4 579 164	31 011	93 499	342	366	143.17	13 716.83
31	4 570 869	34 272	105 055	683	870	393.75	15 350.54

Table 30.1: *E. coli* K-12 MG1655 under unitig contraction followed by bundle-chain contraction. “branch” is the number of repeated k -mers (Lemma 30.2.1) and equals the vertex count of the compacted graph. All structural entries exact; $\log_{10} |\mathcal{K}(\text{Sk})|$ by float LU, and exact as a group at $k = 150, 100$ (Table 30.2).

k	$\mathcal{K}(\text{Sk})$, the arrangement group
150	$(\mathbb{Z}/2)^{17} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/4)^{12} \oplus (\mathbb{Z}/5)^{10} \oplus (\mathbb{Z}/7)^3 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^3 \oplus (\mathbb{Z}/16)^3$ $\oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/21308303$
100	$(\mathbb{Z}/2)^{29} \oplus (\mathbb{Z}/3)^{16} \oplus (\mathbb{Z}/4)^{15} \oplus (\mathbb{Z}/5)^{15} \oplus (\mathbb{Z}/7)^2 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^2 \oplus (\mathbb{Z}/16)^6$ $\oplus \mathbb{Z}/25 \oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/81 \oplus \mathbb{Z}/128 \oplus \mathbb{Z}/512 \oplus \mathbb{Z}/3448649$

Table 30.2: The arrangement group, exactly, where the skeleton is small enough for the p -adic Smith form (check (K)). At $k = 75, 51, 31$ the skeleton has 261, 366 and 870 vertices and only the order is given; a 100- to 400-digit determinant could not be factored in any case.

Remark 30.3.1 (The shape of the answer). Three things are worth reading off Table 30.1.

The skeleton is small. At $k = 150$ the entire arrangement ambiguity of a 4.6 Mbp genome is carried by a digraph on 115 vertices: $4641652 \rightarrow 23933$ by unitig contraction, $23933 \rightarrow 115$ by bundle-chain contraction. The second step is the one the theory supplies, and it is the larger reduction in proportion.

The split is lopsided. At $k = 150$ the length-driven summand has order $10^{10439.18}$ and the arrangement summand $10^{49.71}$: 99.53% of the exponent of $|\mathcal{K}(G_{150})|$ is removable by reading longer k -mers, and 0.47% is not removable by any k . The percentage is not the interesting part — $10^{49.7}$ is still an enormous number of globally distinct threadings, and it is the part that only long reads, mate pairs or a reference can resolve.

The irreducible part grows fast as k falls. From $k = 150$ to $k = 31$ the arrangement summand grows from $10^{49.71}$ to $10^{393.75}$, a factor of 10^{344} , while the skeleton grows from 115 to 870 vertices. Shortening the read length does not merely leave more length-driven ambiguity; it creates genuinely new arrangement ambiguity, because repeats shorter than the old k enter the graph and connect parts of the skeleton that were previously unrelated.

By the BEST theorem the number of circular sequences with exactly the k -mer spectrum of this genome is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$, which is $10^{20735.14}$ at $k = 150$ and $10^{31748.76}$ at $k = 31$.

30.4 The ribosomal operons, and a correction

E. coli K-12 has seven ribosomal RNA operons, each about 5 kb, and the natural expectation — the one the task that prompted this note recorded — is a repeat family of multiplicity seven. *No such family exists, at any k .*

The reason is structural, not numerical. The NCBI feature table places five operons on the + strand (at 223 771, 3 941 808, 4 035 531, 4 166 659, 4 208 147) and two on the – strand (at 2 726 069 and 3 423 423). The theory of [Chs. 26, 27, 28, 29] is single-strand: two copies are a repeat of $G_k(S)$ only if they are co-oriented in S . The seven operons therefore cannot form one family. They split by strand, and the computation confirms the split exactly (check (R)):

- the **longest bundle chain in the whole genome**, at every k tested, is the multiplicity-*two* spine of the two minus-strand operons — $s = 3881$ at $k = 150$ (span 4030 bp), $s = 4412$ at $k = 51$ (span 4462 bp) — annotated 16S, 23S and 5S ribosomal RNA. Two copies annotated on the same (minus) strand are co-oriented with each other, hence a direct repeat in S ;
- the five plus-strand operons give the multiplicity-*five* family, whose longest chain is the 16S gene (span 775 bp), with copies at 223 997, 3 942 034, 4 035 757, 4 166 885 and 4 208 373 — all five operons;
- the five are not identical to one another either, so the rRNA region additionally throws off chains of multiplicity three and four, one per subset of copies that agree.

Multiplicity seven does occur in this genome, but it is IS5A, not rRNA (five chains, 88 k -mers at $k = 31$), as are multiplicities eight and nine. The other long chains are all insertion sequences: IS186A ($\times 3$, 1345 bp), IS2D ($\times 3$, 1331 bp), IS3B ($\times 2$, 1260 bp), IS3A ($\times 3$, 1255 bp), IS5U ($\times 3$, 1202 bp). At $k = 31$ the highest multiplicities are 22 and 23 with spans of 32 to 35 bp: the REP/BIME extragenic palindromic elements, which are shorter than 51 and so vanish completely from the graph at every larger k tested.

Remark 30.4.1 (What this measures). A bidirected theory — the open problem of [Ch. 27, Open (a)] — would see all seven operons as one family, and would see the IS copies that sit in inverted orientation. The $5 + 2$ split is a quantitative statement of what the single-strand restriction costs on a real genome: not a footnote, but the dominant feature of the largest repeat family present. We do not have the bidirected theory and claim nothing about what it would give; we can now say precisely what it would have to explain.

30.5 Verification

The script `ecoli_sandpile.py` runs the six checks of Table 30.3 in about 13 seconds under half a gigabyte; its output is archived as `ecoli_sandpile_output.txt`. The genome and the feature table are cached beside it and re-fetched from the NCBI E-utilities API if absent.

key	check	mode
(V)	the fast path reproduces the published φ X174 decomposition, $k = 9, \dots, 12$	exact
(H)	hashing finds candidates, strings decide; no collision survives	exact
(C)	the compacted graph is balanced with $d^+(v) = \text{mult}(v)$	exact
(B)	maximal bundle chains in one pass; Lemma 6 injectivity asserted on 1541 chains	exact
(K)	the arrangement group, exactly for skeletons of ≤ 200 vertices	exact/numeric
(R)	no rRNA chain has multiplicity 7; the longest chain is the minus-strand pair	exact

Table 30.3: The verification battery, 6/6.

Check (V) deserves a word. The pipeline used here is not the one used in [Ch. 28]: it never builds G_k , it contracts chains in a single pass instead of iteratively, and it takes the Smith form p -adically instead of by Euclidean elimination. Run on φ X174 it returns, at $k = 9, 10, 11, 12$, the same branch counts (2, 10, 47, 146), chain counts (1, 2, 8, 28), skeleton sizes (1, 8, 37, 110), length-driven exponents and the same exact skeleton groups — including $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$ at $k = 10$ and the seventeen elementary divisors at $k = 9$. Three independent implementations of the same theorem agreeing on a genome is the only protection available when the target genome is too large to check by any other route.

30.6 Scope

Exact. Every structural quantity — repeated k -mers, branch counts, the multiset of (r_c, ℓ_c) , skeleton sizes, genomic coordinates, the length-driven summand — is exact integer work, as is $\mathcal{K}(\text{Sk})$ at $k = 150$ and $k = 100$ and the location of every repeat family against the feature table.

Numeric. $\log_{10} |\mathcal{K}(\text{Sk})|$ at $k = 75, 51, 31$ is a float LU determinant (`slogdet`) of a 260- to 869-dimensional matrix; the relative error in the logarithm is at the level of machine precision times the dimension, but it is not an exact integer and is not claimed to be.

Single-strand. As in [Chs. 26, 27, 28, 29]. Section 30.4 is the measurement of what that costs here.

Open. (a) The two exact arrangement groups each carry one large prime factor (21 308 303 at $k = 150$, 3 448 649 at $k = 100$) that we do not interpret; as in [Ch. 28] that is what the sandpile group of a generic digraph looks like, and no topological reading is claimed. (b) The 870-vertex skeleton at $k = 31$ is out of exact reach; a modular Smith form would settle it, and nothing in principle prevents that. (c) Whether the 10^{344} growth of the arrangement summand between $k = 150$ and $k = 31$ has a law in terms of the repeat-length distribution.

Provenance of this chapter

Unrefereed preprint. Lemma 30.2.1 is elementary and is stated because it is what makes the computation possible, not because it is deep; everything else in this note is computation on public data. The genome is NC_000913.3 and the annotation is the NCBI feature table for the same accession, both downloaded by

the script and cached with it. The expectation of a multiplicity-seven rRNA family, which Section 30.4 corrects, was the premise of the task that prompted this note, and is recorded here because the correction is the useful part.

Chapter 31

Double strands without a new lemma: the reverse-complement double cover of a de Bruijn graph

Summary. The single-strand theory of the companion notes sees a repeat only between co-oriented copies, so the seven ribosomal operons of *E. coli* — five on one strand, two on the other — appear as a five-copy family and a two-copy family and never as a seven-copy one. Repairing this was expected to need a sandpile theory for bidirected graphs, which does not exist. It does not need one. Let $D_k(S)$ be the multidigraph whose arcs are the $(k+1)$ -mer occurrences in S and in its reverse complement $\bar{S}$. Then $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$ and $d^+(v) = d^-(v) = \text{mult}_D(v)$: D_k is balanced, so the unitig lemma, the bundle-chain lemma and the splitting theorem apply to it verbatim. Three consequences. *One*: multiplicities add across strands, so a family with a forward and b reverse copies has multiplicity $a + b$ in D_k ; for *E. coli* the longest multiplicity-seven chain has span 728 bp and its seven occurrences are the seven operons, one each, the two minus-strand ones on the reverse strand of D_k . *Two*: D_k is connected exactly when S contains an inverted repeat of length $\geq k$, so the inverted repeats are precisely what glues the strands; ϕ X174 already qualifies at $k = 12$, through ATTAAGCTCATT at 441 and its reverse complement at 1448. *Three*: the involution $\rho(v) = \bar{v}$ reverses arcs, hence carries D_k to its opposite; since $\Delta^{\text{op}} = \Delta^T$ for a balanced digraph and the two Bowen–Franks cokernels are dual, ρ dualises $\mathcal{K}(D_k)$ instead of splitting it — which is what the warning against ± 1 -eigenspace decompositions was pointing at. The phage’s celebrated single global bit is a single-strand phenomenon: $\mathcal{K}(G_{12}) = \mathbb{Z}/2$ but $\mathcal{K}(D_{12}) = \mathbb{Z}/33$. Six checks, all exact. What is *not* done: D_k is the double cover, not the bidirected quotient, and $\mathcal{K}(D_k)$ does not count double-stranded reconstructions; that problem is left exactly where it was.

31.1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ has the distinct k -mers as vertices and the occurrences of $(k+1)$ -mers as arcs, and $\mathcal{K}(G_k(S)) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk})$ by the bundle-chain splitting theorem [Ch. 28]. The theory is single-strand throughout [Chs. 26, 27, 28, 29], and [Ch. 30] measured what that costs on a real genome: of the seven *rrn* operons of *E. coli* K-12, five lie on the $+$ strand and two on the $-$, and since a repeat of $G_k(S)$ requires co-oriented copies, the operons split into a multiplicity-five family and a multiplicity-two family, with multiplicity seven occurring nowhere.

The obvious repair is to work with the bidirected graph that assemblers actually build, whose vertices are the classes $\{v, \bar{v}\}$; [Chs. 26, 27] left this open and warned against the tempting shortcut of decomposing $\mathcal{K}$ into ± 1 -eigenspaces of the reverse-complement involution, since an involution splits an abelian group

only after 2 is inverted and the 2-torsion is where the interesting phenomena live. That warning stands. What this note observes is that most of what one wants from the bidirected picture is already available, without any new lemma, from a balanced digraph.

31.2 The double cover

Definition 31.2.1 (Reverse-complement double cover). Let $\bar{S}$ be the reverse complement of S , again circular of length N . Let $D_k = D_k(S)$ be the multidigraph whose vertices are the k -mers occurring in S or in $\bar{S}$ and whose arcs are the occurrences of $(k+1)$ -mers in S together with those in $\bar{S}$; it has $2N$ arcs. Write $\rho(v) = \bar{v}$.

Lemma 31.2.2 (Balance). $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$ and $d^+(v) = d^-(v) = \text{mult}_D(v)$ for every vertex. Hence D_k is a balanced multidigraph.

Proof. $\text{mult}_{\bar{S}}(v) = \text{mult}_S(\bar{v})$, because a copy of v in $\bar{S}$ is a copy of $\bar{v}$ in S ; the first formula follows. In each of the two circular sequences separately, every occurrence of v contributes one arc out of v and one arc into v , as in [Ch. 26, Lem. 4]; summing the two gives the second. $\square$

Corollary 31.2.3 (Nothing new is needed). *The unitig contraction lemma [Ch. 26, Lem. 4], the bundle-chain lemma [Ch. 28, Lem. 2], the canonicity of the skeleton [Ch. 28, Lem. 6] and the splitting theorem [Ch. 28, Thm. 5] apply to D_k verbatim, and a vertex of D_k survives unitig contraction if and only if $\text{mult}_S(v) + \text{mult}_S(\bar{v}) \geq 2$ [Ch. 30, Lem. 2.1].*

That last clause is the practical point: the branch vertices of D_k are the k -mers that are repeated *when both strands are counted*, so the entire fast pipeline of [Ch. 30] runs on D_k unchanged, with the positions of $\bar{S}$ appended to those of S .

Proposition 31.2.4 (The strands are glued by the inverted repeats). *D_k is connected if and only if some k -mer w occurs in S together with $\bar{w}$; otherwise $D_k = G_k(S) \sqcup G_k(\bar{S})$, two isomorphic components.*

Proof. The arcs coming from S form a closed walk through every k -mer of S , so those vertices lie in one component, and likewise for $\bar{S}$; the two vertex sets are the k -mers of S and their reverse complements. They meet exactly when some w and $\bar{w}$ both occur in S . $\square$

So the double cover is interesting precisely to the extent that the genome has inverted repeats of length at least k . Both of our genomes qualify: ϕ X174 at $k = 12$ through ATTAAGCTCATT at position 441, whose reverse complement sits at 1448; *E. coli* at $k = 150$ through a 150-mer at 19 796 whose reverse complement is at 3 584 046 (check (C)).

31.3 Multiplicities add across strands

Theorem 31.3.1 (Strand merging). *Let W be a sequence occurring a times on the forward strand of S and b times on the reverse, with $a + b \geq 1$ and W not equal to $\bar{W}$. In $G_k(S)$ the k -mers of W have multiplicity a and those of $\bar{W}$ multiplicity b . In D_k both have multiplicity $a + b$, and the two families are exchanged by ρ . If, moreover, the $a + b$ copies agree over a window of $s + k - 1$ letters, that window contributes a bundle chain of multiplicity $a + b$ and length s to D_k , hence a summand $(\mathbb{Z}/(a + b))^{s-1}$ to $\mathcal{K}(D_k)$, and another from its ρ -image.*

Proof. A reverse-strand copy of W is a forward occurrence of $\bar{W}$, so $\text{mult}_S(w) = a$ and $\text{mult}_S(\bar{w}) = b$ for a k -mer w of W in general position; by Lemma 31.2.2, $\text{mult}_D(w) = a + b$, and symmetrically for $\bar{w}$. The last sentence is Lemma 31.2.2 followed by [Ch. 28, Lem. 2] applied to the balanced digraph D_k . $\square$

Corollary 31.3.2 (The seven operons). *For E. coli K-12 the seven rrn operons carry, in D_k , bundle chains of multiplicity seven. At $k = 150, 100, 51$ the longest such chain has span 728 bp and its seven occurrences are the seven operons, one each:*

$$\begin{aligned} &+224\,044 \rightarrow 223\,771, & -2\,730\,735 \rightarrow 2\,726\,069, & -3\,428\,340 \rightarrow 3\,423\,423, \\ &+3\,942\,081 \rightarrow 3\,941\,808, & +4\,035\,804 \rightarrow 4\,035\,531, & +4\,166\,932 \rightarrow 4\,166\,659, \\ &+4\,208\,420 \rightarrow 4\,208\,147, \end{aligned}$$

the sign giving the strand of D_k on which the occurrence lies and the arrow pointing to the operon it meets. The multiplicity-seven chains carry more k -mers than the chains of any other multiplicity over the operons: 4154 at $k = 150$, 5254 at $k = 100$, 6660 at $k = 51$ (check (M)).

Remark 31.3.3 (What was wrong with $5 + 2$, exactly). Nothing was wrong with it; it was the correct answer to a single-strand question. The content of Corollary 31.3.2 is that the question has a balanced-digraph answer too, and that the object which answers it is not the bidirected quotient but the double cover, on which the existing theorems already hold. The residual multiplicities 2, 3, 4, 5, 6 that still appear over the operons in D_k are the sub-families on which only some of the seven copies agree — the operons are not identical to one another, on either strand.

31.4 The involution dualises, it does not split

Proposition 31.4.1 (ρ and the opposite digraph). *ρ is an involution of the vertex set of D_k carrying the arc for a $(k+1)$ -mer x to the arc for $\bar{x}$ with its direction reversed. Hence ρ is an isomorphism $D_k \rightarrow D_k^{\text{op}}$, and it induces an isomorphism*

$$\mathcal{K}(D_k) \xrightarrow{\sim} \mathcal{K}(D_k)^\vee$$

onto the Pontryagin dual, i.e. a perfect pairing on $\mathcal{K}(D_k)$.

Proof. The prefix of $\bar{x}$ is the reverse complement of the suffix of x and vice versa, so the arc of $\bar{x}$ runs $\rho(\text{suffix}) \rightarrow \rho(\text{prefix})$: direction reverses, and ρ is an isomorphism onto the opposite digraph. For a balanced digraph $\Delta^{\text{op}} = D^+ - A^\top = \Delta^\top$, so $\mathcal{K}(D_k^{\text{op}}) = \text{Tor coker } \Delta_s$ while $\mathcal{K}(D_k) = \text{Tor coker } \Delta_s^\top$, and these two are canonically dual by the Bowen–Franks pairing [183, Lem. 1.5]. Composing gives the stated isomorphism. $\square$

Remark 31.4.2 (Against the eigenspace decomposition). [Ch. 26, Open (c)] warned that $\mathcal{K}$ should not be split into ± 1 -eigenspaces of ρ , because that discards the 2-torsion. Proposition 31.4.1 is what happens instead: an involution that reverses arcs cannot act on $\mathcal{K}(D_k)$ at all — it does not preserve the presentation — but it identifies $\mathcal{K}(D_k)$ with its dual. No inversion of 2 occurs anywhere, and the 2-torsion is untouched. We do not determine whether the resulting pairing is symmetric or alternating; that question, and whether it is independent of the choice of sink, are left open.

31.5 Computations

k	graph	branch	chains	skeleton	$\mathcal{K}(\text{Sk})$	length-driven
12	G_k	2	1	1	0	$\mathbb{Z}/2$
	D_k	7	0	7	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	0
11	G_k	10	2	8	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	$(\mathbb{Z}/2)^2$
	D_k	30	7	23	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/49451$	$(\mathbb{Z}/2)^7$
10	G_k	47	8	37	$\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$	$(\mathbb{Z}/2)^{10}$
	D_k	125	23	97	$(\mathbb{Z}/4)^2 \oplus \mathbb{Z}/79624452132275189516228869$	$(\mathbb{Z}/2)^{28}$

Table 31.1: φX174 : the single-strand graph and its double cover. All entries exact (check (K)); the single-strand rows reproduce [Ch. 28, Tab. 1].

Remark 31.5.1 (The phage’s single bit is single-strand). $\mathcal{K}(G_{12}) = \mathbb{Z}/2$: the 12-mer spectrum of φ X174 leaves exactly two global reconstructions, and [Ch. 27] located the bit as a pair of interleaved 12-mers. On the double-stranded spectrum the answer is different: $\mathcal{K}(D_{12}) = \mathbb{Z}/3 \oplus \mathbb{Z}/11 = \mathbb{Z}/33$, and it is entirely arrangement — D_{12} has no bundle chain at all, so the length-driven summand is trivial. The two statements do not conflict; they answer different questions, and the contrast is a fair measure of how much of the single-strand picture is an artefact of ignoring one strand.

k	branch	chains	skeleton	$\log_{10}$ length-driven	$\log_{10} \mathcal{K}(\text{Sk}) $	$\log_{10} \mathcal{K}(D_k) $
150	39 768	278	312	20 987.91	156.51	21 144.42
100	44 808	396	434	23 394.31	208.92	23 603.24
51	53 380	798	886	27 405.96	395.80	27 801.75

Table 31.2: *E. coli* K-12 MG1655, the double cover. Structural entries exact; $\log_{10} |\mathcal{K}(\text{Sk})|$ by float LU. Compare [Ch. 30, Tab. 1]: on one strand the same k give skeletons of 115, 175 and 366 vertices and arrangement exponents 49.71, 71.11 and 143.17.

Two readings of Table 31.2. The skeleton roughly triples ($115 \rightarrow 312$ at $k = 150$) and the arrangement exponent does too ($49.71 \rightarrow 156.51$): inverted repeats are a large part of the irreducible ambiguity of this genome and the single-strand theory sees none of them. And $\log_{10} |\mathcal{K}(D_k)|$ exceeds twice the single-strand value by 166.6, 222.1 and 368.1 at $k = 150, 100, 51$ — a crude but honest measure of what double-strandedness adds beyond simply having two strands to reconstruct.

31.6 Verification

key	check	mode
(B)	Lemma 31.2.2: D_k balanced, multiplicities add across strands	exact
(C)	Proposition 31.2.4: connected iff an inverted repeat of length $\geq k$, with witness	exact
(R)	Proposition 31.4.1: ρ is an involution with $A(\rho u, \rho v) = A(v, u)$	exact
(M)	Corollary 31.3.2: the multiplicity-seven chains and their seven operons	exact
(K)	Table 31.1: $\mathcal{K}(D_k)$ exactly, beside $\mathcal{K}(G_k)$	exact
(E)	Table 31.2	exact/numeric

Table 31.3: The verification battery (`rc_double_cover.py`), 6/6, about 33 seconds.

31.7 Scope

Proved. Lemma 31.2.2, Corollary 31.2.3, Proposition 31.2.4, Theorem 31.3.1, Corollary 31.3.2 and Proposition 31.4.1. All are short; the work is in noticing that the double cover is balanced and that this is enough.

What this is not. D_k is the *double cover*, not the bidirected quotient. The two are different objects and only the first is a digraph. In particular:

- $\mathcal{K}(D_k)$ does *not* count double-stranded reconstructions. By BEST, $|\mathcal{K}(D_k)| \prod_v (d^+(v) - 1)!$ is the number of arc-rooted Eulerian circuits of D_k , and such a circuit traverses all $2N$ arcs in one closed walk, which is not a genome. A double-stranded circular molecule corresponds instead to a decomposition of D_k into two ρ -conjugate circuits. Counting those is not solved here.
- No sandpile theory for the bidirected quotient is offered. The tools [153] named in [Chs. 26, 27] remain the place to start, and the obstruction noted there — that reverse complementation reverses arc direction, so the quotient map is not a covering of digraphs and [151] does not apply — is untouched.

- The pairing of Proposition 31.4.1 is not identified as symmetric or alternating, and we have not checked that it is independent of the sink.

Numeric. $\log_{10} |\mathcal{K}(\text{Sk})|$ in Table 31.2 is a float LU determinant of a 311- to 885-dimensional matrix; everything else is exact integer work.

Open. (a) The two questions just listed. (b) Whether Theorem 31.3.1 has a converse: is every multiplicity- r family of D_k a strand-split family of S , and can a and b be recovered from D_k alone? (c) The 26-digit prime in Table 31.1 we do not interpret, as in [Ch. 28].

Provenance of this chapter

Unrefereed preprint. Every statement above is proved in full and every number is produced by `rc_double_cover.py` from the two public genomes. The task that prompted this note asked for a bidirected bundle-chain lemma yielding summands $(\mathbb{Z}/2r)^{s-1}$; no such lemma is needed for the double cover, and Theorem 31.3.1 shows the correct multiplicity is $a + b$, the number of copies on both strands together, not $2r$. The same task asked for “the algebraic proof that the seven operons participate in the skeleton as a single seven-copy family”; that is Corollary 31.3.2, with the caveat of Remark 31.3.3 — they do so in the double cover, and there are two such families exchanged by ρ , not one.

Chapter 32

The reverse-complement quotient is a signed graph: what Zaslavsky's theory gives, and what it cannot

Summary. The companion note built the reverse-complement double cover $D_k(S)$ of a de Bruijn multidigraph and found that the involution $\rho(v) = \bar{v}$ carries D_k to its *opposite* digraph, so that ρ induces no endomorphism of $\mathcal{K}(D_k)$ and the cover does not descend. The obvious repair is to pass to the quotient $\Sigma_k = D_k/\rho$ and to use Zaslavsky's signed Laplacian there. Half of that works, and the half that fails is the interesting half. *Works:* ρ is free exactly when no repeated k -mer is its own reverse complement, which is automatic for odd k and genuinely fails for even k (ϕ X174 has three palindromic repeated 10-mers and one palindromic repeated 12-mer); every bundle of Σ_k is monochromatic, so Σ_k carries honest r -bundles; Σ_k is *unbalanced* precisely when S carries an inverted repeat of length $\geq k$, which is Zaslavsky balance read through the connectivity criterion of the companion note; and the Reiner–Tseng sequence $0 \rightarrow \mathcal{K}(\Sigma_k) \rightarrow \mathcal{K}(|D_k|) \rightarrow \mathcal{K}(|\Sigma_k|) \rightarrow 0$ holds on the nose, with a non-split 2-primary part. *Fails, twice.* First, if P is the permutation matrix of ρ then $P\Delta(D_k)P = \Delta(D_k)^\top$ and never $\Delta(D_k)$: the deck involution is an *anti*-automorphism of the Laplacian, the ± 1 eigenlattices of P are not Δ -stable, and the *directed* sandpile group — the one the BEST theorem counts reconstructions with — lies in no covering sequence. What it gives instead is a perfect *symmetric* linking form $\mathcal{K}(D_k) \times \mathcal{K}(D_k) \rightarrow \mathbb{Q}/\mathbb{Z}$, so the relevant notion is a metabolizer rather than a Lagrangian; and since $|\mathcal{K}(D_{11})| = 25\,318\,912$ is not a square, ϕ X174 has none. Second, a signed r -bundle chain does not contract: the presentation column at the chain's endpoint differs between Σ and Σ/C by exactly $sr d_0$, the series law, so the splitting theorem that makes the genome-scale computation feasible on one strand has no bidirected analogue. What does survive is the r -rank. The seven ribosomal operons of *E. coli* K-12 merge into genuine 7-bundles of Σ_k whose longest block spans 728 bp at $k = 31, 51$ and 75 alike — the same block the companion note matched one-to-one onto the seven operons — and the only trace the quotient keeps of them is the $\mathbb{F}_7$ -nullity of the signed Laplacian.

32.1 What was asked, and what is true

The single-strand theory of these notes assigns to a sequence S and a word length k the de Bruijn multidigraph $G_k(S)$, whose vertices are the repeated k -mers and whose arcs are the $(k+1)$ -mer occurrences, and reads the reconstruction count off the critical group $\mathcal{K}(G_k)$ through the BEST theorem. Double-strandedness was handled in the companion note by passing to the *double cover* $D_k(S)$: the multidigraph whose arcs are the $(k+1)$ -mer occurrences in S and in $\bar{S}$. That graph is balanced, the unitig lemma and the splitting theorem

k	parity	$ V(D_k) $	ρ -fixed	A_σ symmetric
9	odd	396	0	yes
10	even	125	3	no
11	odd	30	0	yes
12	even	7	1	no

Table 32.1: ϕ X174 (NC_001422.1). Palindromic repeated k -mers exist at $k = 10$ and $k = 12$, and at exactly those k the quotient adjacency matrix fails to be symmetric, so no signed graph is defined. Check (P).

apply to it verbatim, and the seven ribosomal operons of *E. coli* — five on one strand, two on the other — finally appear as one seven-copy family instead of a five-copy and a two-copy family.

What the companion note could not do was quotient. The map $\rho(v) = \bar{v}$ satisfies

$$\rho: D_k \longrightarrow D_k^{\text{op}}, \quad \rho(u \rightarrow v) = \bar{v} \rightarrow \bar{u},$$

so it reverses arcs; the quotient of a digraph by a direction-reversing involution is not a digraph, and ρ induces a map $\mathcal{K}(D_k) \rightarrow \mathcal{K}(D_k^{\text{op}}) \cong \mathcal{K}(D_k)^\vee$ rather than an endomorphism. The standard repair in combinatorics is to forget the directions, to record instead a *sign* on each edge of the quotient, and to work with Zaslavsky’s signed Laplacian $\Delta_\Sigma = D - A_\sigma$ [190]. This note carries that out, and reports both what it yields and the two places where it provably cannot yield what was wanted.

The one-paragraph summary is this. The quotient is a bona fide signed graph, its critical group is a switching invariant of the genome, its balance has an exact biological meaning, and the integral covering theory of Reiner and Tseng [189] applies to it word for word. But that theory computes an invariant of the *undirected shadow* $|D_k|$, which by the matrix-tree theorem counts spanning trees; the quantity of interest counts Eulerian circuits and lives on the digraph, and Theorem 32.5.1 says the digraph admits no such theory. Separately, Theorem 32.6.1 says the chain contraction that reduces a bacterial genome from tens of thousands of vertices to a few hundred has no signed analogue. Together these explain why the “unified bidirected sandpile group” that one would like to write down does not exist, and what the correct residue of the seven operons is instead.

Throughout, S is a circular sequence of length N over $\{A, C, G, T\}$, $\bar{S}$ its reverse complement, and $\bar{w}$ the reverse complement of a word w . All computations are exact integer arithmetic; the script `bidirected_sandpile.py` reproduces every number below and prints a seven-line pass/fail battery.

32.2 The quotient, and when it exists

Definition 32.2.1. $D_k = D_k(S)$ is the multidigraph of the companion note: one vertex per k -mer occurring at least twice in the concatenation $S\bar{S}$, one arc $u \rightarrow v$ per occurrence of a $(k+1)$ -mer w with prefix u and suffix v in S or in $\bar{S}$. The involution ρ sends the vertex w to $\bar{w}$ and the arc $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$.

The first question is whether ρ acts freely on vertices, because the whole signed-graph dictionary presupposes a free $\mathbb{Z}/2$ action.

Proposition 32.2.2. ρ fixes a vertex of D_k if and only if some repeated k -mer w satisfies $\bar{w} = w$. For odd k no such word exists over any alphabet with a fixed-point-free complement, so ρ is free. For even k such words exist and do occur.

Proof. A word w of odd length k with $\bar{w} = w$ would have its middle letter equal to its own complement, and complementation on $\{A, C, G, T\}$ has no fixed point. For even k the condition $\bar{w} = w$ is the condition that w be a palindromic restriction-site-like word, and these are common. $\square$

Table 32.1 makes the point concretely and shows that the obstruction is not hypothetical: it is precisely at the even k with palindromic repeats that the construction below breaks, and it breaks visibly, in the symmetry of the adjacency matrix. From here on k is odd, or even with no palindromic repeated k -mer.

Remark 32.2.3 (What the fixed points are, topologically). The failure at even k is not a technicality but a change of category. A vertex $w = \bar{w}$ is a branch point of the projection $D_k \rightarrow D_k/\rho$, so the quotient is not a covering space and the fibre over w is a single point rather than two. In Zaslavsky's language the image acquires a *half-edge* — an edge with one endpoint — at w : an arc $u \rightarrow w$ and its partner $w \rightarrow \bar{u}$ descend to the same edge of the quotient but attach to $[w]$ with opposite end-signs, since $\varepsilon(w) = +1$ forces the two signs $\varepsilon(u)\varepsilon(w)$ and $\varepsilon(w)\varepsilon(\bar{u})$ to differ. That is exactly why the quotient adjacency matrix in Table 32.1 fails to be symmetric: A_σ records one sign per edge, and a half-edge has two. Reiner and Tseng do treat signed graphs with half-edges [189, §9], so the object is not outside the theory, but its critical group is no longer the one computed below, and the double cover it belongs to is branched. Biologically the fixed points are the palindromic restriction-site-like words, so the branch locus of the quotient is precisely the repeated restriction sites of the genome.

Definition 32.2.4. Choose a representative of each ρ -orbit $\{v, \bar{v}\}$ and set $\varepsilon(v) = +1$ if v is the chosen one and -1 otherwise. The signed graph $\Sigma_k = D_k/\rho$ has one vertex per ρ -orbit of vertices, one edge per ρ -orbit of arcs, and the edge coming from $u \rightarrow v$ carries the sign $\varepsilon(u)\varepsilon(v)$.

The arcs do fall into orbits of size two, and here no parity hypothesis is needed. An arc of D_k is an *occurrence*: a position in S or a position in $\bar{S}$. The involution ρ carries occurrences in S to occurrences in $\bar{S}$ and back, and no occurrence lies in both, so ρ is free on arcs outright. The script nevertheless constructs the pairing explicitly and fails loudly if any arc lacks a partner.

Two facts make Definition 32.2.4 usable.

Lemma 32.2.5. *Every bundle of parallel edges of Σ_k is monochromatic.*

Proof. The sign $\varepsilon(u)\varepsilon(v)$ depends only on the endpoints of the arc, not on the arc. All r arcs $u \rightarrow v$ therefore descend to r edges of the same sign. $\square$

This is what licenses the phrase “ r -bundle” below, and it is not automatic for a general signed multigraph.

Lemma 32.2.6. *Changing the choice of representatives is a Zaslavsky switching, and Δ_{Σ_k} changes by conjugation by a diagonal ± 1 matrix. Hence $\text{coker } \Delta_{\Sigma_k}$, and with it the critical group of Section 32.4, depends only on (S, k) .*

Replacing the representative at one orbit multiplies ε by -1 there, which multiplies the corresponding row and column of A_σ by -1 and leaves D alone. The script verifies invariance directly on eight random representative sets at $k = 9$ and $k = 11$, obtaining for instance

$$\mathcal{K}(\Sigma_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481$$

every time. Check (W).

32.3 Balance is an inverted repeat

Theorem 32.3.1. *Σ_k is unbalanced if and only if S carries an inverted repeat of length at least k .*

Proof. A signed graph is balanced if and only if its signed double cover is disconnected; for the $\mathbb{Z}/2$ -voltage double cover this is Reiner and Tseng, Propositions 11.4 and 11.5 [189]. The underlying multigraph of D_k is that double cover, by Definition 32.2.4. The companion note proves that D_k is connected exactly when S carries an inverted repeat of length at least k , the two components otherwise being the plus-strand and minus-strand halves. $\square$

For ϕ X174 the longest inverted repeat has length 12, realised by ATTAAGCTCATT at position 441 against its reverse complement at 1448. The script confirms the trichotomy at $k = 9, 11, 13, 15$: unbalanced and connected at 9 and 11, balanced and disconnected at 13 and 15. Check (B).

Theorem 32.3.1 is worth stating because it converts a purely combinatorial dichotomy into a statement about the sequence. A genome with no inverted repeat as long as k gives a balanced Σ_k ; balanced means switching-equivalent to all-positive, which means the signs carry no information, which means the whole construction has collapsed back to the single-strand one. The sign structure is *exactly* the record of inverted repeats, nothing more and nothing less.

32.4 The covering theory that does exist

Reiner and Tseng define the critical group of a signed graph not as $\text{coker } \Delta_\Sigma$ but as $\mathcal{K}(\Sigma) = \text{im } \partial / \text{im } \Delta_\Sigma$, where ∂ is Zaslavsky's signed incidence matrix [189, eq. (1.3), §9]. For a connected unbalanced Σ one has $\text{im } \partial = \{x \in \mathbb{Z}^V : \sum_v x_v \text{ even}\}$, an index-two sublattice of $\mathbb{Z}^V$, so $\mathcal{K}(\Sigma)$ is an index-two subgroup of $\text{coker } \Delta_\Sigma$; ignoring the distinction doubles every order and corrupts the 2-primary part. (The competing convention of Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [186] deletes a sink row and column instead, which is the chip-firing-natural choice but a different group.) We use the Reiner–Tseng convention, computed from the explicit basis $2e_0, e_0 + e_1, \dots, e_0 + e_{n-1}$ of $\text{im } \partial$.

Theorem 32.4.1 (Reiner–Tseng [189], Thm. 1.2, verified here). *For every odd k with Σ_k unbalanced there is a short exact sequence*

$$0 \longrightarrow \mathcal{K}(\Sigma_k) \longrightarrow \mathcal{K}(|D_k|) \longrightarrow \mathcal{K}(|\Sigma_k|) \longrightarrow 0,$$

split on p -primary components for odd p and in general not split at $p = 2$.

The ϕ X174 instance at $k = 11$ is a textbook illustration:

$$\begin{aligned} \mathcal{K}(\Sigma_{11}) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481, & \text{order } 16\,817\,924, \\ \mathcal{K}(|D_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757 \oplus \mathbb{Z}/4204481, & \text{order } 19\,738\,255\,595\,056, \\ \mathcal{K}(|\Sigma_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757, & \text{order } 1\,173\,644. \end{aligned}$$

The orders multiply exactly. The odd parts split visibly: 4204481 comes from the signed graph, $23 \cdot 12757$ from the base. The 2-primary parts do *not*: an extension of $(\mathbb{Z}/2)^2$ by $(\mathbb{Z}/2)^2$ has produced $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4$, of the right order 16 and the wrong isomorphism type. This is precisely the failure Reiner and Tseng exhibit with $0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 0$, arising here from a real bacteriophage. Check (R).

32.5 The deck involution is an anti-automorphism

Theorem 32.4.1 is a statement about $|D_k|$, the undirected shadow. The quantity the single-strand theory actually computes is $\mathcal{K}(D_k)$, the sandpile group of the *digraph*, because that is what the BEST theorem multiplies by $\prod_v (d^+(v) - 1)!$ to count reconstructions. The following says the two are unrelated, and says why.

Theorem 32.5.1. *Let $\Delta = \Delta(D_k)$ be the out-degree Laplacian of D_k and P the permutation matrix of ρ . Then*

$$P \Delta P = \Delta^\top,$$

and $P \Delta P \neq \Delta$ whenever D_k has an arc that is not part of a ρ -symmetric digon. Consequently Δ does not commute with P , the ± 1 eigenlattices of P are not Δ -stable, and there is no decomposition of Δ over $\mathbb{Z}$, or over $\mathbb{Z}[1/2]$, or over any ring, into a part on the invariants and a part on the anti-invariants.

Proof. ρ is a bijection from the arc set of D_k to itself carrying $u \rightarrow v$ to $\bar{v} \rightarrow \bar{u}$. Hence the adjacency matrix satisfies $A_{\bar{v}\bar{u}} = A_{uv}$, that is $PAP = A^\top$, and $d^+(\bar{v}) = d^-(v) = d^+(v)$ since D_k is balanced, so $PDP = D$. Subtracting gives $P \Delta P = \Delta^\top$. If $P \Delta P$ were also Δ then $\Delta = \Delta^\top$, forcing $A = A^\top$, i.e. every arc paired with a reverse arc of the same multiplicity.

For the consequence: a decomposition along the eigenlattices of P requires $\Delta P = P\Delta$. Here $\Delta P = P\Delta^\top$, and $\Delta^\top \neq \Delta$. $\square$

This is the structural reason the companion note's obstruction was not an accident of its proof. The classical covering-graph factorisation — of Bilu and Linial [184], in the form used for signed graphs — decomposes the Laplacian of a double cover into an unsigned and a signed part on the ± 1 eigenspaces of the deck transformation. That decomposition needs the deck transformation to *commute* with the Laplacian. Here it conjugates the Laplacian to its transpose instead, and the two eigenlattices are exchanged rather than preserved.

The numbers are unambiguous. For $\phi X174$ at $k = 11$,

$$|\mathcal{K}(D_{11})| = 25\,318\,912, \quad |\mathcal{K}(|D_{11}|)| = 19\,738\,255\,595\,056,$$

and the second is not a multiple of the first; at $k = 9$ the two orders are 124- and 205-digit integers, again with no divisibility. The directed group is not an extension of anything by $\mathcal{K}(\Sigma_k)$ and is not a quotient of anything by it. Check (N).

Remark 32.5.2 (What ρ gives instead: a symmetric linking form). The honest formulation is that ρ makes D_k into a digraph with a *duality*, not a digraph with a symmetry. Bowen–Franks duality (Paper VIII, Lem. 1.5 of the main series) says coker M and coker $M^\top$ are canonically dual for a square integer matrix M with finite cokernel, and Theorem 32.5.1 identifies $\Delta^\top$ with $P\Delta P$. Composing gives a perfect pairing on the sandpile group itself:

$$\langle u, v \rangle = (Pu)^\top \Delta^{-\top} v \bmod \mathbb{Z}, \quad \mathcal{K}(D_k) \times \mathcal{K}(D_k) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

It is well defined — if $u = \Delta^\top z$ then $Pu = P\Delta^\top P(Pz) = \Delta(Pz)$, so $\langle u, v \rangle = (Pz)^\top v \in \mathbb{Z}$ — and it is *symmetric*, not alternating: inverting $P\Delta P = \Delta^\top$ gives $P\Delta^{-\top} = \Delta^{-1}P$, whence $\langle u, v \rangle = u^\top P\Delta^{-\top} v = u^\top \Delta^{-1} P v = \langle v, u \rangle$. Verified exactly for $\phi X174$ at $k = 11$ and $k = 12$; check (D).

This matters for how one should look for a double-stranded count. A symmetric linking form has no Lagrangians in the symplectic sense; the relevant notion is a *metabolizer*, a subgroup L with $L = L^\perp$, and a metabolizer forces $|\mathcal{K}(D_k)| = |L|^2$. For $\phi X174$ that already fails: $|\mathcal{K}(D_{12})| = 33$ and $|\mathcal{K}(D_{11})| = 25\,318\,912$, neither a square. So whatever counts double-stranded reconstructions, it is not the order of a self-orthogonal subgroup of $\mathcal{K}(D_k)$, and the symmetry of the form — not merely its existence — is the first thing a candidate theory has to respect.

One further negative is worth recording, because it removes the most natural-looking formulation. The sandpile group is a torsor on the arborescences *converging* on a fixed root, and those are what the BEST theorem counts with. One would like to say that a double-stranded count is a sum over the ρ -orbits of arborescences. There are none: ρ reverses arcs, so it carries an arborescence converging on r to one *diverging* from $\rho(r)$, and the two families are disjoint. What ρ gives at tree level is therefore a *bijection*

$$\rho: \{\text{arborescences converging on } r\} \xrightarrow{\sim} \{\text{arborescences diverging from } \rho(r)\},$$

both families having $|\mathcal{K}(D_k)|$ elements — the tree-level shadow of the duality, not an action. Verified for $\phi X174$ at $k = 12$ at three roots, where each family has 33 members and the image lands in the diverging family every time and in the converging one never; check (D). A double-stranded theory must be built from that bijection.

32.6 Signed bundle chains do not contract

The single-strand computation for a bacterial genome is only possible because of the splitting theorem: a *chain* $C = (c_1, \dots, c_s)$ of vertices each of which sends all r of its out-arcs to the next splits off a factor $(\mathbb{Z}/r)^{s-1}$ and can be contracted, reducing *E. coli* at $k = 75$ from 28 579 vertices to a skeleton of 261. The question is whether this survives in Σ_k . It does not, and the reason is elementary enough to state exactly.

r	s	$\mathcal{K}(\Sigma)$	$\mathcal{K}(\Sigma/C)$	ratio	r^s
3	1	$(\mathbb{Z}/3)^2 \oplus \mathbb{Z}/8$	$\mathbb{Z}/2 \oplus \mathbb{Z}/9$	4.00	3
3	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^3 \oplus \mathbb{Z}/8$	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/9$	12.00	9
3	3	$(\mathbb{Z}/3)^3 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/9$	$\mathbb{Z}/2 \oplus \mathbb{Z}/9$	54.00	27
5	2	$(\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/5)^2 \oplus \mathbb{Z}/17$	$\mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/7$	30.36	25
5	3	$\mathbb{Z}/4 \oplus (\mathbb{Z}/5)^3 \oplus \mathbb{Z}/11$	$\mathbb{Z}/2 \oplus \mathbb{Z}/13$	211.54	125
7	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/7)^2 \oplus \mathbb{Z}/23$	$\mathbb{Z}/2 \oplus \mathbb{Z}/17$	66.29	49
7	3	$\mathbb{Z}/4 \oplus (\mathbb{Z}/7)^3 \oplus \mathbb{Z}/13$	$\mathbb{Z}/2 \oplus \mathbb{Z}/17$	524.59	343
11	2	$\mathbb{Z}/2 \oplus (\mathbb{Z}/11)^2 \oplus \mathbb{Z}/64$	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/29$	133.52	121

Table 32.2: An r -bundle chain of s internal vertices grafted into a fixed unbalanced base. The r -rank rises by exactly s in all eight rows — the $(\mathbb{Z}/r)^s$ of Theorem 32.6.1(a) is visible — but the order ratio never equals r^s , and is not even an integer in five rows. Check (C).

Theorem 32.6.1. *Let Σ be a signed graph containing a chain $C = (c_1, \dots, c_s)$: vertices of degree exactly $2r$, joined by r -bundles to c_{t-1} and c_{t+1} , where $c_0 = a$ and $c_{s+1} = b$ lie outside C . Switch so that every chain bundle is positive, and write $d_t = x_{c_t} - x_{c_{t+1}}$ for $t = 0, \dots, s$ in coker Δ_Σ , where x_v is the class of the basis vector at v . Then*

- (a) *the presentation columns at $c_1, \dots, c_s$ say exactly $r(d_{t-1} - d_t) = 0$ for $t = 1, \dots, s$; in particular the s second differences $\delta_t = x_{c_{t-1}} - 2x_{c_t} + x_{c_{t+1}}$ are killed by r , and $rd_0 = rd_1 = \dots = rd_s$;*
- (b) *the column at a in Σ and the column at a in the contraction Σ/C — which replaces C by a single r -bundle $a-b$ of the product sign — differ by exactly $sr d_0$.*

Consequently $\mathcal{K}(\Sigma) \not\cong (\mathbb{Z}/r)^s \oplus \mathcal{K}(\Sigma/C)$ in general.

Proof. (a) Δ_Σ is symmetric and its column at c_t has entry $2r$ at c_t and $-r$ at each of c_{t-1}, c_{t+1} , all other entries zero. The relation it imposes is $2r x_{c_t} - r x_{c_{t-1}} - r x_{c_{t+1}} = 0$, which is $-r(d_{t-1} - d_t) = 0$. Chaining the s relations gives $rd_0 = \dots = rd_s$.

(b) The column at a in Σ contains $-r x_{c_1}$ where the column at a in Σ/C contains $-r x_b$; all other entries agree, since $\deg a$ is r in both. The difference is $r(x_b - x_{c_1}) = -r \sum_{t=1}^s d_t$, and by (a) each rd_t equals rd_0 , so this is $-sr d_0$. $\square$

Contrast the directed case. There the column at c_t is $r(x_{c_t} - x_{c_{t+1}}) = rd_t$, which kills each *first* difference outright; the chain contributes $(\mathbb{Z}/r)^{s-1}$ as a direct summand and contracts cleanly. In the signed case only second differences die. The residual first difference d_0 is not torsion; it is coupled to the rest of the graph, and the coupling is the series law of electrical networks — $s+1$ bundles of conductance r in series behave like one bundle of conductance $r/(s+1)$, which is not an integer multiplicity, so there is no contracted signed graph to compare to.

Table 32.2 separates the two halves of the theorem. What survives contraction is the r -rank: the number of invariant factors divisible by r grows by exactly s , in every case tested. What does not survive is the order and the isomorphism type. This is a partial and negative data point for Question 5.5 of [186], which asks how $\mathcal{K}(\Sigma_\varphi)$ depends on $\mathcal{K}(|\Sigma|)$ as the signature varies and observes only that the number of invariant factors seems to be preserved.

Corollary 32.6.2. *The reduction that makes $\mathcal{K}(G_k)$ computable for a bacterial genome — contract every maximal chain, compute on the skeleton, multiply back — has no analogue for $\mathcal{K}(\Sigma_k)$.*

32.7 *E. coli* K-12: what is left of the seven operons

The construction does what was asked of it structurally. ρ is free at all three odd word lengths, Σ_k is a signed graph, and the seven operons appear as genuine 7-bundles: 4919 of them at $k = 31$, organised into

k	$ V(D_k) $	$ V(\Sigma_k) $	ρ -fixed	7-bundles	longest 7-chain	span (bp)	$\dim_{\mathbb{F}_7} \ker \Delta_{\Sigma_k}$
31	60 548	30 274	0	4919	698	728	7441
51	53 380	26 690	0	4352	678	728	6581
75	48 498	24 249	0	3814	654	728	5864

Table 32.3: *E. coli* K-12 MG1655, NC_000913.3, 4 641 652 bp, at the three odd word lengths of the single-strand study. The longest 7-bundle chain spans the same 728 base pairs at all three k — $698 + 31 - 1 = 678 + 51 - 1 = 654 + 75 - 1$ — which is the seven-copy block the companion note matched one-to-one onto the seven ribosomal operons. Check (E).

chains of which the longest has 698 vertices. That chain spans $698 + 31 - 1 = 728$ base pairs, and the corresponding numbers at $k = 51$ and $k = 75$ give 728 as well. The agreement with the companion note is literal and not merely numerical: working on D_{51} rather than on Σ_{51} , and identifying the chain’s seven occurrences against the RefSeq annotation, it reports a longest multiplicity-seven chain of $s = 678$ spanning 728 bp whose occurrences hit the seven **rrn** operons one each. The quotient neither shortens that chain nor merges it with its own reverse complement, so 678 appears again here. On the structural question — do the five plus-strand and two minus-strand copies become one seven-copy object in the quotient — the answer is yes.

On the quantitative question the answer is no, for the two reasons already proved. Corollary 32.6.2 says $\mathcal{K}(\Sigma_k)$ cannot be computed by contraction at this size, and Theorem 32.5.1 says that even if it could, it would not count reconstructions. What is computable, and what is the right residue of the operons, is the $\mathbb{F}_7$ -nullity of the signed Laplacian. The reason it is the right residue is immediate: modulo 7 a 7-bundle contributes $7 \equiv 0$ to the degree and $\pm 7 \equiv 0$ to the adjacency, so *every 7-bundle is invisible over $\mathbb{F}_7$* and every vertex interior to a 7-chain yields a zero row. The $\mathbb{F}_7$ -nullity therefore measures the total length of the seven-copy repeat structure, and decays from 7441 to 5864 as k rises from 31 to 75 and short seven-copy blocks fall below the word length. The corresponding single-strand figures are 3496, 3201 and 2996: roughly half, as one expects from a construction that cannot see five copies and two copies as seven.

32.8 Scope, provenance, and what is open

Scope. Everything proved here concerns the *quotient* Σ_k and its undirected shadow. Proposition 32.2.2, Lemma 32.2.5, Lemma 32.2.6, Theorem 32.5.1 and Theorem 32.6.1 are proved in full generality. Theorem 32.3.1 is a combination of Reiner–Tseng Prop. 11.4–11.5 with the connectivity criterion of the companion note. Theorem 32.4.1 is *not* ours: it is Reiner and Tseng’s Theorem 1.2, and what is new here is only that it is exhibited on a reverse-complement de Bruijn graph, including a non-split 2-primary extension arising from a real genome. Part (b) of Theorem 32.6.1 — that the r -rank gain is exactly s rather than merely at least s — is verified on eight synthetic pairs (r, s) and on the *E. coli* graphs, and is *not* proved.

Provenance. The signed-graph formalism, the signed incidence and Laplacian matrices, balance and switching are Zaslavsky’s [190]; the dictionary between bidirected graphs and oriented signed graphs is Edmonds’, as recorded in Zaslavsky’s glossary [191]. The critical group of a signed graph and the covering sequence used in Section 32.4 are Reiner and Tseng’s [189]; the competing rooted convention and the chip-firing model for signed graphs are due to Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [186], building on the invertible-matrix chip-firing of Guzmán and Klivans [187]. The spectral form of the double-cover decomposition, which Theorem 32.5.1 shows is unavailable here, is Bilu and Linial’s [184]. The bidirected-graph model of double-stranded assembly is Medvedev, Georgiou, Myers and Brudno’s [188]. Sandpile groups of ordinary de Bruijn digraphs are treated by Chan, Hollmann and Pasechnik [114]. We are not aware of prior work computing critical groups of reverse-complement de Bruijn graphs, signed or otherwise; a targeted search found none.

Open. Three things. *First*, Theorem 32.6.1(b) in general: prove that a signed r -bundle chain of s internal vertices raises the r -rank by exactly s , or find a counterexample. A proof would give a clean partial answer to Question 5.5 of [186]. *Second*, the series law suggests working with weighted signed graphs over $\mathbb{Z}[1/(s+1)]$, where the contraction does exist; whether the resulting group has any combinatorial meaning — whether it counts anything — is unknown to us. *Third*, and most important, Theorem 32.5.1 closes the quotient route to a double-stranded reconstruction count but does not close the problem. What survives is the symmetric linking form of Remark 32.5.2, and that is the structure a correct double-stranded theory should use. Two things are already known about it and both are constraints rather than encouragements: the form is symmetric, so there are no Lagrangians in the symplectic sense; and for $\phi X174$ the group order is not a square, so there is no metabolizer either. A count modulo strand exchange must therefore be a weighted sum over isotropic subgroups, or over the orbits of the form's isometry group, rather than the order of a single distinguished subgroup. We have not attempted it, and we note that the naive guess — that the answer is $\sqrt{|\mathcal{K}(D_k)|}$ — is already excluded.

Part VI

The state of the programme

Chapter 33

Solved, open, impossible

This chapter consolidates the closing remarks of the chapters. Verdicts are quoted, not re-argued; the chapter and statement numbers point to where the argument is.

Four of the open problems of the first edition of this list have since been settled, three of them affirmatively and one by a no-go theorem, and a fifth has been settled in a way that was not on the menu at all. Composite q is proved (Chapter 14); the rank- d unitarity mechanism is proved, and then made unconditional (Chapters 20, 21); the search for a cyclic class computing the L -invariant is closed by an obstruction (Chapter 15); and the bidirected sandpile theory asked for by the assembly chapters turns out not to exist, for a reason that is itself a theorem (Chapter 32).

33.1 Corrections made by later chapters to earlier ones

- The untwisted boundary type is $\text{III}_{1/q}$ outright, not a β -family: Chapter 5, Remark 1.6 corrects Chapter 2, Theorem 2.4(iii). The applied note of Chapter 23 sharpens this further: III_{1/q^2} when the graph is bipartite.
- The line-digraph sandpile recursion is $|\text{Jac}(Y_{k+1})| = q^{n_k(q-1)-1} |\text{Jac}(Y_k)|$: Chapter 5, Theorem 2.1 corrects the exponent 2^{n_k-1} of Chapter 3, Remark 5.2 (the two agree for $q = 2$). This recursion should be compared with Levine’s theorem on sandpile groups of line digraphs (J. Combin. Theory Ser. A 118 (2011)); the series flags the comparison and does not claim priority.
- The naive $\mathbb{Z}_p[[T]]$ conjecture for the congruence tower (Chapter 3, Conjecture 3.6(ii-b)) is withdrawn in the same chapter and in Chapter 5; three independent certificates against it are $\lambda = -1$ (Ch. 5), $A_k A_k^\top \neq A_k^\top A_k$ (Ch. 9), and the torsion-freeness of the p -part of the pro-module with $\mu > 0$ (Ch. 12).
- The shadow of a q -regular level needs $q - 1$ return arcs, not q : Chapter 12, Remark 1.4.
- The geodesic edge flow must be defined by “not in a common chamber”, not “not adjacent”: Chapter 19, Remark 1.2 corrects Chapter 17, Definition 3.1; the two agree only on flag complexes, and of the five test complexes only Y_{168} is a flag complex.
- The p -primary hypothesis in the tail-norm theorem was an artefact of the proof, not of the statement: Chapter 14, Theorem 14.2.1 removes it for every $q \geq 2$, because the rank of a line-digraph adjacency matrix is n over *every* field (Ch. 14, Lemma 14.1.1), which is what the counting argument needs.
- The invariant subspace left open at Chapter 20, Remark 4.2(a) does exist: Chapter 21, Corollary 21.3.3. The candidate intertwiner recorded there was the right one, and the identity it needs is a single flag count.
- The expectation, stated in the assembly chapters, that double-strandedness would be handled by a sandpile theory on a quotient graph is withdrawn: Chapter 32, Theorem 32.5.1. Chapter 31 had already shown that no new lemma is *needed*, by passing to the double cover instead; Chapter 32 shows that the quotient route is not merely unnecessary but unavailable.

- Cross-references between the original papers drifted as sections were renumbered (for instance “[V, Prob. 5.1]” is Problem 4.1 of Chapter 9; “[I, Prop. 1.7(iii)]” is Proposition 1.9(iii) of Chapter 2). The concordance and the chapter-wise numbering of this book are authoritative.
- The type III framing of “holographic failure” and the claim that the control positive-realization problem inherits the trace obstruction were both wrong and are retracted in the applied chapters themselves (Chapters 23, 25).

33.2 Proved

- The boundary channel is arithmetically blind: $K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$ (Ch. 2, Thm. 3.7).
- The trace obstruction: no non-negative matrix, of any size, is shift equivalent to the Ihara companion (Ch. 3, Thms. 1.2, 1.5).
- The shadow realization: $K_0(\mathcal{O}_{B(Y)}) \cong \mathbb{Z} \oplus \text{Jac}(Y)$, $K_1 \cong \mathbb{Z}$, functorially and at every level of the tower (Ch. 4, Thm. 2.1; Ch. 12, Thm. 1.3); the signed-shadow lemma for any integer matrix with positive diagonal (Ch. 16, Lem. 5.1).
- The abelian limit module $\Lambda^n/D(1+T)$ with exact control (Ch. 6, Thm. 2.1); the exceptional zero of order two and the formula $\frac{1}{2}\Theta''(0) = -\kappa(Y_0)\|h_\alpha\|^2$ (Ch. 7, Thm. 1.2); the five faces of the L -invariant (Ch. 8, Cor. 7.2).
- The degeneracy calculus and the Euler rank of the degeneracy complex (Ch. 9, Prop. 1.2, Thm. 2.1); the tail norm and its pro-module (Ch. 9, Thm. 3.1); vanishing of the upper Hecke operator on the norm kernel and the snake factorization (Ch. 10, Lem. 1.1, Thm. 2.1); the Eisenstein boundary sequence $0 \rightarrow \mathbb{Z}/n_k \rightarrow \text{coker } \Delta_k^0 \rightarrow \text{Jac}(Y_k) \rightarrow 0$ and its non-splitting on K_4 (Ch. 11, Thm. 3.2).
- **For every** $q \geq 2$, $\ker(\tau_*| \text{Jac}(Y_{k+1})) = \text{Jac}(Y_{k+1})[q] \cong (\mathbb{Z}/q)^{r_k}$, and every invariant factor divisible by a prime $p \mid q$ is divisible by $p^{v_p(q)}$ (Ch. 14, Thm. 14.2.1); hence the pro-module has no q -torsion (Ch. 14, Cor. 14.2.4) and the sequence $0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0(\mathcal{O}_\infty) \rightarrow \mathbb{Z}[1/q] \rightarrow 0$ splits for every q (Ch. 14, Thm. 14.3.1). The p -prime case is Ch. 12, Thm. 2.8; the inductive limit is Kirchberg with $\text{Tor } K_0$ dual to the pro-module (Ch. 12, Thms. 3.2, 3.3); the head correspondence with K_0 -index η_* and the Hecke round trips (Ch. 13, Thms. 2.3, 3.1, 4.1).
- In rank two: no cubic trace obstruction (Ch. 16, Thm. 4.2); $b_1 = 0$ for thick quotients, hence no abelian towers (Ch. 18, Thm. 1.1); the covering norm theorem and the incidence calculus (Ch. 18, Thms. 2.1, 3.1); the edge-flow intertwiner $L_E \Psi = \Psi C'$ (Ch. 17, Thm. 3.3); the two-sided reduction $\ell^2(E) = \text{im } \Psi \oplus W$ and the unitarity of the remainder, $RR^\top = qI$ (Ch. 19, Thms. 2.1, 2.2, 3.1).
- **In every rank** $d \geq 2$: the unitarity identity $L_E L_E^\top = q^{d-1}I + q^{d-1}(q-1)T^\top T$ and its dual (Ch. 20, Thm. 20.2.1), whose whole content is that two hyperplanes of $\text{PG}(d, q)$ always meet in codimension two; exactly $|E| - n$ singular values equal $q^{(d-1)/2}$ and $L_E : \ker S \rightarrow \ker T$ is a bijective isometry (Ch. 20, Cor. 20.2.4). The invariant subspace that makes the critical circle unconditional is supplied in Ch. 21: the number of complete flags of $\text{PG}(d, q)$ with $F_{i+1} = V$ and F_1 inside a hyperplane H takes exactly two values according as $V \subseteq H$ or not (Thm. 21.2.1), which is the intertwining $L_E N_i^\top = q^i(N_0^\top A_{i+1} - N_{i+1}^\top)$ (Thm. 21.3.2); hence $W_d = (\text{im } \Psi_d)^\perp$ is two-sided invariant and every eigenvalue of $L_E|_{W_d}$ has modulus $q^{(d-1)/2}$ with no hypothesis whatever (Cor. 21.3.3). The companion matrix is explicit in every rank and its characteristic polynomial is the local Hecke polynomial of PGL_{d+1} (Thm. 21.4.1).
- In the applications: the bundle-chain splitting theorem $\mathcal{K}(G) \cong \bigoplus (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk})$ and the canonicity of the skeleton (Ch. 28); the two-family theorem $\mathcal{K}(D_w) = \mathbb{Z}/c$ (Ch. 29); the power-sum bound on hidden eigenvalues (Ch. 25); the sign of λ as a normality certificate (Ch. 24).
- In double-stranded assembly: the reverse-complement double cover is balanced, so the whole single-strand apparatus applies to it verbatim and the five plus-strand and two minus-strand ribosomal operons of *E. coli* merge into one seven-copy family (Ch. 31, Thm. 31.3.1, Cor. 31.3.2); the quotient is

a signed graph exactly when no repeated k -mer is its own reverse complement, which is automatic for odd k (Ch. 32, Prop. 32.2.2); it is unbalanced exactly when the sequence carries an inverted repeat of length at least k (Thm. 32.3.1); and the Reiner–Tseng covering sequence holds on it, with a non-split 2-primary part arising from a real bacteriophage (Thm. 32.4.1).

33.3 Verified computationally, not proved

The parahoric factorization on Y_{168} (Ch. 17, Cor. 3.5) was numeric before Ch. 19 proved it in general; the Ramanujan property of the five $\tilde{A}_2$ test complexes is numeric to 10^{-6} (Ch. 16, Thm. 4.5(i)); the character mechanism for kernel orders along finite $\tilde{A}_2$ chains is verified to 10^{-6} (Ch. 18, Thm. 2.1(iii)); finiteness of $K_0(\mathcal{O}_{L_E})$ rests on $1 \notin \text{spec } L_E$ on Y_{168} (Ch. 17, Thm. 3.7); the non-commutativity certificate $A_k A_k^\top \neq A_k^\top A_k$ is checked on K_4 only (Ch. 9, Prop. 3.4); the exhaustive statements about chord diagrams hold for $m \leq 5$ (Ch. 28, Rem. 9) and are known to fail at $m = 6$.

Two items are new. The corank of Ψ_d is at least $d(d+1)$ by Ch. 21, Theorem 21.4.2, and equals it on all three thick $\tilde{A}_2$ complexes tested; equality for $d \geq 3$ is unproved, and with it the exact value of $\dim W_d$. And a signed r -bundle chain of s internal vertices raises the r -rank by exactly s in every case tested (Ch. 32, Thm. 32.6.1); only the inequality is proved.

33.4 Open

1. **The main conjecture for the degeneracy algebra.** Construct, from L -values alone, an element $L_H \in K_1(H_\Sigma)$ equal to the characteristic class $[\Theta_H]$ modulo the image of $K_1(H)$ (Ch. 11, Rem. 5.1); before that, determine the image of $K_1(H)$, i.e. how non-unique the class is, and whether $H \rightarrow H_\Sigma$ is stably flat (Ch. 11, Rem. 5.2). This is Problem 4.1 of Ch. 9 in its final form, and it is now the oldest unsettled question in the programme.
2. **Rank two realization.** A non-negative integer matrix shift equivalent over $\mathbb{Z}$ to the graded companion $C^{(3)}$ — an instance of the weak Generalized Spectral Conjecture of Boyle–Handelman (Ch. 17, Rem. 2.4).
3. **Congruence towers in rank two.** The growth law of Jac^H along a genuine infinite congruence tower of $\tilde{A}_2$ complexes; no second level has been computed (Ch. 18, Rem. 6.1).
4. **The size of the unitary part.** Whether $\text{corank } \Psi_d = d(d+1)$ for $d \geq 3$, and whether W_d is the *largest* two-sided invariant subspace inside $\ker S \cap \ker T$ (Ch. 21, Rem. 21.5.2). Also a thick $\tilde{A}_3$ complex small enough to compute with, and an exact certificate of the vertex Ramanujan property of the test complexes (Ch. 19, Rem. 6.1(c)).
5. **Counting double-stranded reconstructions.** The reverse-complement involution induces a perfect *symmetric* linking form on $\mathcal{K}(D_k)$ (Ch. 32, Rem. 32.5.2). At tree level it is a bijection from the arborescences converging on a root to those diverging from the image root, not an action, so there are no orbits to sum over; and for $\phi X174$ the group order is not a square, so there is no metabolizer either. What a double-stranded count is built from is open.
6. **Non-pairwise arrangement invariants.** Any presentation of the arrangement group for repeats with three or more copies that is not built from pairwise data (Ch. 29, Open (a)); and the even- k case of the quotient, which is a signed graph with half-edges branched over the palindromic repeated k -mers (Ch. 32, Rem. 32.2.3).

33.5 Impossible

No non-negative matrix or dilation realizes the Ihara companion; no Hecke-dynamical C^* -algebra carries $\mathbb{Z} \oplus \text{Jac}(Y)$ as K_0 with the Ihara flow as gauge dynamics; no separable C^* -algebra has $K_0 \supseteq \mathfrak{K}_\infty$; no $\mathbb{Z}_p[[T]]$ -module produces $\lambda = -1$; no unital $*$ -homomorphism $\mathcal{O}_{k+1} \rightarrow \mathcal{O}_k$ induces η_* ; no cyclic cocycle on

the commutative Hecke–Dirac triple computes the L -invariant; no L -invariant exists for the Iwahori tower; no $\mathbb{Z}_p$ -tower of thick $\tilde{A}_2$ complexes exists; no cubic trace obstruction exists; no invariant of the unsigned chord-crossing graph computes the arrangement group of two-copy repeats, and no pairwise invariant at all computes it for three-copy repeats; no coset structure underlies long-read constraints in the reconstruction torsor.

Two entries are added here, and both were open problems on the previous list.

No odd cyclic class computes the graph L -invariant by a linear construction. The crossed product proposed for the job is Morita-trivial — the deck action on a connected $\mathbb{Z}$ -cover is free, so $C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z} \cong \bigoplus_V \mathcal{K}$ and $K_1 = 0$ (Ch. 15, Prop. 15.1.1). In the Floquet commutant, where K_1 is not zero, the natural element does not exist, because the pencil degenerates exactly at the point one wants to evaluate at; what the odd pairing reads there is the *order* of the exceptional zero, which is two for every voltage class, whereas $\kappa_0 \|h_\alpha\|^2$ is its leading *coefficient* (Ch. 15, Prop. 15.2.1, Cor. 15.2.2). And in any algebra whose K_1 has rank one, a pairing linear in each slot is a product of two linear forms and therefore vanishes on a subspace, while $\kappa_0 \|h_\alpha\|^2$ is positive definite; so no such construction can work once $b_1 \geq 2$ (Ch. 15, Thm. 15.3.1). What is *not* excluded is an algebra with rank $K_1 \geq b_1$: the reduced group C^* -algebra of $\pi_1(Y)$ is free of rank b_1 and sits exactly at that threshold.

The directed sandpile group of a reverse-complement de Bruijn graph has no covering theory, and its signed quotient does not contract. With P the permutation matrix of the involution, $P\Delta P = \Delta^\top$ and never Δ : the deck involution is an anti-automorphism of the Laplacian, so the ± 1 eigenlattices are not Δ -stable and the classical covering factorization is unavailable (Ch. 32, Thm. 32.5.1). The Reiner–Tseng sequence that does hold computes an invariant of the undirected shadow, which by the matrix-tree theorem counts spanning trees; the quantity of interest counts Eulerian circuits and lives on the digraph. Separately, a signed r -bundle chain obeys the series law — the presentation column at the chain’s endpoint differs from the contracted graph by $sr d_0$ — so $\mathcal{K}(\Sigma) \not\cong (\mathbb{Z}/r)^s \oplus \mathcal{K}(\Sigma/C)$, and the chain contraction that makes a bacterial genome computable on one strand has no bidirected analogue (Ch. 32, Thm. 32.6.1, Cor. 32.6.2).

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